WORLD AS A SYSTEM

1000 Reasoning Problems in Geography, History, Civics, and Environmental Studies

Grades 1–4 • Final English Edition

The child is not asked to know hidden facts.
The child is asked to reason from explicit facts, rules, relationships, and constraints.

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How to Use This Book

Purpose. Each problem is a small model of the world. It introduces the background concept needed for the task, states all facts and rules required for solution, and then asks the learner to derive a conclusion. Memory matters for vocabulary and concepts, but the central skill is reasoning.

Problem structure. Every problem contains: Knowledge context → Given facts → Rules → Task → Step-by-step solution → Answer → Knowledge takeaway → Formal model. The final line shows how the same problem could be represented symbolically in code.

Progression. Grade 1 uses short chains, visible constraints, and concrete relations. Grade 2 combines several conditions and small calculations. Grade 3 adds incomplete information, conflicts, graphs, uncertainty, and required justification. Grade 4 adds optimization, robustness, counterfactuals, multi-constraint decisions, and explicit reasoning about evidence.

Design principle. No solution depends on an unstated historical date, obscure place name, or political fact. Real-world concepts are taught inside the problem and then used. The aim is to transfer knowledge by reasoning with it.

Answer discipline. A correct choice without a reason is incomplete. Learners should be able to reproduce the chain from facts and rules to conclusion, and to say when the data are insufficient.

Map of the 50 Reasoning Families

Code	Domain / family	Core reasoning
G1	Positions and orientation	vector composition
G2	Roads and routes	shortest path & constraints
G3	Rivers: upstream and downstream	directed reachability
G4	Neighbours and borders	adjacency graph
G5	Maps, legends, and clues	set intersection
G6	Choosing a settlement site	constraint + score
G7	Relief and movement cost	weighted path
G8	Climate from data	threshold classification
G9	Exchange, routes, and capacities	flow & bottleneck
G10	Regions, sets, and membership	hierarchical transitivity
H1	Ordering events	partial order
H2	Dates, durations, and centuries	date arithmetic
H3	Sources and supported claims	claim–evidence matching
H4	Conflicting sources	source comparison
H5	Causes, conditions, and consequences	forward chaining
H6	Travel through time and space	time-distance arithmetic
H7	Change and continuity	set comparison
H8	Layers and dating objects	relative dating
C1	Rules and exceptions	exception logic
C2	Rights as limits on decisions	rights + procedure
C3	Voting, majority, and quorum	threshold voting
C4	Representation and seats	apportionment
C5	Institutions and roles	role matching
C6	Public budgets and priorities	knapsack optimization
C7	Collective choice and preferences	preference aggregation
N1	Scale, proportion, and distance	proportion
N2	Coordinates and constraint-based location	coordinate CSP
N3	Earth, Sun, seasons, and local time	time-zone/season rules
N4	Stocks, flows, and conservation	conservation
N5	Density, capacity, and crowding	density ratio
N6	Land use and spatial compatibility	spatial CSP

N7	Natural risk: hazard, exposure, protection	risk function
N8	Ecosystems and indirect effects	dependency propagation
N9	Renewable resources and sustainable use	renewal balance
N10	Migration and population balance	population balance
N11	Time intervals and overlapping events	interval intersection
N12	Genealogy and generations	kinship graph
N13	Anachronisms and temporal compatibility	temporal compatibility
N14	Provenance and chain of custody	custody path
N15	Perspective, interest, and source credibility	credibility analysis
N16	Uncertain data, intervals, and limits	interval robustness
N17	Counterfactual reasoning	causal intervention
N18	Public goods and common contributions	threshold public good
N19	Common resources and avoiding depletion	commons balance
N20	Fairness: equal, proportional, and need-based	fair allocation
N21	Negotiation, coalitions, and agreement	coalition game
N22	Due process, evidence, and right of reply	procedural conjunction
N23	Samples, surveys, and public opinion	sampling bias
N24	Matching, scheduling, and coverage	bipartite matching
N25	Meta-reasoning: robustness, information, causality	meta-reasoning

Editorial Review of the Romanian Draft and Corrections in This English Edition

The Romanian draft did reach the numbering 1–1000, but it relied too heavily on repeated family templates and sometimes varied only names or numerical values. This English edition keeps the intended curriculum coverage while repairing the main weakness: every problem instance is generated from a complete data model, the answer is recomputed from that model, and the learner is required to identify the logical operation that justifies the result. The second half broadens both knowledge and reasoning with scale, coordinates, Earth–Sun relations, conservation, density, spatial compatibility, risk, ecosystems, sustainability, migration, temporal intervals, genealogy, anachronism, provenance, uncertainty, counterfactuals, public goods, commons, fairness, coalitions, due process, sampling, matching, and meta-reasoning.

GRADE 1

Difficulty increases through the number of relations and reasoning steps, not through hidden knowledge.

Part A — Foundational reasoning about the world

G1. Positions and orientation

Problem 1. Position chain from Cedar to Alder

Knowledge context. A map can be modeled with relative positions. When several moves are given, their effects can be combined rather than memorized as a picture.

Given facts. Pine is 1 map unit(s) east of Cedar; Alder is 1 map unit(s) east of Pine.

Rules. Treat east as +x, west as −x, north as +y, and south as −y. Add the moves in order. A fact that does not affect any coordinate is irrelevant.

Task. Where is Alder relative to Cedar? What is the straight grid displacement measured as $|x|+|y|$?

Step-by-step solution. 1. Start Cedar at (0,0). 2. Translate each relation into a coordinate move. 3. Adding the moves gives (2,0). 4. That coordinate is east of Cedar; the grid displacement is $|2|+|0|=2$.

Answer. Alder is east of Cedar; grid displacement 2 unit(s).

Knowledge takeaway. Cardinal directions become easier to reason about when they are treated as relations that can be composed.

Formal model. Grid coordinates; each statement adds a vector; sum the vectors and classify the final coordinate.

Problem 2. Position chain from Iris to Pine

Knowledge context. A map can be modeled with relative positions. When several moves are given, their effects can be combined rather than memorized as a picture.

Given facts. Fern is 1 map unit(s) north of Iris; Pine is 1 map unit(s) north of Fern.

Rules. Treat east as +x, west as −x, north as +y, and south as −y. Add the moves in order. A fact that does not affect any coordinate is irrelevant.

Task. Where is Pine relative to Iris? What is the straight grid displacement measured as $|x|+|y|$?

Step-by-step solution. 1. Start Iris at (0,0). 2. Translate each relation into a coordinate move. 3. Adding the moves gives (0,2). 4. That coordinate is north of Iris; the grid displacement is $|0|+|2|=2$.

Answer. Pine is north of Iris; grid displacement 2 unit(s).

Knowledge takeaway. Cardinal directions become easier to reason about when they are treated as relations that can be composed.

Formal model. Grid coordinates; each statement adds a vector; sum the vectors and classify the final coordinate.

Problem 3. Position chain from Northfield to Cedar

Knowledge context. A map can be modeled with relative positions. When several moves are given, their effects can be combined rather than memorized as a picture.

Given facts. Juniper is 1 map unit(s) east of Northfield; Cedar is 1 map unit(s) north of Juniper.

Rules. Treat east as +x, west as −x, north as +y, and south as −y. Add the moves in order. A fact that does not affect any coordinate is irrelevant.

Task. Where is Cedar relative to Northfield? What is the straight grid displacement measured as $|x|+|y|$?

Step-by-step solution. 1. Start Northfield at (0,0). 2. Translate each relation into a coordinate move. 3. Adding the moves gives (1,1). 4. That coordinate is north-east of Northfield; the grid displacement is $|1|+|1|=2$.

Answer. Cedar is north-east of Northfield; grid displacement 2 unit(s).

Knowledge takeaway. Cardinal directions become easier to reason about when they are treated as relations that can be composed.

Formal model. Grid coordinates; each statement adds a vector; sum the vectors and classify the final coordinate.

Problem 4. Position chain from Linden to Willow

Knowledge context. A map can be modeled with relative positions. When several moves are given, their effects can be combined rather than memorized as a picture.

Given facts. Yarrow is 1 map unit(s) west of Linden; Willow is 1 map unit(s) south of Yarrow.

Rules. Treat east as +x, west as −x, north as +y, and south as −y. Add the moves in order. A fact that does not affect any coordinate is irrelevant.

Task. Where is Willow relative to Linden? What is the straight grid displacement measured as |x|+|y|?

Step-by-step solution. 1. Start Linden at (0,0). 2. Translate each relation into a coordinate move. 3. Adding the moves gives (-1,-1). 4. That coordinate is south-west of Linden; the grid displacement is |-1|+|-1|=2.

Answer. Willow is south-west of Linden; grid displacement 2 unit(s).

Knowledge takeaway. Cardinal directions become easier to reason about when they are treated as relations that can be composed.

Formal model. Grid coordinates; each statement adds a vector; sum the vectors and classify the final coordinate.

Problem 5. Position chain from Birch to Juniper

Knowledge context. A map can be modeled with relative positions. When several moves are given, their effects can be combined rather than memorized as a picture.

Given facts. Kite is 1 map unit(s) north of Birch; Alder is 1 map unit(s) east of Kite; Juniper is 1 map unit(s) south of Alder.

Rules. Treat east as +x, west as −x, north as +y, and south as −y. Add the moves in order. A fact that does not affect any coordinate is irrelevant.

Task. Where is Juniper relative to Birch? What is the straight grid displacement measured as |x|+|y|?

Step-by-step solution. 1. Start Birch at (0,0). 2. Translate each relation into a coordinate move. 3. Adding the moves gives (1,0). 4. That coordinate is east of Birch; the grid displacement is |1|+|0|=1.

Answer. Juniper is east of Birch; grid displacement 1 unit(s).

Knowledge takeaway. Cardinal directions become easier to reason about when they are treated as relations that can be composed.

Formal model. Grid coordinates; each statement adds a vector; sum the vectors and classify the final coordinate.

G2. Roads and routes

Problem 6. The best route from Willow to Iris

Knowledge context. A road map can be modeled as a graph. The best route depends on the stated criterion, not on how short it looks on paper.

Given facts. Willow–Kite–Iris has segment costs [3, 4] (total 7); Willow–Oak–Northfield–Iris has segment costs [2, 2, 3] (total 7); Willow–Cedar–Iris has segment costs [6, 1] (total 7).

Rules. A route is feasible only if all of its roads are open. The cost of a route is the sum of its segment costs. Choose the feasible route with the smallest total.

Task. Which route should a traveler use from Willow to Iris, and why?

Step-by-step solution. 1. List the candidate routes and their segment costs. 2. All listed routes are open. 3. Add the segment costs for every feasible route. 4. The smallest feasible total is 7, on Willow–Kite–Iris.

Answer. Use Willow–Kite–Iris; total cost 7.

Knowledge takeaway. A shortest path is a reasoning result: first respect constraints, then compare the remaining costs.

Formal model. Weighted graph; delete forbidden edges; minimize the sum of edge weights.

Problem 7. The best route from Kite to Dover

Knowledge context. A road map can be modeled as a graph. The best route depends on the stated criterion, not on how short it looks on paper.

Given facts. Kite–Meadow–Dover has segment costs [4, 5] (total 9); Kite–Pine–Iris–Dover has segment costs [3, 3, 3] (total 9); Kite–Quartz–Dover has segment costs [7, 2] (total 9).

Rules. A route is feasible only if all of its roads are open. The cost of a route is the sum of its segment costs. Choose the feasible route with the smallest total.

Task. Which route should a traveler use from Kite to Dover, and why?

Step-by-step solution. 1. List the candidate routes and their segment costs. 2. All listed routes are open. 3. Add the segment costs for every feasible route. 4. The smallest feasible total is 9, on Kite–Meadow–Dover.

Answer. Use Kite–Meadow–Dover; total cost 9.

Knowledge takeaway. A shortest path is a reasoning result: first respect constraints, then compare the remaining costs.

Formal model. Weighted graph; delete forbidden edges; minimize the sum of edge weights.

Problem 8. The best route from Alder to Oak

Knowledge context. A road map can be modeled as a graph. The best route depends on the stated criterion, not on how short it looks on paper.

Given facts. Alder–Northfield–Oak has segment costs [5, 6] (total 11); Alder–Stone–Meadow–Oak has segment costs [4, 4, 3] (total 11); Alder–Willow–Oak has segment costs [8, 3] (total 11).

Rules. A route is feasible only if all of its roads are open. The cost of a route is the sum of its segment costs. Choose the feasible route with the smallest total.

Task. Which route should a traveler use from Alder to Oak, and why?

Step-by-step solution. 1. List the candidate routes and their segment costs. 2. All listed routes are open. 3. Add the segment costs for every feasible route. 4. The smallest feasible total is 11, on Alder–Northfield–Oak.

Answer. Use Alder–Northfield–Oak; total cost 11.

Knowledge takeaway. A shortest path is a reasoning result: first respect constraints, then compare the remaining costs.

Formal model. Weighted graph; delete forbidden edges; minimize the sum of edge weights.

Problem 9. The best route from Fern to Elm

Knowledge context. A road map can be modeled as a graph. The best route depends on the stated criterion, not on how short it looks on paper.

Given facts. Fern–Alder–Elm has segment costs [6, 7] (total 13); Fern–Cedar–Harbor–Elm has segment costs [5, 5, 3] (total 13); Fern–Grove–Elm has segment costs [9, 4] (total 13).

Rules. A route is feasible only if all of its roads are open. The cost of a route is the sum of its segment costs. Choose the feasible route with the smallest total.

Task. Which route should a traveler use from Fern to Elm, and why?

Step-by-step solution. 1. List the candidate routes and their segment costs. 2. All listed routes are open. 3. Add the segment costs for every feasible route. 4. The smallest feasible total is 13, on Fern–Alder–Elm.

Answer. Use Fern–Alder–Elm; total cost 13.

Knowledge takeaway. A shortest path is a reasoning result: first respect constraints, then compare the remaining costs.

Formal model. Weighted graph; delete forbidden edges; minimize the sum of edge weights.

Problem 10. The best route from Riverbend to Fern

Knowledge context. A road map can be modeled as a graph. The best route depends on the stated criterion, not on how short it looks on paper.

Given facts. Riverbend–Juniper–Fern has segment costs [7, 8] (total 15); Riverbend–Grove–Harbor–Fern has segment costs [6, 6, 3] (total 15); Riverbend–Oak–Fern has segment costs [10, 5] (total 15).

Rules. A route is feasible only if all of its roads are open. The cost of a route is the sum of its segment costs. Choose the feasible route with the smallest total.

Task. Which route should a traveler use from Riverbend to Fern, and why?

Step-by-step solution. 1. List the candidate routes and their segment costs. 2. All listed routes are open. 3. Add the segment costs for every feasible route. 4. The smallest feasible total is 15, on Riverbend–Juniper–Fern.

Answer. Use Riverbend–Juniper–Fern; total cost 15.

Knowledge takeaway. A shortest path is a reasoning result: first respect constraints, then compare the remaining costs.

Formal model. Weighted graph; delete forbidden edges; minimize the sum of edge weights.

G3. Rivers: upstream and downstream

Problem 11. Following water from North Fork

Knowledge context. River systems form directed networks: water has a direction, tributaries join larger streams, and upstream/downstream are relational concepts.

Given facts. North Fork flows into Meadow River; Red Brook flows into Meadow River; Meadow River flows into Blue Creek.

Rules. If river X flows into river Y, X is upstream of Y. Upstream relations can be followed through several confluences. Downstream is the reverse direction of flow.

Task. Is North Fork upstream of Blue Creek? Can water from North Fork reach Blue Creek through the stated network?

Step-by-step solution. 1. Rewrite the facts as arrows: North Fork→Meadow River, Red Brook→Meadow River, Meadow River→Blue Creek. 2. Follow arrows starting from North Fork. 3. An arrow path reaches Blue Creek, so North Fork is upstream of Blue Creek.

Answer. Yes; water from North Fork can reach Blue Creek.

Knowledge takeaway. A river network can be reasoned about like any directed network; following the arrows is more reliable than guessing from names.

Formal model. Directed graph; edge X→Y means water can flow from X to Y; query reachability.

Problem 12. Following water from Pine River

Knowledge context. River systems form directed networks: water has a direction, tributaries join larger streams, and upstream/downstream are relational concepts.

Given facts. North Fork flows into Clearwater; Pine River flows into Clearwater; Clearwater flows into Meadow River.

Rules. If river X flows into river Y, X is upstream of Y. Upstream relations can be followed through several confluences. Downstream is the reverse direction of flow.

Task. Is Pine River upstream of Meadow River? Can water from Pine River reach Meadow River through the stated network?

Step-by-step solution. 1. Rewrite the facts as arrows: North Fork→Clearwater, Pine River→Clearwater, Clearwater→Meadow River. 2. Follow arrows starting from Pine River. 3. An arrow path reaches Meadow River, so Pine River is upstream of Meadow River.

Answer. Yes; water from Pine River can reach Meadow River.

Knowledge takeaway. A river network can be reasoned about like any directed network; following the arrows is more reliable than guessing from names.

Formal model. Directed graph; edge X→Y means water can flow from X to Y; query reachability.

Problem 13. Following water from Willow Stream

Knowledge context. River systems form directed networks: water has a direction, tributaries join larger streams, and upstream/downstream are relational concepts.

Given facts. Willow Stream flows into Blue Creek; Clearwater flows into Blue Creek; Blue Creek flows into Silver River.

Rules. If river X flows into river Y, X is upstream of Y. Upstream relations can be followed through several confluences. Downstream is the reverse direction of flow.

Task. Is Willow Stream upstream of Blue Creek? Can water from Willow Stream reach Blue Creek through the stated network?

Step-by-step solution. 1. Rewrite the facts as arrows: Willow Stream→Blue Creek, Clearwater→Blue Creek, Blue Creek→Silver River. 2. Follow arrows starting from Willow Stream. 3. An arrow path reaches Blue Creek, so Willow Stream is upstream of Blue Creek.

Answer. Yes; water from Willow Stream can reach Blue Creek.

Knowledge takeaway. A river network can be reasoned about like any directed network; following the arrows is more reliable than guessing from names.

Formal model. Directed graph; edge X→Y means water can flow from X to Y; query reachability.

Problem 14. Following water from Clearwater

Knowledge context. River systems form directed networks: water has a direction, tributaries join larger streams, and upstream/downstream are relational concepts.

Given facts. North Fork flows into Clearwater; Meadow River flows into Clearwater; Clearwater flows into Willow Stream.

Rules. If river X flows into river Y, X is upstream of Y. Upstream relations can be followed through several confluences. Downstream is the reverse direction of flow.

Task. Is Clearwater upstream of Willow Stream? Can water from Clearwater reach Willow Stream through the stated network?

Step-by-step solution. 1. Rewrite the facts as arrows: North Fork→Clearwater, Meadow River→Clearwater, Clearwater→Willow Stream. 2. Follow arrows starting from Clearwater. 3. An arrow path reaches Willow Stream, so Clearwater is upstream of Willow Stream.

Answer. Yes; water from Clearwater can reach Willow Stream.

Knowledge takeaway. A river network can be reasoned about like any directed network; following the arrows is more reliable than guessing from names.

Formal model. Directed graph; edge X→Y means water can flow from X to Y; query reachability.

Problem 15. Following water from North Fork

Knowledge context. River systems form directed networks: water has a direction, tributaries join larger streams, and upstream/downstream are relational concepts.

Given facts. Willow Stream flows into Clearwater; North Fork flows into Clearwater; Clearwater flows into Pine River.

Rules. If river X flows into river Y, X is upstream of Y. Upstream relations can be followed through several confluences. Downstream is the reverse direction of flow.

Task. Is North Fork upstream of Clearwater? Can water from North Fork reach Clearwater through the stated network?

Step-by-step solution. 1. Rewrite the facts as arrows: Willow Stream→Clearwater, North Fork→Clearwater, Clearwater→Pine River. 2. Follow arrows starting from North Fork. 3. An arrow path reaches Clearwater, so North Fork is upstream of Clearwater.

Answer. Yes; water from North Fork can reach Clearwater.

Knowledge takeaway. A river network can be reasoned about like any directed network; following the arrows is more reliable than guessing from names.

Formal model. Directed graph; edge X→Y means water can flow from X to Y; query reachability.

G4. Neighbours and borders

Problem 16. Crossing the fictional border network of Birch

Knowledge context. Political maps can be treated as adjacency networks. This lets us reason about border crossings without relying on memorized real-world geography.

Given facts. Countries are represented by the fictional states Birch, Cedar, Northfield, Pine, Alder, Willow. Shared borders: Birch–Cedar, Cedar–Northfield, Northfield–Pine, Birch–Alder, Alder–Willow, Willow–Pine.

Rules. Crossing one border corresponds to traversing one graph edge. A route may use only listed borders. A state is unavoidable only if every feasible route passes through it.

Task. What is the fewest number of border crossings needed to travel from Birch to Pine? Is Cedar necessarily crossed?

Step-by-step solution. 1. Turn each shared border into an undirected link. 2. A shortest-link search gives distance 3 from Birch to Pine. 3. Remove Cedar and check whether Pine is still reachable from Birch. 4. It is still reachable, so Cedar is not unavoidable.

Answer. 3 border crossing(s); Cedar is not unavoidable.

Knowledge takeaway. Neighbourhood on a map is a graph relation; shortest routes and unavoidable transit points are different questions.

Formal model. Undirected graph; shortest path length plus reachability after deleting a candidate vertex.

Problem 17. Crossing the fictional border network of Elm

Knowledge context. Political maps can be treated as adjacency networks. This lets us reason about border crossings without relying on memorized real-world geography.

Given facts. Countries are represented by the fictional states Elm, Iris, Alder, Yarrow, Cedar, Harbor. Shared borders: Elm–Iris, Iris–Alder, Alder–Yarrow, Elm–Cedar, Cedar–Harbor, Harbor–Yarrow.

Rules. Crossing one border corresponds to traversing one graph edge. A route may use only listed borders. A state is unavoidable only if every feasible route passes through it.

Task. What is the fewest number of border crossings needed to travel from Elm to Yarrow? Is Iris necessarily crossed?

Step-by-step solution. 1. Turn each shared border into an undirected link. 2. A shortest-link search gives distance 3 from Elm to Yarrow. 3. Remove Iris and check whether Yarrow is still reachable from Elm. 4. It is still reachable, so Iris is not unavoidable.

Answer. 3 border crossing(s); Iris is not unavoidable.

Knowledge takeaway. Neighbourhood on a map is a graph relation; shortest routes and unavoidable transit points are different questions.

Formal model. Undirected graph; shortest path length plus reachability after deleting a candidate vertex.

Problem 18. Crossing the fictional border network of Riverbend

Knowledge context. Political maps can be treated as adjacency networks. This lets us reason about border crossings without relying on memorized real-world geography.

Given facts. Countries are represented by the fictional states Riverbend, Kite, Fern, Oak, Cedar, Northfield. Shared borders: Riverbend–Kite, Kite–Fern, Fern–Oak, Riverbend–Cedar, Cedar–Northfield, Northfield–Oak.

Rules. Crossing one border corresponds to traversing one graph edge. A route may use only listed borders. A state is unavoidable only if every feasible route passes through it.

Task. What is the fewest number of border crossings needed to travel from Riverbend to Oak? Is Kite necessarily crossed?

Step-by-step solution. 1. Turn each shared border into an undirected link. 2. A shortest-link search gives distance 3 from Riverbend to Oak. 3. Remove Kite and check whether Oak is still reachable from Riverbend. 4. It is still reachable, so Kite is not unavoidable.

Answer. 3 border crossing(s); Kite is not unavoidable.

Knowledge takeaway. Neighbourhood on a map is a graph relation; shortest routes and unavoidable transit points are different questions.

Formal model. Undirected graph; shortest path length plus reachability after deleting a candidate vertex.

Problem 19. Crossing the fictional border network of Oak

Knowledge context. Political maps can be treated as adjacency networks. This lets us reason about border crossings without relying on memorized real-world geography.

Given facts. Countries are represented by the fictional states Oak, Juniper, Yarrow, Elm, Grove, Northfield. Shared borders: Oak–Juniper, Juniper–Yarrow, Yarrow–Elm, Oak–Grove, Grove–Northfield, Northfield–Elm.

Rules. Crossing one border corresponds to traversing one graph edge. A route may use only listed borders. A state is unavoidable only if every feasible route passes through it.

Task. What is the fewest number of border crossings needed to travel from Oak to Elm? Is Juniper necessarily crossed?

Step-by-step solution. 1. Turn each shared border into an undirected link. 2. A shortest-link search gives distance 3 from Oak to Elm. 3. Remove Juniper and check whether Elm is still reachable from Oak. 4. It is still reachable, so Juniper is not unavoidable.

Answer. 3 border crossing(s); Juniper is not unavoidable.

Knowledge takeaway. Neighbourhood on a map is a graph relation; shortest routes and unavoidable transit points are different questions.

Formal model. Undirected graph; shortest path length plus reachability after deleting a candidate vertex.

Problem 20. Crossing the fictional border network of Linden

Knowledge context. Political maps can be treated as adjacency networks. This lets us reason about border crossings without relying on memorized real-world geography.

Given facts. Countries are represented by the fictional states Linden, Cedar, Northfield, Juniper, Alder, Willow. Shared borders: Linden–Cedar, Cedar–Northfield, Northfield–Juniper, Linden–Alder, Alder–Willow, Willow–Juniper.

Rules. Crossing one border corresponds to traversing one graph edge. A route may use only listed borders. A state is unavoidable only if every feasible route passes through it.

Task. What is the fewest number of border crossings needed to travel from Linden to Juniper? Is Cedar necessarily crossed?

Step-by-step solution. 1. Turn each shared border into an undirected link. 2. A shortest-link search gives distance 3 from Linden to Juniper. 3. Remove Cedar and check whether Juniper is still reachable from Linden. 4. It is still reachable, so Cedar is not unavoidable.

Answer. 3 border crossing(s); Cedar is not unavoidable.

Knowledge takeaway. Neighbourhood on a map is a graph relation; shortest routes and unavoidable transit points are different questions.

Formal model. Undirected graph; shortest path length plus reachability after deleting a candidate vertex.

G5. Maps, legends, and clues

Problem 21. Reading a layered map legend (A1)

Knowledge context. A map legend translates symbols into properties. One location may have several properties at the same time.

Given facts. Legend: tree=forest, wavy line=river, double line=road, arch=bridge, square=village, triangle=hill. Cells: A1 has forest, road; A2 has bridge, river; B1 has forest, river, road; B2 has road, village.

Rules. A cell satisfies the task only if it has every required attribute. Extra attributes do not disqualify it unless the task says so.

Task. Which cell or cells satisfy all of these conditions: forest, road? Explain by intersecting the clues.

Step-by-step solution. 1. Convert the requested symbols into the attribute set ['forest', 'road']. 2. Check each cell for every required attribute. 3. The matching set is ['A1', 'B1'].

Answer. A1, B1

Knowledge takeaway. Map reading often means intersecting clues rather than recognizing one symbol in isolation.

Formal model. Set model; each cell has an attribute set; return cells whose set is a superset of the required set.

Problem 22. Reading a layered map legend (A2)

Knowledge context. A map legend translates symbols into properties. One location may have several properties at the same time.

Given facts. Legend: tree=forest, wavy line=river, double line=road, arch=bridge, square=village, triangle=hill. Cells: A1 has forest, road; A2 has bridge, river, village; B1 has forest, river, road; B2 has road, village.

Rules. A cell satisfies the task only if it has every required attribute. Extra attributes do not disqualify it unless the task says so.

Task. Which cell or cells satisfy all of these conditions: bridge, river? Explain by intersecting the clues.

Step-by-step solution. 1. Convert the requested symbols into the attribute set ['bridge', 'river']. 2. Check each cell for every required attribute. 3. The matching set is ['A2'].

Answer. A2

Knowledge takeaway. Map reading often means intersecting clues rather than recognizing one symbol in isolation.

Formal model. Set model; each cell has an attribute set; return cells whose set is a superset of the required set.

Problem 23. Reading a layered map legend (B1)

Knowledge context. A map legend translates symbols into properties. One location may have several properties at the same time.

Given facts. Legend: tree=forest, wavy line=river, double line=road, arch=bridge, square=village, triangle=hill. Cells: A1 has forest, road; A2 has bridge, river; B1 has forest, river, road; B2 has forest, road, village.

Rules. A cell satisfies the task only if it has every required attribute. Extra attributes do not disqualify it unless the task says so.

Task. Which cell or cells satisfy all of these conditions: forest, river? Explain by intersecting the clues.

Step-by-step solution. 1. Convert the requested symbols into the attribute set ['forest', 'river']. 2. Check each cell for every required attribute. 3. The matching set is ['B1'].

Answer. B1

Knowledge takeaway. Map reading often means intersecting clues rather than recognizing one symbol in isolation.

Formal model. Set model; each cell has an attribute set; return cells whose set is a superset of the required set.

Problem 24. Reading a layered map legend (B2)

Knowledge context. A map legend translates symbols into properties. One location may have several properties at the same time.

Given facts. Legend: tree=forest, wavy line=river, double line=road, arch=bridge, square=village, triangle=hill. Cells: A1 has forest, hill, road; A2 has bridge, river; B1 has forest, river, road; B2 has road, village.

Rules. A cell satisfies the task only if it has every required attribute. Extra attributes do not disqualify it unless the task says so.

Task. Which cell or cells satisfy all of these conditions: hill, road? Explain by intersecting the clues.

Step-by-step solution. 1. Convert the requested symbols into the attribute set ['hill', 'road']. 2. Check each cell for every required attribute. 3. The matching set is ['A1'].

Answer. A1

Knowledge takeaway. Map reading often means intersecting clues rather than recognizing one symbol in isolation.

Formal model. Set model; each cell has an attribute set; return cells whose set is a superset of the required set.

Problem 25. Reading a layered map legend (A1)

Knowledge context. A map legend translates symbols into properties. One location may have several properties at the same time.

Given facts. Legend: tree=forest, wavy line=river, double line=road, arch=bridge, square=village, triangle=hill. Cells: A1 has forest, road; A2 has bridge, river; B1 has bridge, forest, river, road; B2 has road, village.

Rules. A cell satisfies the task only if it has every required attribute. Extra attributes do not disqualify it unless the task says so.

Task. Which cell or cells satisfy all of these conditions: river, road? Explain by intersecting the clues.

Step-by-step solution. 1. Convert the requested symbols into the attribute set ['river', 'road']. 2. Check each cell for every required attribute. 3. The matching set is ['B1'].

Answer. B1

Knowledge takeaway. Map reading often means intersecting clues rather than recognizing one symbol in isolation.

Formal model. Set model; each cell has an attribute set; return cells whose set is a superset of the required set.

G6. Choosing a settlement site

Problem 26. Choosing a safe settlement site: case 1

Knowledge context. Human settlements depend on several factors at once. Some conditions may be compulsory while others are preferences.

Given facts. Candidate sites: Site A: water=yes, road=yes, flood-zone=no, nearby farmland score=2; Site B: water=yes, road=no, flood-zone=no, nearby farmland score=4; Site C: water=yes, road=yes, flood-zone=yes, nearby farmland score=4; Site D: water=no, road=yes, flood-zone=no, nearby farmland score=6.

Rules. A valid site must have water, road access, and must not be in the flood zone. Among valid sites, choose the one with the highest farmland score.

Task. Which site should be selected under the stated mandatory conditions and preference rule?

Step-by-step solution. 1. Separate mandatory conditions from the preference. 2. Filtering by water, road, and flood safety leaves ['Site A']. 3. Compare only the farmland scores of feasible sites: Site A=2. 4. Site A has the best score among valid sites.

Answer. Site A.

Knowledge takeaway. Good decisions first satisfy hard constraints and only then optimize among the remaining options.

Formal model. Constraint-satisfaction problem followed by a score maximization over feasible options.

Problem 27. Choosing a safe settlement site: case 2

Knowledge context. Human settlements depend on several factors at once. Some conditions may be compulsory while others are preferences.

Given facts. Candidate sites: Site A: water=yes, road=yes, flood-zone=no, nearby farmland score=3; Site B: water=yes, road=no, flood-zone=no, nearby farmland score=3; Site C: water=yes, road=yes, flood-zone=yes, nearby farmland score=5; Site D: water=no, road=yes, flood-zone=no, nearby farmland score=5.

Rules. A valid site must have water, road access, and must not be in the flood zone. Among valid sites, choose the one with the highest farmland score.

Task. Which site should be selected under the stated mandatory conditions and preference rule?

Step-by-step solution. 1. Separate mandatory conditions from the preference. 2. Filtering by water, road, and flood safety leaves ['Site A']. 3. Compare only the farmland scores of feasible sites: Site A=3. 4. Site A has the best score among valid sites.

Answer. Site A.

Knowledge takeaway. Good decisions first satisfy hard constraints and only then optimize among the remaining options.

Formal model. Constraint-satisfaction problem followed by a score maximization over feasible options.

Problem 28. Choosing a safe settlement site: case 3

Knowledge context. Human settlements depend on several factors at once. Some conditions may be compulsory while others are preferences.

Given facts. Candidate sites: Site A: water=yes, road=yes, flood-zone=no, nearby farmland score=2; Site B: water=yes, road=no, flood-zone=no, nearby farmland score=4; Site C: water=yes, road=yes, flood-zone=yes, nearby farmland score=4; Site D: water=no, road=yes, flood-zone=no, nearby farmland score=6. Cross-domain check: on a field sketch, 1 cm represents 5 km and two checkpoints are 2 cm apart.

Rules. A valid site must have water, road access, and must not be in the flood zone. Among valid sites, choose the one with the highest farmland score. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Which site should be selected under the stated mandatory conditions and preference rule? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. Separate mandatory conditions from the preference. 2. Filtering by water, road, and flood safety leaves ['Site A']. 3. Compare only the farmland scores of feasible sites: Site A=2. 4. Site A has the best score among valid sites. Cross-domain check: 2×5=10 km.

Answer. Site A. Cross-domain answer: 10 km.

Knowledge takeaway. Good decisions first satisfy hard constraints and only then optimize among the remaining options.

Formal model. Constraint-satisfaction problem followed by a score maximization over feasible options.

Problem 29. Choosing a safe settlement site: case 4

Knowledge context. Human settlements depend on several factors at once. Some conditions may be compulsory while others are preferences.

Given facts. Candidate sites: Site A: water=yes, road=yes, flood-zone=no, nearby farmland score=3; Site B: water=yes, road=no, flood-zone=no, nearby farmland score=3; Site C: water=yes, road=yes, flood-zone=yes, nearby farmland score=5; Site D: water=no, road=yes, flood-zone=no, nearby farmland score=5. Cross-domain check: a survey team starts work at 12:00 and works for 2 hour(s) without a break.

Rules. A valid site must have water, road access, and must not be in the flood zone. Among valid sites, choose the one with the highest farmland score. For the cross-domain check, finishing time = starting time + duration.

Task. Which site should be selected under the stated mandatory conditions and preference rule? Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. Separate mandatory conditions from the preference. 2. Filtering by water, road, and flood safety leaves ['Site A']. 3. Compare only the farmland scores of feasible sites: Site A=3. 4. Site A has the best score among valid sites. Cross-domain check: 12+2=14, so the team finishes at 14:00.

Answer. Site A. Cross-domain answer: 14:00.

Knowledge takeaway. Good decisions first satisfy hard constraints and only then optimize among the remaining options.

Formal model. Constraint-satisfaction problem followed by a score maximization over feasible options.

Problem 30. Choosing a safe settlement site: case 5

Knowledge context. Human settlements depend on several factors at once. Some conditions may be compulsory while others are preferences.

Given facts. Candidate sites: Site A: water=yes, road=yes, flood-zone=no, nearby farmland score=2; Site B: water=yes, road=no, flood-zone=no, nearby farmland score=4; Site C: water=yes, road=yes, flood-zone=yes, nearby farmland score=4; Site D: water=no, road=yes, flood-zone=no, nearby farmland score=6. Cross-domain check: a local committee has 11 members, requires at least 6 present for quorum, and 11 are present.

Rules. A valid site must have water, road access, and must not be in the flood zone. Among valid sites, choose the one with the highest farmland score. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. Which site should be selected under the stated mandatory conditions and preference rule? Cross-domain check: is quorum met?

Step-by-step solution. 1. Separate mandatory conditions from the preference. 2. Filtering by water, road, and flood safety leaves ['Site A']. 3. Compare only the farmland scores of feasible sites: Site A=2. 4. Site A has the best score among valid sites. Cross-domain check: compare 11 with 6; $11 \geq 6$ is true.

Answer. Site A. Cross-domain answer: quorum is met.

Knowledge takeaway. Good decisions first satisfy hard constraints and only then optimize among the remaining options.

Formal model. Constraint-satisfaction problem followed by a score maximization over feasible options.

G7. Relief and movement cost

Problem 31. Terrain and travel cost from Harbor

Knowledge context. Equal-looking distances can require different effort because terrain differs. A simple cost model makes that effect explicit.

Given facts. Terrain energy costs per segment: {'plain': 1, 'forest': 2, 'hill': 3, 'marsh': 4}. Three routes from Harbor to Juniper: R1=['plain', 'hill', 'plain']; R2=['forest', 'forest', 'plain']; R3=['plain', 'marsh']. The traveler can spend at most 8 energy units.

Rules. The energy of a route is the sum of its terrain costs. A route above the energy limit is infeasible. Among feasible routes, minimize energy.

Task. Which route is feasible and uses the least energy?

Step-by-step solution. 1. Compute route totals: R1=5, R2=5, R3=5. 2. Discard any total above 8. 3. The smallest remaining total is R1=5.

Answer. Route R1, using 5 energy units.

Knowledge takeaway. Relief changes the cost of movement; “shortest” and “easiest” are not automatically the same.

Formal model. Weighted paths with a feasibility budget; sum costs and choose the minimum feasible path.

Problem 32. Terrain and travel cost from Willow

Knowledge context. Equal-looking distances can require different effort because terrain differs. A simple cost model makes that effect explicit.

Given facts. Terrain energy costs per segment: {'plain': 1, 'forest': 2, 'hill': 3, 'marsh': 4}. Three routes from Willow to Alder: R1=['plain', 'hill', 'plain']; R2=['plain', 'forest', 'plain', 'plain']; R3=['plain', 'marsh']. The traveler can spend at most 8 energy units.

Rules. The energy of a route is the sum of its terrain costs. A route above the energy limit is infeasible. Among feasible routes, minimize energy.

Task. Which route is feasible and uses the least energy?

Step-by-step solution. 1. Compute route totals: R1=5, R2=5, R3=5. 2. Discard any total above 8. 3. The smallest remaining total is R1=5.

Answer. Route R1, using 5 energy units.

Knowledge takeaway. Relief changes the cost of movement; “shortest” and “easiest” are not automatically the same.

Formal model. Weighted paths with a feasibility budget; sum costs and choose the minimum feasible path.

Problem 33. Terrain and travel cost from Pine

Knowledge context. Equal-looking distances can require different effort because terrain differs. A simple cost model makes that effect explicit.

Given facts. Terrain energy costs per segment: {'plain': 1, 'forest': 2, 'hill': 3, 'marsh': 4}. Three routes from Pine to Kite: R1=['plain', 'hill', 'plain']; R2=['forest', 'forest', 'plain']; R3=['plain', 'marsh']. The traveler can spend at most 8 energy units.

Rules. The energy of a route is the sum of its terrain costs. A route above the energy limit is infeasible. Among feasible routes, minimize energy.

Task. Which route is feasible and uses the least energy?

Step-by-step solution. 1. Compute route totals: R1=5, R2=5, R3=5. 2. Discard any total above 8. 3. The smallest remaining total is R1=5.

Answer. Route R1, using 5 energy units.

Knowledge takeaway. Relief changes the cost of movement; “shortest” and “easiest” are not automatically the same.

Formal model. Weighted paths with a feasibility budget; sum costs and choose the minimum feasible path.

Problem 34. Terrain and travel cost from Iris

Knowledge context. Equal-looking distances can require different effort because terrain differs. A simple cost model makes that effect explicit.

Given facts. Terrain energy costs per segment: {'plain': 1, 'forest': 2, 'hill': 3, 'marsh': 4}. Three routes from Iris to Linden: R1=['plain', 'hill', 'plain']; R2=['plain', 'forest', 'plain', 'plain']; R3=['plain', 'marsh']. The traveler can spend at most 8 energy units.

Rules. The energy of a route is the sum of its terrain costs. A route above the energy limit is infeasible. Among feasible routes, minimize energy.

Task. Which route is feasible and uses the least energy?

Step-by-step solution. 1. Compute route totals: R1=5, R2=5, R3=5. 2. Discard any total above 8. 3. The smallest remaining total is R1=5.

Answer. Route R1, using 5 energy units.

Knowledge takeaway. Relief changes the cost of movement; “shortest” and “easiest” are not automatically the same.

Formal model. Weighted paths with a feasibility budget; sum costs and choose the minimum feasible path.

Problem 35. Terrain and travel cost from Grove

Knowledge context. Equal-looking distances can require different effort because terrain differs. A simple cost model makes that effect explicit.

Given facts. Terrain energy costs per segment: {'plain': 1, 'forest': 2, 'hill': 3, 'marsh': 4}. Three routes from Grove to Quartz: R1=['plain', 'hill', 'plain']; R2=['forest', 'forest', 'plain']; R3=['plain', 'marsh']. The traveler can spend at most 8 energy units.

Rules. The energy of a route is the sum of its terrain costs. A route above the energy limit is infeasible. Among feasible routes, minimize energy.

Task. Which route is feasible and uses the least energy?

Step-by-step solution. 1. Compute route totals: R1=5, R2=5, R3=5. 2. Discard any total above 8. 3. The smallest remaining total is R1=5.

Answer. Route R1, using 5 energy units.

Knowledge takeaway. Relief changes the cost of movement; “shortest” and “easiest” are not automatically the same.

Formal model. Weighted paths with a feasibility budget; sum costs and choose the minimum feasible path.

G8. Climate from data

Problem 36. Classifying climate data: Station A

Knowledge context. Climate descriptions summarize long-term patterns. In a reasoning problem, explicit thresholds let us classify numerical data without hidden knowledge.

Given facts. Use this classroom classification: cool if mean temperature <12°C; mild if 12–17°C; warm if >17°C. Dry if annual precipitation <500 mm; wet if ≥ 500 mm. Data: Station A: T=8°C, P=430 mm; Station B: T=16°C, P=720 mm; Station C: T=12°C, P=520 mm.

Rules. Apply only the stated thresholds. Temperature and precipitation are separate dimensions, so each station receives two labels.

Task. Classify Station A. Then identify whether any other station has the same two-label class.

Step-by-step solution. 1. Compare T=8 with the temperature thresholds: it is cool. 2. Compare P=430 with 500 mm: it is dry. 3. Compare the combined label with the other stations; matches: none.

Answer. Station A is cool and dry; no other station has the same class.

Knowledge takeaway. Classification is the application of clear predicates to data; the rule must be stated before the label is inferred.

Formal model. Numeric predicates with thresholds; output the conjunction of the temperature and precipitation categories.

Problem 37. Classifying climate data: Station B

Knowledge context. Climate descriptions summarize long-term patterns. In a reasoning problem, explicit thresholds let us classify numerical data without hidden knowledge.

Given facts. Use this classroom classification: cool if mean temperature <12°C; mild if 12–17°C; warm if >17°C. Dry if annual precipitation <500 mm; wet if ≥ 500 mm. Data: Station A: T=9°C, P=430 mm; Station B: T=17°C, P=720 mm; Station C: T=13°C, P=530 mm.

Rules. Apply only the stated thresholds. Temperature and precipitation are separate dimensions, so each station receives two labels.

Task. Classify Station B. Then identify whether any other station has the same two-label class.

Step-by-step solution. 1. Compare T=17 with the temperature thresholds: it is mild. 2. Compare P=720 with 500 mm: it is wet. 3. Compare the combined label with the other stations; matches: ['Station C'].

Answer. Station B is mild and wet; same class: Station C.

Knowledge takeaway. Classification is the application of clear predicates to data; the rule must be stated before the label is inferred.

Formal model. Numeric predicates with thresholds; output the conjunction of the temperature and precipitation categories.

Problem 38. Classifying climate data: Station C

Knowledge context. Climate descriptions summarize long-term patterns. In a reasoning problem, explicit thresholds let us classify numerical data without hidden knowledge.

Given facts. Use this classroom classification: cool if mean temperature <12°C; mild if 12–17°C; warm if >17°C. Dry if annual precipitation <500 mm; wet if ≥ 500 mm. Data: Station A: T=10°C, P=430 mm; Station B: T=18°C, P=720 mm; Station C: T=14°C, P=540 mm.

Rules. Apply only the stated thresholds. Temperature and precipitation are separate dimensions, so each station receives two labels.

Task. Classify Station C. Then identify whether any other station has the same two-label class.

Step-by-step solution. 1. Compare T=14 with the temperature thresholds: it is mild. 2. Compare P=540 with 500 mm: it is wet. 3. Compare the combined label with the other stations; matches: none.

Answer. Station C is mild and wet; no other station has the same class.

Knowledge takeaway. Classification is the application of clear predicates to data; the rule must be stated before the label is inferred.

Formal model. Numeric predicates with thresholds; output the conjunction of the temperature and precipitation categories.

Problem 39. Classifying climate data: Station A

Knowledge context. Climate descriptions summarize long-term patterns. In a reasoning problem, explicit thresholds let us classify numerical data without hidden knowledge.

Given facts. Use this classroom classification: cool if mean temperature <12°C; mild if 12–17°C; warm if >17°C. Dry if annual precipitation <500 mm; wet if ≥ 500 mm. Data: Station A: T=11°C, P=430 mm; Station B: T=19°C, P=720 mm; Station C: T=15°C, P=550 mm.

Rules. Apply only the stated thresholds. Temperature and precipitation are separate dimensions, so each station receives two labels.

Task. Classify Station A. Then identify whether any other station has the same two-label class.

Step-by-step solution. 1. Compare T=11 with the temperature thresholds: it is cool. 2. Compare P=430 with 500 mm: it is dry. 3. Compare the combined label with the other stations; matches: none.

Answer. Station A is cool and dry; no other station has the same class.

Knowledge takeaway. Classification is the application of clear predicates to data; the rule must be stated before the label is inferred.

Formal model. Numeric predicates with thresholds; output the conjunction of the temperature and precipitation categories.

Problem 40. Classifying climate data: Station B

Knowledge context. Climate descriptions summarize long-term patterns. In a reasoning problem, explicit thresholds let us classify numerical data without hidden knowledge.

Given facts. Use this classroom classification: cool if mean temperature <12°C; mild if 12–17°C; warm if >17°C. Dry if annual precipitation <500 mm; wet if ≥ 500 mm. Data: Station A: T=12°C, P=430 mm; Station B: T=20°C, P=720 mm; Station C: T=16°C, P=560 mm.

Rules. Apply only the stated thresholds. Temperature and precipitation are separate dimensions, so each station receives two labels.

Task. Classify Station B. Then identify whether any other station has the same two-label class.

Step-by-step solution. 1. Compare T=20 with the temperature thresholds: it is warm. 2. Compare P=720 with 500 mm: it is wet. 3. Compare the combined label with the other stations; matches: none.

Answer. Station B is warm and wet; no other station has the same class.

Knowledge takeaway. Classification is the application of clear predicates to data; the rule must be stated before the label is inferred.

Formal model. Numeric predicates with thresholds; output the conjunction of the temperature and precipitation categories.

G9. Exchange, routes, and capacities

Problem 41. Bottlenecks in the Yarrow–Elm exchange network

Knowledge context. Trade and transport networks have capacities. A single narrow connection can limit a much larger network.

Given facts. Goods move from Yarrow to Elm along two branches: Yarrow→Zephyr capacity 4, Zephyr→Elm capacity 3; Yarrow→Kite capacity 2, Kite→Elm capacity 5.

Rules. A branch cannot carry more than its smallest edge capacity. Independent branch capacities can be added, but a shared destination capacity can cap the total.

Task. What is the maximum number of units that can reach Elm in one period? Identify the bottleneck(s).

Step-by-step solution. 1. Branch 1 is limited by min(4,3)=3. 2. Branch 2 is limited by min(2,5)=2. 3. Together the branches could carry 5 units. 4. No extra shared cap is given, so the maximum is 5.

Answer. 5 units.

Knowledge takeaway. A bottleneck is a local limit that constrains the total possible flow.

Formal model. Flow network; branch capacity is a minimum along a path; total is the sum of independent branches subject to shared caps.

Problem 42. Bottlenecks in the Grove–Iris exchange network

Knowledge context. Trade and transport networks have capacities. A single narrow connection can limit a much larger network.

Given facts. Goods move from Grove to Iris along two branches: Grove→Zephyr capacity 5, Zephyr→Iris capacity 3; Grove→Kite capacity 3, Kite→Iris capacity 5.

Rules. A branch cannot carry more than its smallest edge capacity. Independent branch capacities can be added, but a shared destination capacity can cap the total.

Task. What is the maximum number of units that can reach Iris in one period? Identify the bottleneck(s).

Step-by-step solution. 1. Branch 1 is limited by min(5,3)=3. 2. Branch 2 is limited by min(3,5)=3. 3. Together the branches could carry 6 units. 4. No extra shared cap is given, so the maximum is 6.

Answer. 6 units.

Knowledge takeaway. A bottleneck is a local limit that constrains the total possible flow.

Formal model. Flow network; branch capacity is a minimum along a path; total is the sum of independent branches subject to shared caps.

Problem 43. Bottlenecks in the Alder–Quartz exchange network

Knowledge context. Trade and transport networks have capacities. A single narrow connection can limit a much larger network.

Given facts. Goods move from Alder to Quartz along two branches: Alder→Oak capacity 6, Oak→Quartz capacity 3; Alder→Dover capacity 4, Dover→Quartz capacity 5.

Rules. A branch cannot carry more than its smallest edge capacity. Independent branch capacities can be added, but a shared destination capacity can cap the total.

Task. What is the maximum number of units that can reach Quartz in one period? Identify the bottleneck(s).

Step-by-step solution. 1. Branch 1 is limited by min(6,3)=3. 2. Branch 2 is limited by min(4,5)=4. 3. Together the branches could carry 7 units. 4. No extra shared cap is given, so the maximum is 7.

Answer. 7 units.

Knowledge takeaway. A bottleneck is a local limit that constrains the total possible flow.

Formal model. Flow network; branch capacity is a minimum along a path; total is the sum of independent branches subject to shared caps.

Problem 44. Bottlenecks in the Willow–Juniper exchange network

Knowledge context. Trade and transport networks have capacities. A single narrow connection can limit a much larger network.

Given facts. Goods move from Willow to Juniper along two branches: Willow→Yarrow capacity 7, Yarrow→Juniper capacity 3; Willow→Elm capacity 5, Elm→Juniper capacity 5.

Rules. A branch cannot carry more than its smallest edge capacity. Independent branch capacities can be added, but a shared destination capacity can cap the total.

Task. What is the maximum number of units that can reach Juniper in one period? Identify the bottleneck(s).

Step-by-step solution. 1. Branch 1 is limited by min(7,3)=3. 2. Branch 2 is limited by min(5,5)=5. 3. Together the branches could carry 8 units. 4. No extra shared cap is given, so the maximum is 8.

Answer. 8 units.

Knowledge takeaway. A bottleneck is a local limit that constrains the total possible flow.

Formal model. Flow network; branch capacity is a minimum along a path; total is the sum of independent branches subject to shared caps.

Problem 45. Bottlenecks in the Birch–Juniper exchange network

Knowledge context. Trade and transport networks have capacities. A single narrow connection can limit a much larger network.

Given facts. Goods move from Birch to Juniper along two branches: Birch→Meadow capacity 8, Meadow→Juniper capacity 3; Birch→Yarrow capacity 6, Yarrow→Juniper capacity 5.

Rules. A branch cannot carry more than its smallest edge capacity. Independent branch capacities can be added, but a shared destination capacity can cap the total.

Task. What is the maximum number of units that can reach Juniper in one period? Identify the bottleneck(s).

Step-by-step solution. 1. Branch 1 is limited by min(8,3)=3. 2. Branch 2 is limited by min(6,5)=5. 3. Together the branches could carry 8 units. 4. No extra shared cap is given, so the maximum is 8.

Answer. 8 units.

Knowledge takeaway. A bottleneck is a local limit that constrains the total possible flow.

Formal model. Flow network; branch capacity is a minimum along a path; total is the sum of independent branches subject to shared caps.

G10. Regions, sets, and membership

Problem 46. Nested places: Northfield, Green Valley, and Republic B

Knowledge context. Geography uses nested categories: a locality can be inside a region, a region inside a state, and a state inside a continent.

Given facts. Northfield is in Green Valley. Green Valley is in Republic B. Republic B is in the continent of Aurora.

Rules. If X is contained in Y and Y is contained in Z, then X is contained in Z. The reverse inference is not valid: knowing that two places are in one country does not prove that they are in the same region.

Task. Is Northfield in Republic B?

Step-by-step solution. 1. Follow the chain city→region→country. 2. Containment is transitive, so the city is in the country.

Answer. Yes, Northfield is in Republic B.

Knowledge takeaway. Hierarchies support upward transitive inference, but they do not justify arbitrary reverse inferences.

Formal model. Containment relation with transitivity; answer membership queries while rejecting invalid converse inference.

Problem 47. Nested places: Harbor, Highridge, and Republic A

Knowledge context. Geography uses nested categories: a locality can be inside a region, a region inside a state, and a state inside a continent.

Given facts. Harbor is in Highridge. Highridge is in Republic A. Republic A is in the continent of Aurora.

Rules. If X is contained in Y and Y is contained in Z, then X is contained in Z. The reverse inference is not valid: knowing that two places are in one country does not prove that they are in the same region.

Task. Is Harbor in Aurora?

Step-by-step solution. 1. Follow city→region→country→continent. 2. Transitivity carries membership through the chain.

Answer. Yes, Harbor is in Aurora.

Knowledge takeaway. Hierarchies support upward transitive inference, but they do not justify arbitrary reverse inferences.

Formal model. Containment relation with transitivity; answer membership queries while rejecting invalid converse inference.

Problem 48. Nested places: Grove, Coastland, and Republic A

Knowledge context. Geography uses nested categories: a locality can be inside a region, a region inside a state, and a state inside a continent.

Given facts. Grove is in Coastland. Coastland is in Republic A. Republic A is in the continent of Meridian.

Rules. If X is contained in Y and Y is contained in Z, then X is contained in Z. The reverse inference is not valid: knowing that two places are in one country does not prove that they are in the same region.

Task. Can we conclude that Republic A is in Grove?

Step-by-step solution. 1. The facts go from smaller units to larger units. 2. No rule allows country→city, so the reverse claim is unsupported.

Answer. No. The containment relation cannot be reversed.

Knowledge takeaway. Hierarchies support upward transitive inference, but they do not justify arbitrary reverse inferences.

Formal model. Containment relation with transitivity; answer membership queries while rejecting invalid converse inference.

Problem 49. Nested places: Iris, Northland, and Republic B

Knowledge context. Geography uses nested categories: a locality can be inside a region, a region inside a state, and a state inside a continent.

Given facts. Iris is in Northland. Northland is in Republic B. Republic B is in the continent of Aurora.

Rules. If X is contained in Y and Y is contained in Z, then X is contained in Z. The reverse inference is not valid: knowing that two places are in one country does not prove that they are in the same region.

Task. Which larger units contain Iris?

Step-by-step solution. 1. Read the direct container: Northland. 2. Propagate upward to Republic B and Aurora.

Answer. Northland, Republic B, and Aurora.

Knowledge takeaway. Hierarchies support upward transitive inference, but they do not justify arbitrary reverse inferences.

Formal model. Containment relation with transitivity; answer membership queries while rejecting invalid converse inference.

Problem 50. Nested places: Cedar, Northland, and Republic D

Knowledge context. Geography uses nested categories: a locality can be inside a region, a region inside a state, and a state inside a continent.

Given facts. Cedar is in Northland. Northland is in Republic D. Republic D is in the continent of Pacifica.

Rules. If X is contained in Y and Y is contained in Z, then X is contained in Z. The reverse inference is not valid: knowing that two places are in one country does not prove that they are in the same region.

Task. Does being in Republic D prove two cities share a region?

Step-by-step solution. 1. Two cities can belong to the same country while belonging to different regions. 2. The region relation is therefore not deducible from country membership alone.

Answer. No. Country membership alone is insufficient to identify the region.

Knowledge takeaway. Hierarchies support upward transitive inference, but they do not justify arbitrary reverse inferences.

Formal model. Containment relation with transitivity; answer membership queries while rejecting invalid converse inference.

H1. Ordering events

Problem 51. Ordering the events of Westmoor

Knowledge context. Historical chronology often starts from partial information: some events are ordered relative to others even when exact dates are not known.

Given facts. Market charter happened before Bridge opening. Bridge opening happened before New school. Flood happened after Market charter.

Rules. “Before” is transitive: if A is before B and B is before C, then A is before C. If two events are not connected by the facts, their relative order may be undetermined.

Task. What must be true about the order of Market charter and New school? Can the exact order of Flood and Bridge opening be determined?

Step-by-step solution. 1. From Market charter before Bridge opening and Bridge opening before New school, infer Market charter before New school. 2. Flood is only constrained to be after Market charter. There is no stated relation between Flood and Bridge opening. 3. Therefore the first comparison is forced but the second is underdetermined.

Answer. Market charter must be before New school. The relative order of Flood and Bridge opening cannot be determined from the given facts.

Knowledge takeaway. A careful reasoner distinguishes what must be true from what merely could be true.

Formal model. Partial order / directed acyclic graph; use transitive closure and detect incomparable events.

Problem 52. Ordering the events of Coastland

Knowledge context. Historical chronology often starts from partial information: some events are ordered relative to others even when exact dates are not known.

Given facts. Founding happened before Market charter. Market charter happened before Flood. Bridge opening happened after Founding.

Rules. “Before” is transitive: if A is before B and B is before C, then A is before C. If two events are not connected by the facts, their relative order may be undetermined.

Task. What must be true about the order of Founding and Flood? Can the exact order of Bridge opening and Market charter be determined?

Step-by-step solution. 1. From Founding before Market charter and Market charter before Flood, infer Founding before Flood. 2. Bridge opening is only constrained to be after Founding. There is no stated relation between Bridge opening and Market charter. 3. Therefore the first comparison is forced but the second is underdetermined.

Answer. Founding must be before Flood. The relative order of Bridge opening and Market charter cannot be determined from the given facts.

Knowledge takeaway. A careful reasoner distinguishes what must be true from what merely could be true.

Formal model. Partial order / directed acyclic graph; use transitive closure and detect incomparable events.

Problem 53. Ordering the events of Coastland

Knowledge context. Historical chronology often starts from partial information: some events are ordered relative to others even when exact dates are not known.

Given facts. New school happened before Flood. Flood happened before Market charter. Bridge opening happened after New school.

Rules. “Before” is transitive: if A is before B and B is before C, then A is before C. If two events are not connected by the facts, their relative order may be undetermined.

Task. What must be true about the order of New school and Market charter? Can the exact order of Bridge opening and Flood be determined?

Step-by-step solution. 1. From New school before Flood and Flood before Market charter, infer New school before Market charter. 2. Bridge opening is only constrained to be after New school. There is no stated relation between Bridge opening and Flood. 3. Therefore the first comparison is forced but the second is underdetermined.

Answer. New school must be before Market charter. The relative order of Bridge opening and Flood cannot be determined from the given facts.

Knowledge takeaway. A careful reasoner distinguishes what must be true from what merely could be true.

Formal model. Partial order / directed acyclic graph; use transitive closure and detect incomparable events.

Problem 54. Ordering the events of Coastland

Knowledge context. Historical chronology often starts from partial information: some events are ordered relative to others even when exact dates are not known.

Given facts. Bridge opening happened before Flood. Flood happened before Market charter. New school happened after Bridge opening.

Rules. “Before” is transitive: if A is before B and B is before C, then A is before C. If two events are not connected by the facts, their relative order may be undetermined.

Task. What must be true about the order of Bridge opening and Market charter? Can the exact order of New school and Flood be determined?

Step-by-step solution. 1. From Bridge opening before Flood and Flood before Market charter, infer Bridge opening before Market charter. 2. New school is only constrained to be after Bridge opening. There is no stated relation between New school and Flood. 3. Therefore the first comparison is forced but the second is underdetermined.

Answer. Bridge opening must be before Market charter. The relative order of New school and Flood cannot be determined from the given facts.

Knowledge takeaway. A careful reasoner distinguishes what must be true from what merely could be true.

Formal model. Partial order / directed acyclic graph; use transitive closure and detect incomparable events.

Problem 55. Ordering the events of Eastvale

Knowledge context. Historical chronology often starts from partial information: some events are ordered relative to others even when exact dates are not known.

Given facts. Market charter happened before Founding. Founding happened before Flood. Bridge opening happened after Market charter.

Rules. “Before” is transitive: if A is before B and B is before C, then A is before C. If two events are not connected by the facts, their relative order may be undetermined.

Task. What must be true about the order of Market charter and Flood? Can the exact order of Bridge opening and Founding be determined?

Step-by-step solution. 1. From Market charter before Founding and Founding before Flood, infer Market charter before Flood. 2. Bridge opening is only constrained to be after Market charter. There is no stated relation between Bridge opening and Founding. 3. Therefore the first comparison is forced but the second is underdetermined.

Answer. Market charter must be before Flood. The relative order of Bridge opening and Founding cannot be determined from the given facts.

Knowledge takeaway. A careful reasoner distinguishes what must be true from what merely could be true.

Formal model. Partial order / directed acyclic graph; use transitive closure and detect incomparable events.

H2. Dates, durations, and centuries

Problem 56. Duration and centuries: 1334–1357

Knowledge context. Dates can be used as numbers, but historical labels such as centuries follow explicit conventions.

Given facts. An institution was founded in 1334 and reorganized in 1357. Use the rule: years 1–100 are the 1st century, 101–200 the 2nd, and so on. Duration is end year minus start year.

Rules. Apply the century rule mechanically; do not infer a century from the first two digits alone. Compute elapsed time by subtraction.

Task. How many years passed between the two events? In which centuries did 1334 and 1357 fall?

Step-by-step solution. 1. Elapsed time: $1357-1334=23$ years. 2. $\text{Century}(1334)=\text{floor}((1334-1)/100)+1=14$. 3. $\text{Century}(1357)=14$.

Answer. 23 years; 1334 is in the 14th century and 1357 in the 14th century.

Knowledge takeaway. Chronological reasoning combines arithmetic with a naming convention; both should be applied explicitly.

Formal model. Integer arithmetic plus century function $c(y)=\text{floor}((y-1)/100)+1$.

Problem 57. Duration and centuries: 1348–1376

Knowledge context. Dates can be used as numbers, but historical labels such as centuries follow explicit conventions.

Given facts. An institution was founded in 1348 and reorganized in 1376. Use the rule: years 1–100 are the 1st century, 101–200 the 2nd, and so on. Duration is end year minus start year.

Rules. Apply the century rule mechanically; do not infer a century from the first two digits alone. Compute elapsed time by subtraction.

Task. How many years passed between the two events? In which centuries did 1348 and 1376 fall?

Step-by-step solution. 1. Elapsed time: 1376−1348=28 years. 2. Century(1348)=floor((1348−1)/100)+1=14. 3. Century(1376)=14.

Answer. 28 years; 1348 is in the 14th century and 1376 in the 14th century.

Knowledge takeaway. Chronological reasoning combines arithmetic with a naming convention; both should be applied explicitly.

Formal model. Integer arithmetic plus century function c(y)=floor((y−1)/100)+1.

Problem 58. Duration and centuries: 1356–1389

Knowledge context. Dates can be used as numbers, but historical labels such as centuries follow explicit conventions.

Given facts. An institution was founded in 1356 and reorganized in 1389. Use the rule: years 1–100 are the 1st century, 101–200 the 2nd, and so on. Duration is end year minus start year.

Rules. Apply the century rule mechanically; do not infer a century from the first two digits alone. Compute elapsed time by subtraction.

Task. How many years passed between the two events? In which centuries did 1356 and 1389 fall?

Step-by-step solution. 1. Elapsed time: 1389−1356=33 years. 2. Century(1356)=floor((1356−1)/100)+1=14. 3. Century(1389)=14.

Answer. 33 years; 1356 is in the 14th century and 1389 in the 14th century.

Knowledge takeaway. Chronological reasoning combines arithmetic with a naming convention; both should be applied explicitly.

Formal model. Integer arithmetic plus century function c(y)=floor((y−1)/100)+1.

Problem 59. Duration and centuries: 1371–1409

Knowledge context. Dates can be used as numbers, but historical labels such as centuries follow explicit conventions.

Given facts. An institution was founded in 1371 and reorganized in 1409. Use the rule: years 1–100 are the 1st century, 101–200 the 2nd, and so on. Duration is end year minus start year.

Rules. Apply the century rule mechanically; do not infer a century from the first two digits alone. Compute elapsed time by subtraction.

Task. How many years passed between the two events? In which centuries did 1371 and 1409 fall?

Step-by-step solution. 1. Elapsed time: 1409−1371=38 years. 2. Century(1371)=floor((1371−1)/100)+1=14. 3. Century(1409)=15.

Answer. 38 years; 1371 is in the 14th century and 1409 in the 15th century.

Knowledge takeaway. Chronological reasoning combines arithmetic with a naming convention; both should be applied explicitly.

Formal model. Integer arithmetic plus century function c(y)=floor((y−1)/100)+1.

Problem 60. Duration and centuries: 1399–1442

Knowledge context. Dates can be used as numbers, but historical labels such as centuries follow explicit conventions.

Given facts. An institution was founded in 1399 and reorganized in 1442. Use the rule: years 1–100 are the 1st century, 101–200 the 2nd, and so on. Duration is end year minus start year.

Rules. Apply the century rule mechanically; do not infer a century from the first two digits alone. Compute elapsed time by subtraction.

Task. How many years passed between the two events? In which centuries did 1399 and 1442 fall?

Step-by-step solution. 1. Elapsed time: 1442−1399=43 years. 2. Century(1399)=floor((1399−1)/100)+1=14. 3. Century(1442)=15.

Answer. 43 years; 1399 is in the 14th century and 1442 in the 15th century.

Knowledge takeaway. Chronological reasoning combines arithmetic with a naming convention; both should be applied explicitly.

Formal model. Integer arithmetic plus century function c(y)=floor((y−1)/100)+1.

H3. Sources and supported claims

Problem 61. What can one map prove?

Knowledge context. Historical sources are evidence, not automatic answers. The strength of a conclusion depends on what the source actually records.

Given facts. A historian has one source: a map. The source directly shows that a road was mapped. A student proposes two claims: C1: a road was mapped; C2: the whole population agreed with the ruler.

Rules. A source supports only claims that can reasonably be inferred from what it contains. Absence of evidence for a broad claim is not proof of that claim.

Task. Which claim is directly supported, and which is not justified by this source alone?

Step-by-step solution. 1. The map directly contains evidence for C1. 2. C2 is much broader and no stated feature of the source measures the opinion of the whole population. 3. Therefore C1 is supported and C2 is not established.

Answer. C1 is supported. C2 is not justified by the source alone.

Knowledge takeaway. Good historical reasoning matches the scope of a claim to the scope of the evidence.

Formal model. Claim-evidence compatibility relation; accept only claims entailed by the source attributes.

Problem 62. What can one tool prove?

Knowledge context. Historical sources are evidence, not automatic answers. The strength of a conclusion depends on what the source actually records.

Given facts. A historian has one source: a tool. The source directly shows that a tool existed. A student proposes two claims: C1: a tool existed; C2: the whole population agreed with the ruler.

Rules. A source supports only claims that can reasonably be inferred from what it contains. Absence of evidence for a broad claim is not proof of that claim.

Task. Which claim is directly supported, and which is not justified by this source alone?

Step-by-step solution. 1. The tool directly contains evidence for C1. 2. C2 is much broader and no stated feature of the source measures the opinion of the whole population. 3. Therefore C1 is supported and C2 is not established.

Answer. C1 is supported. C2 is not justified by the source alone.

Knowledge takeaway. Good historical reasoning matches the scope of a claim to the scope of the evidence.

Formal model. Claim-evidence compatibility relation; accept only claims entailed by the source attributes.

Problem 63. What can one coin prove?

Knowledge context. Historical sources are evidence, not automatic answers. The strength of a conclusion depends on what the source actually records.

Given facts. A historian has one source: a coin. The source directly shows that people used metal objects. A student proposes two claims: C1: people used metal objects; C2: the whole population agreed with the ruler.

Rules. A source supports only claims that can reasonably be inferred from what it contains. Absence of evidence for a broad claim is not proof of that claim.

Task. Which claim is directly supported, and which is not justified by this source alone?

Step-by-step solution. 1. The coin directly contains evidence for C1. 2. C2 is much broader and no stated feature of the source measures the opinion of the whole population. 3. Therefore C1 is supported and C2 is not established.

Answer. C1 is supported. C2 is not justified by the source alone.

Knowledge takeaway. Good historical reasoning matches the scope of a claim to the scope of the evidence.

Formal model. Claim-evidence compatibility relation; accept only claims entailed by the source attributes.

Problem 64. What can one map prove?

Knowledge context. Historical sources are evidence, not automatic answers. The strength of a conclusion depends on what the source actually records.

Given facts. A historian has one source: a map. The source directly shows that a road was mapped. A student proposes two claims: C1: a road was mapped; C2: the whole population agreed with the ruler. Cross-domain check: on a field sketch, 1 cm represents 6 km and two checkpoints are 3 cm apart.

Rules. A source supports only claims that can reasonably be inferred from what it contains. Absence of evidence for a broad claim is not proof of that claim. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Which claim is directly supported, and which is not justified by this source alone? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. The map directly contains evidence for C1. 2. C2 is much broader and no stated feature of the source measures the opinion of the whole population. 3. Therefore C1 is supported and C2 is not established. Cross-domain check: 3×6=18 km.

Answer. C1 is supported. C2 is not justified by the source alone. Cross-domain answer: 18 km.

Knowledge takeaway. Good historical reasoning matches the scope of a claim to the scope of the evidence.

Formal model. Claim-evidence compatibility relation; accept only claims entailed by the source attributes.

Problem 65. What can one tax register prove?

Knowledge context. Historical sources are evidence, not automatic answers. The strength of a conclusion depends on what the source actually records.

Given facts. A historian has one source: a tax register. The source directly shows that taxes were recorded. A student proposes two claims: C1: taxes were recorded; C2: the whole population agreed with the ruler.

Rules. A source supports only claims that can reasonably be inferred from what it contains. Absence of evidence for a broad claim is not proof of that claim.

Task. Which claim is directly supported, and which is not justified by this source alone?

Step-by-step solution. 1. The tax register directly contains evidence for C1. 2. C2 is much broader and no stated feature of the source measures the opinion of the whole population. 3. Therefore C1 is supported and C2 is not established.

Answer. C1 is supported. C2 is not justified by the source alone.

Knowledge takeaway. Good historical reasoning matches the scope of a claim to the scope of the evidence.

Formal model. Claim-evidence compatibility relation; accept only claims entailed by the source attributes.

H4. Conflicting sources

Problem 66. Resolving conflicting sources: case 1

Knowledge context. Historical accounts can disagree. Reasoning requires evaluating how a source knows what it claims, not simply counting statements.

Given facts. Source A, written on the day of an event, says 40 carts arrived. Source B, copied 80 years later from an unknown account, says 90 carts arrived. Source C is an independent warehouse record from the same week and lists 42 carts.

Rules. When sources conflict, compare proximity to the event, independence, and whether a source has a reason or mechanism to record the specific fact. Agreement between independent relevant sources increases support but does not create absolute certainty.

Task. Which account is better supported by the evidence given, and why?

Step-by-step solution. 1. A is contemporary and reports 40. 2. C is an independent operational record and reports 42, close to A. 3. B is much later and has an unknown chain of transmission. 4. The cluster near 40–42 is therefore better supported than 90.

Answer. An arrival count near 40–42 carts is better supported than 90.

Knowledge takeaway. Independent, relevant, well-positioned evidence usually deserves more weight than late repetition with unknown provenance.

Formal model. Evidence graph with source reliability/independence attributes; compare support for competing hypotheses.

Problem 67. Resolving conflicting sources: case 2

Knowledge context. Historical accounts can disagree. Reasoning requires evaluating how a source knows what it claims, not simply counting statements.

Given facts. Source A, an eyewitness diary, says the bridge opened on Monday. Source B, a later textbook, says Tuesday. Source C, an independent dated toll record, begins charging on Monday.

Rules. When sources conflict, compare proximity to the event, independence, and whether a source has a reason or mechanism to record the specific fact. Agreement between independent relevant sources increases support but does not create absolute certainty.

Task. Which account is better supported by the evidence given, and why?

Step-by-step solution. 1. A and C are close to the event and independent. 2. Both point to Monday. 3. B is later and conflicts with two stronger, independent pieces of evidence.

Answer. Monday is better supported, while remaining a historical conclusion rather than absolute proof.

Knowledge takeaway. Independent, relevant, well-positioned evidence usually deserves more weight than late repetition with unknown provenance.

Formal model. Evidence graph with source reliability/independence attributes; compare support for competing hypotheses.

Problem 68. Resolving conflicting sources: case 3

Knowledge context. Historical accounts can disagree. Reasoning requires evaluating how a source knows what it claims, not simply counting statements.

Given facts. Source A, written on the day of an event, says 40 carts arrived. Source B, copied 80 years later from an unknown account, says 90 carts arrived. Source C is an independent warehouse record from the same week and lists 42 carts. Cross-domain check: on a field sketch, 1 cm represents 5 km and two checkpoints are 7 cm apart.

Rules. When sources conflict, compare proximity to the event, independence, and whether a source has a reason or mechanism to record the specific fact. Agreement between independent relevant sources increases support but does not create absolute certainty. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Which account is better supported by the evidence given, and why? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. A is contemporary and reports 40. 2. C is an independent operational record and reports 42, close to A. 3. B is much later and has an unknown chain of transmission. 4. The cluster near 40–42 is therefore better supported than 90. Cross-domain check: 7×5=35 km.

Answer. An arrival count near 40–42 carts is better supported than 90. Cross-domain answer: 35 km.

Knowledge takeaway. Independent, relevant, well-positioned evidence usually deserves more weight than late repetition with unknown provenance.

Formal model. Evidence graph with source reliability/independence attributes; compare support for competing hypotheses.

Problem 69. Resolving conflicting sources: case 4

Knowledge context. Historical accounts can disagree. Reasoning requires evaluating how a source knows what it claims, not simply counting statements.

Given facts. Source A, an eyewitness diary, says the bridge opened on Monday. Source B, a later textbook, says Tuesday. Source C, an independent dated toll record, begins charging on Monday. Cross-domain check: a survey team starts work at 12:00 and works for 2 hour(s) without a break.

Rules. When sources conflict, compare proximity to the event, independence, and whether a source has a reason or mechanism to record the specific fact. Agreement between independent relevant sources increases support but does not create absolute certainty. For the cross-domain check, finishing time = starting time + duration.

Task. Which account is better supported by the evidence given, and why? Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. A and C are close to the event and independent. 2. Both point to Monday. 3. B is later and conflicts with two stronger, independent pieces of evidence. Cross-domain check: 12+2=14, so the team finishes at 14:00.

Answer. Monday is better supported, while remaining a historical conclusion rather than absolute proof. Cross-domain answer: 14:00.

Knowledge takeaway. Independent, relevant, well-positioned evidence usually deserves more weight than late repetition with unknown provenance.

Formal model. Evidence graph with source reliability/independence attributes; compare support for competing hypotheses.

Problem 70. Resolving conflicting sources: case 5

Knowledge context. Historical accounts can disagree. Reasoning requires evaluating how a source knows what it claims, not simply counting statements.

Given facts. Source A, written on the day of an event, says 40 carts arrived. Source B, copied 80 years later from an unknown account, says 90 carts arrived. Source C is an independent warehouse record from the same week and lists 42 carts. Cross-domain check: a local committee has 15 members, requires at least 8 present for quorum, and 14 are present.

Rules. When sources conflict, compare proximity to the event, independence, and whether a source has a reason or mechanism to record the specific fact. Agreement between independent relevant sources increases support but does not create absolute certainty. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. Which account is better supported by the evidence given, and why? Cross-domain check: is quorum met?

Step-by-step solution. 1. A is contemporary and reports 40. 2. C is an independent operational record and reports 42, close to A. 3. B is much later and has an unknown chain of transmission. 4. The cluster near 40–42 is therefore better supported than 90. Cross-domain check: compare 14 with 8; $14 \geq 8$ is true.

Answer. An arrival count near 40–42 carts is better supported than 90. Cross-domain answer: quorum is met.

Knowledge takeaway. Independent, relevant, well-positioned evidence usually deserves more weight than late repetition with unknown provenance.

Formal model. Evidence graph with source reliability/independence attributes; compare support for competing hypotheses.

H5. Causes, conditions, and consequences

Problem 71. Cause-and-consequence chain: case 1

Knowledge context. Many explanations can be represented as condition-and-consequence rules. The logical form matters: necessary, sufficient, AND, and OR are not interchangeable.

Given facts. Facts: Rain occurred; The river was already high; The warning siren worked; The gate was closed. Rules: If rain occurs AND the river is already high, flooding occurs. If flooding occurs, the riverside road closes. If the siren works, residents receive a warning.

Rules. Use the rules exactly as written. AND requires all listed conditions; OR requires at least one. A one-way implication cannot automatically be reversed.

Task. Which consequences can be deduced by applying the rules exactly? Do not assume the converse of any rule.

Step-by-step solution. 1. Rain and high river satisfy both conditions of the first rule. 2. Infer flooding. 3. Flooding implies road closure. 4. The working siren independently implies a warning.

Answer. Flooding, road closure, and a warning can be deduced.

Knowledge takeaway. Causal reasoning becomes clearer when every link is stated and the direction of implication is respected.

Formal model. Forward-chaining rule system over Boolean facts; apply implications until no new facts appear.

Problem 72. Cause-and-consequence chain: case 2

Knowledge context. Many explanations can be represented as condition-and-consequence rules. The logical form matters: necessary, sufficient, AND, and OR are not interchangeable.

Given facts. Facts: Rain occurred; The river was not high; The warning siren worked; The gate was closed. Rules: If rain occurs AND the river is already high, flooding occurs. If flooding occurs, the riverside road closes. If the siren works, residents receive a warning.

Rules. Use the rules exactly as written. AND requires all listed conditions; OR requires at least one. A one-way implication cannot automatically be reversed.

Task. Which consequences can be deduced by applying the rules exactly? Do not assume the converse of any rule.

Step-by-step solution. 1. Rain is true but the high-river condition is false. 2. The AND rule for flooding therefore does not fire. 3. Without flooding, road closure is not inferred. 4. The siren rule still gives a warning.

Answer. A warning can be deduced, but flooding and road closure cannot be deduced.

Knowledge takeaway. Causal reasoning becomes clearer when every link is stated and the direction of implication is respected.

Formal model. Forward-chaining rule system over Boolean facts; apply implications until no new facts appear.

Problem 73. Cause-and-consequence chain: case 3

Knowledge context. Many explanations can be represented as condition-and-consequence rules. The logical form matters: necessary, sufficient, AND, and OR are not interchangeable.

Given facts. Facts: Rain occurred; The river was already high; The warning siren worked; The gate was closed; Flooding occurred. Rules: If rain occurs AND the river is already high, flooding occurs. If flooding occurs, the riverside road closes. If the siren works, residents receive a warning.

Rules. Use the rules exactly as written. AND requires all listed conditions; OR requires at least one. A one-way implication cannot automatically be reversed.

Task. Which consequences can be deduced by applying the rules exactly? Do not assume the converse of any rule.

Step-by-step solution. 1. Flooding is stated directly, so apply the flooding→road-closure rule. 2. Apply siren→warning. 3. Do not use flooding to infer rain or a high river; that would reverse a rule.

Answer. Road closure and a warning can be deduced; flooding is already given as a fact.

Knowledge takeaway. Causal reasoning becomes clearer when every link is stated and the direction of implication is respected.

Formal model. Forward-chaining rule system over Boolean facts; apply implications until no new facts appear.

Problem 74. Cause-and-consequence chain: case 4

Knowledge context. Many explanations can be represented as condition-and-consequence rules. The logical form matters: necessary, sufficient, AND, and OR are not interchangeable.

Given facts. Facts: Rain occurred; The river was already high; The warning siren worked; The gate was closed. Rules: If rain occurs OR the river is already high, the field becomes muddy. If flooding occurs, the riverside road closes. If the siren works, residents receive a warning.

Rules. Use the rules exactly as written. AND requires all listed conditions; OR requires at least one. A one-way implication cannot automatically be reversed.

Task. Which consequences can be deduced by applying the rules exactly? Do not assume the converse of any rule.

Step-by-step solution. 1. The mud rule uses OR, so rain alone is enough. 2. Rain is given, therefore mud follows. 3. The siren rule gives a warning.

Answer. The field becomes muddy, and the warning follows from the working siren.

Knowledge takeaway. Causal reasoning becomes clearer when every link is stated and the direction of implication is respected.

Formal model. Forward-chaining rule system over Boolean facts; apply implications until no new facts appear.

Problem 75. Cause-and-consequence chain: case 5

Knowledge context. Many explanations can be represented as condition-and-consequence rules. The logical form matters: necessary, sufficient, AND, and OR are not interchangeable.

Given facts. Facts: Rain occurred; The river was already high. Rules: If rain AND high river, flooding occurs. If flooding, road closes. If road closes, the bus reroutes.

Rules. Use the rules exactly as written. AND requires all listed conditions; OR requires at least one. A one-way implication cannot automatically be reversed.

Task. Which consequences can be deduced by applying the rules exactly? Do not assume the converse of any rule.

Step-by-step solution. 1. Rain and high river jointly imply flooding. 2. Flooding implies road closure. 3. Road closure implies bus rerouting.

Answer. Flooding, road closure, and bus rerouting can all be deduced.

Knowledge takeaway. Causal reasoning becomes clearer when every link is stated and the direction of implication is respected.

Formal model. Forward-chaining rule system over Boolean facts; apply implications until no new facts appear.

H6. Travel through time and space

Problem 76. Courier journey from Willow to Pine

Knowledge context. Historical travel can be analyzed with ordinary distance, speed, and time reasoning when the needed values are supplied.

Given facts. A courier leaves Willow for Pine at 7:00. Distance is 60 km and the stated average speed is 15 km/h. A council meeting in Pine begins at 10.0:00. Ignore stops and use travel time = distance ÷ speed.

Rules. Arrival time equals departure time plus travel time. To attend an event from its beginning, arrival must be no later than the event start.

Task. When does the courier arrive, and can the courier attend the meeting from its beginning?

Step-by-step solution. 1. Travel time=60÷15=4.0 hours. 2. Arrival=7.0+4.0=11.0. 3. Compare 11.0 with meeting time 10.0.

Answer. Arrival at about 11.0:00; no, the meeting has already begun.

Knowledge takeaway. Chronology and geography interact: being able to reach a place in time is a constraint, not a fact to memorize.

Formal model. Arithmetic time model; t_arrive=t_depart+d/v; compare with event time.

Problem 77. Courier journey from Iris to Stone

Knowledge context. Historical travel can be analyzed with ordinary distance, speed, and time reasoning when the needed values are supplied.

Given facts. A courier leaves Iris for Stone at 8:00. Distance is 70 km and the stated average speed is 15 km/h. A council meeting in Stone begins at 13.7:00. Ignore stops and use travel time = distance ÷ speed.

Rules. Arrival time equals departure time plus travel time. To attend an event from its beginning, arrival must be no later than the event start.

Task. When does the courier arrive, and can the courier attend the meeting from its beginning?

Step-by-step solution. 1. Travel time=70÷15=4.7 hours. 2. Arrival=8.0+4.7=12.7. 3. Compare 12.7 with meeting time 13.7.

Answer. Arrival at about 12.7:00; yes, on time.

Knowledge takeaway. Chronology and geography interact: being able to reach a place in time is a constraint, not a fact to memorize.

Formal model. Arithmetic time model; t_arrive=t_depart+d/v; compare with event time.

Problem 78. Courier journey from Grove to Cedar

Knowledge context. Historical travel can be analyzed with ordinary distance, speed, and time reasoning when the needed values are supplied.

Given facts. A courier leaves Grove for Cedar at 9:00. Distance is 80 km and the stated average speed is 15 km/h. A council meeting in Cedar begins at 13.3:00. Ignore stops and use travel time = distance ÷ speed.

Rules. Arrival time equals departure time plus travel time. To attend an event from its beginning, arrival must be no later than the event start.

Task. When does the courier arrive, and can the courier attend the meeting from its beginning?

Step-by-step solution. 1. Travel time=80÷15=5.3 hours. 2. Arrival=9.0+5.3=14.3. 3. Compare 14.3 with meeting time 13.3.

Answer. Arrival at about 14.3:00; no, the meeting has already begun.

Knowledge takeaway. Chronology and geography interact: being able to reach a place in time is a constraint, not a fact to memorize.

Formal model. Arithmetic time model; t_arrive=t_depart+d/v; compare with event time.

Problem 79. Courier journey from Willow to Harbor

Knowledge context. Historical travel can be analyzed with ordinary distance, speed, and time reasoning when the needed values are supplied.

Given facts. A courier leaves Willow for Harbor at 10:00. Distance is 90 km and the stated average speed is 15 km/h. A council meeting in Harbor begins at 17.0:00. Ignore stops and use travel time = distance ÷ speed.

Rules. Arrival time equals departure time plus travel time. To attend an event from its beginning, arrival must be no later than the event start.

Task. When does the courier arrive, and can the courier attend the meeting from its beginning?

Step-by-step solution. 1. Travel time=90÷15=6.0 hours. 2. Arrival=10.0+6.0=16.0. 3. Compare 16.0 with meeting time 17.0.

Answer. Arrival at about 16.0:00; yes, on time.

Knowledge takeaway. Chronology and geography interact: being able to reach a place in time is a constraint, not a fact to memorize.

Formal model. Arithmetic time model; t_arrive=t_depart+d/v; compare with event time.

Problem 80. Courier journey from Northfield to Dover

Knowledge context. Historical travel can be analyzed with ordinary distance, speed, and time reasoning when the needed values are supplied.

Given facts. A courier leaves Northfield for Dover at 11:00. Distance is 100 km and the stated average speed is 15 km/h. A council meeting in Dover begins at 16.7:00. Ignore stops and use travel time = distance ÷ speed.

Rules. Arrival time equals departure time plus travel time. To attend an event from its beginning, arrival must be no later than the event start.

Task. When does the courier arrive, and can the courier attend the meeting from its beginning?

Step-by-step solution. 1. Travel time=100÷15=6.7 hours. 2. Arrival=11.0+6.7=17.7. 3. Compare 17.7 with meeting time 16.7.

Answer. Arrival at about 17.7:00; no, the meeting has already begun.

Knowledge takeaway. Chronology and geography interact: being able to reach a place in time is a constraint, not a fact to memorize.

Formal model. Arithmetic time model; t_arrive=t_depart+d/v; compare with event time.

H7. Change and continuity

Problem 81. Change and continuity in a settlement: case 1

Knowledge context. History studies both change and continuity. Comparing two snapshots is a systematic way to separate what persisted from what appeared or disappeared.

Given facts. Period 1 features: footpath, market, well, wooden bridge. Period 2 features: market, road, stone bridge, well.

Rules. A continuity is a feature present in both periods. A disappearance is present only in Period 1. An appearance is present only in Period 2. A replacement can be described when two differently named features serve the same stated role.

Task. Identify the continuities, disappearances, and appearances. Which conclusions are direct set comparisons rather than interpretations?

Step-by-step solution. 1. Intersection gives continuities: ['market', 'well']. 2. Period1−Period2 gives disappearances: ['footpath', 'wooden bridge']. 3. Period2−Period1 gives appearances: ['road', 'stone bridge']. 4. Only role-based replacement would require extra information; the set comparisons are direct.

Answer. Continuities: ['market', 'well']; disappearances: ['footpath', 'wooden bridge']; appearances: ['road', 'stone bridge'].

Knowledge takeaway. Change is not the opposite of continuity; both can occur in the same community at the same time.

Formal model. Set operations: intersection for continuity, set difference for appearance/disappearance.

Problem 82. Change and continuity in a settlement: case 2

Knowledge context. History studies both change and continuity. Comparing two snapshots is a systematic way to separate what persisted from what appeared or disappeared.

Given facts. Period 1 features: footpath, market, well, wooden bridge. Period 2 features: market, road, school, stone bridge, well.

Rules. A continuity is a feature present in both periods. A disappearance is present only in Period 1. An appearance is present only in Period 2. A replacement can be described when two differently named features serve the same stated role.

Task. Identify the continuities, disappearances, and appearances. Which conclusions are direct set comparisons rather than interpretations?

Step-by-step solution. 1. Intersection gives continuities: ['market', 'well']. 2. Period1−Period2 gives disappearances: ['footpath', 'wooden bridge']. 3. Period2−Period1 gives appearances: ['road', 'school', 'stone bridge']. 4. Only role-based replacement would require extra information; the set comparisons are direct.

Answer. Continuities: ['market', 'well']; disappearances: ['footpath', 'wooden bridge']; appearances: ['road', 'school', 'stone bridge'].

Knowledge takeaway. Change is not the opposite of continuity; both can occur in the same community at the same time.

Formal model. Set operations: intersection for continuity, set difference for appearance/disappearance.

Problem 83. Change and continuity in a settlement: case 3

Knowledge context. History studies both change and continuity. Comparing two snapshots is a systematic way to separate what persisted from what appeared or disappeared.

Given facts. Period 1 features: footpath, market, mill, well, wooden bridge. Period 2 features: market, mill, road, stone bridge, well.

Rules. A continuity is a feature present in both periods. A disappearance is present only in Period 1. An appearance is present only in Period 2. A replacement can be described when two differently named features serve the same stated role.

Task. Identify the continuities, disappearances, and appearances. Which conclusions are direct set comparisons rather than interpretations?

Step-by-step solution. 1. Intersection gives continuities: ['market', 'mill', 'well']. 2. Period1−Period2 gives disappearances: ['footpath', 'wooden bridge']. 3. Period2−Period1 gives appearances: ['road', 'stone bridge']. 4. Only role-based replacement would require extra information; the set comparisons are direct.

Answer. Continuities: ['market', 'mill', 'well']; disappearances: ['footpath', 'wooden bridge']; appearances: ['road', 'stone bridge'].

Knowledge takeaway. Change is not the opposite of continuity; both can occur in the same community at the same time.

Formal model. Set operations: intersection for continuity, set difference for appearance/disappearance.

Problem 84. Change and continuity in a settlement: case 4

Knowledge context. History studies both change and continuity. Comparing two snapshots is a systematic way to separate what persisted from what appeared or disappeared.

Given facts. Period 1 features: footpath, market, well, wooden bridge. Period 2 features: market, road, stone bridge.

Rules. A continuity is a feature present in both periods. A disappearance is present only in Period 1. An appearance is present only in Period 2. A replacement can be described when two differently named features serve the same stated role.

Task. Identify the continuities, disappearances, and appearances. Which conclusions are direct set comparisons rather than interpretations?

Step-by-step solution. 1. Intersection gives continuities: ['market']. 2. Period1−Period2 gives disappearances: ['footpath', 'well', 'wooden bridge']. 3. Period2−Period1 gives appearances: ['road', 'stone bridge']. 4. Only role-based replacement would require extra information; the set comparisons are direct.

Answer. Continuities: ['market']; disappearances: ['footpath', 'well', 'wooden bridge']; appearances: ['road', 'stone bridge'].

Knowledge takeaway. Change is not the opposite of continuity; both can occur in the same community at the same time.

Formal model. Set operations: intersection for continuity, set difference for appearance/disappearance.

Problem 85. Change and continuity in a settlement: case 5

Knowledge context. History studies both change and continuity. Comparing two snapshots is a systematic way to separate what persisted from what appeared or disappeared.

Given facts. Period 1 features: footpath, market, wall, well, wooden bridge. Period 2 features: market, park, road, stone bridge, well.

Rules. A continuity is a feature present in both periods. A disappearance is present only in Period 1. An appearance is present only in Period 2. A replacement can be described when two differently named features serve the same stated role.

Task. Identify the continuities, disappearances, and appearances. Which conclusions are direct set comparisons rather than interpretations?

Step-by-step solution. 1. Intersection gives continuities: ['market', 'well']. 2. Period1−Period2 gives disappearances: ['footpath', 'wall', 'wooden bridge']. 3. Period2−Period1 gives appearances: ['park', 'road', 'stone bridge']. 4. Only role-based replacement would require extra information; the set comparisons are direct.

Answer. Continuities: ['market', 'well']; disappearances: ['footpath', 'wall', 'wooden bridge']; appearances: ['park', 'road', 'stone bridge'].

Knowledge takeaway. Change is not the opposite of continuity; both can occur in the same community at the same time.

Formal model. Set operations: intersection for continuity, set difference for appearance/disappearance.

H8. Layers and dating objects

Problem 86. Reading an archaeological sequence: case 1

Knowledge context. Archaeological layers provide relative chronological information. Dates from objects add constraints but do not automatically make every date exact.

Given facts. An undisturbed archaeological site has four layers from top to bottom: A, B, C, D. In an undisturbed sequence, lower layers were deposited earlier. D contains an object dated 1300, C contains one dated 1450, and B contains one dated 1600.

Rules. In an undisturbed sequence, a lower layer is older than a layer above it. A securely dated object provides a constraint on the layer that contains it, but one object does not date every other object exactly.

Task. Which is older, Layer C or Layer B? Can the exact year of deposition of Layer A be deduced?

Step-by-step solution. 1. C lies below B, so C is older than B by the stratigraphic rule. 2. No exact dated object or deposition year is given for A. 3. Therefore only relative ordering, not an exact year for A, is justified.

Answer. Layer C is older than Layer B. The exact deposition year of Layer A cannot be deduced.

Knowledge takeaway. Relative dating and absolute dating answer different questions and should not be confused.

Formal model. Partial order from vertical position plus numeric constraints from dated objects.

Problem 87. Reading an archaeological sequence: case 2

Knowledge context. Archaeological layers provide relative chronological information. Dates from objects add constraints but do not automatically make every date exact.

Given facts. An undisturbed archaeological site has four layers from top to bottom: A, B, C, D. In an undisturbed sequence, lower layers were deposited earlier. D contains an object dated 1305, C contains one dated 1465, and B contains one dated 1620.

Rules. In an undisturbed sequence, a lower layer is older than a layer above it. A securely dated object provides a constraint on the layer that contains it, but one object does not date every other object exactly.

Task. Which is older, Layer C or Layer B? Can the exact year of deposition of Layer A be deduced?

Step-by-step solution. 1. C lies below B, so C is older than B by the stratigraphic rule. 2. No exact dated object or deposition year is given for A. 3. Therefore only relative ordering, not an exact year for A, is justified.

Answer. Layer C is older than Layer B. The exact deposition year of Layer A cannot be deduced.

Knowledge takeaway. Relative dating and absolute dating answer different questions and should not be confused.

Formal model. Partial order from vertical position plus numeric constraints from dated objects.

Problem 88. Reading an archaeological sequence: case 3

Knowledge context. Archaeological layers provide relative chronological information. Dates from objects add constraints but do not automatically make every date exact.

Given facts. An undisturbed archaeological site has four layers from top to bottom: A, B, C, D. In an undisturbed sequence, lower layers were deposited earlier. D contains an object dated 1310, C contains one dated 1480, and B contains one dated 1640.

Rules. In an undisturbed sequence, a lower layer is older than a layer above it. A securely dated object provides a constraint on the layer that contains it, but one object does not date every other object exactly.

Task. Which is older, Layer C or Layer B? Can the exact year of deposition of Layer A be deduced?

Step-by-step solution. 1. C lies below B, so C is older than B by the stratigraphic rule. 2. No exact dated object or deposition year is given for A. 3. Therefore only relative ordering, not an exact year for A, is justified.

Answer. Layer C is older than Layer B. The exact deposition year of Layer A cannot be deduced.

Knowledge takeaway. Relative dating and absolute dating answer different questions and should not be confused.

Formal model. Partial order from vertical position plus numeric constraints from dated objects.

Problem 89. Reading an archaeological sequence: case 4

Knowledge context. Archaeological layers provide relative chronological information. Dates from objects add constraints but do not automatically make every date exact.

Given facts. An undisturbed archaeological site has four layers from top to bottom: A, B, C, D. In an undisturbed sequence, lower layers were deposited earlier. D contains an object dated 1315, C contains one dated 1495, and B contains one dated 1660.

Rules. In an undisturbed sequence, a lower layer is older than a layer above it. A securely dated object provides a constraint on the layer that contains it, but one object does not date every other object exactly.

Task. Which is older, Layer C or Layer B? Can the exact year of deposition of Layer A be deduced?

Step-by-step solution. 1. C lies below B, so C is older than B by the stratigraphic rule. 2. No exact dated object or deposition year is given for A. 3. Therefore only relative ordering, not an exact year for A, is justified.

Answer. Layer C is older than Layer B. The exact deposition year of Layer A cannot be deduced.

Knowledge takeaway. Relative dating and absolute dating answer different questions and should not be confused.

Formal model. Partial order from vertical position plus numeric constraints from dated objects.

Problem 90. Reading an archaeological sequence: case 5

Knowledge context. Archaeological layers provide relative chronological information. Dates from objects add constraints but do not automatically make every date exact.

Given facts. An undisturbed archaeological site has four layers from top to bottom: A, B, C, D. In an undisturbed sequence, lower layers were deposited earlier. D contains an object dated 1320, C contains one dated 1510, and B contains one dated 1680.

Rules. In an undisturbed sequence, a lower layer is older than a layer above it. A securely dated object provides a constraint on the layer that contains it, but one object does not date every other object exactly.

Task. Which is older, Layer C or Layer B? Can the exact year of deposition of Layer A be deduced?

Step-by-step solution. 1. C lies below B, so C is older than B by the stratigraphic rule. 2. No exact dated object or deposition year is given for A. 3. Therefore only relative ordering, not an exact year for A, is justified.

Answer. Layer C is older than Layer B. The exact deposition year of Layer A cannot be deduced.

Knowledge takeaway. Relative dating and absolute dating answer different questions and should not be confused.

Formal model. Partial order from vertical position plus numeric constraints from dated objects.

C1. Rules and exceptions

Problem 91. Applying a rule with an exception: Owen

Knowledge context. Rules often have conditions and exceptions. Logical reading prevents us from applying a general rule when an overriding exception is active.

Given facts. Owen is 10 years old. A library rule says: a child may enter the archive room if age $\geq$ 11 AND an adult permit is present. Exception: during an evacuation drill, nobody may enter. Permit present: yes. Evacuation drill: no.

Rules. Check exceptions first if they override the general rule. Otherwise every condition joined by AND must be true.

Task. May Owen enter the archive room under these facts? Show which conditions succeed or fail.

Step-by-step solution. 1. Age test: 10 ≥ 11 is False. 2. Permit test is True. 3. Because the general rule uses AND, both tests must pass. 4. Therefore entry is not allowed.

Answer. No, Owen may not enter.

Knowledge takeaway. The order of reasoning matters when exceptions override ordinary rules.

Formal model. Boolean rule with priority: exception predicate first, then conjunction of eligibility conditions.

Problem 92. Applying a rule with an exception: Sara

Knowledge context. Rules often have conditions and exceptions. Logical reading prevents us from applying a general rule when an overriding exception is active.

Given facts. Sara is 11 years old. A library rule says: a child may enter the archive room if age $\geq$ 11 AND an adult permit is present. Exception: during an evacuation drill, nobody may enter. Permit present: no. Evacuation drill: no.

Rules. Check exceptions first if they override the general rule. Otherwise every condition joined by AND must be true.

Task. May Sara enter the archive room under these facts? Show which conditions succeed or fail.

Step-by-step solution. 1. Age test: 11 ≥ 11 is True. 2. Permit test is False. 3. Because the general rule uses AND, both tests must pass. 4. Therefore entry is not allowed.

Answer. No, Sara may not enter.

Knowledge takeaway. The order of reasoning matters when exceptions override ordinary rules.

Formal model. Boolean rule with priority: exception predicate first, then conjunction of eligibility conditions.

Problem 93. Applying a rule with an exception: Theo

Knowledge context. Rules often have conditions and exceptions. Logical reading prevents us from applying a general rule when an overriding exception is active.

Given facts. Theo is 12 years old. A library rule says: a child may enter the archive room if age ≥ 11 AND an adult permit is present. Exception: during an evacuation drill, nobody may enter. Permit present: yes. Evacuation drill: no.

Rules. Check exceptions first if they override the general rule. Otherwise every condition joined by AND must be true.

Task. May Theo enter the archive room under these facts? Show which conditions succeed or fail.

Step-by-step solution. 1. Age test: 12 ≥ 11 is True. 2. Permit test is True. 3. Because the general rule uses AND, both tests must pass. 4. Therefore entry is allowed.

Answer. Yes, Theo may enter.

Knowledge takeaway. The order of reasoning matters when exceptions override ordinary rules.

Formal model. Boolean rule with priority: exception predicate first, then conjunction of eligibility conditions.

Problem 94. Applying a rule with an exception: Ivy

Knowledge context. Rules often have conditions and exceptions. Logical reading prevents us from applying a general rule when an overriding exception is active.

Given facts. Ivy is 13 years old. A library rule says: a child may enter the archive room if age ≥ 11 AND an adult permit is present. Exception: during an evacuation drill, nobody may enter. Permit present: no. Evacuation drill: no.

Rules. Check exceptions first if they override the general rule. Otherwise every condition joined by AND must be true.

Task. May Ivy enter the archive room under these facts? Show which conditions succeed or fail.

Step-by-step solution. 1. Age test: 13 ≥ 11 is True. 2. Permit test is False. 3. Because the general rule uses AND, both tests must pass. 4. Therefore entry is not allowed.

Answer. No, Ivy may not enter.

Knowledge takeaway. The order of reasoning matters when exceptions override ordinary rules.

Formal model. Boolean rule with priority: exception predicate first, then conjunction of eligibility conditions.

Problem 95. Applying a rule with an exception: Leo

Knowledge context. Rules often have conditions and exceptions. Logical reading prevents us from applying a general rule when an overriding exception is active.

Given facts. Leo is 14 years old. A library rule says: a child may enter the archive room if age ≥ 11 AND an adult permit is present. Exception: during an evacuation drill, nobody may enter. Permit present: yes. Evacuation drill: yes.

Rules. Check exceptions first if they override the general rule. Otherwise every condition joined by AND must be true.

Task. May Leo enter the archive room under these facts? Show which conditions succeed or fail.

Step-by-step solution. 1. The evacuation exception applies and overrides ordinary eligibility. 2. Therefore entry is not allowed.

Answer. No, Leo may not enter.

Knowledge takeaway. The order of reasoning matters when exceptions override ordinary rules.

Formal model. Boolean rule with priority: exception predicate first, then conjunction of eligibility conditions.

C2. Rights as limits on decisions

Problem 96. Majority rule and a protected right: case 1

Knowledge context. Democratic procedures use voting, but constitutional systems can place some rights beyond ordinary majority power.

Given facts. A council of 18 members votes 13 to 5 for a proposal that would forbid one named group from the right to speak at a peaceful meeting. The classroom constitution states: (1) ordinary proposals pass by simple majority; (2) an ordinary majority may not remove the listed equal rights from a group.

Rules. A decision must satisfy both the voting procedure and the higher rights constraint. Passing a vote is not sufficient if the content violates a protected right.

Task. Did the proposal receive a majority? May it take effect under the stated constitution?

Step-by-step solution. 1. Vote check: 13>5, so the proposal has a simple majority. 2. Content check: the proposal removes a protected equal right (speak at a peaceful meeting) from a group. 3. The higher rights rule therefore blocks the proposal even though the vote passed.

Answer. It received a majority, but it may not take effect under the stated rights rule.

Knowledge takeaway. A majority answers who won a vote; it does not by itself answer whether every possible decision is lawful or fair under higher rules.

Formal model. Two-stage validity test: procedural predicate AND rights-compliance predicate.

Problem 97. Majority rule and a protected right: case 2

Knowledge context. Democratic procedures use voting, but constitutional systems can place some rights beyond ordinary majority power.

Given facts. A council of 20 members votes 14 to 6 for a proposal that would forbid one named group from the right to use the public library equally. The classroom constitution states: (1) ordinary proposals pass by simple majority; (2) an ordinary majority may not remove the listed equal rights from a group.

Rules. A decision must satisfy both the voting procedure and the higher rights constraint. Passing a vote is not sufficient if the content violates a protected right.

Task. Did the proposal receive a majority? May it take effect under the stated constitution?

Step-by-step solution. 1. Vote check: 14>6, so the proposal has a simple majority. 2. Content check: the proposal removes a protected equal right (use the public library equally) from a group. 3. The higher rights rule therefore blocks the proposal even though the vote passed.

Answer. It received a majority, but it may not take effect under the stated rights rule.

Knowledge takeaway. A majority answers who won a vote; it does not by itself answer whether every possible decision is lawful or fair under higher rules.

Formal model. Two-stage validity test: procedural predicate AND rights-compliance predicate.

Problem 98. Majority rule and a protected right: case 3

Knowledge context. Democratic procedures use voting, but constitutional systems can place some rights beyond ordinary majority power.

Given facts. A council of 22 members votes 15 to 7 for a proposal that would forbid one named group from the right to receive a fair hearing. The classroom constitution states: (1) ordinary proposals pass by simple majority; (2) an ordinary majority may not remove the listed equal rights from a group.

Rules. A decision must satisfy both the voting procedure and the higher rights constraint. Passing a vote is not sufficient if the content violates a protected right.

Task. Did the proposal receive a majority? May it take effect under the stated constitution?

Step-by-step solution. 1. Vote check: 15>7, so the proposal has a simple majority. 2. Content check: the proposal removes a protected equal right (receive a fair hearing) from a group. 3. The higher rights rule therefore blocks the proposal even though the vote passed.

Answer. It received a majority, but it may not take effect under the stated rights rule.

Knowledge takeaway. A majority answers who won a vote; it does not by itself answer whether every possible decision is lawful or fair under higher rules.

Formal model. Two-stage validity test: procedural predicate AND rights-compliance predicate.

Problem 99. Majority rule and a protected right: case 4

Knowledge context. Democratic procedures use voting, but constitutional systems can place some rights beyond ordinary majority power.

Given facts. A council of 24 members votes 16 to 8 for a proposal that would forbid one named group from the right to practice a lawful belief. The classroom constitution states: (1) ordinary proposals pass by simple majority; (2) an ordinary majority may not remove the listed equal rights from a group.

Rules. A decision must satisfy both the voting procedure and the higher rights constraint. Passing a vote is not sufficient if the content violates a protected right.

Task. Did the proposal receive a majority? May it take effect under the stated constitution?

Step-by-step solution. 1. Vote check: 16>8, so the proposal has a simple majority. 2. Content check: the proposal removes a protected equal right (practice a lawful belief) from a group. 3. The higher rights rule therefore blocks the proposal even though the vote passed.

Answer. It received a majority, but it may not take effect under the stated rights rule.

Knowledge takeaway. A majority answers who won a vote; it does not by itself answer whether every possible decision is lawful or fair under higher rules.

Formal model. Two-stage validity test: procedural predicate AND rights-compliance predicate.

Problem 100. Majority rule and a protected right: case 5

Knowledge context. Democratic procedures use voting, but constitutional systems can place some rights beyond ordinary majority power.

Given facts. A council of 26 members votes 17 to 9 for a proposal that would forbid one named group from the right to be protected from arbitrary punishment. The classroom constitution states: (1) ordinary proposals pass by simple majority; (2) an ordinary majority may not remove the listed equal rights from a group.

Rules. A decision must satisfy both the voting procedure and the higher rights constraint. Passing a vote is not sufficient if the content violates a protected right.

Task. Did the proposal receive a majority? May it take effect under the stated constitution?

Step-by-step solution. 1. Vote check: 17>9, so the proposal has a simple majority. 2. Content check: the proposal removes a protected equal right (be protected from arbitrary punishment) from a group. 3. The higher rights rule therefore blocks the proposal even though the vote passed.

Answer. It received a majority, but it may not take effect under the stated rights rule.

Knowledge takeaway. A majority answers who won a vote; it does not by itself answer whether every possible decision is lawful or fair under higher rules.

Formal model. Two-stage validity test: procedural predicate AND rights-compliance predicate.

C3. Voting, majority, and quorum

Problem 101. Quorum and majority in a 17-member committee

Knowledge context. Voting rules often contain more than one threshold. Quorum concerns participation; majority concerns the votes cast according to the stated rule.

Given facts. A committee has 17 members. 17 are present. Votes: yes=8, no=8, abstain=1. Rule: quorum is at least 60% of all members; if quorum exists, a motion passes when yes votes are strictly more than no votes.

Rules. First test quorum using all members as the denominator. Only then compare yes and no; abstentions count for presence but not as no votes under this rule.

Task. Is quorum met, and does the motion pass?

Step-by-step solution. 1. Quorum threshold=ceil(0.60×17)=11. 2. Present=17, so quorum is met. 3. Yes=8, no=8; yes>no is False. 4. Combine the two tests.

Answer. Quorum is met; the motion does not pass.

Knowledge takeaway. Procedural reasoning requires applying tests in the right order and using the correct denominator.

Formal model. Threshold predicates: quorum(present,total) and majority(yes,no); pass = quorum ∧ majority.

Problem 102. Quorum and majority in a 18-member committee

Knowledge context. Voting rules often contain more than one threshold. Quorum concerns participation; majority concerns the votes cast according to the stated rule.

Given facts. A committee has 18 members. 17 are present. Votes: yes=9, no=7, abstain=1. Rule: quorum is at least 60% of all members; if quorum exists, a motion passes when yes votes are strictly more than no votes.

Rules. First test quorum using all members as the denominator. Only then compare yes and no; abstentions count for presence but not as no votes under this rule.

Task. Is quorum met, and does the motion pass?

Step-by-step solution. 1. Quorum threshold=ceil(0.60×18)=11. 2. Present=17, so quorum is met. 3. Yes=9, no=7; yes>no is True. 4. Combine the two tests.

Answer. Quorum is met; the motion passes.

Knowledge takeaway. Procedural reasoning requires applying tests in the right order and using the correct denominator.

Formal model. Threshold predicates: quorum(present,total) and majority(yes,no); pass = quorum ∧ majority.

Problem 103. Quorum and majority in a 19-member committee

Knowledge context. Voting rules often contain more than one threshold. Quorum concerns participation; majority concerns the votes cast according to the stated rule.

Given facts. A committee has 19 members. 17 are present. Votes: yes=10, no=6, abstain=1. Rule: quorum is at least 60% of all members; if quorum exists, a motion passes when yes votes are strictly more than no votes.

Rules. First test quorum using all members as the denominator. Only then compare yes and no; abstentions count for presence but not as no votes under this rule.

Task. Is quorum met, and does the motion pass?

Step-by-step solution. 1. Quorum threshold=ceil(0.60×19)=12. 2. Present=17, so quorum is met. 3. Yes=10, no=6; yes>no is True. 4. Combine the two tests.

Answer. Quorum is met; the motion passes.

Knowledge takeaway. Procedural reasoning requires applying tests in the right order and using the correct denominator.

Formal model. Threshold predicates: quorum(present,total) and majority(yes,no); pass = quorum ∧ majority.

Problem 104. Quorum and majority in a 20-member committee

Knowledge context. Voting rules often contain more than one threshold. Quorum concerns participation; majority concerns the votes cast according to the stated rule.

Given facts. A committee has 20 members. 20 are present. Votes: yes=11, no=8, abstain=1. Rule: quorum is at least 60% of all members; if quorum exists, a motion passes when yes votes are strictly more than no votes.

Rules. First test quorum using all members as the denominator. Only then compare yes and no; abstentions count for presence but not as no votes under this rule.

Task. Is quorum met, and does the motion pass?

Step-by-step solution. 1. Quorum threshold=ceil(0.60×20)=12. 2. Present=20, so quorum is met. 3. Yes=11, no=8; yes>no is True. 4. Combine the two tests.

Answer. Quorum is met; the motion passes.

Knowledge takeaway. Procedural reasoning requires applying tests in the right order and using the correct denominator.

Formal model. Threshold predicates: quorum(present,total) and majority(yes,no); pass = quorum ∧ majority.

Problem 105. Quorum and majority in a 21-member committee

Knowledge context. Voting rules often contain more than one threshold. Quorum concerns participation; majority concerns the votes cast according to the stated rule.

Given facts. A committee has 21 members. 20 are present. Votes: yes=12, no=7, abstain=1. Rule: quorum is at least 60% of all members; if quorum exists, a motion passes when yes votes are strictly more than no votes.

Rules. First test quorum using all members as the denominator. Only then compare yes and no; abstentions count for presence but not as no votes under this rule.

Task. Is quorum met, and does the motion pass?

Step-by-step solution. 1. Quorum threshold=ceil(0.60×21)=13. 2. Present=20, so quorum is met. 3. Yes=12, no=7; yes>no is True. 4. Combine the two tests.

Answer. Quorum is met; the motion passes.

Knowledge takeaway. Procedural reasoning requires applying tests in the right order and using the correct denominator.

Formal model. Threshold predicates: quorum(present,total) and majority(yes,no); pass = quorum ∧ majority.

C4. Representation and seats

Problem 106. Representing three districts: case 1

Knowledge context. Representation can be modeled by rules that convert population into seats. Different allocation rules can produce different results, so the rule must be stated.

Given facts. Three districts have populations A=120, B=90, C=60. They must share 11 seats. Use this stated method: compute each exact quota population/total×seats; give each district the whole-number part; assign remaining seats to the largest fractional remainders.

Rules. Use only the stated largest-remainder method. Seat totals must add exactly to the available number.

Task. How many seats does each district receive?

Step-by-step solution. 1. Total population=270. 2. Exact quotas: A=4.89, B=3.67, C=2.44. 3. Initial whole seats: [4, 3, 2], leaving 2. 4. Give the remaining seat(s) to the largest remainder(s). 5. Final allocation=[5, 4, 2], which sums to 11.

Answer. A=5, B=4, C=2 seats.

Knowledge takeaway. Fair representation is not just arithmetic; it depends on an explicit apportionment procedure.

Formal model. Largest-remainder apportionment: quotas, floors, then rank fractional remainders.

Problem 107. Representing three districts: case 2

Knowledge context. Representation can be modeled by rules that convert population into seats. Different allocation rules can produce different results, so the rule must be stated.

Given facts. Three districts have populations A=140, B=90, C=75. They must share 11 seats. Use this stated method: compute each exact quota population/total×seats; give each district the whole-number part; assign remaining seats to the largest fractional remainders.

Rules. Use only the stated largest-remainder method. Seat totals must add exactly to the available number.

Task. How many seats does each district receive?

Step-by-step solution. 1. Total population=305. 2. Exact quotas: A=5.05, B=3.25, C=2.70. 3. Initial whole seats: [5, 3, 2], leaving 1. 4. Give the remaining seat(s) to the largest remainder(s). 5. Final allocation=[5, 3, 3], which sums to 11.

Answer. A=5, B=3, C=3 seats.

Knowledge takeaway. Fair representation is not just arithmetic; it depends on an explicit apportionment procedure.

Formal model. Largest-remainder apportionment: quotas, floors, then rank fractional remainders.

Problem 108. Representing three districts: case 3

Knowledge context. Representation can be modeled by rules that convert population into seats. Different allocation rules can produce different results, so the rule must be stated.

Given facts. Three districts have populations A=160, B=90, C=90. They must share 11 seats. Use this stated method: compute each exact quota population/total×seats; give each district the whole-number part; assign remaining seats to the largest fractional remainders.

Rules. Use only the stated largest-remainder method. Seat totals must add exactly to the available number.

Task. How many seats does each district receive?

Step-by-step solution. 1. Total population=340. 2. Exact quotas: A=5.18, B=2.91, C=2.91. 3. Initial whole seats: [5, 2, 2], leaving 2. 4. Give the remaining seat(s) to the largest remainder(s). 5. Final allocation=[5, 3, 3], which sums to 11.

Answer. A=5, B=3, C=3 seats.

Knowledge takeaway. Fair representation is not just arithmetic; it depends on an explicit apportionment procedure.

Formal model. Largest-remainder apportionment: quotas, floors, then rank fractional remainders.

Problem 109. Representing three districts: case 4

Knowledge context. Representation can be modeled by rules that convert population into seats. Different allocation rules can produce different results, so the rule must be stated.

Given facts. Three districts have populations A=180, B=90, C=105. They must share 11 seats. Use this stated method: compute each exact quota population/total×seats; give each district the whole-number part; assign remaining seats to the largest fractional remainders.

Rules. Use only the stated largest-remainder method. Seat totals must add exactly to the available number.

Task. How many seats does each district receive?

Step-by-step solution. 1. Total population=375. 2. Exact quotas: A=5.28, B=2.64, C=3.08. 3. Initial whole seats: [5, 2, 3], leaving 1. 4. Give the remaining seat(s) to the largest remainder(s). 5. Final allocation=[5, 3, 3], which sums to 11.

Answer. A=5, B=3, C=3 seats.

Knowledge takeaway. Fair representation is not just arithmetic; it depends on an explicit apportionment procedure.

Formal model. Largest-remainder apportionment: quotas, floors, then rank fractional remainders.

Problem 110. Representing three districts: case 5

Knowledge context. Representation can be modeled by rules that convert population into seats. Different allocation rules can produce different results, so the rule must be stated.

Given facts. Three districts have populations A=200, B=90, C=120. They must share 11 seats. Use this stated method: compute each exact quota population/total×seats; give each district the whole-number part; assign remaining seats to the largest fractional remainders.

Rules. Use only the stated largest-remainder method. Seat totals must add exactly to the available number.

Task. How many seats does each district receive?

Step-by-step solution. 1. Total population=410. 2. Exact quotas: A=5.37, B=2.41, C=3.22. 3. Initial whole seats: [5, 2, 3], leaving 1. 4. Give the remaining seat(s) to the largest remainder(s). 5. Final allocation=[5, 3, 3], which sums to 11.

Answer. A=5, B=3, C=3 seats.

Knowledge takeaway. Fair representation is not just arithmetic; it depends on an explicit apportionment procedure.

Formal model. Largest-remainder apportionment: quotas, floors, then rank fractional remainders.

C5. Institutions and roles

Problem 111. Which institution has the role? Repair a broken public road

Knowledge context. Civic institutions divide responsibilities. A problem can be solved by matching an action to a clearly defined function.

Given facts. Roles are defined as follows: Town Council makes local policy and approves the budget; Public Works Office maintains roads; Local Court decides legal disputes; Election Committee administers elections; School Board sets school policies. A resident asks which institution should repair a broken public road.

Rules. Match the requested action to the stated role. Do not choose an institution merely because it is powerful or familiar.

Task. Should the request go first to the Public Works Office or the School Board? Why?

Step-by-step solution. 1. Identify the action: repair a broken public road. 2. Find the role whose definition contains that function: Public Works Office. 3. School Board has a different stated role, so it is not the first match.

Answer. The request should go first to the Public Works Office.

Knowledge takeaway. Institutional reasoning is role matching, not guessing which body is “most important.”

Formal model. Bipartite relation between actions and authorized institutional roles.

Problem 112. Which institution has the role? Decide a dispute under law

Knowledge context. Civic institutions divide responsibilities. A problem can be solved by matching an action to a clearly defined function.

Given facts. Roles are defined as follows: Town Council makes local policy and approves the budget; Public Works Office maintains roads; Local Court decides legal disputes; Election Committee administers elections; School Board sets school policies. A resident asks which institution should decide a dispute under law.

Rules. Match the requested action to the stated role. Do not choose an institution merely because it is powerful or familiar.

Task. Should the request go first to the Local Court or the Election Committee? Why?

Step-by-step solution. 1. Identify the action: decide a dispute under law. 2. Find the role whose definition contains that function: Local Court. 3. Election Committee has a different stated role, so it is not the first match.

Answer. The request should go first to the Local Court.

Knowledge takeaway. Institutional reasoning is role matching, not guessing which body is “most important.”

Formal model. Bipartite relation between actions and authorized institutional roles.

Problem 113. Which institution has the role? Approve the town budget

Knowledge context. Civic institutions divide responsibilities. A problem can be solved by matching an action to a clearly defined function.

Given facts. Roles are defined as follows: Town Council makes local policy and approves the budget; Public Works Office maintains roads; Local Court decides legal disputes; Election Committee administers elections; School Board sets school policies. A resident asks which institution should approve the town budget.

Rules. Match the requested action to the stated role. Do not choose an institution merely because it is powerful or familiar.

Task. Should the request go first to the Town Council or the Public Works Office? Why?

Step-by-step solution. 1. Identify the action: approve the town budget. 2. Find the role whose definition contains that function: Town Council. 3. Public Works Office has a different stated role, so it is not the first match.

Answer. The request should go first to the Town Council.

Knowledge takeaway. Institutional reasoning is role matching, not guessing which body is “most important.”

Formal model. Bipartite relation between actions and authorized institutional roles.

Problem 114. Which institution has the role? Organize the ballot count

Knowledge context. Civic institutions divide responsibilities. A problem can be solved by matching an action to a clearly defined function.

Given facts. Roles are defined as follows: Town Council makes local policy and approves the budget; Public Works Office maintains roads; Local Court decides legal disputes; Election Committee administers elections; School Board sets school policies. A resident asks which institution should organize the ballot count.

Rules. Match the requested action to the stated role. Do not choose an institution merely because it is powerful or familiar.

Task. Should the request go first to the Election Committee or the Health Office? Why?

Step-by-step solution. 1. Identify the action: organize the ballot count. 2. Find the role whose definition contains that function: Election Committee. 3. Health Office has a different stated role, so it is not the first match.

Answer. The request should go first to the Election Committee.

Knowledge takeaway. Institutional reasoning is role matching, not guessing which body is “most important.”

Formal model. Bipartite relation between actions and authorized institutional roles.

Problem 115. Which institution has the role? Set school calendar rules

Knowledge context. Civic institutions divide responsibilities. A problem can be solved by matching an action to a clearly defined function.

Given facts. Roles are defined as follows: Town Council makes local policy and approves the budget; Public Works Office maintains roads; Local Court decides legal disputes; Election Committee administers elections; School Board sets school policies. A resident asks which institution should set school calendar rules.

Rules. Match the requested action to the stated role. Do not choose an institution merely because it is powerful or familiar.

Task. Should the request go first to the School Board or the Town Council? Why?

Step-by-step solution. 1. Identify the action: set school calendar rules. 2. Find the role whose definition contains that function: School Board. 3. Town Council has a different stated role, so it is not the first match.

Answer. The request should go first to the School Board.

Knowledge takeaway. Institutional reasoning is role matching, not guessing which body is “most important.”

Formal model. Bipartite relation between actions and authorized institutional roles.

C6. Public budgets and priorities

Problem 116. Choosing a public budget package: case 1

Knowledge context. Public budgets require choices under limited resources. A transparent model separates costs, benefits, and mandatory obligations.

Given facts. A town has a budget of 12 units. Projects (cost, benefit): park=(4,5); bridge repair=(6,9); library books=(3,4); drainage=(5,8).

Rules. Choose a subset whose total cost does not exceed the budget. Maximize total benefit. If a mandatory project is stated, every feasible plan must include it.

Task. Which set of projects gives the greatest total benefit while satisfying every constraint?

Step-by-step solution. 1. Budget limit is 12. 2. Evaluate feasible combinations (and keep the mandatory project if any). 3. The best feasible combination has cost 11 and benefit 17: ['bridge repair', 'drainage']. 4. No higher-benefit feasible combination exists under the stated costs.

Answer. Choose bridge repair, drainage; cost 11, benefit 17.

Knowledge takeaway. Optimization means comparing feasible alternatives under an explicit objective, not simply choosing the cheapest item.

Formal model. 0/1 knapsack with optional mandatory-item constraint; maximize benefit subject to total cost $\leq$ budget.

Problem 117. Choosing a public budget package: case 2

Knowledge context. Public budgets require choices under limited resources. A transparent model separates costs, benefits, and mandatory obligations.

Given facts. A town has a budget of 13 units. Projects (cost, benefit): tree planting=(5,7); bridge repair=(6,9); library books=(3,4); drainage=(5,8).

Rules. Choose a subset whose total cost does not exceed the budget. Maximize total benefit. If a mandatory project is stated, every feasible plan must include it.

Task. Which set of projects gives the greatest total benefit while satisfying every constraint?

Step-by-step solution. 1. Budget limit is 13. 2. Evaluate feasible combinations (and keep the mandatory project if any). 3. The best feasible combination has cost 13 and benefit 19: ['tree planting', 'library books', 'drainage']. 4. No higher-benefit feasible combination exists under the stated costs.

Answer. Choose tree planting, library books, drainage; cost 13, benefit 19.

Knowledge takeaway. Optimization means comparing feasible alternatives under an explicit objective, not simply choosing the cheapest item.

Formal model. 0/1 knapsack with optional mandatory-item constraint; maximize benefit subject to total cost $\leq$ budget.

Problem 118. Choosing a public budget package: case 3

Knowledge context. Public budgets require choices under limited resources. A transparent model separates costs, benefits, and mandatory obligations.

Given facts. A town has a budget of 14 units. Projects (cost, benefit): park=(4,5); bridge repair=(6,9); library books=(3,4); drainage=(5,8).

Rules. Choose a subset whose total cost does not exceed the budget. Maximize total benefit. If a mandatory project is stated, every feasible plan must include it.

Task. Which set of projects gives the greatest total benefit while satisfying every constraint?

Step-by-step solution. 1. Budget limit is 14. 2. Evaluate feasible combinations (and keep the mandatory project if any). 3. The best feasible combination has cost 14 and benefit 21: ['bridge repair', 'library books', 'drainage']. 4. No higher-benefit feasible combination exists under the stated costs.

Answer. Choose bridge repair, library books, drainage; cost 14, benefit 21.

Knowledge takeaway. Optimization means comparing feasible alternatives under an explicit objective, not simply choosing the cheapest item.

Formal model. 0/1 knapsack with optional mandatory-item constraint; maximize benefit subject to total cost $\leq$ budget.

Problem 119. Choosing a public budget package: case 4

Knowledge context. Public budgets require choices under limited resources. A transparent model separates costs, benefits, and mandatory obligations.

Given facts. A town has a budget of 15 units. Projects (cost, benefit): tree planting=(5,7); bridge repair=(6,9); library books=(3,4); drainage=(5,8).

Rules. Choose a subset whose total cost does not exceed the budget. Maximize total benefit. If a mandatory project is stated, every feasible plan must include it.

Task. Which set of projects gives the greatest total benefit while satisfying every constraint?

Step-by-step solution. 1. Budget limit is 15. 2. Evaluate feasible combinations (and keep the mandatory project if any). 3. The best feasible combination has cost 14 and benefit 21: ['bridge repair', 'library books', 'drainage']. 4. No higher-benefit feasible combination exists under the stated costs.

Answer. Choose bridge repair, library books, drainage; cost 14, benefit 21.

Knowledge takeaway. Optimization means comparing feasible alternatives under an explicit objective, not simply choosing the cheapest item.

Formal model. 0/1 knapsack with optional mandatory-item constraint; maximize benefit subject to total cost $\leq$ budget.

Problem 120. Choosing a public budget package: case 5

Knowledge context. Public budgets require choices under limited resources. A transparent model separates costs, benefits, and mandatory obligations.

Given facts. A town has a budget of 16 units. Projects (cost, benefit): park=(4,5); bridge repair=(6,9); library books=(3,4); drainage=(5,8).

Rules. Choose a subset whose total cost does not exceed the budget. Maximize total benefit. If a mandatory project is stated, every feasible plan must include it.

Task. Which set of projects gives the greatest total benefit while satisfying every constraint?

Step-by-step solution. 1. Budget limit is 16. 2. Evaluate feasible combinations (and keep the mandatory project if any). 3. The best feasible combination has cost 15 and benefit 22: ['park', 'bridge repair', 'drainage']. 4. No higher-benefit feasible combination exists under the stated costs.

Answer. Choose park, bridge repair, drainage; cost 15, benefit 22.

Knowledge takeaway. Optimization means comparing feasible alternatives under an explicit objective, not simply choosing the cheapest item.

Formal model. 0/1 knapsack with optional mandatory-item constraint; maximize benefit subject to total cost ≤ budget.

C7. Collective choice and preferences

Problem 121. Pairwise collective choice: Park versus Library

Knowledge context. Group preferences can be aggregated in different ways. Pairwise majority asks a specific question about two options at a time.

Given facts. Three public projects are considered: Park, Library, Sports Hall. Voter groups and rankings: 4 voters rank Park>Library>Sports Hall; 4 voters rank Library>Sports Hall>Park; 2 voters rank Sports Hall>Park>Library. Use pairwise majority: compare two options at a time; each voter supports the one ranked higher.

Rules. For a pairwise contest, count how many voters prefer each of the two options. Ignore the third option for that comparison.

Task. In a pairwise vote between Park and Library, which option wins?

Step-by-step solution. 1. For each group, check whether Park or Library is ranked higher. 2. Totals: Park=6, Library=4. 3. Therefore the result is Park.

Answer. Park wins 6 to 4.

Knowledge takeaway. A collective decision rule must be explicit; different rules can summarize the same rankings differently.

Formal model. Preference rankings; for pair (a,b), sum weights of voters ranking a>b versus b>a.

Problem 122. Pairwise collective choice: Library versus Sports Hall

Knowledge context. Group preferences can be aggregated in different ways. Pairwise majority asks a specific question about two options at a time.

Given facts. Three public projects are considered: Park, Library, Sports Hall. Voter groups and rankings: 5 voters rank Park>Library>Sports Hall; 4 voters rank Library>Sports Hall>Park; 3 voters rank Sports Hall>Park>Library. Use pairwise majority: compare two options at a time; each voter supports the one ranked higher.

Rules. For a pairwise contest, count how many voters prefer each of the two options. Ignore the third option for that comparison.

Task. In a pairwise vote between Library and Sports Hall, which option wins?

Step-by-step solution. 1. For each group, check whether Library or Sports Hall is ranked higher. 2. Totals: Library=9, Sports Hall=3. 3. Therefore the result is Library.

Answer. Library wins 9 to 3.

Knowledge takeaway. A collective decision rule must be explicit; different rules can summarize the same rankings differently.

Formal model. Preference rankings; for pair (a,b), sum weights of voters ranking a>b versus b>a.

Problem 123. Pairwise collective choice: Park versus Sports Hall

Knowledge context. Group preferences can be aggregated in different ways. Pairwise majority asks a specific question about two options at a time.

Given facts. Three public projects are considered: Park, Library, Sports Hall. Voter groups and rankings: 6 voters rank Park>Library>Sports Hall; 4 voters rank Library>Sports Hall>Park; 4 voters rank Sports Hall>Park>Library. Use pairwise majority: compare two options at a time; each voter supports the one ranked higher.

Rules. For a pairwise contest, count how many voters prefer each of the two options. Ignore the third option for that comparison.

Task. In a pairwise vote between Park and Sports Hall, which option wins?

Step-by-step solution. 1. For each group, check whether Park or Sports Hall is ranked higher. 2. Totals: Park=6, Sports Hall=8. 3. Therefore the result is Sports Hall.

Answer. Sports Hall wins 8 to 6.

Knowledge takeaway. A collective decision rule must be explicit; different rules can summarize the same rankings differently.

Formal model. Preference rankings; for pair (a,b), sum weights of voters ranking a>b versus b>a.

Problem 124. Pairwise collective choice: Library versus Park

Knowledge context. Group preferences can be aggregated in different ways. Pairwise majority asks a specific question about two options at a time.

Given facts. Three public projects are considered: Park, Library, Sports Hall. Voter groups and rankings: 7 voters rank Park>Library>Sports Hall; 4 voters rank Library>Sports Hall>Park; 5 voters rank Sports Hall>Park>Library. Use pairwise majority: compare two options at a time; each voter supports the one ranked higher.

Rules. For a pairwise contest, count how many voters prefer each of the two options. Ignore the third option for that comparison.

Task. In a pairwise vote between Library and Park, which option wins?

Step-by-step solution. 1. For each group, check whether Library or Park is ranked higher. 2. Totals: Library=4, Park=12. 3. Therefore the result is Park.

Answer. Park wins 12 to 4.

Knowledge takeaway. A collective decision rule must be explicit; different rules can summarize the same rankings differently.

Formal model. Preference rankings; for pair (a,b), sum weights of voters ranking a>b versus b>a.

Problem 125. Pairwise collective choice: Sports Hall versus Park

Knowledge context. Group preferences can be aggregated in different ways. Pairwise majority asks a specific question about two options at a time.

Given facts. Three public projects are considered: Park, Library, Sports Hall. Voter groups and rankings: 8 voters rank Park>Library>Sports Hall; 4 voters rank Library>Sports Hall>Park; 6 voters rank Sports Hall>Park>Library. Use pairwise majority: compare two options at a time; each voter supports the one ranked higher.

Rules. For a pairwise contest, count how many voters prefer each of the two options. Ignore the third option for that comparison.

Task. In a pairwise vote between Sports Hall and Park, which option wins?

Step-by-step solution. 1. For each group, check whether Sports Hall or Park is ranked higher. 2. Totals: Sports Hall=10, Park=8. 3. Therefore the result is Sports Hall.

Answer. Sports Hall wins 10 to 8.

Knowledge takeaway. A collective decision rule must be explicit; different rules can summarize the same rankings differently.

Formal model. Preference rankings; for pair (a,b), sum weights of voters ranking a>b versus b>a.

Part B — New and mixed reasoning domains

N1. Scale, proportion, and distance

Problem 126. Map scale 1:10000

Knowledge context. Map scale links a model to the real world through proportional reasoning.

Given facts. A map has scale 1:10000. The measured map distance between two points is 3 cm. Rule: 1 cm on the map represents 10000 cm in reality; 100,000 cm = 1 km.

Rules. Multiply map length by the scale denominator, then convert units. A scale is a ratio, not an added distance.

Task. What is the real straight-line distance in kilometres?

Step-by-step solution. 1. Real centimetres=3×10000=30000. 2. Convert to kilometres: 30000÷100,000=0.3.

Answer. 0.3 km.

Knowledge takeaway. Scale problems are ratio problems: keep units explicit and conversion becomes mechanical.

Formal model. Proportion d_real=d_map×denominator, followed by unit conversion.

Problem 127. Map scale 1:25000

Knowledge context. Map scale links a model to the real world through proportional reasoning.

Given facts. A map has scale 1:25000. The measured map distance between two points is 4 cm. Rule: 1 cm on the map represents 25000 cm in reality; 100,000 cm = 1 km.

Rules. Multiply map length by the scale denominator, then convert units. A scale is a ratio, not an added distance.

Task. What is the real straight-line distance in kilometres?

Step-by-step solution. 1. Real centimetres=4×25000=100000. 2. Convert to kilometres: 100000÷100,000=1.

Answer. 1 km.

Knowledge takeaway. Scale problems are ratio problems: keep units explicit and conversion becomes mechanical.

Formal model. Proportion d_real=d_map×denominator, followed by unit conversion.

Problem 128. Map scale 1:50000

Knowledge context. Map scale links a model to the real world through proportional reasoning.

Given facts. A map has scale 1:50000. The measured map distance between two points is 5 cm. Rule: 1 cm on the map represents 50000 cm in reality; 100,000 cm = 1 km.

Rules. Multiply map length by the scale denominator, then convert units. A scale is a ratio, not an added distance.

Task. What is the real straight-line distance in kilometres?

Step-by-step solution. 1. Real centimetres=5×50000=250000. 2. Convert to kilometres: 250000÷100,000=2.5.

Answer. 2.5 km.

Knowledge takeaway. Scale problems are ratio problems: keep units explicit and conversion becomes mechanical.

Formal model. Proportion d_real=d_map×denominator, followed by unit conversion.

Problem 129. Map scale 1:100000

Knowledge context. Map scale links a model to the real world through proportional reasoning.

Given facts. A map has scale 1:100000. The measured map distance between two points is 6 cm. Rule: 1 cm on the map represents 100000 cm in reality; 100,000 cm = 1 km.

Rules. Multiply map length by the scale denominator, then convert units. A scale is a ratio, not an added distance.

Task. What is the real straight-line distance in kilometres?

Step-by-step solution. 1. Real centimetres=6×100000=600000. 2. Convert to kilometres: 600000÷100,000=6.

Answer. 6 km.

Knowledge takeaway. Scale problems are ratio problems: keep units explicit and conversion becomes mechanical.

Formal model. Proportion d_real=d_map×denominator, followed by unit conversion.

Problem 130. Map scale 1:200000

Knowledge context. Map scale links a model to the real world through proportional reasoning.

Given facts. A map has scale 1:200000. The measured map distance between two points is 7 cm. Rule: 1 cm on the map represents 200000 cm in reality; 100,000 cm = 1 km.

Rules. Multiply map length by the scale denominator, then convert units. A scale is a ratio, not an added distance.

Task. What is the real straight-line distance in kilometres?

Step-by-step solution. 1. Real centimetres=7×200000=1400000. 2. Convert to kilometres: 1400000÷100,000=14.

Answer. 14 km.

Knowledge takeaway. Scale problems are ratio problems: keep units explicit and conversion becomes mechanical.

Formal model. Proportion d_real=d_map×denominator, followed by unit conversion.

N2. Coordinates and constraint-based location

Problem 131. Locating a point by constraints: case 1

Knowledge context. Coordinates turn location into numbers. Multiple spatial clues can be treated as simultaneous constraints.

Given facts. A point P lies on an integer grid with x,y from 0 to 5. It satisfies x+y=2 and x−y=0.

Rules. A location is valid only if it satisfies every stated equation and inequality at the same time.

Task. Which coordinate(s) can P occupy? Is the solution unique?

Step-by-step solution. 1. List pairs satisfying x+y=2. 2. Keep only those also satisfying x−y=0. 3. The remaining set is [(1, 1)].

Answer. P=(1, 1); the solution is unique.

Knowledge takeaway. A constraint problem is solved by satisfying all conditions together, not one at a time independently.

Formal model. Constraint satisfaction over integer pairs (x,y).

Problem 132. Locating a point by constraints: case 2

Knowledge context. Coordinates turn location into numbers. Multiple spatial clues can be treated as simultaneous constraints.

Given facts. A point P lies on an integer grid with x,y from 0 to 5. It satisfies x+y=5 and x−y=-1.

Rules. A location is valid only if it satisfies every stated equation and inequality at the same time.

Task. Which coordinate(s) can P occupy? Is the solution unique?

Step-by-step solution. 1. List pairs satisfying x+y=5. 2. Keep only those also satisfying x−y=-1. 3. The remaining set is [(2, 3)].

Answer. P=(2, 3); the solution is unique.

Knowledge takeaway. A constraint problem is solved by satisfying all conditions together, not one at a time independently.

Formal model. Constraint satisfaction over integer pairs (x,y).

Problem 133. Locating a point by constraints: case 3

Knowledge context. Coordinates turn location into numbers. Multiple spatial clues can be treated as simultaneous constraints.

Given facts. A point P lies on an integer grid with x,y from 0 to 5. It satisfies x+y=8 and x−y=-2.

Rules. A location is valid only if it satisfies every stated equation and inequality at the same time.

Task. Which coordinate(s) can P occupy? Is the solution unique?

Step-by-step solution. 1. List pairs satisfying x+y=8. 2. Keep only those also satisfying x−y=-2. 3. The remaining set is [(3, 5)].

Answer. P=(3, 5); the solution is unique.

Knowledge takeaway. A constraint problem is solved by satisfying all conditions together, not one at a time independently.

Formal model. Constraint satisfaction over integer pairs (x,y).

Problem 134. Locating a point by constraints: case 4

Knowledge context. Coordinates turn location into numbers. Multiple spatial clues can be treated as simultaneous constraints.

Given facts. A point P lies on an integer grid with x,y from 0 to 5. It satisfies x+y=5 and x−y=3.

Rules. A location is valid only if it satisfies every stated equation and inequality at the same time.

Task. Which coordinate(s) can P occupy? Is the solution unique?

Step-by-step solution. 1. List pairs satisfying x+y=5. 2. Keep only those also satisfying x−y=3. 3. The remaining set is [(4, 1)].

Answer. P=(4, 1); the solution is unique.

Knowledge takeaway. A constraint problem is solved by satisfying all conditions together, not one at a time independently.

Formal model. Constraint satisfaction over integer pairs (x,y).

Problem 135. Locating a point by constraints: case 5

Knowledge context. Coordinates turn location into numbers. Multiple spatial clues can be treated as simultaneous constraints.

Given facts. A point P lies on an integer grid with x,y from 0 to 5. It satisfies x+y=8 and x−y=2.

Rules. A location is valid only if it satisfies every stated equation and inequality at the same time.

Task. Which coordinate(s) can P occupy? Is the solution unique?

Step-by-step solution. 1. List pairs satisfying x+y=8. 2. Keep only those also satisfying x−y=2. 3. The remaining set is [(5, 3)].

Answer. P=(5, 3); the solution is unique.

Knowledge takeaway. A constraint problem is solved by satisfying all conditions together, not one at a time independently.

Formal model. Constraint satisfaction over integer pairs (x,y).

N3. Earth, Sun, seasons, and local time

Problem 136. Earth, sunlight, and local time: case 1

Knowledge context. Earth–Sun relationships explain broad patterns such as day length and local time, but a problem should state the simplified model it expects children to use.

Given facts. For this classroom model, Eastport is 1 time zone(s) east of Westport. Each zone east is one hour later. At Westport it is 9:00.

Rules. Moving east adds one hour per stated time zone; moving west subtracts. Use the given model only.

Task. What local time is it in Eastport?

Step-by-step solution. 1. Time difference is +1 hour(s) because Eastport is east. 2. 9+1=10 (mod 24).

Answer. 10:00.

Knowledge takeaway. A model supports only the conclusions encoded in its rules; seasonal light does not automatically determine daily weather.

Formal model. Rule-based astronomical/time-zone model; apply signed offsets or seasonal relation.

Problem 137. Earth, sunlight, and local time: case 2

Knowledge context. Earth–Sun relationships explain broad patterns such as day length and local time, but a problem should state the simplified model it expects children to use.

Given facts. For this classroom model, Eastport is 2 time zone(s) east of Westport. Each zone east is one hour later. At Westport it is 9:00.

Rules. Moving east adds one hour per stated time zone; moving west subtracts. Use the given model only.

Task. What local time is it in Eastport?

Step-by-step solution. 1. Time difference is +2 hour(s) because Eastport is east. 2. 9+2=11 (mod 24).

Answer. 11:00.

Knowledge takeaway. A model supports only the conclusions encoded in its rules; seasonal light does not automatically determine daily weather.

Formal model. Rule-based astronomical/time-zone model; apply signed offsets or seasonal relation.

Problem 138. Earth, sunlight, and local time: case 3

Knowledge context. Earth–Sun relationships explain broad patterns such as day length and local time, but a problem should state the simplified model it expects children to use.

Given facts. For this classroom model, Eastport is 3 time zone(s) east of Westport. Each zone east is one hour later. At Westport it is 9:00.

Rules. Moving east adds one hour per stated time zone; moving west subtracts. Use the given model only.

Task. What local time is it in Eastport?

Step-by-step solution. 1. Time difference is +3 hour(s) because Eastport is east. 2. 9+3=12 (mod 24).

Answer. 12:00.

Knowledge takeaway. A model supports only the conclusions encoded in its rules; seasonal light does not automatically determine daily weather.

Formal model. Rule-based astronomical/time-zone model; apply signed offsets or seasonal relation.

Problem 139. Earth, sunlight, and local time: case 4

Knowledge context. Earth–Sun relationships explain broad patterns such as day length and local time, but a problem should state the simplified model it expects children to use.

Given facts. Use this simplified model: Earth's axis is tilted. In the Northern Hemisphere, days are generally longer around June than around December. Compare June and December for the same place.

Rules. Use the stated seasonal relation; do not infer daily weather from season.

Task. Which of the two months has the longer daylight period in this model, and can that fact alone tell us whether it will rain?

Step-by-step solution. 1. Apply the stated tilt-season relation. 2. June has the longer daylight period in the Northern Hemisphere. 3. The rule contains no information about rainfall, so rainfall cannot be deduced.

Answer. June has longer daylight; rainfall cannot be determined from this fact alone.

Knowledge takeaway. A model supports only the conclusions encoded in its rules; seasonal light does not automatically determine daily weather.

Formal model. Rule-based astronomical/time-zone model; apply signed offsets or seasonal relation.

Problem 140. Earth, sunlight, and local time: case 5

Knowledge context. Earth–Sun relationships explain broad patterns such as day length and local time, but a problem should state the simplified model it expects children to use.

Given facts. Use this simplified model: Earth's axis is tilted. In the Northern Hemisphere, days are generally longer around June than around December. Compare June and December for the same place. Cross-domain check: on a field sketch, 1 cm represents 2 km and two checkpoints are 2 cm apart.

Rules. Use the stated seasonal relation; do not infer daily weather from season. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Which of the two months has the longer daylight period in this model, and can that fact alone tell us whether it will rain? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. Apply the stated tilt-season relation. 2. June has the longer daylight period in the Northern Hemisphere. 3. The rule contains no information about rainfall, so rainfall cannot be deduced. Cross-domain check: 2×2=4 km.

Answer. June has longer daylight; rainfall cannot be determined from this fact alone. Cross-domain answer: 4 km.

Knowledge takeaway. A model supports only the conclusions encoded in its rules; seasonal light does not automatically determine daily weather.

Formal model. Rule-based astronomical/time-zone model; apply signed offsets or seasonal relation.

N4. Stocks, flows, and conservation

Problem 141. Reservoir stock and flow: case 1

Knowledge context. Many geographic and civic quantities are stocks and flows: water in a reservoir, people in a city, or goods in a warehouse.

Given facts. A reservoir starts with 25 water units. Each day 5 units flow in and 4 units flow out. Assume these rates stay constant for 3 days.

Rules. Stock at the end = starting stock + total inflow − total outflow. Flow is an amount per time period; stock is the amount currently stored.

Task. How much water is stored after 3 days? Is the stock increasing or decreasing each day?

Step-by-step solution. 1. Daily net change=5−4=1. 2. Over 3 days, net change=3×1=3. 3. End stock=25+(3)=28.

Answer. 28 units; the stock is increasing.

Knowledge takeaway. Do not confuse a stored amount with a rate of change; conservation links them.

Formal model. Difference equation S(t+1)=S(t)+inflow−outflow.

Problem 142. Reservoir stock and flow: case 2

Knowledge context. Many geographic and civic quantities are stocks and flows: water in a reservoir, people in a city, or goods in a warehouse.

Given facts. A reservoir starts with 28 water units. Each day 6 units flow in and 5 units flow out. Assume these rates stay constant for 3 days.

Rules. Stock at the end = starting stock + total inflow − total outflow. Flow is an amount per time period; stock is the amount currently stored.

Task. How much water is stored after 3 days? Is the stock increasing or decreasing each day?

Step-by-step solution. 1. Daily net change=6−5=1. 2. Over 3 days, net change=3×1=3. 3. End stock=28+(3)=31.

Answer. 31 units; the stock is increasing.

Knowledge takeaway. Do not confuse a stored amount with a rate of change; conservation links them.

Formal model. Difference equation S(t+1)=S(t)+inflow−outflow.

Problem 143. Reservoir stock and flow: case 3

Knowledge context. Many geographic and civic quantities are stocks and flows: water in a reservoir, people in a city, or goods in a warehouse.

Given facts. A reservoir starts with 31 water units. Each day 7 units flow in and 4 units flow out. Assume these rates stay constant for 3 days.

Rules. Stock at the end = starting stock + total inflow − total outflow. Flow is an amount per time period; stock is the amount currently stored.

Task. How much water is stored after 3 days? Is the stock increasing or decreasing each day?

Step-by-step solution. 1. Daily net change=7−4=3. 2. Over 3 days, net change=3×3=9. 3. End stock=31+(9)=40.

Answer. 40 units; the stock is increasing.

Knowledge takeaway. Do not confuse a stored amount with a rate of change; conservation links them.

Formal model. Difference equation S(t+1)=S(t)+inflow−outflow.

Problem 144. Reservoir stock and flow: case 4

Knowledge context. Many geographic and civic quantities are stocks and flows: water in a reservoir, people in a city, or goods in a warehouse.

Given facts. A reservoir starts with 34 water units. Each day 8 units flow in and 5 units flow out. Assume these rates stay constant for 3 days.

Rules. Stock at the end = starting stock + total inflow − total outflow. Flow is an amount per time period; stock is the amount currently stored.

Task. How much water is stored after 3 days? Is the stock increasing or decreasing each day?

Step-by-step solution. 1. Daily net change=8−5=3. 2. Over 3 days, net change=3×3=9. 3. End stock=34+(9)=43.

Answer. 43 units; the stock is increasing.

Knowledge takeaway. Do not confuse a stored amount with a rate of change; conservation links them.

Formal model. Difference equation S(t+1)=S(t)+inflow−outflow.

Problem 145. Reservoir stock and flow: case 5

Knowledge context. Many geographic and civic quantities are stocks and flows: water in a reservoir, people in a city, or goods in a warehouse.

Given facts. A reservoir starts with 37 water units. Each day 9 units flow in and 4 units flow out. Assume these rates stay constant for 3 days.

Rules. Stock at the end = starting stock + total inflow − total outflow. Flow is an amount per time period; stock is the amount currently stored.

Task. How much water is stored after 3 days? Is the stock increasing or decreasing each day?

Step-by-step solution. 1. Daily net change=9−4=5. 2. Over 3 days, net change=3×5=15. 3. End stock=37+(15)=52.

Answer. 52 units; the stock is increasing.

Knowledge takeaway. Do not confuse a stored amount with a rate of change; conservation links them.

Formal model. Difference equation S(t+1)=S(t)+inflow−outflow.

N5. Density, capacity, and crowding

Problem 146. Density and capacity in a public space: case 1

Knowledge context. Density compares an amount with the space available. It can reveal crowding even when two places have similar total populations.

Given facts. A park zone covers 4 hectares and currently has 25 visitors. For this exercise, comfortable density is at most 6 visitors per hectare.

Rules. Density = number ÷ area. A zone exceeds comfortable capacity if density is greater than the stated threshold.

Task. What is the visitor density, and is the zone above the stated comfortable capacity?

Step-by-step solution. 1. Density=25÷4=6.25 visitors/hectare. 2. Compare 6.25 with the threshold 6.

Answer. 6.25 visitors/hectare; above capacity.

Knowledge takeaway. Totals and densities answer different questions; always identify the denominator.

Formal model. Ratio density=N/A followed by threshold comparison.

Problem 147. Density and capacity in a public space: case 2

Knowledge context. Density compares an amount with the space available. It can reveal crowding even when two places have similar total populations.

Given facts. A park zone covers 5 hectares and currently has 29 visitors. For this exercise, comfortable density is at most 6 visitors per hectare.

Rules. Density = number ÷ area. A zone exceeds comfortable capacity if density is greater than the stated threshold.

Task. What is the visitor density, and is the zone above the stated comfortable capacity?

Step-by-step solution. 1. Density=29÷5=5.80 visitors/hectare. 2. Compare 5.80 with the threshold 6.

Answer. 5.80 visitors/hectare; within capacity.

Knowledge takeaway. Totals and densities answer different questions; always identify the denominator.

Formal model. Ratio density=N/A followed by threshold comparison.

Problem 148. Density and capacity in a public space: case 3

Knowledge context. Density compares an amount with the space available. It can reveal crowding even when two places have similar total populations.

Given facts. A park zone covers 6 hectares and currently has 33 visitors. For this exercise, comfortable density is at most 6 visitors per hectare.

Rules. Density = number ÷ area. A zone exceeds comfortable capacity if density is greater than the stated threshold.

Task. What is the visitor density, and is the zone above the stated comfortable capacity?

Step-by-step solution. 1. Density=33÷6=5.50 visitors/hectare. 2. Compare 5.50 with the threshold 6.

Answer. 5.50 visitors/hectare; within capacity.

Knowledge takeaway. Totals and densities answer different questions; always identify the denominator.

Formal model. Ratio density=N/A followed by threshold comparison.

Problem 149. Density and capacity in a public space: case 4

Knowledge context. Density compares an amount with the space available. It can reveal crowding even when two places have similar total populations.

Given facts. A park zone covers 7 hectares and currently has 37 visitors. For this exercise, comfortable density is at most 6 visitors per hectare.

Rules. Density = number ÷ area. A zone exceeds comfortable capacity if density is greater than the stated threshold.

Task. What is the visitor density, and is the zone above the stated comfortable capacity?

Step-by-step solution. 1. Density=37÷7=5.29 visitors/hectare. 2. Compare 5.29 with the threshold 6.

Answer. 5.29 visitors/hectare; within capacity.

Knowledge takeaway. Totals and densities answer different questions; always identify the denominator.

Formal model. Ratio density=N/A followed by threshold comparison.

Problem 150. Density and capacity in a public space: case 5

Knowledge context. Density compares an amount with the space available. It can reveal crowding even when two places have similar total populations.

Given facts. A park zone covers 8 hectares and currently has 41 visitors. For this exercise, comfortable density is at most 6 visitors per hectare.

Rules. Density = number ÷ area. A zone exceeds comfortable capacity if density is greater than the stated threshold.

Task. What is the visitor density, and is the zone above the stated comfortable capacity?

Step-by-step solution. 1. Density=41÷8=5.12 visitors/hectare. 2. Compare 5.12 with the threshold 6.

Answer. 5.12 visitors/hectare; within capacity.

Knowledge takeaway. Totals and densities answer different questions; always identify the denominator.

Formal model. Ratio density=N/A followed by threshold comparison.

N6. Land use and spatial compatibility

Problem 151. Land-use compatibility on four plots: case 1

Knowledge context. Land-use planning combines spatial relationships with compatibility rules.

Given facts. Four plots lie in a line A–B–C–D. Assign one different use to each plot: school, factory, park, homes. Constraints: the factory may not be adjacent to the school; the park must be adjacent to the homes.

Rules. An assignment is valid only if every use is placed exactly once and all adjacency constraints hold.

Task. Give one valid assignment and explain why it satisfies every constraint.

Step-by-step solution. 1. Try the assignment A=school, B=park, C=homes, D=factory. 2. Factory and school are not adjacent. 3. Park and homes are adjacent. 4. Each use appears exactly once.

Answer. A=school, B=park, C=homes, D=factory is a valid assignment.

Knowledge takeaway. A place can be unsuitable not because of its own properties, but because of what would be next to it.

Formal model. Permutation CSP with all-different and adjacency/non-adjacency constraints.

Problem 152. Land-use compatibility on four plots: case 2

Knowledge context. Land-use planning combines spatial relationships with compatibility rules.

Given facts. Four plots lie in a line A–B–C–D. Assign one different use to each plot: school, factory, park, homes. Constraints: the factory may not be adjacent to the school; the park must be adjacent to the homes.

Rules. An assignment is valid only if every use is placed exactly once and all adjacency constraints hold.

Task. Give one valid assignment and explain why it satisfies every constraint.

Step-by-step solution. 1. Try the assignment A=school, B=homes, C=park, D=factory. 2. Factory and school are not adjacent. 3. Park and homes are adjacent. 4. Each use appears exactly once.

Answer. A=school, B=homes, C=park, D=factory is a valid assignment.

Knowledge takeaway. A place can be unsuitable not because of its own properties, but because of what would be next to it.

Formal model. Permutation CSP with all-different and adjacency/non-adjacency constraints.

Problem 153. Land-use compatibility on four plots: case 3

Knowledge context. Land-use planning combines spatial relationships with compatibility rules.

Given facts. Four plots lie in a line A–B–C–D. Assign one different use to each plot: school, factory, park, homes. Constraints: the factory may not be adjacent to the school; the park must be adjacent to the homes.

Rules. An assignment is valid only if every use is placed exactly once and all adjacency constraints hold.

Task. Give one valid assignment and explain why it satisfies every constraint.

Step-by-step solution. 1. Try the assignment A=factory, B=park, C=homes, D=school. 2. Factory and school are not adjacent. 3. Park and homes are adjacent. 4. Each use appears exactly once.

Answer. A=factory, B=park, C=homes, D=school is a valid assignment.

Knowledge takeaway. A place can be unsuitable not because of its own properties, but because of what would be next to it.

Formal model. Permutation CSP with all-different and adjacency/non-adjacency constraints.

Problem 154. Land-use compatibility on four plots: case 4

Knowledge context. Land-use planning combines spatial relationships with compatibility rules.

Given facts. Four plots lie in a line A–B–C–D. Assign one different use to each plot: school, factory, park, homes. Constraints: the factory may not be adjacent to the school; the park must be adjacent to the homes.

Rules. An assignment is valid only if every use is placed exactly once and all adjacency constraints hold.

Task. Give one valid assignment and explain why it satisfies every constraint.

Step-by-step solution. 1. Try the assignment A=factory, B=homes, C=park, D=school. 2. Factory and school are not adjacent. 3. Park and homes are adjacent. 4. Each use appears exactly once.

Answer. A=factory, B=homes, C=park, D=school is a valid assignment.

Knowledge takeaway. A place can be unsuitable not because of its own properties, but because of what would be next to it.

Formal model. Permutation CSP with all-different and adjacency/non-adjacency constraints.

Problem 155. Land-use compatibility on four plots: case 5

Knowledge context. Land-use planning combines spatial relationships with compatibility rules.

Given facts. Four plots lie in a line A–B–C–D. Assign one different use to each plot: school, factory, park, homes. Constraints: the factory may not be adjacent to the school; the park must be adjacent to the homes. Cross-domain check: researchers collected 11 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. An assignment is valid only if every use is placed exactly once and all adjacency constraints hold. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. Give one valid assignment and explain why it satisfies every constraint. Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. Try the assignment A=school, B=park, C=homes, D=factory. 2. Factory and school are not adjacent. 3. Park and homes are adjacent. 4. Each use appears exactly once. Cross-domain check: 11−4=7 independent reports remain.

Answer. A=school, B=park, C=homes, D=factory is a valid assignment. Cross-domain answer: 7 independent reports.

Knowledge takeaway. A place can be unsuitable not because of its own properties, but because of what would be next to it.

Formal model. Permutation CSP with all-different and adjacency/non-adjacency constraints.

N7. Natural risk: hazard, exposure, protection

Problem 156. Hazard, exposure, and protection: case 1

Knowledge context. Natural risk depends on more than the physical hazard. Exposure and protective measures also affect outcomes.

Given facts. For this classroom model, risk index = hazard × exposure ÷ protection. Data: Site A: hazard=2, exposure=4, protection=1; Site B: hazard=4, exposure=2, protection=2; Site C: hazard=4, exposure=3, protection=1.

Rules. Use the formula exactly as stated. A higher hazard does not necessarily mean higher risk if exposure or protection differs.

Task. Compute the risk index for each site. Which site has the lowest modeled risk?

Step-by-step solution. 1. Compute hazard×exposure÷protection for each site. 2. Results: A=8.00, B=4.00, C=12.00. 3. The lowest value is Site B.

Answer. A=8.00; B=4.00; C=12.00. Lowest: Site B.

Knowledge takeaway. Separating hazard from risk prevents the mistaken idea that the most dangerous process always creates the greatest overall risk.

Formal model. Explicit risk-index function over three variables; compare computed values.

Problem 157. Hazard, exposure, and protection: case 2

Knowledge context. Natural risk depends on more than the physical hazard. Exposure and protective measures also affect outcomes.

Given facts. For this classroom model, risk index = hazard × exposure ÷ protection. Data: Site A: hazard=3, exposure=4, protection=1; Site B: hazard=4, exposure=3, protection=2; Site C: hazard=4, exposure=3, protection=1.

Rules. Use the formula exactly as stated. A higher hazard does not necessarily mean higher risk if exposure or protection differs.

Task. Compute the risk index for each site. Which site has the lowest modeled risk?

Step-by-step solution. 1. Compute hazard×exposure÷protection for each site. 2. Results: A=12.00, B=6.00, C=12.00. 3. The lowest value is Site B.

Answer. A=12.00; B=6.00; C=12.00. Lowest: Site B.

Knowledge takeaway. Separating hazard from risk prevents the mistaken idea that the most dangerous process always creates the greatest overall risk.

Formal model. Explicit risk-index function over three variables; compare computed values.

Problem 158. Hazard, exposure, and protection: case 3

Knowledge context. Natural risk depends on more than the physical hazard. Exposure and protective measures also affect outcomes.

Given facts. For this classroom model, risk index = hazard × exposure ÷ protection. Data: Site A: hazard=4, exposure=4, protection=1; Site B: hazard=4, exposure=4, protection=2; Site C: hazard=4, exposure=3, protection=1.

Rules. Use the formula exactly as stated. A higher hazard does not necessarily mean higher risk if exposure or protection differs.

Task. Compute the risk index for each site. Which site has the lowest modeled risk?

Step-by-step solution. 1. Compute hazard×exposure÷protection for each site. 2. Results: A=16.00, B=8.00, C=12.00. 3. The lowest value is Site B.

Answer. A=16.00; B=8.00; C=12.00. Lowest: Site B.

Knowledge takeaway. Separating hazard from risk prevents the mistaken idea that the most dangerous process always creates the greatest overall risk.

Formal model. Explicit risk-index function over three variables; compare computed values.

Problem 159. Hazard, exposure, and protection: case 4

Knowledge context. Natural risk depends on more than the physical hazard. Exposure and protective measures also affect outcomes.

Given facts. For this classroom model, risk index = hazard × exposure ÷ protection. Data: Site A: hazard=5, exposure=4, protection=1; Site B: hazard=4, exposure=5, protection=2; Site C: hazard=4, exposure=3, protection=1.

Rules. Use the formula exactly as stated. A higher hazard does not necessarily mean higher risk if exposure or protection differs.

Task. Compute the risk index for each site. Which site has the lowest modeled risk?

Step-by-step solution. 1. Compute hazard×exposure÷protection for each site. 2. Results: A=20.00, B=10.00, C=12.00. 3. The lowest value is Site B.

Answer. A=20.00; B=10.00; C=12.00. Lowest: Site B.

Knowledge takeaway. Separating hazard from risk prevents the mistaken idea that the most dangerous process always creates the greatest overall risk.

Formal model. Explicit risk-index function over three variables; compare computed values.

Problem 160. Hazard, exposure, and protection: case 5

Knowledge context. Natural risk depends on more than the physical hazard. Exposure and protective measures also affect outcomes.

Given facts. For this classroom model, risk index = hazard × exposure ÷ protection. Data: Site A: hazard=6, exposure=4, protection=1; Site B: hazard=4, exposure=6, protection=2; Site C: hazard=4, exposure=3, protection=1.

Rules. Use the formula exactly as stated. A higher hazard does not necessarily mean higher risk if exposure or protection differs.

Task. Compute the risk index for each site. Which site has the lowest modeled risk?

Step-by-step solution. 1. Compute hazard×exposure÷protection for each site. 2. Results: A=24.00, B=12.00, C=12.00. 3. The lowest value is Site B.

Answer. A=24.00; B=12.00; C=12.00. Lowest: Site B.

Knowledge takeaway. Separating hazard from risk prevents the mistaken idea that the most dangerous process always creates the greatest overall risk.

Formal model. Explicit risk-index function over three variables; compare computed values.

N8. Ecosystems and indirect effects

Problem 161. Indirect effects in a food chain: case 1

Knowledge context. Ecosystems contain dependency networks. A change at one point can propagate indirectly through several links.

Given facts. Food relation: rabbit eats grass; fox eats rabbit; eagle eats fox. Classroom rule: if a food source sharply decreases and no alternative food is given, the consumer may also decrease.

Rules. Follow dependency arrows from food to consumer. The rule supports possible directional effects, not exact population numbers.

Task. If rabbit sharply decreases, which later populations may be indirectly affected according to the rule?

Step-by-step solution. 1. fox depends directly on rabbit, so it may decrease. 2. eagle depends on fox, so it may also be indirectly affected. 3. grass is food for rabbit; the stated rule does not imply it decreases for the same reason.

Answer. fox may decrease directly and eagle may be affected indirectly.

Knowledge takeaway. Indirect effects require following the direction of dependency and respecting the limits of the stated model.

Formal model. Directed dependency graph; propagate a qualitative decrease along outgoing dependency links.

Problem 162. Indirect effects in a food chain: case 2

Knowledge context. Ecosystems contain dependency networks. A change at one point can propagate indirectly through several links.

Given facts. Food relation: small fish eats algae; large fish eats small fish; heron eats large fish. Classroom rule: if a food source sharply decreases and no alternative food is given, the consumer may also decrease.

Rules. Follow dependency arrows from food to consumer. The rule supports possible directional effects, not exact population numbers.

Task. If small fish sharply decreases, which later populations may be indirectly affected according to the rule?

Step-by-step solution. 1. large fish depends directly on small fish, so it may decrease. 2. heron depends on large fish, so it may also be indirectly affected. 3. algae is food for small fish; the stated rule does not imply it decreases for the same reason.

Answer. large fish may decrease directly and heron may be affected indirectly.

Knowledge takeaway. Indirect effects require following the direction of dependency and respecting the limits of the stated model.

Formal model. Directed dependency graph; propagate a qualitative decrease along outgoing dependency links.

Problem 163. Indirect effects in a food chain: case 3

Knowledge context. Ecosystems contain dependency networks. A change at one point can propagate indirectly through several links.

Given facts. Food relation: rabbit eats grass; fox eats rabbit; eagle eats fox. Classroom rule: if a food source sharply decreases and no alternative food is given, the consumer may also decrease. Cross-domain check: researchers collected 9 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. Follow dependency arrows from food to consumer. The rule supports possible directional effects, not exact population numbers. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. If rabbit sharply decreases, which later populations may be indirectly affected according to the rule? Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. fox depends directly on rabbit, so it may decrease. 2. eagle depends on fox, so it may also be indirectly affected. 3. grass is food for rabbit; the stated rule does not imply it decreases for the same reason. Cross-domain check: 9−4=5 independent reports remain.

Answer. fox may decrease directly and eagle may be affected indirectly. Cross-domain answer: 5 independent reports.

Knowledge takeaway. Indirect effects require following the direction of dependency and respecting the limits of the stated model.

Formal model. Directed dependency graph; propagate a qualitative decrease along outgoing dependency links.

Problem 164. Indirect effects in a food chain: case 4

Knowledge context. Ecosystems contain dependency networks. A change at one point can propagate indirectly through several links.

Given facts. Food relation: small fish eats algae; large fish eats small fish; heron eats large fish. Classroom rule: if a food source sharply decreases and no alternative food is given, the consumer may also decrease. Cross-domain check: on a field sketch, 1 cm represents 6 km and two checkpoints are 5 cm apart.

Rules. Follow dependency arrows from food to consumer. The rule supports possible directional effects, not exact population numbers. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. If small fish sharply decreases, which later populations may be indirectly affected according to the rule? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. large fish depends directly on small fish, so it may decrease. 2. heron depends on large fish, so it may also be indirectly affected. 3. algae is food for small fish; the stated rule does not imply it decreases for the same reason. Cross-domain check: 5×6=30 km.

Answer. large fish may decrease directly and heron may be affected indirectly. Cross-domain answer: 30 km.

Knowledge takeaway. Indirect effects require following the direction of dependency and respecting the limits of the stated model.

Formal model. Directed dependency graph; propagate a qualitative decrease along outgoing dependency links.

Problem 165. Indirect effects in a food chain: case 5

Knowledge context. Ecosystems contain dependency networks. A change at one point can propagate indirectly through several links.

Given facts. Food relation: rabbit eats grass; fox eats rabbit; eagle eats fox. Classroom rule: if a food source sharply decreases and no alternative food is given, the consumer may also decrease. Cross-domain check: a survey team starts work at 12:00 and works for 2 hour(s) without a break.

Rules. Follow dependency arrows from food to consumer. The rule supports possible directional effects, not exact population numbers. For the cross-domain check, finishing time = starting time + duration.

Task. If rabbit sharply decreases, which later populations may be indirectly affected according to the rule? Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. fox depends directly on rabbit, so it may decrease. 2. eagle depends on fox, so it may also be indirectly affected. 3. grass is food for rabbit; the stated rule does not imply it decreases for the same reason. Cross-domain check: 12+2=14, so the team finishes at 14:00.

Answer. fox may decrease directly and eagle may be affected indirectly. Cross-domain answer: 14:00.

Knowledge takeaway. Indirect effects require following the direction of dependency and respecting the limits of the stated model.

Formal model. Directed dependency graph; propagate a qualitative decrease along outgoing dependency links.

N9. Renewable resources and sustainable use

Problem 166. Regeneration and harvest: case 1

Knowledge context. Renewable does not mean unlimited. A resource can decline when use exceeds regeneration.

Given facts. A forest resource starts at 110 units. Each year it regenerates 12 units and people harvest 9 units. Assume these amounts stay constant for 3 years and no other changes occur.

Rules. Annual net change = regeneration − harvest. In this simplified model, use is sustainable over the period if stock does not decline.

Task. What is the stock after 3 years, and is the harvest sustainable under this classroom definition?

Step-by-step solution. 1. Annual net change=12−9=3. 2. Three-year change=9. 3. Final stock=110+(9)=119. 4. Compare final stock with starting stock.

Answer. 119 units; sustainable under the stated definition.

Knowledge takeaway. Sustainability is a balance between renewal and extraction over time.

Formal model. Stock-flow model with regeneration and harvest; test non-decreasing stock.

Problem 167. Regeneration and harvest: case 2

Knowledge context. Renewable does not mean unlimited. A resource can decline when use exceeds regeneration.

Given facts. A forest resource starts at 110 units. Each year it regenerates 14 units and people harvest 12 units. Assume these amounts stay constant for 3 years and no other changes occur.

Rules. Annual net change = regeneration − harvest. In this simplified model, use is sustainable over the period if stock does not decline.

Task. What is the stock after 3 years, and is the harvest sustainable under this classroom definition?

Step-by-step solution. 1. Annual net change=14−12=2. 2. Three-year change=6. 3. Final stock=110+(6)=116. 4. Compare final stock with starting stock.

Answer. 116 units; sustainable under the stated definition.

Knowledge takeaway. Sustainability is a balance between renewal and extraction over time.

Formal model. Stock-flow model with regeneration and harvest; test non-decreasing stock.

Problem 168. Regeneration and harvest: case 3

Knowledge context. Renewable does not mean unlimited. A resource can decline when use exceeds regeneration.

Given facts. A forest resource starts at 110 units. Each year it regenerates 16 units and people harvest 15 units. Assume these amounts stay constant for 3 years and no other changes occur.

Rules. Annual net change = regeneration − harvest. In this simplified model, use is sustainable over the period if stock does not decline.

Task. What is the stock after 3 years, and is the harvest sustainable under this classroom definition?

Step-by-step solution. 1. Annual net change=16−15=1. 2. Three-year change=3. 3. Final stock=110+(3)=113. 4. Compare final stock with starting stock.

Answer. 113 units; sustainable under the stated definition.

Knowledge takeaway. Sustainability is a balance between renewal and extraction over time.

Formal model. Stock-flow model with regeneration and harvest; test non-decreasing stock.

Problem 169. Regeneration and harvest: case 4

Knowledge context. Renewable does not mean unlimited. A resource can decline when use exceeds regeneration.

Given facts. A forest resource starts at 110 units. Each year it regenerates 18 units and people harvest 18 units. Assume these amounts stay constant for 3 years and no other changes occur.

Rules. Annual net change = regeneration − harvest. In this simplified model, use is sustainable over the period if stock does not decline.

Task. What is the stock after 3 years, and is the harvest sustainable under this classroom definition?

Step-by-step solution. 1. Annual net change=18−18=0. 2. Three-year change=0. 3. Final stock=110+(0)=110. 4. Compare final stock with starting stock.

Answer. 110 units; sustainable under the stated definition.

Knowledge takeaway. Sustainability is a balance between renewal and extraction over time.

Formal model. Stock-flow model with regeneration and harvest; test non-decreasing stock.

Problem 170. Regeneration and harvest: case 5

Knowledge context. Renewable does not mean unlimited. A resource can decline when use exceeds regeneration.

Given facts. A forest resource starts at 110 units. Each year it regenerates 20 units and people harvest 21 units. Assume these amounts stay constant for 3 years and no other changes occur.

Rules. Annual net change = regeneration − harvest. In this simplified model, use is sustainable over the period if stock does not decline.

Task. What is the stock after 3 years, and is the harvest sustainable under this classroom definition?

Step-by-step solution. 1. Annual net change=20−21=-1. 2. Three-year change=-3. 3. Final stock=110+(-3)=107. 4. Compare final stock with starting stock.

Answer. 107 units; not sustainable under the stated definition.

Knowledge takeaway. Sustainability is a balance between renewal and extraction over time.

Formal model. Stock-flow model with regeneration and harvest; test non-decreasing stock.

N10. Migration and population balance

Problem 171. Population balance in a town: case 1

Knowledge context. Population changes through births, deaths, immigration, and emigration.

Given facts. A town begins the year with 1100 residents. During the year: births=20, deaths=11, people moving in=32, people moving out=25.

Rules. End population = start + births − deaths + in-migration − out-migration. Natural change and migration change are separate components.

Task. Compute natural change, migration change, and the year-end population.

Step-by-step solution. 1. Natural change=20−11=9. 2. Migration change=32−25=7. 3. Total change=9+7=16. 4. End population=1100+16=1116.

Answer. Natural change +9; migration change +7; end population 1116.

Knowledge takeaway. Breaking total change into components helps explain why a population grew or shrank.

Formal model. Balance equation P1=P0+births−deaths+in−out.

Problem 172. Population balance in a town: case 2

Knowledge context. Population changes through births, deaths, immigration, and emigration.

Given facts. A town begins the year with 1150 residents. During the year: births=23, deaths=11, people moving in=32, people moving out=29.

Rules. End population = start + births − deaths + in-migration − out-migration. Natural change and migration change are separate components.

Task. Compute natural change, migration change, and the year-end population.

Step-by-step solution. 1. Natural change=23−11=12. 2. Migration change=32−29=3. 3. Total change=12+3=15. 4. End population=1150+15=1165.

Answer. Natural change +12; migration change +3; end population 1165.

Knowledge takeaway. Breaking total change into components helps explain why a population grew or shrank.

Formal model. Balance equation P1=P0+births−deaths+in−out.

Problem 173. Population balance in a town: case 3

Knowledge context. Population changes through births, deaths, immigration, and emigration.

Given facts. A town begins the year with 1200 residents. During the year: births=26, deaths=11, people moving in=32, people moving out=33.

Rules. End population = start + births − deaths + in-migration − out-migration. Natural change and migration change are separate components.

Task. Compute natural change, migration change, and the year-end population.

Step-by-step solution. 1. Natural change=26−11=15. 2. Migration change=32−33=-1. 3. Total change=15+-1=14. 4. End population=1200+14=1214.

Answer. Natural change +15; migration change -1; end population 1214.

Knowledge takeaway. Breaking total change into components helps explain why a population grew or shrank.

Formal model. Balance equation P1=P0+births−deaths+in−out.

Problem 174. Population balance in a town: case 4

Knowledge context. Population changes through births, deaths, immigration, and emigration.

Given facts. A town begins the year with 1250 residents. During the year: births=29, deaths=11, people moving in=32, people moving out=37.

Rules. End population = start + births − deaths + in-migration − out-migration. Natural change and migration change are separate components.

Task. Compute natural change, migration change, and the year-end population.

Step-by-step solution. 1. Natural change=29−11=18. 2. Migration change=32−37=-5. 3. Total change=18+-5=13. 4. End population=1250+13=1263.

Answer. Natural change +18; migration change -5; end population 1263.

Knowledge takeaway. Breaking total change into components helps explain why a population grew or shrank.

Formal model. Balance equation P1=P0+births−deaths+in−out.

Problem 175. Population balance in a town: case 5

Knowledge context. Population changes through births, deaths, immigration, and emigration.

Given facts. A town begins the year with 1300 residents. During the year: births=32, deaths=11, people moving in=32, people moving out=41.

Rules. End population = start + births − deaths + in-migration − out-migration. Natural change and migration change are separate components.

Task. Compute natural change, migration change, and the year-end population.

Step-by-step solution. 1. Natural change=32−11=21. 2. Migration change=32−41=-9. 3. Total change=21+-9=12. 4. End population=1300+12=1312.

Answer. Natural change +21; migration change -9; end population 1312.

Knowledge takeaway. Breaking total change into components helps explain why a population grew or shrank.

Formal model. Balance equation P1=P0+births−deaths+in−out.

N11. Time intervals and overlapping events

Problem 176. Overlapping historical intervals: case 1

Knowledge context. Historical periods often overlap rather than forming a single neat sequence.

Given facts. Event A lasted from year 100 to 135. Event B lasted from 100 to 128. Treat intervals as continuous for this exercise.

Rules. Two intervals overlap when max(start times) < min(end times). Overlap length = min(end times) − max(start times), if positive.

Task. Did the events overlap? If so, for how many years?

Step-by-step solution. 1. Later start=max(100,100)=100. 2. Earlier end=min(135,128)=128. 3. Difference=128−100=28.

Answer. Yes, overlap 28 years.

Knowledge takeaway. Interval reasoning distinguishes “before/after” from “overlapping in time.”

Formal model. Interval intersection; compute max starts and min ends.

Problem 177. Overlapping historical intervals: case 2

Knowledge context. Historical periods often overlap rather than forming a single neat sequence.

Given facts. Event A lasted from year 120 to 155. Event B lasted from 130 to 158. Treat intervals as continuous for this exercise.

Rules. Two intervals overlap when max(start times) < min(end times). Overlap length = min(end times) − max(start times), if positive.

Task. Did the events overlap? If so, for how many years?

Step-by-step solution. 1. Later start=max(120,130)=130. 2. Earlier end=min(155,158)=155. 3. Difference=155−130=25.

Answer. Yes, overlap 25 years.

Knowledge takeaway. Interval reasoning distinguishes “before/after” from “overlapping in time.”

Formal model. Interval intersection; compute max starts and min ends.

Problem 178. Overlapping historical intervals: case 3

Knowledge context. Historical periods often overlap rather than forming a single neat sequence.

Given facts. Event A lasted from year 140 to 175. Event B lasted from 160 to 188. Treat intervals as continuous for this exercise.

Rules. Two intervals overlap when max(start times) < min(end times). Overlap length = min(end times) − max(start times), if positive.

Task. Did the events overlap? If so, for how many years?

Step-by-step solution. 1. Later start=max(140,160)=160. 2. Earlier end=min(175,188)=175. 3. Difference=175−160=15.

Answer. Yes, overlap 15 years.

Knowledge takeaway. Interval reasoning distinguishes “before/after” from “overlapping in time.”

Formal model. Interval intersection; compute max starts and min ends.

Problem 179. Overlapping historical intervals: case 4

Knowledge context. Historical periods often overlap rather than forming a single neat sequence.

Given facts. Event A lasted from year 160 to 195. Event B lasted from 160 to 188. Treat intervals as continuous for this exercise.

Rules. Two intervals overlap when max(start times) < min(end times). Overlap length = min(end times) − max(start times), if positive.

Task. Did the events overlap? If so, for how many years?

Step-by-step solution. 1. Later start=max(160,160)=160. 2. Earlier end=min(195,188)=188. 3. Difference=188−160=28.

Answer. Yes, overlap 28 years.

Knowledge takeaway. Interval reasoning distinguishes “before/after” from “overlapping in time.”

Formal model. Interval intersection; compute max starts and min ends.

Problem 180. Overlapping historical intervals: case 5

Knowledge context. Historical periods often overlap rather than forming a single neat sequence.

Given facts. Event A lasted from year 180 to 215. Event B lasted from 190 to 218. Treat intervals as continuous for this exercise.

Rules. Two intervals overlap when max(start times) < min(end times). Overlap length = min(end times) − max(start times), if positive.

Task. Did the events overlap? If so, for how many years?

Step-by-step solution. 1. Later start=max(180,190)=190. 2. Earlier end=min(215,218)=215. 3. Difference=215−190=25.

Answer. Yes, overlap 25 years.

Knowledge takeaway. Interval reasoning distinguishes “before/after” from “overlapping in time.”

Formal model. Interval intersection; compute max starts and min ends.

N12. Genealogy and generations

Problem 181. Reasoning across generations: case 1

Knowledge context. Genealogies are relational graphs. Generation distance can be counted through parent links.

Given facts. Maya is a parent of Noah. Noah is a parent of Lina. Maya is also a parent of Owen. Owen is a parent of Sara.

Rules. A parent is one generation above a child. A grandparent is two parent-links above. People who share a parent are siblings under this simplified model.

Task. What family relation between Maya and Lina follows from the stated links?

Step-by-step solution. 1. Maya→Noah is one parent link. 2. Noah→Lina is a second parent link. 3. Two links give the grandparent relation.

Answer. Maya is a grandparent of Lina.

Knowledge takeaway. Family relations can be derived step by step from a small set of explicit links.

Formal model. Directed parent graph; kinship predicates derived from path length and shared ancestors.

Problem 182. Reasoning across generations: case 2

Knowledge context. Genealogies are relational graphs. Generation distance can be counted through parent links.

Given facts. Maya is a parent of Noah. Noah is a parent of Lina. Maya is also a parent of Owen. Owen is a parent of Sara.

Rules. A parent is one generation above a child. A grandparent is two parent-links above. People who share a parent are siblings under this simplified model.

Task. What family relation between Noah and Lina follows from the stated links?

Step-by-step solution. 1. The relation is stated directly.

Answer. Noah is a parent of Lina.

Knowledge takeaway. Family relations can be derived step by step from a small set of explicit links.

Formal model. Directed parent graph; kinship predicates derived from path length and shared ancestors.

Problem 183. Reasoning across generations: case 3

Knowledge context. Genealogies are relational graphs. Generation distance can be counted through parent links.

Given facts. Maya is a parent of Noah. Noah is a parent of Lina. Maya is also a parent of Owen. Owen is a parent of Sara.

Rules. A parent is one generation above a child. A grandparent is two parent-links above. People who share a parent are siblings under this simplified model.

Task. What family relation between Noah and Owen follows from the stated links?

Step-by-step solution. 1. Maya is parent of both Noah and Owen. 2. Sharing a parent satisfies the simplified sibling rule.

Answer. Noah and Owen are siblings.

Knowledge takeaway. Family relations can be derived step by step from a small set of explicit links.

Formal model. Directed parent graph; kinship predicates derived from path length and shared ancestors.

Problem 184. Reasoning across generations: case 4

Knowledge context. Genealogies are relational graphs. Generation distance can be counted through parent links.

Given facts. Maya is a parent of Noah. Noah is a parent of Lina. Maya is also a parent of Owen. Owen is a parent of Sara.

Rules. A parent is one generation above a child. A grandparent is two parent-links above. People who share a parent are siblings under this simplified model.

Task. What family relation between Maya and Sara follows from the stated links?

Step-by-step solution. 1. Maya→Owen and Owen→Sara form two parent links.

Answer. Maya is a grandparent of Sara.

Knowledge takeaway. Family relations can be derived step by step from a small set of explicit links.

Formal model. Directed parent graph; kinship predicates derived from path length and shared ancestors.

Problem 185. Reasoning across generations: case 5

Knowledge context. Genealogies are relational graphs. Generation distance can be counted through parent links.

Given facts. Maya is a parent of Noah. Noah is a parent of Lina. Maya is also a parent of Owen. Owen is a parent of Sara.

Rules. A parent is one generation above a child. A grandparent is two parent-links above. People who share a parent are siblings under this simplified model.

Task. What family relation between Lina and Sara follows from the stated links?

Step-by-step solution. 1. Noah and Owen are siblings because they share parent Maya. 2. Lina is child of Noah; Sara is child of Owen. 3. Children of siblings are cousins in this model.

Answer. Lina and Sara are cousins under the implied family tree.

Knowledge takeaway. Family relations can be derived step by step from a small set of explicit links.

Formal model. Directed parent graph; kinship predicates derived from path length and shared ancestors.

N13. Anachronisms and temporal compatibility

Problem 186. Temporal compatibility in a story set in 1400

Knowledge context. Anachronism can be tested as a compatibility problem when the relevant date ranges are supplied.

Given facts. For this problem, availability intervals are: Technology A: 1200–1600; Institution B: 1500–1900; Tool C: 1700–2000; Practice D: 1300–1450. A story is set in 1400 and mentions Technology A and Tool C together.

Rules. An item is temporally compatible with a year only if the year lies inside its stated availability interval. An anachronism is an item placed outside its given interval.

Task. Which mentioned item, if any, is anachronistic in the story?

Step-by-step solution. 1. Check 1400 against Technology A interval 1200–1600. 2. Check 1400 against Tool C interval 1700–2000. 3. Items outside their intervals: ['Tool C'].

Answer. Anachronistic: Tool C.

Knowledge takeaway. The reasoning does not require memorizing invention dates if the intervals are part of the problem.

Formal model. Interval membership test for each object-event pair.

Problem 187. Temporal compatibility in a story set in 1550

Knowledge context. Anachronism can be tested as a compatibility problem when the relevant date ranges are supplied.

Given facts. For this problem, availability intervals are: Technology A: 1200–1600; Institution B: 1500–1900; Tool C: 1700–2000; Practice D: 1300–1450. A story is set in 1550 and mentions Technology A and Tool C together.

Rules. An item is temporally compatible with a year only if the year lies inside its stated availability interval. An anachronism is an item placed outside its given interval.

Task. Which mentioned item, if any, is anachronistic in the story?

Step-by-step solution. 1. Check 1550 against Technology A interval 1200–1600. 2. Check 1550 against Tool C interval 1700–2000. 3. Items outside their intervals: ['Tool C'].

Answer. Anachronistic: Tool C.

Knowledge takeaway. The reasoning does not require memorizing invention dates if the intervals are part of the problem.

Formal model. Interval membership test for each object-event pair.

Problem 188. Temporal compatibility in a story set in 1750

Knowledge context. Anachronism can be tested as a compatibility problem when the relevant date ranges are supplied.

Given facts. For this problem, availability intervals are: Technology A: 1200–1600; Institution B: 1500–1900; Tool C: 1700–2000; Practice D: 1300–1450. A story is set in 1750 and mentions Technology A and Tool C together.

Rules. An item is temporally compatible with a year only if the year lies inside its stated availability interval. An anachronism is an item placed outside its given interval.

Task. Which mentioned item, if any, is anachronistic in the story?

Step-by-step solution. 1. Check 1750 against Technology A interval 1200–1600. 2. Check 1750 against Tool C interval 1700–2000. 3. Items outside their intervals: ['Technology A'].

Answer. Anachronistic: Technology A.

Knowledge takeaway. The reasoning does not require memorizing invention dates if the intervals are part of the problem.

Formal model. Interval membership test for each object-event pair.

Problem 189. Temporal compatibility in a story set in 1850

Knowledge context. Anachronism can be tested as a compatibility problem when the relevant date ranges are supplied.

Given facts. For this problem, availability intervals are: Technology A: 1200–1600; Institution B: 1500–1900; Tool C: 1700–2000; Practice D: 1300–1450. A story is set in 1850 and mentions Technology A and Tool C together.

Rules. An item is temporally compatible with a year only if the year lies inside its stated availability interval. An anachronism is an item placed outside its given interval.

Task. Which mentioned item, if any, is anachronistic in the story?

Step-by-step solution. 1. Check 1850 against Technology A interval 1200–1600. 2. Check 1850 against Tool C interval 1700–2000. 3. Items outside their intervals: ['Technology A'].

Answer. Anachronistic: Technology A.

Knowledge takeaway. The reasoning does not require memorizing invention dates if the intervals are part of the problem.

Formal model. Interval membership test for each object-event pair.

Problem 190. Temporal compatibility in a story set in 1500

Knowledge context. Anachronism can be tested as a compatibility problem when the relevant date ranges are supplied.

Given facts. For this problem, availability intervals are: Technology A: 1200–1600; Institution B: 1500–1900; Tool C: 1700–2000; Practice D: 1300–1450. A story is set in 1500 and mentions Technology A and Tool C together.

Rules. An item is temporally compatible with a year only if the year lies inside its stated availability interval. An anachronism is an item placed outside its given interval.

Task. Which mentioned item, if any, is anachronistic in the story?

Step-by-step solution. 1. Check 1500 against Technology A interval 1200–1600. 2. Check 1500 against Tool C interval 1700–2000. 3. Items outside their intervals: ['Tool C'].

Answer. Anachronistic: Tool C.

Knowledge takeaway. The reasoning does not require memorizing invention dates if the intervals are part of the problem.

Formal model. Interval membership test for each object-event pair.

N14. Provenance and chain of custody

Problem 191. Provenance and chain of custody: case 1

Knowledge context. Provenance records where an object came from and how it moved between custodians.

Given facts. An artifact record states: found by Excavator on day 1; transferred to Museum A on day 2; sent from Museum A to Research Lab on day 5; returned by Research Lab to Museum B on day 8. Every transfer must name both sender and receiver.

Rules. A continuous chain of custody requires each handoff to connect the previous holder to the next. A missing sender or receiver creates a gap that limits certainty about continuous custody.

Task. Is the stated chain of custody complete under the rule? If not, identify the gap.

Step-by-step solution. 1. Read the transfers in time order. 2. Check that every transfer links the previous holder to the next. 3. Every handoff names both sides and connects continuously.

Answer. The chain is complete under the stated rule.

Knowledge takeaway. A strong provenance claim depends on an unbroken sequence of documented handoffs.

Formal model. Time-ordered path through holders; verify continuity of sender/receiver pairs.

Problem 192. Provenance and chain of custody: case 2

Knowledge context. Provenance records where an object came from and how it moved between custodians.

Given facts. An artifact record states: found by Excavator on day 1; transferred to Museum A on day 2; sent from Museum A to Research Lab on day 5; returned by Research Lab to Museum B on day 8. Every transfer must name both sender and receiver. The record for day 5 names Research Lab as receiver but does not name the sender.

Rules. A continuous chain of custody requires each handoff to connect the previous holder to the next. A missing sender or receiver creates a gap that limits certainty about continuous custody.

Task. Is the stated chain of custody complete under the rule? If not, identify the gap.

Step-by-step solution. 1. Read the transfers in time order. 2. Check that every transfer links the previous holder to the next. 3. The day-5 record lacks a sender, so the chain has a gap.

Answer. The chain is incomplete because the day-5 sender is missing.

Knowledge takeaway. A strong provenance claim depends on an unbroken sequence of documented handoffs.

Formal model. Time-ordered path through holders; verify continuity of sender/receiver pairs.

Problem 193. Provenance and chain of custody: case 3

Knowledge context. Provenance records where an object came from and how it moved between custodians.

Given facts. An artifact record states: found by Excavator on day 1; transferred to Museum A on day 2; sent from Museum A to Research Lab on day 5; returned by Research Lab to Museum B on day 8. Every transfer must name both sender and receiver. Cross-domain check: a survey team starts work at 8:00 and works for 2 hour(s) without a break.

Rules. A continuous chain of custody requires each handoff to connect the previous holder to the next. A missing sender or receiver creates a gap that limits certainty about continuous custody. For the cross-domain check, finishing time = starting time + duration.

Task. Is the stated chain of custody complete under the rule? If not, identify the gap. Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. Read the transfers in time order. 2. Check that every transfer links the previous holder to the next. 3. Every handoff names both sides and connects continuously. Cross-domain check: 8+2=10, so the team finishes at 10:00.

Answer. The chain is complete under the stated rule. Cross-domain answer: 10:00.

Knowledge takeaway. A strong provenance claim depends on an unbroken sequence of documented handoffs.

Formal model. Time-ordered path through holders; verify continuity of sender/receiver pairs.

Problem 194. Provenance and chain of custody: case 4

Knowledge context. Provenance records where an object came from and how it moved between custodians.

Given facts. An artifact record states: found by Excavator on day 1; transferred to Museum A on day 2; sent from Museum A to Research Lab on day 5; returned by Research Lab to Museum B on day 8. Every transfer must name both sender and receiver. The record for day 5 names Research Lab as receiver but does not name the sender. Cross-domain check: a local committee has 13 members, requires at least 7 present for quorum, and 11 are present.

Rules. A continuous chain of custody requires each handoff to connect the previous holder to the next. A missing sender or receiver creates a gap that limits certainty about continuous custody. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. Is the stated chain of custody complete under the rule? If not, identify the gap. Cross-domain check: is quorum met?

Step-by-step solution. 1. Read the transfers in time order. 2. Check that every transfer links the previous holder to the next. 3. The day-5 record lacks a sender, so the chain has a gap. Cross-domain check: compare 11 with 7; $11 \geq 7$ is true.

Answer. The chain is incomplete because the day-5 sender is missing. Cross-domain answer: quorum is met.

Knowledge takeaway. A strong provenance claim depends on an unbroken sequence of documented handoffs.

Formal model. Time-ordered path through holders; verify continuity of sender/receiver pairs.

Problem 195. Provenance and chain of custody: case 5

Knowledge context. Provenance records where an object came from and how it moved between custodians.

Given facts. An artifact record states: found by Excavator on day 1; transferred to Museum A on day 2; sent from Museum A to Research Lab on day 5; returned by Research Lab to Museum B on day 8. Every transfer must name both sender and receiver. Cross-domain check: researchers collected 11 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. A continuous chain of custody requires each handoff to connect the previous holder to the next. A missing sender or receiver creates a gap that limits certainty about continuous custody. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. Is the stated chain of custody complete under the rule? If not, identify the gap. Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. Read the transfers in time order. 2. Check that every transfer links the previous holder to the next. 3. Every handoff names both sides and connects continuously. Cross-domain check: 11−4=7 independent reports remain.

Answer. The chain is complete under the stated rule. Cross-domain answer: 7 independent reports.

Knowledge takeaway. A strong provenance claim depends on an unbroken sequence of documented handoffs.

Formal model. Time-ordered path through holders; verify continuity of sender/receiver pairs.

N15. Perspective, interest, and source credibility

Problem 196. Perspective, interest, and source quality: case 1

Knowledge context. Sources have perspectives and interests. Credibility is not a simple label; it depends on the claim being evaluated and how the source could know it.

Given facts. Three sources describe a proposed market tax. Source A is a merchant who would pay the tax and argues it is harmful. Source B is a council report that estimates revenue and lists its calculation. Source C is a traveler repeating a rumor heard in another town.

Rules. Perspective and interest do not automatically make a source false. Evaluate relevance, access to information, explicit evidence, and independence. A source with an interest may still provide useful facts, but its claims should be checked.

Task. Which source is best positioned to estimate the amount of revenue, and which source should be treated most cautiously for that numerical claim?

Step-by-step solution. 1. Revenue estimation requires access to tax rates and expected transactions. 2. Source B explicitly provides a calculation and is directly about the policy. 3. Source C is a secondhand rumor from elsewhere and has the weakest access to the specific numerical question. 4. Source A has an interest, but interest alone is not enough to discard it.

Answer. Source B is best positioned for the revenue estimate; Source C should be treated most cautiously for that numerical claim.

Knowledge takeaway. Bias is a reason to inspect evidence, not a magic switch that makes every statement true or false.

Formal model. Source-claim assessment using access, relevance, independence, and evidentiary support.

Problem 197. Perspective, interest, and source quality: case 2

Knowledge context. Sources have perspectives and interests. Credibility is not a simple label; it depends on the claim being evaluated and how the source could know it.

Given facts. Three sources describe a proposed market tax. Source A is a merchant who would pay the tax and argues it is harmful. Source B is a council report that estimates revenue and lists its calculation. Source C is a traveler repeating a rumor heard in another town. Cross-domain check: a survey team starts work at 12:00 and works for 2 hour(s) without a break.

Rules. Perspective and interest do not automatically make a source false. Evaluate relevance, access to information, explicit evidence, and independence. A source with an interest may still provide useful facts, but its claims should be checked. For the cross-domain check, finishing time = starting time + duration.

Task. Which source is best positioned to estimate the amount of revenue, and which source should be treated most cautiously for that numerical claim? Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. Revenue estimation requires access to tax rates and expected transactions. 2. Source B explicitly provides a calculation and is directly about the policy. 3. Source C is a secondhand rumor from elsewhere and has the weakest access to the specific numerical question. 4. Source A has an interest, but interest alone is not enough to discard it. Cross-domain check: 12+2=14, so the team finishes at 14:00.

Answer. Source B is best positioned for the revenue estimate; Source C should be treated most cautiously for that numerical claim. Cross-domain answer: 14:00.

Knowledge takeaway. Bias is a reason to inspect evidence, not a magic switch that makes every statement true or false.

Formal model. Source-claim assessment using access, relevance, independence, and evidentiary support.

Problem 198. Perspective, interest, and source quality: case 3

Knowledge context. Sources have perspectives and interests. Credibility is not a simple label; it depends on the claim being evaluated and how the source could know it.

Given facts. Three sources describe a proposed market tax. Source A is a merchant who would pay the tax and argues it is harmful. Source B is a council report that estimates revenue and lists its calculation. Source C is a traveler repeating a rumor heard in another town. Cross-domain check: a local committee has 8 members, requires at least 5 present for quorum, and 8 are present.

Rules. Perspective and interest do not automatically make a source false. Evaluate relevance, access to information, explicit evidence, and independence. A source with an interest may still provide useful facts, but its claims should be checked. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. Which source is best positioned to estimate the amount of revenue, and which source should be treated most cautiously for that numerical claim? Cross-domain check: is quorum met?

Step-by-step solution. 1. Revenue estimation requires access to tax rates and expected transactions. 2. Source B explicitly provides a calculation and is directly about the policy. 3. Source C is a secondhand rumor from elsewhere and has the weakest access to the specific numerical question. 4. Source A has an interest, but interest alone is not enough to discard it. Cross-domain check: compare 8 with 5; $8 \geq 5$ is true.

Answer. Source B is best positioned for the revenue estimate; Source C should be treated most cautiously for that numerical claim. Cross-domain answer: quorum is met.

Knowledge takeaway. Bias is a reason to inspect evidence, not a magic switch that makes every statement true or false.

Formal model. Source-claim assessment using access, relevance, independence, and evidentiary support.

Problem 199. Perspective, interest, and source quality: case 4

Knowledge context. Sources have perspectives and interests. Credibility is not a simple label; it depends on the claim being evaluated and how the source could know it.

Given facts. Three sources describe a proposed market tax. Source A is a merchant who would pay the tax and argues it is harmful. Source B is a council report that estimates revenue and lists its calculation. Source C is a traveler repeating a rumor heard in another town. Cross-domain check: researchers collected 15 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. Perspective and interest do not automatically make a source false. Evaluate relevance, access to information, explicit evidence, and independence. A source with an interest may still provide useful facts, but its claims should be checked. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. Which source is best positioned to estimate the amount of revenue, and which source should be treated most cautiously for that numerical claim? Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. Revenue estimation requires access to tax rates and expected transactions. 2. Source B explicitly provides a calculation and is directly about the policy. 3. Source C is a secondhand rumor from elsewhere and has the weakest access to the specific numerical question. 4. Source A has an interest, but interest alone is not enough to discard it. Cross-domain check: $15-4=11$ independent reports remain.

Answer. Source B is best positioned for the revenue estimate; Source C should be treated most cautiously for that numerical claim. Cross-domain answer: 11 independent reports.

Knowledge takeaway. Bias is a reason to inspect evidence, not a magic switch that makes every statement true or false.

Formal model. Source-claim assessment using access, relevance, independence, and evidentiary support.

Problem 200. Perspective, interest, and source quality: case 5

Knowledge context. Sources have perspectives and interests. Credibility is not a simple label; it depends on the claim being evaluated and how the source could know it.

Given facts. Three sources describe a proposed market tax. Source A is a merchant who would pay the tax and argues it is harmful. Source B is a council report that estimates revenue and lists its calculation. Source C is a traveler repeating a rumor heard in another town. Cross-domain check: on a field sketch, 1 cm represents 2 km and two checkpoints are 6 cm apart.

Rules. Perspective and interest do not automatically make a source false. Evaluate relevance, access to information, explicit evidence, and independence. A source with an interest may still provide useful facts, but its claims should be checked. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Which source is best positioned to estimate the amount of revenue, and which source should be treated most cautiously for that numerical claim? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. Revenue estimation requires access to tax rates and expected transactions. 2. Source B explicitly provides a calculation and is directly about the policy. 3. Source C is a secondhand rumor from elsewhere and has the weakest access to the specific numerical question. 4. Source A has an interest, but interest alone is not enough to discard it. Cross-domain check: $6\times2=12$ km.

Answer. Source B is best positioned for the revenue estimate; Source C should be treated most cautiously for that numerical claim. Cross-domain answer: 12 km.

Knowledge takeaway. Bias is a reason to inspect evidence, not a magic switch that makes every statement true or false.

Formal model. Source-claim assessment using access, relevance, independence, and evidentiary support.

N16. Uncertain data, intervals, and limits

Problem 201. Reasoning with uncertain intervals: case 1

Knowledge context. Measurements and historical dates can be uncertain. An interval represents a range of possibilities rather than one exact number.

Given facts. Two measurements are uncertain intervals. A is between 8 and 12. B is between 11 and 15. A statement “A is definitely greater than B” is valid only if A’s minimum is greater than B’s maximum.

Rules. For a definite comparison under interval uncertainty, test the worst cases. Overlapping intervals usually prevent a strict guaranteed ordering.

Task. Can we conclude that A is definitely greater than B? What comparison is justified?

Step-by-step solution. 1. A minimum=8, maximum=12; B minimum=11, maximum=15. 2. Test A_min > B_max: 8>15 is False. 3. If false, at least one allowed pair of values defeats the “definitely greater” claim.

Answer. No. The intervals do not justify saying A is definitely greater than B.

Knowledge takeaway. Robust conclusions must remain true across all values allowed by the uncertainty.

Formal model. Interval arithmetic / worst-case comparison.

Problem 202. Reasoning with uncertain intervals: case 2

Knowledge context. Measurements and historical dates can be uncertain. An interval represents a range of possibilities rather than one exact number.

Given facts. Two measurements are uncertain intervals. A is between 9 and 13. B is between 11 and 15. A statement “A is definitely greater than B” is valid only if A’s minimum is greater than B’s maximum.

Rules. For a definite comparison under interval uncertainty, test the worst cases. Overlapping intervals usually prevent a strict guaranteed ordering.

Task. Can we conclude that A is definitely greater than B? What comparison is justified?

Step-by-step solution. 1. A minimum=9, maximum=13; B minimum=11, maximum=15. 2. Test A_min > B_max: 9>15 is False. 3. If false, at least one allowed pair of values defeats the “definitely greater” claim.

Answer. No. The intervals do not justify saying A is definitely greater than B.

Knowledge takeaway. Robust conclusions must remain true across all values allowed by the uncertainty.

Formal model. Interval arithmetic / worst-case comparison.

Problem 203. Reasoning with uncertain intervals: case 3

Knowledge context. Measurements and historical dates can be uncertain. An interval represents a range of possibilities rather than one exact number.

Given facts. Two measurements are uncertain intervals. A is between 10 and 14. B is between 11 and 15. A statement “A is definitely greater than B” is valid only if A’s minimum is greater than B’s maximum.

Rules. For a definite comparison under interval uncertainty, test the worst cases. Overlapping intervals usually prevent a strict guaranteed ordering.

Task. Can we conclude that A is definitely greater than B? What comparison is justified?

Step-by-step solution. 1. A minimum=10, maximum=14; B minimum=11, maximum=15. 2. Test A_min > B_max: 10>15 is False. 3. If false, at least one allowed pair of values defeats the “definitely greater” claim.

Answer. No. The intervals do not justify saying A is definitely greater than B.

Knowledge takeaway. Robust conclusions must remain true across all values allowed by the uncertainty.

Formal model. Interval arithmetic / worst-case comparison.

Problem 204. Reasoning with uncertain intervals: case 4

Knowledge context. Measurements and historical dates can be uncertain. An interval represents a range of possibilities rather than one exact number.

Given facts. Two measurements are uncertain intervals. A is between 11 and 15. B is between 11 and 15. A statement “A is definitely greater than B” is valid only if A’s minimum is greater than B’s maximum.

Rules. For a definite comparison under interval uncertainty, test the worst cases. Overlapping intervals usually prevent a strict guaranteed ordering.

Task. Can we conclude that A is definitely greater than B? What comparison is justified?

Step-by-step solution. 1. A minimum=11, maximum=15; B minimum=11, maximum=15. 2. Test A_min > B_max: 11>15 is False. 3. If false, at least one allowed pair of values defeats the “definitely greater” claim.

Answer. No. The intervals do not justify saying A is definitely greater than B.

Knowledge takeaway. Robust conclusions must remain true across all values allowed by the uncertainty.

Formal model. Interval arithmetic / worst-case comparison.

Problem 205. Reasoning with uncertain intervals: case 5

Knowledge context. Measurements and historical dates can be uncertain. An interval represents a range of possibilities rather than one exact number.

Given facts. Two measurements are uncertain intervals. A is between 12 and 16. B is between 11 and 15. A statement “A is definitely greater than B” is valid only if A’s minimum is greater than B’s maximum.

Rules. For a definite comparison under interval uncertainty, test the worst cases. Overlapping intervals usually prevent a strict guaranteed ordering.

Task. Can we conclude that A is definitely greater than B? What comparison is justified?

Step-by-step solution. 1. A minimum=12, maximum=16; B minimum=11, maximum=15. 2. Test A_min > B_max: 12>15 is False. 3. If false, at least one allowed pair of values defeats the “definitely greater” claim.

Answer. No. The intervals do not justify saying A is definitely greater than B.

Knowledge takeaway. Robust conclusions must remain true across all values allowed by the uncertainty.

Formal model. Interval arithmetic / worst-case comparison.

N17. Counterfactual reasoning

Problem 206. Counterfactual intervention: close the gate

Knowledge context. Counterfactual reasoning asks what the model predicts if one condition were changed while the rest of the causal structure is held fixed.

Given facts. Causal rules for a simplified town model: heavy rain → high river; high river AND unclosed gate → flooded square; flooded square → market closes. Actual facts: heavy rain occurred and the gate was not closed.

Rules. A counterfactual intervention changes one specified fact and then recomputes consequences from the rules. Keep all other stated rules the same unless the intervention changes them.

Task. Under the intervention “close the gate”, what changes in the consequences?

Step-by-step solution. 1. Actual chain gives high river, then flood, then market closure. 2. Intervene: set gate=closed. 3. The flood rule requires an unclosed gate, so flooding no longer follows. 4. Without flooding, market closure does not follow.

Answer. With the gate closed, the model no longer implies flooding or market closure.

Knowledge takeaway. A counterfactual is not ordinary storytelling; it is a controlled change followed by recomputation.

Formal model. Structural causal rules with a do-style intervention on one variable.

Problem 207. Counterfactual intervention: remove heavy rain

Knowledge context. Counterfactual reasoning asks what the model predicts if one condition were changed while the rest of the causal structure is held fixed.

Given facts. Causal rules for a simplified town model: heavy rain → high river; high river AND unclosed gate → flooded square; flooded square → market closes. Actual facts: heavy rain occurred and the gate was not closed.

Rules. A counterfactual intervention changes one specified fact and then recomputes consequences from the rules. Keep all other stated rules the same unless the intervention changes them.

Task. Under the intervention “remove heavy rain”, what changes in the consequences?

Step-by-step solution. 1. Intervene: heavy rain=false. 2. Then high river is not generated by the first rule. 3. Without high river, flooding and market closure are not generated.

Answer. Without heavy rain, the model generates neither high river, flood, nor market closure.

Knowledge takeaway. A counterfactual is not ordinary storytelling; it is a controlled change followed by recomputation.

Formal model. Structural causal rules with a do-style intervention on one variable.

Problem 208. Counterfactual intervention: keep gate open

Knowledge context. Counterfactual reasoning asks what the model predicts if one condition were changed while the rest of the causal structure is held fixed.

Given facts. Causal rules for a simplified town model: heavy rain → high river; high river AND unclosed gate → flooded square; flooded square → market closes. Actual facts: heavy rain occurred and the gate was not closed.

Rules. A counterfactual intervention changes one specified fact and then recomputes consequences from the rules. Keep all other stated rules the same unless the intervention changes them.

Task. Under the intervention “keep gate open”, what changes in the consequences?

Step-by-step solution. 1. Gate already remains open. 2. Heavy rain gives high river. 3. High river plus open gate gives flood, then market closure.

Answer. The original chain remains: flood and market closure occur.

Knowledge takeaway. A counterfactual is not ordinary storytelling; it is a controlled change followed by recomputation.

Formal model. Structural causal rules with a do-style intervention on one variable.

Problem 209. Counterfactual intervention: prevent high river

Knowledge context. Counterfactual reasoning asks what the model predicts if one condition were changed while the rest of the causal structure is held fixed.

Given facts. Causal rules for a simplified town model: heavy rain → high river; high river AND unclosed gate → flooded square; flooded square → market closes. Actual facts: heavy rain occurred and the gate was not closed.

Rules. A counterfactual intervention changes one specified fact and then recomputes consequences from the rules. Keep all other stated rules the same unless the intervention changes them.

Task. Under the intervention “prevent high river”, what changes in the consequences?

Step-by-step solution. 1. Intervene directly: high river=false. 2. Even with rain, the counterfactual intervention fixes high river as prevented for this question. 3. Flood requires high river, so it does not follow.

Answer. Preventing the high river blocks the flood and market closure in this model.

Knowledge takeaway. A counterfactual is not ordinary storytelling; it is a controlled change followed by recomputation.

Formal model. Structural causal rules with a do-style intervention on one variable.

Problem 210. Counterfactual intervention: keep market open

Knowledge context. Counterfactual reasoning asks what the model predicts if one condition were changed while the rest of the causal structure is held fixed.

Given facts. Causal rules for a simplified town model: heavy rain → high river; high river AND unclosed gate → flooded square; flooded square → market closes. Actual facts: heavy rain occurred and the gate was not closed.

Rules. A counterfactual intervention changes one specified fact and then recomputes consequences from the rules. Keep all other stated rules the same unless the intervention changes them.

Task. Under the intervention “keep market open”, what changes in the consequences?

Step-by-step solution. 1. The proposed intervention changes the market outcome directly, not the upstream causes. 2. Rain still gives high river and flood. 3. Setting market=open does not erase the flood unless a rule says so.

Answer. The square still floods; the market is held open only by the direct intervention.

Knowledge takeaway. A counterfactual is not ordinary storytelling; it is a controlled change followed by recomputation.

Formal model. Structural causal rules with a do-style intervention on one variable.

N18. Public goods and common contributions

Problem 211. Funding a shared public project: case 1

Knowledge context. Public goods can benefit many people at once, which creates coordination problems about contributions.

Given facts. A community project is built only if total voluntary contributions reach 10 tokens. Contributions from 5 households are [1, 2, 3, 4, 1]. Everyone can use the project if it is built, including households that contribute zero.

Rules. Add contributions and compare with the threshold. A public-good threshold concerns whether the project is produced; it does not by itself decide whether every contribution pattern is fair.

Task. Is the project built? How many additional tokens, if any, are needed?

Step-by-step solution. 1. Total contribution=1+2+3+4+1=11. 2. Compare 11 with threshold 10. 3. Additional amount needed=max(0,10−11)=0.

Answer. The project is built. Additional tokens needed: 0.

Knowledge takeaway. Threshold provision and fairness are separate questions; first determine whether enough resources exist.

Formal model. Sum contributions; compare with provision threshold; compute shortfall.

Problem 212. Funding a shared public project: case 2

Knowledge context. Public goods can benefit many people at once, which creates coordination problems about contributions.

Given facts. A community project is built only if total voluntary contributions reach 10 tokens. Contributions from 5 households are [2, 3, 4, 1, 2]. Everyone can use the project if it is built, including households that contribute zero.

Rules. Add contributions and compare with the threshold. A public-good threshold concerns whether the project is produced; it does not by itself decide whether every contribution pattern is fair.

Task. Is the project built? How many additional tokens, if any, are needed?

Step-by-step solution. 1. Total contribution=2+3+4+1+2=12. 2. Compare 12 with threshold 10. 3. Additional amount needed=max(0,10−12)=0.

Answer. The project is built. Additional tokens needed: 0.

Knowledge takeaway. Threshold provision and fairness are separate questions; first determine whether enough resources exist.

Formal model. Sum contributions; compare with provision threshold; compute shortfall.

Problem 213. Funding a shared public project: case 3

Knowledge context. Public goods can benefit many people at once, which creates coordination problems about contributions.

Given facts. A community project is built only if total voluntary contributions reach 10 tokens. Contributions from 5 households are [3, 4, 1, 2, 3]. Everyone can use the project if it is built, including households that contribute zero.

Rules. Add contributions and compare with the threshold. A public-good threshold concerns whether the project is produced; it does not by itself decide whether every contribution pattern is fair.

Task. Is the project built? How many additional tokens, if any, are needed?

Step-by-step solution. 1. Total contribution=3+4+1+2+3=13. 2. Compare 13 with threshold 10. 3. Additional amount needed=max(0,10−13)=0.

Answer. The project is built. Additional tokens needed: 0.

Knowledge takeaway. Threshold provision and fairness are separate questions; first determine whether enough resources exist.

Formal model. Sum contributions; compare with provision threshold; compute shortfall.

Problem 214. Funding a shared public project: case 4

Knowledge context. Public goods can benefit many people at once, which creates coordination problems about contributions.

Given facts. A community project is built only if total voluntary contributions reach 10 tokens. Contributions from 5 households are [4, 1, 2, 3, 4]. Everyone can use the project if it is built, including households that contribute zero.

Rules. Add contributions and compare with the threshold. A public-good threshold concerns whether the project is produced; it does not by itself decide whether every contribution pattern is fair.

Task. Is the project built? How many additional tokens, if any, are needed?

Step-by-step solution. 1. Total contribution=4+1+2+3+4=14. 2. Compare 14 with threshold 10. 3. Additional amount needed=max(0,10−14)=0.

Answer. The project is built. Additional tokens needed: 0.

Knowledge takeaway. Threshold provision and fairness are separate questions; first determine whether enough resources exist.

Formal model. Sum contributions; compare with provision threshold; compute shortfall.

Problem 215. Funding a shared public project: case 5

Knowledge context. Public goods can benefit many people at once, which creates coordination problems about contributions.

Given facts. A community project is built only if total voluntary contributions reach 10 tokens. Contributions from 5 households are [1, 2, 3, 4, 1]. Everyone can use the project if it is built, including households that contribute zero. Cross-domain check: researchers collected 11 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. Add contributions and compare with the threshold. A public-good threshold concerns whether the project is produced; it does not by itself decide whether every contribution pattern is fair. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. Is the project built? How many additional tokens, if any, are needed? Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. Total contribution=1+2+3+4+1=11. 2. Compare 11 with threshold 10. 3. Additional amount needed=max(0,10−11)=0. Cross-domain check: 11−4=7 independent reports remain.

Answer. The project is built. Additional tokens needed: 0. Cross-domain answer: 7 independent reports.

Knowledge takeaway. Threshold provision and fairness are separate questions; first determine whether enough resources exist.

Formal model. Sum contributions; compare with provision threshold; compute shortfall.

N19. Common resources and avoiding depletion

Problem 216. Using a shared renewable resource: case 1

Knowledge context. A common resource can be overused when many individually reasonable withdrawals add up to more than regeneration.

Given facts. A shared fishery has 70 fish at the start of the period and regenerates 10. There are 4 users, each planning to take 3. For this simplified exercise, the stock should not decline below its starting level if use is to be called non-depleting.

Rules. End stock = start + regeneration − total harvest. Total harvest is number of users × harvest per user.

Task. What is the end stock? Is the planned use non-depleting under the stated definition? What equal per-user harvest would exactly match regeneration?

Step-by-step solution. 1. Total harvest=4×3=12. 2. End stock=70+10−12=68. 3. To match regeneration exactly, equal harvest per user would be 10÷4=2.50.

Answer. End stock 68; depleting. Equal regeneration-matching share: 2.50 each.

Knowledge takeaway. The shared-resource problem appears at the total level, so individual plans must be aggregated.

Formal model. Commons stock equation plus equal-share benchmark.

Problem 217. Using a shared renewable resource: case 2

Knowledge context. A common resource can be overused when many individually reasonable withdrawals add up to more than regeneration.

Given facts. A shared fishery has 70 fish at the start of the period and regenerates 11. There are 4 users, each planning to take 4. For this simplified exercise, the stock should not decline below its starting level if use is to be called non-depleting.

Rules. End stock = start + regeneration − total harvest. Total harvest is number of users × harvest per user.

Task. What is the end stock? Is the planned use non-depleting under the stated definition? What equal per-user harvest would exactly match regeneration?

Step-by-step solution. 1. Total harvest=4×4=16. 2. End stock=70+11−16=65. 3. To match regeneration exactly, equal harvest per user would be 11÷4=2.75.

Answer. End stock 65; depleting. Equal regeneration-matching share: 2.75 each.

Knowledge takeaway. The shared-resource problem appears at the total level, so individual plans must be aggregated.

Formal model. Commons stock equation plus equal-share benchmark.

Problem 218. Using a shared renewable resource: case 3

Knowledge context. A common resource can be overused when many individually reasonable withdrawals add up to more than regeneration.

Given facts. A shared fishery has 70 fish at the start of the period and regenerates 12. There are 4 users, each planning to take 5. For this simplified exercise, the stock should not decline below its starting level if use is to be called non-depleting.

Rules. End stock = start + regeneration − total harvest. Total harvest is number of users × harvest per user.

Task. What is the end stock? Is the planned use non-depleting under the stated definition? What equal per-user harvest would exactly match regeneration?

Step-by-step solution. 1. Total harvest=4×5=20. 2. End stock=70+12−20=62. 3. To match regeneration exactly, equal harvest per user would be 12÷4=3.00.

Answer. End stock 62; depleting. Equal regeneration-matching share: 3.00 each.

Knowledge takeaway. The shared-resource problem appears at the total level, so individual plans must be aggregated.

Formal model. Commons stock equation plus equal-share benchmark.

Problem 219. Using a shared renewable resource: case 4

Knowledge context. A common resource can be overused when many individually reasonable withdrawals add up to more than regeneration.

Given facts. A shared fishery has 70 fish at the start of the period and regenerates 13. There are 4 users, each planning to take 6. For this simplified exercise, the stock should not decline below its starting level if use is to be called non-depleting.

Rules. End stock = start + regeneration − total harvest. Total harvest is number of users × harvest per user.

Task. What is the end stock? Is the planned use non-depleting under the stated definition? What equal per-user harvest would exactly match regeneration?

Step-by-step solution. 1. Total harvest=4×6=24. 2. End stock=70+13−24=59. 3. To match regeneration exactly, equal harvest per user would be 13÷4=3.25.

Answer. End stock 59; depleting. Equal regeneration-matching share: 3.25 each.

Knowledge takeaway. The shared-resource problem appears at the total level, so individual plans must be aggregated.

Formal model. Commons stock equation plus equal-share benchmark.

Problem 220. Using a shared renewable resource: case 5

Knowledge context. A common resource can be overused when many individually reasonable withdrawals add up to more than regeneration.

Given facts. A shared fishery has 70 fish at the start of the period and regenerates 14. There are 4 users, each planning to take 7. For this simplified exercise, the stock should not decline below its starting level if use is to be called non-depleting.

Rules. End stock = start + regeneration − total harvest. Total harvest is number of users × harvest per user.

Task. What is the end stock? Is the planned use non-depleting under the stated definition? What equal per-user harvest would exactly match regeneration?

Step-by-step solution. 1. Total harvest=4×7=28. 2. End stock=70+14−28=56. 3. To match regeneration exactly, equal harvest per user would be 14÷4=3.50.

Answer. End stock 56; depleting. Equal regeneration-matching share: 3.50 each.

Knowledge takeaway. The shared-resource problem appears at the total level, so individual plans must be aggregated.

Formal model. Commons stock equation plus equal-share benchmark.

N20. Fairness: equal, proportional, and need-based

Problem 221. Fair allocation by the rule: equal

Knowledge context. Fairness is not one formula. Equal shares, proportional shares, and need-based shares embody different principles.

Given facts. A relief fund has 35 units for three districts A, B, C. Populations are [10, 20, 30]; need scores are [1, 3, 3]. Three fairness principles are defined: equal = same amount each; population-proportional = shares in ratio 10:20:30; need-proportional = shares in ratio 1:3:3.

Rules. Different fairness principles answer different questions. Apply the principle named in the task; do not mix them unless asked to compare.

Task. Allocate the fund equally.

Step-by-step solution. 1. Equal share=35÷3=11.67. 2. Allocation result: A=11.67, B=11.67, C=11.67.

Answer. A=11.67, B=11.67, C=11.67.

Knowledge takeaway. A sound argument states the fairness rule first and then applies it consistently.

Formal model. Allocation function parameterized by a fairness principle and weights.

Problem 222. Fair allocation by the rule: proportional to population

Knowledge context. Fairness is not one formula. Equal shares, proportional shares, and need-based shares embody different principles.

Given facts. A relief fund has 35 units for three districts A, B, C. Populations are [10, 20, 30]; need scores are [2, 3, 4]. Three fairness principles are defined: equal = same amount each; population-proportional = shares in ratio 10:20:30; need-proportional = shares in ratio 2:3:4.

Rules. Different fairness principles answer different questions. Apply the principle named in the task; do not mix them unless asked to compare.

Task. Allocate in proportion to population.

Step-by-step solution. 1. Total population weight=60. 2. Multiply the fund by each population fraction. 3. Allocation result: A=5.83, B=11.67, C=17.50.

Answer. A=5.83, B=11.67, C=17.50.

Knowledge takeaway. A sound argument states the fairness rule first and then applies it consistently.

Formal model. Allocation function parameterized by a fairness principle and weights.

Problem 223. Fair allocation by the rule: proportional to need

Knowledge context. Fairness is not one formula. Equal shares, proportional shares, and need-based shares embody different principles.

Given facts. A relief fund has 35 units for three districts A, B, C. Populations are [10, 20, 30]; need scores are [3, 3, 5]. Three fairness principles are defined: equal = same amount each; population-proportional = shares in ratio 10:20:30; need-proportional = shares in ratio 3:3:5.

Rules. Different fairness principles answer different questions. Apply the principle named in the task; do not mix them unless asked to compare.

Task. Allocate in proportion to need score.

Step-by-step solution. 1. Total need weight=11. 2. Multiply the fund by each need fraction. 3. Allocation result: A=9.55, B=9.55, C=15.91.

Answer. A=9.55, B=9.55, C=15.91.

Knowledge takeaway. A sound argument states the fairness rule first and then applies it consistently.

Formal model. Allocation function parameterized by a fairness principle and weights.

Problem 224. Fair allocation by the rule: minimum guarantee then equal remainder

Knowledge context. Fairness is not one formula. Equal shares, proportional shares, and need-based shares embody different principles.

Given facts. A relief fund has 35 units for three districts A, B, C. Populations are [10, 20, 30]; need scores are [4, 3, 6]. Three fairness principles are defined: equal = same amount each; population-proportional = shares in ratio 10:20:30; need-proportional = shares in ratio 4:3:6.

Rules. Different fairness principles answer different questions. Apply the principle named in the task; do not mix them unless asked to compare.

Task. Give each district a 5-unit minimum guarantee, then divide the remainder equally.

Step-by-step solution. 1. Minimum guarantees use 15 units. 2. Remainder=35-15=20. 3. Each gets an additional 6.67. 4. Allocation result: A=11.67, B=11.67, C=11.67.

Answer. A=11.67, B=11.67, C=11.67.

Knowledge takeaway. A sound argument states the fairness rule first and then applies it consistently.

Formal model. Allocation function parameterized by a fairness principle and weights.

Problem 225. Fair allocation by the rule: compare methods

Knowledge context. Fairness is not one formula. Equal shares, proportional shares, and need-based shares embody different principles.

Given facts. A relief fund has 35 units for three districts A, B, C. Populations are [10, 20, 30]; need scores are [5, 3, 7]. Three fairness principles are defined: equal = same amount each; population-proportional = shares in ratio 10:20:30; need-proportional = shares in ratio 5:3:7.

Rules. Different fairness principles answer different questions. Apply the principle named in the task; do not mix them unless asked to compare.

Task. Compare equal allocation with need-proportional allocation: which gives more to the highest-need district?

Step-by-step solution. 1. Equal allocation gives C 11.67. 2. Need-proportional allocation gives C 16.33. 3. Compare those amounts. 4. Allocation result: A=11.67, B=7.00, C=16.33.

Answer. A=11.67, B=7.00, C=16.33.

Knowledge takeaway. A sound argument states the fairness rule first and then applies it consistently.

Formal model. Allocation function parameterized by a fairness principle and weights.

N21. Negotiation, coalitions, and agreement

Problem 226. Coalition formation: case 1

Knowledge context. In representative bodies, groups may need to cooperate to reach a decision threshold.

Given facts. Three groups have seats A=5, B=3, C=3. A governing coalition needs at least 6 seats. Any two groups may cooperate.

Rules. A coalition is winning if the sum of its members' seats reaches the threshold. Minimal winning means it is winning but removing any member makes it lose.

Task. Which coalitions are minimal winning coalitions?

Step-by-step solution. 1. Majority threshold=6. 2. Add seats for each possible coalition. 3. Minimal winning coalitions: [('AB', 8), ('AC', 8), ('BC', 6)].

Answer. AB with 8 seats; AC with 8 seats; BC with 6 seats.

Knowledge takeaway. A coalition can be winning without being minimal; removing redundant members tests how necessary each partner is.

Formal model. Weighted coalition game; enumerate subsets, test threshold and minimality.

Problem 227. Coalition formation: case 2

Knowledge context. In representative bodies, groups may need to cooperate to reach a decision threshold.

Given facts. Three groups have seats A=5, B=4, C=3. A governing coalition needs at least 7 seats. Any two groups may cooperate.

Rules. A coalition is winning if the sum of its members' seats reaches the threshold. Minimal winning means it is winning but removing any member makes it lose.

Task. Which coalitions are minimal winning coalitions?

Step-by-step solution. 1. Majority threshold=7. 2. Add seats for each possible coalition. 3. Minimal winning coalitions: [('AB', 9), ('AC', 8), ('BC', 7)].

Answer. AB with 9 seats; AC with 8 seats; BC with 7 seats.

Knowledge takeaway. A coalition can be winning without being minimal; removing redundant members tests how necessary each partner is.

Formal model. Weighted coalition game; enumerate subsets, test threshold and minimality.

Problem 228. Coalition formation: case 3

Knowledge context. In representative bodies, groups may need to cooperate to reach a decision threshold.

Given facts. Three groups have seats A=5, B=5, C=3. A governing coalition needs at least 7 seats. Any two groups may cooperate.

Rules. A coalition is winning if the sum of its members' seats reaches the threshold. Minimal winning means it is winning but removing any member makes it lose.

Task. Which coalitions are minimal winning coalitions?

Step-by-step solution. 1. Majority threshold=7. 2. Add seats for each possible coalition. 3. Minimal winning coalitions: [('AB', 10), ('AC', 8), ('BC', 8)].

Answer. AB with 10 seats; AC with 8 seats; BC with 8 seats.

Knowledge takeaway. A coalition can be winning without being minimal; removing redundant members tests how necessary each partner is.

Formal model. Weighted coalition game; enumerate subsets, test threshold and minimality.

Problem 229. Coalition formation: case 4

Knowledge context. In representative bodies, groups may need to cooperate to reach a decision threshold.

Given facts. Three groups have seats A=5, B=6, C=3. A governing coalition needs at least 8 seats. Any two groups may cooperate.

Rules. A coalition is winning if the sum of its members' seats reaches the threshold. Minimal winning means it is winning but removing any member makes it lose.

Task. Which coalitions are minimal winning coalitions?

Step-by-step solution. 1. Majority threshold=8. 2. Add seats for each possible coalition. 3. Minimal winning coalitions: [('AB', 11), ('AC', 8), ('BC', 9)].

Answer. AB with 11 seats; AC with 8 seats; BC with 9 seats.

Knowledge takeaway. A coalition can be winning without being minimal; removing redundant members tests how necessary each partner is.

Formal model. Weighted coalition game; enumerate subsets, test threshold and minimality.

Problem 230. Coalition formation: case 5

Knowledge context. In representative bodies, groups may need to cooperate to reach a decision threshold.

Given facts. Three groups have seats A=5, B=7, C=3. A governing coalition needs at least 8 seats. Any two groups may cooperate.

Rules. A coalition is winning if the sum of its members' seats reaches the threshold. Minimal winning means it is winning but removing any member makes it lose.

Task. Which coalitions are minimal winning coalitions?

Step-by-step solution. 1. Majority threshold=8. 2. Add seats for each possible coalition. 3. Minimal winning coalitions: [('AB', 12), ('AC', 8), ('BC', 10)].

Answer. AB with 12 seats; AC with 8 seats; BC with 10 seats.

Knowledge takeaway. A coalition can be winning without being minimal; removing redundant members tests how necessary each partner is.

Formal model. Weighted coalition game; enumerate subsets, test threshold and minimality.

N22. Due process, evidence, and right of reply

Problem 231. Procedure, evidence, and a right to respond: case 1

Knowledge context. Fair procedures often combine evidentiary standards with an opportunity for the affected person to answer the case.

Given facts. A school rule says a student may receive a serious penalty only if: (1) at least two independent pieces of evidence support the allegation, and (2) the student is given a chance to respond before the final decision. In a case, one camera record and one independent witness support the allegation, but the student was not allowed to respond.

Rules. Independence concerns distinct information sources, not merely the number of people repeating one source. Every required procedural condition must be satisfied.

Task. May the serious penalty be imposed under the stated procedure? Identify the condition that succeeds or fails.

Step-by-step solution. 1. Camera and witness are independent, so the evidence condition is met. 2. The student was not given a chance to respond. 3. The procedure requires both conditions, so the penalty is not authorized.

Answer. No. The right-to-respond condition is not met.

Knowledge takeaway. A correct outcome does not excuse skipping a required procedure when the rule explicitly makes procedure part of validity.

Formal model. Conjunctive procedural predicate over evidence independence/count and response opportunity.

Problem 232. Procedure, evidence, and a right to respond: case 2

Knowledge context. Fair procedures often combine evidentiary standards with an opportunity for the affected person to answer the case.

Given facts. A school rule says a serious penalty requires two independent pieces of evidence and a chance to respond. In this case there are two friends repeating the same rumor from one original speaker, plus a chance to respond.

Rules. Independence concerns distinct information sources, not merely the number of people repeating one source. Every required procedural condition must be satisfied.

Task. May the serious penalty be imposed under the stated procedure? Identify the condition that succeeds or fails.

Step-by-step solution. 1. The two friends trace to the same original rumor, so they are not two independent pieces under the rule. 2. The response condition is met. 3. Because the evidence condition fails, the serious penalty is not authorized.

Answer. No. The evidence requirement is not met.

Knowledge takeaway. A correct outcome does not excuse skipping a required procedure when the rule explicitly makes procedure part of validity.

Formal model. Conjunctive procedural predicate over evidence independence/count and response opportunity.

Problem 233. Procedure, evidence, and a right to respond: case 3

Knowledge context. Fair procedures often combine evidentiary standards with an opportunity for the affected person to answer the case.

Given facts. A school rule says a student may receive a serious penalty only if: (1) at least two independent pieces of evidence support the allegation, and (2) the student is given a chance to respond before the final decision. In a case, one camera record and one independent witness support the allegation, but the student was not allowed to respond. Cross-domain check: a survey team starts work at 8:00 and works for 2 hour(s) without a break.

Rules. Independence concerns distinct information sources, not merely the number of people repeating one source. Every required procedural condition must be satisfied. For the cross-domain check, finishing time = starting time + duration.

Task. May the serious penalty be imposed under the stated procedure? Identify the condition that succeeds or fails. Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. Camera and witness are independent, so the evidence condition is met. 2. The student was not given a chance to respond. 3. The procedure requires both conditions, so the penalty is not authorized. Cross-domain check: 8+2=10, so the team finishes at 10:00.

Answer. No. The right-to-respond condition is not met. Cross-domain answer: 10:00.

Knowledge takeaway. A correct outcome does not excuse skipping a required procedure when the rule explicitly makes procedure part of validity.

Formal model. Conjunctive procedural predicate over evidence independence/count and response opportunity.

Problem 234. Procedure, evidence, and a right to respond: case 4

Knowledge context. Fair procedures often combine evidentiary standards with an opportunity for the affected person to answer the case.

Given facts. A school rule says a serious penalty requires two independent pieces of evidence and a chance to respond. In this case there are two friends repeating the same rumor from one original speaker, plus a chance to respond. Cross-domain check: a local committee has 8 members, requires at least 5 present for quorum, and 8 are present.

Rules. Independence concerns distinct information sources, not merely the number of people repeating one source. Every required procedural condition must be satisfied. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. May the serious penalty be imposed under the stated procedure? Identify the condition that succeeds or fails. Cross-domain check: is quorum met?

Step-by-step solution. 1. The two friends trace to the same original rumor, so they are not two independent pieces under the rule. 2. The response condition is met. 3. Because the evidence condition fails, the serious penalty is not authorized. Cross-domain check: compare 8 with 5; $8 \geq 5$ is true.

Answer. No. The evidence requirement is not met. Cross-domain answer: quorum is met.

Knowledge takeaway. A correct outcome does not excuse skipping a required procedure when the rule explicitly makes procedure part of validity.

Formal model. Conjunctive procedural predicate over evidence independence/count and response opportunity.

Problem 235. Procedure, evidence, and a right to respond: case 5

Knowledge context. Fair procedures often combine evidentiary standards with an opportunity for the affected person to answer the case.

Given facts. A school rule says a student may receive a serious penalty only if: (1) at least two independent pieces of evidence support the allegation, and (2) the student is given a chance to respond before the final decision. In a case, one camera record and one independent witness support the allegation, but the student was not allowed to respond. Cross-domain check: researchers collected 11 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. Independence concerns distinct information sources, not merely the number of people repeating one source. Every required procedural condition must be satisfied. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. May the serious penalty be imposed under the stated procedure? Identify the condition that succeeds or fails. Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. Camera and witness are independent, so the evidence condition is met. 2. The student was not given a chance to respond. 3. The procedure requires both conditions, so the penalty is not authorized. Cross-domain check: 11−4=7 independent reports remain.

Answer. No. The right-to-respond condition is not met. Cross-domain answer: 7 independent reports.

Knowledge takeaway. A correct outcome does not excuse skipping a required procedure when the rule explicitly makes procedure part of validity.

Formal model. Conjunctive procedural predicate over evidence independence/count and response opportunity.

N23. Samples, surveys, and public opinion

Problem 236. Sampling and opinion: case 1

Knowledge context. Surveys use samples to learn about larger populations. The selection process matters as much as the number of answers.

Given facts. A town wants to estimate how many residents use buses. It surveys only people waiting at the bus station.

Rules. A sample is more representative when selection does not systematically favor people connected to the answer. Random or broad systematic sampling can reduce selection bias; voluntary or topic-specific locations may create bias.

Task. Is the sampling method likely to be biased for the stated question? Explain the direction of the possible bias.

Step-by-step solution. 1. Identify the target population. 2. Compare the selection method with that target. 3. People already at the bus station are more likely than average to use buses. 4. Therefore the method is likely biased.

Answer. Likely biased. People already at the bus station are more likely than average to use buses.

Knowledge takeaway. A large biased sample can still give a misleading estimate; representativeness is a reasoning question about selection.

Formal model. Sampling-frame analysis: compare inclusion mechanism with target population and likely outcome-related selection.

Problem 237. Sampling and opinion: case 2

Knowledge context. Surveys use samples to learn about larger populations. The selection process matters as much as the number of answers.

Given facts. A school wants opinions on lunch. It randomly selects 10 students from each grade.

Rules. A sample is more representative when selection does not systematically favor people connected to the answer. Random or broad systematic sampling can reduce selection bias; voluntary or topic-specific locations may create bias.

Task. Is the sampling method likely to be biased for the stated question? Explain the direction of the possible bias.

Step-by-step solution. 1. Identify the target population. 2. Compare the selection method with that target. 3. Random selection within every grade gives broad grade coverage; no obvious topic-specific selection is stated. 4. Therefore the method is not obviously biased for the stated target, within the limits given.

Answer. Not obviously biased under the stated design. Random selection within every grade gives broad grade coverage; no obvious topic-specific selection is stated.

Knowledge takeaway. A large biased sample can still give a misleading estimate; representativeness is a reasoning question about selection.

Formal model. Sampling-frame analysis: compare inclusion mechanism with target population and likely outcome-related selection.

Problem 238. Sampling and opinion: case 3

Knowledge context. Surveys use samples to learn about larger populations. The selection process matters as much as the number of answers.

Given facts. A village asks only members of the sports club whether to build a sports field.

Rules. A sample is more representative when selection does not systematically favor people connected to the answer. Random or broad systematic sampling can reduce selection bias; voluntary or topic-specific locations may create bias.

Task. Is the sampling method likely to be biased for the stated question? Explain the direction of the possible bias.

Step-by-step solution. 1. Identify the target population. 2. Compare the selection method with that target. 3. Sports-club members are likely to be more favorable to a sports field than the whole village. 4. Therefore the method is likely biased.

Answer. Likely biased. Sports-club members are likely to be more favorable to a sports field than the whole village.

Knowledge takeaway. A large biased sample can still give a misleading estimate; representativeness is a reasoning question about selection.

Formal model. Sampling-frame analysis: compare inclusion mechanism with target population and likely outcome-related selection.

Problem 239. Sampling and opinion: case 4

Knowledge context. Surveys use samples to learn about larger populations. The selection process matters as much as the number of answers.

Given facts. A library surveys every 20th visitor entering over a full week.

Rules. A sample is more representative when selection does not systematically favor people connected to the answer. Random or broad systematic sampling can reduce selection bias; voluntary or topic-specific locations may create bias.

Task. Is the sampling method likely to be biased for the stated question? Explain the direction of the possible bias.

Step-by-step solution. 1. Identify the target population. 2. Compare the selection method with that target. 3. Sampling every 20th visitor across a full week is systematic within the visitor population; it may represent visitors reasonably, though not non-visitors. 4. Therefore the method is not obviously biased for the stated target, within the limits given.

Answer. Not obviously biased under the stated design. Sampling every 20th visitor across a full week is systematic within the visitor population; it may represent visitors reasonably, though not non-visitors.

Knowledge takeaway. A large biased sample can still give a misleading estimate; representativeness is a reasoning question about selection.

Formal model. Sampling-frame analysis: compare inclusion mechanism with target population and likely outcome-related selection.

Problem 240. Sampling and opinion: case 5

Knowledge context. Surveys use samples to learn about larger populations. The selection process matters as much as the number of answers.

Given facts. An online poll is open to anyone who chooses to click it.

Rules. A sample is more representative when selection does not systematically favor people connected to the answer. Random or broad systematic sampling can reduce selection bias; voluntary or topic-specific locations may create bias.

Task. Is the sampling method likely to be biased for the stated question? Explain the direction of the possible bias.

Step-by-step solution. 1. Identify the target population. 2. Compare the selection method with that target. 3. People who choose to click may differ systematically from those who ignore the poll. 4. Therefore the method is likely biased.

Answer. Likely biased. People who choose to click may differ systematically from those who ignore the poll.

Knowledge takeaway. A large biased sample can still give a misleading estimate; representativeness is a reasoning question about selection.

Formal model. Sampling-frame analysis: compare inclusion mechanism with target population and likely outcome-related selection.

N24. Matching, scheduling, and coverage

Problem 241. Matching people to tasks: case 1

Knowledge context. Scheduling and assignment problems connect people, tasks, times, and constraints.

Given facts. Four students must each receive one different task. Allowed assignments: Ava: ['Map', 'Timeline']; Ben: ['Budget', 'Survey']; Cara: ['Map', 'Survey']; Dion: ['Budget', 'Timeline'].

Rules. Each student gets exactly one allowed task, and each task is used exactly once. A complete assignment must satisfy both conditions.

Task. Find one complete valid assignment. If none exists, explain why.

Step-by-step solution. 1. Start with the most restricted student if one exists. 2. Assign an allowed task, then remove that task from the remaining choices. 3. One complete assignment is {'Ava': 'Map', 'Ben': 'Budget', 'Cara': 'Survey', 'Dion': 'Timeline'}.

Answer. Valid assignment: Ava→Map, Ben→Budget, Cara→Survey, Dion→Timeline.

Knowledge takeaway. A local choice can block a later assignment, so the whole matching must be checked.

Formal model. Bipartite matching / all-different constraint satisfaction.

Problem 242. Matching people to tasks: case 2

Knowledge context. Scheduling and assignment problems connect people, tasks, times, and constraints.

Given facts. Four students must each receive one different task. Allowed assignments: Ava: ['Map', 'Timeline']; Ben: ['Budget', 'Survey']; Cara: ['Survey']; Dion: ['Budget', 'Timeline'].

Rules. Each student gets exactly one allowed task, and each task is used exactly once. A complete assignment must satisfy both conditions.

Task. Find one complete valid assignment. If none exists, explain why.

Step-by-step solution. 1. Start with the most restricted student if one exists. 2. Assign an allowed task, then remove that task from the remaining choices. 3. One complete assignment is {'Ava': 'Map', 'Ben': 'Budget', 'Cara': 'Survey', 'Dion': 'Timeline'}.

Answer. Valid assignment: Ava→Map, Ben→Budget, Cara→Survey, Dion→Timeline.

Knowledge takeaway. A local choice can block a later assignment, so the whole matching must be checked.

Formal model. Bipartite matching / all-different constraint satisfaction.

Problem 243. Matching people to tasks: case 3

Knowledge context. Scheduling and assignment problems connect people, tasks, times, and constraints.

Given facts. Four students must each receive one different task. Allowed assignments: Ava: ['Map', 'Timeline']; Ben: ['Budget', 'Survey']; Cara: ['Map', 'Survey']; Dion: ['Budget'].

Rules. Each student gets exactly one allowed task, and each task is used exactly once. A complete assignment must satisfy both conditions.

Task. Find one complete valid assignment. If none exists, explain why.

Step-by-step solution. 1. Start with the most restricted student if one exists. 2. Assign an allowed task, then remove that task from the remaining choices. 3. One complete assignment is {'Ava': 'Timeline', 'Ben': 'Survey', 'Cara': 'Map', 'Dion': 'Budget'}.

Answer. Valid assignment: Ava→Timeline, Ben→Survey, Cara→Map, Dion→Budget.

Knowledge takeaway. A local choice can block a later assignment, so the whole matching must be checked.

Formal model. Bipartite matching / all-different constraint satisfaction.

Problem 244. Matching people to tasks: case 4

Knowledge context. Scheduling and assignment problems connect people, tasks, times, and constraints.

Given facts. Four students must each receive one different task. Allowed assignments: Ava: ['Map']; Ben: ['Budget', 'Survey']; Cara: ['Map', 'Survey']; Dion: ['Budget', 'Timeline'].

Rules. Each student gets exactly one allowed task, and each task is used exactly once. A complete assignment must satisfy both conditions.

Task. Find one complete valid assignment. If none exists, explain why.

Step-by-step solution. 1. Start with the most restricted student if one exists. 2. Assign an allowed task, then remove that task from the remaining choices. 3. One complete assignment is {'Ava': 'Map', 'Ben': 'Budget', 'Cara': 'Survey', 'Dion': 'Timeline'}.

Answer. Valid assignment: Ava→Map, Ben→Budget, Cara→Survey, Dion→Timeline.

Knowledge takeaway. A local choice can block a later assignment, so the whole matching must be checked.

Formal model. Bipartite matching / all-different constraint satisfaction.

Problem 245. Matching people to tasks: case 5

Knowledge context. Scheduling and assignment problems connect people, tasks, times, and constraints.

Given facts. Four students must each receive one different task. Allowed assignments: Ava: ['Map', 'Timeline']; Ben: ['Budget', 'Survey']; Cara: ['Map', 'Survey']; Dion: ['Budget', 'Timeline']. Cross-domain check: a survey team starts work at 12:00 and works for 2 hour(s) without a break.

Rules. Each student gets exactly one allowed task, and each task is used exactly once. A complete assignment must satisfy both conditions. For the cross-domain check, finishing time = starting time + duration.

Task. Find one complete valid assignment. If none exists, explain why. Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. Start with the most restricted student if one exists. 2. Assign an allowed task, then remove that task from the remaining choices. 3. One complete assignment is {'Ava': 'Map', 'Ben': 'Budget', 'Cara': 'Survey', 'Dion': 'Timeline'}. Cross-domain check: 12+2=14, so the team finishes at 14:00.

Answer. Valid assignment: Ava→Map, Ben→Budget, Cara→Survey, Dion→Timeline. Cross-domain answer: 14:00.

Knowledge takeaway. A local choice can block a later assignment, so the whole matching must be checked.

Formal model. Bipartite matching / all-different constraint satisfaction.

N25. Meta-reasoning: robustness, information, causality

Problem 246. Meta-reasoning: case 1

Knowledge context. Good reasoning includes checking how informative, robust, falsifiable, causal, or dominated a conclusion is.

Given facts. Four equally likely locations A,B,C,D remain possible. Question Q1 asks “Is it A?”; Q2 asks “Is it A or B?” A yes/no question is more balanced when its two answer branches contain more nearly equal numbers of possibilities.

Rules. Use branch sizes as the information criterion in this exercise.

Task. Which question is more balanced and therefore more informative by the stated criterion?

Step-by-step solution. 1. Q1 splits the four possibilities 1 versus 3. 2. Q2 splits them 2 versus 2. 3. The 2–2 split is more balanced.

Answer. Q2.

Knowledge takeaway. A result can be numerically correct yet epistemically weak; meta-reasoning evaluates the quality of the inference itself.

Formal model. Meta-level predicate over hypotheses or decisions: information balance, robustness, counterexample, causality, or dominance.

Problem 247. Meta-reasoning: case 2

Knowledge context. Good reasoning includes checking how informative, robust, falsifiable, causal, or dominated a conclusion is.

Given facts. Option A scores between 49 and 53 after measurement error; option B scores between 50 and 52. A ranking is robust only if every allowed value of A remains above every allowed value of B.

Rules. Test worst cases: robust A>B requires min(A)>max(B).

Task. Is the ranking A>B robust?

Step-by-step solution. 1. min(A)=49; max(B)=52. 2. 49>52 is false. 3. Some allowed values reverse the ranking.

Answer. No; the ranking is sensitive to the uncertainty.

Knowledge takeaway. A result can be numerically correct yet epistemically weak; meta-reasoning evaluates the quality of the inference itself.

Formal model. Meta-level predicate over hypotheses or decisions: information balance, robustness, counterexample, causality, or dominance.

Problem 248. Meta-reasoning: case 3

Knowledge context. Good reasoning includes checking how informative, robust, falsifiable, causal, or dominated a conclusion is.

Given facts. Claim: “Every town with more roads has shorter average travel time.” Town X has 6 roads and average travel time 35 minutes. Town Y has 4 roads and average travel time 20 minutes.

Rules. A universal claim is disproved by one case that satisfies its condition but not its conclusion.

Task. Do X and Y provide a counterexample to the claim?

Step-by-step solution. 1. X has more roads than Y. 2. Yet X has longer, not shorter, travel time. 3. Therefore the universal claim fails on this pair.

Answer. Yes. X and Y form a counterexample.

Knowledge takeaway. A result can be numerically correct yet epistemically weak; meta-reasoning evaluates the quality of the inference itself.

Formal model. Meta-level predicate over hypotheses or decisions: information balance, robustness, counterexample, causality, or dominance.

Problem 249. Meta-reasoning: case 4

Knowledge context. Good reasoning includes checking how informative, robust, falsifiable, causal, or dominated a conclusion is.

Given facts. Across several months, ice-cream sales and sunburn cases rise together. Temperature also rises in those months. No intervention is described.

Rules. Correlation between X and Y does not by itself prove X causes Y; a third variable may influence both.

Task. Can we conclude that ice-cream sales cause sunburn?

Step-by-step solution. 1. Sales and sunburn are correlated. 2. Temperature is a plausible common factor stated in the data. 3. Without an intervention or stronger causal evidence, direct causation is not established.

Answer. No. The association alone does not establish that ice-cream sales cause sunburn.

Knowledge takeaway. A result can be numerically correct yet epistemically weak; meta-reasoning evaluates the quality of the inference itself.

Formal model. Meta-level predicate over hypotheses or decisions: information balance, robustness, counterexample, causality, or dominance.

Problem 250. Meta-reasoning: case 5

Knowledge context. Good reasoning includes checking how informative, robust, falsifiable, causal, or dominated a conclusion is.

Given facts. Two transport plans are compared. A costs 8, takes 20 minutes, and has safety score 9. B costs 10, takes 25 minutes, and has safety score 9. Lower cost/time and higher safety are better.

Rules. A Pareto-dominates B if A is at least as good on every stated criterion and strictly better on at least one.

Task. Does one plan dominate the other?

Step-by-step solution. 1. A is cheaper than B. 2. A is faster than B. 3. Safety is equal. 4. A is no worse anywhere and better on two criteria.

Answer. A Pareto-dominates B.

Knowledge takeaway. A result can be numerically correct yet epistemically weak; meta-reasoning evaluates the quality of the inference itself.

Formal model. Meta-level predicate over hypotheses or decisions: information balance, robustness, counterexample, causality, or dominance.

GRADE 2

Difficulty increases through the number of relations and reasoning steps, not through hidden knowledge.

Part A — Foundational reasoning about the world

G1. Positions and orientation

Problem 251. Position chain from Fern to Kite

Knowledge context. A map can be modeled with relative positions. When several moves are given, their effects can be combined rather than memorized as a picture.

Given facts. Cedar is 1 map unit(s) east of Fern; Kite is 1 map unit(s) east of Cedar.

Rules. Treat east as +x, west as −x, north as +y, and south as −y. Add the moves in order. A fact that does not affect any coordinate is irrelevant.

Task. Where is Kite relative to Fern? What is the straight grid displacement measured as |x|+|y|?

Step-by-step solution. 1. Start Fern at (0,0). 2. Translate each relation into a coordinate move. 3. Adding the moves gives (2,0). 4. That coordinate is east of Fern; the grid displacement is |2|+|0|=2.

Answer. Kite is east of Fern; grid displacement 2 unit(s).

Knowledge takeaway. Cardinal directions become easier to reason about when they are treated as relations that can be composed.

Formal model. Grid coordinates; each statement adds a vector; sum the vectors and classify the final coordinate.

Problem 252. Position chain from Zephyr to Juniper

Knowledge context. A map can be modeled with relative positions. When several moves are given, their effects can be combined rather than memorized as a picture.

Given facts. Willow is 2 map unit(s) north of Zephyr; Juniper is 2 map unit(s) north of Willow.

Rules. Treat east as +x, west as −x, north as +y, and south as −y. Add the moves in order. A fact that does not affect any coordinate is irrelevant.

Task. Where is Juniper relative to Zephyr? What is the straight grid displacement measured as |x|+|y|?

Step-by-step solution. 1. Start Zephyr at (0,0). 2. Translate each relation into a coordinate move. 3. Adding the moves gives (0,4). 4. That coordinate is north of Zephyr; the grid displacement is |0|+|4|=4.

Answer. Juniper is north of Zephyr; grid displacement 4 unit(s).

Knowledge takeaway. Cardinal directions become easier to reason about when they are treated as relations that can be composed.

Formal model. Grid coordinates; each statement adds a vector; sum the vectors and classify the final coordinate.

Problem 253. Position chain from Alder to Dover

Knowledge context. A map can be modeled with relative positions. When several moves are given, their effects can be combined rather than memorized as a picture.

Given facts. Quartz is 1 map unit(s) east of Alder; Dover is 1 map unit(s) north of Quartz.

Rules. Treat east as +x, west as −x, north as +y, and south as −y. Add the moves in order. A fact that does not affect any coordinate is irrelevant.

Task. Where is Dover relative to Alder? What is the straight grid displacement measured as |x|+|y|?

Step-by-step solution. 1. Start Alder at (0,0). 2. Translate each relation into a coordinate move. 3. Adding the moves gives (1,1). 4. That coordinate is north-east of Alder; the grid displacement is |1|+|1|=2.

Answer. Dover is north-east of Alder; grid displacement 2 unit(s).

Knowledge takeaway. Cardinal directions become easier to reason about when they are treated as relations that can be composed.

Formal model. Grid coordinates; each statement adds a vector; sum the vectors and classify the final coordinate.

Problem 254. Position chain from Willow to Stone

Knowledge context. A map can be modeled with relative positions. When several moves are given, their effects can be combined rather than memorized as a picture.

Given facts. Quartz is 2 map unit(s) west of Willow; Stone is 2 map unit(s) south of Quartz.

Rules. Treat east as +x, west as −x, north as +y, and south as −y. Add the moves in order. A fact that does not affect any coordinate is irrelevant.

Task. Where is Stone relative to Willow? What is the straight grid displacement measured as |x|+|y|?

Step-by-step solution. 1. Start Willow at (0,0). 2. Translate each relation into a coordinate move. 3. Adding the moves gives (-2,-2). 4. That coordinate is south-west of Willow; the grid displacement is |-2|+|-2|=4.

Answer. Stone is south-west of Willow; grid displacement 4 unit(s).

Knowledge takeaway. Cardinal directions become easier to reason about when they are treated as relations that can be composed.

Formal model. Grid coordinates; each statement adds a vector; sum the vectors and classify the final coordinate.

Problem 255. Position chain from Juniper to Birch

Knowledge context. A map can be modeled with relative positions. When several moves are given, their effects can be combined rather than memorized as a picture.

Given facts. Alder is 1 map unit(s) north of Juniper; Iris is 1 map unit(s) east of Alder; Birch is 1 map unit(s) south of Iris.

Rules. Treat east as +x, west as −x, north as +y, and south as −y. Add the moves in order. A fact that does not affect any coordinate is irrelevant.

Task. Where is Birch relative to Juniper? What is the straight grid displacement measured as |x|+|y|?

Step-by-step solution. 1. Start Juniper at (0,0). 2. Translate each relation into a coordinate move. 3. Adding the moves gives (1,0). 4. That coordinate is east of Juniper; the grid displacement is |1|+|0|=1.

Answer. Birch is east of Juniper; grid displacement 1 unit(s).

Knowledge takeaway. Cardinal directions become easier to reason about when they are treated as relations that can be composed.

Formal model. Grid coordinates; each statement adds a vector; sum the vectors and classify the final coordinate.

G2. Roads and routes

Problem 256. The best route from Iris to Zephyr

Knowledge context. A road map can be modeled as a graph. The best route depends on the stated criterion, not on how short it looks on paper.

Given facts. Iris–Quartz–Zephyr has segment costs [4, 5] (total 9); Iris–Riverbend–Yarrow–Zephyr has segment costs [3, 2, 4] (total 9); Iris–Birch–Zephyr has segment costs [7, 2] (total 9).

Rules. A route is feasible only if all of its roads are open. The cost of a route is the sum of its segment costs. Choose the feasible route with the smallest total.

Task. Which route should a traveler use from Iris to Zephyr, and why?

Step-by-step solution. 1. List the candidate routes and their segment costs. 2. All listed routes are open. 3. Add the segment costs for every feasible route. 4. The smallest feasible total is 9, on Iris–Quartz–Zephyr.

Answer. Use Iris–Quartz–Zephyr; total cost 9.

Knowledge takeaway. A shortest path is a reasoning result: first respect constraints, then compare the remaining costs.

Formal model. Weighted graph; delete forbidden edges; minimize the sum of edge weights.

Problem 257. The best route from Iris to Zephyr

Knowledge context. A road map can be modeled as a graph. The best route depends on the stated criterion, not on how short it looks on paper.

Given facts. Iris–Riverbend–Zephyr has segment costs [5, 6] (total 11); Iris–Pine–Linden–Zephyr has segment costs [4, 3, 4] (total 11); Iris–Fern–Zephyr has segment costs [8, 3] (total 11).

Rules. A route is feasible only if all of its roads are open. The cost of a route is the sum of its segment costs. Choose the feasible route with the smallest total.

Task. Which route should a traveler use from Iris to Zephyr, and why?

Step-by-step solution. 1. List the candidate routes and their segment costs. 2. All listed routes are open. 3. Add the segment costs for every feasible route. 4. The smallest feasible total is 11, on Iris–Riverbend–Zephyr.

Answer. Use Iris–Riverbend–Zephyr; total cost 11.

Knowledge takeaway. A shortest path is a reasoning result: first respect constraints, then compare the remaining costs.

Formal model. Weighted graph; delete forbidden edges; minimize the sum of edge weights.

Problem 258. The best route from Yarrow to Linden

Knowledge context. A road map can be modeled as a graph. The best route depends on the stated criterion, not on how short it looks on paper.

Given facts. Yarrow–Quartz–Linden has segment costs [6, 7] (total 13); Yarrow–Willow–Iris–Linden has segment costs [5, 4, 4] (total 13); Yarrow–Riverbend–Linden has segment costs [9, 4] (total 13).

Rules. A route is feasible only if all of its roads are open. The cost of a route is the sum of its segment costs. Choose the feasible route with the smallest total.

Task. Which route should a traveler use from Yarrow to Linden, and why?

Step-by-step solution. 1. List the candidate routes and their segment costs. 2. All listed routes are open. 3. Add the segment costs for every feasible route. 4. The smallest feasible total is 13, on Yarrow–Quartz–Linden.

Answer. Use Yarrow–Quartz–Linden; total cost 13.

Knowledge takeaway. A shortest path is a reasoning result: first respect constraints, then compare the remaining costs.

Formal model. Weighted graph; delete forbidden edges; minimize the sum of edge weights.

Problem 259. The best route from Zephyr to Birch

Knowledge context. A road map can be modeled as a graph. The best route depends on the stated criterion, not on how short it looks on paper.

Given facts. Zephyr–Yarrow–Birch has segment costs [7, 8] (total 15); Zephyr–Cedar–Grove–Birch has segment costs [6, 5, 4] (total 15); Zephyr–Linden–Birch has segment costs [10, 5] (total 15).

Rules. A route is feasible only if all of its roads are open. The cost of a route is the sum of its segment costs. Choose the feasible route with the smallest total.

Task. Which route should a traveler use from Zephyr to Birch, and why?

Step-by-step solution. 1. List the candidate routes and their segment costs. 2. All listed routes are open. 3. Add the segment costs for every feasible route. 4. The smallest feasible total is 15, on Zephyr–Yarrow–Birch.

Answer. Use Zephyr–Yarrow–Birch; total cost 15.

Knowledge takeaway. A shortest path is a reasoning result: first respect constraints, then compare the remaining costs.

Formal model. Weighted graph; delete forbidden edges; minimize the sum of edge weights.

Problem 260. The best route from Fern to Elm

Knowledge context. A road map can be modeled as a graph. The best route depends on the stated criterion, not on how short it looks on paper.

Given facts. Fern–Harbor–Elm has segment costs [8, 9] (total 17); Fern–Oak–Dover–Elm has segment costs [7, 6, 4] (total 17); Fern–Iris–Elm has segment costs [11, 6] (total 17).

Rules. A route is feasible only if all of its roads are open. The cost of a route is the sum of its segment costs. Choose the feasible route with the smallest total.

Task. Which route should a traveler use from Fern to Elm, and why?

Step-by-step solution. 1. List the candidate routes and their segment costs. 2. All listed routes are open. 3. Add the segment costs for every feasible route. 4. The smallest feasible total is 17, on Fern–Harbor–Elm.

Answer. Use Fern–Harbor–Elm; total cost 17.

Knowledge takeaway. A shortest path is a reasoning result: first respect constraints, then compare the remaining costs.

Formal model. Weighted graph; delete forbidden edges; minimize the sum of edge weights.

G3. Rivers: upstream and downstream

Problem 261. Following water from North Fork

Knowledge context. River systems form directed networks: water has a direction, tributaries join larger streams, and upstream/downstream are relational concepts.

Given facts. North Fork flows into Willow Stream; Blue Creek flows into Willow Stream; Willow Stream flows into Clearwater.

Rules. If river X flows into river Y, X is upstream of Y. Upstream relations can be followed through several confluences. Downstream is the reverse direction of flow.

Task. Is North Fork upstream of Clearwater? Can water from North Fork reach Clearwater through the stated network?

Step-by-step solution. 1. Rewrite the facts as arrows: North Fork→Willow Stream, Blue Creek→Willow Stream, Willow Stream→Clearwater. 2. Follow arrows starting from North Fork. 3. An arrow path reaches Clearwater, so North Fork is upstream of Clearwater.

Answer. Yes; water from North Fork can reach Clearwater.

Knowledge takeaway. A river network can be reasoned about like any directed network; following the arrows is more reliable than guessing from names.

Formal model. Directed graph; edge X→Y means water can flow from X to Y; query reachability.

Problem 262. Following water from Pine River

Knowledge context. River systems form directed networks: water has a direction, tributaries join larger streams, and upstream/downstream are relational concepts.

Given facts. North Fork flows into Willow Stream; Pine River flows into Willow Stream; Willow Stream flows into Blue Creek.

Rules. If river X flows into river Y, X is upstream of Y. Upstream relations can be followed through several confluences. Downstream is the reverse direction of flow.

Task. Is Pine River upstream of Blue Creek? Can water from Pine River reach Blue Creek through the stated network?

Step-by-step solution. 1. Rewrite the facts as arrows: North Fork→Willow Stream, Pine River→Willow Stream, Willow Stream→Blue Creek. 2. Follow arrows starting from Pine River. 3. An arrow path reaches Blue Creek, so Pine River is upstream of Blue Creek.

Answer. Yes; water from Pine River can reach Blue Creek.

Knowledge takeaway. A river network can be reasoned about like any directed network; following the arrows is more reliable than guessing from names.

Formal model. Directed graph; edge X→Y means water can flow from X to Y; query reachability.

Problem 263. Following water from Meadow River

Knowledge context. River systems form directed networks: water has a direction, tributaries join larger streams, and upstream/downstream are relational concepts.

Given facts. Meadow River flows into Pine River; Clearwater flows into Pine River; Pine River flows into Red Brook.

Rules. If river X flows into river Y, X is upstream of Y. Upstream relations can be followed through several confluences. Downstream is the reverse direction of flow.

Task. Is Meadow River upstream of Pine River? Can water from Meadow River reach Pine River through the stated network?

Step-by-step solution. 1. Rewrite the facts as arrows: Meadow River→Pine River, Clearwater→Pine River, Pine River→Red Brook. 2. Follow arrows starting from Meadow River. 3. An arrow path reaches Pine River, so Meadow River is upstream of Pine River.

Answer. Yes; water from Meadow River can reach Pine River.

Knowledge takeaway. A river network can be reasoned about like any directed network; following the arrows is more reliable than guessing from names.

Formal model. Directed graph; edge X→Y means water can flow from X to Y; query reachability.

Problem 264. Following water from North Fork

Knowledge context. River systems form directed networks: water has a direction, tributaries join larger streams, and upstream/downstream are relational concepts.

Given facts. Blue Creek flows into North Fork; Pine River flows into North Fork; North Fork flows into Clearwater.

Rules. If river X flows into river Y, X is upstream of Y. Upstream relations can be followed through several confluences. Downstream is the reverse direction of flow.

Task. Is North Fork upstream of Clearwater? Can water from North Fork reach Clearwater through the stated network?

Step-by-step solution. 1. Rewrite the facts as arrows: Blue Creek→North Fork, Pine River→North Fork, North Fork→Clearwater. 2. Follow arrows starting from North Fork. 3. An arrow path reaches Clearwater, so North Fork is upstream of Clearwater.

Answer. Yes; water from North Fork can reach Clearwater.

Knowledge takeaway. A river network can be reasoned about like any directed network; following the arrows is more reliable than guessing from names.

Formal model. Directed graph; edge X→Y means water can flow from X to Y; query reachability.

Problem 265. Following water from Pine River

Knowledge context. River systems form directed networks: water has a direction, tributaries join larger streams, and upstream/downstream are relational concepts.

Given facts. North Fork flows into Blue Creek; Pine River flows into Blue Creek; Blue Creek flows into Willow Stream.

Rules. If river X flows into river Y, X is upstream of Y. Upstream relations can be followed through several confluences. Downstream is the reverse direction of flow.

Task. Is Pine River upstream of Blue Creek? Can water from Pine River reach Blue Creek through the stated network?

Step-by-step solution. 1. Rewrite the facts as arrows: North Fork→Blue Creek, Pine River→Blue Creek, Blue Creek→Willow Stream. 2. Follow arrows starting from Pine River. 3. An arrow path reaches Blue Creek, so Pine River is upstream of Blue Creek.

Answer. Yes; water from Pine River can reach Blue Creek.

Knowledge takeaway. A river network can be reasoned about like any directed network; following the arrows is more reliable than guessing from names.

Formal model. Directed graph; edge X→Y means water can flow from X to Y; query reachability.

G4. Neighbours and borders

Problem 266. Crossing the fictional border network of Oak

Knowledge context. Political maps can be treated as adjacency networks. This lets us reason about border crossings without relying on memorized real-world geography.

Given facts. Countries are represented by the fictional states Oak, Riverbend, Meadow, Willow, Northfield, Cedar. Shared borders: Oak–Riverbend, Riverbend–Meadow, Meadow–Willow, Oak–Northfield, Northfield–Cedar, Cedar–Willow.

Rules. Crossing one border corresponds to traversing one graph edge. A route may use only listed borders. A state is unavoidable only if every feasible route passes through it.

Task. What is the fewest number of border crossings needed to travel from Oak to Willow? Is Riverbend necessarily crossed?

Step-by-step solution. 1. Turn each shared border into an undirected link. 2. A shortest-link search gives distance 3 from Oak to Willow. 3. Remove Riverbend and check whether Willow is still reachable from Oak. 4. It is still reachable, so Riverbend is not unavoidable.

Answer. 3 border crossing(s); Riverbend is not unavoidable.

Knowledge takeaway. Neighbourhood on a map is a graph relation; shortest routes and unavoidable transit points are different questions.

Formal model. Undirected graph; shortest path length plus reachability after deleting a candidate vertex.

Problem 267. Crossing the fictional border network of Pine

Knowledge context. Political maps can be treated as adjacency networks. This lets us reason about border crossings without relying on memorized real-world geography.

Given facts. Countries are represented by the fictional states Pine, Elm, Juniper, Stone, Quartz, Yarrow. Shared borders: Pine–Elm, Elm–Juniper, Juniper–Stone, Pine–Quartz, Quartz–Yarrow, Yarrow–Stone.

Rules. Crossing one border corresponds to traversing one graph edge. A route may use only listed borders. A state is unavoidable only if every feasible route passes through it.

Task. What is the fewest number of border crossings needed to travel from Pine to Stone? Is Elm necessarily crossed?

Step-by-step solution. 1. Turn each shared border into an undirected link. 2. A shortest-link search gives distance 3 from Pine to Stone. 3. Remove Elm and check whether Stone is still reachable from Pine. 4. It is still reachable, so Elm is not unavoidable.

Answer. 3 border crossing(s); Elm is not unavoidable.

Knowledge takeaway. Neighbourhood on a map is a graph relation; shortest routes and unavoidable transit points are different questions.

Formal model. Undirected graph; shortest path length plus reachability after deleting a candidate vertex.

Problem 268. Crossing the fictional border network of Quartz

Knowledge context. Political maps can be treated as adjacency networks. This lets us reason about border crossings without relying on memorized real-world geography.

Given facts. Countries are represented by the fictional states Quartz, Grove, Fern, Zephyr, Elm, Juniper. Shared borders: Quartz–Grove, Grove–Fern, Fern–Zephyr, Quartz–Elm, Elm–Juniper, Juniper–Zephyr.

Rules. Crossing one border corresponds to traversing one graph edge. A route may use only listed borders. A state is unavoidable only if every feasible route passes through it.

Task. What is the fewest number of border crossings needed to travel from Quartz to Zephyr? Is Grove necessarily crossed?

Step-by-step solution. 1. Turn each shared border into an undirected link. 2. A shortest-link search gives distance 3 from Quartz to Zephyr. 3. Remove Grove and check whether Zephyr is still reachable from Quartz. 4. It is still reachable, so Grove is not unavoidable.

Answer. 3 border crossing(s); Grove is not unavoidable.

Knowledge takeaway. Neighbourhood on a map is a graph relation; shortest routes and unavoidable transit points are different questions.

Formal model. Undirected graph; shortest path length plus reachability after deleting a candidate vertex.

Problem 269. Crossing the fictional border network of Stone

Knowledge context. Political maps can be treated as adjacency networks. This lets us reason about border crossings without relying on memorized real-world geography.

Given facts. Countries are represented by the fictional states Stone, Grove, Pine, Willow, Alder, Birch. Shared borders: Stone–Grove, Grove–Pine, Pine–Willow, Stone–Alder, Alder–Birch, Birch–Willow.

Rules. Crossing one border corresponds to traversing one graph edge. A route may use only listed borders. A state is unavoidable only if every feasible route passes through it.

Task. What is the fewest number of border crossings needed to travel from Stone to Willow? Is Grove necessarily crossed?

Step-by-step solution. 1. Turn each shared border into an undirected link. 2. A shortest-link search gives distance 3 from Stone to Willow. 3. Remove Grove and check whether Willow is still reachable from Stone. 4. It is still reachable, so Grove is not unavoidable.

Answer. 3 border crossing(s); Grove is not unavoidable.

Knowledge takeaway. Neighbourhood on a map is a graph relation; shortest routes and unavoidable transit points are different questions.

Formal model. Undirected graph; shortest path length plus reachability after deleting a candidate vertex.

Problem 270. Crossing the fictional border network of Pine

Knowledge context. Political maps can be treated as adjacency networks. This lets us reason about border crossings without relying on memorized real-world geography.

Given facts. Countries are represented by the fictional states Pine, Zephyr, Meadow, Dover, Riverbend, Iris. Shared borders: Pine–Zephyr, Zephyr–Meadow, Meadow–Dover, Pine–Riverbend, Riverbend–Iris, Iris–Dover.

Rules. Crossing one border corresponds to traversing one graph edge. A route may use only listed borders. A state is unavoidable only if every feasible route passes through it.

Task. What is the fewest number of border crossings needed to travel from Pine to Dover? Is Zephyr necessarily crossed?

Step-by-step solution. 1. Turn each shared border into an undirected link. 2. A shortest-link search gives distance 3 from Pine to Dover. 3. Remove Zephyr and check whether Dover is still reachable from Pine. 4. It is still reachable, so Zephyr is not unavoidable.

Answer. 3 border crossing(s); Zephyr is not unavoidable.

Knowledge takeaway. Neighbourhood on a map is a graph relation; shortest routes and unavoidable transit points are different questions.

Formal model. Undirected graph; shortest path length plus reachability after deleting a candidate vertex.

G5. Maps, legends, and clues

Problem 271. Reading a layered map legend (A1)

Knowledge context. A map legend translates symbols into properties. One location may have several properties at the same time.

Given facts. Legend: tree=forest, wavy line=river, double line=road, arch=bridge, square=village, triangle=hill. Cells: A1 has forest, road; A2 has bridge, river; B1 has forest, river, road; B2 has road, village. Cross-domain check: researchers collected 7 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. A cell satisfies the task only if it has every required attribute. Extra attributes do not disqualify it unless the task says so. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. Which cell or cells satisfy all of these conditions: forest, road? Explain by intersecting the clues. Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. Convert the requested symbols into the attribute set ['forest', 'road']. 2. Check each cell for every required attribute. 3. The matching set is ['A1', 'B1']. Cross-domain check: 7−4=3 independent reports remain.

Answer. A1, B1 Cross-domain answer: 3 independent reports.

Knowledge takeaway. Map reading often means intersecting clues rather than recognizing one symbol in isolation.

Formal model. Set model; each cell has an attribute set; return cells whose set is a superset of the required set.

Problem 272. Reading a layered map legend (A2)

Knowledge context. A map legend translates symbols into properties. One location may have several properties at the same time.

Given facts. Legend: tree=forest, wavy line=river, double line=road, arch=bridge, square=village, triangle=hill. Cells: A1 has forest, road; A2 has bridge, river, village; B1 has forest, river, road; B2 has road, village. Cross-domain check: on a field sketch, 1 cm represents 4 km and two checkpoints are 8 cm apart.

Rules. A cell satisfies the task only if it has every required attribute. Extra attributes do not disqualify it unless the task says so. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Which cell or cells satisfy all of these conditions: bridge, river? Explain by intersecting the clues. Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. Convert the requested symbols into the attribute set ['bridge', 'river']. 2. Check each cell for every required attribute. 3. The matching set is ['A2']. Cross-domain check: 8×4=32 km.

Answer. A2 Cross-domain answer: 32 km.

Knowledge takeaway. Map reading often means intersecting clues rather than recognizing one symbol in isolation.

Formal model. Set model; each cell has an attribute set; return cells whose set is a superset of the required set.

Problem 273. Reading a layered map legend (B1)

Knowledge context. A map legend translates symbols into properties. One location may have several properties at the same time.

Given facts. Legend: tree=forest, wavy line=river, double line=road, arch=bridge, square=village, triangle=hill. Cells: A1 has forest, road; A2 has bridge, river; B1 has forest, river, road; B2 has forest, road, village. Cross-domain check: a survey team starts work at 8:00 and works for 2 hour(s) without a break.

Rules. A cell satisfies the task only if it has every required attribute. Extra attributes do not disqualify it unless the task says so. For the cross-domain check, finishing time = starting time + duration.

Task. Which cell or cells satisfy all of these conditions: forest, river? Explain by intersecting the clues. Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. Convert the requested symbols into the attribute set ['forest', 'river']. 2. Check each cell for every required attribute. 3. The matching set is ['B1']. Cross-domain check: 8+2=10, so the team finishes at 10:00.

Answer. B1 Cross-domain answer: 10:00.

Knowledge takeaway. Map reading often means intersecting clues rather than recognizing one symbol in isolation.

Formal model. Set model; each cell has an attribute set; return cells whose set is a superset of the required set.

Problem 274. Reading a layered map legend (B2)

Knowledge context. A map legend translates symbols into properties. One location may have several properties at the same time.

Given facts. Legend: tree=forest, wavy line=river, double line=road, arch=bridge, square=village, triangle=hill. Cells: A1 has forest, hill, road; A2 has bridge, river; B1 has forest, river, road; B2 has road, village. Cross-domain check: a local committee has 12 members, requires at least 7 present for quorum, and 11 are present.

Rules. A cell satisfies the task only if it has every required attribute. Extra attributes do not disqualify it unless the task says so. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. Which cell or cells satisfy all of these conditions: hill, road? Explain by intersecting the clues. Cross-domain check: is quorum met?

Step-by-step solution. 1. Convert the requested symbols into the attribute set ['hill', 'road']. 2. Check each cell for every required attribute. 3. The matching set is ['A1']. Cross-domain check: compare 11 with 7; 11 ≥ 7 is true.

Answer. A1 Cross-domain answer: quorum is met.

Knowledge takeaway. Map reading often means intersecting clues rather than recognizing one symbol in isolation.

Formal model. Set model; each cell has an attribute set; return cells whose set is a superset of the required set.

Problem 275. Reading a layered map legend (A1)

Knowledge context. A map legend translates symbols into properties. One location may have several properties at the same time.

Given facts. Legend: tree=forest, wavy line=river, double line=road, arch=bridge, square=village, triangle=hill. Cells: A1 has forest, road; A2 has bridge, river; B1 has bridge, forest, river, road; B2 has road, village. Cross-domain check: researchers collected 11 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. A cell satisfies the task only if it has every required attribute. Extra attributes do not disqualify it unless the task says so. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. Which cell or cells satisfy all of these conditions: river, road? Explain by intersecting the clues. Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. Convert the requested symbols into the attribute set ['river', 'road']. 2. Check each cell for every required attribute. 3. The matching set is ['B1']. Cross-domain check: 11−4=7 independent reports remain.

Answer. B1 Cross-domain answer: 7 independent reports.

Knowledge takeaway. Map reading often means intersecting clues rather than recognizing one symbol in isolation.

Formal model. Set model; each cell has an attribute set; return cells whose set is a superset of the required set.

G6. Choosing a settlement site

Problem 276. Choosing a safe settlement site: case 1

Knowledge context. Human settlements depend on several factors at once. Some conditions may be compulsory while others are preferences.

Given facts. Candidate sites: Site A: water=yes, road=yes, flood-zone=no, nearby farmland score=2; Site B: water=yes, road=no, flood-zone=no, nearby farmland score=4; Site C: water=yes, road=yes, flood-zone=yes, nearby farmland score=4; Site D: water=no, road=yes, flood-zone=no, nearby farmland score=6. Cross-domain check: on a field sketch, 1 cm represents 3 km and two checkpoints are 5 cm apart.

Rules. A valid site must have water, road access, and must not be in the flood zone. Among valid sites, choose the one with the highest farmland score. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Which site should be selected under the stated mandatory conditions and preference rule? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. Separate mandatory conditions from the preference. 2. Filtering by water, road, and flood safety leaves ['Site A']. 3. Compare only the farmland scores of feasible sites: Site A=2. 4. Site A has the best score among valid sites. Cross-domain check: 5×3=15 km.

Answer. Site A. Cross-domain answer: 15 km.

Knowledge takeaway. Good decisions first satisfy hard constraints and only then optimize among the remaining options.

Formal model. Constraint-satisfaction problem followed by a score maximization over feasible options.

Problem 277. Choosing a safe settlement site: case 2

Knowledge context. Human settlements depend on several factors at once. Some conditions may be compulsory while others are preferences.

Given facts. Candidate sites: Site A: water=yes, road=yes, flood-zone=no, nearby farmland score=3; Site B: water=yes, road=no, flood-zone=no, nearby farmland score=3; Site C: water=yes, road=yes, flood-zone=yes, nearby farmland score=5; Site D: water=no, road=yes, flood-zone=no, nearby farmland score=5. Cross-domain check: a survey team starts work at 12:00 and works for 2 hour(s) without a break. Mixed-domain verification: a regional archive has 29 map sheets and receives 8 additional sheets during the exercise.

Rules. A valid site must have water, road access, and must not be in the flood zone. Among valid sites, choose the one with the highest farmland score. For the cross-domain check, finishing time = starting time + duration. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. Which site should be selected under the stated mandatory conditions and preference rule? Cross-domain check: at what time does the team finish? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. Separate mandatory conditions from the preference. 2. Filtering by water, road, and flood safety leaves ['Site A']. 3. Compare only the farmland scores of feasible sites: Site A=3. 4. Site A has the best score among valid sites. Cross-domain check: 12+2=14, so the team finishes at 14:00. Mixed-domain verification: 29+8=37.

Answer. Site A. Cross-domain answer: 14:00. Mixed-domain answer: 37 map sheets.

Knowledge takeaway. Good decisions first satisfy hard constraints and only then optimize among the remaining options.

Formal model. Constraint-satisfaction problem followed by a score maximization over feasible options.

Problem 278. Choosing a safe settlement site: case 3

Knowledge context. Human settlements depend on several factors at once. Some conditions may be compulsory while others are preferences.

Given facts. Candidate sites: Site A: water=yes, road=yes, flood-zone=no, nearby farmland score=2; Site B: water=yes, road=no, flood-zone=no, nearby farmland score=4; Site C: water=yes, road=yes, flood-zone=yes, nearby farmland score=4;

Site D: water=no, road=yes, flood-zone=no, nearby farmland score=6. Cross-domain check: a local committee has 16 members, requires at least 9 present for quorum, and 14 are present.

Rules. A valid site must have water, road access, and must not be in the flood zone. Among valid sites, choose the one with the highest farmland score. For the cross-domain check, quorum exists when present members ≥ the stated threshold.

Task. Which site should be selected under the stated mandatory conditions and preference rule? Cross-domain check: is quorum met?

Step-by-step solution. 1. Separate mandatory conditions from the preference. 2. Filtering by water, road, and flood safety leaves ['Site A']. 3. Compare only the farmland scores of feasible sites: Site A=2. 4. Site A has the best score among valid sites. Cross-domain check: compare 14 with 9; 14 ≥ 9 is true.

Answer. Site A. Cross-domain answer: quorum is met.

Knowledge takeaway. Good decisions first satisfy hard constraints and only then optimize among the remaining options.

Formal model. Constraint-satisfaction problem followed by a score maximization over feasible options.

Problem 279. Choosing a safe settlement site: case 4

Knowledge context. Human settlements depend on several factors at once. Some conditions may be compulsory while others are preferences.

Given facts. Candidate sites: Site A: water=yes, road=yes, flood-zone=no, nearby farmland score=3; Site B: water=yes, road=no, flood-zone=no, nearby farmland score=3; Site C: water=yes, road=yes, flood-zone=yes, nearby farmland score=5; Site D: water=no, road=yes, flood-zone=no, nearby farmland score=5. Cross-domain check: researchers collected 15 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. A valid site must have water, road access, and must not be in the flood zone. Among valid sites, choose the one with the highest farmland score. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. Which site should be selected under the stated mandatory conditions and preference rule? Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. Separate mandatory conditions from the preference. 2. Filtering by water, road, and flood safety leaves ['Site A']. 3. Compare only the farmland scores of feasible sites: Site A=3. 4. Site A has the best score among valid sites. Cross-domain check: 15−4=11 independent reports remain.

Answer. Site A. Cross-domain answer: 11 independent reports.

Knowledge takeaway. Good decisions first satisfy hard constraints and only then optimize among the remaining options.

Formal model. Constraint-satisfaction problem followed by a score maximization over feasible options.

Problem 280. Choosing a safe settlement site: case 5

Knowledge context. Human settlements depend on several factors at once. Some conditions may be compulsory while others are preferences.

Given facts. Candidate sites: Site A: water=yes, road=yes, flood-zone=no, nearby farmland score=2; Site B: water=yes, road=no, flood-zone=no, nearby farmland score=4; Site C: water=yes, road=yes, flood-zone=yes, nearby farmland score=4; Site D: water=no, road=yes, flood-zone=no, nearby farmland score=6. Cross-domain check: on a field sketch, 1 cm represents 2 km and two checkpoints are 2 cm apart.

Rules. A valid site must have water, road access, and must not be in the flood zone. Among valid sites, choose the one with the highest farmland score. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Which site should be selected under the stated mandatory conditions and preference rule? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. Separate mandatory conditions from the preference. 2. Filtering by water, road, and flood safety leaves ['Site A']. 3. Compare only the farmland scores of feasible sites: Site A=2. 4. Site A has the best score among valid sites. Cross-domain check: 2×2=4 km.

Answer. Site A. Cross-domain answer: 4 km.

Knowledge takeaway. Good decisions first satisfy hard constraints and only then optimize among the remaining options.

Formal model. Constraint-satisfaction problem followed by a score maximization over feasible options.

G7. Relief and movement cost

Problem 281. Terrain and travel cost from Linden

Knowledge context. Equal-looking distances can require different effort because terrain differs. A simple cost model makes that effect explicit.

Given facts. Terrain energy costs per segment: {'plain': 1, 'forest': 2, 'hill': 3, 'marsh': 4}. Three routes from Linden to Stone: R1=['plain', 'hill', 'plain']; R2=['forest', 'forest', 'plain']; R3=['plain', 'marsh']. The traveler can spend at most 9 energy units.

Rules. The energy of a route is the sum of its terrain costs. A route above the energy limit is infeasible. Among feasible routes, minimize energy.

Task. Which route is feasible and uses the least energy?

Step-by-step solution. 1. Compute route totals: R1=5, R2=5, R3=5. 2. Discard any total above 9. 3. The smallest remaining total is R1=5.

Answer. Route R1, using 5 energy units.

Knowledge takeaway. Relief changes the cost of movement; “shortest” and “easiest” are not automatically the same.

Formal model. Weighted paths with a feasibility budget; sum costs and choose the minimum feasible path.

Problem 282. Terrain and travel cost from Northfield

Knowledge context. Equal-looking distances can require different effort because terrain differs. A simple cost model makes that effect explicit.

Given facts. Terrain energy costs per segment: {'plain': 1, 'forest': 2, 'hill': 3, 'marsh': 4}. Three routes from Northfield to Oak: R1=['plain', 'hill', 'plain']; R2=['plain', 'forest', 'plain', 'plain']; R3=['plain', 'marsh']. The traveler can spend at most 9 energy units.

Rules. The energy of a route is the sum of its terrain costs. A route above the energy limit is infeasible. Among feasible routes, minimize energy.

Task. Which route is feasible and uses the least energy?

Step-by-step solution. 1. Compute route totals: R1=5, R2=5, R3=5. 2. Discard any total above 9. 3. The smallest remaining total is R1=5.

Answer. Route R1, using 5 energy units.

Knowledge takeaway. Relief changes the cost of movement; “shortest” and “easiest” are not automatically the same.

Formal model. Weighted paths with a feasibility budget; sum costs and choose the minimum feasible path.

Problem 283. Terrain and travel cost from Linden

Knowledge context. Equal-looking distances can require different effort because terrain differs. A simple cost model makes that effect explicit.

Given facts. Terrain energy costs per segment: {'plain': 1, 'forest': 2, 'hill': 3, 'marsh': 4}. Three routes from Linden to Juniper: R1=['plain', 'hill', 'plain']; R2=['forest', 'forest', 'plain']; R3=['plain', 'marsh']. The traveler can spend at most 9 energy units.

Rules. The energy of a route is the sum of its terrain costs. A route above the energy limit is infeasible. Among feasible routes, minimize energy.

Task. Which route is feasible and uses the least energy?

Step-by-step solution. 1. Compute route totals: R1=5, R2=5, R3=5. 2. Discard any total above 9. 3. The smallest remaining total is R1=5.

Answer. Route R1, using 5 energy units.

Knowledge takeaway. Relief changes the cost of movement; “shortest” and “easiest” are not automatically the same.

Formal model. Weighted paths with a feasibility budget; sum costs and choose the minimum feasible path.

Problem 284. Terrain and travel cost from Grove

Knowledge context. Equal-looking distances can require different effort because terrain differs. A simple cost model makes that effect explicit.

Given facts. Terrain energy costs per segment: {'plain': 1, 'forest': 2, 'hill': 3, 'marsh': 4}. Three routes from Grove to Cedar: R1=['plain', 'hill', 'plain']; R2=['plain', 'forest', 'plain', 'plain']; R3=['plain', 'marsh']. The traveler can spend at most 9 energy units.

Rules. The energy of a route is the sum of its terrain costs. A route above the energy limit is infeasible. Among feasible routes, minimize energy.

Task. Which route is feasible and uses the least energy?

Step-by-step solution. 1. Compute route totals: R1=5, R2=5, R3=5. 2. Discard any total above 9. 3. The smallest remaining total is R1=5.

Answer. Route R1, using 5 energy units.

Knowledge takeaway. Relief changes the cost of movement; “shortest” and “easiest” are not automatically the same.

Formal model. Weighted paths with a feasibility budget; sum costs and choose the minimum feasible path.

Problem 285. Terrain and travel cost from Cedar

Knowledge context. Equal-looking distances can require different effort because terrain differs. A simple cost model makes that effect explicit.

Given facts. Terrain energy costs per segment: {'plain': 1, 'forest': 2, 'hill': 3, 'marsh': 4}. Three routes from Cedar to Pine: R1=['plain', 'hill', 'plain']; R2=['forest', 'forest', 'plain']; R3=['plain', 'marsh']. The traveler can spend at most 9 energy units.

Rules. The energy of a route is the sum of its terrain costs. A route above the energy limit is infeasible. Among feasible routes, minimize energy.

Task. Which route is feasible and uses the least energy?

Step-by-step solution. 1. Compute route totals: R1=5, R2=5, R3=5. 2. Discard any total above 9. 3. The smallest remaining total is R1=5.

Answer. Route R1, using 5 energy units.

Knowledge takeaway. Relief changes the cost of movement; “shortest” and “easiest” are not automatically the same.

Formal model. Weighted paths with a feasibility budget; sum costs and choose the minimum feasible path.

G8. Climate from data

Problem 286. Classifying climate data: Station A

Knowledge context. Climate descriptions summarize long-term patterns. In a reasoning problem, explicit thresholds let us classify numerical data without hidden knowledge.

Given facts. Use this classroom classification: cool if mean temperature <12°C; mild if 12–17°C; warm if >17°C. Dry if annual precipitation <500 mm; wet if ≥ 500 mm. Data: Station A: T=8°C, P=460 mm; Station B: T=16°C, P=740 mm; Station C: T=12°C, P=520 mm.

Rules. Apply only the stated thresholds. Temperature and precipitation are separate dimensions, so each station receives two labels.

Task. Classify Station A. Then identify whether any other station has the same two-label class.

Step-by-step solution. 1. Compare T=8 with the temperature thresholds: it is cool. 2. Compare P=460 with 500 mm: it is dry. 3. Compare the combined label with the other stations; matches: none.

Answer. Station A is cool and dry; no other station has the same class.

Knowledge takeaway. Classification is the application of clear predicates to data; the rule must be stated before the label is inferred.

Formal model. Numeric predicates with thresholds; output the conjunction of the temperature and precipitation categories.

Problem 287. Classifying climate data: Station B

Knowledge context. Climate descriptions summarize long-term patterns. In a reasoning problem, explicit thresholds let us classify numerical data without hidden knowledge.

Given facts. Use this classroom classification: cool if mean temperature <12°C; mild if 12–17°C; warm if >17°C. Dry if annual precipitation <500 mm; wet if ≥ 500 mm. Data: Station A: T=9°C, P=460 mm; Station B: T=17°C, P=740 mm; Station C: T=13°C, P=530 mm.

Rules. Apply only the stated thresholds. Temperature and precipitation are separate dimensions, so each station receives two labels.

Task. Classify Station B. Then identify whether any other station has the same two-label class.

Step-by-step solution. 1. Compare T=17 with the temperature thresholds: it is mild. 2. Compare P=740 with 500 mm: it is wet. 3. Compare the combined label with the other stations; matches: ['Station C'].

Answer. Station B is mild and wet; same class: Station C.

Knowledge takeaway. Classification is the application of clear predicates to data; the rule must be stated before the label is inferred.

Formal model. Numeric predicates with thresholds; output the conjunction of the temperature and precipitation categories.

Problem 288. Classifying climate data: Station C

Knowledge context. Climate descriptions summarize long-term patterns. In a reasoning problem, explicit thresholds let us classify numerical data without hidden knowledge.

Given facts. Use this classroom classification: cool if mean temperature <12°C; mild if 12–17°C; warm if >17°C. Dry if annual precipitation <500 mm; wet if ≥ 500 mm. Data: Station A: T=10°C, P=460 mm; Station B: T=18°C, P=740 mm; Station C: T=14°C, P=540 mm.

Rules. Apply only the stated thresholds. Temperature and precipitation are separate dimensions, so each station receives two labels.

Task. Classify Station C. Then identify whether any other station has the same two-label class.

Step-by-step solution. 1. Compare T=14 with the temperature thresholds: it is mild. 2. Compare P=540 with 500 mm: it is wet. 3. Compare the combined label with the other stations; matches: none.

Answer. Station C is mild and wet; no other station has the same class.

Knowledge takeaway. Classification is the application of clear predicates to data; the rule must be stated before the label is inferred.

Formal model. Numeric predicates with thresholds; output the conjunction of the temperature and precipitation categories.

Problem 289. Classifying climate data: Station A

Knowledge context. Climate descriptions summarize long-term patterns. In a reasoning problem, explicit thresholds let us classify numerical data without hidden knowledge.

Given facts. Use this classroom classification: cool if mean temperature <12°C; mild if 12–17°C; warm if >17°C. Dry if annual precipitation <500 mm; wet if ≥ 500 mm. Data: Station A: T=11°C, P=460 mm; Station B: T=19°C, P=740 mm; Station C: T=15°C, P=550 mm.

Rules. Apply only the stated thresholds. Temperature and precipitation are separate dimensions, so each station receives two labels.

Task. Classify Station A. Then identify whether any other station has the same two-label class.

Step-by-step solution. 1. Compare T=11 with the temperature thresholds: it is cool. 2. Compare P=460 with 500 mm: it is dry. 3. Compare the combined label with the other stations; matches: none.

Answer. Station A is cool and dry; no other station has the same class.

Knowledge takeaway. Classification is the application of clear predicates to data; the rule must be stated before the label is inferred.

Formal model. Numeric predicates with thresholds; output the conjunction of the temperature and precipitation categories.

Problem 290. Classifying climate data: Station B

Knowledge context. Climate descriptions summarize long-term patterns. In a reasoning problem, explicit thresholds let us classify numerical data without hidden knowledge.

Given facts. Use this classroom classification: cool if mean temperature <12°C; mild if 12–17°C; warm if >17°C. Dry if annual precipitation <500 mm; wet if ≥ 500 mm. Data: Station A: T=12°C, P=460 mm; Station B: T=20°C, P=740 mm; Station C: T=16°C, P=560 mm.

Rules. Apply only the stated thresholds. Temperature and precipitation are separate dimensions, so each station receives two labels.

Task. Classify Station B. Then identify whether any other station has the same two-label class.

Step-by-step solution. 1. Compare T=20 with the temperature thresholds: it is warm. 2. Compare P=740 with 500 mm: it is wet. 3. Compare the combined label with the other stations; matches: none.

Answer. Station B is warm and wet; no other station has the same class.

Knowledge takeaway. Classification is the application of clear predicates to data; the rule must be stated before the label is inferred.

Formal model. Numeric predicates with thresholds; output the conjunction of the temperature and precipitation categories.

G9. Exchange, routes, and capacities

Problem 291. Bottlenecks in the Meadow–Fern exchange network

Knowledge context. Trade and transport networks have capacities. A single narrow connection can limit a much larger network.

Given facts. Goods move from Meadow to Fern along two branches: Meadow→Riverbend capacity 5, Riverbend→Fern capacity 4; Meadow→Cedar capacity 2, Cedar→Fern capacity 6.

Rules. A branch cannot carry more than its smallest edge capacity. Independent branch capacities can be added, but a shared destination capacity can cap the total.

Task. What is the maximum number of units that can reach Fern in one period? Identify the bottleneck(s).

Step-by-step solution. 1. Branch 1 is limited by min(5,4)=4. 2. Branch 2 is limited by min(2,6)=2. 3. Together the branches could carry 6 units. 4. No extra shared cap is given, so the maximum is 6.

Answer. 6 units.

Knowledge takeaway. A bottleneck is a local limit that constrains the total possible flow.

Formal model. Flow network; branch capacity is a minimum along a path; total is the sum of independent branches subject to shared caps.

Problem 292. Bottlenecks in the Zephyr–Harbor exchange network

Knowledge context. Trade and transport networks have capacities. A single narrow connection can limit a much larger network.

Given facts. Goods move from Zephyr to Harbor along two branches: Zephyr→Stone capacity 6, Stone→Harbor capacity 4; Zephyr→Dover capacity 3, Dover→Harbor capacity 6.

Rules. A branch cannot carry more than its smallest edge capacity. Independent branch capacities can be added, but a shared destination capacity can cap the total.

Task. What is the maximum number of units that can reach Harbor in one period? Identify the bottleneck(s).

Step-by-step solution. 1. Branch 1 is limited by min(6,4)=4. 2. Branch 2 is limited by min(3,6)=3. 3. Together the branches could carry 7 units. 4. No extra shared cap is given, so the maximum is 7.

Answer. 7 units.

Knowledge takeaway. A bottleneck is a local limit that constrains the total possible flow.

Formal model. Flow network; branch capacity is a minimum along a path; total is the sum of independent branches subject to shared caps.

Problem 293. Bottlenecks in the Grove–Pine exchange network

Knowledge context. Trade and transport networks have capacities. A single narrow connection can limit a much larger network.

Given facts. Goods move from Grove to Pine along two branches: Grove→Alder capacity 7, Alder→Pine capacity 4; Grove→Kite capacity 4, Kite→Pine capacity 6.

Rules. A branch cannot carry more than its smallest edge capacity. Independent branch capacities can be added, but a shared destination capacity can cap the total.

Task. What is the maximum number of units that can reach Pine in one period? Identify the bottleneck(s).

Step-by-step solution. 1. Branch 1 is limited by min(7,4)=4. 2. Branch 2 is limited by min(4,6)=4. 3. Together the branches could carry 8 units. 4. No extra shared cap is given, so the maximum is 8.

Answer. 8 units.

Knowledge takeaway. A bottleneck is a local limit that constrains the total possible flow.

Formal model. Flow network; branch capacity is a minimum along a path; total is the sum of independent branches subject to shared caps.

Problem 294. Bottlenecks in the Zephyr–Harbor exchange network

Knowledge context. Trade and transport networks have capacities. A single narrow connection can limit a much larger network.

Given facts. Goods move from Zephyr to Harbor along two branches: Zephyr→Dover capacity 8, Dover→Harbor capacity 4; Zephyr→Oak capacity 5, Oak→Harbor capacity 6.

Rules. A branch cannot carry more than its smallest edge capacity. Independent branch capacities can be added, but a shared destination capacity can cap the total.

Task. What is the maximum number of units that can reach Harbor in one period? Identify the bottleneck(s).

Step-by-step solution. 1. Branch 1 is limited by min(8,4)=4. 2. Branch 2 is limited by min(5,6)=5. 3. Together the branches could carry 9 units. 4. No extra shared cap is given, so the maximum is 9.

Answer. 9 units.

Knowledge takeaway. A bottleneck is a local limit that constrains the total possible flow.

Formal model. Flow network; branch capacity is a minimum along a path; total is the sum of independent branches subject to shared caps.

Problem 295. Bottlenecks in the Yarrow–Linden exchange network

Knowledge context. Trade and transport networks have capacities. A single narrow connection can limit a much larger network.

Given facts. Goods move from Yarrow to Linden along two branches: Yarrow→Kite capacity 9, Kite→Linden capacity 4; Yarrow→Birch capacity 6, Birch→Linden capacity 6.

Rules. A branch cannot carry more than its smallest edge capacity. Independent branch capacities can be added, but a shared destination capacity can cap the total.

Task. What is the maximum number of units that can reach Linden in one period? Identify the bottleneck(s).

Step-by-step solution. 1. Branch 1 is limited by min(9,4)=4. 2. Branch 2 is limited by min(6,6)=6. 3. Together the branches could carry 10 units. 4. No extra shared cap is given, so the maximum is 10.

Answer. 10 units.

Knowledge takeaway. A bottleneck is a local limit that constrains the total possible flow.

Formal model. Flow network; branch capacity is a minimum along a path; total is the sum of independent branches subject to shared caps.

G10. Regions, sets, and membership

Problem 296. Nested places: Fern, Coastland, and Republic C

Knowledge context. Geography uses nested categories: a locality can be inside a region, a region inside a state, and a state inside a continent.

Given facts. Fern is in Coastland. Coastland is in Republic C. Republic C is in the continent of Meridian.

Rules. If X is contained in Y and Y is contained in Z, then X is contained in Z. The reverse inference is not valid: knowing that two places are in one country does not prove that they are in the same region.

Task. Is Fern in Republic C?

Step-by-step solution. 1. Follow the chain city→region→country. 2. Containment is transitive, so the city is in the country.

Answer. Yes, Fern is in Republic C.

Knowledge takeaway. Hierarchies support upward transitive inference, but they do not justify arbitrary reverse inferences.

Formal model. Containment relation with transitivity; answer membership queries while rejecting invalid converse inference.

Problem 297. Nested places: Zephyr, Highridge, and Republic D

Knowledge context. Geography uses nested categories: a locality can be inside a region, a region inside a state, and a state inside a continent.

Given facts. Zephyr is in Highridge. Highridge is in Republic D. Republic D is in the continent of Pacifica.

Rules. If X is contained in Y and Y is contained in Z, then X is contained in Z. The reverse inference is not valid: knowing that two places are in one country does not prove that they are in the same region.

Task. Is Zephyr in Pacifica?

Step-by-step solution. 1. Follow city→region→country→continent. 2. Transitivity carries membership through the chain.

Answer. Yes, Zephyr is in Pacifica.

Knowledge takeaway. Hierarchies support upward transitive inference, but they do not justify arbitrary reverse inferences.

Formal model. Containment relation with transitivity; answer membership queries while rejecting invalid converse inference.

Problem 298. Nested places: Meadow, Eastvale, and Republic B

Knowledge context. Geography uses nested categories: a locality can be inside a region, a region inside a state, and a state inside a continent.

Given facts. Meadow is in Eastvale. Eastvale is in Republic B. Republic B is in the continent of Aurora.

Rules. If X is contained in Y and Y is contained in Z, then X is contained in Z. The reverse inference is not valid: knowing that two places are in one country does not prove that they are in the same region.

Task. Can we conclude that Republic B is in Meadow?

Step-by-step solution. 1. The facts go from smaller units to larger units. 2. No rule allows country→city, so the reverse claim is unsupported.

Answer. No. The containment relation cannot be reversed.

Knowledge takeaway. Hierarchies support upward transitive inference, but they do not justify arbitrary reverse inferences.

Formal model. Containment relation with transitivity; answer membership queries while rejecting invalid converse inference.

Problem 299. Nested places: Linden, Lake District, and Republic A

Knowledge context. Geography uses nested categories: a locality can be inside a region, a region inside a state, and a state inside a continent.

Given facts. Linden is in Lake District. Lake District is in Republic A. Republic A is in the continent of Meridian.

Rules. If X is contained in Y and Y is contained in Z, then X is contained in Z. The reverse inference is not valid: knowing that two places are in one country does not prove that they are in the same region.

Task. Which larger units contain Linden?

Step-by-step solution. 1. Read the direct container: Lake District. 2. Propagate upward to Republic A and Meridian.

Answer. Lake District, Republic A, and Meridian.

Knowledge takeaway. Hierarchies support upward transitive inference, but they do not justify arbitrary reverse inferences.

Formal model. Containment relation with transitivity; answer membership queries while rejecting invalid converse inference.

Problem 300. Nested places: Riverbend, Northland, and Republic D

Knowledge context. Geography uses nested categories: a locality can be inside a region, a region inside a state, and a state inside a continent.

Given facts. Riverbend is in Northland. Northland is in Republic D. Republic D is in the continent of Meridian.

Rules. If X is contained in Y and Y is contained in Z, then X is contained in Z. The reverse inference is not valid: knowing that two places are in one country does not prove that they are in the same region.

Task. Does being in Republic D prove two cities share a region?

Step-by-step solution. 1. Two cities can belong to the same country while belonging to different regions. 2. The region relation is therefore not deducible from country membership alone.

Answer. No. Country membership alone is insufficient to identify the region.

Knowledge takeaway. Hierarchies support upward transitive inference, but they do not justify arbitrary reverse inferences.

Formal model. Containment relation with transitivity; answer membership queries while rejecting invalid converse inference.

H1. Ordering events

Problem 301. Ordering the events of Eastvale

Knowledge context. Historical chronology often starts from partial information: some events are ordered relative to others even when exact dates are not known.

Given facts. Bridge opening happened before Founding. Founding happened before New school. Market charter happened after Bridge opening. Flood happened after New school.

Rules. “Before” is transitive: if A is before B and B is before C, then A is before C. If two events are not connected by the facts, their relative order may be undetermined.

Task. What must be true about the order of Bridge opening and New school? Can the exact order of Market charter and Founding be determined?

Step-by-step solution. 1. From Bridge opening before Founding and Founding before New school, infer Bridge opening before New school. 2. Market charter is only constrained to be after Bridge opening. There is no stated relation between Market charter and Founding. 3. Therefore the first comparison is forced but the second is underdetermined.

Answer. Bridge opening must be before New school. The relative order of Market charter and Founding cannot be determined from the given facts.

Knowledge takeaway. A careful reasoner distinguishes what must be true from what merely could be true.

Formal model. Partial order / directed acyclic graph; use transitive closure and detect incomparable events.

Problem 302. Ordering the events of Coastland

Knowledge context. Historical chronology often starts from partial information: some events are ordered relative to others even when exact dates are not known.

Given facts. Market charter happened before Bridge opening. Bridge opening happened before Founding. Flood happened after Market charter. New school happened after Founding.

Rules. “Before” is transitive: if A is before B and B is before C, then A is before C. If two events are not connected by the facts, their relative order may be undetermined.

Task. What must be true about the order of Market charter and Founding? Can the exact order of Flood and Bridge opening be determined?

Step-by-step solution. 1. From Market charter before Bridge opening and Bridge opening before Founding, infer Market charter before Founding. 2. Flood is only constrained to be after Market charter. There is no stated relation between Flood and Bridge opening. 3. Therefore the first comparison is forced but the second is underdetermined.

Answer. Market charter must be before Founding. The relative order of Flood and Bridge opening cannot be determined from the given facts.

Knowledge takeaway. A careful reasoner distinguishes what must be true from what merely could be true.

Formal model. Partial order / directed acyclic graph; use transitive closure and detect incomparable events.

Problem 303. Ordering the events of Highridge

Knowledge context. Historical chronology often starts from partial information: some events are ordered relative to others even when exact dates are not known.

Given facts. New school happened before Flood. Flood happened before Founding. Market charter happened after New school. Bridge opening happened after Founding.

Rules. “Before” is transitive: if A is before B and B is before C, then A is before C. If two events are not connected by the facts, their relative order may be undetermined.

Task. What must be true about the order of New school and Founding? Can the exact order of Market charter and Flood be determined?

Step-by-step solution. 1. From New school before Flood and Flood before Founding, infer New school before Founding. 2. Market charter is only constrained to be after New school. There is no stated relation between Market charter and Flood. 3. Therefore the first comparison is forced but the second is underdetermined.

Answer. New school must be before Founding. The relative order of Market charter and Flood cannot be determined from the given facts.

Knowledge takeaway. A careful reasoner distinguishes what must be true from what merely could be true.

Formal model. Partial order / directed acyclic graph; use transitive closure and detect incomparable events.

Problem 304. Ordering the events of Eastvale

Knowledge context. Historical chronology often starts from partial information: some events are ordered relative to others even when exact dates are not known.

Given facts. New school happened before Bridge opening. Bridge opening happened before Market charter. Founding happened after New school. Flood happened after Market charter.

Rules. “Before” is transitive: if A is before B and B is before C, then A is before C. If two events are not connected by the facts, their relative order may be undetermined.

Task. What must be true about the order of New school and Market charter? Can the exact order of Founding and Bridge opening be determined?

Step-by-step solution. 1. From New school before Bridge opening and Bridge opening before Market charter, infer New school before Market charter. 2. Founding is only constrained to be after New school. There is no stated relation between Founding and Bridge opening. 3. Therefore the first comparison is forced but the second is underdetermined.

Answer. New school must be before Market charter. The relative order of Founding and Bridge opening cannot be determined from the given facts.

Knowledge takeaway. A careful reasoner distinguishes what must be true from what merely could be true.

Formal model. Partial order / directed acyclic graph; use transitive closure and detect incomparable events.

Problem 305. Ordering the events of Eastvale

Knowledge context. Historical chronology often starts from partial information: some events are ordered relative to others even when exact dates are not known.

Given facts. Founding happened before Bridge opening. Bridge opening happened before Market charter. New school happened after Founding. Flood happened after Market charter.

Rules. “Before” is transitive: if A is before B and B is before C, then A is before C. If two events are not connected by the facts, their relative order may be undetermined.

Task. What must be true about the order of Founding and Market charter? Can the exact order of New school and Bridge opening be determined?

Step-by-step solution. 1. From Founding before Bridge opening and Bridge opening before Market charter, infer Founding before Market charter. 2. New school is only constrained to be after Founding. There is no stated relation between New school and Bridge opening. 3. Therefore the first comparison is forced but the second is underdetermined.

Answer. Founding must be before Market charter. The relative order of New school and Bridge opening cannot be determined from the given facts.

Knowledge takeaway. A careful reasoner distinguishes what must be true from what merely could be true.

Formal model. Partial order / directed acyclic graph; use transitive closure and detect incomparable events.

H2. Dates, durations, and centuries

Problem 306. Duration and centuries: 1448–1474

Knowledge context. Dates can be used as numbers, but historical labels such as centuries follow explicit conventions.

Given facts. An institution was founded in 1448 and reorganized in 1474. Use the rule: years 1–100 are the 1st century, 101–200 the 2nd, and so on. Duration is end year minus start year.

Rules. Apply the century rule mechanically; do not infer a century from the first two digits alone. Compute elapsed time by subtraction.

Task. How many years passed between the two events? In which centuries did 1448 and 1474 fall?

Step-by-step solution. 1. Elapsed time: 1474−1448=26 years. 2. Century(1448)=floor((1448−1)/100)+1=15. 3. Century(1474)=15.

Answer. 26 years; 1448 is in the 15th century and 1474 in the 15th century.

Knowledge takeaway. Chronological reasoning combines arithmetic with a naming convention; both should be applied explicitly.

Formal model. Integer arithmetic plus century function c(y)=floor((y−1)/100)+1.

Problem 307. Duration and centuries: 1465–1496

Knowledge context. Dates can be used as numbers, but historical labels such as centuries follow explicit conventions.

Given facts. An institution was founded in 1465 and reorganized in 1496. Use the rule: years 1–100 are the 1st century, 101–200 the 2nd, and so on. Duration is end year minus start year.

Rules. Apply the century rule mechanically; do not infer a century from the first two digits alone. Compute elapsed time by subtraction.

Task. How many years passed between the two events? In which centuries did 1465 and 1496 fall?

Step-by-step solution. 1. Elapsed time: 1496−1465=31 years. 2. Century(1465)=floor((1465−1)/100)+1=15. 3. Century(1496)=15.

Answer. 31 years; 1465 is in the 15th century and 1496 in the 15th century.

Knowledge takeaway. Chronological reasoning combines arithmetic with a naming convention; both should be applied explicitly.

Formal model. Integer arithmetic plus century function c(y)=floor((y−1)/100)+1.

Problem 308. Duration and centuries: 1480–1516

Knowledge context. Dates can be used as numbers, but historical labels such as centuries follow explicit conventions.

Given facts. An institution was founded in 1480 and reorganized in 1516. Use the rule: years 1–100 are the 1st century, 101–200 the 2nd, and so on. Duration is end year minus start year.

Rules. Apply the century rule mechanically; do not infer a century from the first two digits alone. Compute elapsed time by subtraction.

Task. How many years passed between the two events? In which centuries did 1480 and 1516 fall?

Step-by-step solution. 1. Elapsed time: 1516−1480=36 years. 2. Century(1480)=floor((1480−1)/100)+1=15. 3. Century(1516)=16.

Answer. 36 years; 1480 is in the 15th century and 1516 in the 16th century.

Knowledge takeaway. Chronological reasoning combines arithmetic with a naming convention; both should be applied explicitly.

Formal model. Integer arithmetic plus century function c(y)=floor((y−1)/100)+1.

Problem 309. Duration and centuries: 1499–1540

Knowledge context. Dates can be used as numbers, but historical labels such as centuries follow explicit conventions.

Given facts. An institution was founded in 1499 and reorganized in 1540. Use the rule: years 1–100 are the 1st century, 101–200 the 2nd, and so on. Duration is end year minus start year.

Rules. Apply the century rule mechanically; do not infer a century from the first two digits alone. Compute elapsed time by subtraction.

Task. How many years passed between the two events? In which centuries did 1499 and 1540 fall?

Step-by-step solution. 1. Elapsed time: 1540−1499=41 years. 2. Century(1499)=floor((1499−1)/100)+1=15. 3. Century(1540)=16.

Answer. 41 years; 1499 is in the 15th century and 1540 in the 16th century.

Knowledge takeaway. Chronological reasoning combines arithmetic with a naming convention; both should be applied explicitly.

Formal model. Integer arithmetic plus century function c(y)=floor((y−1)/100)+1.

Problem 310. Duration and centuries: 1509–1555

Knowledge context. Dates can be used as numbers, but historical labels such as centuries follow explicit conventions.

Given facts. An institution was founded in 1509 and reorganized in 1555. Use the rule: years 1–100 are the 1st century, 101–200 the 2nd, and so on. Duration is end year minus start year.

Rules. Apply the century rule mechanically; do not infer a century from the first two digits alone. Compute elapsed time by subtraction.

Task. How many years passed between the two events? In which centuries did 1509 and 1555 fall?

Step-by-step solution. 1. Elapsed time: 1555−1509=46 years. 2. Century(1509)=floor((1509−1)/100)+1=16. 3. Century(1555)=16.

Answer. 46 years; 1509 is in the 16th century and 1555 in the 16th century.

Knowledge takeaway. Chronological reasoning combines arithmetic with a naming convention; both should be applied explicitly.

Formal model. Integer arithmetic plus century function c(y)=floor((y−1)/100)+1.

H3. Sources and supported claims

Problem 311. What can one letter prove?

Knowledge context. Historical sources are evidence, not automatic answers. The strength of a conclusion depends on what the source actually records.

Given facts. A historian has one source: a letter. The source directly shows that a named person wrote a message. A student proposes two claims: C1: a named person wrote a message; C2: the whole population agreed with the ruler.

Rules. A source supports only claims that can reasonably be inferred from what it contains. Absence of evidence for a broad claim is not proof of that claim.

Task. Which claim is directly supported, and which is not justified by this source alone?

Step-by-step solution. 1. The letter directly contains evidence for C1. 2. C2 is much broader and no stated feature of the source measures the opinion of the whole population. 3. Therefore C1 is supported and C2 is not established.

Answer. C1 is supported. C2 is not justified by the source alone.

Knowledge takeaway. Good historical reasoning matches the scope of a claim to the scope of the evidence.

Formal model. Claim-evidence compatibility relation; accept only claims entailed by the source attributes.

Problem 312. What can one photograph prove?

Knowledge context. Historical sources are evidence, not automatic answers. The strength of a conclusion depends on what the source actually records.

Given facts. A historian has one source: a photograph. The source directly shows that a scene was photographed. A student proposes two claims: C1: a scene was photographed; C2: the whole population agreed with the ruler.

Rules. A source supports only claims that can reasonably be inferred from what it contains. Absence of evidence for a broad claim is not proof of that claim.

Task. Which claim is directly supported, and which is not justified by this source alone?

Step-by-step solution. 1. The photograph directly contains evidence for C1. 2. C2 is much broader and no stated feature of the source measures the opinion of the whole population. 3. Therefore C1 is supported and C2 is not established.

Answer. C1 is supported. C2 is not justified by the source alone.

Knowledge takeaway. Good historical reasoning matches the scope of a claim to the scope of the evidence.

Formal model. Claim-evidence compatibility relation; accept only claims entailed by the source attributes.

Problem 313. What can one coin prove?

Knowledge context. Historical sources are evidence, not automatic answers. The strength of a conclusion depends on what the source actually records.

Given facts. A historian has one source: a coin. The source directly shows that people used metal objects. A student proposes two claims: C1: people used metal objects; C2: the whole population agreed with the ruler. Cross-domain check: a survey team starts work at 8:00 and works for 2 hour(s) without a break.

Rules. A source supports only claims that can reasonably be inferred from what it contains. Absence of evidence for a broad claim is not proof of that claim. For the cross-domain check, finishing time = starting time + duration.

Task. Which claim is directly supported, and which is not justified by this source alone? Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. The coin directly contains evidence for C1. 2. C2 is much broader and no stated feature of the source measures the opinion of the whole population. 3. Therefore C1 is supported and C2 is not established. Cross-domain check: 8+2=10, so the team finishes at 10:00.

Answer. C1 is supported. C2 is not justified by the source alone. Cross-domain answer: 10:00.

Knowledge takeaway. Good historical reasoning matches the scope of a claim to the scope of the evidence.

Formal model. Claim-evidence compatibility relation; accept only claims entailed by the source attributes.

Problem 314. What can one letter prove?

Knowledge context. Historical sources are evidence, not automatic answers. The strength of a conclusion depends on what the source actually records.

Given facts. A historian has one source: a letter. The source directly shows that a named person wrote a message. A student proposes two claims: C1: a named person wrote a message; C2: the whole population agreed with the ruler. Cross-domain check: a local committee has 16 members, requires at least 9 present for quorum, and 14 are present.

Rules. A source supports only claims that can reasonably be inferred from what it contains. Absence of evidence for a broad claim is not proof of that claim. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. Which claim is directly supported, and which is not justified by this source alone? Cross-domain check: is quorum met?

Step-by-step solution. 1. The letter directly contains evidence for C1. 2. C2 is much broader and no stated feature of the source measures the opinion of the whole population. 3. Therefore C1 is supported and C2 is not established. Cross-domain check: compare 14 with 9; $14 \geq 9$ is true.

Answer. C1 is supported. C2 is not justified by the source alone. Cross-domain answer: quorum is met.

Knowledge takeaway. Good historical reasoning matches the scope of a claim to the scope of the evidence.

Formal model. Claim-evidence compatibility relation; accept only claims entailed by the source attributes.

Problem 315. What can one tool prove?

Knowledge context. Historical sources are evidence, not automatic answers. The strength of a conclusion depends on what the source actually records.

Given facts. A historian has one source: a tool. The source directly shows that a tool existed. A student proposes two claims: C1: a tool existed; C2: the whole population agreed with the ruler. Cross-domain check: researchers collected 11 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. A source supports only claims that can reasonably be inferred from what it contains. Absence of evidence for a broad claim is not proof of that claim. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. Which claim is directly supported, and which is not justified by this source alone? Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. The tool directly contains evidence for C1. 2. C2 is much broader and no stated feature of the source measures the opinion of the whole population. 3. Therefore C1 is supported and C2 is not established. Cross-domain check: 11−4=7 independent reports remain.

Answer. C1 is supported. C2 is not justified by the source alone. Cross-domain answer: 7 independent reports.

Knowledge takeaway. Good historical reasoning matches the scope of a claim to the scope of the evidence.

Formal model. Claim-evidence compatibility relation; accept only claims entailed by the source attributes.

H4. Conflicting sources

Problem 316. Resolving conflicting sources: case 1

Knowledge context. Historical accounts can disagree. Reasoning requires evaluating how a source knows what it claims, not simply counting statements.

Given facts. Source A, written on the day of an event, says 40 carts arrived. Source B, copied 80 years later from an unknown account, says 90 carts arrived. Source C is an independent warehouse record from the same week and lists 42 carts. Cross-domain check: on a field sketch, 1 cm represents 3 km and two checkpoints are 3 cm apart.

Rules. When sources conflict, compare proximity to the event, independence, and whether a source has a reason or mechanism to record the specific fact. Agreement between independent relevant sources increases support but does not create absolute certainty. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Which account is better supported by the evidence given, and why? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. A is contemporary and reports 40. 2. C is an independent operational record and reports 42, close to A. 3. B is much later and has an unknown chain of transmission. 4. The cluster near 40–42 is therefore better supported than 90. Cross-domain check: 3×3=9 km.

Answer. An arrival count near 40–42 carts is better supported than 90. Cross-domain answer: 9 km.

Knowledge takeaway. Independent, relevant, well-positioned evidence usually deserves more weight than late repetition with unknown provenance.

Formal model. Evidence graph with source reliability/independence attributes; compare support for competing hypotheses.

Problem 317. Resolving conflicting sources: case 2

Knowledge context. Historical accounts can disagree. Reasoning requires evaluating how a source knows what it claims, not simply counting statements.

Given facts. Source A, an eyewitness diary, says the bridge opened on Monday. Source B, a later textbook, says Tuesday. Source C, an independent dated toll record, begins charging on Monday. Cross-domain check: a survey team starts work at 12:00 and works for 2 hour(s) without a break. Mixed-domain verification: a regional archive has 33 map sheets and receives 8 additional sheets during the exercise.

Rules. When sources conflict, compare proximity to the event, independence, and whether a source has a reason or mechanism to record the specific fact. Agreement between independent relevant sources increases support but does not create absolute certainty. For the cross-domain check, finishing time = starting time + duration. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. Which account is better supported by the evidence given, and why? Cross-domain check: at what time does the team finish? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. A and C are close to the event and independent. 2. Both point to Monday. 3. B is later and conflicts with two stronger, independent pieces of evidence. Cross-domain check: 12+2=14, so the team finishes at 14:00. Mixed-domain verification: 33+8=41.

Answer. Monday is better supported, while remaining a historical conclusion rather than absolute proof. Cross-domain answer: 14:00. Mixed-domain answer: 41 map sheets.

Knowledge takeaway. Independent, relevant, well-positioned evidence usually deserves more weight than late repetition with unknown provenance.

Formal model. Evidence graph with source reliability/independence attributes; compare support for competing hypotheses.

Problem 318. Resolving conflicting sources: case 3

Knowledge context. Historical accounts can disagree. Reasoning requires evaluating how a source knows what it claims, not simply counting statements.

Given facts. Source A, written on the day of an event, says 40 carts arrived. Source B, copied 80 years later from an unknown account, says 90 carts arrived. Source C is an independent warehouse record from the same week and lists 42 carts. Cross-domain check: a local committee has 11 members, requires at least 6 present for quorum, and 11 are present.

Rules. When sources conflict, compare proximity to the event, independence, and whether a source has a reason or mechanism to record the specific fact. Agreement between independent relevant sources increases support but does not create absolute certainty. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. Which account is better supported by the evidence given, and why? Cross-domain check: is quorum met?

Step-by-step solution. 1. A is contemporary and reports 40. 2. C is an independent operational record and reports 42, close to A. 3. B is much later and has an unknown chain of transmission. 4. The cluster near 40–42 is therefore better supported than 90. Cross-domain check: compare 11 with 6; 11 ≥ 6 is true.

Answer. An arrival count near 40–42 carts is better supported than 90. Cross-domain answer: quorum is met.

Knowledge takeaway. Independent, relevant, well-positioned evidence usually deserves more weight than late repetition with unknown provenance.

Formal model. Evidence graph with source reliability/independence attributes; compare support for competing hypotheses.

Problem 319. Resolving conflicting sources: case 4

Knowledge context. Historical accounts can disagree. Reasoning requires evaluating how a source knows what it claims, not simply counting statements.

Given facts. Source A, an eyewitness diary, says the bridge opened on Monday. Source B, a later textbook, says Tuesday. Source C, an independent dated toll record, begins charging on Monday. Cross-domain check: researchers collected 15 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. When sources conflict, compare proximity to the event, independence, and whether a source has a reason or mechanism to record the specific fact. Agreement between independent relevant sources increases support but does not create absolute certainty. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. Which account is better supported by the evidence given, and why? Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. A and C are close to the event and independent. 2. Both point to Monday. 3. B is later and conflicts with two stronger, independent pieces of evidence. Cross-domain check: 15−4=11 independent reports remain.

Answer. Monday is better supported, while remaining a historical conclusion rather than absolute proof. Cross-domain answer: 11 independent reports.

Knowledge takeaway. Independent, relevant, well-positioned evidence usually deserves more weight than late repetition with unknown provenance.

Formal model. Evidence graph with source reliability/independence attributes; compare support for competing hypotheses.

Problem 320. Resolving conflicting sources: case 5

Knowledge context. Historical accounts can disagree. Reasoning requires evaluating how a source knows what it claims, not simply counting statements.

Given facts. Source A, written on the day of an event, says 40 carts arrived. Source B, copied 80 years later from an unknown account, says 90 carts arrived. Source C is an independent warehouse record from the same week and lists 42 carts. Cross-domain check: on a field sketch, 1 cm represents 2 km and two checkpoints are 7 cm apart.

Rules. When sources conflict, compare proximity to the event, independence, and whether a source has a reason or mechanism to record the specific fact. Agreement between independent relevant sources increases support but does not create absolute certainty. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Which account is better supported by the evidence given, and why? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. A is contemporary and reports 40. 2. C is an independent operational record and reports 42, close to A. 3. B is much later and has an unknown chain of transmission. 4. The cluster near 40–42 is therefore better supported than 90. Cross-domain check: 7×2=14 km.

Answer. An arrival count near 40–42 carts is better supported than 90. Cross-domain answer: 14 km.

Knowledge takeaway. Independent, relevant, well-positioned evidence usually deserves more weight than late repetition with unknown provenance.

Formal model. Evidence graph with source reliability/independence attributes; compare support for competing hypotheses.

H5. Causes, conditions, and consequences

Problem 321. Cause-and-consequence chain: case 1

Knowledge context. Many explanations can be represented as condition-and-consequence rules. The logical form matters: necessary, sufficient, AND, and OR are not interchangeable.

Given facts. Facts: Rain occurred; The river was already high; The warning siren worked; The gate was closed. Rules: If rain occurs AND the river is already high, flooding occurs. If flooding occurs, the riverside road closes. If the siren works, residents receive a warning. Cross-domain check: a survey team starts work at 8:00 and works for 2 hour(s) without a break.

Rules. Use the rules exactly as written. AND requires all listed conditions; OR requires at least one. A one-way implication cannot automatically be reversed. For the cross-domain check, finishing time = starting time + duration.

Task. Which consequences can be deduced by applying the rules exactly? Do not assume the converse of any rule. Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. Rain and high river satisfy both conditions of the first rule. 2. Infer flooding. 3. Flooding implies road closure. 4. The working siren independently implies a warning. Cross-domain check: 8+2=10, so the team finishes at 10:00.

Answer. Flooding, road closure, and a warning can be deduced. Cross-domain answer: 10:00.

Knowledge takeaway. Causal reasoning becomes clearer when every link is stated and the direction of implication is respected.

Formal model. Forward-chaining rule system over Boolean facts; apply implications until no new facts appear.

Problem 322. Cause-and-consequence chain: case 2

Knowledge context. Many explanations can be represented as condition-and-consequence rules. The logical form matters: necessary, sufficient, AND, and OR are not interchangeable.

Given facts. Facts: Rain occurred; The river was not high; The warning siren worked; The gate was closed. Rules: If rain occurs AND the river is already high, flooding occurs. If flooding occurs, the riverside road closes. If the siren works, residents receive a warning. Cross-domain check: a local committee has 15 members, requires at least 8 present for quorum, and 14 are present.

Rules. Use the rules exactly as written. AND requires all listed conditions; OR requires at least one. A one-way implication cannot automatically be reversed. For the cross-domain check, quorum exists when present members ≥ the stated threshold.

Task. Which consequences can be deduced by applying the rules exactly? Do not assume the converse of any rule. Cross-domain check: is quorum met?

Step-by-step solution. 1. Rain is true but the high-river condition is false. 2. The AND rule for flooding therefore does not fire. 3. Without flooding, road closure is not inferred. 4. The siren rule still gives a warning. Cross-domain check: compare 14 with 8; 14 ≥ 8 is true.

Answer. A warning can be deduced, but flooding and road closure cannot be deduced. Cross-domain answer: quorum is met.

Knowledge takeaway. Causal reasoning becomes clearer when every link is stated and the direction of implication is respected.

Formal model. Forward-chaining rule system over Boolean facts; apply implications until no new facts appear.

Problem 323. Cause-and-consequence chain: case 3

Knowledge context. Many explanations can be represented as condition-and-consequence rules. The logical form matters: necessary, sufficient, AND, and OR are not interchangeable.

Given facts. Facts: Rain occurred; The river was already high; The warning siren worked; The gate was closed; Flooding occurred. Rules: If rain occurs AND the river is already high, flooding occurs. If flooding occurs, the riverside road closes. If the siren works, residents receive a warning. Cross-domain check: researchers collected 9 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. Use the rules exactly as written. AND requires all listed conditions; OR requires at least one. A one-way implication cannot automatically be reversed. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. Which consequences can be deduced by applying the rules exactly? Do not assume the converse of any rule. Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. Flooding is stated directly, so apply the flooding→road-closure rule. 2. Apply siren→warning. 3. Do not use flooding to infer rain or a high river; that would reverse a rule. Cross-domain check: 9−4=5 independent reports remain.

Answer. Road closure and a warning can be deduced; flooding is already given as a fact. Cross-domain answer: 5 independent reports.

Knowledge takeaway. Causal reasoning becomes clearer when every link is stated and the direction of implication is respected.

Formal model. Forward-chaining rule system over Boolean facts; apply implications until no new facts appear.

Problem 324. Cause-and-consequence chain: case 4

Knowledge context. Many explanations can be represented as condition-and-consequence rules. The logical form matters: necessary, sufficient, AND, and OR are not interchangeable.

Given facts. Facts: Rain occurred; The river was already high; The warning siren worked; The gate was closed. Rules: If rain occurs OR the river is already high, the field becomes muddy. If flooding occurs, the riverside road closes. If the siren works, residents receive a warning. Cross-domain check: on a field sketch, 1 cm represents 6 km and two checkpoints are 4 cm apart.

Rules. Use the rules exactly as written. AND requires all listed conditions; OR requires at least one. A one-way implication cannot automatically be reversed. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Which consequences can be deduced by applying the rules exactly? Do not assume the converse of any rule. Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. The mud rule uses OR, so rain alone is enough. 2. Rain is given, therefore mud follows. 3. The siren rule gives a warning. Cross-domain check: 4×6=24 km.

Answer. The field becomes muddy, and the warning follows from the working siren. Cross-domain answer: 24 km.

Knowledge takeaway. Causal reasoning becomes clearer when every link is stated and the direction of implication is respected.

Formal model. Forward-chaining rule system over Boolean facts; apply implications until no new facts appear.

Problem 325. Cause-and-consequence chain: case 5

Knowledge context. Many explanations can be represented as condition-and-consequence rules. The logical form matters: necessary, sufficient, AND, and OR are not interchangeable.

Given facts. Facts: Rain occurred; The river was already high. Rules: If rain AND high river, flooding occurs. If flooding, road closes. If road closes, the bus reroutes. Cross-domain check: a survey team starts work at 12:00 and works for 2 hour(s) without a break.

Rules. Use the rules exactly as written. AND requires all listed conditions; OR requires at least one. A one-way implication cannot automatically be reversed. For the cross-domain check, finishing time = starting time + duration.

Task. Which consequences can be deduced by applying the rules exactly? Do not assume the converse of any rule. Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. Rain and high river jointly imply flooding. 2. Flooding implies road closure. 3. Road closure implies bus rerouting. Cross-domain check: 12+2=14, so the team finishes at 14:00.

Answer. Flooding, road closure, and bus rerouting can all be deduced. Cross-domain answer: 14:00.

Knowledge takeaway. Causal reasoning becomes clearer when every link is stated and the direction of implication is respected.

Formal model. Forward-chaining rule system over Boolean facts; apply implications until no new facts appear.

H6. Travel through time and space

Problem 326. Courier journey from Stone to Oak

Knowledge context. Historical travel can be analyzed with ordinary distance, speed, and time reasoning when the needed values are supplied.

Given facts. A courier leaves Stone for Oak at 7:00. Distance is 80 km and the stated average speed is 20 km/h. A council meeting in Oak begins at 10.0:00. Ignore stops and use travel time = distance ÷ speed.

Rules. Arrival time equals departure time plus travel time. To attend an event from its beginning, arrival must be no later than the event start.

Task. When does the courier arrive, and can the courier attend the meeting from its beginning?

Step-by-step solution. 1. Travel time=80÷20=4.0 hours. 2. Arrival=7.0+4.0=11.0. 3. Compare 11.0 with meeting time 10.0.

Answer. Arrival at about 11.0:00; no, the meeting has already begun.

Knowledge takeaway. Chronology and geography interact: being able to reach a place in time is a constraint, not a fact to memorize.

Formal model. Arithmetic time model; t_arrive=t_depart+d/v; compare with event time.

Problem 327. Courier journey from Pine to Iris

Knowledge context. Historical travel can be analyzed with ordinary distance, speed, and time reasoning when the needed values are supplied.

Given facts. A courier leaves Pine for Iris at 8:00. Distance is 90 km and the stated average speed is 20 km/h. A council meeting in Iris begins at 13.5:00. Ignore stops and use travel time = distance ÷ speed.

Rules. Arrival time equals departure time plus travel time. To attend an event from its beginning, arrival must be no later than the event start.

Task. When does the courier arrive, and can the courier attend the meeting from its beginning?

Step-by-step solution. 1. Travel time=90÷20=4.5 hours. 2. Arrival=8.0+4.5=12.5. 3. Compare 12.5 with meeting time 13.5.

Answer. Arrival at about 12.5:00; yes, on time.

Knowledge takeaway. Chronology and geography interact: being able to reach a place in time is a constraint, not a fact to memorize.

Formal model. Arithmetic time model; t_arrive=t_depart+d/v; compare with event time.

Problem 328. Courier journey from Grove to Cedar

Knowledge context. Historical travel can be analyzed with ordinary distance, speed, and time reasoning when the needed values are supplied.

Given facts. A courier leaves Grove for Cedar at 9:00. Distance is 100 km and the stated average speed is 20 km/h. A council meeting in Cedar begins at 13.0:00. Ignore stops and use travel time = distance ÷ speed.

Rules. Arrival time equals departure time plus travel time. To attend an event from its beginning, arrival must be no later than the event start.

Task. When does the courier arrive, and can the courier attend the meeting from its beginning?

Step-by-step solution. 1. Travel time=100÷20=5.0 hours. 2. Arrival=9.0+5.0=14.0. 3. Compare 14.0 with meeting time 13.0.

Answer. Arrival at about 14.0:00; no, the meeting has already begun.

Knowledge takeaway. Chronology and geography interact: being able to reach a place in time is a constraint, not a fact to memorize.

Formal model. Arithmetic time model; t_arrive=t_depart+d/v; compare with event time.

Problem 329. Courier journey from Yarrow to Cedar

Knowledge context. Historical travel can be analyzed with ordinary distance, speed, and time reasoning when the needed values are supplied.

Given facts. A courier leaves Yarrow for Cedar at 10:00. Distance is 110 km and the stated average speed is 20 km/h. A council meeting in Cedar begins at 16.5:00. Ignore stops and use travel time = distance ÷ speed.

Rules. Arrival time equals departure time plus travel time. To attend an event from its beginning, arrival must be no later than the event start.

Task. When does the courier arrive, and can the courier attend the meeting from its beginning?

Step-by-step solution. 1. Travel time=110÷20=5.5 hours. 2. Arrival=10.0+5.5=15.5. 3. Compare 15.5 with meeting time 16.5.

Answer. Arrival at about 15.5:00; yes, on time.

Knowledge takeaway. Chronology and geography interact: being able to reach a place in time is a constraint, not a fact to memorize.

Formal model. Arithmetic time model; t_arrive=t_depart+d/v; compare with event time.

Problem 330. Courier journey from Kite to Linden

Knowledge context. Historical travel can be analyzed with ordinary distance, speed, and time reasoning when the needed values are supplied.

Given facts. A courier leaves Kite for Linden at 11:00. Distance is 120 km and the stated average speed is 20 km/h. A council meeting in Linden begins at 16.0:00. Ignore stops and use travel time = distance ÷ speed.

Rules. Arrival time equals departure time plus travel time. To attend an event from its beginning, arrival must be no later than the event start.

Task. When does the courier arrive, and can the courier attend the meeting from its beginning?

Step-by-step solution. 1. Travel time=120÷20=6.0 hours. 2. Arrival=11.0+6.0=17.0. 3. Compare 17.0 with meeting time 16.0.

Answer. Arrival at about 17.0:00; no, the meeting has already begun.

Knowledge takeaway. Chronology and geography interact: being able to reach a place in time is a constraint, not a fact to memorize.

Formal model. Arithmetic time model; t_arrive=t_depart+d/v; compare with event time.

H7. Change and continuity

Problem 331. Change and continuity in a settlement: case 1

Knowledge context. History studies both change and continuity. Comparing two snapshots is a systematic way to separate what persisted from what appeared or disappeared.

Given facts. Period 1 features: footpath, market, well, wooden bridge. Period 2 features: market, road, stone bridge, well. Cross-domain check: researchers collected 7 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. A continuity is a feature present in both periods. A disappearance is present only in Period 1. An appearance is present only in Period 2. A replacement can be described when two differently named features serve the same stated role. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. Identify the continuities, disappearances, and appearances. Which conclusions are direct set comparisons rather than interpretations? Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. Intersection gives continuities: ['market', 'well']. 2. Period1−Period2 gives disappearances: ['footpath', 'wooden bridge']. 3. Period2−Period1 gives appearances: ['road', 'stone bridge']. 4. Only role-based replacement would require extra information; the set comparisons are direct. Cross-domain check: 7−4=3 independent reports remain.

Answer. Continuities: ['market', 'well']; disappearances: ['footpath', 'wooden bridge']; appearances: ['road', 'stone bridge']. Cross-domain answer: 3 independent reports.

Knowledge takeaway. Change is not the opposite of continuity; both can occur in the same community at the same time.

Formal model. Set operations: intersection for continuity, set difference for appearance/disappearance.

Problem 332. Change and continuity in a settlement: case 2

Knowledge context. History studies both change and continuity. Comparing two snapshots is a systematic way to separate what persisted from what appeared or disappeared.

Given facts. Period 1 features: footpath, market, well, wooden bridge. Period 2 features: market, road, school, stone bridge, well. Cross-domain check: on a field sketch, 1 cm represents 4 km and two checkpoints are 5 cm apart.

Rules. A continuity is a feature present in both periods. A disappearance is present only in Period 1. An appearance is present only in Period 2. A replacement can be described when two differently named features serve the same stated role. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Identify the continuities, disappearances, and appearances. Which conclusions are direct set comparisons rather than interpretations? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. Intersection gives continuities: ['market', 'well']. 2. Period1−Period2 gives disappearances: ['footpath', 'wooden bridge']. 3. Period2−Period1 gives appearances: ['road', 'school', 'stone bridge']. 4. Only role-based replacement would require extra information; the set comparisons are direct. Cross-domain check: 5×4=20 km.

Answer. Continuities: ['market', 'well']; disappearances: ['footpath', 'wooden bridge']; appearances: ['road', 'school', 'stone bridge']. Cross-domain answer: 20 km.

Knowledge takeaway. Change is not the opposite of continuity; both can occur in the same community at the same time.

Formal model. Set operations: intersection for continuity, set difference for appearance/disappearance.

Problem 333. Change and continuity in a settlement: case 3

Knowledge context. History studies both change and continuity. Comparing two snapshots is a systematic way to separate what persisted from what appeared or disappeared.

Given facts. Period 1 features: footpath, market, mill, well, wooden bridge. Period 2 features: market, mill, road, stone bridge, well. Cross-domain check: a survey team starts work at 12:00 and works for 2 hour(s) without a break.

Rules. A continuity is a feature present in both periods. A disappearance is present only in Period 1. An appearance is present only in Period 2. A replacement can be described when two differently named features serve the same stated role. For the cross-domain check, finishing time = starting time + duration.

Task. Identify the continuities, disappearances, and appearances. Which conclusions are direct set comparisons rather than interpretations? Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. Intersection gives continuities: ['market', 'mill', 'well']. 2. Period1−Period2 gives disappearances: ['footpath', 'wooden bridge']. 3. Period2−Period1 gives appearances: ['road', 'stone bridge']. 4. Only role-based replacement would require extra information; the set comparisons are direct. Cross-domain check: 12+2=14, so the team finishes at 14:00.

Answer. Continuities: ['market', 'mill', 'well']; disappearances: ['footpath', 'wooden bridge']; appearances: ['road', 'stone bridge']. Cross-domain answer: 14:00.

Knowledge takeaway. Change is not the opposite of continuity; both can occur in the same community at the same time.

Formal model. Set operations: intersection for continuity, set difference for appearance/disappearance.

Problem 334. Change and continuity in a settlement: case 4

Knowledge context. History studies both change and continuity. Comparing two snapshots is a systematic way to separate what persisted from what appeared or disappeared.

Given facts. Period 1 features: footpath, market, well, wooden bridge. Period 2 features: market, road, stone bridge. Cross-domain check: a local committee has 9 members, requires at least 5 present for quorum, and 8 are present.

Rules. A continuity is a feature present in both periods. A disappearance is present only in Period 1. An appearance is present only in Period 2. A replacement can be described when two differently named features serve the same stated role. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. Identify the continuities, disappearances, and appearances. Which conclusions are direct set comparisons rather than interpretations? Cross-domain check: is quorum met?

Step-by-step solution. 1. Intersection gives continuities: ['market']. 2. Period1−Period2 gives disappearances: ['footpath', 'well', 'wooden bridge']. 3. Period2−Period1 gives appearances: ['road', 'stone bridge']. 4. Only role-based replacement would require extra information; the set comparisons are direct. Cross-domain check: compare 8 with 5; $8 \geq 5$ is true.

Answer. Continuities: ['market']; disappearances: ['footpath', 'well', 'wooden bridge']; appearances: ['road', 'stone bridge']. Cross-domain answer: quorum is met.

Knowledge takeaway. Change is not the opposite of continuity; both can occur in the same community at the same time.

Formal model. Set operations: intersection for continuity, set difference for appearance/disappearance.

Problem 335. Change and continuity in a settlement: case 5

Knowledge context. History studies both change and continuity. Comparing two snapshots is a systematic way to separate what persisted from what appeared or disappeared.

Given facts. Period 1 features: footpath, market, wall, well, wooden bridge. Period 2 features: market, park, road, stone bridge, well. Cross-domain check: researchers collected 11 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. A continuity is a feature present in both periods. A disappearance is present only in Period 1. An appearance is present only in Period 2. A replacement can be described when two differently named features serve the same stated role. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. Identify the continuities, disappearances, and appearances. Which conclusions are direct set comparisons rather than interpretations? Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. Intersection gives continuities: ['market', 'well']. 2. Period1−Period2 gives disappearances: ['footpath', 'wall', 'wooden bridge']. 3. Period2−Period1 gives appearances: ['park', 'road', 'stone bridge']. 4. Only role-based replacement would require extra information; the set comparisons are direct. Cross-domain check: 11−4=7 independent reports remain.

Answer. Continuities: ['market', 'well']; disappearances: ['footpath', 'wall', 'wooden bridge']; appearances: ['park', 'road', 'stone bridge']. Cross-domain answer: 7 independent reports.

Knowledge takeaway. Change is not the opposite of continuity; both can occur in the same community at the same time.

Formal model. Set operations: intersection for continuity, set difference for appearance/disappearance.

H8. Layers and dating objects

Problem 336. Reading an archaeological sequence: case 1

Knowledge context. Archaeological layers provide relative chronological information. Dates from objects add constraints but do not automatically make every date exact.

Given facts. An undisturbed archaeological site has four layers from top to bottom: A, B, C, D. In an undisturbed sequence, lower layers were deposited earlier. D contains an object dated 1300, C contains one dated 1450, and B contains one dated 1600. Cross-domain check: on a field sketch, 1 cm represents 3 km and two checkpoints are 2 cm apart.

Rules. In an undisturbed sequence, a lower layer is older than a layer above it. A securely dated object provides a constraint on the layer that contains it, but one object does not date every other object exactly. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Which is older, Layer C or Layer B? Can the exact year of deposition of Layer A be deduced? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. C lies below B, so C is older than B by the stratigraphic rule. 2. No exact dated object or deposition year is given for A. 3. Therefore only relative ordering, not an exact year for A, is justified. Cross-domain check: 2×3=6 km.

Answer. Layer C is older than Layer B. The exact deposition year of Layer A cannot be deduced. Cross-domain answer: 6 km.

Knowledge takeaway. Relative dating and absolute dating answer different questions and should not be confused.

Formal model. Partial order from vertical position plus numeric constraints from dated objects.

Problem 337. Reading an archaeological sequence: case 2

Knowledge context. Archaeological layers provide relative chronological information. Dates from objects add constraints but do not automatically make every date exact.

Given facts. An undisturbed archaeological site has four layers from top to bottom: A, B, C, D. In an undisturbed sequence, lower layers were deposited earlier. D contains an object dated 1305, C contains one dated 1465, and B contains one dated 1620. Cross-domain check: a survey team starts work at 8:00 and works for 2 hour(s) without a break.

Rules. In an undisturbed sequence, a lower layer is older than a layer above it. A securely dated object provides a constraint on the layer that contains it, but one object does not date every other object exactly. For the cross-domain check, finishing time = starting time + duration.

Task. Which is older, Layer C or Layer B? Can the exact year of deposition of Layer A be deduced? Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. C lies below B, so C is older than B by the stratigraphic rule. 2. No exact dated object or deposition year is given for A. 3. Therefore only relative ordering, not an exact year for A, is justified. Cross-domain check: 8+2=10, so the team finishes at 10:00.

Answer. Layer C is older than Layer B. The exact deposition year of Layer A cannot be deduced. Cross-domain answer: 10:00.

Knowledge takeaway. Relative dating and absolute dating answer different questions and should not be confused.

Formal model. Partial order from vertical position plus numeric constraints from dated objects.

Problem 338. Reading an archaeological sequence: case 3

Knowledge context. Archaeological layers provide relative chronological information. Dates from objects add constraints but do not automatically make every date exact.

Given facts. An undisturbed archaeological site has four layers from top to bottom: A, B, C, D. In an undisturbed sequence, lower layers were deposited earlier. D contains an object dated 1310, C contains one dated 1480, and B contains one dated 1640. Cross-domain check: a local committee has 13 members, requires at least 7 present for quorum, and 11 are present.

Rules. In an undisturbed sequence, a lower layer is older than a layer above it. A securely dated object provides a constraint on the layer that contains it, but one object does not date every other object exactly. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. Which is older, Layer C or Layer B? Can the exact year of deposition of Layer A be deduced? Cross-domain check: is quorum met?

Step-by-step solution. 1. C lies below B, so C is older than B by the stratigraphic rule. 2. No exact dated object or deposition year is given for A. 3. Therefore only relative ordering, not an exact year for A, is justified. Cross-domain check: compare 11 with 7; $11 \geq 7$ is true.

Answer. Layer C is older than Layer B. The exact deposition year of Layer A cannot be deduced. Cross-domain answer: quorum is met.

Knowledge takeaway. Relative dating and absolute dating answer different questions and should not be confused.

Formal model. Partial order from vertical position plus numeric constraints from dated objects.

Problem 339. Reading an archaeological sequence: case 4

Knowledge context. Archaeological layers provide relative chronological information. Dates from objects add constraints but do not automatically make every date exact.

Given facts. An undisturbed archaeological site has four layers from top to bottom: A, B, C, D. In an undisturbed sequence, lower layers were deposited earlier. D contains an object dated 1315, C contains one dated 1495, and B contains one dated 1660. Cross-domain check: researchers collected 15 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. In an undisturbed sequence, a lower layer is older than a layer above it. A securely dated object provides a constraint on the layer that contains it, but one object does not date every other object exactly. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. Which is older, Layer C or Layer B? Can the exact year of deposition of Layer A be deduced? Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. C lies below B, so C is older than B by the stratigraphic rule. 2. No exact dated object or deposition year is given for A. 3. Therefore only relative ordering, not an exact year for A, is justified. Cross-domain check: 15−4=11 independent reports remain.

Answer. Layer C is older than Layer B. The exact deposition year of Layer A cannot be deduced. Cross-domain answer: 11 independent reports.

Knowledge takeaway. Relative dating and absolute dating answer different questions and should not be confused.

Formal model. Partial order from vertical position plus numeric constraints from dated objects.

Problem 340. Reading an archaeological sequence: case 5

Knowledge context. Archaeological layers provide relative chronological information. Dates from objects add constraints but do not automatically make every date exact.

Given facts. An undisturbed archaeological site has four layers from top to bottom: A, B, C, D. In an undisturbed sequence, lower layers were deposited earlier. D contains an object dated 1320, C contains one dated 1510, and B contains one dated 1680. Cross-domain check: on a field sketch, 1 cm represents 2 km and two checkpoints are 6 cm apart.

Rules. In an undisturbed sequence, a lower layer is older than a layer above it. A securely dated object provides a constraint on the layer that contains it, but one object does not date every other object exactly. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Which is older, Layer C or Layer B? Can the exact year of deposition of Layer A be deduced? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. C lies below B, so C is older than B by the stratigraphic rule. 2. No exact dated object or deposition year is given for A. 3. Therefore only relative ordering, not an exact year for A, is justified. Cross-domain check: 6×2=12 km.

Answer. Layer C is older than Layer B. The exact deposition year of Layer A cannot be deduced. Cross-domain answer: 12 km.

Knowledge takeaway. Relative dating and absolute dating answer different questions and should not be confused.

Formal model. Partial order from vertical position plus numeric constraints from dated objects.

C1. Rules and exceptions

Problem 341. Applying a rule with an exception: Ivy

Knowledge context. Rules often have conditions and exceptions. Logical reading prevents us from applying a general rule when an overriding exception is active.

Given facts. Ivy is 11 years old. A library rule says: a child may enter the archive room if age ≥ 12 AND an adult permit is present. Exception: during an evacuation drill, nobody may enter. Permit present: yes. Evacuation drill: no.

Rules. Check exceptions first if they override the general rule. Otherwise every condition joined by AND must be true.

Task. May Ivy enter the archive room under these facts? Show which conditions succeed or fail.

Step-by-step solution. 1. Age test: 11 ≥ 12 is False. 2. Permit test is True. 3. Because the general rule uses AND, both tests must pass. 4. Therefore entry is not allowed.

Answer. No, Ivy may not enter.

Knowledge takeaway. The order of reasoning matters when exceptions override ordinary rules.

Formal model. Boolean rule with priority: exception predicate first, then conjunction of eligibility conditions.

Problem 342. Applying a rule with an exception: Leo

Knowledge context. Rules often have conditions and exceptions. Logical reading prevents us from applying a general rule when an overriding exception is active.

Given facts. Leo is 12 years old. A library rule says: a child may enter the archive room if age ≥ 12 AND an adult permit is present. Exception: during an evacuation drill, nobody may enter. Permit present: no. Evacuation drill: no.

Rules. Check exceptions first if they override the general rule. Otherwise every condition joined by AND must be true.

Task. May Leo enter the archive room under these facts? Show which conditions succeed or fail.

Step-by-step solution. 1. Age test: 12 ≥ 12 is True. 2. Permit test is False. 3. Because the general rule uses AND, both tests must pass. 4. Therefore entry is not allowed.

Answer. No, Leo may not enter.

Knowledge takeaway. The order of reasoning matters when exceptions override ordinary rules.

Formal model. Boolean rule with priority: exception predicate first, then conjunction of eligibility conditions.

Problem 343. Applying a rule with an exception: Nora

Knowledge context. Rules often have conditions and exceptions. Logical reading prevents us from applying a general rule when an overriding exception is active.

Given facts. Nora is 13 years old. A library rule says: a child may enter the archive room if age ≥ 12 AND an adult permit is present. Exception: during an evacuation drill, nobody may enter. Permit present: yes. Evacuation drill: no.

Rules. Check exceptions first if they override the general rule. Otherwise every condition joined by AND must be true.

Task. May Nora enter the archive room under these facts? Show which conditions succeed or fail.

Step-by-step solution. 1. Age test: 13 ≥ 12 is True. 2. Permit test is True. 3. Because the general rule uses AND, both tests must pass. 4. Therefore entry is allowed.

Answer. Yes, Nora may enter.

Knowledge takeaway. The order of reasoning matters when exceptions override ordinary rules.

Formal model. Boolean rule with priority: exception predicate first, then conjunction of eligibility conditions.

Problem 344. Applying a rule with an exception: Eli

Knowledge context. Rules often have conditions and exceptions. Logical reading prevents us from applying a general rule when an overriding exception is active.

Given facts. Eli is 14 years old. A library rule says: a child may enter the archive room if age ≥ 12 AND an adult permit is present. Exception: during an evacuation drill, nobody may enter. Permit present: no. Evacuation drill: no.

Rules. Check exceptions first if they override the general rule. Otherwise every condition joined by AND must be true.

Task. May Eli enter the archive room under these facts? Show which conditions succeed or fail.

Step-by-step solution. 1. Age test: 14 ≥ 12 is True. 2. Permit test is False. 3. Because the general rule uses AND, both tests must pass. 4. Therefore entry is not allowed.

Answer. No, Eli may not enter.

Knowledge takeaway. The order of reasoning matters when exceptions override ordinary rules.

Formal model. Boolean rule with priority: exception predicate first, then conjunction of eligibility conditions.

Problem 345. Applying a rule with an exception: Mina

Knowledge context. Rules often have conditions and exceptions. Logical reading prevents us from applying a general rule when an overriding exception is active.

Given facts. Mina is 15 years old. A library rule says: a child may enter the archive room if age ≥ 12 AND an adult permit is present. Exception: during an evacuation drill, nobody may enter. Permit present: yes. Evacuation drill: yes.

Rules. Check exceptions first if they override the general rule. Otherwise every condition joined by AND must be true.

Task. May Mina enter the archive room under these facts? Show which conditions succeed or fail.

Step-by-step solution. 1. The evacuation exception applies and overrides ordinary eligibility. 2. Therefore entry is not allowed.

Answer. No, Mina may not enter.

Knowledge takeaway. The order of reasoning matters when exceptions override ordinary rules.

Formal model. Boolean rule with priority: exception predicate first, then conjunction of eligibility conditions.

C2. Rights as limits on decisions

Problem 346. Majority rule and a protected right: case 1

Knowledge context. Democratic procedures use voting, but constitutional systems can place some rights beyond ordinary majority power.

Given facts. A council of 19 members votes 14 to 5 for a proposal that would forbid one named group from the right to speak at a peaceful meeting. The classroom constitution states: (1) ordinary proposals pass by simple majority; (2) an ordinary majority may not remove the listed equal rights from a group.

Rules. A decision must satisfy both the voting procedure and the higher rights constraint. Passing a vote is not sufficient if the content violates a protected right.

Task. Did the proposal receive a majority? May it take effect under the stated constitution?

Step-by-step solution. 1. Vote check: 14>5, so the proposal has a simple majority. 2. Content check: the proposal removes a protected equal right (speak at a peaceful meeting) from a group. 3. The higher rights rule therefore blocks the proposal even though the vote passed.

Answer. It received a majority, but it may not take effect under the stated rights rule.

Knowledge takeaway. A majority answers who won a vote; it does not by itself answer whether every possible decision is lawful or fair under higher rules.

Formal model. Two-stage validity test: procedural predicate AND rights-compliance predicate.

Problem 347. Majority rule and a protected right: case 2

Knowledge context. Democratic procedures use voting, but constitutional systems can place some rights beyond ordinary majority power.

Given facts. A council of 21 members votes 15 to 6 for a proposal that would forbid one named group from the right to use the public library equally. The classroom constitution states: (1) ordinary proposals pass by simple majority; (2) an ordinary majority may not remove the listed equal rights from a group.

Rules. A decision must satisfy both the voting procedure and the higher rights constraint. Passing a vote is not sufficient if the content violates a protected right.

Task. Did the proposal receive a majority? May it take effect under the stated constitution?

Step-by-step solution. 1. Vote check: 15>6, so the proposal has a simple majority. 2. Content check: the proposal removes a protected equal right (use the public library equally) from a group. 3. The higher rights rule therefore blocks the proposal even though the vote passed.

Answer. It received a majority, but it may not take effect under the stated rights rule.

Knowledge takeaway. A majority answers who won a vote; it does not by itself answer whether every possible decision is lawful or fair under higher rules.

Formal model. Two-stage validity test: procedural predicate AND rights-compliance predicate.

Problem 348. Majority rule and a protected right: case 3

Knowledge context. Democratic procedures use voting, but constitutional systems can place some rights beyond ordinary majority power.

Given facts. A council of 23 members votes 16 to 7 for a proposal that would forbid one named group from the right to receive a fair hearing. The classroom constitution states: (1) ordinary proposals pass by simple majority; (2) an ordinary majority may not remove the listed equal rights from a group.

Rules. A decision must satisfy both the voting procedure and the higher rights constraint. Passing a vote is not sufficient if the content violates a protected right.

Task. Did the proposal receive a majority? May it take effect under the stated constitution?

Step-by-step solution. 1. Vote check: 16>7, so the proposal has a simple majority. 2. Content check: the proposal removes a protected equal right (receive a fair hearing) from a group. 3. The higher rights rule therefore blocks the proposal even though the vote passed.

Answer. It received a majority, but it may not take effect under the stated rights rule.

Knowledge takeaway. A majority answers who won a vote; it does not by itself answer whether every possible decision is lawful or fair under higher rules.

Formal model. Two-stage validity test: procedural predicate AND rights-compliance predicate.

Problem 349. Majority rule and a protected right: case 4

Knowledge context. Democratic procedures use voting, but constitutional systems can place some rights beyond ordinary majority power.

Given facts. A council of 25 members votes 17 to 8 for a proposal that would forbid one named group from the right to practice a lawful belief. The classroom constitution states: (1) ordinary proposals pass by simple majority; (2) an ordinary majority may not remove the listed equal rights from a group.

Rules. A decision must satisfy both the voting procedure and the higher rights constraint. Passing a vote is not sufficient if the content violates a protected right.

Task. Did the proposal receive a majority? May it take effect under the stated constitution?

Step-by-step solution. 1. Vote check: 17>8, so the proposal has a simple majority. 2. Content check: the proposal removes a protected equal right (practice a lawful belief) from a group. 3. The higher rights rule therefore blocks the proposal even though the vote passed.

Answer. It received a majority, but it may not take effect under the stated rights rule.

Knowledge takeaway. A majority answers who won a vote; it does not by itself answer whether every possible decision is lawful or fair under higher rules.

Formal model. Two-stage validity test: procedural predicate AND rights-compliance predicate.

Problem 350. Majority rule and a protected right: case 5

Knowledge context. Democratic procedures use voting, but constitutional systems can place some rights beyond ordinary majority power.

Given facts. A council of 27 members votes 18 to 9 for a proposal that would forbid one named group from the right to be protected from arbitrary punishment. The classroom constitution states: (1) ordinary proposals pass by simple majority; (2) an ordinary majority may not remove the listed equal rights from a group.

Rules. A decision must satisfy both the voting procedure and the higher rights constraint. Passing a vote is not sufficient if the content violates a protected right.

Task. Did the proposal receive a majority? May it take effect under the stated constitution?

Step-by-step solution. 1. Vote check: 18>9, so the proposal has a simple majority. 2. Content check: the proposal removes a protected equal right (be protected from arbitrary punishment) from a group. 3. The higher rights rule therefore blocks the proposal even though the vote passed.

Answer. It received a majority, but it may not take effect under the stated rights rule.

Knowledge takeaway. A majority answers who won a vote; it does not by itself answer whether every possible decision is lawful or fair under higher rules.

Formal model. Two-stage validity test: procedural predicate AND rights-compliance predicate.

C3. Voting, majority, and quorum

Problem 351. Quorum and majority in a 19-member committee

Knowledge context. Voting rules often contain more than one threshold. Quorum concerns participation; majority concerns the votes cast according to the stated rule.

Given facts. A committee has 19 members. 19 are present. Votes: yes=9, no=9, abstain=1. Rule: quorum is at least 60% of all members; if quorum exists, a motion passes when yes votes are strictly more than no votes.

Rules. First test quorum using all members as the denominator. Only then compare yes and no; abstentions count for presence but not as no votes under this rule.

Task. Is quorum met, and does the motion pass?

Step-by-step solution. 1. Quorum threshold=ceil(0.60×19)=12. 2. Present=19, so quorum is met. 3. Yes=9, no=9; yes>no is False. 4. Combine the two tests.

Answer. Quorum is met; the motion does not pass.

Knowledge takeaway. Procedural reasoning requires applying tests in the right order and using the correct denominator.

Formal model. Threshold predicates: quorum(present,total) and majority(yes,no); pass = quorum ∧ majority.

Problem 352. Quorum and majority in a 20-member committee

Knowledge context. Voting rules often contain more than one threshold. Quorum concerns participation; majority concerns the votes cast according to the stated rule.

Given facts. A committee has 20 members. 19 are present. Votes: yes=10, no=8, abstain=1. Rule: quorum is at least 60% of all members; if quorum exists, a motion passes when yes votes are strictly more than no votes.

Rules. First test quorum using all members as the denominator. Only then compare yes and no; abstentions count for presence but not as no votes under this rule.

Task. Is quorum met, and does the motion pass?

Step-by-step solution. 1. Quorum threshold=ceil(0.60×20)=12. 2. Present=19, so quorum is met. 3. Yes=10, no=8; yes>no is True. 4. Combine the two tests.

Answer. Quorum is met; the motion passes.

Knowledge takeaway. Procedural reasoning requires applying tests in the right order and using the correct denominator.

Formal model. Threshold predicates: quorum(present,total) and majority(yes,no); pass = quorum ∧ majority.

Problem 353. Quorum and majority in a 21-member committee

Knowledge context. Voting rules often contain more than one threshold. Quorum concerns participation; majority concerns the votes cast according to the stated rule.

Given facts. A committee has 21 members. 19 are present. Votes: yes=11, no=7, abstain=1. Rule: quorum is at least 60% of all members; if quorum exists, a motion passes when yes votes are strictly more than no votes.

Rules. First test quorum using all members as the denominator. Only then compare yes and no; abstentions count for presence but not as no votes under this rule.

Task. Is quorum met, and does the motion pass?

Step-by-step solution. 1. Quorum threshold=ceil(0.60×21)=13. 2. Present=19, so quorum is met. 3. Yes=11, no=7; yes>no is True. 4. Combine the two tests.

Answer. Quorum is met; the motion passes.

Knowledge takeaway. Procedural reasoning requires applying tests in the right order and using the correct denominator.

Formal model. Threshold predicates: quorum(present,total) and majority(yes,no); pass = quorum ∧ majority.

Problem 354. Quorum and majority in a 22-member committee

Knowledge context. Voting rules often contain more than one threshold. Quorum concerns participation; majority concerns the votes cast according to the stated rule.

Given facts. A committee has 22 members. 22 are present. Votes: yes=12, no=9, abstain=1. Rule: quorum is at least 60% of all members; if quorum exists, a motion passes when yes votes are strictly more than no votes.

Rules. First test quorum using all members as the denominator. Only then compare yes and no; abstentions count for presence but not as no votes under this rule.

Task. Is quorum met, and does the motion pass?

Step-by-step solution. 1. Quorum threshold=ceil(0.60×22)=14. 2. Present=22, so quorum is met. 3. Yes=12, no=9; yes>no is True. 4. Combine the two tests.

Answer. Quorum is met; the motion passes.

Knowledge takeaway. Procedural reasoning requires applying tests in the right order and using the correct denominator.

Formal model. Threshold predicates: quorum(present,total) and majority(yes,no); pass = quorum ∧ majority.

Problem 355. Quorum and majority in a 23-member committee

Knowledge context. Voting rules often contain more than one threshold. Quorum concerns participation; majority concerns the votes cast according to the stated rule.

Given facts. A committee has 23 members. 22 are present. Votes: yes=13, no=8, abstain=1. Rule: quorum is at least 60% of all members; if quorum exists, a motion passes when yes votes are strictly more than no votes.

Rules. First test quorum using all members as the denominator. Only then compare yes and no; abstentions count for presence but not as no votes under this rule.

Task. Is quorum met, and does the motion pass?

Step-by-step solution. 1. Quorum threshold=ceil(0.60×23)=14. 2. Present=22, so quorum is met. 3. Yes=13, no=8; yes>no is True. 4. Combine the two tests.

Answer. Quorum is met; the motion passes.

Knowledge takeaway. Procedural reasoning requires applying tests in the right order and using the correct denominator.

Formal model. Threshold predicates: quorum(present,total) and majority(yes,no); pass = quorum ∧ majority.

C4. Representation and seats

Problem 356. Representing three districts: case 1

Knowledge context. Representation can be modeled by rules that convert population into seats. Different allocation rules can produce different results, so the rule must be stated.

Given facts. Three districts have populations A=120, B=100, C=60. They must share 12 seats. Use this stated method: compute each exact quota population/total×seats; give each district the whole-number part; assign remaining seats to the largest fractional remainders.

Rules. Use only the stated largest-remainder method. Seat totals must add exactly to the available number.

Task. How many seats does each district receive?

Step-by-step solution. 1. Total population=280. 2. Exact quotas: A=5.14, B=4.29, C=2.57. 3. Initial whole seats: [5, 4, 2], leaving 1. 4. Give the remaining seat(s) to the largest remainder(s). 5. Final allocation=[5, 4, 3], which sums to 12.

Answer. A=5, B=4, C=3 seats.

Knowledge takeaway. Fair representation is not just arithmetic; it depends on an explicit apportionment procedure.

Formal model. Largest-remainder apportionment: quotas, floors, then rank fractional remainders.

Problem 357. Representing three districts: case 2

Knowledge context. Representation can be modeled by rules that convert population into seats. Different allocation rules can produce different results, so the rule must be stated.

Given facts. Three districts have populations A=140, B=100, C=75. They must share 12 seats. Use this stated method: compute each exact quota population/total×seats; give each district the whole-number part; assign remaining seats to the largest fractional remainders.

Rules. Use only the stated largest-remainder method. Seat totals must add exactly to the available number.

Task. How many seats does each district receive?

Step-by-step solution. 1. Total population=315. 2. Exact quotas: A=5.33, B=3.81, C=2.86. 3. Initial whole seats: [5, 3, 2], leaving 2. 4. Give the remaining seat(s) to the largest remainder(s). 5. Final allocation=[5, 4, 3], which sums to 12.

Answer. A=5, B=4, C=3 seats.

Knowledge takeaway. Fair representation is not just arithmetic; it depends on an explicit apportionment procedure.

Formal model. Largest-remainder apportionment: quotas, floors, then rank fractional remainders.

Problem 358. Representing three districts: case 3

Knowledge context. Representation can be modeled by rules that convert population into seats. Different allocation rules can produce different results, so the rule must be stated.

Given facts. Three districts have populations A=160, B=100, C=90. They must share 12 seats. Use this stated method: compute each exact quota population/total×seats; give each district the whole-number part; assign remaining seats to the largest fractional remainders.

Rules. Use only the stated largest-remainder method. Seat totals must add exactly to the available number.

Task. How many seats does each district receive?

Step-by-step solution. 1. Total population=350. 2. Exact quotas: A=5.49, B=3.43, C=3.09. 3. Initial whole seats: [5, 3, 3], leaving 1. 4. Give the remaining seat(s) to the largest remainder(s). 5. Final allocation=[6, 3, 3], which sums to 12.

Answer. A=6, B=3, C=3 seats.

Knowledge takeaway. Fair representation is not just arithmetic; it depends on an explicit apportionment procedure.

Formal model. Largest-remainder apportionment: quotas, floors, then rank fractional remainders.

Problem 359. Representing three districts: case 4

Knowledge context. Representation can be modeled by rules that convert population into seats. Different allocation rules can produce different results, so the rule must be stated.

Given facts. Three districts have populations A=180, B=100, C=105. They must share 12 seats. Use this stated method: compute each exact quota population/total×seats; give each district the whole-number part; assign remaining seats to the largest fractional remainders.

Rules. Use only the stated largest-remainder method. Seat totals must add exactly to the available number.

Task. How many seats does each district receive?

Step-by-step solution. 1. Total population=385. 2. Exact quotas: A=5.61, B=3.12, C=3.27. 3. Initial whole seats: [5, 3, 3], leaving 1. 4. Give the remaining seat(s) to the largest remainder(s). 5. Final allocation=[6, 3, 3], which sums to 12.

Answer. A=6, B=3, C=3 seats.

Knowledge takeaway. Fair representation is not just arithmetic; it depends on an explicit apportionment procedure.

Formal model. Largest-remainder apportionment: quotas, floors, then rank fractional remainders.

Problem 360. Representing three districts: case 5

Knowledge context. Representation can be modeled by rules that convert population into seats. Different allocation rules can produce different results, so the rule must be stated.

Given facts. Three districts have populations A=200, B=100, C=120. They must share 12 seats. Use this stated method: compute each exact quota population/total×seats; give each district the whole-number part; assign remaining seats to the largest fractional remainders.

Rules. Use only the stated largest-remainder method. Seat totals must add exactly to the available number.

Task. How many seats does each district receive?

Step-by-step solution. 1. Total population=420. 2. Exact quotas: A=5.71, B=2.86, C=3.43. 3. Initial whole seats: [5, 2, 3], leaving 2. 4. Give the remaining seat(s) to the largest remainder(s). 5. Final allocation=[6, 3, 3], which sums to 12.

Answer. A=6, B=3, C=3 seats.

Knowledge takeaway. Fair representation is not just arithmetic; it depends on an explicit apportionment procedure.

Formal model. Largest-remainder apportionment: quotas, floors, then rank fractional remainders.

C5. Institutions and roles

Problem 361. Which institution has the role? Repair a broken public road

Knowledge context. Civic institutions divide responsibilities. A problem can be solved by matching an action to a clearly defined function.

Given facts. Roles are defined as follows: Town Council makes local policy and approves the budget; Public Works Office maintains roads; Local Court decides legal disputes; Election Committee administers elections; School Board sets school policies. A resident asks which institution should repair a broken public road.

Rules. Match the requested action to the stated role. Do not choose an institution merely because it is powerful or familiar.

Task. Should the request go first to the Public Works Office or the Health Office? Why?

Step-by-step solution. 1. Identify the action: repair a broken public road. 2. Find the role whose definition contains that function: Public Works Office. 3. Health Office has a different stated role, so it is not the first match.

Answer. The request should go first to the Public Works Office.

Knowledge takeaway. Institutional reasoning is role matching, not guessing which body is “most important.”

Formal model. Bipartite relation between actions and authorized institutional roles.

Problem 362. Which institution has the role? Decide a dispute under law

Knowledge context. Civic institutions divide responsibilities. A problem can be solved by matching an action to a clearly defined function.

Given facts. Roles are defined as follows: Town Council makes local policy and approves the budget; Public Works Office maintains roads; Local Court decides legal disputes; Election Committee administers elections; School Board sets school policies. A resident asks which institution should decide a dispute under law.

Rules. Match the requested action to the stated role. Do not choose an institution merely because it is powerful or familiar.

Task. Should the request go first to the Local Court or the Parks Department? Why?

Step-by-step solution. 1. Identify the action: decide a dispute under law. 2. Find the role whose definition contains that function: Local Court. 3. Parks Department has a different stated role, so it is not the first match.

Answer. The request should go first to the Local Court.

Knowledge takeaway. Institutional reasoning is role matching, not guessing which body is “most important.”

Formal model. Bipartite relation between actions and authorized institutional roles.

Problem 363. Which institution has the role? Approve the town budget

Knowledge context. Civic institutions divide responsibilities. A problem can be solved by matching an action to a clearly defined function.

Given facts. Roles are defined as follows: Town Council makes local policy and approves the budget; Public Works Office maintains roads; Local Court decides legal disputes; Election Committee administers elections; School Board sets school policies. A resident asks which institution should approve the town budget.

Rules. Match the requested action to the stated role. Do not choose an institution merely because it is powerful or familiar.

Task. Should the request go first to the Town Council or the Health Office? Why?

Step-by-step solution. 1. Identify the action: approve the town budget. 2. Find the role whose definition contains that function: Town Council. 3. Health Office has a different stated role, so it is not the first match.

Answer. The request should go first to the Town Council.

Knowledge takeaway. Institutional reasoning is role matching, not guessing which body is “most important.”

Formal model. Bipartite relation between actions and authorized institutional roles.

Problem 364. Which institution has the role? Organize the ballot count

Knowledge context. Civic institutions divide responsibilities. A problem can be solved by matching an action to a clearly defined function.

Given facts. Roles are defined as follows: Town Council makes local policy and approves the budget; Public Works Office maintains roads; Local Court decides legal disputes; Election Committee administers elections; School Board sets school policies. A resident asks which institution should organize the ballot count.

Rules. Match the requested action to the stated role. Do not choose an institution merely because it is powerful or familiar.

Task. Should the request go first to the Election Committee or the Public Works Office? Why?

Step-by-step solution. 1. Identify the action: organize the ballot count. 2. Find the role whose definition contains that function: Election Committee. 3. Public Works Office has a different stated role, so it is not the first match.

Answer. The request should go first to the Election Committee.

Knowledge takeaway. Institutional reasoning is role matching, not guessing which body is “most important.”

Formal model. Bipartite relation between actions and authorized institutional roles.

Problem 365. Which institution has the role? Set school calendar rules

Knowledge context. Civic institutions divide responsibilities. A problem can be solved by matching an action to a clearly defined function.

Given facts. Roles are defined as follows: Town Council makes local policy and approves the budget; Public Works Office maintains roads; Local Court decides legal disputes; Election Committee administers elections; School Board sets school policies. A resident asks which institution should set school calendar rules.

Rules. Match the requested action to the stated role. Do not choose an institution merely because it is powerful or familiar.

Task. Should the request go first to the School Board or the Public Works Office? Why?

Step-by-step solution. 1. Identify the action: set school calendar rules. 2. Find the role whose definition contains that function: School Board. 3. Public Works Office has a different stated role, so it is not the first match.

Answer. The request should go first to the School Board.

Knowledge takeaway. Institutional reasoning is role matching, not guessing which body is “most important.”

Formal model. Bipartite relation between actions and authorized institutional roles.

C6. Public budgets and priorities

Problem 366. Choosing a public budget package: case 1

Knowledge context. Public budgets require choices under limited resources. A transparent model separates costs, benefits, and mandatory obligations.

Given facts. A town has a budget of 14 units. Projects (cost, benefit): park=(4,5); bridge repair=(6,9); library books=(3,4); drainage=(5,8). Cross-domain check: a local committee has 14 members, requires at least 8 present for quorum, and 14 are present.

Rules. Choose a subset whose total cost does not exceed the budget. Maximize total benefit. If a mandatory project is stated, every feasible plan must include it. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. Which set of projects gives the greatest total benefit while satisfying every constraint? Cross-domain check: is quorum met?

Step-by-step solution. 1. Budget limit is 14. 2. Evaluate feasible combinations (and keep the mandatory project if any). 3. The best feasible combination has cost 14 and benefit 21: ['bridge repair', 'library books', 'drainage']. 4. No higher-benefit feasible combination exists under the stated costs. Cross-domain check: compare 14 with 8; $14 \geq 8$ is true.

Answer. Choose bridge repair, library books, drainage; cost 14, benefit 21. Cross-domain answer: quorum is met.

Knowledge takeaway. Optimization means comparing feasible alternatives under an explicit objective, not simply choosing the cheapest item.

Formal model. 0/1 knapsack with optional mandatory-item constraint; maximize benefit subject to total cost $\leq$ budget.

Problem 367. Choosing a public budget package: case 2

Knowledge context. Public budgets require choices under limited resources. A transparent model separates costs, benefits, and mandatory obligations.

Given facts. A town has a budget of 15 units. Projects (cost, benefit): tree planting=(5,7); bridge repair=(6,9); library books=(3,4); drainage=(5,8). Cross-domain check: researchers collected 13 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. Choose a subset whose total cost does not exceed the budget. Maximize total benefit. If a mandatory project is stated, every feasible plan must include it. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. Which set of projects gives the greatest total benefit while satisfying every constraint? Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. Budget limit is 15. 2. Evaluate feasible combinations (and keep the mandatory project if any). 3. The best feasible combination has cost 14 and benefit 21: ['bridge repair', 'library books', 'drainage']. 4. No higher-benefit feasible combination exists under the stated costs. Cross-domain check: 13−4=9 independent reports remain.

Answer. Choose bridge repair, library books, drainage; cost 14, benefit 21. Cross-domain answer: 9 independent reports.

Knowledge takeaway. Optimization means comparing feasible alternatives under an explicit objective, not simply choosing the cheapest item.

Formal model. 0/1 knapsack with optional mandatory-item constraint; maximize benefit subject to total cost $\leq$ budget.

Problem 368. Choosing a public budget package: case 3

Knowledge context. Public budgets require choices under limited resources. A transparent model separates costs, benefits, and mandatory obligations.

Given facts. A town has a budget of 16 units. Projects (cost, benefit): park=(4,5); bridge repair=(6,9); library books=(3,4); drainage=(5,8). Cross-domain check: on a field sketch, 1 cm represents 5 km and two checkpoints are 6 cm apart.

Rules. Choose a subset whose total cost does not exceed the budget. Maximize total benefit. If a mandatory project is stated, every feasible plan must include it. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Which set of projects gives the greatest total benefit while satisfying every constraint? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. Budget limit is 16. 2. Evaluate feasible combinations (and keep the mandatory project if any). 3. The best feasible combination has cost 15 and benefit 22: ['park', 'bridge repair', 'drainage']. 4. No higher-benefit feasible combination exists under the stated costs. Cross-domain check: 6×5=30 km.

Answer. Choose park, bridge repair, drainage; cost 15, benefit 22. Cross-domain answer: 30 km.

Knowledge takeaway. Optimization means comparing feasible alternatives under an explicit objective, not simply choosing the cheapest item.

Formal model. 0/1 knapsack with optional mandatory-item constraint; maximize benefit subject to total cost ≤ budget.

Problem 369. Choosing a public budget package: case 4

Knowledge context. Public budgets require choices under limited resources. A transparent model separates costs, benefits, and mandatory obligations.

Given facts. A town has a budget of 17 units. Projects (cost, benefit): tree planting=(5,7); bridge repair=(6,9); library books=(3,4); drainage=(5,8).

Rules. Choose a subset whose total cost does not exceed the budget. Maximize total benefit. If a mandatory project is stated, every feasible plan must include it.

Task. Which set of projects gives the greatest total benefit while satisfying every constraint?

Step-by-step solution. 1. Budget limit is 17. 2. Evaluate feasible combinations (and keep the mandatory project if any). 3. The best feasible combination has cost 16 and benefit 24: ['tree planting', 'bridge repair', 'drainage']. 4. No higher-benefit feasible combination exists under the stated costs.

Answer. Choose tree planting, bridge repair, drainage; cost 16, benefit 24.

Knowledge takeaway. Optimization means comparing feasible alternatives under an explicit objective, not simply choosing the cheapest item.

Formal model. 0/1 knapsack with optional mandatory-item constraint; maximize benefit subject to total cost ≤ budget.

Problem 370. Choosing a public budget package: case 5

Knowledge context. Public budgets require choices under limited resources. A transparent model separates costs, benefits, and mandatory obligations.

Given facts. A town has a budget of 18 units. Projects (cost, benefit): park=(4,5); bridge repair=(6,9); library books=(3,4); drainage=(5,8).

Rules. Choose a subset whose total cost does not exceed the budget. Maximize total benefit. If a mandatory project is stated, every feasible plan must include it.

Task. Which set of projects gives the greatest total benefit while satisfying every constraint?

Step-by-step solution. 1. Budget limit is 18. 2. Evaluate feasible combinations (and keep the mandatory project if any). 3. The best feasible combination has cost 18 and benefit 26: ['park', 'bridge repair', 'library books', 'drainage']. 4. No higher-benefit feasible combination exists under the stated costs.

Answer. Choose park, bridge repair, library books, drainage; cost 18, benefit 26.

Knowledge takeaway. Optimization means comparing feasible alternatives under an explicit objective, not simply choosing the cheapest item.

Formal model. 0/1 knapsack with optional mandatory-item constraint; maximize benefit subject to total cost ≤ budget.

C7. Collective choice and preferences

Problem 371. Pairwise collective choice: Park versus Library

Knowledge context. Group preferences can be aggregated in different ways. Pairwise majority asks a specific question about two options at a time.

Given facts. Three public projects are considered: Park, Library, Sports Hall. Voter groups and rankings: 4 voters rank Park>Library>Sports Hall; 5 voters rank Library>Sports Hall>Park; 2 voters rank Sports Hall>Park>Library. Use pairwise majority: compare two options at a time; each voter supports the one ranked higher.

Rules. For a pairwise contest, count how many voters prefer each of the two options. Ignore the third option for that comparison.

Task. In a pairwise vote between Park and Library, which option wins?

Step-by-step solution. 1. For each group, check whether Park or Library is ranked higher. 2. Totals: Park=6, Library=5. 3. Therefore the result is Park.

Answer. Park wins 6 to 5.

Knowledge takeaway. A collective decision rule must be explicit; different rules can summarize the same rankings differently.

Formal model. Preference rankings; for pair (a,b), sum weights of voters ranking a>b versus b>a.

Problem 372. Pairwise collective choice: Library versus Sports Hall

Knowledge context. Group preferences can be aggregated in different ways. Pairwise majority asks a specific question about two options at a time.

Given facts. Three public projects are considered: Park, Library, Sports Hall. Voter groups and rankings: 5 voters rank Park>Library>Sports Hall; 5 voters rank Library>Sports Hall>Park; 3 voters rank Sports Hall>Park>Library. Use pairwise majority: compare two options at a time; each voter supports the one ranked higher.

Rules. For a pairwise contest, count how many voters prefer each of the two options. Ignore the third option for that comparison.

Task. In a pairwise vote between Library and Sports Hall, which option wins?

Step-by-step solution. 1. For each group, check whether Library or Sports Hall is ranked higher. 2. Totals: Library=10, Sports Hall=3. 3. Therefore the result is Library.

Answer. Library wins 10 to 3.

Knowledge takeaway. A collective decision rule must be explicit; different rules can summarize the same rankings differently.

Formal model. Preference rankings; for pair (a,b), sum weights of voters ranking a>b versus b>a.

Problem 373. Pairwise collective choice: Park versus Sports Hall

Knowledge context. Group preferences can be aggregated in different ways. Pairwise majority asks a specific question about two options at a time.

Given facts. Three public projects are considered: Park, Library, Sports Hall. Voter groups and rankings: 6 voters rank Park>Library>Sports Hall; 5 voters rank Library>Sports Hall>Park; 4 voters rank Sports Hall>Park>Library. Use pairwise majority: compare two options at a time; each voter supports the one ranked higher.

Rules. For a pairwise contest, count how many voters prefer each of the two options. Ignore the third option for that comparison.

Task. In a pairwise vote between Park and Sports Hall, which option wins?

Step-by-step solution. 1. For each group, check whether Park or Sports Hall is ranked higher. 2. Totals: Park=6, Sports Hall=9. 3. Therefore the result is Sports Hall.

Answer. Sports Hall wins 9 to 6.

Knowledge takeaway. A collective decision rule must be explicit; different rules can summarize the same rankings differently.

Formal model. Preference rankings; for pair (a,b), sum weights of voters ranking a>b versus b>a.

Problem 374. Pairwise collective choice: Library versus Park

Knowledge context. Group preferences can be aggregated in different ways. Pairwise majority asks a specific question about two options at a time.

Given facts. Three public projects are considered: Park, Library, Sports Hall. Voter groups and rankings: 7 voters rank Park>Library>Sports Hall; 5 voters rank Library>Sports Hall>Park; 5 voters rank Sports Hall>Park>Library. Use pairwise majority: compare two options at a time; each voter supports the one ranked higher.

Rules. For a pairwise contest, count how many voters prefer each of the two options. Ignore the third option for that comparison.

Task. In a pairwise vote between Library and Park, which option wins?

Step-by-step solution. 1. For each group, check whether Library or Park is ranked higher. 2. Totals: Library=5, Park=12. 3. Therefore the result is Park.

Answer. Park wins 12 to 5.

Knowledge takeaway. A collective decision rule must be explicit; different rules can summarize the same rankings differently.

Formal model. Preference rankings; for pair (a,b), sum weights of voters ranking a>b versus b>a.

Problem 375. Pairwise collective choice: Sports Hall versus Park

Knowledge context. Group preferences can be aggregated in different ways. Pairwise majority asks a specific question about two options at a time.

Given facts. Three public projects are considered: Park, Library, Sports Hall. Voter groups and rankings: 8 voters rank Park>Library>Sports Hall; 5 voters rank Library>Sports Hall>Park; 6 voters rank Sports Hall>Park>Library. Use pairwise majority: compare two options at a time; each voter supports the one ranked higher.

Rules. For a pairwise contest, count how many voters prefer each of the two options. Ignore the third option for that comparison.

Task. In a pairwise vote between Sports Hall and Park, which option wins?

Step-by-step solution. 1. For each group, check whether Sports Hall or Park is ranked higher. 2. Totals: Sports Hall=11, Park=8. 3. Therefore the result is Sports Hall.

Answer. Sports Hall wins 11 to 8.

Knowledge takeaway. A collective decision rule must be explicit; different rules can summarize the same rankings differently.

Formal model. Preference rankings; for pair (a,b), sum weights of voters ranking a>b versus b>a.

Part B — New and mixed reasoning domains

N1. Scale, proportion, and distance

Problem 376. Map scale 1:10000

Knowledge context. Map scale links a model to the real world through proportional reasoning.

Given facts. A map has scale 1:10000. The measured map distance between two points is 4 cm. Rule: 1 cm on the map represents 10000 cm in reality; 100,000 cm = 1 km.

Rules. Multiply map length by the scale denominator, then convert units. A scale is a ratio, not an added distance.

Task. What is the real straight-line distance in kilometres?

Step-by-step solution. 1. Real centimetres=4×10000=40000. 2. Convert to kilometres: 40000÷100,000=0.4.

Answer. 0.4 km.

Knowledge takeaway. Scale problems are ratio problems: keep units explicit and conversion becomes mechanical.

Formal model. Proportion d_real=d_map×denominator, followed by unit conversion.

Problem 377. Map scale 1:25000

Knowledge context. Map scale links a model to the real world through proportional reasoning.

Given facts. A map has scale 1:25000. The measured map distance between two points is 5 cm. Rule: 1 cm on the map represents 25000 cm in reality; 100,000 cm = 1 km.

Rules. Multiply map length by the scale denominator, then convert units. A scale is a ratio, not an added distance.

Task. What is the real straight-line distance in kilometres?

Step-by-step solution. 1. Real centimetres=5×25000=125000. 2. Convert to kilometres: 125000÷100,000=1.25.

Answer. 1.25 km.

Knowledge takeaway. Scale problems are ratio problems: keep units explicit and conversion becomes mechanical.

Formal model. Proportion d_real=d_map×denominator, followed by unit conversion.

Problem 378. Map scale 1:50000

Knowledge context. Map scale links a model to the real world through proportional reasoning.

Given facts. A map has scale 1:50000. The measured map distance between two points is 6 cm. Rule: 1 cm on the map represents 50000 cm in reality; 100,000 cm = 1 km.

Rules. Multiply map length by the scale denominator, then convert units. A scale is a ratio, not an added distance.

Task. What is the real straight-line distance in kilometres?

Step-by-step solution. 1. Real centimetres=6×50000=300000. 2. Convert to kilometres: 300000÷100,000=3.

Answer. 3 km.

Knowledge takeaway. Scale problems are ratio problems: keep units explicit and conversion becomes mechanical.

Formal model. Proportion d_real=d_map×denominator, followed by unit conversion.

Problem 379. Map scale 1:100000

Knowledge context. Map scale links a model to the real world through proportional reasoning.

Given facts. A map has scale 1:100000. The measured map distance between two points is 7 cm. Rule: 1 cm on the map represents 100000 cm in reality; 100,000 cm = 1 km.

Rules. Multiply map length by the scale denominator, then convert units. A scale is a ratio, not an added distance.

Task. What is the real straight-line distance in kilometres?

Step-by-step solution. 1. Real centimetres=7×100000=700000. 2. Convert to kilometres: 700000÷100,000=7.

Answer. 7 km.

Knowledge takeaway. Scale problems are ratio problems: keep units explicit and conversion becomes mechanical.

Formal model. Proportion d_real=d_map×denominator, followed by unit conversion.

Problem 380. Map scale 1:200000

Knowledge context. Map scale links a model to the real world through proportional reasoning.

Given facts. A map has scale 1:200000. The measured map distance between two points is 8 cm. Rule: 1 cm on the map represents 200000 cm in reality; 100,000 cm = 1 km.

Rules. Multiply map length by the scale denominator, then convert units. A scale is a ratio, not an added distance.

Task. What is the real straight-line distance in kilometres?

Step-by-step solution. 1. Real centimetres=8×200000=1600000. 2. Convert to kilometres: 1600000÷100,000=16.

Answer. 16 km.

Knowledge takeaway. Scale problems are ratio problems: keep units explicit and conversion becomes mechanical.

Formal model. Proportion d_real=d_map×denominator, followed by unit conversion.

N2. Coordinates and constraint-based location

Problem 381. Locating a point by constraints: case 1

Knowledge context. Coordinates turn location into numbers. Multiple spatial clues can be treated as simultaneous constraints.

Given facts. A point P lies on an integer grid with x,y from 0 to 6. It satisfies x+y=4 and x−y=0.

Rules. A location is valid only if it satisfies every stated equation and inequality at the same time.

Task. Which coordinate(s) can P occupy? Is the solution unique?

Step-by-step solution. 1. List pairs satisfying x+y=4. 2. Keep only those also satisfying x−y=0. 3. The remaining set is [(2, 2)].

Answer. P=(2, 2); the solution is unique.

Knowledge takeaway. A constraint problem is solved by satisfying all conditions together, not one at a time independently.

Formal model. Constraint satisfaction over integer pairs (x,y).

Problem 382. Locating a point by constraints: case 2

Knowledge context. Coordinates turn location into numbers. Multiple spatial clues can be treated as simultaneous constraints.

Given facts. A point P lies on an integer grid with x,y from 0 to 6. It satisfies x+y=7 and x−y=-1.

Rules. A location is valid only if it satisfies every stated equation and inequality at the same time.

Task. Which coordinate(s) can P occupy? Is the solution unique?

Step-by-step solution. 1. List pairs satisfying x+y=7. 2. Keep only those also satisfying x−y=-1. 3. The remaining set is [(3, 4)].

Answer. P=(3, 4); the solution is unique.

Knowledge takeaway. A constraint problem is solved by satisfying all conditions together, not one at a time independently.

Formal model. Constraint satisfaction over integer pairs (x,y).

Problem 383. Locating a point by constraints: case 3

Knowledge context. Coordinates turn location into numbers. Multiple spatial clues can be treated as simultaneous constraints.

Given facts. A point P lies on an integer grid with x,y from 0 to 6. It satisfies x+y=10 and x−y=-2.

Rules. A location is valid only if it satisfies every stated equation and inequality at the same time.

Task. Which coordinate(s) can P occupy? Is the solution unique?

Step-by-step solution. 1. List pairs satisfying x+y=10. 2. Keep only those also satisfying x−y=-2. 3. The remaining set is [(4, 6)].

Answer. P=(4, 6); the solution is unique.

Knowledge takeaway. A constraint problem is solved by satisfying all conditions together, not one at a time independently.

Formal model. Constraint satisfaction over integer pairs (x,y).

Problem 384. Locating a point by constraints: case 4

Knowledge context. Coordinates turn location into numbers. Multiple spatial clues can be treated as simultaneous constraints.

Given facts. A point P lies on an integer grid with x,y from 0 to 6. It satisfies x+y=6 and x−y=4.

Rules. A location is valid only if it satisfies every stated equation and inequality at the same time.

Task. Which coordinate(s) can P occupy? Is the solution unique?

Step-by-step solution. 1. List pairs satisfying x+y=6. 2. Keep only those also satisfying x−y=4. 3. The remaining set is [(5, 1)].

Answer. P=(5, 1); the solution is unique.

Knowledge takeaway. A constraint problem is solved by satisfying all conditions together, not one at a time independently.

Formal model. Constraint satisfaction over integer pairs (x,y).

Problem 385. Locating a point by constraints: case 5

Knowledge context. Coordinates turn location into numbers. Multiple spatial clues can be treated as simultaneous constraints.

Given facts. A point P lies on an integer grid with x,y from 0 to 6. It satisfies x+y=9 and x−y=3.

Rules. A location is valid only if it satisfies every stated equation and inequality at the same time.

Task. Which coordinate(s) can P occupy? Is the solution unique?

Step-by-step solution. 1. List pairs satisfying x+y=9. 2. Keep only those also satisfying x−y=3. 3. The remaining set is [(6, 3)].

Answer. P=(6, 3); the solution is unique.

Knowledge takeaway. A constraint problem is solved by satisfying all conditions together, not one at a time independently.

Formal model. Constraint satisfaction over integer pairs (x,y).

N3. Earth, Sun, seasons, and local time

Problem 386. Earth, sunlight, and local time: case 1

Knowledge context. Earth–Sun relationships explain broad patterns such as day length and local time, but a problem should state the simplified model it expects children to use.

Given facts. For this classroom model, Eastport is 1 time zone(s) east of Westport. Each zone east is one hour later. At Westport it is 10:00.

Rules. Moving east adds one hour per stated time zone; moving west subtracts. Use the given model only.

Task. What local time is it in Eastport?

Step-by-step solution. 1. Time difference is +1 hour(s) because Eastport is east. 2. 10+1=11 (mod 24).

Answer. 11:00.

Knowledge takeaway. A model supports only the conclusions encoded in its rules; seasonal light does not automatically determine daily weather.

Formal model. Rule-based astronomical/time-zone model; apply signed offsets or seasonal relation.

Problem 387. Earth, sunlight, and local time: case 2

Knowledge context. Earth–Sun relationships explain broad patterns such as day length and local time, but a problem should state the simplified model it expects children to use.

Given facts. For this classroom model, Eastport is 2 time zone(s) east of Westport. Each zone east is one hour later. At Westport it is 10:00.

Rules. Moving east adds one hour per stated time zone; moving west subtracts. Use the given model only.

Task. What local time is it in Eastport?

Step-by-step solution. 1. Time difference is +2 hour(s) because Eastport is east. 2. 10+2=12 (mod 24).

Answer. 12:00.

Knowledge takeaway. A model supports only the conclusions encoded in its rules; seasonal light does not automatically determine daily weather.

Formal model. Rule-based astronomical/time-zone model; apply signed offsets or seasonal relation.

Problem 388. Earth, sunlight, and local time: case 3

Knowledge context. Earth–Sun relationships explain broad patterns such as day length and local time, but a problem should state the simplified model it expects children to use.

Given facts. For this classroom model, Eastport is 3 time zone(s) east of Westport. Each zone east is one hour later. At Westport it is 10:00.

Rules. Moving east adds one hour per stated time zone; moving west subtracts. Use the given model only.

Task. What local time is it in Eastport?

Step-by-step solution. 1. Time difference is +3 hour(s) because Eastport is east. 2. 10+3=13 (mod 24).

Answer. 13:00.

Knowledge takeaway. A model supports only the conclusions encoded in its rules; seasonal light does not automatically determine daily weather.

Formal model. Rule-based astronomical/time-zone model; apply signed offsets or seasonal relation.

Problem 389. Earth, sunlight, and local time: case 4

Knowledge context. Earth–Sun relationships explain broad patterns such as day length and local time, but a problem should state the simplified model it expects children to use.

Given facts. Use this simplified model: Earth's axis is tilted. In the Northern Hemisphere, days are generally longer around June than around December. Compare June and December for the same place. Cross-domain check: a survey team starts work at 12:00 and works for 2 hour(s) without a break.

Rules. Use the stated seasonal relation; do not infer daily weather from season. For the cross-domain check, finishing time = starting time + duration.

Task. Which of the two months has the longer daylight period in this model, and can that fact alone tell us whether it will rain? Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. Apply the stated tilt-season relation. 2. June has the longer daylight period in the Northern Hemisphere. 3. The rule contains no information about rainfall, so rainfall cannot be deduced. Cross-domain check: 12+2=14, so the team finishes at 14:00.

Answer. June has longer daylight; rainfall cannot be determined from this fact alone. Cross-domain answer: 14:00.

Knowledge takeaway. A model supports only the conclusions encoded in its rules; seasonal light does not automatically determine daily weather.

Formal model. Rule-based astronomical/time-zone model; apply signed offsets or seasonal relation.

Problem 390. Earth, sunlight, and local time: case 5

Knowledge context. Earth–Sun relationships explain broad patterns such as day length and local time, but a problem should state the simplified model it expects children to use.

Given facts. Use this simplified model: Earth's axis is tilted. In the Northern Hemisphere, days are generally longer around June than around December. Compare June and December for the same place. Cross-domain check: a local committee has 11 members, requires at least 6 present for quorum, and 11 are present.

Rules. Use the stated seasonal relation; do not infer daily weather from season. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. Which of the two months has the longer daylight period in this model, and can that fact alone tell us whether it will rain? Cross-domain check: is quorum met?

Step-by-step solution. 1. Apply the stated tilt-season relation. 2. June has the longer daylight period in the Northern Hemisphere. 3. The rule contains no information about rainfall, so rainfall cannot be deduced. Cross-domain check: compare 11 with 6; $11 \geq 6$ is true.

Answer. June has longer daylight; rainfall cannot be determined from this fact alone. Cross-domain answer: quorum is met.

Knowledge takeaway. A model supports only the conclusions encoded in its rules; seasonal light does not automatically determine daily weather.

Formal model. Rule-based astronomical/time-zone model; apply signed offsets or seasonal relation.

N4. Stocks, flows, and conservation

Problem 391. Reservoir stock and flow: case 1

Knowledge context. Many geographic and civic quantities are stocks and flows: water in a reservoir, people in a city, or goods in a warehouse.

Given facts. A reservoir starts with 30 water units. Each day 5 units flow in and 5 units flow out. Assume these rates stay constant for 4 days.

Rules. Stock at the end = starting stock + total inflow − total outflow. Flow is an amount per time period; stock is the amount currently stored.

Task. How much water is stored after 4 days? Is the stock increasing or decreasing each day?

Step-by-step solution. 1. Daily net change=5−5=0. 2. Over 4 days, net change=4×0=0. 3. End stock=30+(0)=30.

Answer. 30 units; the stock is unchanged.

Knowledge takeaway. Do not confuse a stored amount with a rate of change; conservation links them.

Formal model. Difference equation S(t+1)=S(t)+inflow−outflow.

Problem 392. Reservoir stock and flow: case 2

Knowledge context. Many geographic and civic quantities are stocks and flows: water in a reservoir, people in a city, or goods in a warehouse.

Given facts. A reservoir starts with 33 water units. Each day 6 units flow in and 6 units flow out. Assume these rates stay constant for 4 days.

Rules. Stock at the end = starting stock + total inflow − total outflow. Flow is an amount per time period; stock is the amount currently stored.

Task. How much water is stored after 4 days? Is the stock increasing or decreasing each day?

Step-by-step solution. 1. Daily net change=6−6=0. 2. Over 4 days, net change=4×0=0. 3. End stock=33+(0)=33.

Answer. 33 units; the stock is unchanged.

Knowledge takeaway. Do not confuse a stored amount with a rate of change; conservation links them.

Formal model. Difference equation S(t+1)=S(t)+inflow−outflow.

Problem 393. Reservoir stock and flow: case 3

Knowledge context. Many geographic and civic quantities are stocks and flows: water in a reservoir, people in a city, or goods in a warehouse.

Given facts. A reservoir starts with 36 water units. Each day 7 units flow in and 5 units flow out. Assume these rates stay constant for 4 days.

Rules. Stock at the end = starting stock + total inflow − total outflow. Flow is an amount per time period; stock is the amount currently stored.

Task. How much water is stored after 4 days? Is the stock increasing or decreasing each day?

Step-by-step solution. 1. Daily net change=7−5=2. 2. Over 4 days, net change=4×2=8. 3. End stock=36+(8)=44.

Answer. 44 units; the stock is increasing.

Knowledge takeaway. Do not confuse a stored amount with a rate of change; conservation links them.

Formal model. Difference equation S(t+1)=S(t)+inflow−outflow.

Problem 394. Reservoir stock and flow: case 4

Knowledge context. Many geographic and civic quantities are stocks and flows: water in a reservoir, people in a city, or goods in a warehouse.

Given facts. A reservoir starts with 39 water units. Each day 8 units flow in and 6 units flow out. Assume these rates stay constant for 4 days.

Rules. Stock at the end = starting stock + total inflow − total outflow. Flow is an amount per time period; stock is the amount currently stored.

Task. How much water is stored after 4 days? Is the stock increasing or decreasing each day?

Step-by-step solution. 1. Daily net change=8−6=2. 2. Over 4 days, net change=4×2=8. 3. End stock=39+(8)=47.

Answer. 47 units; the stock is increasing.

Knowledge takeaway. Do not confuse a stored amount with a rate of change; conservation links them.

Formal model. Difference equation S(t+1)=S(t)+inflow−outflow.

Problem 395. Reservoir stock and flow: case 5

Knowledge context. Many geographic and civic quantities are stocks and flows: water in a reservoir, people in a city, or goods in a warehouse.

Given facts. A reservoir starts with 42 water units. Each day 9 units flow in and 5 units flow out. Assume these rates stay constant for 4 days.

Rules. Stock at the end = starting stock + total inflow − total outflow. Flow is an amount per time period; stock is the amount currently stored.

Task. How much water is stored after 4 days? Is the stock increasing or decreasing each day?

Step-by-step solution. 1. Daily net change=9−5=4. 2. Over 4 days, net change=4×4=16. 3. End stock=42+(16)=58.

Answer. 58 units; the stock is increasing.

Knowledge takeaway. Do not confuse a stored amount with a rate of change; conservation links them.

Formal model. Difference equation S(t+1)=S(t)+inflow−outflow.

N5. Density, capacity, and crowding

Problem 396. Density and capacity in a public space: case 1

Knowledge context. Density compares an amount with the space available. It can reveal crowding even when two places have similar total populations.

Given facts. A park zone covers 4 hectares and currently has 30 visitors. For this exercise, comfortable density is at most 7 visitors per hectare.

Rules. Density = number ÷ area. A zone exceeds comfortable capacity if density is greater than the stated threshold.

Task. What is the visitor density, and is the zone above the stated comfortable capacity?

Step-by-step solution. 1. Density=30÷4=7.50 visitors/hectare. 2. Compare 7.50 with the threshold 7.

Answer. 7.50 visitors/hectare; above capacity.

Knowledge takeaway. Totals and densities answer different questions; always identify the denominator.

Formal model. Ratio density=N/A followed by threshold comparison.

Problem 397. Density and capacity in a public space: case 2

Knowledge context. Density compares an amount with the space available. It can reveal crowding even when two places have similar total populations.

Given facts. A park zone covers 5 hectares and currently has 34 visitors. For this exercise, comfortable density is at most 7 visitors per hectare.

Rules. Density = number ÷ area. A zone exceeds comfortable capacity if density is greater than the stated threshold.

Task. What is the visitor density, and is the zone above the stated comfortable capacity?

Step-by-step solution. 1. Density=34÷5=6.80 visitors/hectare. 2. Compare 6.80 with the threshold 7.

Answer. 6.80 visitors/hectare; within capacity.

Knowledge takeaway. Totals and densities answer different questions; always identify the denominator.

Formal model. Ratio density=N/A followed by threshold comparison.

Problem 398. Density and capacity in a public space: case 3

Knowledge context. Density compares an amount with the space available. It can reveal crowding even when two places have similar total populations.

Given facts. A park zone covers 6 hectares and currently has 38 visitors. For this exercise, comfortable density is at most 7 visitors per hectare.

Rules. Density = number ÷ area. A zone exceeds comfortable capacity if density is greater than the stated threshold.

Task. What is the visitor density, and is the zone above the stated comfortable capacity?

Step-by-step solution. 1. Density=38÷6=6.33 visitors/hectare. 2. Compare 6.33 with the threshold 7.

Answer. 6.33 visitors/hectare; within capacity.

Knowledge takeaway. Totals and densities answer different questions; always identify the denominator.

Formal model. Ratio density=N/A followed by threshold comparison.

Problem 399. Density and capacity in a public space: case 4

Knowledge context. Density compares an amount with the space available. It can reveal crowding even when two places have similar total populations.

Given facts. A park zone covers 7 hectares and currently has 42 visitors. For this exercise, comfortable density is at most 7 visitors per hectare.

Rules. Density = number ÷ area. A zone exceeds comfortable capacity if density is greater than the stated threshold.

Task. What is the visitor density, and is the zone above the stated comfortable capacity?

Step-by-step solution. 1. Density=42÷7=6.00 visitors/hectare. 2. Compare 6.00 with the threshold 7.

Answer. 6.00 visitors/hectare; within capacity.

Knowledge takeaway. Totals and densities answer different questions; always identify the denominator.

Formal model. Ratio density=N/A followed by threshold comparison.

Problem 400. Density and capacity in a public space: case 5

Knowledge context. Density compares an amount with the space available. It can reveal crowding even when two places have similar total populations.

Given facts. A park zone covers 8 hectares and currently has 46 visitors. For this exercise, comfortable density is at most 7 visitors per hectare.

Rules. Density = number ÷ area. A zone exceeds comfortable capacity if density is greater than the stated threshold.

Task. What is the visitor density, and is the zone above the stated comfortable capacity?

Step-by-step solution. 1. Density=46÷8=5.75 visitors/hectare. 2. Compare 5.75 with the threshold 7.

Answer. 5.75 visitors/hectare; within capacity.

Knowledge takeaway. Totals and densities answer different questions; always identify the denominator.

Formal model. Ratio density=N/A followed by threshold comparison.

N6. Land use and spatial compatibility

Problem 401. Land-use compatibility on four plots: case 1

Knowledge context. Land-use planning combines spatial relationships with compatibility rules.

Given facts. Four plots lie in a line A–B–C–D. Assign one different use to each plot: school, factory, park, homes. Constraints: the factory may not be adjacent to the school; the park must be adjacent to the homes. Cross-domain check: a survey team starts work at 8:00 and works for 2 hour(s) without a break.

Rules. An assignment is valid only if every use is placed exactly once and all adjacency constraints hold. For the cross-domain check, finishing time = starting time + duration.

Task. Give one valid assignment and explain why it satisfies every constraint. Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. Try the assignment A=school, B=park, C=homes, D=factory. 2. Factory and school are not adjacent. 3. Park and homes are adjacent. 4. Each use appears exactly once. Cross-domain check: 8+2=10, so the team finishes at 10:00.

Answer. A=school, B=park, C=homes, D=factory is a valid assignment. Cross-domain answer: 10:00.

Knowledge takeaway. A place can be unsuitable not because of its own properties, but because of what would be next to it.

Formal model. Permutation CSP with all-different and adjacency/non-adjacency constraints.

Problem 402. Land-use compatibility on four plots: case 2

Knowledge context. Land-use planning combines spatial relationships with compatibility rules.

Given facts. Four plots lie in a line A–B–C–D. Assign one different use to each plot: school, factory, park, homes. Constraints: the factory may not be adjacent to the school; the park must be adjacent to the homes. Cross-domain check: a local committee has 14 members, requires at least 8 present for quorum, and 14 are present.

Rules. An assignment is valid only if every use is placed exactly once and all adjacency constraints hold. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. Give one valid assignment and explain why it satisfies every constraint. Cross-domain check: is quorum met?

Step-by-step solution. 1. Try the assignment A=school, B=homes, C=park, D=factory. 2. Factory and school are not adjacent. 3. Park and homes are adjacent. 4. Each use appears exactly once. Cross-domain check: compare 14 with 8; $14 \geq 8$ is true.

Answer. A=school, B=homes, C=park, D=factory is a valid assignment. Cross-domain answer: quorum is met.

Knowledge takeaway. A place can be unsuitable not because of its own properties, but because of what would be next to it.

Formal model. Permutation CSP with all-different and adjacency/non-adjacency constraints.

Problem 403. Land-use compatibility on four plots: case 3

Knowledge context. Land-use planning combines spatial relationships with compatibility rules.

Given facts. Four plots lie in a line A–B–C–D. Assign one different use to each plot: school, factory, park, homes. Constraints: the factory may not be adjacent to the school; the park must be adjacent to the homes. Cross-domain check: researchers collected 9 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. An assignment is valid only if every use is placed exactly once and all adjacency constraints hold. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. Give one valid assignment and explain why it satisfies every constraint. Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. Try the assignment A=factory, B=park, C=homes, D=school. 2. Factory and school are not adjacent. 3. Park and homes are adjacent. 4. Each use appears exactly once. Cross-domain check: 9−4=5 independent reports remain.

Answer. A=factory, B=park, C=homes, D=school is a valid assignment. Cross-domain answer: 5 independent reports.

Knowledge takeaway. A place can be unsuitable not because of its own properties, but because of what would be next to it.

Formal model. Permutation CSP with all-different and adjacency/non-adjacency constraints.

Problem 404. Land-use compatibility on four plots: case 4

Knowledge context. Land-use planning combines spatial relationships with compatibility rules.

Given facts. Four plots lie in a line A–B–C–D. Assign one different use to each plot: school, factory, park, homes. Constraints: the factory may not be adjacent to the school; the park must be adjacent to the homes. Cross-domain check: on a field sketch, 1 cm represents 6 km and two checkpoints are 7 cm apart.

Rules. An assignment is valid only if every use is placed exactly once and all adjacency constraints hold. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Give one valid assignment and explain why it satisfies every constraint. Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. Try the assignment A=factory, B=homes, C=park, D=school. 2. Factory and school are not adjacent. 3. Park and homes are adjacent. 4. Each use appears exactly once. Cross-domain check: 7×6=42 km.

Answer. A=factory, B=homes, C=park, D=school is a valid assignment. Cross-domain answer: 42 km.

Knowledge takeaway. A place can be unsuitable not because of its own properties, but because of what would be next to it.

Formal model. Permutation CSP with all-different and adjacency/non-adjacency constraints.

Problem 405. Land-use compatibility on four plots: case 5

Knowledge context. Land-use planning combines spatial relationships with compatibility rules.

Given facts. Four plots lie in a line A–B–C–D. Assign one different use to each plot: school, factory, park, homes. Constraints: the factory may not be adjacent to the school; the park must be adjacent to the homes. Cross-domain check: a survey team starts work at 12:00 and works for 2 hour(s) without a break.

Rules. An assignment is valid only if every use is placed exactly once and all adjacency constraints hold. For the cross-domain check, finishing time = starting time + duration.

Task. Give one valid assignment and explain why it satisfies every constraint. Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. Try the assignment A=school, B=park, C=homes, D=factory. 2. Factory and school are not adjacent. 3. Park and homes are adjacent. 4. Each use appears exactly once. Cross-domain check: 12+2=14, so the team finishes at 14:00.

Answer. A=school, B=park, C=homes, D=factory is a valid assignment. Cross-domain answer: 14:00.

Knowledge takeaway. A place can be unsuitable not because of its own properties, but because of what would be next to it.

Formal model. Permutation CSP with all-different and adjacency/non-adjacency constraints.

N7. Natural risk: hazard, exposure, protection

Problem 406. Hazard, exposure, and protection: case 1

Knowledge context. Natural risk depends on more than the physical hazard. Exposure and protective measures also affect outcomes.

Given facts. For this classroom model, risk index = hazard × exposure ÷ protection. Data: Site A: hazard=2, exposure=4, protection=1; Site B: hazard=4, exposure=2, protection=2; Site C: hazard=3, exposure=3, protection=2.

Rules. Use the formula exactly as stated. A higher hazard does not necessarily mean higher risk if exposure or protection differs.

Task. Compute the risk index for each site. Which site has the lowest modeled risk?

Step-by-step solution. 1. Compute hazard×exposure÷protection for each site. 2. Results: A=8.00, B=4.00, C=4.50. 3. The lowest value is Site B.

Answer. A=8.00; B=4.00; C=4.50. Lowest: Site B.

Knowledge takeaway. Separating hazard from risk prevents the mistaken idea that the most dangerous process always creates the greatest overall risk.

Formal model. Explicit risk-index function over three variables; compare computed values.

Problem 407. Hazard, exposure, and protection: case 2

Knowledge context. Natural risk depends on more than the physical hazard. Exposure and protective measures also affect outcomes.

Given facts. For this classroom model, risk index = hazard × exposure ÷ protection. Data: Site A: hazard=3, exposure=4, protection=1; Site B: hazard=4, exposure=3, protection=2; Site C: hazard=3, exposure=3, protection=2.

Rules. Use the formula exactly as stated. A higher hazard does not necessarily mean higher risk if exposure or protection differs.

Task. Compute the risk index for each site. Which site has the lowest modeled risk?

Step-by-step solution. 1. Compute hazard×exposure÷protection for each site. 2. Results: A=12.00, B=6.00, C=4.50. 3. The lowest value is Site C.

Answer. A=12.00; B=6.00; C=4.50. Lowest: Site C.

Knowledge takeaway. Separating hazard from risk prevents the mistaken idea that the most dangerous process always creates the greatest overall risk.

Formal model. Explicit risk-index function over three variables; compare computed values.

Problem 408. Hazard, exposure, and protection: case 3

Knowledge context. Natural risk depends on more than the physical hazard. Exposure and protective measures also affect outcomes.

Given facts. For this classroom model, risk index = hazard × exposure ÷ protection. Data: Site A: hazard=4, exposure=4, protection=1; Site B: hazard=4, exposure=4, protection=2; Site C: hazard=3, exposure=3, protection=2.

Rules. Use the formula exactly as stated. A higher hazard does not necessarily mean higher risk if exposure or protection differs.

Task. Compute the risk index for each site. Which site has the lowest modeled risk?

Step-by-step solution. 1. Compute hazard×exposure÷protection for each site. 2. Results: A=16.00, B=8.00, C=4.50. 3. The lowest value is Site C.

Answer. A=16.00; B=8.00; C=4.50. Lowest: Site C.

Knowledge takeaway. Separating hazard from risk prevents the mistaken idea that the most dangerous process always creates the greatest overall risk.

Formal model. Explicit risk-index function over three variables; compare computed values.

Problem 409. Hazard, exposure, and protection: case 4

Knowledge context. Natural risk depends on more than the physical hazard. Exposure and protective measures also affect outcomes.

Given facts. For this classroom model, risk index = hazard × exposure ÷ protection. Data: Site A: hazard=5, exposure=4, protection=1; Site B: hazard=4, exposure=5, protection=2; Site C: hazard=3, exposure=3, protection=2.

Rules. Use the formula exactly as stated. A higher hazard does not necessarily mean higher risk if exposure or protection differs.

Task. Compute the risk index for each site. Which site has the lowest modeled risk?

Step-by-step solution. 1. Compute hazard×exposure÷protection for each site. 2. Results: A=20.00, B=10.00, C=4.50. 3. The lowest value is Site C.

Answer. A=20.00; B=10.00; C=4.50. Lowest: Site C.

Knowledge takeaway. Separating hazard from risk prevents the mistaken idea that the most dangerous process always creates the greatest overall risk.

Formal model. Explicit risk-index function over three variables; compare computed values.

Problem 410. Hazard, exposure, and protection: case 5

Knowledge context. Natural risk depends on more than the physical hazard. Exposure and protective measures also affect outcomes.

Given facts. For this classroom model, risk index = hazard × exposure ÷ protection. Data: Site A: hazard=6, exposure=4, protection=1; Site B: hazard=4, exposure=6, protection=2; Site C: hazard=3, exposure=3, protection=2.

Rules. Use the formula exactly as stated. A higher hazard does not necessarily mean higher risk if exposure or protection differs.

Task. Compute the risk index for each site. Which site has the lowest modeled risk?

Step-by-step solution. 1. Compute hazard×exposure÷protection for each site. 2. Results: A=24.00, B=12.00, C=4.50. 3. The lowest value is Site C.

Answer. A=24.00; B=12.00; C=4.50. Lowest: Site C.

Knowledge takeaway. Separating hazard from risk prevents the mistaken idea that the most dangerous process always creates the greatest overall risk.

Formal model. Explicit risk-index function over three variables; compare computed values.

N8. Ecosystems and indirect effects

Problem 411. Indirect effects in a food chain: case 1

Knowledge context. Ecosystems contain dependency networks. A change at one point can propagate indirectly through several links.

Given facts. Food relation: rabbit eats grass; fox eats rabbit; eagle eats fox. Classroom rule: if a food source sharply decreases and no alternative food is given, the consumer may also decrease. Cross-domain check: researchers collected 7 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. Follow dependency arrows from food to consumer. The rule supports possible directional effects, not exact population numbers. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. If rabbit sharply decreases, which later populations may be indirectly affected according to the rule? Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. fox depends directly on rabbit, so it may decrease. 2. eagle depends on fox, so it may also be indirectly affected. 3. grass is food for rabbit; the stated rule does not imply it decreases for the same reason. Cross-domain check: 7−4=3 independent reports remain.

Answer. fox may decrease directly and eagle may be affected indirectly. Cross-domain answer: 3 independent reports.

Knowledge takeaway. Indirect effects require following the direction of dependency and respecting the limits of the stated model.

Formal model. Directed dependency graph; propagate a qualitative decrease along outgoing dependency links.

Problem 412. Indirect effects in a food chain: case 2

Knowledge context. Ecosystems contain dependency networks. A change at one point can propagate indirectly through several links.

Given facts. Food relation: small fish eats algae; large fish eats small fish; heron eats large fish. Classroom rule: if a food source sharply decreases and no alternative food is given, the consumer may also decrease. Cross-domain check: on a field sketch, 1 cm represents 4 km and two checkpoints are 8 cm apart.

Rules. Follow dependency arrows from food to consumer. The rule supports possible directional effects, not exact population numbers. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. If small fish sharply decreases, which later populations may be indirectly affected according to the rule? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. large fish depends directly on small fish, so it may decrease. 2. heron depends on large fish, so it may also be indirectly affected. 3. algae is food for small fish; the stated rule does not imply it decreases for the same reason. Cross-domain check: 8×4=32 km.

Answer. large fish may decrease directly and heron may be affected indirectly. Cross-domain answer: 32 km.

Knowledge takeaway. Indirect effects require following the direction of dependency and respecting the limits of the stated model.

Formal model. Directed dependency graph; propagate a qualitative decrease along outgoing dependency links.

Problem 413. Indirect effects in a food chain: case 3

Knowledge context. Ecosystems contain dependency networks. A change at one point can propagate indirectly through several links.

Given facts. Food relation: rabbit eats grass; fox eats rabbit; eagle eats fox. Classroom rule: if a food source sharply decreases and no alternative food is given, the consumer may also decrease. Cross-domain check: a survey team starts

work at 12:00 and works for 2 hour(s) without a break. Mixed-domain verification: a regional archive has 43 map sheets and receives 4 additional sheets during the exercise.

Rules. Follow dependency arrows from food to consumer. The rule supports possible directional effects, not exact population numbers. For the cross-domain check, finishing time = starting time + duration. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. If rabbit sharply decreases, which later populations may be indirectly affected according to the rule? Cross-domain check: at what time does the team finish? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. fox depends directly on rabbit, so it may decrease. 2. eagle depends on fox, so it may also be indirectly affected. 3. grass is food for rabbit; the stated rule does not imply it decreases for the same reason. Cross-domain check: 12+2=14, so the team finishes at 14:00. Mixed-domain verification: 43+4=47.

Answer. fox may decrease directly and eagle may be affected indirectly. Cross-domain answer: 14:00. Mixed-domain answer: 47 map sheets.

Knowledge takeaway. Indirect effects require following the direction of dependency and respecting the limits of the stated model.

Formal model. Directed dependency graph; propagate a qualitative decrease along outgoing dependency links.

Problem 414. Indirect effects in a food chain: case 4

Knowledge context. Ecosystems contain dependency networks. A change at one point can propagate indirectly through several links.

Given facts. Food relation: small fish eats algae; large fish eats small fish; heron eats large fish. Classroom rule: if a food source sharply decreases and no alternative food is given, the consumer may also decrease. Cross-domain check: a local committee has 8 members, requires at least 5 present for quorum, and 8 are present.

Rules. Follow dependency arrows from food to consumer. The rule supports possible directional effects, not exact population numbers. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. If small fish sharply decreases, which later populations may be indirectly affected according to the rule? Cross-domain check: is quorum met?

Step-by-step solution. 1. large fish depends directly on small fish, so it may decrease. 2. heron depends on large fish, so it may also be indirectly affected. 3. algae is food for small fish; the stated rule does not imply it decreases for the same reason. Cross-domain check: compare 8 with 5; $8 \geq 5$ is true.

Answer. large fish may decrease directly and heron may be affected indirectly. Cross-domain answer: quorum is met.

Knowledge takeaway. Indirect effects require following the direction of dependency and respecting the limits of the stated model.

Formal model. Directed dependency graph; propagate a qualitative decrease along outgoing dependency links.

Problem 415. Indirect effects in a food chain: case 5

Knowledge context. Ecosystems contain dependency networks. A change at one point can propagate indirectly through several links.

Given facts. Food relation: rabbit eats grass; fox eats rabbit; eagle eats fox. Classroom rule: if a food source sharply decreases and no alternative food is given, the consumer may also decrease. Cross-domain check: researchers collected 11 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. Follow dependency arrows from food to consumer. The rule supports possible directional effects, not exact population numbers. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. If rabbit sharply decreases, which later populations may be indirectly affected according to the rule? Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. fox depends directly on rabbit, so it may decrease. 2. eagle depends on fox, so it may also be indirectly affected. 3. grass is food for rabbit; the stated rule does not imply it decreases for the same reason. Cross-domain check: 11−4=7 independent reports remain.

Answer. fox may decrease directly and eagle may be affected indirectly. Cross-domain answer: 7 independent reports.

Knowledge takeaway. Indirect effects require following the direction of dependency and respecting the limits of the stated model.

Formal model. Directed dependency graph; propagate a qualitative decrease along outgoing dependency links.

N9. Renewable resources and sustainable use

Problem 416. Regeneration and harvest: case 1

Knowledge context. Renewable does not mean unlimited. A resource can decline when use exceeds regeneration.

Given facts. A forest resource starts at 120 units. Each year it regenerates 12 units and people harvest 10 units. Assume these amounts stay constant for 3 years and no other changes occur.

Rules. Annual net change = regeneration − harvest. In this simplified model, use is sustainable over the period if stock does not decline.

Task. What is the stock after 3 years, and is the harvest sustainable under this classroom definition?

Step-by-step solution. 1. Annual net change=12−10=2. 2. Three-year change=6. 3. Final stock=120+(6)=126. 4. Compare final stock with starting stock.

Answer. 126 units; sustainable under the stated definition.

Knowledge takeaway. Sustainability is a balance between renewal and extraction over time.

Formal model. Stock-flow model with regeneration and harvest; test non-decreasing stock.

Problem 417. Regeneration and harvest: case 2

Knowledge context. Renewable does not mean unlimited. A resource can decline when use exceeds regeneration.

Given facts. A forest resource starts at 120 units. Each year it regenerates 14 units and people harvest 13 units. Assume these amounts stay constant for 3 years and no other changes occur.

Rules. Annual net change = regeneration − harvest. In this simplified model, use is sustainable over the period if stock does not decline.

Task. What is the stock after 3 years, and is the harvest sustainable under this classroom definition?

Step-by-step solution. 1. Annual net change=14−13=1. 2. Three-year change=3. 3. Final stock=120+(3)=123. 4. Compare final stock with starting stock.

Answer. 123 units; sustainable under the stated definition.

Knowledge takeaway. Sustainability is a balance between renewal and extraction over time.

Formal model. Stock-flow model with regeneration and harvest; test non-decreasing stock.

Problem 418. Regeneration and harvest: case 3

Knowledge context. Renewable does not mean unlimited. A resource can decline when use exceeds regeneration.

Given facts. A forest resource starts at 120 units. Each year it regenerates 16 units and people harvest 16 units. Assume these amounts stay constant for 3 years and no other changes occur.

Rules. Annual net change = regeneration − harvest. In this simplified model, use is sustainable over the period if stock does not decline.

Task. What is the stock after 3 years, and is the harvest sustainable under this classroom definition?

Step-by-step solution. 1. Annual net change=16−16=0. 2. Three-year change=0. 3. Final stock=120+(0)=120. 4. Compare final stock with starting stock.

Answer. 120 units; sustainable under the stated definition.

Knowledge takeaway. Sustainability is a balance between renewal and extraction over time.

Formal model. Stock-flow model with regeneration and harvest; test non-decreasing stock.

Problem 419. Regeneration and harvest: case 4

Knowledge context. Renewable does not mean unlimited. A resource can decline when use exceeds regeneration.

Given facts. A forest resource starts at 120 units. Each year it regenerates 18 units and people harvest 19 units. Assume these amounts stay constant for 3 years and no other changes occur.

Rules. Annual net change = regeneration − harvest. In this simplified model, use is sustainable over the period if stock does not decline.

Task. What is the stock after 3 years, and is the harvest sustainable under this classroom definition?

Step-by-step solution. 1. Annual net change=18−19=-1. 2. Three-year change=-3. 3. Final stock=120+(-3)=117. 4. Compare final stock with starting stock.

Answer. 117 units; not sustainable under the stated definition.

Knowledge takeaway. Sustainability is a balance between renewal and extraction over time.

Formal model. Stock-flow model with regeneration and harvest; test non-decreasing stock.

Problem 420. Regeneration and harvest: case 5

Knowledge context. Renewable does not mean unlimited. A resource can decline when use exceeds regeneration.

Given facts. A forest resource starts at 120 units. Each year it regenerates 20 units and people harvest 22 units. Assume these amounts stay constant for 3 years and no other changes occur.

Rules. Annual net change = regeneration − harvest. In this simplified model, use is sustainable over the period if stock does not decline.

Task. What is the stock after 3 years, and is the harvest sustainable under this classroom definition?

Step-by-step solution. 1. Annual net change=20−22=-2. 2. Three-year change=-6. 3. Final stock=120+(-6)=114. 4. Compare final stock with starting stock.

Answer. 114 units; not sustainable under the stated definition.

Knowledge takeaway. Sustainability is a balance between renewal and extraction over time.

Formal model. Stock-flow model with regeneration and harvest; test non-decreasing stock.

N10. Migration and population balance

Problem 421. Population balance in a town: case 1

Knowledge context. Population changes through births, deaths, immigration, and emigration.

Given facts. A town begins the year with 1200 residents. During the year: births=20, deaths=12, people moving in=34, people moving out=25.

Rules. End population = start + births − deaths + in-migration − out-migration. Natural change and migration change are separate components.

Task. Compute natural change, migration change, and the year-end population.

Step-by-step solution. 1. Natural change=20−12=8. 2. Migration change=34−25=9. 3. Total change=8+9=17. 4. End population=1200+17=1217.

Answer. Natural change +8; migration change +9; end population 1217.

Knowledge takeaway. Breaking total change into components helps explain why a population grew or shrank.

Formal model. Balance equation P1=P0+births−deaths+in−out.

Problem 422. Population balance in a town: case 2

Knowledge context. Population changes through births, deaths, immigration, and emigration.

Given facts. A town begins the year with 1250 residents. During the year: births=23, deaths=12, people moving in=34, people moving out=29.

Rules. End population = start + births − deaths + in-migration − out-migration. Natural change and migration change are separate components.

Task. Compute natural change, migration change, and the year-end population.

Step-by-step solution. 1. Natural change=23−12=11. 2. Migration change=34−29=5. 3. Total change=11+5=16. 4. End population=1250+16=1266.

Answer. Natural change +11; migration change +5; end population 1266.

Knowledge takeaway. Breaking total change into components helps explain why a population grew or shrank.

Formal model. Balance equation P1=P0+births−deaths+in−out.

Problem 423. Population balance in a town: case 3

Knowledge context. Population changes through births, deaths, immigration, and emigration.

Given facts. A town begins the year with 1300 residents. During the year: births=26, deaths=12, people moving in=34, people moving out=33.

Rules. End population = start + births − deaths + in-migration − out-migration. Natural change and migration change are separate components.

Task. Compute natural change, migration change, and the year-end population.

Step-by-step solution. 1. Natural change=26−12=14. 2. Migration change=34−33=1. 3. Total change=14+1=15. 4. End population=1300+15=1315.

Answer. Natural change +14; migration change +1; end population 1315.

Knowledge takeaway. Breaking total change into components helps explain why a population grew or shrank.

Formal model. Balance equation P1=P0+births−deaths+in−out.

Problem 424. Population balance in a town: case 4

Knowledge context. Population changes through births, deaths, immigration, and emigration.

Given facts. A town begins the year with 1350 residents. During the year: births=29, deaths=12, people moving in=34, people moving out=37.

Rules. End population = start + births − deaths + in-migration − out-migration. Natural change and migration change are separate components.

Task. Compute natural change, migration change, and the year-end population.

Step-by-step solution. 1. Natural change=29−12=17. 2. Migration change=34−37=-3. 3. Total change=17+-3=14. 4. End population=1350+14=1364.

Answer. Natural change +17; migration change -3; end population 1364.

Knowledge takeaway. Breaking total change into components helps explain why a population grew or shrank.

Formal model. Balance equation P1=P0+births−deaths+in−out.

Problem 425. Population balance in a town: case 5

Knowledge context. Population changes through births, deaths, immigration, and emigration.

Given facts. A town begins the year with 1400 residents. During the year: births=32, deaths=12, people moving in=34, people moving out=41.

Rules. End population = start + births − deaths + in-migration − out-migration. Natural change and migration change are separate components.

Task. Compute natural change, migration change, and the year-end population.

Step-by-step solution. 1. Natural change=32−12=20. 2. Migration change=34−41=-7. 3. Total change=20+-7=13. 4. End population=1400+13=1413.

Answer. Natural change +20; migration change -7; end population 1413.

Knowledge takeaway. Breaking total change into components helps explain why a population grew or shrank.

Formal model. Balance equation P1=P0+births−deaths+in−out.

N11. Time intervals and overlapping events

Problem 426. Overlapping historical intervals: case 1

Knowledge context. Historical periods often overlap rather than forming a single neat sequence.

Given facts. Event A lasted from year 100 to 140. Event B lasted from 100 to 131. Treat intervals as continuous for this exercise.

Rules. Two intervals overlap when max(start times) < min(end times). Overlap length = min(end times) − max(start times), if positive.

Task. Did the events overlap? If so, for how many years?

Step-by-step solution. 1. Later start=max(100,100)=100. 2. Earlier end=min(140,131)=131. 3. Difference=131−100=31.

Answer. Yes, overlap 31 years.

Knowledge takeaway. Interval reasoning distinguishes “before/after” from “overlapping in time.”

Formal model. Interval intersection; compute max starts and min ends.

Problem 427. Overlapping historical intervals: case 2

Knowledge context. Historical periods often overlap rather than forming a single neat sequence.

Given facts. Event A lasted from year 120 to 160. Event B lasted from 130 to 161. Treat intervals as continuous for this exercise.

Rules. Two intervals overlap when max(start times) < min(end times). Overlap length = min(end times) − max(start times), if positive.

Task. Did the events overlap? If so, for how many years?

Step-by-step solution. 1. Later start=max(120,130)=130. 2. Earlier end=min(160,161)=160. 3. Difference=160−130=30.

Answer. Yes, overlap 30 years.

Knowledge takeaway. Interval reasoning distinguishes “before/after” from “overlapping in time.”

Formal model. Interval intersection; compute max starts and min ends.

Problem 428. Overlapping historical intervals: case 3

Knowledge context. Historical periods often overlap rather than forming a single neat sequence.

Given facts. Event A lasted from year 140 to 180. Event B lasted from 160 to 191. Treat intervals as continuous for this exercise.

Rules. Two intervals overlap when max(start times) < min(end times). Overlap length = min(end times) − max(start times), if positive.

Task. Did the events overlap? If so, for how many years?

Step-by-step solution. 1. Later start=max(140,160)=160. 2. Earlier end=min(180,191)=180. 3. Difference=180−160=20.

Answer. Yes, overlap 20 years.

Knowledge takeaway. Interval reasoning distinguishes “before/after” from “overlapping in time.”

Formal model. Interval intersection; compute max starts and min ends.

Problem 429. Overlapping historical intervals: case 4

Knowledge context. Historical periods often overlap rather than forming a single neat sequence.

Given facts. Event A lasted from year 160 to 200. Event B lasted from 160 to 191. Treat intervals as continuous for this exercise.

Rules. Two intervals overlap when max(start times) < min(end times). Overlap length = min(end times) − max(start times), if positive.

Task. Did the events overlap? If so, for how many years?

Step-by-step solution. 1. Later start=max(160,160)=160. 2. Earlier end=min(200,191)=191. 3. Difference=191−160=31.

Answer. Yes, overlap 31 years.

Knowledge takeaway. Interval reasoning distinguishes “before/after” from “overlapping in time.”

Formal model. Interval intersection; compute max starts and min ends.

Problem 430. Overlapping historical intervals: case 5

Knowledge context. Historical periods often overlap rather than forming a single neat sequence.

Given facts. Event A lasted from year 180 to 220. Event B lasted from 190 to 221. Treat intervals as continuous for this exercise.

Rules. Two intervals overlap when max(start times) < min(end times). Overlap length = min(end times) − max(start times), if positive.

Task. Did the events overlap? If so, for how many years?

Step-by-step solution. 1. Later start=max(180,190)=190. 2. Earlier end=min(220,221)=220. 3. Difference=220−190=30.

Answer. Yes, overlap 30 years.

Knowledge takeaway. Interval reasoning distinguishes “before/after” from “overlapping in time.”

Formal model. Interval intersection; compute max starts and min ends.

N12. Genealogy and generations

Problem 431. Reasoning across generations: case 1

Knowledge context. Genealogies are relational graphs. Generation distance can be counted through parent links.

Given facts. Maya is a parent of Noah. Noah is a parent of Lina. Maya is also a parent of Owen. Owen is a parent of Sara. Cross-domain check: researchers collected 7 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. A parent is one generation above a child. A grandparent is two parent-links above. People who share a parent are siblings under this simplified model. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. What family relation between Maya and Lina follows from the stated links? Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. Maya→Noah is one parent link. 2. Noah→Lina is a second parent link. 3. Two links give the grandparent relation. Cross-domain check: 7−4=3 independent reports remain.

Answer. Maya is a grandparent of Lina. Cross-domain answer: 3 independent reports.

Knowledge takeaway. Family relations can be derived step by step from a small set of explicit links.

Formal model. Directed parent graph; kinship predicates derived from path length and shared ancestors.

Problem 432. Reasoning across generations: case 2

Knowledge context. Genealogies are relational graphs. Generation distance can be counted through parent links.

Given facts. Maya is a parent of Noah. Noah is a parent of Lina. Maya is also a parent of Owen. Owen is a parent of Sara. Cross-domain check: on a field sketch, 1 cm represents 4 km and two checkpoints are 7 cm apart.

Rules. A parent is one generation above a child. A grandparent is two parent-links above. People who share a parent are siblings under this simplified model. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. What family relation between Noah and Lina follows from the stated links? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. The relation is stated directly. Cross-domain check: 7×4=28 km.

Answer. Noah is a parent of Lina. Cross-domain answer: 28 km.

Knowledge takeaway. Family relations can be derived step by step from a small set of explicit links.

Formal model. Directed parent graph; kinship predicates derived from path length and shared ancestors.

Problem 433. Reasoning across generations: case 3

Knowledge context. Genealogies are relational graphs. Generation distance can be counted through parent links.

Given facts. Maya is a parent of Noah. Noah is a parent of Lina. Maya is also a parent of Owen. Owen is a parent of Sara. Cross-domain check: a survey team starts work at 8:00 and works for 2 hour(s) without a break.

Rules. A parent is one generation above a child. A grandparent is two parent-links above. People who share a parent are siblings under this simplified model. For the cross-domain check, finishing time = starting time + duration.

Task. What family relation between Noah and Owen follows from the stated links? Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. Maya is parent of both Noah and Owen. 2. Sharing a parent satisfies the simplified sibling rule. Cross-domain check: 8+2=10, so the team finishes at 10:00.

Answer. Noah and Owen are siblings. Cross-domain answer: 10:00.

Knowledge takeaway. Family relations can be derived step by step from a small set of explicit links.

Formal model. Directed parent graph; kinship predicates derived from path length and shared ancestors.

Problem 434. Reasoning across generations: case 4

Knowledge context. Genealogies are relational graphs. Generation distance can be counted through parent links.

Given facts. Maya is a parent of Noah. Noah is a parent of Lina. Maya is also a parent of Owen. Owen is a parent of Sara. Cross-domain check: a local committee has 10 members, requires at least 6 present for quorum, and 8 are present.

Rules. A parent is one generation above a child. A grandparent is two parent-links above. People who share a parent are siblings under this simplified model. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. What family relation between Maya and Sara follows from the stated links? Cross-domain check: is quorum met?

Step-by-step solution. 1. Maya→Owen and Owen→Sara form two parent links. Cross-domain check: compare 8 with 6; $8 \geq 6$ is true.

Answer. Maya is a grandparent of Sara. Cross-domain answer: quorum is met.

Knowledge takeaway. Family relations can be derived step by step from a small set of explicit links.

Formal model. Directed parent graph; kinship predicates derived from path length and shared ancestors.

Problem 435. Reasoning across generations: case 5

Knowledge context. Genealogies are relational graphs. Generation distance can be counted through parent links.

Given facts. Maya is a parent of Noah. Noah is a parent of Lina. Maya is also a parent of Owen. Owen is a parent of Sara. Cross-domain check: researchers collected 11 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. A parent is one generation above a child. A grandparent is two parent-links above. People who share a parent are siblings under this simplified model. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. What family relation between Lina and Sara follows from the stated links? Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. Noah and Owen are siblings because they share parent Maya. 2. Lina is child of Noah; Sara is child of Owen. 3. Children of siblings are cousins in this model. Cross-domain check: 11−4=7 independent reports remain.

Answer. Lina and Sara are cousins under the implied family tree. Cross-domain answer: 7 independent reports.

Knowledge takeaway. Family relations can be derived step by step from a small set of explicit links.

Formal model. Directed parent graph; kinship predicates derived from path length and shared ancestors.

N13. Anachronisms and temporal compatibility

Problem 436. Temporal compatibility in a story set in 1400

Knowledge context. Anachronism can be tested as a compatibility problem when the relevant date ranges are supplied.

Given facts. For this problem, availability intervals are: Technology A: 1200–1600; Institution B: 1500–1900; Tool C: 1700–2000; Practice D: 1300–1450. A story is set in 1400 and mentions Technology A and Tool C together. Cross-domain check: on a field sketch, 1 cm represents 3 km and two checkpoints are 4 cm apart.

Rules. An item is temporally compatible with a year only if the year lies inside its stated availability interval. An anachronism is an item placed outside its given interval. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Which mentioned item, if any, is anachronistic in the story? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. Check 1400 against Technology A interval 1200–1600. 2. Check 1400 against Tool C interval 1700–2000. 3. Items outside their intervals: ['Tool C']. Cross-domain check: 4×3=12 km.

Answer. Anachronistic: Tool C. Cross-domain answer: 12 km.

Knowledge takeaway. The reasoning does not require memorizing invention dates if the intervals are part of the problem.

Formal model. Interval membership test for each object-event pair.

Problem 437. Temporal compatibility in a story set in 1550

Knowledge context. Anachronism can be tested as a compatibility problem when the relevant date ranges are supplied.

Given facts. For this problem, availability intervals are: Technology A: 1200–1600; Institution B: 1500–1900; Tool C: 1700–2000; Practice D: 1300–1450. A story is set in 1550 and mentions Technology A and Tool C together. Cross-domain check: a survey team starts work at 12:00 and works for 2 hour(s) without a break.

Rules. An item is temporally compatible with a year only if the year lies inside its stated availability interval. An anachronism is an item placed outside its given interval. For the cross-domain check, finishing time = starting time + duration.

Task. Which mentioned item, if any, is anachronistic in the story? Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. Check 1550 against Technology A interval 1200–1600. 2. Check 1550 against Tool C interval 1700–2000. 3. Items outside their intervals: ['Tool C']. Cross-domain check: 12+2=14, so the team finishes at 14:00.

Answer. Anachronistic: Tool C. Cross-domain answer: 14:00.

Knowledge takeaway. The reasoning does not require memorizing invention dates if the intervals are part of the problem.

Formal model. Interval membership test for each object-event pair.

Problem 438. Temporal compatibility in a story set in 1750

Knowledge context. Anachronism can be tested as a compatibility problem when the relevant date ranges are supplied.

Given facts. For this problem, availability intervals are: Technology A: 1200–1600; Institution B: 1500–1900; Tool C: 1700–2000; Practice D: 1300–1450. A story is set in 1750 and mentions Technology A and Tool C together. Cross-domain check: a local committee has 14 members, requires at least 8 present for quorum, and 14 are present.

Rules. An item is temporally compatible with a year only if the year lies inside its stated availability interval. An anachronism is an item placed outside its given interval. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. Which mentioned item, if any, is anachronistic in the story? Cross-domain check: is quorum met?

Step-by-step solution. 1. Check 1750 against Technology A interval 1200–1600. 2. Check 1750 against Tool C interval 1700–2000. 3. Items outside their intervals: ['Technology A']. Cross-domain check: compare 14 with 8; $14 \geq 8$ is true.

Answer. Anachronistic: Technology A. Cross-domain answer: quorum is met.

Knowledge takeaway. The reasoning does not require memorizing invention dates if the intervals are part of the problem.

Formal model. Interval membership test for each object-event pair.

Problem 439. Temporal compatibility in a story set in 1850

Knowledge context. Anachronism can be tested as a compatibility problem when the relevant date ranges are supplied.

Given facts. For this problem, availability intervals are: Technology A: 1200–1600; Institution B: 1500–1900; Tool C: 1700–2000; Practice D: 1300–1450. A story is set in 1850 and mentions Technology A and Tool C together. Cross-domain check: researchers collected 15 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. An item is temporally compatible with a year only if the year lies inside its stated availability interval. An anachronism is an item placed outside its given interval. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. Which mentioned item, if any, is anachronistic in the story? Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. Check 1850 against Technology A interval 1200–1600. 2. Check 1850 against Tool C interval 1700–2000. 3. Items outside their intervals: ['Technology A']. Cross-domain check: 15−4=11 independent reports remain.

Answer. Anachronistic: Technology A. Cross-domain answer: 11 independent reports.

Knowledge takeaway. The reasoning does not require memorizing invention dates if the intervals are part of the problem.

Formal model. Interval membership test for each object-event pair.

Problem 440. Temporal compatibility in a story set in 1500

Knowledge context. Anachronism can be tested as a compatibility problem when the relevant date ranges are supplied.

Given facts. For this problem, availability intervals are: Technology A: 1200–1600; Institution B: 1500–1900; Tool C: 1700–2000; Practice D: 1300–1450. A story is set in 1500 and mentions Technology A and Tool C together. Cross-domain check: on a field sketch, 1 cm represents 2 km and two checkpoints are 8 cm apart.

Rules. An item is temporally compatible with a year only if the year lies inside its stated availability interval. An anachronism is an item placed outside its given interval. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Which mentioned item, if any, is anachronistic in the story? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. Check 1500 against Technology A interval 1200–1600. 2. Check 1500 against Tool C interval 1700–2000. 3. Items outside their intervals: ['Tool C']. Cross-domain check: 8×2=16 km.

Answer. Anachronistic: Tool C. Cross-domain answer: 16 km.

Knowledge takeaway. The reasoning does not require memorizing invention dates if the intervals are part of the problem.

Formal model. Interval membership test for each object-event pair.

N14. Provenance and chain of custody

Problem 441. Provenance and chain of custody: case 1

Knowledge context. Provenance records where an object came from and how it moved between custodians.

Given facts. An artifact record states: found by Excavator on day 1; transferred to Museum A on day 2; sent from Museum A to Research Lab on day 5; returned by Research Lab to Museum B on day 8. Every transfer must name both sender and receiver. Cross-domain check: a survey team starts work at 8:00 and works for 2 hour(s) without a break. Mixed-domain verification: a regional archive has 46 map sheets and receives 2 additional sheets during the exercise.

Rules. A continuous chain of custody requires each handoff to connect the previous holder to the next. A missing sender or receiver creates a gap that limits certainty about continuous custody. For the cross-domain check, finishing time = starting time + duration. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. Is the stated chain of custody complete under the rule? If not, identify the gap. Cross-domain check: at what time does the team finish? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. Read the transfers in time order. 2. Check that every transfer links the previous holder to the next. 3. Every handoff names both sides and connects continuously. Cross-domain check: 8+2=10, so the team finishes at 10:00. Mixed-domain verification: 46+2=48.

Answer. The chain is complete under the stated rule. Cross-domain answer: 10:00. Mixed-domain answer: 48 map sheets.

Knowledge takeaway. A strong provenance claim depends on an unbroken sequence of documented handoffs.

Formal model. Time-ordered path through holders; verify continuity of sender/receiver pairs.

Problem 442. Provenance and chain of custody: case 2

Knowledge context. Provenance records where an object came from and how it moved between custodians.

Given facts. An artifact record states: found by Excavator on day 1; transferred to Museum A on day 2; sent from Museum A to Research Lab on day 5; returned by Research Lab to Museum B on day 8. Every transfer must name both sender and receiver. The record for day 5 names Research Lab as receiver but does not name the sender. Cross-domain check: a local committee has 9 members, requires at least 5 present for quorum, and 8 are present.

Rules. A continuous chain of custody requires each handoff to connect the previous holder to the next. A missing sender or receiver creates a gap that limits certainty about continuous custody. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. Is the stated chain of custody complete under the rule? If not, identify the gap. Cross-domain check: is quorum met?

Step-by-step solution. 1. Read the transfers in time order. 2. Check that every transfer links the previous holder to the next. 3. The day-5 record lacks a sender, so the chain has a gap. Cross-domain check: compare 8 with 5; $8 \geq 5$ is true.

Answer. The chain is incomplete because the day-5 sender is missing. Cross-domain answer: quorum is met.

Knowledge takeaway. A strong provenance claim depends on an unbroken sequence of documented handoffs.

Formal model. Time-ordered path through holders; verify continuity of sender/receiver pairs.

Problem 443. Provenance and chain of custody: case 3

Knowledge context. Provenance records where an object came from and how it moved between custodians.

Given facts. An artifact record states: found by Excavator on day 1; transferred to Museum A on day 2; sent from Museum A to Research Lab on day 5; returned by Research Lab to Museum B on day 8. Every transfer must name both sender and receiver. Cross-domain check: researchers collected 9 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. A continuous chain of custody requires each handoff to connect the previous holder to the next. A missing sender or receiver creates a gap that limits certainty about continuous custody. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. Is the stated chain of custody complete under the rule? If not, identify the gap. Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. Read the transfers in time order. 2. Check that every transfer links the previous holder to the next. 3. Every handoff names both sides and connects continuously. Cross-domain check: 9−4=5 independent reports remain.

Answer. The chain is complete under the stated rule. Cross-domain answer: 5 independent reports.

Knowledge takeaway. A strong provenance claim depends on an unbroken sequence of documented handoffs.

Formal model. Time-ordered path through holders; verify continuity of sender/receiver pairs.

Problem 444. Provenance and chain of custody: case 4

Knowledge context. Provenance records where an object came from and how it moved between custodians.

Given facts. An artifact record states: found by Excavator on day 1; transferred to Museum A on day 2; sent from Museum A to Research Lab on day 5; returned by Research Lab to Museum B on day 8. Every transfer must name both sender and receiver. The record for day 5 names Research Lab as receiver but does not name the sender. Cross-domain check: on a field sketch, 1 cm represents 6 km and two checkpoints are 5 cm apart.

Rules. A continuous chain of custody requires each handoff to connect the previous holder to the next. A missing sender or receiver creates a gap that limits certainty about continuous custody. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Is the stated chain of custody complete under the rule? If not, identify the gap. Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. Read the transfers in time order. 2. Check that every transfer links the previous holder to the next. 3. The day-5 record lacks a sender, so the chain has a gap. Cross-domain check: 5×6=30 km.

Answer. The chain is incomplete because the day-5 sender is missing. Cross-domain answer: 30 km.

Knowledge takeaway. A strong provenance claim depends on an unbroken sequence of documented handoffs.

Formal model. Time-ordered path through holders; verify continuity of sender/receiver pairs.

Problem 445. Provenance and chain of custody: case 5

Knowledge context. Provenance records where an object came from and how it moved between custodians.

Given facts. An artifact record states: found by Excavator on day 1; transferred to Museum A on day 2; sent from Museum A to Research Lab on day 5; returned by Research Lab to Museum B on day 8. Every transfer must name both sender and receiver. Cross-domain check: a survey team starts work at 12:00 and works for 2 hour(s) without a break.

Rules. A continuous chain of custody requires each handoff to connect the previous holder to the next. A missing sender or receiver creates a gap that limits certainty about continuous custody. For the cross-domain check, finishing time = starting time + duration.

Task. Is the stated chain of custody complete under the rule? If not, identify the gap. Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. Read the transfers in time order. 2. Check that every transfer links the previous holder to the next. 3. Every handoff names both sides and connects continuously. Cross-domain check: 12+2=14, so the team finishes at 14:00.

Answer. The chain is complete under the stated rule. Cross-domain answer: 14:00.

Knowledge takeaway. A strong provenance claim depends on an unbroken sequence of documented handoffs.

Formal model. Time-ordered path through holders; verify continuity of sender/receiver pairs.

N15. Perspective, interest, and source credibility

Problem 446. Perspective, interest, and source quality: case 1

Knowledge context. Sources have perspectives and interests. Credibility is not a simple label; it depends on the claim being evaluated and how the source could know it.

Given facts. Three sources describe a proposed market tax. Source A is a merchant who would pay the tax and argues it is harmful. Source B is a council report that estimates revenue and lists its calculation. Source C is a traveler repeating a rumor heard in another town. Cross-domain check: a local committee has 13 members, requires at least 7 present for quorum, and 11 are present.

Rules. Perspective and interest do not automatically make a source false. Evaluate relevance, access to information, explicit evidence, and independence. A source with an interest may still provide useful facts, but its claims should be checked. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. Which source is best positioned to estimate the amount of revenue, and which source should be treated most cautiously for that numerical claim? Cross-domain check: is quorum met?

Step-by-step solution. 1. Revenue estimation requires access to tax rates and expected transactions. 2. Source B explicitly provides a calculation and is directly about the policy. 3. Source C is a secondhand rumor from elsewhere and has the weakest access to the specific numerical question. 4. Source A has an interest, but interest alone is not enough to discard it. Cross-domain check: compare 11 with 7; $11 \geq 7$ is true.

Answer. Source B is best positioned for the revenue estimate; Source C should be treated most cautiously for that numerical claim. Cross-domain answer: quorum is met.

Knowledge takeaway. Bias is a reason to inspect evidence, not a magic switch that makes every statement true or false.

Formal model. Source-claim assessment using access, relevance, independence, and evidentiary support.

Problem 447. Perspective, interest, and source quality: case 2

Knowledge context. Sources have perspectives and interests. Credibility is not a simple label; it depends on the claim being evaluated and how the source could know it.

Given facts. Three sources describe a proposed market tax. Source A is a merchant who would pay the tax and argues it is harmful. Source B is a council report that estimates revenue and lists its calculation. Source C is a traveler repeating a rumor heard in another town. Cross-domain check: researchers collected 13 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. Perspective and interest do not automatically make a source false. Evaluate relevance, access to information, explicit evidence, and independence. A source with an interest may still provide useful facts, but its claims should be checked. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. Which source is best positioned to estimate the amount of revenue, and which source should be treated most cautiously for that numerical claim? Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. Revenue estimation requires access to tax rates and expected transactions. 2. Source B explicitly provides a calculation and is directly about the policy. 3. Source C is a secondhand rumor from elsewhere and has the weakest access to the specific numerical question. 4. Source A has an interest, but interest alone is not enough to discard it. Cross-domain check: 13−4=9 independent reports remain.

Answer. Source B is best positioned for the revenue estimate; Source C should be treated most cautiously for that numerical claim. Cross-domain answer: 9 independent reports.

Knowledge takeaway. Bias is a reason to inspect evidence, not a magic switch that makes every statement true or false.

Formal model. Source-claim assessment using access, relevance, independence, and evidentiary support.

Problem 448. Perspective, interest, and source quality: case 3

Knowledge context. Sources have perspectives and interests. Credibility is not a simple label; it depends on the claim being evaluated and how the source could know it.

Given facts. Three sources describe a proposed market tax. Source A is a merchant who would pay the tax and argues it is harmful. Source B is a council report that estimates revenue and lists its calculation. Source C is a traveler repeating a rumor heard in another town. Cross-domain check: on a field sketch, 1 cm represents 5 km and two checkpoints are 2 cm apart.

Rules. Perspective and interest do not automatically make a source false. Evaluate relevance, access to information, explicit evidence, and independence. A source with an interest may still provide useful facts, but its claims should be checked. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Which source is best positioned to estimate the amount of revenue, and which source should be treated most cautiously for that numerical claim? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. Revenue estimation requires access to tax rates and expected transactions. 2. Source B explicitly provides a calculation and is directly about the policy. 3. Source C is a secondhand rumor from elsewhere and has the weakest access to the specific numerical question. 4. Source A has an interest, but interest alone is not enough to discard it. Cross-domain check: 2×5=10 km.

Answer. Source B is best positioned for the revenue estimate; Source C should be treated most cautiously for that numerical claim. Cross-domain answer: 10 km.

Knowledge takeaway. Bias is a reason to inspect evidence, not a magic switch that makes every statement true or false.

Formal model. Source-claim assessment using access, relevance, independence, and evidentiary support.

Problem 449. Perspective, interest, and source quality: case 4

Knowledge context. Sources have perspectives and interests. Credibility is not a simple label; it depends on the claim being evaluated and how the source could know it.

Given facts. Three sources describe a proposed market tax. Source A is a merchant who would pay the tax and argues it is harmful. Source B is a council report that estimates revenue and lists its calculation. Source C is a traveler repeating a rumor heard in another town. Cross-domain check: a survey team starts work at 8:00 and works for 2 hour(s) without a break.

Rules. Perspective and interest do not automatically make a source false. Evaluate relevance, access to information, explicit evidence, and independence. A source with an interest may still provide useful facts, but its claims should be checked. For the cross-domain check, finishing time = starting time + duration.

Task. Which source is best positioned to estimate the amount of revenue, and which source should be treated most cautiously for that numerical claim? Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. Revenue estimation requires access to tax rates and expected transactions. 2. Source B explicitly provides a calculation and is directly about the policy. 3. Source C is a secondhand rumor from elsewhere and has the weakest access to the specific numerical question. 4. Source A has an interest, but interest alone is not enough to discard it. Cross-domain check: 8+2=10, so the team finishes at 10:00.

Answer. Source B is best positioned for the revenue estimate; Source C should be treated most cautiously for that numerical claim. Cross-domain answer: 10:00.

Knowledge takeaway. Bias is a reason to inspect evidence, not a magic switch that makes every statement true or false.

Formal model. Source-claim assessment using access, relevance, independence, and evidentiary support.

Problem 450. Perspective, interest, and source quality: case 5

Knowledge context. Sources have perspectives and interests. Credibility is not a simple label; it depends on the claim being evaluated and how the source could know it.

Given facts. Three sources describe a proposed market tax. Source A is a merchant who would pay the tax and argues it is harmful. Source B is a council report that estimates revenue and lists its calculation. Source C is a traveler repeating a rumor heard in another town. Cross-domain check: a local committee has 8 members, requires at least 5 present for quorum, and 8 are present. Mixed-domain verification: a regional archive has 47 map sheets and receives 1 additional sheets during the exercise.

Rules. Perspective and interest do not automatically make a source false. Evaluate relevance, access to information, explicit evidence, and independence. A source with an interest may still provide useful facts, but its claims should be checked. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. Which source is best positioned to estimate the amount of revenue, and which source should be treated most cautiously for that numerical claim? Cross-domain check: is quorum met? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. Revenue estimation requires access to tax rates and expected transactions. 2. Source B explicitly provides a calculation and is directly about the policy. 3. Source C is a secondhand rumor from elsewhere and has the weakest access to the specific numerical question. 4. Source A has an interest, but interest alone is not enough to discard it. Cross-domain check: compare 8 with 5; $8 \geq 5$ is true. Mixed-domain verification: 47+1=48.

Answer. Source B is best positioned for the revenue estimate; Source C should be treated most cautiously for that numerical claim. Cross-domain answer: quorum is met. Mixed-domain answer: 48 map sheets.

Knowledge takeaway. Bias is a reason to inspect evidence, not a magic switch that makes every statement true or false.

Formal model. Source-claim assessment using access, relevance, independence, and evidentiary support.

N16. Uncertain data, intervals, and limits

Problem 451. Reasoning with uncertain intervals: case 1

Knowledge context. Measurements and historical dates can be uncertain. An interval represents a range of possibilities rather than one exact number.

Given facts. Two measurements are uncertain intervals. A is between 8 and 12. B is between 12 and 16. A statement “A is definitely greater than B” is valid only if A’s minimum is greater than B’s maximum.

Rules. For a definite comparison under interval uncertainty, test the worst cases. Overlapping intervals usually prevent a strict guaranteed ordering.

Task. Can we conclude that A is definitely greater than B? What comparison is justified?

Step-by-step solution. 1. A minimum=8, maximum=12; B minimum=12, maximum=16. 2. Test A_min > B_max: 8>16 is False. 3. If false, at least one allowed pair of values defeats the “definitely greater” claim.

Answer. No. The intervals do not justify saying A is definitely greater than B.

Knowledge takeaway. Robust conclusions must remain true across all values allowed by the uncertainty.

Formal model. Interval arithmetic / worst-case comparison.

Problem 452. Reasoning with uncertain intervals: case 2

Knowledge context. Measurements and historical dates can be uncertain. An interval represents a range of possibilities rather than one exact number.

Given facts. Two measurements are uncertain intervals. A is between 9 and 13. B is between 12 and 16. A statement “A is definitely greater than B” is valid only if A’s minimum is greater than B’s maximum.

Rules. For a definite comparison under interval uncertainty, test the worst cases. Overlapping intervals usually prevent a strict guaranteed ordering.

Task. Can we conclude that A is definitely greater than B? What comparison is justified?

Step-by-step solution. 1. A minimum=9, maximum=13; B minimum=12, maximum=16. 2. Test A_min > B_max: 9>16 is False. 3. If false, at least one allowed pair of values defeats the “definitely greater” claim.

Answer. No. The intervals do not justify saying A is definitely greater than B.

Knowledge takeaway. Robust conclusions must remain true across all values allowed by the uncertainty.

Formal model. Interval arithmetic / worst-case comparison.

Problem 453. Reasoning with uncertain intervals: case 3

Knowledge context. Measurements and historical dates can be uncertain. An interval represents a range of possibilities rather than one exact number.

Given facts. Two measurements are uncertain intervals. A is between 10 and 14. B is between 12 and 16. A statement “A is definitely greater than B” is valid only if A’s minimum is greater than B’s maximum.

Rules. For a definite comparison under interval uncertainty, test the worst cases. Overlapping intervals usually prevent a strict guaranteed ordering.

Task. Can we conclude that A is definitely greater than B? What comparison is justified?

Step-by-step solution. 1. A minimum=10, maximum=14; B minimum=12, maximum=16. 2. Test A_min > B_max: 10>16 is False. 3. If false, at least one allowed pair of values defeats the “definitely greater” claim.

Answer. No. The intervals do not justify saying A is definitely greater than B.

Knowledge takeaway. Robust conclusions must remain true across all values allowed by the uncertainty.

Formal model. Interval arithmetic / worst-case comparison.

Problem 454. Reasoning with uncertain intervals: case 4

Knowledge context. Measurements and historical dates can be uncertain. An interval represents a range of possibilities rather than one exact number.

Given facts. Two measurements are uncertain intervals. A is between 11 and 15. B is between 12 and 16. A statement “A is definitely greater than B” is valid only if A’s minimum is greater than B’s maximum.

Rules. For a definite comparison under interval uncertainty, test the worst cases. Overlapping intervals usually prevent a strict guaranteed ordering.

Task. Can we conclude that A is definitely greater than B? What comparison is justified?

Step-by-step solution. 1. A minimum=11, maximum=15; B minimum=12, maximum=16. 2. Test A_min > B_max: 11>16 is False. 3. If false, at least one allowed pair of values defeats the “definitely greater” claim.

Answer. No. The intervals do not justify saying A is definitely greater than B.

Knowledge takeaway. Robust conclusions must remain true across all values allowed by the uncertainty.

Formal model. Interval arithmetic / worst-case comparison.

Problem 455. Reasoning with uncertain intervals: case 5

Knowledge context. Measurements and historical dates can be uncertain. An interval represents a range of possibilities rather than one exact number.

Given facts. Two measurements are uncertain intervals. A is between 12 and 16. B is between 12 and 16. A statement “A is definitely greater than B” is valid only if A’s minimum is greater than B’s maximum.

Rules. For a definite comparison under interval uncertainty, test the worst cases. Overlapping intervals usually prevent a strict guaranteed ordering.

Task. Can we conclude that A is definitely greater than B? What comparison is justified?

Step-by-step solution. 1. A minimum=12, maximum=16; B minimum=12, maximum=16. 2. Test A_min > B_max: 12>16 is False. 3. If false, at least one allowed pair of values defeats the “definitely greater” claim.

Answer. No. The intervals do not justify saying A is definitely greater than B.

Knowledge takeaway. Robust conclusions must remain true across all values allowed by the uncertainty.

Formal model. Interval arithmetic / worst-case comparison.

N17. Counterfactual reasoning

Problem 456. Counterfactual intervention: close the gate

Knowledge context. Counterfactual reasoning asks what the model predicts if one condition were changed while the rest of the causal structure is held fixed.

Given facts. Causal rules for a simplified town model: heavy rain → high river; high river AND unclosed gate → flooded square; flooded square → market closes. Actual facts: heavy rain occurred and the gate was not closed. Cross-domain check: on a field sketch, 1 cm represents 3 km and two checkpoints are 3 cm apart.

Rules. A counterfactual intervention changes one specified fact and then recomputes consequences from the rules. Keep all other stated rules the same unless the intervention changes them. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Under the intervention “close the gate”, what changes in the consequences? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. Actual chain gives high river, then flood, then market closure. 2. Intervene: set gate=closed. 3. The flood rule requires an unclosed gate, so flooding no longer follows. 4. Without flooding, market closure does not follow. Cross-domain check: 3×3=9 km.

Answer. With the gate closed, the model no longer implies flooding or market closure. Cross-domain answer: 9 km.

Knowledge takeaway. A counterfactual is not ordinary storytelling; it is a controlled change followed by recomputation.

Formal model. Structural causal rules with a do-style intervention on one variable.

Problem 457. Counterfactual intervention: remove heavy rain

Knowledge context. Counterfactual reasoning asks what the model predicts if one condition were changed while the rest of the causal structure is held fixed.

Given facts. Causal rules for a simplified town model: heavy rain → high river; high river AND unclosed gate → flooded square; flooded square → market closes. Actual facts: heavy rain occurred and the gate was not closed. Cross-domain check: a survey team starts work at 8:00 and works for 2 hour(s) without a break.

Rules. A counterfactual intervention changes one specified fact and then recomputes consequences from the rules. Keep all other stated rules the same unless the intervention changes them. For the cross-domain check, finishing time = starting time + duration.

Task. Under the intervention “remove heavy rain”, what changes in the consequences? Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. Intervene: heavy rain=false. 2. Then high river is not generated by the first rule. 3. Without high river, flooding and market closure are not generated. Cross-domain check: 8+2=10, so the team finishes at 10:00.

Answer. Without heavy rain, the model generates neither high river, flood, nor market closure. Cross-domain answer: 10:00.

Knowledge takeaway. A counterfactual is not ordinary storytelling; it is a controlled change followed by recomputation.

Formal model. Structural causal rules with a do-style intervention on one variable.

Problem 458. Counterfactual intervention: keep gate open

Knowledge context. Counterfactual reasoning asks what the model predicts if one condition were changed while the rest of the causal structure is held fixed.

Given facts. Causal rules for a simplified town model: heavy rain → high river; high river AND unclosed gate → flooded square; flooded square → market closes. Actual facts: heavy rain occurred and the gate was not closed. Cross-domain check: a local committee has 16 members, requires at least 9 present for quorum, and 14 are present.

Rules. A counterfactual intervention changes one specified fact and then recomputes consequences from the rules. Keep all other stated rules the same unless the intervention changes them. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. Under the intervention “keep gate open”, what changes in the consequences? Cross-domain check: is quorum met?

Step-by-step solution. 1. Gate already remains open. 2. Heavy rain gives high river. 3. High river plus open gate gives flood, then market closure. Cross-domain check: compare 14 with 9; $14 \geq 9$ is true.

Answer. The original chain remains: flood and market closure occur. Cross-domain answer: quorum is met.

Knowledge takeaway. A counterfactual is not ordinary storytelling; it is a controlled change followed by recomputation.

Formal model. Structural causal rules with a do-style intervention on one variable.

Problem 459. Counterfactual intervention: prevent high river

Knowledge context. Counterfactual reasoning asks what the model predicts if one condition were changed while the rest of the causal structure is held fixed.

Given facts. Causal rules for a simplified town model: heavy rain → high river; high river AND unclosed gate → flooded square; flooded square → market closes. Actual facts: heavy rain occurred and the gate was not closed. Cross-domain check: researchers collected 15 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. A counterfactual intervention changes one specified fact and then recomputes consequences from the rules. Keep all other stated rules the same unless the intervention changes them. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. Under the intervention “prevent high river”, what changes in the consequences? Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. Intervene directly: high river=false. 2. Even with rain, the counterfactual intervention fixes high river as prevented for this question. 3. Flood requires high river, so it does not follow. Cross-domain check: 15−4=11 independent reports remain.

Answer. Preventing the high river blocks the flood and market closure in this model. Cross-domain answer: 11 independent reports.

Knowledge takeaway. A counterfactual is not ordinary storytelling; it is a controlled change followed by recomputation.

Formal model. Structural causal rules with a do-style intervention on one variable.

Problem 460. Counterfactual intervention: keep market open

Knowledge context. Counterfactual reasoning asks what the model predicts if one condition were changed while the rest of the causal structure is held fixed.

Given facts. Causal rules for a simplified town model: heavy rain → high river; high river AND unclosed gate → flooded square; flooded square → market closes. Actual facts: heavy rain occurred and the gate was not closed. Cross-domain check: on a field sketch, 1 cm represents 2 km and two checkpoints are 7 cm apart.

Rules. A counterfactual intervention changes one specified fact and then recomputes consequences from the rules. Keep all other stated rules the same unless the intervention changes them. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Under the intervention “keep market open”, what changes in the consequences? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. The proposed intervention changes the market outcome directly, not the upstream causes. 2. Rain still gives high river and flood. 3. Setting market=open does not erase the flood unless a rule says so. Cross-domain check: 7×2=14 km.

Answer. The square still floods; the market is held open only by the direct intervention. Cross-domain answer: 14 km.

Knowledge takeaway. A counterfactual is not ordinary storytelling; it is a controlled change followed by recomputation.

Formal model. Structural causal rules with a do-style intervention on one variable.

N18. Public goods and common contributions

Problem 461. Funding a shared public project: case 1

Knowledge context. Public goods can benefit many people at once, which creates coordination problems about contributions.

Given facts. A community project is built only if total voluntary contributions reach 12 tokens. Contributions from 6 households are [1, 2, 3, 4, 1, 2]. Everyone can use the project if it is built, including households that contribute zero.

Rules. Add contributions and compare with the threshold. A public-good threshold concerns whether the project is produced; it does not by itself decide whether every contribution pattern is fair.

Task. Is the project built? How many additional tokens, if any, are needed?

Step-by-step solution. 1. Total contribution=1+2+3+4+1+2=13. 2. Compare 13 with threshold 12. 3. Additional amount needed=max(0,12−13)=0.

Answer. The project is built. Additional tokens needed: 0.

Knowledge takeaway. Threshold provision and fairness are separate questions; first determine whether enough resources exist.

Formal model. Sum contributions; compare with provision threshold; compute shortfall.

Problem 462. Funding a shared public project: case 2

Knowledge context. Public goods can benefit many people at once, which creates coordination problems about contributions.

Given facts. A community project is built only if total voluntary contributions reach 12 tokens. Contributions from 6 households are [2, 3, 4, 1, 2, 3]. Everyone can use the project if it is built, including households that contribute zero.

Rules. Add contributions and compare with the threshold. A public-good threshold concerns whether the project is produced; it does not by itself decide whether every contribution pattern is fair.

Task. Is the project built? How many additional tokens, if any, are needed?

Step-by-step solution. 1. Total contribution=2+3+4+1+2+3=15. 2. Compare 15 with threshold 12. 3. Additional amount needed=max(0,12−15)=0.

Answer. The project is built. Additional tokens needed: 0.

Knowledge takeaway. Threshold provision and fairness are separate questions; first determine whether enough resources exist.

Formal model. Sum contributions; compare with provision threshold; compute shortfall.

Problem 463. Funding a shared public project: case 3

Knowledge context. Public goods can benefit many people at once, which creates coordination problems about contributions.

Given facts. A community project is built only if total voluntary contributions reach 12 tokens. Contributions from 6 households are [3, 4, 1, 2, 3, 4]. Everyone can use the project if it is built, including households that contribute zero.

Rules. Add contributions and compare with the threshold. A public-good threshold concerns whether the project is produced; it does not by itself decide whether every contribution pattern is fair.

Task. Is the project built? How many additional tokens, if any, are needed?

Step-by-step solution. 1. Total contribution=3+4+1+2+3+4=17. 2. Compare 17 with threshold 12. 3. Additional amount needed=max(0,12−17)=0.

Answer. The project is built. Additional tokens needed: 0.

Knowledge takeaway. Threshold provision and fairness are separate questions; first determine whether enough resources exist.

Formal model. Sum contributions; compare with provision threshold; compute shortfall.

Problem 464. Funding a shared public project: case 4

Knowledge context. Public goods can benefit many people at once, which creates coordination problems about contributions.

Given facts. A community project is built only if total voluntary contributions reach 12 tokens. Contributions from 6 households are [4, 1, 2, 3, 4, 1]. Everyone can use the project if it is built, including households that contribute zero.

Rules. Add contributions and compare with the threshold. A public-good threshold concerns whether the project is produced; it does not by itself decide whether every contribution pattern is fair.

Task. Is the project built? How many additional tokens, if any, are needed?

Step-by-step solution. 1. Total contribution=4+1+2+3+4+1=15. 2. Compare 15 with threshold 12. 3. Additional amount needed=max(0,12−15)=0.

Answer. The project is built. Additional tokens needed: 0.

Knowledge takeaway. Threshold provision and fairness are separate questions; first determine whether enough resources exist.

Formal model. Sum contributions; compare with provision threshold; compute shortfall.

Problem 465. Funding a shared public project: case 5

Knowledge context. Public goods can benefit many people at once, which creates coordination problems about contributions.

Given facts. A community project is built only if total voluntary contributions reach 12 tokens. Contributions from 6 households are [1, 2, 3, 4, 1, 2]. Everyone can use the project if it is built, including households that contribute zero. Cross-domain check: a survey team starts work at 8:00 and works for 2 hour(s) without a break.

Rules. Add contributions and compare with the threshold. A public-good threshold concerns whether the project is produced; it does not by itself decide whether every contribution pattern is fair. For the cross-domain check, finishing time = starting time + duration.

Task. Is the project built? How many additional tokens, if any, are needed? Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. Total contribution=1+2+3+4+1+2=13. 2. Compare 13 with threshold 12. 3. Additional amount needed=max(0,12−13)=0. Cross-domain check: 8+2=10, so the team finishes at 10:00.

Answer. The project is built. Additional tokens needed: 0. Cross-domain answer: 10:00.

Knowledge takeaway. Threshold provision and fairness are separate questions; first determine whether enough resources exist.

Formal model. Sum contributions; compare with provision threshold; compute shortfall.

N19. Common resources and avoiding depletion

Problem 466. Using a shared renewable resource: case 1

Knowledge context. A common resource can be overused when many individually reasonable withdrawals add up to more than regeneration.

Given facts. A shared fishery has 80 fish at the start of the period and regenerates 10. There are 5 users, each planning to take 3. For this simplified exercise, the stock should not decline below its starting level if use is to be called non-depleting.

Rules. End stock = start + regeneration − total harvest. Total harvest is number of users × harvest per user.

Task. What is the end stock? Is the planned use non-depleting under the stated definition? What equal per-user harvest would exactly match regeneration?

Step-by-step solution. 1. Total harvest=5×3=15. 2. End stock=80+10−15=75. 3. To match regeneration exactly, equal harvest per user would be 10÷5=2.00.

Answer. End stock 75; depleting. Equal regeneration-matching share: 2.00 each.

Knowledge takeaway. The shared-resource problem appears at the total level, so individual plans must be aggregated.

Formal model. Commons stock equation plus equal-share benchmark.

Problem 467. Using a shared renewable resource: case 2

Knowledge context. A common resource can be overused when many individually reasonable withdrawals add up to more than regeneration.

Given facts. A shared fishery has 80 fish at the start of the period and regenerates 11. There are 5 users, each planning to take 4. For this simplified exercise, the stock should not decline below its starting level if use is to be called non-depleting.

Rules. End stock = start + regeneration − total harvest. Total harvest is number of users × harvest per user.

Task. What is the end stock? Is the planned use non-depleting under the stated definition? What equal per-user harvest would exactly match regeneration?

Step-by-step solution. 1. Total harvest=5×4=20. 2. End stock=80+11−20=71. 3. To match regeneration exactly, equal harvest per user would be 11÷5=2.20.

Answer. End stock 71; depleting. Equal regeneration-matching share: 2.20 each.

Knowledge takeaway. The shared-resource problem appears at the total level, so individual plans must be aggregated.

Formal model. Commons stock equation plus equal-share benchmark.

Problem 468. Using a shared renewable resource: case 3

Knowledge context. A common resource can be overused when many individually reasonable withdrawals add up to more than regeneration.

Given facts. A shared fishery has 80 fish at the start of the period and regenerates 12. There are 5 users, each planning to take 5. For this simplified exercise, the stock should not decline below its starting level if use is to be called non-depleting.

Rules. End stock = start + regeneration − total harvest. Total harvest is number of users × harvest per user.

Task. What is the end stock? Is the planned use non-depleting under the stated definition? What equal per-user harvest would exactly match regeneration?

Step-by-step solution. 1. Total harvest=5×5=25. 2. End stock=80+12−25=67. 3. To match regeneration exactly, equal harvest per user would be 12÷5=2.40.

Answer. End stock 67; depleting. Equal regeneration-matching share: 2.40 each.

Knowledge takeaway. The shared-resource problem appears at the total level, so individual plans must be aggregated.

Formal model. Commons stock equation plus equal-share benchmark.

Problem 469. Using a shared renewable resource: case 4

Knowledge context. A common resource can be overused when many individually reasonable withdrawals add up to more than regeneration.

Given facts. A shared fishery has 80 fish at the start of the period and regenerates 13. There are 5 users, each planning to take 6. For this simplified exercise, the stock should not decline below its starting level if use is to be called non-depleting.

Rules. End stock = start + regeneration − total harvest. Total harvest is number of users × harvest per user.

Task. What is the end stock? Is the planned use non-depleting under the stated definition? What equal per-user harvest would exactly match regeneration?

Step-by-step solution. 1. Total harvest=5×6=30. 2. End stock=80+13−30=63. 3. To match regeneration exactly, equal harvest per user would be 13÷5=2.60.

Answer. End stock 63; depleting. Equal regeneration-matching share: 2.60 each.

Knowledge takeaway. The shared-resource problem appears at the total level, so individual plans must be aggregated.

Formal model. Commons stock equation plus equal-share benchmark.

Problem 470. Using a shared renewable resource: case 5

Knowledge context. A common resource can be overused when many individually reasonable withdrawals add up to more than regeneration.

Given facts. A shared fishery has 80 fish at the start of the period and regenerates 14. There are 5 users, each planning to take 7. For this simplified exercise, the stock should not decline below its starting level if use is to be called non-depleting.

Rules. End stock = start + regeneration − total harvest. Total harvest is number of users × harvest per user.

Task. What is the end stock? Is the planned use non-depleting under the stated definition? What equal per-user harvest would exactly match regeneration?

Step-by-step solution. 1. Total harvest=5×7=35. 2. End stock=80+14−35=59. 3. To match regeneration exactly, equal harvest per user would be 14÷5=2.80.

Answer. End stock 59; depleting. Equal regeneration-matching share: 2.80 each.

Knowledge takeaway. The shared-resource problem appears at the total level, so individual plans must be aggregated.

Formal model. Commons stock equation plus equal-share benchmark.

N20. Fairness: equal, proportional, and need-based

Problem 471. Fair allocation by the rule: equal

Knowledge context. Fairness is not one formula. Equal shares, proportional shares, and need-based shares embody different principles.

Given facts. A relief fund has 40 units for three districts A, B, C. Populations are [10, 20, 30]; need scores are [1, 4, 3]. Three fairness principles are defined: equal = same amount each; population-proportional = shares in ratio 10:20:30; need-proportional = shares in ratio 1:4:3.

Rules. Different fairness principles answer different questions. Apply the principle named in the task; do not mix them unless asked to compare.

Task. Allocate the fund equally.

Step-by-step solution. 1. Equal share=40÷3=13.33. 2. Allocation result: A=13.33, B=13.33, C=13.33.

Answer. A=13.33, B=13.33, C=13.33.

Knowledge takeaway. A sound argument states the fairness rule first and then applies it consistently.

Formal model. Allocation function parameterized by a fairness principle and weights.

Problem 472. Fair allocation by the rule: proportional to population

Knowledge context. Fairness is not one formula. Equal shares, proportional shares, and need-based shares embody different principles.

Given facts. A relief fund has 40 units for three districts A, B, C. Populations are [10, 20, 30]; need scores are [2, 4, 4]. Three fairness principles are defined: equal = same amount each; population-proportional = shares in ratio 10:20:30; need-proportional = shares in ratio 2:4:4.

Rules. Different fairness principles answer different questions. Apply the principle named in the task; do not mix them unless asked to compare.

Task. Allocate in proportion to population.

Step-by-step solution. 1. Total population weight=60. 2. Multiply the fund by each population fraction. 3. Allocation result: A=6.67, B=13.33, C=20.00.

Answer. A=6.67, B=13.33, C=20.00.

Knowledge takeaway. A sound argument states the fairness rule first and then applies it consistently.

Formal model. Allocation function parameterized by a fairness principle and weights.

Problem 473. Fair allocation by the rule: proportional to need

Knowledge context. Fairness is not one formula. Equal shares, proportional shares, and need-based shares embody different principles.

Given facts. A relief fund has 40 units for three districts A, B, C. Populations are [10, 20, 30]; need scores are [3, 4, 5]. Three fairness principles are defined: equal = same amount each; population-proportional = shares in ratio 10:20:30; need-proportional = shares in ratio 3:4:5.

Rules. Different fairness principles answer different questions. Apply the principle named in the task; do not mix them unless asked to compare.

Task. Allocate in proportion to need score.

Step-by-step solution. 1. Total need weight=12. 2. Multiply the fund by each need fraction. 3. Allocation result: A=10.00, B=13.33, C=16.67.

Answer. A=10.00, B=13.33, C=16.67.

Knowledge takeaway. A sound argument states the fairness rule first and then applies it consistently.

Formal model. Allocation function parameterized by a fairness principle and weights.

Problem 474. Fair allocation by the rule: minimum guarantee then equal remainder

Knowledge context. Fairness is not one formula. Equal shares, proportional shares, and need-based shares embody different principles.

Given facts. A relief fund has 40 units for three districts A, B, C. Populations are [10, 20, 30]; need scores are [4, 4, 6]. Three fairness principles are defined: equal = same amount each; population-proportional = shares in ratio 10:20:30; need-proportional = shares in ratio 4:4:6.

Rules. Different fairness principles answer different questions. Apply the principle named in the task; do not mix them unless asked to compare.

Task. Give each district a 5-unit minimum guarantee, then divide the remainder equally.

Step-by-step solution. 1. Minimum guarantees use 15 units. 2. Remainder=40-15=25. 3. Each gets an additional 8.33. 4. Allocation result: A=13.33, B=13.33, C=13.33.

Answer. A=13.33, B=13.33, C=13.33.

Knowledge takeaway. A sound argument states the fairness rule first and then applies it consistently.

Formal model. Allocation function parameterized by a fairness principle and weights.

Problem 475. Fair allocation by the rule: compare methods

Knowledge context. Fairness is not one formula. Equal shares, proportional shares, and need-based shares embody different principles.

Given facts. A relief fund has 40 units for three districts A, B, C. Populations are [10, 20, 30]; need scores are [5, 4, 7]. Three fairness principles are defined: equal = same amount each; population-proportional = shares in ratio 10:20:30; need-proportional = shares in ratio 5:4:7.

Rules. Different fairness principles answer different questions. Apply the principle named in the task; do not mix them unless asked to compare.

Task. Compare equal allocation with need-proportional allocation: which gives more to the highest-need district?

Step-by-step solution. 1. Equal allocation gives C 13.33. 2. Need-proportional allocation gives C 17.50. 3. Compare those amounts. 4. Allocation result: A=12.50, B=10.00, C=17.50.

Answer. A=12.50, B=10.00, C=17.50.

Knowledge takeaway. A sound argument states the fairness rule first and then applies it consistently.

Formal model. Allocation function parameterized by a fairness principle and weights.

N21. Negotiation, coalitions, and agreement

Problem 476. Coalition formation: case 1

Knowledge context. In representative bodies, groups may need to cooperate to reach a decision threshold.

Given facts. Three groups have seats A=6, B=3, C=2. A governing coalition needs at least 6 seats. Any two groups may cooperate.

Rules. A coalition is winning if the sum of its members' seats reaches the threshold. Minimal winning means it is winning but removing any member makes it lose.

Task. Which coalitions are minimal winning coalitions?

Step-by-step solution. 1. Majority threshold=6. 2. Add seats for each possible coalition. 3. Minimal winning coalitions: [('A', 6)].

Answer. A with 6 seats.

Knowledge takeaway. A coalition can be winning without being minimal; removing redundant members tests how necessary each partner is.

Formal model. Weighted coalition game; enumerate subsets, test threshold and minimality.

Problem 477. Coalition formation: case 2

Knowledge context. In representative bodies, groups may need to cooperate to reach a decision threshold.

Given facts. Three groups have seats A=6, B=4, C=2. A governing coalition needs at least 7 seats. Any two groups may cooperate.

Rules. A coalition is winning if the sum of its members' seats reaches the threshold. Minimal winning means it is winning but removing any member makes it lose.

Task. Which coalitions are minimal winning coalitions?

Step-by-step solution. 1. Majority threshold=7. 2. Add seats for each possible coalition. 3. Minimal winning coalitions: [('AB', 10), ('AC', 8)].

Answer. AB with 10 seats; AC with 8 seats.

Knowledge takeaway. A coalition can be winning without being minimal; removing redundant members tests how necessary each partner is.

Formal model. Weighted coalition game; enumerate subsets, test threshold and minimality.

Problem 478. Coalition formation: case 3

Knowledge context. In representative bodies, groups may need to cooperate to reach a decision threshold.

Given facts. Three groups have seats A=6, B=5, C=2. A governing coalition needs at least 7 seats. Any two groups may cooperate.

Rules. A coalition is winning if the sum of its members' seats reaches the threshold. Minimal winning means it is winning but removing any member makes it lose.

Task. Which coalitions are minimal winning coalitions?

Step-by-step solution. 1. Majority threshold=7. 2. Add seats for each possible coalition. 3. Minimal winning coalitions: [('AB', 11), ('AC', 8), ('BC', 7)].

Answer. AB with 11 seats; AC with 8 seats; BC with 7 seats.

Knowledge takeaway. A coalition can be winning without being minimal; removing redundant members tests how necessary each partner is.

Formal model. Weighted coalition game; enumerate subsets, test threshold and minimality.

Problem 479. Coalition formation: case 4

Knowledge context. In representative bodies, groups may need to cooperate to reach a decision threshold.

Given facts. Three groups have seats A=6, B=6, C=2. A governing coalition needs at least 8 seats. Any two groups may cooperate.

Rules. A coalition is winning if the sum of its members' seats reaches the threshold. Minimal winning means it is winning but removing any member makes it lose.

Task. Which coalitions are minimal winning coalitions?

Step-by-step solution. 1. Majority threshold=8. 2. Add seats for each possible coalition. 3. Minimal winning coalitions: [('AB', 12), ('AC', 8), ('BC', 8)].

Answer. AB with 12 seats; AC with 8 seats; BC with 8 seats.

Knowledge takeaway. A coalition can be winning without being minimal; removing redundant members tests how necessary each partner is.

Formal model. Weighted coalition game; enumerate subsets, test threshold and minimality.

Problem 480. Coalition formation: case 5

Knowledge context. In representative bodies, groups may need to cooperate to reach a decision threshold.

Given facts. Three groups have seats A=6, B=7, C=2. A governing coalition needs at least 8 seats. Any two groups may cooperate.

Rules. A coalition is winning if the sum of its members' seats reaches the threshold. Minimal winning means it is winning but removing any member makes it lose.

Task. Which coalitions are minimal winning coalitions?

Step-by-step solution. 1. Majority threshold=8. 2. Add seats for each possible coalition. 3. Minimal winning coalitions: [('AB', 13), ('AC', 8), ('BC', 9)].

Answer. AB with 13 seats; AC with 8 seats; BC with 9 seats.

Knowledge takeaway. A coalition can be winning without being minimal; removing redundant members tests how necessary each partner is.

Formal model. Weighted coalition game; enumerate subsets, test threshold and minimality.

N22. Due process, evidence, and right of reply

Problem 481. Procedure, evidence, and a right to respond: case 1

Knowledge context. Fair procedures often combine evidentiary standards with an opportunity for the affected person to answer the case.

Given facts. A school rule says a student may receive a serious penalty only if: (1) at least two independent pieces of evidence support the allegation, and (2) the student is given a chance to respond before the final decision. In a case, one camera record and one independent witness support the allegation, but the student was not allowed to respond. Cross-domain check: a survey team starts work at 8:00 and works for 2 hour(s) without a break. Mixed-domain verification: a regional archive has 50 map sheets and receives 2 additional sheets during the exercise.

Rules. Independence concerns distinct information sources, not merely the number of people repeating one source. Every required procedural condition must be satisfied. For the cross-domain check, finishing time = starting time + duration. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. May the serious penalty be imposed under the stated procedure? Identify the condition that succeeds or fails. Cross-domain check: at what time does the team finish? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. Camera and witness are independent, so the evidence condition is met. 2. The student was not given a chance to respond. 3. The procedure requires both conditions, so the penalty is not authorized. Cross-domain check: 8+2=10, so the team finishes at 10:00. Mixed-domain verification: 50+2=52.

Answer. No. The right-to-respond condition is not met. Cross-domain answer: 10:00. Mixed-domain answer: 52 map sheets.

Knowledge takeaway. A correct outcome does not excuse skipping a required procedure when the rule explicitly makes procedure part of validity.

Formal model. Conjunctive procedural predicate over evidence independence/count and response opportunity.

Problem 482. Procedure, evidence, and a right to respond: case 2

Knowledge context. Fair procedures often combine evidentiary standards with an opportunity for the affected person to answer the case.

Given facts. A school rule says a serious penalty requires two independent pieces of evidence and a chance to respond. In this case there are two friends repeating the same rumor from one original speaker, plus a chance to respond. Cross-domain check: a local committee has 13 members, requires at least 7 present for quorum, and 11 are present.

Rules. Independence concerns distinct information sources, not merely the number of people repeating one source. Every required procedural condition must be satisfied. For the cross-domain check, quorum exists when present members ≥ the stated threshold.

Task. May the serious penalty be imposed under the stated procedure? Identify the condition that succeeds or fails. Cross-domain check: is quorum met?

Step-by-step solution. 1. The two friends trace to the same original rumor, so they are not two independent pieces under the rule. 2. The response condition is met. 3. Because the evidence condition fails, the serious penalty is not authorized. Cross-domain check: compare 11 with 7; 11 ≥ 7 is true.

Answer. No. The evidence requirement is not met. Cross-domain answer: quorum is met.

Knowledge takeaway. A correct outcome does not excuse skipping a required procedure when the rule explicitly makes procedure part of validity.

Formal model. Conjunctive procedural predicate over evidence independence/count and response opportunity.

Problem 483. Procedure, evidence, and a right to respond: case 3

Knowledge context. Fair procedures often combine evidentiary standards with an opportunity for the affected person to answer the case.

Given facts. A school rule says a student may receive a serious penalty only if: (1) at least two independent pieces of evidence support the allegation, and (2) the student is given a chance to respond before the final decision. In a case, one camera record and one independent witness support the allegation, but the student was not allowed to respond. Cross-domain check: researchers collected 9 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. Independence concerns distinct information sources, not merely the number of people repeating one source. Every required procedural condition must be satisfied. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. May the serious penalty be imposed under the stated procedure? Identify the condition that succeeds or fails. Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. Camera and witness are independent, so the evidence condition is met. 2. The student was not given a chance to respond. 3. The procedure requires both conditions, so the penalty is not authorized. Cross-domain check: 9−4=5 independent reports remain.

Answer. No. The right-to-respond condition is not met. Cross-domain answer: 5 independent reports.

Knowledge takeaway. A correct outcome does not excuse skipping a required procedure when the rule explicitly makes procedure part of validity.

Formal model. Conjunctive procedural predicate over evidence independence/count and response opportunity.

Problem 484. Procedure, evidence, and a right to respond: case 4

Knowledge context. Fair procedures often combine evidentiary standards with an opportunity for the affected person to answer the case.

Given facts. A school rule says a serious penalty requires two independent pieces of evidence and a chance to respond. In this case there are two friends repeating the same rumor from one original speaker, plus a chance to respond. Cross-domain check: on a field sketch, 1 cm represents 6 km and two checkpoints are 3 cm apart.

Rules. Independence concerns distinct information sources, not merely the number of people repeating one source. Every required procedural condition must be satisfied. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. May the serious penalty be imposed under the stated procedure? Identify the condition that succeeds or fails. Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. The two friends trace to the same original rumor, so they are not two independent pieces under the rule. 2. The response condition is met. 3. Because the evidence condition fails, the serious penalty is not authorized. Cross-domain check: 3×6=18 km.

Answer. No. The evidence requirement is not met. Cross-domain answer: 18 km.

Knowledge takeaway. A correct outcome does not excuse skipping a required procedure when the rule explicitly makes procedure part of validity.

Formal model. Conjunctive procedural predicate over evidence independence/count and response opportunity.

Problem 485. Procedure, evidence, and a right to respond: case 5

Knowledge context. Fair procedures often combine evidentiary standards with an opportunity for the affected person to answer the case.

Given facts. A school rule says a student may receive a serious penalty only if: (1) at least two independent pieces of evidence support the allegation, and (2) the student is given a chance to respond before the final decision. In a case, one camera record and one independent witness support the allegation, but the student was not allowed to respond. Cross-domain check: a survey team starts work at 12:00 and works for 2 hour(s) without a break.

Rules. Independence concerns distinct information sources, not merely the number of people repeating one source. Every required procedural condition must be satisfied. For the cross-domain check, finishing time = starting time + duration.

Task. May the serious penalty be imposed under the stated procedure? Identify the condition that succeeds or fails. Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. Camera and witness are independent, so the evidence condition is met. 2. The student was not given a chance to respond. 3. The procedure requires both conditions, so the penalty is not authorized. Cross-domain check: 12+2=14, so the team finishes at 14:00.

Answer. No. The right-to-respond condition is not met. Cross-domain answer: 14:00.

Knowledge takeaway. A correct outcome does not excuse skipping a required procedure when the rule explicitly makes procedure part of validity.

Formal model. Conjunctive procedural predicate over evidence independence/count and response opportunity.

N23. Samples, surveys, and public opinion

Problem 486. Sampling and opinion: case 1

Knowledge context. Surveys use samples to learn about larger populations. The selection process matters as much as the number of answers.

Given facts. A town wants to estimate how many residents use buses. It surveys only people waiting at the bus station. Cross-domain check: a local committee has 8 members, requires at least 5 present for quorum, and 8 are present.

Rules. A sample is more representative when selection does not systematically favor people connected to the answer. Random or broad systematic sampling can reduce selection bias; voluntary or topic-specific locations may create bias. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. Is the sampling method likely to be biased for the stated question? Explain the direction of the possible bias. Cross-domain check: is quorum met?

Step-by-step solution. 1. Identify the target population. 2. Compare the selection method with that target. 3. People already at the bus station are more likely than average to use buses. 4. Therefore the method is likely biased. Cross-domain check: compare 8 with 5; $8 \geq 5$ is true.

Answer. Likely biased. People already at the bus station are more likely than average to use buses. Cross-domain answer: quorum is met.

Knowledge takeaway. A large biased sample can still give a misleading estimate; representativeness is a reasoning question about selection.

Formal model. Sampling-frame analysis: compare inclusion mechanism with target population and likely outcome-related selection.

Problem 487. Sampling and opinion: case 2

Knowledge context. Surveys use samples to learn about larger populations. The selection process matters as much as the number of answers.

Given facts. A school wants opinions on lunch. It randomly selects 10 students from each grade. Cross-domain check: researchers collected 13 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. A sample is more representative when selection does not systematically favor people connected to the answer. Random or broad systematic sampling can reduce selection bias; voluntary or topic-specific locations may create bias. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. Is the sampling method likely to be biased for the stated question? Explain the direction of the possible bias. Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. Identify the target population. 2. Compare the selection method with that target. 3. Random selection within every grade gives broad grade coverage; no obvious topic-specific selection is stated. 4. Therefore the method is not obviously biased for the stated target, within the limits given. Cross-domain check: 13−4=9 independent reports remain.

Answer. Not obviously biased under the stated design. Random selection within every grade gives broad grade coverage; no obvious topic-specific selection is stated. Cross-domain answer: 9 independent reports.

Knowledge takeaway. A large biased sample can still give a misleading estimate; representativeness is a reasoning question about selection.

Formal model. Sampling-frame analysis: compare inclusion mechanism with target population and likely outcome-related selection.

Problem 488. Sampling and opinion: case 3

Knowledge context. Surveys use samples to learn about larger populations. The selection process matters as much as the number of answers.

Given facts. A village asks only members of the sports club whether to build a sports field. Cross-domain check: on a field sketch, 1 cm represents 5 km and two checkpoints are 7 cm apart.

Rules. A sample is more representative when selection does not systematically favor people connected to the answer. Random or broad systematic sampling can reduce selection bias; voluntary or topic-specific locations may create bias. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Is the sampling method likely to be biased for the stated question? Explain the direction of the possible bias. Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. Identify the target population. 2. Compare the selection method with that target. 3. Sports-club members are likely to be more favorable to a sports field than the whole village. 4. Therefore the method is likely biased. Cross-domain check: 7×5=35 km.

Answer. Likely biased. Sports-club members are likely to be more favorable to a sports field than the whole village. Cross-domain answer: 35 km.

Knowledge takeaway. A large biased sample can still give a misleading estimate; representativeness is a reasoning question about selection.

Formal model. Sampling-frame analysis: compare inclusion mechanism with target population and likely outcome-related selection.

Problem 489. Sampling and opinion: case 4

Knowledge context. Surveys use samples to learn about larger populations. The selection process matters as much as the number of answers.

Given facts. A library surveys every 20th visitor entering over a full week. Cross-domain check: a survey team starts work at 8:00 and works for 2 hour(s) without a break.

Rules. A sample is more representative when selection does not systematically favor people connected to the answer. Random or broad systematic sampling can reduce selection bias; voluntary or topic-specific locations may create bias. For the cross-domain check, finishing time = starting time + duration.

Task. Is the sampling method likely to be biased for the stated question? Explain the direction of the possible bias. Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. Identify the target population. 2. Compare the selection method with that target. 3. Sampling every 20th visitor across a full week is systematic within the visitor population; it may represent visitors reasonably, though not non-visitors. 4. Therefore the method is not obviously biased for the stated target, within the limits given. Cross-domain check: 8+2=10, so the team finishes at 10:00.

Answer. Not obviously biased under the stated design. Sampling every 20th visitor across a full week is systematic within the visitor population; it may represent visitors reasonably, though not non-visitors. Cross-domain answer: 10:00.

Knowledge takeaway. A large biased sample can still give a misleading estimate; representativeness is a reasoning question about selection.

Formal model. Sampling-frame analysis: compare inclusion mechanism with target population and likely outcome-related selection.

Problem 490. Sampling and opinion: case 5

Knowledge context. Surveys use samples to learn about larger populations. The selection process matters as much as the number of answers.

Given facts. An online poll is open to anyone who chooses to click it. Cross-domain check: a local committee has 12 members, requires at least 7 present for quorum, and 11 are present.

Rules. A sample is more representative when selection does not systematically favor people connected to the answer. Random or broad systematic sampling can reduce selection bias; voluntary or topic-specific locations may create bias. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. Is the sampling method likely to be biased for the stated question? Explain the direction of the possible bias. Cross-domain check: is quorum met?

Step-by-step solution. 1. Identify the target population. 2. Compare the selection method with that target. 3. People who choose to click may differ systematically from those who ignore the poll. 4. Therefore the method is likely biased. Cross-domain check: compare 11 with 7; $11 \geq 7$ is true.

Answer. Likely biased. People who choose to click may differ systematically from those who ignore the poll. Cross-domain answer: quorum is met.

Knowledge takeaway. A large biased sample can still give a misleading estimate; representativeness is a reasoning question about selection.

Formal model. Sampling-frame analysis: compare inclusion mechanism with target population and likely outcome-related selection.

N24. Matching, scheduling, and coverage

Problem 491. Matching people to tasks: case 1

Knowledge context. Scheduling and assignment problems connect people, tasks, times, and constraints.

Given facts. Four students must each receive one different task. Allowed assignments: Ava: ['Map', 'Timeline']; Ben: ['Budget', 'Survey']; Cara: ['Map', 'Survey']; Dion: ['Budget', 'Timeline']. Cross-domain check: researchers collected 7 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. Each student gets exactly one allowed task, and each task is used exactly once. A complete assignment must satisfy both conditions. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. Find one complete valid assignment. If none exists, explain why. Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. Start with the most restricted student if one exists. 2. Assign an allowed task, then remove that task from the remaining choices. 3. One complete assignment is {'Ava': 'Map', 'Ben': 'Budget', 'Cara': 'Survey', 'Dion': 'Timeline'}. Cross-domain check: 7−4=3 independent reports remain.

Answer. Valid assignment: Ava→Map, Ben→Budget, Cara→Survey, Dion→Timeline. Cross-domain answer: 3 independent reports.

Knowledge takeaway. A local choice can block a later assignment, so the whole matching must be checked.

Formal model. Bipartite matching / all-different constraint satisfaction.

Problem 492. Matching people to tasks: case 2

Knowledge context. Scheduling and assignment problems connect people, tasks, times, and constraints.

Given facts. Four students must each receive one different task. Allowed assignments: Ava: ['Map', 'Timeline']; Ben: ['Budget', 'Survey']; Cara: ['Survey']; Dion: ['Budget', 'Timeline']. Cross-domain check: on a field sketch, 1 cm represents 4 km and two checkpoints are 4 cm apart.

Rules. Each student gets exactly one allowed task, and each task is used exactly once. A complete assignment must satisfy both conditions. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Find one complete valid assignment. If none exists, explain why. Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. Start with the most restricted student if one exists. 2. Assign an allowed task, then remove that task from the remaining choices. 3. One complete assignment is {'Ava': 'Map', 'Ben': 'Budget', 'Cara': 'Survey', 'Dion': 'Timeline'}. Cross-domain check: 4×4=16 km.

Answer. Valid assignment: Ava→Map, Ben→Budget, Cara→Survey, Dion→Timeline. Cross-domain answer: 16 km.

Knowledge takeaway. A local choice can block a later assignment, so the whole matching must be checked.

Formal model. Bipartite matching / all-different constraint satisfaction.

Problem 493. Matching people to tasks: case 3

Knowledge context. Scheduling and assignment problems connect people, tasks, times, and constraints.

Given facts. Four students must each receive one different task. Allowed assignments: Ava: ['Map', 'Timeline']; Ben: ['Budget', 'Survey']; Cara: ['Map', 'Survey']; Dion: ['Budget']. Cross-domain check: a survey team starts work at 12:00 and works for 2 hour(s) without a break.

Rules. Each student gets exactly one allowed task, and each task is used exactly once. A complete assignment must satisfy both conditions. For the cross-domain check, finishing time = starting time + duration.

Task. Find one complete valid assignment. If none exists, explain why. Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. Start with the most restricted student if one exists. 2. Assign an allowed task, then remove that task from the remaining choices. 3. One complete assignment is {'Ava': 'Timeline', 'Ben': 'Survey', 'Cara': 'Map', 'Dion': 'Budget'}. Cross-domain check: 12+2=14, so the team finishes at 14:00.

Answer. Valid assignment: Ava→Timeline, Ben→Survey, Cara→Map, Dion→Budget. Cross-domain answer: 14:00.

Knowledge takeaway. A local choice can block a later assignment, so the whole matching must be checked.

Formal model. Bipartite matching / all-different constraint satisfaction.

Problem 494. Matching people to tasks: case 4

Knowledge context. Scheduling and assignment problems connect people, tasks, times, and constraints.

Given facts. Four students must each receive one different task. Allowed assignments: Ava: ['Map']; Ben: ['Budget', 'Survey']; Cara: ['Map', 'Survey']; Dion: ['Budget', 'Timeline']. Cross-domain check: a local committee has 16 members, requires at least 9 present for quorum, and 14 are present.

Rules. Each student gets exactly one allowed task, and each task is used exactly once. A complete assignment must satisfy both conditions. For the cross-domain check, quorum exists when present members ≥ the stated threshold.

Task. Find one complete valid assignment. If none exists, explain why. Cross-domain check: is quorum met?

Step-by-step solution. 1. Start with the most restricted student if one exists. 2. Assign an allowed task, then remove that task from the remaining choices. 3. One complete assignment is {'Ava': 'Map', 'Ben': 'Budget', 'Cara': 'Survey', 'Dion': 'Timeline'}. Cross-domain check: compare 14 with 9; 14 ≥ 9 is true.

Answer. Valid assignment: Ava→Map, Ben→Budget, Cara→Survey, Dion→Timeline. Cross-domain answer: quorum is met.

Knowledge takeaway. A local choice can block a later assignment, so the whole matching must be checked.

Formal model. Bipartite matching / all-different constraint satisfaction.

Problem 495. Matching people to tasks: case 5

Knowledge context. Scheduling and assignment problems connect people, tasks, times, and constraints.

Given facts. Four students must each receive one different task. Allowed assignments: Ava: ['Map', 'Timeline']; Ben: ['Budget', 'Survey']; Cara: ['Map', 'Survey']; Dion: ['Budget', 'Timeline']. Cross-domain check: researchers collected 11 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. Each student gets exactly one allowed task, and each task is used exactly once. A complete assignment must satisfy both conditions. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. Find one complete valid assignment. If none exists, explain why. Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. Start with the most restricted student if one exists. 2. Assign an allowed task, then remove that task from the remaining choices. 3. One complete assignment is {'Ava': 'Map', 'Ben': 'Budget', 'Cara': 'Survey', 'Dion': 'Timeline'}. Cross-domain check: 11−4=7 independent reports remain.

Answer. Valid assignment: Ava→Map, Ben→Budget, Cara→Survey, Dion→Timeline. Cross-domain answer: 7 independent reports.

Knowledge takeaway. A local choice can block a later assignment, so the whole matching must be checked.

Formal model. Bipartite matching / all-different constraint satisfaction.

N25. Meta-reasoning: robustness, information, causality

Problem 496. Meta-reasoning: case 1

Knowledge context. Good reasoning includes checking how informative, robust, falsifiable, causal, or dominated a conclusion is.

Given facts. Four equally likely locations A,B,C,D remain possible. Question Q1 asks “Is it A?”; Q2 asks “Is it A or B?” A yes/no question is more balanced when its two answer branches contain more nearly equal numbers of possibilities. Cross-domain check: on a field sketch, 1 cm represents 3 km and two checkpoints are 8 cm apart.

Rules. Use branch sizes as the information criterion in this exercise. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Which question is more balanced and therefore more informative by the stated criterion? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. Q1 splits the four possibilities 1 versus 3. 2. Q2 splits them 2 versus 2. 3. The 2–2 split is more balanced. Cross-domain check: 8×3=24 km.

Answer. Q2. Cross-domain answer: 24 km.

Knowledge takeaway. A result can be numerically correct yet epistemically weak; meta-reasoning evaluates the quality of the inference itself.

Formal model. Meta-level predicate over hypotheses or decisions: information balance, robustness, counterexample, causality, or dominance.

Problem 497. Meta-reasoning: case 2

Knowledge context. Good reasoning includes checking how informative, robust, falsifiable, causal, or dominated a conclusion is.

Given facts. Option A scores between 49 and 53 after measurement error; option B scores between 50 and 52. A ranking is robust only if every allowed value of A remains above every allowed value of B. Cross-domain check: a survey team starts work at 8:00 and works for 2 hour(s) without a break.

Rules. Test worst cases: robust A>B requires min(A)>max(B). For the cross-domain check, finishing time = starting time + duration.

Task. Is the ranking A>B robust? Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. min(A)=49; max(B)=52. 2. 49>52 is false. 3. Some allowed values reverse the ranking. Cross-domain check: 8+2=10, so the team finishes at 10:00.

Answer. No; the ranking is sensitive to the uncertainty. Cross-domain answer: 10:00.

Knowledge takeaway. A result can be numerically correct yet epistemically weak; meta-reasoning evaluates the quality of the inference itself.

Formal model. Meta-level predicate over hypotheses or decisions: information balance, robustness, counterexample, causality, or dominance.

Problem 498. Meta-reasoning: case 3

Knowledge context. Good reasoning includes checking how informative, robust, falsifiable, causal, or dominated a conclusion is.

Given facts. Claim: “Every town with more roads has shorter average travel time.” Town X has 6 roads and average travel time 35 minutes. Town Y has 4 roads and average travel time 20 minutes. Cross-domain check: a local committee has 11 members, requires at least 6 present for quorum, and 11 are present.

Rules. A universal claim is disproved by one case that satisfies its condition but not its conclusion. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. Do X and Y provide a counterexample to the claim? Cross-domain check: is quorum met?

Step-by-step solution. 1. X has more roads than Y. 2. Yet X has longer, not shorter, travel time. 3. Therefore the universal claim fails on this pair. Cross-domain check: compare 11 with 6; $11 \geq 6$ is true.

Answer. Yes. X and Y form a counterexample. Cross-domain answer: quorum is met.

Knowledge takeaway. A result can be numerically correct yet epistemically weak; meta-reasoning evaluates the quality of the inference itself.

Formal model. Meta-level predicate over hypotheses or decisions: information balance, robustness, counterexample, causality, or dominance.

Problem 499. Meta-reasoning: case 4

Knowledge context. Good reasoning includes checking how informative, robust, falsifiable, causal, or dominated a conclusion is.

Given facts. Across several months, ice-cream sales and sunburn cases rise together. Temperature also rises in those months. No intervention is described. Cross-domain check: researchers collected 15 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. Correlation between X and Y does not by itself prove X causes Y; a third variable may influence both. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. Can we conclude that ice-cream sales cause sunburn? Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. Sales and sunburn are correlated. 2. Temperature is a plausible common factor stated in the data. 3. Without an intervention or stronger causal evidence, direct causation is not established. Cross-domain check: 15−4=11 independent reports remain.

Answer. No. The association alone does not establish that ice-cream sales cause sunburn. Cross-domain answer: 11 independent reports.

Knowledge takeaway. A result can be numerically correct yet epistemically weak; meta-reasoning evaluates the quality of the inference itself.

Formal model. Meta-level predicate over hypotheses or decisions: information balance, robustness, counterexample, causality, or dominance.

Problem 500. Meta-reasoning: case 5

Knowledge context. Good reasoning includes checking how informative, robust, falsifiable, causal, or dominated a conclusion is.

Given facts. Two transport plans are compared. A costs 8, takes 20 minutes, and has safety score 9. B costs 10, takes 25 minutes, and has safety score 9. Lower cost/time and higher safety are better. Cross-domain check: on a field sketch, 1 cm represents 2 km and two checkpoints are 5 cm apart.

Rules. A Pareto-dominates B if A is at least as good on every stated criterion and strictly better on at least one. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Does one plan dominate the other? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. A is cheaper than B. 2. A is faster than B. 3. Safety is equal. 4. A is no worse anywhere and better on two criteria. Cross-domain check: 5×2=10 km.

Answer. A Pareto-dominates B. Cross-domain answer: 10 km.

Knowledge takeaway. A result can be numerically correct yet epistemically weak; meta-reasoning evaluates the quality of the inference itself.

Formal model. Meta-level predicate over hypotheses or decisions: information balance, robustness, counterexample, causality, or dominance.

GRADE 3

Difficulty increases through the number of relations and reasoning steps, not through hidden knowledge.

Part A — Foundational reasoning about the world

G1. Positions and orientation

Problem 501. Position chain from Yarrow to Alder

Knowledge context. A map can be modeled with relative positions. When several moves are given, their effects can be combined rather than memorized as a picture.

Given facts. Riverbend is 2 map unit(s) east of Yarrow; Alder is 2 map unit(s) east of Riverbend. A museum in Iris opens on Saturday.

Rules. Treat east as +x, west as −x, north as +y, and south as −y. Add the moves in order. A fact that does not affect any coordinate is irrelevant.

Task. Where is Alder relative to Yarrow? What is the straight grid displacement measured as |x|+|y|? Identify any irrelevant fact.

Step-by-step solution. 1. Start Yarrow at (0,0). 2. Translate each relation into a coordinate move. 3. Adding the moves gives (4,0). 4. That coordinate is east of Yarrow; the grid displacement is |4|+|0|=4. 5. The museum sentence changes no coordinate, so it is a distractor.

Answer. Alder is east of Yarrow; grid displacement 4 unit(s). The museum fact is irrelevant.

Knowledge takeaway. Cardinal directions become easier to reason about when they are treated as relations that can be composed.

Formal model. Grid coordinates; each statement adds a vector; sum the vectors and classify the final coordinate.

Problem 502. Position chain from Juniper to Iris

Knowledge context. A map can be modeled with relative positions. When several moves are given, their effects can be combined rather than memorized as a picture.

Given facts. Meadow is 1 map unit(s) north of Juniper; Iris is 1 map unit(s) north of Meadow. A museum in Elm opens on Saturday.

Rules. Treat east as +x, west as −x, north as +y, and south as −y. Add the moves in order. A fact that does not affect any coordinate is irrelevant.

Task. Where is Iris relative to Juniper? What is the straight grid displacement measured as |x|+|y|? Identify any irrelevant fact.

Step-by-step solution. 1. Start Juniper at (0,0). 2. Translate each relation into a coordinate move. 3. Adding the moves gives (0,2). 4. That coordinate is north of Juniper; the grid displacement is |0|+|2|=2. 5. The museum sentence changes no coordinate, so it is a distractor.

Answer. Iris is north of Juniper; grid displacement 2 unit(s). The museum fact is irrelevant.

Knowledge takeaway. Cardinal directions become easier to reason about when they are treated as relations that can be composed.

Formal model. Grid coordinates; each statement adds a vector; sum the vectors and classify the final coordinate.

Problem 503. Position chain from Northfield to Pine

Knowledge context. A map can be modeled with relative positions. When several moves are given, their effects can be combined rather than memorized as a picture.

Given facts. Dover is 2 map unit(s) east of Northfield; Riverbend is 2 map unit(s) north of Dover; Pine is 1 map unit(s) west of Riverbend. A museum in Oak opens on Saturday.

Rules. Treat east as +x, west as −x, north as +y, and south as −y. Add the moves in order. A fact that does not affect any coordinate is irrelevant.

Task. Where is Pine relative to Northfield? What is the straight grid displacement measured as |x|+|y|? Identify any irrelevant fact.

Step-by-step solution. 1. Start Northfield at (0,0). 2. Translate each relation into a coordinate move. 3. Adding the moves gives (1,2). 4. That coordinate is north-east of Northfield; the grid displacement is |1|+|2|=3. 5. The museum sentence changes no coordinate, so it is a distractor.

Answer. Pine is north-east of Northfield; grid displacement 3 unit(s). The museum fact is irrelevant.

Knowledge takeaway. Cardinal directions become easier to reason about when they are treated as relations that can be composed.

Formal model. Grid coordinates; each statement adds a vector; sum the vectors and classify the final coordinate.

Problem 504. Position chain from Riverbend to Dover

Knowledge context. A map can be modeled with relative positions. When several moves are given, their effects can be combined rather than memorized as a picture.

Given facts. Harbor is 1 map unit(s) west of Riverbend; Dover is 1 map unit(s) south of Harbor. A museum in Quartz opens on Saturday.

Rules. Treat east as +x, west as −x, north as +y, and south as −y. Add the moves in order. A fact that does not affect any coordinate is irrelevant.

Task. Where is Dover relative to Riverbend? What is the straight grid displacement measured as |x|+|y|? Identify any irrelevant fact.

Step-by-step solution. 1. Start Riverbend at (0,0). 2. Translate each relation into a coordinate move. 3. Adding the moves gives (-1,-1). 4. That coordinate is south-west of Riverbend; the grid displacement is |-1|+|-1|=2. 5. The museum sentence changes no coordinate, so it is a distractor.

Answer. Dover is south-west of Riverbend; grid displacement 2 unit(s). The museum fact is irrelevant.

Knowledge takeaway. Cardinal directions become easier to reason about when they are treated as relations that can be composed.

Formal model. Grid coordinates; each statement adds a vector; sum the vectors and classify the final coordinate.

Problem 505. Position chain from Grove to Oak

Knowledge context. A map can be modeled with relative positions. When several moves are given, their effects can be combined rather than memorized as a picture.

Given facts. Fern is 2 map unit(s) north of Grove; Kite is 2 map unit(s) east of Fern; Yarrow is 2 map unit(s) south of Kite; Oak is 1 map unit(s) west of Yarrow. A museum in Yarrow opens on Saturday.

Rules. Treat east as +x, west as −x, north as +y, and south as −y. Add the moves in order. A fact that does not affect any coordinate is irrelevant.

Task. Where is Oak relative to Grove? What is the straight grid displacement measured as |x|+|y|? Identify any irrelevant fact.

Step-by-step solution. 1. Start Grove at (0,0). 2. Translate each relation into a coordinate move. 3. Adding the moves gives (1,0). 4. That coordinate is east of Grove; the grid displacement is |1|+|0|=1. 5. The museum sentence changes no coordinate, so it is a distractor.

Answer. Oak is east of Grove; grid displacement 1 unit(s). The museum fact is irrelevant.

Knowledge takeaway. Cardinal directions become easier to reason about when they are treated as relations that can be composed.

Formal model. Grid coordinates; each statement adds a vector; sum the vectors and classify the final coordinate.

G2. Roads and routes

Problem 506. The best route from Stone to Cedar

Knowledge context. A road map can be modeled as a graph. The best route depends on the stated criterion, not on how short it looks on paper.

Given facts. Stone–Dover–Cedar has segment costs [5, 6] (total 11); Stone–Oak–Meadow–Cedar has segment costs [4, 2, 5] (total 11); Stone–Linden–Cedar has segment costs [8, 3] (total 11).

Rules. A route is feasible only if all of its roads are open. The cost of a route is the sum of its segment costs. Choose the feasible route with the smallest total.

Task. Which route should a traveler use from Stone to Cedar, and why?

Step-by-step solution. 1. List the candidate routes and their segment costs. 2. All listed routes are open. 3. Add the segment costs for every feasible route. 4. The smallest feasible total is 11, on Stone–Dover–Cedar.

Answer. Use Stone–Dover–Cedar; total cost 11.

Knowledge takeaway. A shortest path is a reasoning result: first respect constraints, then compare the remaining costs.

Formal model. Weighted graph; delete forbidden edges; minimize the sum of edge weights.

Problem 507. The best route from Iris to Oak

Knowledge context. A road map can be modeled as a graph. The best route depends on the stated criterion, not on how short it looks on paper.

Given facts. Iris–Meadow–Oak has segment costs [6, 7] (total 13); Iris–Zephyr–Linden–Oak has segment costs [5, 3, 5] (total 13); Iris–Riverbend–Oak has segment costs [9, 4] (total 13). The road Iris–Zephyr is closed.

Rules. A route is feasible only if all of its roads are open. The cost of a route is the sum of its segment costs. Choose the feasible route with the smallest total.

Task. Which route should a traveler use from Iris to Oak, and why?

Step-by-step solution. 1. List the candidate routes and their segment costs. 2. Remove any route containing a closed road. 3. Add the segment costs for every feasible route. 4. The smallest feasible total is 13, on Iris–Meadow–Oak.

Answer. Use Iris–Meadow–Oak; total cost 13.

Knowledge takeaway. A shortest path is a reasoning result: first respect constraints, then compare the remaining costs.

Formal model. Weighted graph; delete forbidden edges; minimize the sum of edge weights.

Problem 508. The best route from Iris to Meadow

Knowledge context. A road map can be modeled as a graph. The best route depends on the stated criterion, not on how short it looks on paper.

Given facts. Iris–Juniper–Meadow has segment costs [7, 8] (total 15); Iris–Linden–Riverbend–Meadow has segment costs [6, 4, 5] (total 15); Iris–Kite–Meadow has segment costs [10, 5] (total 15).

Rules. A route is feasible only if all of its roads are open. The cost of a route is the sum of its segment costs. Choose the feasible route with the smallest total.

Task. Which route should a traveler use from Iris to Meadow, and why?

Step-by-step solution. 1. List the candidate routes and their segment costs. 2. All listed routes are open. 3. Add the segment costs for every feasible route. 4. The smallest feasible total is 15, on Iris–Juniper–Meadow.

Answer. Use Iris–Juniper–Meadow; total cost 15.

Knowledge takeaway. A shortest path is a reasoning result: first respect constraints, then compare the remaining costs.

Formal model. Weighted graph; delete forbidden edges; minimize the sum of edge weights.

Problem 509. The best route from Elm to Dover

Knowledge context. A road map can be modeled as a graph. The best route depends on the stated criterion, not on how short it looks on paper.

Given facts. Elm–Meadow–Dover has segment costs [8, 9] (total 17); Elm–Iris–Harbor–Dover has segment costs [7, 5, 5] (total 17); Elm–Yarrow–Dover has segment costs [11, 6] (total 17). The road Elm–Iris is closed.

Rules. A route is feasible only if all of its roads are open. The cost of a route is the sum of its segment costs. Choose the feasible route with the smallest total.

Task. Which route should a traveler use from Elm to Dover, and why?

Step-by-step solution. 1. List the candidate routes and their segment costs. 2. Remove any route containing a closed road. 3. Add the segment costs for every feasible route. 4. The smallest feasible total is 17, on Elm–Meadow–Dover.

Answer. Use Elm–Meadow–Dover; total cost 17.

Knowledge takeaway. A shortest path is a reasoning result: first respect constraints, then compare the remaining costs.

Formal model. Weighted graph; delete forbidden edges; minimize the sum of edge weights.

Problem 510. The best route from Riverbend to Stone

Knowledge context. A road map can be modeled as a graph. The best route depends on the stated criterion, not on how short it looks on paper.

Given facts. Riverbend–Kite–Stone has segment costs [9, 10] (total 19); Riverbend–Iris–Birch–Stone has segment costs [8, 6, 5] (total 19); Riverbend–Oak–Stone has segment costs [12, 7] (total 19).

Rules. A route is feasible only if all of its roads are open. The cost of a route is the sum of its segment costs. Choose the feasible route with the smallest total.

Task. Which route should a traveler use from Riverbend to Stone, and why?

Step-by-step solution. 1. List the candidate routes and their segment costs. 2. All listed routes are open. 3. Add the segment costs for every feasible route. 4. The smallest feasible total is 19, on Riverbend–Kite–Stone.

Answer. Use Riverbend–Kite–Stone; total cost 19.

Knowledge takeaway. A shortest path is a reasoning result: first respect constraints, then compare the remaining costs.

Formal model. Weighted graph; delete forbidden edges; minimize the sum of edge weights.

G3. Rivers: upstream and downstream

Problem 511. Following water from Clearwater

Knowledge context. River systems form directed networks: water has a direction, tributaries join larger streams, and upstream/downstream are relational concepts.

Given facts. Clearwater flows into North Fork; Blue Creek flows into North Fork; North Fork flows into Silver River. A canal diverts some water from Blue Creek to a reservoir, but the remaining flow still reaches North Fork.

Rules. If river X flows into river Y, X is upstream of Y. Upstream relations can be followed through several confluences. Downstream is the reverse direction of flow.

Task. Is Clearwater upstream of Silver River? Can water from Clearwater reach Silver River through the stated network?

Step-by-step solution. 1. Rewrite the facts as arrows: Clearwater→North Fork, Blue Creek→North Fork, North Fork→Silver River. 2. Follow arrows starting from Clearwater. 3. An arrow path reaches Silver River, so Clearwater is upstream of Silver River.

Answer. Yes; water from Clearwater can reach Silver River.

Knowledge takeaway. A river network can be reasoned about like any directed network; following the arrows is more reliable than guessing from names.

Formal model. Directed graph; edge X→Y means water can flow from X to Y; query reachability.

Problem 512. Following water from North Fork

Knowledge context. River systems form directed networks: water has a direction, tributaries join larger streams, and upstream/downstream are relational concepts.

Given facts. Red Brook flows into Willow Stream; North Fork flows into Willow Stream; Willow Stream flows into Silver River. A canal diverts some water from North Fork to a reservoir, but the remaining flow still reaches Willow Stream.

Rules. If river X flows into river Y, X is upstream of Y. Upstream relations can be followed through several confluences. Downstream is the reverse direction of flow.

Task. Is North Fork upstream of Silver River? Can water from North Fork reach Silver River through the stated network?

Step-by-step solution. 1. Rewrite the facts as arrows: Red Brook→Willow Stream, North Fork→Willow Stream, Willow Stream→Silver River. 2. Follow arrows starting from North Fork. 3. An arrow path reaches Silver River, so North Fork is upstream of Silver River.

Answer. Yes; water from North Fork can reach Silver River.

Knowledge takeaway. A river network can be reasoned about like any directed network; following the arrows is more reliable than guessing from names.

Formal model. Directed graph; edge X→Y means water can flow from X to Y; query reachability.

Problem 513. Following water from Clearwater

Knowledge context. River systems form directed networks: water has a direction, tributaries join larger streams, and upstream/downstream are relational concepts.

Given facts. Clearwater flows into Silver River; North Fork flows into Silver River; Silver River flows into Blue Creek. A canal diverts some water from North Fork to a reservoir, but the remaining flow still reaches Silver River.

Rules. If river X flows into river Y, X is upstream of Y. Upstream relations can be followed through several confluences. Downstream is the reverse direction of flow.

Task. Is Clearwater upstream of Silver River? Can water from Clearwater reach Silver River through the stated network?

Step-by-step solution. 1. Rewrite the facts as arrows: Clearwater→Silver River, North Fork→Silver River, Silver River→Blue Creek. 2. Follow arrows starting from Clearwater. 3. An arrow path reaches Silver River, so Clearwater is upstream of Silver River.

Answer. Yes; water from Clearwater can reach Silver River.

Knowledge takeaway. A river network can be reasoned about like any directed network; following the arrows is more reliable than guessing from names.

Formal model. Directed graph; edge X→Y means water can flow from X to Y; query reachability.

Problem 514. Following water from Blue Creek

Knowledge context. River systems form directed networks: water has a direction, tributaries join larger streams, and upstream/downstream are relational concepts.

Given facts. Willow Stream flows into Blue Creek; Pine River flows into Blue Creek; Blue Creek flows into Red Brook. A canal diverts some water from Pine River to a reservoir, but the remaining flow still reaches Blue Creek.

Rules. If river X flows into river Y, X is upstream of Y. Upstream relations can be followed through several confluences. Downstream is the reverse direction of flow.

Task. Is Blue Creek upstream of Red Brook? Can water from Blue Creek reach Red Brook through the stated network?

Step-by-step solution. 1. Rewrite the facts as arrows: Willow Stream→Blue Creek, Pine River→Blue Creek, Blue Creek→Red Brook. 2. Follow arrows starting from Blue Creek. 3. An arrow path reaches Red Brook, so Blue Creek is upstream of Red Brook.

Answer. Yes; water from Blue Creek can reach Red Brook.

Knowledge takeaway. A river network can be reasoned about like any directed network; following the arrows is more reliable than guessing from names.

Formal model. Directed graph; edge X→Y means water can flow from X to Y; query reachability.

Problem 515. Following water from Pine River

Knowledge context. River systems form directed networks: water has a direction, tributaries join larger streams, and upstream/downstream are relational concepts.

Given facts. Willow Stream flows into Blue Creek; Pine River flows into Blue Creek; Blue Creek flows into Silver River. A canal diverts some water from Pine River to a reservoir, but the remaining flow still reaches Blue Creek.

Rules. If river X flows into river Y, X is upstream of Y. Upstream relations can be followed through several confluences. Downstream is the reverse direction of flow.

Task. Is Pine River upstream of Blue Creek? Can water from Pine River reach Blue Creek through the stated network?

Step-by-step solution. 1. Rewrite the facts as arrows: Willow Stream→Blue Creek, Pine River→Blue Creek, Blue Creek→Silver River. 2. Follow arrows starting from Pine River. 3. An arrow path reaches Blue Creek, so Pine River is upstream of Blue Creek.

Answer. Yes; water from Pine River can reach Blue Creek.

Knowledge takeaway. A river network can be reasoned about like any directed network; following the arrows is more reliable than guessing from names.

Formal model. Directed graph; edge X→Y means water can flow from X to Y; query reachability.

G4. Neighbours and borders

Problem 516. Crossing the fictional border network of Elm

Knowledge context. Political maps can be treated as adjacency networks. This lets us reason about border crossings without relying on memorized real-world geography.

Given facts. Countries are represented by the fictional states Elm, Kite, Willow, Stone, Oak, Fern. Shared borders: Elm–Kite, Kite–Willow, Willow–Stone, Elm–Oak, Oak–Fern, Fern–Stone.

Rules. Crossing one border corresponds to traversing one graph edge. A route may use only listed borders. A state is unavoidable only if every feasible route passes through it.

Task. What is the fewest number of border crossings needed to travel from Elm to Stone? Is Kite necessarily crossed?

Step-by-step solution. 1. Turn each shared border into an undirected link. 2. A shortest-link search gives distance 3 from Elm to Stone. 3. Remove Kite and check whether Stone is still reachable from Elm. 4. It is still reachable, so Kite is not unavoidable.

Answer. 3 border crossing(s); Kite is not unavoidable.

Knowledge takeaway. Neighbourhood on a map is a graph relation; shortest routes and unavoidable transit points are different questions.

Formal model. Undirected graph; shortest path length plus reachability after deleting a candidate vertex.

Problem 517. Crossing the fictional border network of Stone

Knowledge context. Political maps can be treated as adjacency networks. This lets us reason about border crossings without relying on memorized real-world geography.

Given facts. Countries are represented by the fictional states Stone, Linden, Fern, Meadow, Riverbend, Alder. Shared borders: Stone–Linden, Linden–Fern, Fern–Meadow, Stone–Riverbend, Alder–Meadow.

Rules. Crossing one border corresponds to traversing one graph edge. A route may use only listed borders. A state is unavoidable only if every feasible route passes through it.

Task. What is the fewest number of border crossings needed to travel from Stone to Meadow? Is Linden necessarily crossed?

Step-by-step solution. 1. Turn each shared border into an undirected link. 2. A shortest-link search gives distance 3 from Stone to Meadow. 3. Remove Linden and check whether Meadow is still reachable from Stone. 4. It is not reachable, so Linden is unavoidable.

Answer. 3 border crossing(s); Linden is unavoidable.

Knowledge takeaway. Neighbourhood on a map is a graph relation; shortest routes and unavoidable transit points are different questions.

Formal model. Undirected graph; shortest path length plus reachability after deleting a candidate vertex.

Problem 518. Crossing the fictional border network of Stone

Knowledge context. Political maps can be treated as adjacency networks. This lets us reason about border crossings without relying on memorized real-world geography.

Given facts. Countries are represented by the fictional states Stone, Kite, Willow, Birch, Cedar, Riverbend. Shared borders: Stone–Kite, Kite–Willow, Willow–Birch, Stone–Cedar, Cedar–Riverbend, Riverbend–Birch.

Rules. Crossing one border corresponds to traversing one graph edge. A route may use only listed borders. A state is unavoidable only if every feasible route passes through it.

Task. What is the fewest number of border crossings needed to travel from Stone to Birch? Is Kite necessarily crossed?

Step-by-step solution. 1. Turn each shared border into an undirected link. 2. A shortest-link search gives distance 3 from Stone to Birch. 3. Remove Kite and check whether Birch is still reachable from Stone. 4. It is still reachable, so Kite is not unavoidable.

Answer. 3 border crossing(s); Kite is not unavoidable.

Knowledge takeaway. Neighbourhood on a map is a graph relation; shortest routes and unavoidable transit points are different questions.

Formal model. Undirected graph; shortest path length plus reachability after deleting a candidate vertex.

Problem 519. Crossing the fictional border network of Zephyr

Knowledge context. Political maps can be treated as adjacency networks. This lets us reason about border crossings without relying on memorized real-world geography.

Given facts. Countries are represented by the fictional states Zephyr, Birch, Fern, Oak, Linden, Harbor. Shared borders: Zephyr–Birch, Birch–Fern, Fern–Oak, Zephyr–Linden, Linden–Harbor, Harbor–Oak.

Rules. Crossing one border corresponds to traversing one graph edge. A route may use only listed borders. A state is unavoidable only if every feasible route passes through it.

Task. What is the fewest number of border crossings needed to travel from Zephyr to Oak? Is Birch necessarily crossed?

Step-by-step solution. 1. Turn each shared border into an undirected link. 2. A shortest-link search gives distance 3 from Zephyr to Oak. 3. Remove Birch and check whether Oak is still reachable from Zephyr. 4. It is still reachable, so Birch is not unavoidable.

Answer. 3 border crossing(s); Birch is not unavoidable.

Knowledge takeaway. Neighbourhood on a map is a graph relation; shortest routes and unavoidable transit points are different questions.

Formal model. Undirected graph; shortest path length plus reachability after deleting a candidate vertex.

Problem 520. Crossing the fictional border network of Birch

Knowledge context. Political maps can be treated as adjacency networks. This lets us reason about border crossings without relying on memorized real-world geography.

Given facts. Countries are represented by the fictional states Birch, Oak, Alder, Riverbend, Dover, Kite. Shared borders: Birch–Oak, Oak–Alder, Alder–Riverbend, Birch–Dover, Kite–Riverbend.

Rules. Crossing one border corresponds to traversing one graph edge. A route may use only listed borders. A state is unavoidable only if every feasible route passes through it.

Task. What is the fewest number of border crossings needed to travel from Birch to Riverbend? Is Oak necessarily crossed?

Step-by-step solution. 1. Turn each shared border into an undirected link. 2. A shortest-link search gives distance 3 from Birch to Riverbend. 3. Remove Oak and check whether Riverbend is still reachable from Birch. 4. It is not reachable, so Oak is unavoidable.

Answer. 3 border crossing(s); Oak is unavoidable.

Knowledge takeaway. Neighbourhood on a map is a graph relation; shortest routes and unavoidable transit points are different questions.

Formal model. Undirected graph; shortest path length plus reachability after deleting a candidate vertex.

G5. Maps, legends, and clues

Problem 521. Reading a layered map legend (A1)

Knowledge context. A map legend translates symbols into properties. One location may have several properties at the same time.

Given facts. Legend: tree=forest, wavy line=river, double line=road, arch=bridge, square=village, triangle=hill. Cells: A1 has forest, road; A2 has bridge, river; B1 has forest, river, road; B2 has road, village. Cross-domain check: a survey team starts work at 8:00 and works for 2 hour(s) without a break.

Rules. A cell satisfies the task only if it has every required attribute. Extra attributes do not disqualify it unless the task says so. For the cross-domain check, finishing time = starting time + duration.

Task. Which cell or cells satisfy all of these conditions: forest, road? Explain by intersecting the clues. Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. Convert the requested symbols into the attribute set ['forest', 'road']. 2. Check each cell for every required attribute. 3. The matching set is ['A1', 'B1']. Cross-domain check: 8+2=10, so the team finishes at 10:00.

Answer. A1, B1 Cross-domain answer: 10:00.

Knowledge takeaway. Map reading often means intersecting clues rather than recognizing one symbol in isolation.

Formal model. Set model; each cell has an attribute set; return cells whose set is a superset of the required set.

Problem 522. Reading a layered map legend (A2)

Knowledge context. A map legend translates symbols into properties. One location may have several properties at the same time.

Given facts. Legend: tree=forest, wavy line=river, double line=road, arch=bridge, square=village, triangle=hill. Cells: A1 has forest, road; A2 has bridge, river, village; B1 has forest, river, road; B2 has road, village. Cross-domain check: a local committee has 8 members, requires at least 5 present for quorum, and 8 are present.

Rules. A cell satisfies the task only if it has every required attribute. Extra attributes do not disqualify it unless the task says so. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. Which cell or cells satisfy all of these conditions: bridge, river? Explain by intersecting the clues. Cross-domain check: is quorum met?

Step-by-step solution. 1. Convert the requested symbols into the attribute set ['bridge', 'river']. 2. Check each cell for every required attribute. 3. The matching set is ['A2']. Cross-domain check: compare 8 with 5; $8 \geq 5$ is true.

Answer. A2 Cross-domain answer: quorum is met.

Knowledge takeaway. Map reading often means intersecting clues rather than recognizing one symbol in isolation.

Formal model. Set model; each cell has an attribute set; return cells whose set is a superset of the required set.

Problem 523. Reading a layered map legend (B1)

Knowledge context. A map legend translates symbols into properties. One location may have several properties at the same time.

Given facts. Legend: tree=forest, wavy line=river, double line=road, arch=bridge, square=village, triangle=hill. Cells: A1 has forest, road; A2 has bridge, river; B1 has forest, river, road; B2 has forest, road, village. Cross-domain check: researchers collected 9 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. A cell satisfies the task only if it has every required attribute. Extra attributes do not disqualify it unless the task says so. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. Which cell or cells satisfy all of these conditions: forest, river? Explain by intersecting the clues. Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. Convert the requested symbols into the attribute set ['forest', 'river']. 2. Check each cell for every required attribute. 3. The matching set is ['B1']. Cross-domain check: 9−4=5 independent reports remain.

Answer. B1 Cross-domain answer: 5 independent reports.

Knowledge takeaway. Map reading often means intersecting clues rather than recognizing one symbol in isolation.

Formal model. Set model; each cell has an attribute set; return cells whose set is a superset of the required set.

Problem 524. Reading a layered map legend (B2)

Knowledge context. A map legend translates symbols into properties. One location may have several properties at the same time.

Given facts. Legend: tree=forest, wavy line=river, double line=road, arch=bridge, square=village, triangle=hill. Cells: A1 has forest, hill, road; A2 has bridge, river; B1 has forest, river, road; B2 has road, village. Cross-domain check: on a field sketch, 1 cm represents 6 km and two checkpoints are 8 cm apart.

Rules. A cell satisfies the task only if it has every required attribute. Extra attributes do not disqualify it unless the task says so. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Which cell or cells satisfy all of these conditions: hill, road? Explain by intersecting the clues. Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. Convert the requested symbols into the attribute set ['hill', 'road']. 2. Check each cell for every required attribute. 3. The matching set is ['A1']. Cross-domain check: 8×6=48 km.

Answer. A1 Cross-domain answer: 48 km.

Knowledge takeaway. Map reading often means intersecting clues rather than recognizing one symbol in isolation.

Formal model. Set model; each cell has an attribute set; return cells whose set is a superset of the required set.

Problem 525. Reading a layered map legend (A1)

Knowledge context. A map legend translates symbols into properties. One location may have several properties at the same time.

Given facts. Legend: tree=forest, wavy line=river, double line=road, arch=bridge, square=village, triangle=hill. Cells: A1 has forest, road; A2 has bridge, river; B1 has bridge, forest, river, road; B2 has road, village. Cross-domain check: a survey team starts work at 12:00 and works for 2 hour(s) without a break.

Rules. A cell satisfies the task only if it has every required attribute. Extra attributes do not disqualify it unless the task says so. For the cross-domain check, finishing time = starting time + duration.

Task. Which cell or cells satisfy all of these conditions: river, road? Explain by intersecting the clues. Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. Convert the requested symbols into the attribute set ['river', 'road']. 2. Check each cell for every required attribute. 3. The matching set is ['B1']. Cross-domain check: 12+2=14, so the team finishes at 14:00.

Answer. B1 Cross-domain answer: 14:00.

Knowledge takeaway. Map reading often means intersecting clues rather than recognizing one symbol in isolation.

Formal model. Set model; each cell has an attribute set; return cells whose set is a superset of the required set.

G6. Choosing a settlement site

Problem 526. Choosing a safe settlement site: case 1

Knowledge context. Human settlements depend on several factors at once. Some conditions may be compulsory while others are preferences.

Given facts. Candidate sites: Site A: water=yes, road=yes, flood-zone=no, nearby farmland score=2; Site B: water=yes, road=no, flood-zone=no, nearby farmland score=4; Site C: water=yes, road=yes, flood-zone=yes, nearby farmland score=4; Site D: water=no, road=yes, flood-zone=no, nearby farmland score=6. Cross-domain check: a local committee has 12 members, requires at least 7 present for quorum, and 11 are present.

Rules. A valid site must have water, road access, and must not be in the flood zone. Among valid sites, choose the one with the highest farmland score. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. Which site should be selected under the stated mandatory conditions and preference rule? Cross-domain check: is quorum met?

Step-by-step solution. 1. Separate mandatory conditions from the preference. 2. Filtering by water, road, and flood safety leaves ['Site A']. 3. Compare only the farmland scores of feasible sites: Site A=2. 4. Site A has the best score among valid sites. Cross-domain check: compare 11 with 7; 11 ≥ 7 is true.

Answer. Site A. Cross-domain answer: quorum is met.

Knowledge takeaway. Good decisions first satisfy hard constraints and only then optimize among the remaining options.

Formal model. Constraint-satisfaction problem followed by a score maximization over feasible options.

Problem 527. Choosing a safe settlement site: case 2

Knowledge context. Human settlements depend on several factors at once. Some conditions may be compulsory while others are preferences.

Given facts. Candidate sites: Site A: water=yes, road=yes, flood-zone=no, nearby farmland score=3; Site B: water=yes, road=no, flood-zone=no, nearby farmland score=3; Site C: water=yes, road=yes, flood-zone=yes, nearby farmland score=5;

Site D: water=no, road=yes, flood-zone=no, nearby farmland score=5. Cross-domain check: researchers collected 13 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. A valid site must have water, road access, and must not be in the flood zone. Among valid sites, choose the one with the highest farmland score. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. Which site should be selected under the stated mandatory conditions and preference rule? Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. Separate mandatory conditions from the preference. 2. Filtering by water, road, and flood safety leaves ['Site A']. 3. Compare only the farmland scores of feasible sites: Site A=3. 4. Site A has the best score among valid sites. Cross-domain check: 13−4=9 independent reports remain.

Answer. Site A. Cross-domain answer: 9 independent reports.

Knowledge takeaway. Good decisions first satisfy hard constraints and only then optimize among the remaining options.

Formal model. Constraint-satisfaction problem followed by a score maximization over feasible options.

Problem 528. Choosing a safe settlement site: case 3

Knowledge context. Human settlements depend on several factors at once. Some conditions may be compulsory while others are preferences.

Given facts. Candidate sites: Site A: water=yes, road=yes, flood-zone=no, nearby farmland score=2; Site B: water=yes, road=no, flood-zone=no, nearby farmland score=4; Site C: water=yes, road=yes, flood-zone=yes, nearby farmland score=4; Site D: water=no, road=yes, flood-zone=no, nearby farmland score=6. Cross-domain check: on a field sketch, 1 cm represents 5 km and two checkpoints are 5 cm apart.

Rules. A valid site must have water, road access, and must not be in the flood zone. Among valid sites, choose the one with the highest farmland score. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Which site should be selected under the stated mandatory conditions and preference rule? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. Separate mandatory conditions from the preference. 2. Filtering by water, road, and flood safety leaves ['Site A']. 3. Compare only the farmland scores of feasible sites: Site A=2. 4. Site A has the best score among valid sites. Cross-domain check: 5×5=25 km.

Answer. Site A. Cross-domain answer: 25 km.

Knowledge takeaway. Good decisions first satisfy hard constraints and only then optimize among the remaining options.

Formal model. Constraint-satisfaction problem followed by a score maximization over feasible options.

Problem 529. Choosing a safe settlement site: case 4

Knowledge context. Human settlements depend on several factors at once. Some conditions may be compulsory while others are preferences.

Given facts. Candidate sites: Site A: water=yes, road=yes, flood-zone=no, nearby farmland score=3; Site B: water=yes, road=no, flood-zone=no, nearby farmland score=3; Site C: water=yes, road=yes, flood-zone=yes, nearby farmland score=5; Site D: water=no, road=yes, flood-zone=no, nearby farmland score=5. Cross-domain check: a survey team starts work at 8:00 and works for 2 hour(s) without a break.

Rules. A valid site must have water, road access, and must not be in the flood zone. Among valid sites, choose the one with the highest farmland score. For the cross-domain check, finishing time = starting time + duration.

Task. Which site should be selected under the stated mandatory conditions and preference rule? Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. Separate mandatory conditions from the preference. 2. Filtering by water, road, and flood safety leaves ['Site A']. 3. Compare only the farmland scores of feasible sites: Site A=3. 4. Site A has the best score among valid sites. Cross-domain check: 8+2=10, so the team finishes at 10:00.

Answer. Site A. Cross-domain answer: 10:00.

Knowledge takeaway. Good decisions first satisfy hard constraints and only then optimize among the remaining options.

Formal model. Constraint-satisfaction problem followed by a score maximization over feasible options.

Problem 530. Choosing a safe settlement site: case 5

Knowledge context. Human settlements depend on several factors at once. Some conditions may be compulsory while others are preferences.

Given facts. Candidate sites: Site A: water=yes, road=yes, flood-zone=no, nearby farmland score=2; Site B: water=yes, road=no, flood-zone=no, nearby farmland score=4; Site C: water=yes, road=yes, flood-zone=yes, nearby farmland score=4; Site D: water=no, road=yes, flood-zone=no, nearby farmland score=6. Cross-domain check: a local committee has 16

members, requires at least 9 present for quorum, and 14 are present. Mixed-domain verification: a regional archive has 55 map sheets and receives 1 additional sheets during the exercise.

Rules. A valid site must have water, road access, and must not be in the flood zone. Among valid sites, choose the one with the highest farmland score. For the cross-domain check, quorum exists when present members ≥ the stated threshold. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. Which site should be selected under the stated mandatory conditions and preference rule? Cross-domain check: is quorum met? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. Separate mandatory conditions from the preference. 2. Filtering by water, road, and flood safety leaves ['Site A']. 3. Compare only the farmland scores of feasible sites: Site A=2. 4. Site A has the best score among valid sites. Cross-domain check: compare 14 with 9; 14 ≥ 9 is true. Mixed-domain verification: 55+1=56.

Answer. Site A. Cross-domain answer: quorum is met. Mixed-domain answer: 56 map sheets.

Knowledge takeaway. Good decisions first satisfy hard constraints and only then optimize among the remaining options.

Formal model. Constraint-satisfaction problem followed by a score maximization over feasible options.

G7. Relief and movement cost

Problem 531. Terrain and travel cost from Riverbend

Knowledge context. Equal-looking distances can require different effort because terrain differs. A simple cost model makes that effect explicit.

Given facts. Terrain energy costs per segment: {'plain': 1, 'forest': 2, 'hill': 3, 'marsh': 4}. Three routes from Riverbend to Alder: R1=['plain', 'hill', 'plain']; R2=['forest', 'forest', 'plain']; R3=['plain', 'marsh']. The traveler can spend at most 10 energy units.

Rules. The energy of a route is the sum of its terrain costs. A route above the energy limit is infeasible. Among feasible routes, minimize energy.

Task. Which route is feasible and uses the least energy?

Step-by-step solution. 1. Compute route totals: R1=5, R2=5, R3=5. 2. Discard any total above 10. 3. The smallest remaining total is R1=5.

Answer. Route R1, using 5 energy units.

Knowledge takeaway. Relief changes the cost of movement; “shortest” and “easiest” are not automatically the same.

Formal model. Weighted paths with a feasibility budget; sum costs and choose the minimum feasible path.

Problem 532. Terrain and travel cost from Meadow

Knowledge context. Equal-looking distances can require different effort because terrain differs. A simple cost model makes that effect explicit.

Given facts. Terrain energy costs per segment: {'plain': 1, 'forest': 2, 'hill': 3, 'marsh': 4}. Three routes from Meadow to Cedar: R1=['plain', 'hill', 'plain']; R2=['plain', 'forest', 'plain', 'plain']; R3=['plain', 'marsh']. The traveler can spend at most 10 energy units.

Rules. The energy of a route is the sum of its terrain costs. A route above the energy limit is infeasible. Among feasible routes, minimize energy.

Task. Which route is feasible and uses the least energy?

Step-by-step solution. 1. Compute route totals: R1=5, R2=5, R3=5. 2. Discard any total above 10. 3. The smallest remaining total is R1=5.

Answer. Route R1, using 5 energy units.

Knowledge takeaway. Relief changes the cost of movement; “shortest” and “easiest” are not automatically the same.

Formal model. Weighted paths with a feasibility budget; sum costs and choose the minimum feasible path.

Problem 533. Terrain and travel cost from Elm

Knowledge context. Equal-looking distances can require different effort because terrain differs. A simple cost model makes that effect explicit.

Given facts. Terrain energy costs per segment: {'plain': 1, 'forest': 2, 'hill': 3, 'marsh': 4}. Three routes from Elm to Stone: R1=['plain', 'hill', 'plain']; R2=['forest', 'forest', 'plain']; R3=['plain', 'marsh']. The traveler can spend at most 10 energy units.

Rules. The energy of a route is the sum of its terrain costs. A route above the energy limit is infeasible. Among feasible routes, minimize energy.

Task. Which route is feasible and uses the least energy?

Step-by-step solution. 1. Compute route totals: R1=5, R2=5, R3=5. 2. Discard any total above 10. 3. The smallest remaining total is R1=5.

Answer. Route R1, using 5 energy units.

Knowledge takeaway. Relief changes the cost of movement; “shortest” and “easiest” are not automatically the same.

Formal model. Weighted paths with a feasibility budget; sum costs and choose the minimum feasible path.

Problem 534. Terrain and travel cost from Quartz

Knowledge context. Equal-looking distances can require different effort because terrain differs. A simple cost model makes that effect explicit.

Given facts. Terrain energy costs per segment: {'plain': 1, 'forest': 2, 'hill': 3, 'marsh': 4}. Three routes from Quartz to Alder: R1=['plain', 'hill', 'plain']; R2=['plain', 'forest', 'plain', 'plain']; R3=['plain', 'marsh']. The traveler can spend at most 10 energy units.

Rules. The energy of a route is the sum of its terrain costs. A route above the energy limit is infeasible. Among feasible routes, minimize energy.

Task. Which route is feasible and uses the least energy?

Step-by-step solution. 1. Compute route totals: R1=5, R2=5, R3=5. 2. Discard any total above 10. 3. The smallest remaining total is R1=5.

Answer. Route R1, using 5 energy units.

Knowledge takeaway. Relief changes the cost of movement; “shortest” and “easiest” are not automatically the same.

Formal model. Weighted paths with a feasibility budget; sum costs and choose the minimum feasible path.

Problem 535. Terrain and travel cost from Northfield

Knowledge context. Equal-looking distances can require different effort because terrain differs. A simple cost model makes that effect explicit.

Given facts. Terrain energy costs per segment: {'plain': 1, 'forest': 2, 'hill': 3, 'marsh': 4}. Three routes from Northfield to Meadow: R1=['plain', 'hill', 'plain']; R2=['forest', 'forest', 'plain']; R3=['plain', 'marsh']. The traveler can spend at most 10 energy units.

Rules. The energy of a route is the sum of its terrain costs. A route above the energy limit is infeasible. Among feasible routes, minimize energy.

Task. Which route is feasible and uses the least energy?

Step-by-step solution. 1. Compute route totals: R1=5, R2=5, R3=5. 2. Discard any total above 10. 3. The smallest remaining total is R1=5.

Answer. Route R1, using 5 energy units.

Knowledge takeaway. Relief changes the cost of movement; “shortest” and “easiest” are not automatically the same.

Formal model. Weighted paths with a feasibility budget; sum costs and choose the minimum feasible path.

G8. Climate from data

Problem 536. Classifying climate data: Station A

Knowledge context. Climate descriptions summarize long-term patterns. In a reasoning problem, explicit thresholds let us classify numerical data without hidden knowledge.

Given facts. Use this classroom classification: cool if mean temperature <12°C; mild if 12–17°C; warm if >17°C. Dry if annual precipitation <500 mm; wet if ≥ 500 mm. Data: Station A: T=8°C, P=490 mm; Station B: T=16°C, P=760 mm; Station C: T=12°C, P=520 mm.

Rules. Apply only the stated thresholds. Temperature and precipitation are separate dimensions, so each station receives two labels.

Task. Classify Station A. Then identify whether any other station has the same two-label class.

Step-by-step solution. 1. Compare T=8 with the temperature thresholds: it is cool. 2. Compare P=490 with 500 mm: it is dry. 3. Compare the combined label with the other stations; matches: none.

Answer. Station A is cool and dry; no other station has the same class.

Knowledge takeaway. Classification is the application of clear predicates to data; the rule must be stated before the label is inferred.

Formal model. Numeric predicates with thresholds; output the conjunction of the temperature and precipitation categories.

Problem 537. Classifying climate data: Station B

Knowledge context. Climate descriptions summarize long-term patterns. In a reasoning problem, explicit thresholds let us classify numerical data without hidden knowledge.

Given facts. Use this classroom classification: cool if mean temperature <12°C; mild if 12–17°C; warm if >17°C. Dry if annual precipitation <500 mm; wet if ≥ 500 mm. Data: Station A: T=9°C, P=490 mm; Station B: T=17°C, P=760 mm; Station C: T=13°C, P=530 mm.

Rules. Apply only the stated thresholds. Temperature and precipitation are separate dimensions, so each station receives two labels.

Task. Classify Station B. Then identify whether any other station has the same two-label class.

Step-by-step solution. 1. Compare T=17 with the temperature thresholds: it is mild. 2. Compare P=760 with 500 mm: it is wet. 3. Compare the combined label with the other stations; matches: ['Station C'].

Answer. Station B is mild and wet; same class: Station C.

Knowledge takeaway. Classification is the application of clear predicates to data; the rule must be stated before the label is inferred.

Formal model. Numeric predicates with thresholds; output the conjunction of the temperature and precipitation categories.

Problem 538. Classifying climate data: Station C

Knowledge context. Climate descriptions summarize long-term patterns. In a reasoning problem, explicit thresholds let us classify numerical data without hidden knowledge.

Given facts. Use this classroom classification: cool if mean temperature <12°C; mild if 12–17°C; warm if >17°C. Dry if annual precipitation <500 mm; wet if ≥ 500 mm. Data: Station A: T=10°C, P=490 mm; Station B: T=18°C, P=760 mm; Station C: T=14°C, P=540 mm.

Rules. Apply only the stated thresholds. Temperature and precipitation are separate dimensions, so each station receives two labels.

Task. Classify Station C. Then identify whether any other station has the same two-label class.

Step-by-step solution. 1. Compare T=14 with the temperature thresholds: it is mild. 2. Compare P=540 with 500 mm: it is wet. 3. Compare the combined label with the other stations; matches: none.

Answer. Station C is mild and wet; no other station has the same class.

Knowledge takeaway. Classification is the application of clear predicates to data; the rule must be stated before the label is inferred.

Formal model. Numeric predicates with thresholds; output the conjunction of the temperature and precipitation categories.

Problem 539. Classifying climate data: Station A

Knowledge context. Climate descriptions summarize long-term patterns. In a reasoning problem, explicit thresholds let us classify numerical data without hidden knowledge.

Given facts. Use this classroom classification: cool if mean temperature <12°C; mild if 12–17°C; warm if >17°C. Dry if annual precipitation <500 mm; wet if ≥ 500 mm. Data: Station A: T=11°C, P=490 mm; Station B: T=19°C, P=760 mm; Station C: T=15°C, P=550 mm.

Rules. Apply only the stated thresholds. Temperature and precipitation are separate dimensions, so each station receives two labels.

Task. Classify Station A. Then identify whether any other station has the same two-label class.

Step-by-step solution. 1. Compare T=11 with the temperature thresholds: it is cool. 2. Compare P=490 with 500 mm: it is dry. 3. Compare the combined label with the other stations; matches: none.

Answer. Station A is cool and dry; no other station has the same class.

Knowledge takeaway. Classification is the application of clear predicates to data; the rule must be stated before the label is inferred.

Formal model. Numeric predicates with thresholds; output the conjunction of the temperature and precipitation categories.

Problem 540. Classifying climate data: Station B

Knowledge context. Climate descriptions summarize long-term patterns. In a reasoning problem, explicit thresholds let us classify numerical data without hidden knowledge.

Given facts. Use this classroom classification: cool if mean temperature <12°C; mild if 12–17°C; warm if >17°C. Dry if annual precipitation <500 mm; wet if ≥ 500 mm. Data: Station A: T=12°C, P=490 mm; Station B: T=20°C, P=760 mm; Station C: T=16°C, P=560 mm.

Rules. Apply only the stated thresholds. Temperature and precipitation are separate dimensions, so each station receives two labels.

Task. Classify Station B. Then identify whether any other station has the same two-label class.

Step-by-step solution. 1. Compare T=20 with the temperature thresholds: it is warm. 2. Compare P=760 with 500 mm: it is wet. 3. Compare the combined label with the other stations; matches: none.

Answer. Station B is warm and wet; no other station has the same class.

Knowledge takeaway. Classification is the application of clear predicates to data; the rule must be stated before the label is inferred.

Formal model. Numeric predicates with thresholds; output the conjunction of the temperature and precipitation categories.

G9. Exchange, routes, and capacities

Problem 541. Bottlenecks in the Iris–Pine exchange network

Knowledge context. Trade and transport networks have capacities. A single narrow connection can limit a much larger network.

Given facts. Goods move from Iris to Pine along two branches: Iris→Kite capacity 6, Kite→Pine capacity 5; Iris→Birch capacity 2, Birch→Pine capacity 7. The final warehouse at Pine can accept at most 9 units.

Rules. A branch cannot carry more than its smallest edge capacity. Independent branch capacities can be added, but a shared destination capacity can cap the total.

Task. What is the maximum number of units that can reach Pine in one period? Identify the bottleneck(s).

Step-by-step solution. 1. Branch 1 is limited by min(6,5)=5. 2. Branch 2 is limited by min(2,7)=2. 3. Together the branches could carry 7 units. 4. The destination accepts at most 9, so the final maximum is min(7,9)=7.

Answer. 7 units.

Knowledge takeaway. A bottleneck is a local limit that constrains the total possible flow.

Formal model. Flow network; branch capacity is a minimum along a path; total is the sum of independent branches subject to shared caps.

Problem 542. Bottlenecks in the Juniper–Dover exchange network

Knowledge context. Trade and transport networks have capacities. A single narrow connection can limit a much larger network.

Given facts. Goods move from Juniper to Dover along two branches: Juniper→Riverbend capacity 7, Riverbend→Dover capacity 5; Juniper→Linden capacity 3, Linden→Dover capacity 7. The final warehouse at Dover can accept at most 9 units.

Rules. A branch cannot carry more than its smallest edge capacity. Independent branch capacities can be added, but a shared destination capacity can cap the total.

Task. What is the maximum number of units that can reach Dover in one period? Identify the bottleneck(s).

Step-by-step solution. 1. Branch 1 is limited by min(7,5)=5. 2. Branch 2 is limited by min(3,7)=3. 3. Together the branches could carry 8 units. 4. The destination accepts at most 9, so the final maximum is min(8,9)=8.

Answer. 8 units.

Knowledge takeaway. A bottleneck is a local limit that constrains the total possible flow.

Formal model. Flow network; branch capacity is a minimum along a path; total is the sum of independent branches subject to shared caps.

Problem 543. Bottlenecks in the Stone–Cedar exchange network

Knowledge context. Trade and transport networks have capacities. A single narrow connection can limit a much larger network.

Given facts. Goods move from Stone to Cedar along two branches: Stone→Willow capacity 8, Willow→Cedar capacity 5; Stone→Juniper capacity 4, Juniper→Cedar capacity 7. The final warehouse at Cedar can accept at most 9 units.

Rules. A branch cannot carry more than its smallest edge capacity. Independent branch capacities can be added, but a shared destination capacity can cap the total.

Task. What is the maximum number of units that can reach Cedar in one period? Identify the bottleneck(s).

Step-by-step solution. 1. Branch 1 is limited by min(8,5)=5. 2. Branch 2 is limited by min(4,7)=4. 3. Together the branches could carry 9 units. 4. The destination accepts at most 9, so the final maximum is min(9,9)=9.

Answer. 9 units.

Knowledge takeaway. A bottleneck is a local limit that constrains the total possible flow.

Formal model. Flow network; branch capacity is a minimum along a path; total is the sum of independent branches subject to shared caps.

Problem 544. Bottlenecks in the Northfield–Willow exchange network

Knowledge context. Trade and transport networks have capacities. A single narrow connection can limit a much larger network.

Given facts. Goods move from Northfield to Willow along two branches: Northfield→Dover capacity 9, Dover→Willow capacity 5; Northfield→Meadow capacity 5, Meadow→Willow capacity 7. The final warehouse at Willow can accept at most 9 units.

Rules. A branch cannot carry more than its smallest edge capacity. Independent branch capacities can be added, but a shared destination capacity can cap the total.

Task. What is the maximum number of units that can reach Willow in one period? Identify the bottleneck(s).

Step-by-step solution. 1. Branch 1 is limited by min(9,5)=5. 2. Branch 2 is limited by min(5,7)=5. 3. Together the branches could carry 10 units. 4. The destination accepts at most 9, so the final maximum is min(10,9)=9.

Answer. 9 units.

Knowledge takeaway. A bottleneck is a local limit that constrains the total possible flow.

Formal model. Flow network; branch capacity is a minimum along a path; total is the sum of independent branches subject to shared caps.

Problem 545. Bottlenecks in the Northfield–Riverbend exchange network

Knowledge context. Trade and transport networks have capacities. A single narrow connection can limit a much larger network.

Given facts. Goods move from Northfield to Riverbend along two branches: Northfield→Linden capacity 10, Linden→Riverbend capacity 5; Northfield→Harbor capacity 6, Harbor→Riverbend capacity 7. The final warehouse at Riverbend can accept at most 9 units.

Rules. A branch cannot carry more than its smallest edge capacity. Independent branch capacities can be added, but a shared destination capacity can cap the total.

Task. What is the maximum number of units that can reach Riverbend in one period? Identify the bottleneck(s).

Step-by-step solution. 1. Branch 1 is limited by min(10,5)=5. 2. Branch 2 is limited by min(6,7)=6. 3. Together the branches could carry 11 units. 4. The destination accepts at most 9, so the final maximum is min(11,9)=9.

Answer. 9 units.

Knowledge takeaway. A bottleneck is a local limit that constrains the total possible flow.

Formal model. Flow network; branch capacity is a minimum along a path; total is the sum of independent branches subject to shared caps.

G10. Regions, sets, and membership

Problem 546. Nested places: Linden, Highridge, and Republic B

Knowledge context. Geography uses nested categories: a locality can be inside a region, a region inside a state, and a state inside a continent.

Given facts. Linden is in Highridge. Highridge is in Republic B. Republic B is in the continent of Pacifica. Another city, Willow, is also in Republic B, but its region is not specified.

Rules. If X is contained in Y and Y is contained in Z, then X is contained in Z. The reverse inference is not valid: knowing that two places are in one country does not prove that they are in the same region.

Task. Is Linden in Republic B?

Step-by-step solution. 1. Follow the chain city→region→country. 2. Containment is transitive, so the city is in the country.

Answer. Yes, Linden is in Republic B.

Knowledge takeaway. Hierarchies support upward transitive inference, but they do not justify arbitrary reverse inferences.

Formal model. Containment relation with transitivity; answer membership queries while rejecting invalid converse inference.

Problem 547. Nested places: Alder, Green Valley, and Republic A

Knowledge context. Geography uses nested categories: a locality can be inside a region, a region inside a state, and a state inside a continent.

Given facts. Alder is in Green Valley. Green Valley is in Republic A. Republic A is in the continent of Aurora. Another city, Fern, is also in Republic A, but its region is not specified.

Rules. If X is contained in Y and Y is contained in Z, then X is contained in Z. The reverse inference is not valid: knowing that two places are in one country does not prove that they are in the same region.

Task. Is Alder in Aurora?

Step-by-step solution. 1. Follow city→region→country→continent. 2. Transitivity carries membership through the chain.

Answer. Yes, Alder is in Aurora.

Knowledge takeaway. Hierarchies support upward transitive inference, but they do not justify arbitrary reverse inferences.

Formal model. Containment relation with transitivity; answer membership queries while rejecting invalid converse inference.

Problem 548. Nested places: Alder, Coastland, and Republic B

Knowledge context. Geography uses nested categories: a locality can be inside a region, a region inside a state, and a state inside a continent.

Given facts. Alder is in Coastland. Coastland is in Republic B. Republic B is in the continent of Meridian. Another city, Pine, is also in Republic B, but its region is not specified.

Rules. If X is contained in Y and Y is contained in Z, then X is contained in Z. The reverse inference is not valid: knowing that two places are in one country does not prove that they are in the same region.

Task. Can we conclude that Republic B is in Alder?

Step-by-step solution. 1. The facts go from smaller units to larger units. 2. No rule allows country→city, so the reverse claim is unsupported.

Answer. No. The containment relation cannot be reversed.

Knowledge takeaway. Hierarchies support upward transitive inference, but they do not justify arbitrary reverse inferences.

Formal model. Containment relation with transitivity; answer membership queries while rejecting invalid converse inference.

Problem 549. Nested places: Zephyr, Highridge, and Republic B

Knowledge context. Geography uses nested categories: a locality can be inside a region, a region inside a state, and a state inside a continent.

Given facts. Zephyr is in Highridge. Highridge is in Republic B. Republic B is in the continent of Meridian. Another city, Juniper, is also in Republic B, but its region is not specified.

Rules. If X is contained in Y and Y is contained in Z, then X is contained in Z. The reverse inference is not valid: knowing that two places are in one country does not prove that they are in the same region.

Task. Which larger units contain Zephyr?

Step-by-step solution. 1. Read the direct container: Highridge. 2. Propagate upward to Republic B and Meridian.

Answer. Highridge, Republic B, and Meridian.

Knowledge takeaway. Hierarchies support upward transitive inference, but they do not justify arbitrary reverse inferences.

Formal model. Containment relation with transitivity; answer membership queries while rejecting invalid converse inference.

Problem 550. Nested places: Zephyr, Coastland, and Republic B

Knowledge context. Geography uses nested categories: a locality can be inside a region, a region inside a state, and a state inside a continent.

Given facts. Zephyr is in Coastland. Coastland is in Republic B. Republic B is in the continent of Meridian. Another city, Pine, is also in Republic B, but its region is not specified.

Rules. If X is contained in Y and Y is contained in Z, then X is contained in Z. The reverse inference is not valid: knowing that two places are in one country does not prove that they are in the same region.

Task. Does being in Republic B prove two cities share a region?

Step-by-step solution. 1. Two cities can belong to the same country while belonging to different regions. 2. The region relation is therefore not deducible from country membership alone.

Answer. No. Country membership alone is insufficient to identify the region.

Knowledge takeaway. Hierarchies support upward transitive inference, but they do not justify arbitrary reverse inferences.

Formal model. Containment relation with transitivity; answer membership queries while rejecting invalid converse inference.

H1. Ordering events

Problem 551. Ordering the events of Westmoor

Knowledge context. Historical chronology often starts from partial information: some events are ordered relative to others even when exact dates are not known.

Given facts. Founding happened before New school. New school happened before Market charter. Flood happened after Founding. Bridge opening happened after Market charter.

Rules. “Before” is transitive: if A is before B and B is before C, then A is before C. If two events are not connected by the facts, their relative order may be undetermined.

Task. What must be true about the order of Founding and Market charter? Can the exact order of Flood and New school be determined?

Step-by-step solution. 1. From Founding before New school and New school before Market charter, infer Founding before Market charter. 2. Flood is only constrained to be after Founding. There is no stated relation between Flood and New school. 3. Therefore the first comparison is forced but the second is underdetermined.

Answer. Founding must be before Market charter. The relative order of Flood and New school cannot be determined from the given facts.

Knowledge takeaway. A careful reasoner distinguishes what must be true from what merely could be true.

Formal model. Partial order / directed acyclic graph; use transitive closure and detect incomparable events.

Problem 552. Ordering the events of Coastland

Knowledge context. Historical chronology often starts from partial information: some events are ordered relative to others even when exact dates are not known.

Given facts. Market charter happened before Bridge opening. Bridge opening happened before Founding. Flood happened after Market charter. New school happened after Founding. Cross-domain check: on a field sketch, 1 cm represents 4 km and two checkpoints are 8 cm apart.

Rules. “Before” is transitive: if A is before B and B is before C, then A is before C. If two events are not connected by the facts, their relative order may be undetermined. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. What must be true about the order of Market charter and Founding? Can the exact order of Flood and Bridge opening be determined? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. From Market charter before Bridge opening and Bridge opening before Founding, infer Market charter before Founding. 2. Flood is only constrained to be after Market charter. There is no stated relation between Flood and Bridge opening. 3. Therefore the first comparison is forced but the second is underdetermined. Cross-domain check: 8×4=32 km.

Answer. Market charter must be before Founding. The relative order of Flood and Bridge opening cannot be determined from the given facts. Cross-domain answer: 32 km.

Knowledge takeaway. A careful reasoner distinguishes what must be true from what merely could be true.

Formal model. Partial order / directed acyclic graph; use transitive closure and detect incomparable events.

Problem 553. Ordering the events of Lake District

Knowledge context. Historical chronology often starts from partial information: some events are ordered relative to others even when exact dates are not known.

Given facts. Market charter happened before New school. New school happened before Bridge opening. Flood happened after Market charter. Founding happened after Bridge opening.

Rules. “Before” is transitive: if A is before B and B is before C, then A is before C. If two events are not connected by the facts, their relative order may be undetermined.

Task. What must be true about the order of Market charter and Bridge opening? Can the exact order of Flood and New school be determined?

Step-by-step solution. 1. From Market charter before New school and New school before Bridge opening, infer Market charter before Bridge opening. 2. Flood is only constrained to be after Market charter. There is no stated relation between Flood and New school. 3. Therefore the first comparison is forced but the second is underdetermined.

Answer. Market charter must be before Bridge opening. The relative order of Flood and New school cannot be determined from the given facts.

Knowledge takeaway. A careful reasoner distinguishes what must be true from what merely could be true.

Formal model. Partial order / directed acyclic graph; use transitive closure and detect incomparable events.

Problem 554. Ordering the events of Eastvale

Knowledge context. Historical chronology often starts from partial information: some events are ordered relative to others even when exact dates are not known.

Given facts. Flood happened before New school. New school happened before Founding. Market charter happened after Flood. Bridge opening happened after Founding.

Rules. “Before” is transitive: if A is before B and B is before C, then A is before C. If two events are not connected by the facts, their relative order may be undetermined.

Task. What must be true about the order of Flood and Founding? Can the exact order of Market charter and New school be determined?

Step-by-step solution. 1. From Flood before New school and New school before Founding, infer Flood before Founding. 2. Market charter is only constrained to be after Flood. There is no stated relation between Market charter and New school. 3. Therefore the first comparison is forced but the second is underdetermined.

Answer. Flood must be before Founding. The relative order of Market charter and New school cannot be determined from the given facts.

Knowledge takeaway. A careful reasoner distinguishes what must be true from what merely could be true.

Formal model. Partial order / directed acyclic graph; use transitive closure and detect incomparable events.

Problem 555. Ordering the events of Green Valley

Knowledge context. Historical chronology often starts from partial information: some events are ordered relative to others even when exact dates are not known.

Given facts. Flood happened before Market charter. Market charter happened before New school. Bridge opening happened after Flood. Founding happened after New school.

Rules. “Before” is transitive: if A is before B and B is before C, then A is before C. If two events are not connected by the facts, their relative order may be undetermined.

Task. What must be true about the order of Flood and New school? Can the exact order of Bridge opening and Market charter be determined?

Step-by-step solution. 1. From Flood before Market charter and Market charter before New school, infer Flood before New school. 2. Bridge opening is only constrained to be after Flood. There is no stated relation between Bridge opening and Market charter. 3. Therefore the first comparison is forced but the second is underdetermined.

Answer. Flood must be before New school. The relative order of Bridge opening and Market charter cannot be determined from the given facts.

Knowledge takeaway. A careful reasoner distinguishes what must be true from what merely could be true.

Formal model. Partial order / directed acyclic graph; use transitive closure and detect incomparable events.

H2. Dates, durations, and centuries

Problem 556. Duration and centuries: 1560–1589

Knowledge context. Dates can be used as numbers, but historical labels such as centuries follow explicit conventions.

Given facts. An institution was founded in 1560 and reorganized in 1589. Use the rule: years 1–100 are the 1st century, 101–200 the 2nd, and so on. Duration is end year minus start year.

Rules. Apply the century rule mechanically; do not infer a century from the first two digits alone. Compute elapsed time by subtraction.

Task. How many years passed between the two events? In which centuries did 1560 and 1589 fall?

Step-by-step solution. 1. Elapsed time: 1589−1560=29 years. 2. Century(1560)=floor((1560−1)/100)+1=16. 3. Century(1589)=16.

Answer. 29 years; 1560 is in the 16th century and 1589 in the 16th century.

Knowledge takeaway. Chronological reasoning combines arithmetic with a naming convention; both should be applied explicitly.

Formal model. Integer arithmetic plus century function c(y)=floor((y−1)/100)+1.

Problem 557. Duration and centuries: 1586–1620

Knowledge context. Dates can be used as numbers, but historical labels such as centuries follow explicit conventions.

Given facts. An institution was founded in 1586 and reorganized in 1620. Use the rule: years 1–100 are the 1st century, 101–200 the 2nd, and so on. Duration is end year minus start year.

Rules. Apply the century rule mechanically; do not infer a century from the first two digits alone. Compute elapsed time by subtraction.

Task. How many years passed between the two events? In which centuries did 1586 and 1620 fall?

Step-by-step solution. 1. Elapsed time: 1620−1586=34 years. 2. Century(1586)=floor((1586−1)/100)+1=16. 3. Century(1620)=17.

Answer. 34 years; 1586 is in the 16th century and 1620 in the 17th century.

Knowledge takeaway. Chronological reasoning combines arithmetic with a naming convention; both should be applied explicitly.

Formal model. Integer arithmetic plus century function c(y)=floor((y−1)/100)+1.

Problem 558. Duration and centuries: 1598–1637

Knowledge context. Dates can be used as numbers, but historical labels such as centuries follow explicit conventions.

Given facts. An institution was founded in 1598 and reorganized in 1637. Use the rule: years 1–100 are the 1st century, 101–200 the 2nd, and so on. Duration is end year minus start year.

Rules. Apply the century rule mechanically; do not infer a century from the first two digits alone. Compute elapsed time by subtraction.

Task. How many years passed between the two events? In which centuries did 1598 and 1637 fall?

Step-by-step solution. 1. Elapsed time: 1637−1598=39 years. 2. Century(1598)=floor((1598−1)/100)+1=16. 3. Century(1637)=17.

Answer. 39 years; 1598 is in the 16th century and 1637 in the 17th century.

Knowledge takeaway. Chronological reasoning combines arithmetic with a naming convention; both should be applied explicitly.

Formal model. Integer arithmetic plus century function c(y)=floor((y−1)/100)+1.

Problem 559. Duration and centuries: 1626–1670

Knowledge context. Dates can be used as numbers, but historical labels such as centuries follow explicit conventions.

Given facts. An institution was founded in 1626 and reorganized in 1670. Use the rule: years 1–100 are the 1st century, 101–200 the 2nd, and so on. Duration is end year minus start year.

Rules. Apply the century rule mechanically; do not infer a century from the first two digits alone. Compute elapsed time by subtraction.

Task. How many years passed between the two events? In which centuries did 1626 and 1670 fall?

Step-by-step solution. 1. Elapsed time: 1670−1626=44 years. 2. Century(1626)=floor((1626−1)/100)+1=17. 3. Century(1670)=17.

Answer. 44 years; 1626 is in the 17th century and 1670 in the 17th century.

Knowledge takeaway. Chronological reasoning combines arithmetic with a naming convention; both should be applied explicitly.

Formal model. Integer arithmetic plus century function c(y)=floor((y−1)/100)+1.

Problem 560. Duration and centuries: 1638–1687

Knowledge context. Dates can be used as numbers, but historical labels such as centuries follow explicit conventions.

Given facts. An institution was founded in 1638 and reorganized in 1687. Use the rule: years 1–100 are the 1st century, 101–200 the 2nd, and so on. Duration is end year minus start year.

Rules. Apply the century rule mechanically; do not infer a century from the first two digits alone. Compute elapsed time by subtraction.

Task. How many years passed between the two events? In which centuries did 1638 and 1687 fall?

Step-by-step solution. 1. Elapsed time: 1687−1638=49 years. 2. Century(1638)=floor((1638−1)/100)+1=17. 3. Century(1687)=17.

Answer. 49 years; 1638 is in the 17th century and 1687 in the 17th century.

Knowledge takeaway. Chronological reasoning combines arithmetic with a naming convention; both should be applied explicitly.

Formal model. Integer arithmetic plus century function c(y)=floor((y−1)/100)+1.

H3. Sources and supported claims

Problem 561. What can one coin prove?

Knowledge context. Historical sources are evidence, not automatic answers. The strength of a conclusion depends on what the source actually records.

Given facts. A historian has one source: a coin. The source directly shows that people used metal objects. A student proposes two claims: C1: people used metal objects; C2: the whole population agreed with the ruler. Cross-domain check: a survey team starts work at 8:00 and works for 2 hour(s) without a break. Mixed-domain verification: a regional archive has 58 map sheets and receives 2 additional sheets during the exercise.

Rules. A source supports only claims that can reasonably be inferred from what it contains. Absence of evidence for a broad claim is not proof of that claim. For the cross-domain check, finishing time = starting time + duration. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. Which claim is directly supported, and which is not justified by this source alone? Cross-domain check: at what time does the team finish? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. The coin directly contains evidence for C1. 2. C2 is much broader and no stated feature of the source measures the opinion of the whole population. 3. Therefore C1 is supported and C2 is not established. Cross-domain check: 8+2=10, so the team finishes at 10:00. Mixed-domain verification: 58+2=60.

Answer. C1 is supported. C2 is not justified by the source alone. Cross-domain answer: 10:00. Mixed-domain answer: 60 map sheets.

Knowledge takeaway. Good historical reasoning matches the scope of a claim to the scope of the evidence.

Formal model. Claim-evidence compatibility relation; accept only claims entailed by the source attributes.

Problem 562. What can one tax register prove?

Knowledge context. Historical sources are evidence, not automatic answers. The strength of a conclusion depends on what the source actually records.

Given facts. A historian has one source: a tax register. The source directly shows that taxes were recorded. A student proposes two claims: C1: taxes were recorded; C2: the whole population agreed with the ruler. Cross-domain check: a local committee has 12 members, requires at least 7 present for quorum, and 11 are present.

Rules. A source supports only claims that can reasonably be inferred from what it contains. Absence of evidence for a broad claim is not proof of that claim. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. Which claim is directly supported, and which is not justified by this source alone? Cross-domain check: is quorum met?

Step-by-step solution. 1. The tax register directly contains evidence for C1. 2. C2 is much broader and no stated feature of the source measures the opinion of the whole population. 3. Therefore C1 is supported and C2 is not established. Cross-domain check: compare 11 with 7; $11 \geq 7$ is true.

Answer. C1 is supported. C2 is not justified by the source alone. Cross-domain answer: quorum is met.

Knowledge takeaway. Good historical reasoning matches the scope of a claim to the scope of the evidence.

Formal model. Claim-evidence compatibility relation; accept only claims entailed by the source attributes.

Problem 563. What can one map prove?

Knowledge context. Historical sources are evidence, not automatic answers. The strength of a conclusion depends on what the source actually records.

Given facts. A historian has one source: a map. The source directly shows that a road was mapped. A student proposes two claims: C1: a road was mapped; C2: the whole population agreed with the ruler. Cross-domain check: researchers collected 9 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. A source supports only claims that can reasonably be inferred from what it contains. Absence of evidence for a broad claim is not proof of that claim. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. Which claim is directly supported, and which is not justified by this source alone? Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. The map directly contains evidence for C1. 2. C2 is much broader and no stated feature of the source measures the opinion of the whole population. 3. Therefore C1 is supported and C2 is not established. Cross-domain check: 9−4=5 independent reports remain.

Answer. C1 is supported. C2 is not justified by the source alone. Cross-domain answer: 5 independent reports.

Knowledge takeaway. Good historical reasoning matches the scope of a claim to the scope of the evidence.

Formal model. Claim-evidence compatibility relation; accept only claims entailed by the source attributes.

Problem 564. What can one tax register prove?

Knowledge context. Historical sources are evidence, not automatic answers. The strength of a conclusion depends on what the source actually records.

Given facts. A historian has one source: a tax register. The source directly shows that taxes were recorded. A student proposes two claims: C1: taxes were recorded; C2: the whole population agreed with the ruler. Cross-domain check: on a field sketch, 1 cm represents 6 km and two checkpoints are 6 cm apart.

Rules. A source supports only claims that can reasonably be inferred from what it contains. Absence of evidence for a broad claim is not proof of that claim. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Which claim is directly supported, and which is not justified by this source alone? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. The tax register directly contains evidence for C1. 2. C2 is much broader and no stated feature of the source measures the opinion of the whole population. 3. Therefore C1 is supported and C2 is not established. Cross-domain check: 6×6=36 km.

Answer. C1 is supported. C2 is not justified by the source alone. Cross-domain answer: 36 km.

Knowledge takeaway. Good historical reasoning matches the scope of a claim to the scope of the evidence.

Formal model. Claim-evidence compatibility relation; accept only claims entailed by the source attributes.

Problem 565. What can one tool prove?

Knowledge context. Historical sources are evidence, not automatic answers. The strength of a conclusion depends on what the source actually records.

Given facts. A historian has one source: a tool. The source directly shows that a tool existed. A student proposes two claims: C1: a tool existed; C2: the whole population agreed with the ruler. Cross-domain check: a survey team starts work at 12:00 and works for 2 hour(s) without a break.

Rules. A source supports only claims that can reasonably be inferred from what it contains. Absence of evidence for a broad claim is not proof of that claim. For the cross-domain check, finishing time = starting time + duration.

Task. Which claim is directly supported, and which is not justified by this source alone? Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. The tool directly contains evidence for C1. 2. C2 is much broader and no stated feature of the source measures the opinion of the whole population. 3. Therefore C1 is supported and C2 is not established. Cross-domain check: 12+2=14, so the team finishes at 14:00.

Answer. C1 is supported. C2 is not justified by the source alone. Cross-domain answer: 14:00.

Knowledge takeaway. Good historical reasoning matches the scope of a claim to the scope of the evidence.

Formal model. Claim-evidence compatibility relation; accept only claims entailed by the source attributes.

H4. Conflicting sources

Problem 566. Resolving conflicting sources: case 1

Knowledge context. Historical accounts can disagree. Reasoning requires evaluating how a source knows what it claims, not simply counting statements.

Given facts. Source A, written on the day of an event, says 40 carts arrived. Source B, copied 80 years later from an unknown account, says 90 carts arrived. Source C is an independent warehouse record from the same week and lists 42 carts. Cross-domain check: a local committee has 16 members, requires at least 9 present for quorum, and 14 are present.

Rules. When sources conflict, compare proximity to the event, independence, and whether a source has a reason or mechanism to record the specific fact. Agreement between independent relevant sources increases support but does not create absolute certainty. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. Which account is better supported by the evidence given, and why? Cross-domain check: is quorum met?

Step-by-step solution. 1. A is contemporary and reports 40. 2. C is an independent operational record and reports 42, close to A. 3. B is much later and has an unknown chain of transmission. 4. The cluster near 40–42 is therefore better supported than 90. Cross-domain check: compare 14 with 9; $14 \geq 9$ is true.

Answer. An arrival count near 40–42 carts is better supported than 90. Cross-domain answer: quorum is met.

Knowledge takeaway. Independent, relevant, well-positioned evidence usually deserves more weight than late repetition with unknown provenance.

Formal model. Evidence graph with source reliability/independence attributes; compare support for competing hypotheses.

Problem 567. Resolving conflicting sources: case 2

Knowledge context. Historical accounts can disagree. Reasoning requires evaluating how a source knows what it claims, not simply counting statements.

Given facts. Source A, an eyewitness diary, says the bridge opened on Monday. Source B, a later textbook, says Tuesday. Source C, an independent dated toll record, begins charging on Monday. Cross-domain check: researchers collected 13 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. When sources conflict, compare proximity to the event, independence, and whether a source has a reason or mechanism to record the specific fact. Agreement between independent relevant sources increases support but does not create absolute certainty. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. Which account is better supported by the evidence given, and why? Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. A and C are close to the event and independent. 2. Both point to Monday. 3. B is later and conflicts with two stronger, independent pieces of evidence. Cross-domain check: 13−4=9 independent reports remain.

Answer. Monday is better supported, while remaining a historical conclusion rather than absolute proof. Cross-domain answer: 9 independent reports.

Knowledge takeaway. Independent, relevant, well-positioned evidence usually deserves more weight than late repetition with unknown provenance.

Formal model. Evidence graph with source reliability/independence attributes; compare support for competing hypotheses.

Problem 568. Resolving conflicting sources: case 3

Knowledge context. Historical accounts can disagree. Reasoning requires evaluating how a source knows what it claims, not simply counting statements.

Given facts. Source A, written on the day of an event, says 40 carts arrived. Source B, copied 80 years later from an unknown account, says 90 carts arrived. Source C is an independent warehouse record from the same week and lists 42 carts. Cross-domain check: on a field sketch, 1 cm represents 5 km and two checkpoints are 3 cm apart.

Rules. When sources conflict, compare proximity to the event, independence, and whether a source has a reason or mechanism to record the specific fact. Agreement between independent relevant sources increases support but does not create absolute certainty. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Which account is better supported by the evidence given, and why? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. A is contemporary and reports 40. 2. C is an independent operational record and reports 42, close to A. 3. B is much later and has an unknown chain of transmission. 4. The cluster near 40–42 is therefore better supported than 90. Cross-domain check: 3×5=15 km.

Answer. An arrival count near 40–42 carts is better supported than 90. Cross-domain answer: 15 km.

Knowledge takeaway. Independent, relevant, well-positioned evidence usually deserves more weight than late repetition with unknown provenance.

Formal model. Evidence graph with source reliability/independence attributes; compare support for competing hypotheses.

Problem 569. Resolving conflicting sources: case 4

Knowledge context. Historical accounts can disagree. Reasoning requires evaluating how a source knows what it claims, not simply counting statements.

Given facts. Source A, an eyewitness diary, says the bridge opened on Monday. Source B, a later textbook, says Tuesday. Source C, an independent dated toll record, begins charging on Monday. Cross-domain check: a survey team starts work at 8:00 and works for 2 hour(s) without a break.

Rules. When sources conflict, compare proximity to the event, independence, and whether a source has a reason or mechanism to record the specific fact. Agreement between independent relevant sources increases support but does not create absolute certainty. For the cross-domain check, finishing time = starting time + duration.

Task. Which account is better supported by the evidence given, and why? Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. A and C are close to the event and independent. 2. Both point to Monday. 3. B is later and conflicts with two stronger, independent pieces of evidence. Cross-domain check: 8+2=10, so the team finishes at 10:00.

Answer. Monday is better supported, while remaining a historical conclusion rather than absolute proof. Cross-domain answer: 10:00.

Knowledge takeaway. Independent, relevant, well-positioned evidence usually deserves more weight than late repetition with unknown provenance.

Formal model. Evidence graph with source reliability/independence attributes; compare support for competing hypotheses.

Problem 570. Resolving conflicting sources: case 5

Knowledge context. Historical accounts can disagree. Reasoning requires evaluating how a source knows what it claims, not simply counting statements.

Given facts. Source A, written on the day of an event, says 40 carts arrived. Source B, copied 80 years later from an unknown account, says 90 carts arrived. Source C is an independent warehouse record from the same week and lists 42

carts. Cross-domain check: a local committee has 11 members, requires at least 6 present for quorum, and 11 are present. Mixed-domain verification: a regional archive has 59 map sheets and receives 1 additional sheets during the exercise.

Rules. When sources conflict, compare proximity to the event, independence, and whether a source has a reason or mechanism to record the specific fact. Agreement between independent relevant sources increases support but does not create absolute certainty. For the cross-domain check, quorum exists when present members ≥ the stated threshold. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. Which account is better supported by the evidence given, and why? Cross-domain check: is quorum met? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. A is contemporary and reports 40. 2. C is an independent operational record and reports 42, close to A. 3. B is much later and has an unknown chain of transmission. 4. The cluster near 40–42 is therefore better supported than 90. Cross-domain check: compare 11 with 6; 11 ≥ 6 is true. Mixed-domain verification: 59+1=60.

Answer. An arrival count near 40–42 carts is better supported than 90. Cross-domain answer: quorum is met. Mixed-domain answer: 60 map sheets.

Knowledge takeaway. Independent, relevant, well-positioned evidence usually deserves more weight than late repetition with unknown provenance.

Formal model. Evidence graph with source reliability/independence attributes; compare support for competing hypotheses.

H5. Causes, conditions, and consequences

Problem 571. Cause-and-consequence chain: case 1

Knowledge context. Many explanations can be represented as condition-and-consequence rules. The logical form matters: necessary, sufficient, AND, and OR are not interchangeable.

Given facts. Facts: Rain occurred; The river was already high; The warning siren worked; The gate was closed. Rules: If rain occurs AND the river is already high, flooding occurs. If flooding occurs, the riverside road closes. If the siren works, residents receive a warning. Cross-domain check: researchers collected 7 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. Use the rules exactly as written. AND requires all listed conditions; OR requires at least one. A one-way implication cannot automatically be reversed. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. Which consequences can be deduced by applying the rules exactly? Do not assume the converse of any rule. Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. Rain and high river satisfy both conditions of the first rule. 2. Infer flooding. 3. Flooding implies road closure. 4. The working siren independently implies a warning. Cross-domain check: 7−4=3 independent reports remain.

Answer. Flooding, road closure, and a warning can be deduced. Cross-domain answer: 3 independent reports.

Knowledge takeaway. Causal reasoning becomes clearer when every link is stated and the direction of implication is respected.

Formal model. Forward-chaining rule system over Boolean facts; apply implications until no new facts appear.

Problem 572. Cause-and-consequence chain: case 2

Knowledge context. Many explanations can be represented as condition-and-consequence rules. The logical form matters: necessary, sufficient, AND, and OR are not interchangeable.

Given facts. Facts: Rain occurred; The river was not high; The warning siren worked; The gate was closed. Rules: If rain occurs AND the river is already high, flooding occurs. If flooding occurs, the riverside road closes. If the siren works, residents receive a warning. Cross-domain check: on a field sketch, 1 cm represents 4 km and two checkpoints are 7 cm apart.

Rules. Use the rules exactly as written. AND requires all listed conditions; OR requires at least one. A one-way implication cannot automatically be reversed. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Which consequences can be deduced by applying the rules exactly? Do not assume the converse of any rule. Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. Rain is true but the high-river condition is false. 2. The AND rule for flooding therefore does not fire. 3. Without flooding, road closure is not inferred. 4. The siren rule still gives a warning. Cross-domain check: 7×4=28 km.

Answer. A warning can be deduced, but flooding and road closure cannot be deduced. Cross-domain answer: 28 km.

Knowledge takeaway. Causal reasoning becomes clearer when every link is stated and the direction of implication is respected.

Formal model. Forward-chaining rule system over Boolean facts; apply implications until no new facts appear.

Problem 573. Cause-and-consequence chain: case 3

Knowledge context. Many explanations can be represented as condition-and-consequence rules. The logical form matters: necessary, sufficient, AND, and OR are not interchangeable.

Given facts. Facts: Rain occurred; The river was already high; The warning siren worked; The gate was closed; Flooding occurred. Rules: If rain occurs AND the river is already high, flooding occurs. If flooding occurs, the riverside road closes. If the siren works, residents receive a warning. Cross-domain check: a survey team starts work at 12:00 and works for 2 hour(s) without a break.

Rules. Use the rules exactly as written. AND requires all listed conditions; OR requires at least one. A one-way implication cannot automatically be reversed. For the cross-domain check, finishing time = starting time + duration.

Task. Which consequences can be deduced by applying the rules exactly? Do not assume the converse of any rule. Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. Flooding is stated directly, so apply the flooding→road-closure rule. 2. Apply siren→warning. 3. Do not use flooding to infer rain or a high river; that would reverse a rule. Cross-domain check: 12+2=14, so the team finishes at 14:00.

Answer. Road closure and a warning can be deduced; flooding is already given as a fact. Cross-domain answer: 14:00.

Knowledge takeaway. Causal reasoning becomes clearer when every link is stated and the direction of implication is respected.

Formal model. Forward-chaining rule system over Boolean facts; apply implications until no new facts appear.

Problem 574. Cause-and-consequence chain: case 4

Knowledge context. Many explanations can be represented as condition-and-consequence rules. The logical form matters: necessary, sufficient, AND, and OR are not interchangeable.

Given facts. Facts: Rain occurred; The river was already high; The warning siren worked; The gate was closed. Rules: If rain occurs OR the river is already high, the field becomes muddy. If flooding occurs, the riverside road closes. If the siren works, residents receive a warning. Cross-domain check: a local committee has 15 members, requires at least 8 present for quorum, and 14 are present.

Rules. Use the rules exactly as written. AND requires all listed conditions; OR requires at least one. A one-way implication cannot automatically be reversed. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. Which consequences can be deduced by applying the rules exactly? Do not assume the converse of any rule. Cross-domain check: is quorum met?

Step-by-step solution. 1. The mud rule uses OR, so rain alone is enough. 2. Rain is given, therefore mud follows. 3. The siren rule gives a warning. Cross-domain check: compare 14 with 8; $14 \geq 8$ is true.

Answer. The field becomes muddy, and the warning follows from the working siren. Cross-domain answer: quorum is met.

Knowledge takeaway. Causal reasoning becomes clearer when every link is stated and the direction of implication is respected.

Formal model. Forward-chaining rule system over Boolean facts; apply implications until no new facts appear.

Problem 575. Cause-and-consequence chain: case 5

Knowledge context. Many explanations can be represented as condition-and-consequence rules. The logical form matters: necessary, sufficient, AND, and OR are not interchangeable.

Given facts. Facts: Rain occurred; The river was already high. Rules: If rain AND high river, flooding occurs. If flooding, road closes. If road closes, the bus reroutes. Cross-domain check: researchers collected 11 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. Use the rules exactly as written. AND requires all listed conditions; OR requires at least one. A one-way implication cannot automatically be reversed. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. Which consequences can be deduced by applying the rules exactly? Do not assume the converse of any rule. Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. Rain and high river jointly imply flooding. 2. Flooding implies road closure. 3. Road closure implies bus rerouting. Cross-domain check: 11−4=7 independent reports remain.

Answer. Flooding, road closure, and bus rerouting can all be deduced. Cross-domain answer: 7 independent reports.

Knowledge takeaway. Causal reasoning becomes clearer when every link is stated and the direction of implication is respected.

Formal model. Forward-chaining rule system over Boolean facts; apply implications until no new facts appear.

H6. Travel through time and space

Problem 576. Courier journey from Iris to Riverbend

Knowledge context. Historical travel can be analyzed with ordinary distance, speed, and time reasoning when the needed values are supplied.

Given facts. A courier leaves Iris for Riverbend at 7:00. Distance is 100 km and the stated average speed is 25 km/h. A council meeting in Riverbend begins at 10.0:00. Ignore stops and use travel time = distance ÷ speed.

Rules. Arrival time equals departure time plus travel time. To attend an event from its beginning, arrival must be no later than the event start.

Task. When does the courier arrive, and can the courier attend the meeting from its beginning?

Step-by-step solution. 1. Travel time=100÷25=4.0 hours. 2. Arrival=7.0+4.0=11.0. 3. Compare 11.0 with meeting time 10.0.

Answer. Arrival at about 11.0:00; no, the meeting has already begun.

Knowledge takeaway. Chronology and geography interact: being able to reach a place in time is a constraint, not a fact to memorize.

Formal model. Arithmetic time model; t_arrive=t_depart+d/v; compare with event time.

Problem 577. Courier journey from Elm to Oak

Knowledge context. Historical travel can be analyzed with ordinary distance, speed, and time reasoning when the needed values are supplied.

Given facts. A courier leaves Elm for Oak at 8:00. Distance is 110 km and the stated average speed is 25 km/h. A council meeting in Oak begins at 13.4:00. Ignore stops and use travel time = distance ÷ speed.

Rules. Arrival time equals departure time plus travel time. To attend an event from its beginning, arrival must be no later than the event start.

Task. When does the courier arrive, and can the courier attend the meeting from its beginning?

Step-by-step solution. 1. Travel time=110÷25=4.4 hours. 2. Arrival=8.0+4.4=12.4. 3. Compare 12.4 with meeting time 13.4.

Answer. Arrival at about 12.4:00; yes, on time.

Knowledge takeaway. Chronology and geography interact: being able to reach a place in time is a constraint, not a fact to memorize.

Formal model. Arithmetic time model; t_arrive=t_depart+d/v; compare with event time.

Problem 578. Courier journey from Linden to Willow

Knowledge context. Historical travel can be analyzed with ordinary distance, speed, and time reasoning when the needed values are supplied.

Given facts. A courier leaves Linden for Willow at 9:00. Distance is 120 km and the stated average speed is 25 km/h. A council meeting in Willow begins at 12.8:00. Ignore stops and use travel time = distance ÷ speed.

Rules. Arrival time equals departure time plus travel time. To attend an event from its beginning, arrival must be no later than the event start.

Task. When does the courier arrive, and can the courier attend the meeting from its beginning?

Step-by-step solution. 1. Travel time=120÷25=4.8 hours. 2. Arrival=9.0+4.8=13.8. 3. Compare 13.8 with meeting time 12.8.

Answer. Arrival at about 13.8:00; no, the meeting has already begun.

Knowledge takeaway. Chronology and geography interact: being able to reach a place in time is a constraint, not a fact to memorize.

Formal model. Arithmetic time model; t_arrive=t_depart+d/v; compare with event time.

Problem 579. Courier journey from Cedar to Grove

Knowledge context. Historical travel can be analyzed with ordinary distance, speed, and time reasoning when the needed values are supplied.

Given facts. A courier leaves Cedar for Grove at 10:00. Distance is 130 km and the stated average speed is 25 km/h. A council meeting in Grove begins at 16.2:00. Ignore stops and use travel time = distance ÷ speed.

Rules. Arrival time equals departure time plus travel time. To attend an event from its beginning, arrival must be no later than the event start.

Task. When does the courier arrive, and can the courier attend the meeting from its beginning?

Step-by-step solution. 1. Travel time=130÷25=5.2 hours. 2. Arrival=10.0+5.2=15.2. 3. Compare 15.2 with meeting time 16.2.

Answer. Arrival at about 15.2:00; yes, on time.

Knowledge takeaway. Chronology and geography interact: being able to reach a place in time is a constraint, not a fact to memorize.

Formal model. Arithmetic time model; t_arrive=t_depart+d/v; compare with event time.

Problem 580. Courier journey from Grove to Iris

Knowledge context. Historical travel can be analyzed with ordinary distance, speed, and time reasoning when the needed values are supplied.

Given facts. A courier leaves Grove for Iris at 11:00. Distance is 140 km and the stated average speed is 25 km/h. A council meeting in Iris begins at 15.6:00. Ignore stops and use travel time = distance ÷ speed.

Rules. Arrival time equals departure time plus travel time. To attend an event from its beginning, arrival must be no later than the event start.

Task. When does the courier arrive, and can the courier attend the meeting from its beginning?

Step-by-step solution. 1. Travel time=140÷25=5.6 hours. 2. Arrival=11.0+5.6=16.6. 3. Compare 16.6 with meeting time 15.6.

Answer. Arrival at about 16.6:00; no, the meeting has already begun.

Knowledge takeaway. Chronology and geography interact: being able to reach a place in time is a constraint, not a fact to memorize.

Formal model. Arithmetic time model; t_arrive=t_depart+d/v; compare with event time.

H7. Change and continuity

Problem 581. Change and continuity in a settlement: case 1

Knowledge context. History studies both change and continuity. Comparing two snapshots is a systematic way to separate what persisted from what appeared or disappeared.

Given facts. Period 1 features: footpath, market, well, wooden bridge. Period 2 features: market, road, stone bridge, well. Cross-domain check: a survey team starts work at 12:00 and works for 2 hour(s) without a break.

Rules. A continuity is a feature present in both periods. A disappearance is present only in Period 1. An appearance is present only in Period 2. A replacement can be described when two differently named features serve the same stated role. For the cross-domain check, finishing time = starting time + duration.

Task. Identify the continuities, disappearances, and appearances. Which conclusions are direct set comparisons rather than interpretations? Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. Intersection gives continuities: ['market', 'well']. 2. Period1−Period2 gives disappearances: ['footpath', 'wooden bridge']. 3. Period2−Period1 gives appearances: ['road', 'stone bridge']. 4. Only role-based replacement would require extra information; the set comparisons are direct. Cross-domain check: 12+2=14, so the team finishes at 14:00.

Answer. Continuities: ['market', 'well']; disappearances: ['footpath', 'wooden bridge']; appearances: ['road', 'stone bridge']. Cross-domain answer: 14:00.

Knowledge takeaway. Change is not the opposite of continuity; both can occur in the same community at the same time.

Formal model. Set operations: intersection for continuity, set difference for appearance/disappearance.

Problem 582. Change and continuity in a settlement: case 2

Knowledge context. History studies both change and continuity. Comparing two snapshots is a systematic way to separate what persisted from what appeared or disappeared.

Given facts. Period 1 features: footpath, market, well, wooden bridge. Period 2 features: market, road, school, stone bridge, well. Cross-domain check: a local committee has 14 members, requires at least 8 present for quorum, and 14 are present.

Rules. A continuity is a feature present in both periods. A disappearance is present only in Period 1. An appearance is present only in Period 2. A replacement can be described when two differently named features serve the same stated role. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. Identify the continuities, disappearances, and appearances. Which conclusions are direct set comparisons rather than interpretations? Cross-domain check: is quorum met?

Step-by-step solution. 1. Intersection gives continuities: ['market', 'well']. 2. Period1−Period2 gives disappearances: ['footpath', 'wooden bridge']. 3. Period2−Period1 gives appearances: ['road', 'school', 'stone bridge']. 4. Only role-based replacement would require extra information; the set comparisons are direct. Cross-domain check: compare 14 with 8; 14 ≥ 8 is true.

Answer. Continuities: ['market', 'well']; disappearances: ['footpath', 'wooden bridge']; appearances: ['road', 'school', 'stone bridge']. Cross-domain answer: quorum is met.

Knowledge takeaway. Change is not the opposite of continuity; both can occur in the same community at the same time.

Formal model. Set operations: intersection for continuity, set difference for appearance/disappearance.

Problem 583. Change and continuity in a settlement: case 3

Knowledge context. History studies both change and continuity. Comparing two snapshots is a systematic way to separate what persisted from what appeared or disappeared.

Given facts. Period 1 features: footpath, market, mill, well, wooden bridge. Period 2 features: market, mill, road, stone bridge, well. Cross-domain check: researchers collected 9 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. A continuity is a feature present in both periods. A disappearance is present only in Period 1. An appearance is present only in Period 2. A replacement can be described when two differently named features serve the same stated role. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. Identify the continuities, disappearances, and appearances. Which conclusions are direct set comparisons rather than interpretations? Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. Intersection gives continuities: ['market', 'mill', 'well']. 2. Period1−Period2 gives disappearances: ['footpath', 'wooden bridge']. 3. Period2−Period1 gives appearances: ['road', 'stone bridge']. 4. Only role-based replacement would require extra information; the set comparisons are direct. Cross-domain check: 9−4=5 independent reports remain.

Answer. Continuities: ['market', 'mill', 'well']; disappearances: ['footpath', 'wooden bridge']; appearances: ['road', 'stone bridge']. Cross-domain answer: 5 independent reports.

Knowledge takeaway. Change is not the opposite of continuity; both can occur in the same community at the same time.

Formal model. Set operations: intersection for continuity, set difference for appearance/disappearance.

Problem 584. Change and continuity in a settlement: case 4

Knowledge context. History studies both change and continuity. Comparing two snapshots is a systematic way to separate what persisted from what appeared or disappeared.

Given facts. Period 1 features: footpath, market, well, wooden bridge. Period 2 features: market, road, stone bridge. Cross-domain check: on a field sketch, 1 cm represents 6 km and two checkpoints are 5 cm apart.

Rules. A continuity is a feature present in both periods. A disappearance is present only in Period 1. An appearance is present only in Period 2. A replacement can be described when two differently named features serve the same stated role. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Identify the continuities, disappearances, and appearances. Which conclusions are direct set comparisons rather than interpretations? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. Intersection gives continuities: ['market']. 2. Period1−Period2 gives disappearances: ['footpath', 'well', 'wooden bridge']. 3. Period2−Period1 gives appearances: ['road', 'stone bridge']. 4. Only role-based replacement would require extra information; the set comparisons are direct. Cross-domain check: 5×6=30 km.

Answer. Continuities: ['market']; disappearances: ['footpath', 'well', 'wooden bridge']; appearances: ['road', 'stone bridge']. Cross-domain answer: 30 km.

Knowledge takeaway. Change is not the opposite of continuity; both can occur in the same community at the same time.

Formal model. Set operations: intersection for continuity, set difference for appearance/disappearance.

Problem 585. Change and continuity in a settlement: case 5

Knowledge context. History studies both change and continuity. Comparing two snapshots is a systematic way to separate what persisted from what appeared or disappeared.

Given facts. Period 1 features: footpath, market, wall, well, wooden bridge. Period 2 features: market, park, road, stone bridge, well. Cross-domain check: a survey team starts work at 8:00 and works for 2 hour(s) without a break.

Rules. A continuity is a feature present in both periods. A disappearance is present only in Period 1. An appearance is present only in Period 2. A replacement can be described when two differently named features serve the same stated role. For the cross-domain check, finishing time = starting time + duration.

Task. Identify the continuities, disappearances, and appearances. Which conclusions are direct set comparisons rather than interpretations? Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. Intersection gives continuities: ['market', 'well']. 2. Period1−Period2 gives disappearances: ['footpath', 'wall', 'wooden bridge']. 3. Period2−Period1 gives appearances: ['park', 'road', 'stone bridge']. 4. Only role-based replacement would require extra information; the set comparisons are direct. Cross-domain check: 8+2=10, so the team finishes at 10:00.

Answer. Continuities: ['market', 'well']; disappearances: ['footpath', 'wall', 'wooden bridge']; appearances: ['park', 'road', 'stone bridge']. Cross-domain answer: 10:00.

Knowledge takeaway. Change is not the opposite of continuity; both can occur in the same community at the same time.

Formal model. Set operations: intersection for continuity, set difference for appearance/disappearance.

H8. Layers and dating objects

Problem 586. Reading an archaeological sequence: case 1

Knowledge context. Archaeological layers provide relative chronological information. Dates from objects add constraints but do not automatically make every date exact.

Given facts. An undisturbed archaeological site has four layers from top to bottom: A, B, C, D. In an undisturbed sequence, lower layers were deposited earlier. D contains an object dated 1300, C contains one dated 1450, and B contains one dated 1600. Cross-domain check: a local committee has 9 members, requires at least 5 present for quorum, and 8 are present.

Rules. In an undisturbed sequence, a lower layer is older than a layer above it. A securely dated object provides a constraint on the layer that contains it, but one object does not date every other object exactly. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. Which is older, Layer C or Layer B? Can the exact year of deposition of Layer A be deduced? Cross-domain check: is quorum met?

Step-by-step solution. 1. C lies below B, so C is older than B by the stratigraphic rule. 2. No exact dated object or deposition year is given for A. 3. Therefore only relative ordering, not an exact year for A, is justified. Cross-domain check: compare 8 with 5; $8 \geq 5$ is true.

Answer. Layer C is older than Layer B. The exact deposition year of Layer A cannot be deduced. Cross-domain answer: quorum is met.

Knowledge takeaway. Relative dating and absolute dating answer different questions and should not be confused.

Formal model. Partial order from vertical position plus numeric constraints from dated objects.

Problem 587. Reading an archaeological sequence: case 2

Knowledge context. Archaeological layers provide relative chronological information. Dates from objects add constraints but do not automatically make every date exact.

Given facts. An undisturbed archaeological site has four layers from top to bottom: A, B, C, D. In an undisturbed sequence, lower layers were deposited earlier. D contains an object dated 1305, C contains one dated 1465, and B contains one dated 1620. Cross-domain check: researchers collected 13 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. In an undisturbed sequence, a lower layer is older than a layer above it. A securely dated object provides a constraint on the layer that contains it, but one object does not date every other object exactly. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. Which is older, Layer C or Layer B? Can the exact year of deposition of Layer A be deduced? Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. C lies below B, so C is older than B by the stratigraphic rule. 2. No exact dated object or deposition year is given for A. 3. Therefore only relative ordering, not an exact year for A, is justified. Cross-domain check: 13−4=9 independent reports remain.

Answer. Layer C is older than Layer B. The exact deposition year of Layer A cannot be deduced. Cross-domain answer: 9 independent reports.

Knowledge takeaway. Relative dating and absolute dating answer different questions and should not be confused.

Formal model. Partial order from vertical position plus numeric constraints from dated objects.

Problem 588. Reading an archaeological sequence: case 3

Knowledge context. Archaeological layers provide relative chronological information. Dates from objects add constraints but do not automatically make every date exact.

Given facts. An undisturbed archaeological site has four layers from top to bottom: A, B, C, D. In an undisturbed sequence, lower layers were deposited earlier. D contains an object dated 1310, C contains one dated 1480, and B contains one dated 1640. Cross-domain check: on a field sketch, 1 cm represents 5 km and two checkpoints are 2 cm apart.

Rules. In an undisturbed sequence, a lower layer is older than a layer above it. A securely dated object provides a constraint on the layer that contains it, but one object does not date every other object exactly. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Which is older, Layer C or Layer B? Can the exact year of deposition of Layer A be deduced? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. C lies below B, so C is older than B by the stratigraphic rule. 2. No exact dated object or deposition year is given for A. 3. Therefore only relative ordering, not an exact year for A, is justified. Cross-domain check: 2×5=10 km.

Answer. Layer C is older than Layer B. The exact deposition year of Layer A cannot be deduced. Cross-domain answer: 10 km.

Knowledge takeaway. Relative dating and absolute dating answer different questions and should not be confused.

Formal model. Partial order from vertical position plus numeric constraints from dated objects.

Problem 589. Reading an archaeological sequence: case 4

Knowledge context. Archaeological layers provide relative chronological information. Dates from objects add constraints but do not automatically make every date exact.

Given facts. An undisturbed archaeological site has four layers from top to bottom: A, B, C, D. In an undisturbed sequence, lower layers were deposited earlier. D contains an object dated 1315, C contains one dated 1495, and B contains one dated 1660. Cross-domain check: a survey team starts work at 12:00 and works for 2 hour(s) without a break.

Rules. In an undisturbed sequence, a lower layer is older than a layer above it. A securely dated object provides a constraint on the layer that contains it, but one object does not date every other object exactly. For the cross-domain check, finishing time = starting time + duration.

Task. Which is older, Layer C or Layer B? Can the exact year of deposition of Layer A be deduced? Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. C lies below B, so C is older than B by the stratigraphic rule. 2. No exact dated object or deposition year is given for A. 3. Therefore only relative ordering, not an exact year for A, is justified. Cross-domain check: 12+2=14, so the team finishes at 14:00.

Answer. Layer C is older than Layer B. The exact deposition year of Layer A cannot be deduced. Cross-domain answer: 14:00.

Knowledge takeaway. Relative dating and absolute dating answer different questions and should not be confused.

Formal model. Partial order from vertical position plus numeric constraints from dated objects.

Problem 590. Reading an archaeological sequence: case 5

Knowledge context. Archaeological layers provide relative chronological information. Dates from objects add constraints but do not automatically make every date exact.

Given facts. An undisturbed archaeological site has four layers from top to bottom: A, B, C, D. In an undisturbed sequence, lower layers were deposited earlier. D contains an object dated 1320, C contains one dated 1510, and B contains one dated 1680. Cross-domain check: a local committee has 13 members, requires at least 7 present for quorum, and 11 are present.

Rules. In an undisturbed sequence, a lower layer is older than a layer above it. A securely dated object provides a constraint on the layer that contains it, but one object does not date every other object exactly. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. Which is older, Layer C or Layer B? Can the exact year of deposition of Layer A be deduced? Cross-domain check: is quorum met?

Step-by-step solution. 1. C lies below B, so C is older than B by the stratigraphic rule. 2. No exact dated object or deposition year is given for A. 3. Therefore only relative ordering, not an exact year for A, is justified. Cross-domain check: compare 11 with 7; $11 \geq 7$ is true.

Answer. Layer C is older than Layer B. The exact deposition year of Layer A cannot be deduced. Cross-domain answer: quorum is met.

Knowledge takeaway. Relative dating and absolute dating answer different questions and should not be confused.

Formal model. Partial order from vertical position plus numeric constraints from dated objects.

C1. Rules and exceptions

Problem 591. Applying a rule with an exception: Eli

Knowledge context. Rules often have conditions and exceptions. Logical reading prevents us from applying a general rule when an overriding exception is active.

Given facts. Eli is 12 years old. A library rule says: a child may enter the archive room if age $\geq$ 13 AND an adult permit is present. Exception: during an evacuation drill, nobody may enter. Permit present: yes. Evacuation drill: no.

Rules. Check exceptions first if they override the general rule. Otherwise every condition joined by AND must be true.

Task. May Eli enter the archive room under these facts? Show which conditions succeed or fail.

Step-by-step solution. 1. Age test: 12 ≥ 13 is False. 2. Permit test is True. 3. Because the general rule uses AND, both tests must pass. 4. Therefore entry is not allowed.

Answer. No, Eli may not enter.

Knowledge takeaway. The order of reasoning matters when exceptions override ordinary rules.

Formal model. Boolean rule with priority: exception predicate first, then conjunction of eligibility conditions.

Problem 592. Applying a rule with an exception: Mina

Knowledge context. Rules often have conditions and exceptions. Logical reading prevents us from applying a general rule when an overriding exception is active.

Given facts. Mina is 13 years old. A library rule says: a child may enter the archive room if age ≥ 13 AND an adult permit is present. Exception: during an evacuation drill, nobody may enter. Permit present: no. Evacuation drill: no.

Rules. Check exceptions first if they override the general rule. Otherwise every condition joined by AND must be true.

Task. May Mina enter the archive room under these facts? Show which conditions succeed or fail.

Step-by-step solution. 1. Age test: 13 ≥ 13 is True. 2. Permit test is False. 3. Because the general rule uses AND, both tests must pass. 4. Therefore entry is not allowed.

Answer. No, Mina may not enter.

Knowledge takeaway. The order of reasoning matters when exceptions override ordinary rules.

Formal model. Boolean rule with priority: exception predicate first, then conjunction of eligibility conditions.

Problem 593. Applying a rule with an exception: Arun

Knowledge context. Rules often have conditions and exceptions. Logical reading prevents us from applying a general rule when an overriding exception is active.

Given facts. Arun is 14 years old. A library rule says: a child may enter the archive room if age ≥ 13 AND an adult permit is present. Exception: during an evacuation drill, nobody may enter. Permit present: yes. Evacuation drill: no.

Rules. Check exceptions first if they override the general rule. Otherwise every condition joined by AND must be true.

Task. May Arun enter the archive room under these facts? Show which conditions succeed or fail.

Step-by-step solution. 1. Age test: 14 ≥ 13 is True. 2. Permit test is True. 3. Because the general rule uses AND, both tests must pass. 4. Therefore entry is allowed.

Answer. Yes, Arun may enter.

Knowledge takeaway. The order of reasoning matters when exceptions override ordinary rules.

Formal model. Boolean rule with priority: exception predicate first, then conjunction of eligibility conditions.

Problem 594. Applying a rule with an exception: Maya

Knowledge context. Rules often have conditions and exceptions. Logical reading prevents us from applying a general rule when an overriding exception is active.

Given facts. Maya is 15 years old. A library rule says: a child may enter the archive room if age ≥ 13 AND an adult permit is present. Exception: during an evacuation drill, nobody may enter. Permit present: no. Evacuation drill: no.

Rules. Check exceptions first if they override the general rule. Otherwise every condition joined by AND must be true.

Task. May Maya enter the archive room under these facts? Show which conditions succeed or fail.

Step-by-step solution. 1. Age test: 15 ≥ 13 is True. 2. Permit test is False. 3. Because the general rule uses AND, both tests must pass. 4. Therefore entry is not allowed.

Answer. No, Maya may not enter.

Knowledge takeaway. The order of reasoning matters when exceptions override ordinary rules.

Formal model. Boolean rule with priority: exception predicate first, then conjunction of eligibility conditions.

Problem 595. Applying a rule with an exception: Noah

Knowledge context. Rules often have conditions and exceptions. Logical reading prevents us from applying a general rule when an overriding exception is active.

Given facts. Noah is 16 years old. A library rule says: a child may enter the archive room if age ≥ 13 AND an adult permit is present. Exception: during an evacuation drill, nobody may enter. Permit present: yes. Evacuation drill: yes.

Rules. Check exceptions first if they override the general rule. Otherwise every condition joined by AND must be true.

Task. May Noah enter the archive room under these facts? Show which conditions succeed or fail.

Step-by-step solution. 1. The evacuation exception applies and overrides ordinary eligibility. 2. Therefore entry is not allowed.

Answer. No, Noah may not enter.

Knowledge takeaway. The order of reasoning matters when exceptions override ordinary rules.

Formal model. Boolean rule with priority: exception predicate first, then conjunction of eligibility conditions.

C2. Rights as limits on decisions

Problem 596. Majority rule and a protected right: case 1

Knowledge context. Democratic procedures use voting, but constitutional systems can place some rights beyond ordinary majority power.

Given facts. A council of 20 members votes 15 to 5 for a proposal that would forbid one named group from the right to speak at a peaceful meeting. The classroom constitution states: (1) ordinary proposals pass by simple majority; (2) an ordinary majority may not remove the listed equal rights from a group.

Rules. A decision must satisfy both the voting procedure and the higher rights constraint. Passing a vote is not sufficient if the content violates a protected right.

Task. Did the proposal receive a majority? May it take effect under the stated constitution?

Step-by-step solution. 1. Vote check: 15>5, so the proposal has a simple majority. 2. Content check: the proposal removes a protected equal right (speak at a peaceful meeting) from a group. 3. The higher rights rule therefore blocks the proposal even though the vote passed.

Answer. It received a majority, but it may not take effect under the stated rights rule.

Knowledge takeaway. A majority answers who won a vote; it does not by itself answer whether every possible decision is lawful or fair under higher rules.

Formal model. Two-stage validity test: procedural predicate AND rights-compliance predicate.

Problem 597. Majority rule and a protected right: case 2

Knowledge context. Democratic procedures use voting, but constitutional systems can place some rights beyond ordinary majority power.

Given facts. A council of 22 members votes 16 to 6 for a proposal that would forbid one named group from the right to use the public library equally. The classroom constitution states: (1) ordinary proposals pass by simple majority; (2) an ordinary majority may not remove the listed equal rights from a group.

Rules. A decision must satisfy both the voting procedure and the higher rights constraint. Passing a vote is not sufficient if the content violates a protected right.

Task. Did the proposal receive a majority? May it take effect under the stated constitution?

Step-by-step solution. 1. Vote check: 16>6, so the proposal has a simple majority. 2. Content check: the proposal removes a protected equal right (use the public library equally) from a group. 3. The higher rights rule therefore blocks the proposal even though the vote passed.

Answer. It received a majority, but it may not take effect under the stated rights rule.

Knowledge takeaway. A majority answers who won a vote; it does not by itself answer whether every possible decision is lawful or fair under higher rules.

Formal model. Two-stage validity test: procedural predicate AND rights-compliance predicate.

Problem 598. Majority rule and a protected right: case 3

Knowledge context. Democratic procedures use voting, but constitutional systems can place some rights beyond ordinary majority power.

Given facts. A council of 24 members votes 17 to 7 for a proposal that would forbid one named group from the right to receive a fair hearing. The classroom constitution states: (1) ordinary proposals pass by simple majority; (2) an ordinary majority may not remove the listed equal rights from a group.

Rules. A decision must satisfy both the voting procedure and the higher rights constraint. Passing a vote is not sufficient if the content violates a protected right.

Task. Did the proposal receive a majority? May it take effect under the stated constitution?

Step-by-step solution. 1. Vote check: 17>7, so the proposal has a simple majority. 2. Content check: the proposal removes a protected equal right (receive a fair hearing) from a group. 3. The higher rights rule therefore blocks the proposal even though the vote passed.

Answer. It received a majority, but it may not take effect under the stated rights rule.

Knowledge takeaway. A majority answers who won a vote; it does not by itself answer whether every possible decision is lawful or fair under higher rules.

Formal model. Two-stage validity test: procedural predicate AND rights-compliance predicate.

Problem 599. Majority rule and a protected right: case 4

Knowledge context. Democratic procedures use voting, but constitutional systems can place some rights beyond ordinary majority power.

Given facts. A council of 26 members votes 18 to 8 for a proposal that would forbid one named group from the right to practice a lawful belief. The classroom constitution states: (1) ordinary proposals pass by simple majority; (2) an ordinary majority may not remove the listed equal rights from a group.

Rules. A decision must satisfy both the voting procedure and the higher rights constraint. Passing a vote is not sufficient if the content violates a protected right.

Task. Did the proposal receive a majority? May it take effect under the stated constitution?

Step-by-step solution. 1. Vote check: 18>8, so the proposal has a simple majority. 2. Content check: the proposal removes a protected equal right (practice a lawful belief) from a group. 3. The higher rights rule therefore blocks the proposal even though the vote passed.

Answer. It received a majority, but it may not take effect under the stated rights rule.

Knowledge takeaway. A majority answers who won a vote; it does not by itself answer whether every possible decision is lawful or fair under higher rules.

Formal model. Two-stage validity test: procedural predicate AND rights-compliance predicate.

Problem 600. Majority rule and a protected right: case 5

Knowledge context. Democratic procedures use voting, but constitutional systems can place some rights beyond ordinary majority power.

Given facts. A council of 28 members votes 19 to 9 for a proposal that would forbid one named group from the right to be protected from arbitrary punishment. The classroom constitution states: (1) ordinary proposals pass by simple majority; (2) an ordinary majority may not remove the listed equal rights from a group.

Rules. A decision must satisfy both the voting procedure and the higher rights constraint. Passing a vote is not sufficient if the content violates a protected right.

Task. Did the proposal receive a majority? May it take effect under the stated constitution?

Step-by-step solution. 1. Vote check: 19>9, so the proposal has a simple majority. 2. Content check: the proposal removes a protected equal right (be protected from arbitrary punishment) from a group. 3. The higher rights rule therefore blocks the proposal even though the vote passed.

Answer. It received a majority, but it may not take effect under the stated rights rule.

Knowledge takeaway. A majority answers who won a vote; it does not by itself answer whether every possible decision is lawful or fair under higher rules.

Formal model. Two-stage validity test: procedural predicate AND rights-compliance predicate.

C3. Voting, majority, and quorum

Problem 601. Quorum and majority in a 21-member committee

Knowledge context. Voting rules often contain more than one threshold. Quorum concerns participation; majority concerns the votes cast according to the stated rule.

Given facts. A committee has 21 members. 21 are present. Votes: yes=10, no=10, abstain=1. Rule: quorum is at least 60% of all members; if quorum exists, a motion passes when yes votes are strictly more than no votes.

Rules. First test quorum using all members as the denominator. Only then compare yes and no; abstentions count for presence but not as no votes under this rule.

Task. Is quorum met, and does the motion pass?

Step-by-step solution. 1. Quorum threshold=ceil(0.60×21)=13. 2. Present=21, so quorum is met. 3. Yes=10, no=10; yes>no is False. 4. Combine the two tests.

Answer. Quorum is met; the motion does not pass.

Knowledge takeaway. Procedural reasoning requires applying tests in the right order and using the correct denominator.

Formal model. Threshold predicates: quorum(present,total) and majority(yes,no); pass = quorum $\wedge$ majority.

Problem 602. Quorum and majority in a 22-member committee

Knowledge context. Voting rules often contain more than one threshold. Quorum concerns participation; majority concerns the votes cast according to the stated rule.

Given facts. A committee has 22 members. 21 are present. Votes: yes=11, no=9, abstain=1. Rule: quorum is at least 60% of all members; if quorum exists, a motion passes when yes votes are strictly more than no votes.

Rules. First test quorum using all members as the denominator. Only then compare yes and no; abstentions count for presence but not as no votes under this rule.

Task. Is quorum met, and does the motion pass?

Step-by-step solution. 1. Quorum threshold=ceil(0.60×22)=14. 2. Present=21, so quorum is met. 3. Yes=11, no=9; yes>no is True. 4. Combine the two tests.

Answer. Quorum is met; the motion passes.

Knowledge takeaway. Procedural reasoning requires applying tests in the right order and using the correct denominator.

Formal model. Threshold predicates: quorum(present,total) and majority(yes,no); pass = quorum ∧ majority.

Problem 603. Quorum and majority in a 23-member committee

Knowledge context. Voting rules often contain more than one threshold. Quorum concerns participation; majority concerns the votes cast according to the stated rule.

Given facts. A committee has 23 members. 21 are present. Votes: yes=12, no=8, abstain=1. Rule: quorum is at least 60% of all members; if quorum exists, a motion passes when yes votes are strictly more than no votes.

Rules. First test quorum using all members as the denominator. Only then compare yes and no; abstentions count for presence but not as no votes under this rule.

Task. Is quorum met, and does the motion pass?

Step-by-step solution. 1. Quorum threshold=ceil(0.60×23)=14. 2. Present=21, so quorum is met. 3. Yes=12, no=8; yes>no is True. 4. Combine the two tests.

Answer. Quorum is met; the motion passes.

Knowledge takeaway. Procedural reasoning requires applying tests in the right order and using the correct denominator.

Formal model. Threshold predicates: quorum(present,total) and majority(yes,no); pass = quorum ∧ majority.

Problem 604. Quorum and majority in a 24-member committee

Knowledge context. Voting rules often contain more than one threshold. Quorum concerns participation; majority concerns the votes cast according to the stated rule.

Given facts. A committee has 24 members. 24 are present. Votes: yes=13, no=10, abstain=1. Rule: quorum is at least 60% of all members; if quorum exists, a motion passes when yes votes are strictly more than no votes.

Rules. First test quorum using all members as the denominator. Only then compare yes and no; abstentions count for presence but not as no votes under this rule.

Task. Is quorum met, and does the motion pass?

Step-by-step solution. 1. Quorum threshold=ceil(0.60×24)=15. 2. Present=24, so quorum is met. 3. Yes=13, no=10; yes>no is True. 4. Combine the two tests.

Answer. Quorum is met; the motion passes.

Knowledge takeaway. Procedural reasoning requires applying tests in the right order and using the correct denominator.

Formal model. Threshold predicates: quorum(present,total) and majority(yes,no); pass = quorum ∧ majority.

Problem 605. Quorum and majority in a 25-member committee

Knowledge context. Voting rules often contain more than one threshold. Quorum concerns participation; majority concerns the votes cast according to the stated rule.

Given facts. A committee has 25 members. 24 are present. Votes: yes=14, no=9, abstain=1. Rule: quorum is at least 60% of all members; if quorum exists, a motion passes when yes votes are strictly more than no votes.

Rules. First test quorum using all members as the denominator. Only then compare yes and no; abstentions count for presence but not as no votes under this rule.

Task. Is quorum met, and does the motion pass?

Step-by-step solution. 1. Quorum threshold=ceil(0.60×25)=15. 2. Present=24, so quorum is met. 3. Yes=14, no=9; yes>no is True. 4. Combine the two tests.

Answer. Quorum is met; the motion passes.

Knowledge takeaway. Procedural reasoning requires applying tests in the right order and using the correct denominator.

Formal model. Threshold predicates: quorum(present,total) and majority(yes,no); pass = quorum ∧ majority.

C4. Representation and seats

Problem 606. Representing three districts: case 1

Knowledge context. Representation can be modeled by rules that convert population into seats. Different allocation rules can produce different results, so the rule must be stated.

Given facts. Three districts have populations A=120, B=110, C=60. They must share 13 seats. Use this stated method: compute each exact quota population/total×seats; give each district the whole-number part; assign remaining seats to the largest fractional remainders.

Rules. Use only the stated largest-remainder method. Seat totals must add exactly to the available number.

Task. How many seats does each district receive?

Step-by-step solution. 1. Total population=290. 2. Exact quotas: A=5.38, B=4.93, C=2.69. 3. Initial whole seats: [5, 4, 2], leaving 2. 4. Give the remaining seat(s) to the largest remainder(s). 5. Final allocation=[5, 5, 3], which sums to 13.

Answer. A=5, B=5, C=3 seats.

Knowledge takeaway. Fair representation is not just arithmetic; it depends on an explicit apportionment procedure.

Formal model. Largest-remainder apportionment: quotas, floors, then rank fractional remainders.

Problem 607. Representing three districts: case 2

Knowledge context. Representation can be modeled by rules that convert population into seats. Different allocation rules can produce different results, so the rule must be stated.

Given facts. Three districts have populations A=140, B=110, C=75. They must share 13 seats. Use this stated method: compute each exact quota population/total×seats; give each district the whole-number part; assign remaining seats to the largest fractional remainders.

Rules. Use only the stated largest-remainder method. Seat totals must add exactly to the available number.

Task. How many seats does each district receive?

Step-by-step solution. 1. Total population=325. 2. Exact quotas: A=5.60, B=4.40, C=3.00. 3. Initial whole seats: [5, 4, 3], leaving 1. 4. Give the remaining seat(s) to the largest remainder(s). 5. Final allocation=[6, 4, 3], which sums to 13.

Answer. A=6, B=4, C=3 seats.

Knowledge takeaway. Fair representation is not just arithmetic; it depends on an explicit apportionment procedure.

Formal model. Largest-remainder apportionment: quotas, floors, then rank fractional remainders.

Problem 608. Representing three districts: case 3

Knowledge context. Representation can be modeled by rules that convert population into seats. Different allocation rules can produce different results, so the rule must be stated.

Given facts. Three districts have populations A=160, B=110, C=90. They must share 13 seats. Use this stated method: compute each exact quota population/total×seats; give each district the whole-number part; assign remaining seats to the largest fractional remainders.

Rules. Use only the stated largest-remainder method. Seat totals must add exactly to the available number.

Task. How many seats does each district receive?

Step-by-step solution. 1. Total population=360. 2. Exact quotas: A=5.78, B=3.97, C=3.25. 3. Initial whole seats: [5, 3, 3], leaving 2. 4. Give the remaining seat(s) to the largest remainder(s). 5. Final allocation=[6, 4, 3], which sums to 13.

Answer. A=6, B=4, C=3 seats.

Knowledge takeaway. Fair representation is not just arithmetic; it depends on an explicit apportionment procedure.

Formal model. Largest-remainder apportionment: quotas, floors, then rank fractional remainders.

Problem 609. Representing three districts: case 4

Knowledge context. Representation can be modeled by rules that convert population into seats. Different allocation rules can produce different results, so the rule must be stated.

Given facts. Three districts have populations A=180, B=110, C=105. They must share 13 seats. Use this stated method: compute each exact quota population/total×seats; give each district the whole-number part; assign remaining seats to the largest fractional remainders.

Rules. Use only the stated largest-remainder method. Seat totals must add exactly to the available number.

Task. How many seats does each district receive?

Step-by-step solution. 1. Total population=395. 2. Exact quotas: A=5.92, B=3.62, C=3.46. 3. Initial whole seats: [5, 3, 3], leaving 2. 4. Give the remaining seat(s) to the largest remainder(s). 5. Final allocation=[6, 4, 3], which sums to 13.

Answer. A=6, B=4, C=3 seats.

Knowledge takeaway. Fair representation is not just arithmetic; it depends on an explicit apportionment procedure.

Formal model. Largest-remainder apportionment: quotas, floors, then rank fractional remainders.

Problem 610. Representing three districts: case 5

Knowledge context. Representation can be modeled by rules that convert population into seats. Different allocation rules can produce different results, so the rule must be stated.

Given facts. Three districts have populations A=200, B=110, C=120. They must share 13 seats. Use this stated method: compute each exact quota population/total×seats; give each district the whole-number part; assign remaining seats to the largest fractional remainders.

Rules. Use only the stated largest-remainder method. Seat totals must add exactly to the available number.

Task. How many seats does each district receive?

Step-by-step solution. 1. Total population=430. 2. Exact quotas: A=6.05, B=3.33, C=3.63. 3. Initial whole seats: [6, 3, 3], leaving 1. 4. Give the remaining seat(s) to the largest remainder(s). 5. Final allocation=[6, 3, 4], which sums to 13.

Answer. A=6, B=3, C=4 seats.

Knowledge takeaway. Fair representation is not just arithmetic; it depends on an explicit apportionment procedure.

Formal model. Largest-remainder apportionment: quotas, floors, then rank fractional remainders.

C5. Institutions and roles

Problem 611. Which institution has the role? Repair a broken public road

Knowledge context. Civic institutions divide responsibilities. A problem can be solved by matching an action to a clearly defined function.

Given facts. Roles are defined as follows: Town Council makes local policy and approves the budget; Public Works Office maintains roads; Local Court decides legal disputes; Election Committee administers elections; School Board sets school policies. A resident asks which institution should repair a broken public road.

Rules. Match the requested action to the stated role. Do not choose an institution merely because it is powerful or familiar.

Task. Should the request go first to the Public Works Office or the Parks Department? Why?

Step-by-step solution. 1. Identify the action: repair a broken public road. 2. Find the role whose definition contains that function: Public Works Office. 3. Parks Department has a different stated role, so it is not the first match.

Answer. The request should go first to the Public Works Office.

Knowledge takeaway. Institutional reasoning is role matching, not guessing which body is “most important.”

Formal model. Bipartite relation between actions and authorized institutional roles.

Problem 612. Which institution has the role? Decide a dispute under law

Knowledge context. Civic institutions divide responsibilities. A problem can be solved by matching an action to a clearly defined function.

Given facts. Roles are defined as follows: Town Council makes local policy and approves the budget; Public Works Office maintains roads; Local Court decides legal disputes; Election Committee administers elections; School Board sets school policies. A resident asks which institution should decide a dispute under law.

Rules. Match the requested action to the stated role. Do not choose an institution merely because it is powerful or familiar.

Task. Should the request go first to the Local Court or the School Board? Why?

Step-by-step solution. 1. Identify the action: decide a dispute under law. 2. Find the role whose definition contains that function: Local Court. 3. School Board has a different stated role, so it is not the first match.

Answer. The request should go first to the Local Court.

Knowledge takeaway. Institutional reasoning is role matching, not guessing which body is “most important.”

Formal model. Bipartite relation between actions and authorized institutional roles.

Problem 613. Which institution has the role? Approve the town budget

Knowledge context. Civic institutions divide responsibilities. A problem can be solved by matching an action to a clearly defined function.

Given facts. Roles are defined as follows: Town Council makes local policy and approves the budget; Public Works Office maintains roads; Local Court decides legal disputes; Election Committee administers elections; School Board sets school policies. A resident asks which institution should approve the town budget. Cross-domain check: a survey team starts work at 12:00 and works for 2 hour(s) without a break.

Rules. Match the requested action to the stated role. Do not choose an institution merely because it is powerful or familiar. For the cross-domain check, finishing time = starting time + duration.

Task. Should the request go first to the Town Council or the Health Office? Why? Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. Identify the action: approve the town budget. 2. Find the role whose definition contains that function: Town Council. 3. Health Office has a different stated role, so it is not the first match. Cross-domain check: 12+2=14, so the team finishes at 14:00.

Answer. The request should go first to the Town Council. Cross-domain answer: 14:00.

Knowledge takeaway. Institutional reasoning is role matching, not guessing which body is “most important.”

Formal model. Bipartite relation between actions and authorized institutional roles.

Problem 614. Which institution has the role? Organize the ballot count

Knowledge context. Civic institutions divide responsibilities. A problem can be solved by matching an action to a clearly defined function.

Given facts. Roles are defined as follows: Town Council makes local policy and approves the budget; Public Works Office maintains roads; Local Court decides legal disputes; Election Committee administers elections; School Board sets school policies. A resident asks which institution should organize the ballot count. Cross-domain check: a local committee has 10 members, requires at least 6 present for quorum, and 8 are present.

Rules. Match the requested action to the stated role. Do not choose an institution merely because it is powerful or familiar. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. Should the request go first to the Election Committee or the Public Works Office? Why? Cross-domain check: is quorum met?

Step-by-step solution. 1. Identify the action: organize the ballot count. 2. Find the role whose definition contains that function: Election Committee. 3. Public Works Office has a different stated role, so it is not the first match. Cross-domain check: compare 8 with 6; $8 \geq 6$ is true.

Answer. The request should go first to the Election Committee. Cross-domain answer: quorum is met.

Knowledge takeaway. Institutional reasoning is role matching, not guessing which body is “most important.”

Formal model. Bipartite relation between actions and authorized institutional roles.

Problem 615. Which institution has the role? Set school calendar rules

Knowledge context. Civic institutions divide responsibilities. A problem can be solved by matching an action to a clearly defined function.

Given facts. Roles are defined as follows: Town Council makes local policy and approves the budget; Public Works Office maintains roads; Local Court decides legal disputes; Election Committee administers elections; School Board sets school policies. A resident asks which institution should set school calendar rules.

Rules. Match the requested action to the stated role. Do not choose an institution merely because it is powerful or familiar.

Task. Should the request go first to the School Board or the Health Office? Why?

Step-by-step solution. 1. Identify the action: set school calendar rules. 2. Find the role whose definition contains that function: School Board. 3. Health Office has a different stated role, so it is not the first match.

Answer. The request should go first to the School Board.

Knowledge takeaway. Institutional reasoning is role matching, not guessing which body is “most important.”

Formal model. Bipartite relation between actions and authorized institutional roles.

C6. Public budgets and priorities

Problem 616. Choosing a public budget package: case 1

Knowledge context. Public budgets require choices under limited resources. A transparent model separates costs, benefits, and mandatory obligations.

Given facts. A town has a budget of 16 units. Projects (cost, benefit): park=(4,5); bridge repair=(6,9); library books=(3,4); drainage=(5,8). Cross-domain check: on a field sketch, 1 cm represents 3 km and two checkpoints are 2 cm apart.

Rules. Choose a subset whose total cost does not exceed the budget. Maximize total benefit. If a mandatory project is stated, every feasible plan must include it. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Which set of projects gives the greatest total benefit while satisfying every constraint? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. Budget limit is 16. 2. Evaluate feasible combinations (and keep the mandatory project if any). 3. The best feasible combination has cost 15 and benefit 22: ['park', 'bridge repair', 'drainage']. 4. No higher-benefit feasible combination exists under the stated costs. Cross-domain check: 2×3=6 km.

Answer. Choose park, bridge repair, drainage; cost 15, benefit 22. Cross-domain answer: 6 km.

Knowledge takeaway. Optimization means comparing feasible alternatives under an explicit objective, not simply choosing the cheapest item.

Formal model. 0/1 knapsack with optional mandatory-item constraint; maximize benefit subject to total cost $\leq$ budget.

Problem 617. Choosing a public budget package: case 2

Knowledge context. Public budgets require choices under limited resources. A transparent model separates costs, benefits, and mandatory obligations.

Given facts. A town has a budget of 17 units. Projects (cost, benefit): tree planting=(5,7); bridge repair=(6,9); library books=(3,4); drainage=(5,8). Cross-domain check: a survey team starts work at 8:00 and works for 2 hour(s) without a break.

Rules. Choose a subset whose total cost does not exceed the budget. Maximize total benefit. If a mandatory project is stated, every feasible plan must include it. For the cross-domain check, finishing time = starting time + duration.

Task. Which set of projects gives the greatest total benefit while satisfying every constraint? Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. Budget limit is 17. 2. Evaluate feasible combinations (and keep the mandatory project if any). 3. The best feasible combination has cost 16 and benefit 24: ['tree planting', 'bridge repair', 'drainage']. 4. No higher-benefit feasible combination exists under the stated costs. Cross-domain check: 8+2=10, so the team finishes at 10:00.

Answer. Choose tree planting, bridge repair, drainage; cost 16, benefit 24. Cross-domain answer: 10:00.

Knowledge takeaway. Optimization means comparing feasible alternatives under an explicit objective, not simply choosing the cheapest item.

Formal model. 0/1 knapsack with optional mandatory-item constraint; maximize benefit subject to total cost $\leq$ budget.

Problem 618. Choosing a public budget package: case 3

Knowledge context. Public budgets require choices under limited resources. A transparent model separates costs, benefits, and mandatory obligations.

Given facts. A town has a budget of 18 units. Projects (cost, benefit): park=(4,5); bridge repair=(6,9); library books=(3,4); drainage=(5,8). Cross-domain check: a local committee has 14 members, requires at least 8 present for quorum, and 14 are present.

Rules. Choose a subset whose total cost does not exceed the budget. Maximize total benefit. If a mandatory project is stated, every feasible plan must include it. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. Which set of projects gives the greatest total benefit while satisfying every constraint? Cross-domain check: is quorum met?

Step-by-step solution. 1. Budget limit is 18. 2. Evaluate feasible combinations (and keep the mandatory project if any). 3. The best feasible combination has cost 18 and benefit 26: ['park', 'bridge repair', 'library books', 'drainage']. 4. No higher-benefit feasible combination exists under the stated costs. Cross-domain check: compare 14 with 8; 14 ≥ 8 is true.

Answer. Choose park, bridge repair, library books, drainage; cost 18, benefit 26. Cross-domain answer: quorum is met.

Knowledge takeaway. Optimization means comparing feasible alternatives under an explicit objective, not simply choosing the cheapest item.

Formal model. 0/1 knapsack with optional mandatory-item constraint; maximize benefit subject to total cost $\leq$ budget.

Problem 619. Choosing a public budget package: case 4

Knowledge context. Public budgets require choices under limited resources. A transparent model separates costs, benefits, and mandatory obligations.

Given facts. A town has a budget of 19 units. Projects (cost, benefit): tree planting=(5,7); bridge repair=(6,9); library books=(3,4); drainage=(5,8).

Rules. Choose a subset whose total cost does not exceed the budget. Maximize total benefit. If a mandatory project is stated, every feasible plan must include it.

Task. Which set of projects gives the greatest total benefit while satisfying every constraint?

Step-by-step solution. 1. Budget limit is 19. 2. Evaluate feasible combinations (and keep the mandatory project if any). 3. The best feasible combination has cost 19 and benefit 28: ['tree planting', 'bridge repair', 'library books', 'drainage']. 4. No higher-benefit feasible combination exists under the stated costs.

Answer. Choose tree planting, bridge repair, library books, drainage; cost 19, benefit 28.

Knowledge takeaway. Optimization means comparing feasible alternatives under an explicit objective, not simply choosing the cheapest item.

Formal model. 0/1 knapsack with optional mandatory-item constraint; maximize benefit subject to total cost $\leq$ budget.

Problem 620. Choosing a public budget package: case 5

Knowledge context. Public budgets require choices under limited resources. A transparent model separates costs, benefits, and mandatory obligations.

Given facts. A town has a budget of 20 units. Projects (cost, benefit): park=(4,5); bridge repair=(6,9); library books=(3,4); drainage=(5,8). Safety rule: bridge repair must be included.

Rules. Choose a subset whose total cost does not exceed the budget. Maximize total benefit. If a mandatory project is stated, every feasible plan must include it.

Task. Which set of projects gives the greatest total benefit while satisfying every constraint?

Step-by-step solution. 1. Budget limit is 20. 2. Evaluate feasible combinations (and keep the mandatory project if any). 3. The best feasible combination has cost 18 and benefit 26: ['park', 'bridge repair', 'library books', 'drainage']. 4. No higher-benefit feasible combination exists under the stated costs.

Answer. Choose park, bridge repair, library books, drainage; cost 18, benefit 26.

Knowledge takeaway. Optimization means comparing feasible alternatives under an explicit objective, not simply choosing the cheapest item.

Formal model. 0/1 knapsack with optional mandatory-item constraint; maximize benefit subject to total cost $\leq$ budget.

C7. Collective choice and preferences

Problem 621. Pairwise collective choice: Park versus Library

Knowledge context. Group preferences can be aggregated in different ways. Pairwise majority asks a specific question about two options at a time.

Given facts. Three public projects are considered: Park, Library, Sports Hall. Voter groups and rankings: 4 voters rank Park>Library>Sports Hall; 6 voters rank Library>Sports Hall>Park; 2 voters rank Sports Hall>Park>Library. Use pairwise majority: compare two options at a time; each voter supports the one ranked higher.

Rules. For a pairwise contest, count how many voters prefer each of the two options. Ignore the third option for that comparison.

Task. In a pairwise vote between Park and Library, which option wins?

Step-by-step solution. 1. For each group, check whether Park or Library is ranked higher. 2. Totals: Park=6, Library=6. 3. Therefore the result is tie.

Answer. tie 6–6.

Knowledge takeaway. A collective decision rule must be explicit; different rules can summarize the same rankings differently.

Formal model. Preference rankings; for pair (a,b), sum weights of voters ranking a>b versus b>a.

Problem 622. Pairwise collective choice: Library versus Sports Hall

Knowledge context. Group preferences can be aggregated in different ways. Pairwise majority asks a specific question about two options at a time.

Given facts. Three public projects are considered: Park, Library, Sports Hall. Voter groups and rankings: 5 voters rank Park>Library>Sports Hall; 6 voters rank Library>Sports Hall>Park; 3 voters rank Sports Hall>Park>Library. Use pairwise majority: compare two options at a time; each voter supports the one ranked higher.

Rules. For a pairwise contest, count how many voters prefer each of the two options. Ignore the third option for that comparison.

Task. In a pairwise vote between Library and Sports Hall, which option wins?

Step-by-step solution. 1. For each group, check whether Library or Sports Hall is ranked higher. 2. Totals: Library=11, Sports Hall=3. 3. Therefore the result is Library.

Answer. Library wins 11 to 3.

Knowledge takeaway. A collective decision rule must be explicit; different rules can summarize the same rankings differently.

Formal model. Preference rankings; for pair (a,b), sum weights of voters ranking a>b versus b>a.

Problem 623. Pairwise collective choice: Park versus Sports Hall

Knowledge context. Group preferences can be aggregated in different ways. Pairwise majority asks a specific question about two options at a time.

Given facts. Three public projects are considered: Park, Library, Sports Hall. Voter groups and rankings: 6 voters rank Park>Library>Sports Hall; 6 voters rank Library>Sports Hall>Park; 4 voters rank Sports Hall>Park>Library. Use pairwise majority: compare two options at a time; each voter supports the one ranked higher.

Rules. For a pairwise contest, count how many voters prefer each of the two options. Ignore the third option for that comparison.

Task. In a pairwise vote between Park and Sports Hall, which option wins?

Step-by-step solution. 1. For each group, check whether Park or Sports Hall is ranked higher. 2. Totals: Park=6, Sports Hall=10. 3. Therefore the result is Sports Hall.

Answer. Sports Hall wins 10 to 6.

Knowledge takeaway. A collective decision rule must be explicit; different rules can summarize the same rankings differently.

Formal model. Preference rankings; for pair (a,b), sum weights of voters ranking a>b versus b>a.

Problem 624. Pairwise collective choice: Library versus Park

Knowledge context. Group preferences can be aggregated in different ways. Pairwise majority asks a specific question about two options at a time.

Given facts. Three public projects are considered: Park, Library, Sports Hall. Voter groups and rankings: 7 voters rank Park>Library>Sports Hall; 6 voters rank Library>Sports Hall>Park; 5 voters rank Sports Hall>Park>Library. Use pairwise majority: compare two options at a time; each voter supports the one ranked higher.

Rules. For a pairwise contest, count how many voters prefer each of the two options. Ignore the third option for that comparison.

Task. In a pairwise vote between Library and Park, which option wins?

Step-by-step solution. 1. For each group, check whether Library or Park is ranked higher. 2. Totals: Library=6, Park=12. 3. Therefore the result is Park.

Answer. Park wins 12 to 6.

Knowledge takeaway. A collective decision rule must be explicit; different rules can summarize the same rankings differently.

Formal model. Preference rankings; for pair (a,b), sum weights of voters ranking a>b versus b>a.

Problem 625. Pairwise collective choice: Sports Hall versus Park

Knowledge context. Group preferences can be aggregated in different ways. Pairwise majority asks a specific question about two options at a time.

Given facts. Three public projects are considered: Park, Library, Sports Hall. Voter groups and rankings: 8 voters rank Park>Library>Sports Hall; 6 voters rank Library>Sports Hall>Park; 6 voters rank Sports Hall>Park>Library. Use pairwise majority: compare two options at a time; each voter supports the one ranked higher.

Rules. For a pairwise contest, count how many voters prefer each of the two options. Ignore the third option for that comparison.

Task. In a pairwise vote between Sports Hall and Park, which option wins?

Step-by-step solution. 1. For each group, check whether Sports Hall or Park is ranked higher. 2. Totals: Sports Hall=12, Park=8. 3. Therefore the result is Sports Hall.

Answer. Sports Hall wins 12 to 8.

Knowledge takeaway. A collective decision rule must be explicit; different rules can summarize the same rankings differently.

Formal model. Preference rankings; for pair (a,b), sum weights of voters ranking a>b versus b>a.

Part B — New and mixed reasoning domains

N1. Scale, proportion, and distance

Problem 626. Map scale 1:10000

Knowledge context. Map scale links a model to the real world through proportional reasoning.

Given facts. A map has scale 1:10000. The measured map distance between two points is 5 cm. Rule: 1 cm on the map represents 10000 cm in reality; 100,000 cm = 1 km.

Rules. Multiply map length by the scale denominator, then convert units. A scale is a ratio, not an added distance.

Task. What is the real straight-line distance in kilometres?

Step-by-step solution. 1. Real centimetres=5×10000=50000. 2. Convert to kilometres: 50000÷100,000=0.5.

Answer. 0.5 km.

Knowledge takeaway. Scale problems are ratio problems: keep units explicit and conversion becomes mechanical.

Formal model. Proportion d_real=d_map×denominator, followed by unit conversion.

Problem 627. Map scale 1:25000

Knowledge context. Map scale links a model to the real world through proportional reasoning.

Given facts. A map has scale 1:25000. The measured map distance between two points is 6 cm. Rule: 1 cm on the map represents 25000 cm in reality; 100,000 cm = 1 km.

Rules. Multiply map length by the scale denominator, then convert units. A scale is a ratio, not an added distance.

Task. What is the real straight-line distance in kilometres?

Step-by-step solution. 1. Real centimetres=6×25000=150000. 2. Convert to kilometres: 150000÷100,000=1.5.

Answer. 1.5 km.

Knowledge takeaway. Scale problems are ratio problems: keep units explicit and conversion becomes mechanical.

Formal model. Proportion d_real=d_map×denominator, followed by unit conversion.

Problem 628. Map scale 1:50000

Knowledge context. Map scale links a model to the real world through proportional reasoning.

Given facts. A map has scale 1:50000. The measured map distance between two points is 7 cm. Rule: 1 cm on the map represents 50000 cm in reality; 100,000 cm = 1 km.

Rules. Multiply map length by the scale denominator, then convert units. A scale is a ratio, not an added distance.

Task. What is the real straight-line distance in kilometres?

Step-by-step solution. 1. Real centimetres=7×50000=350000. 2. Convert to kilometres: 350000÷100,000=3.5.

Answer. 3.5 km.

Knowledge takeaway. Scale problems are ratio problems: keep units explicit and conversion becomes mechanical.

Formal model. Proportion d_real=d_map×denominator, followed by unit conversion.

Problem 629. Map scale 1:100000

Knowledge context. Map scale links a model to the real world through proportional reasoning.

Given facts. A map has scale 1:100000. The measured map distance between two points is 8 cm. Rule: 1 cm on the map represents 100000 cm in reality; 100,000 cm = 1 km.

Rules. Multiply map length by the scale denominator, then convert units. A scale is a ratio, not an added distance.

Task. What is the real straight-line distance in kilometres?

Step-by-step solution. 1. Real centimetres=8×100000=800000. 2. Convert to kilometres: 800000÷100,000=8.

Answer. 8 km.

Knowledge takeaway. Scale problems are ratio problems: keep units explicit and conversion becomes mechanical.

Formal model. Proportion d_real=d_map×denominator, followed by unit conversion.

Problem 630. Map scale 1:200000

Knowledge context. Map scale links a model to the real world through proportional reasoning.

Given facts. A map has scale 1:200000. The measured map distance between two points is 9 cm. Rule: 1 cm on the map represents 200000 cm in reality; 100,000 cm = 1 km.

Rules. Multiply map length by the scale denominator, then convert units. A scale is a ratio, not an added distance.

Task. What is the real straight-line distance in kilometres?

Step-by-step solution. 1. Real centimetres=9×200000=1800000. 2. Convert to kilometres: 1800000÷100,000=18.

Answer. 18 km.

Knowledge takeaway. Scale problems are ratio problems: keep units explicit and conversion becomes mechanical.

Formal model. Proportion d_real=d_map×denominator, followed by unit conversion.

N2. Coordinates and constraint-based location

Problem 631. Locating a point by constraints: case 1

Knowledge context. Coordinates turn location into numbers. Multiple spatial clues can be treated as simultaneous constraints.

Given facts. A point P lies on an integer grid with x,y from 0 to 7. It satisfies x+y=6 and x−y=0. It also satisfies x ≥ 2.

Rules. A location is valid only if it satisfies every stated equation and inequality at the same time.

Task. Which coordinate(s) can P occupy? Is the solution unique?

Step-by-step solution. 1. List pairs satisfying x+y=6. 2. Keep only those also satisfying x−y=0. 3. The remaining set is [(3, 3)].

Answer. P=(3, 3); the solution is unique.

Knowledge takeaway. A constraint problem is solved by satisfying all conditions together, not one at a time independently.

Formal model. Constraint satisfaction over integer pairs (x,y).

Problem 632. Locating a point by constraints: case 2

Knowledge context. Coordinates turn location into numbers. Multiple spatial clues can be treated as simultaneous constraints.

Given facts. A point P lies on an integer grid with x,y from 0 to 7. It satisfies x+y=9 and x−y=-1. It also satisfies x ≥ 3.

Rules. A location is valid only if it satisfies every stated equation and inequality at the same time.

Task. Which coordinate(s) can P occupy? Is the solution unique?

Step-by-step solution. 1. List pairs satisfying x+y=9. 2. Keep only those also satisfying x−y=-1. 3. The remaining set is [(4, 5)].

Answer. P=(4, 5); the solution is unique.

Knowledge takeaway. A constraint problem is solved by satisfying all conditions together, not one at a time independently.

Formal model. Constraint satisfaction over integer pairs (x,y).

Problem 633. Locating a point by constraints: case 3

Knowledge context. Coordinates turn location into numbers. Multiple spatial clues can be treated as simultaneous constraints.

Given facts. A point P lies on an integer grid with x,y from 0 to 7. It satisfies x+y=12 and x−y=-2. It also satisfies x ≥ 4.

Rules. A location is valid only if it satisfies every stated equation and inequality at the same time.

Task. Which coordinate(s) can P occupy? Is the solution unique?

Step-by-step solution. 1. List pairs satisfying x+y=12. 2. Keep only those also satisfying x−y=-2. 3. The remaining set is [(5, 7)].

Answer. P=(5, 7); the solution is unique.

Knowledge takeaway. A constraint problem is solved by satisfying all conditions together, not one at a time independently.

Formal model. Constraint satisfaction over integer pairs (x,y).

Problem 634. Locating a point by constraints: case 4

Knowledge context. Coordinates turn location into numbers. Multiple spatial clues can be treated as simultaneous constraints.

Given facts. A point P lies on an integer grid with x,y from 0 to 7. It satisfies x+y=7 and x−y=5. It also satisfies x ≥ 5.

Rules. A location is valid only if it satisfies every stated equation and inequality at the same time.

Task. Which coordinate(s) can P occupy? Is the solution unique?

Step-by-step solution. 1. List pairs satisfying x+y=7. 2. Keep only those also satisfying x−y=5. 3. The remaining set is [(6, 1)].

Answer. P=(6, 1); the solution is unique.

Knowledge takeaway. A constraint problem is solved by satisfying all conditions together, not one at a time independently.

Formal model. Constraint satisfaction over integer pairs (x,y).

Problem 635. Locating a point by constraints: case 5

Knowledge context. Coordinates turn location into numbers. Multiple spatial clues can be treated as simultaneous constraints.

Given facts. A point P lies on an integer grid with x,y from 0 to 7. It satisfies x+y=10 and x−y=4. It also satisfies x ≥ 6.

Rules. A location is valid only if it satisfies every stated equation and inequality at the same time.

Task. Which coordinate(s) can P occupy? Is the solution unique?

Step-by-step solution. 1. List pairs satisfying x+y=10. 2. Keep only those also satisfying x−y=4. 3. The remaining set is [(7, 3)].

Answer. P=(7, 3); the solution is unique.

Knowledge takeaway. A constraint problem is solved by satisfying all conditions together, not one at a time independently.

Formal model. Constraint satisfaction over integer pairs (x,y).

N3. Earth, Sun, seasons, and local time

Problem 636. Earth, sunlight, and local time: case 1

Knowledge context. Earth–Sun relationships explain broad patterns such as day length and local time, but a problem should state the simplified model it expects children to use.

Given facts. For this classroom model, Eastport is 1 time zone(s) east of Westport. Each zone east is one hour later. At Westport it is 11:00.

Rules. Moving east adds one hour per stated time zone; moving west subtracts. Use the given model only.

Task. What local time is it in Eastport?

Step-by-step solution. 1. Time difference is +1 hour(s) because Eastport is east. 2. 11+1=12 (mod 24).

Answer. 12:00.

Knowledge takeaway. A model supports only the conclusions encoded in its rules; seasonal light does not automatically determine daily weather.

Formal model. Rule-based astronomical/time-zone model; apply signed offsets or seasonal relation.

Problem 637. Earth, sunlight, and local time: case 2

Knowledge context. Earth–Sun relationships explain broad patterns such as day length and local time, but a problem should state the simplified model it expects children to use.

Given facts. For this classroom model, Eastport is 2 time zone(s) east of Westport. Each zone east is one hour later. At Westport it is 11:00.

Rules. Moving east adds one hour per stated time zone; moving west subtracts. Use the given model only.

Task. What local time is it in Eastport?

Step-by-step solution. 1. Time difference is +2 hour(s) because Eastport is east. 2. 11+2=13 (mod 24).

Answer. 13:00.

Knowledge takeaway. A model supports only the conclusions encoded in its rules; seasonal light does not automatically determine daily weather.

Formal model. Rule-based astronomical/time-zone model; apply signed offsets or seasonal relation.

Problem 638. Earth, sunlight, and local time: case 3

Knowledge context. Earth–Sun relationships explain broad patterns such as day length and local time, but a problem should state the simplified model it expects children to use.

Given facts. For this classroom model, Eastport is 3 time zone(s) east of Westport. Each zone east is one hour later. At Westport it is 11:00.

Rules. Moving east adds one hour per stated time zone; moving west subtracts. Use the given model only.

Task. What local time is it in Eastport?

Step-by-step solution. 1. Time difference is +3 hour(s) because Eastport is east. 2. 11+3=14 (mod 24).

Answer. 14:00.

Knowledge takeaway. A model supports only the conclusions encoded in its rules; seasonal light does not automatically determine daily weather.

Formal model. Rule-based astronomical/time-zone model; apply signed offsets or seasonal relation.

Problem 639. Earth, sunlight, and local time: case 4

Knowledge context. Earth–Sun relationships explain broad patterns such as day length and local time, but a problem should state the simplified model it expects children to use.

Given facts. Use this simplified model: Earth's axis is tilted. In the Northern Hemisphere, days are generally longer around June than around December. Compare June and December for the same place. Cross-domain check: researchers collected 15 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. Use the stated seasonal relation; do not infer daily weather from season. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. Which of the two months has the longer daylight period in this model, and can that fact alone tell us whether it will rain? Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. Apply the stated tilt-season relation. 2. June has the longer daylight period in the Northern Hemisphere. 3. The rule contains no information about rainfall, so rainfall cannot be deduced. Cross-domain check: 15−4=11 independent reports remain.

Answer. June has longer daylight; rainfall cannot be determined from this fact alone. Cross-domain answer: 11 independent reports.

Knowledge takeaway. A model supports only the conclusions encoded in its rules; seasonal light does not automatically determine daily weather.

Formal model. Rule-based astronomical/time-zone model; apply signed offsets or seasonal relation.

Problem 640. Earth, sunlight, and local time: case 5

Knowledge context. Earth–Sun relationships explain broad patterns such as day length and local time, but a problem should state the simplified model it expects children to use.

Given facts. Use this simplified model: Earth's axis is tilted. In the Northern Hemisphere, days are generally longer around June than around December. Compare June and December for the same place. Cross-domain check: on a field sketch, 1 cm represents 2 km and two checkpoints are 5 cm apart.

Rules. Use the stated seasonal relation; do not infer daily weather from season. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Which of the two months has the longer daylight period in this model, and can that fact alone tell us whether it will rain? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. Apply the stated tilt-season relation. 2. June has the longer daylight period in the Northern Hemisphere. 3. The rule contains no information about rainfall, so rainfall cannot be deduced. Cross-domain check: 5×2=10 km.

Answer. June has longer daylight; rainfall cannot be determined from this fact alone. Cross-domain answer: 10 km.

Knowledge takeaway. A model supports only the conclusions encoded in its rules; seasonal light does not automatically determine daily weather.

Formal model. Rule-based astronomical/time-zone model; apply signed offsets or seasonal relation.

N4. Stocks, flows, and conservation

Problem 641. Reservoir stock and flow: case 1

Knowledge context. Many geographic and civic quantities are stocks and flows: water in a reservoir, people in a city, or goods in a warehouse.

Given facts. A reservoir starts with 35 water units. Each day 5 units flow in and 6 units flow out. Assume these rates stay constant for 5 days.

Rules. Stock at the end = starting stock + total inflow − total outflow. Flow is an amount per time period; stock is the amount currently stored.

Task. How much water is stored after 5 days? Is the stock increasing or decreasing each day?

Step-by-step solution. 1. Daily net change=5−6=-1. 2. Over 5 days, net change=5×-1=-5. 3. End stock=35+(-5)=30.

Answer. 30 units; the stock is decreasing.

Knowledge takeaway. Do not confuse a stored amount with a rate of change; conservation links them.

Formal model. Difference equation S(t+1)=S(t)+inflow−outflow.

Problem 642. Reservoir stock and flow: case 2

Knowledge context. Many geographic and civic quantities are stocks and flows: water in a reservoir, people in a city, or goods in a warehouse.

Given facts. A reservoir starts with 38 water units. Each day 6 units flow in and 7 units flow out. Assume these rates stay constant for 5 days.

Rules. Stock at the end = starting stock + total inflow − total outflow. Flow is an amount per time period; stock is the amount currently stored.

Task. How much water is stored after 5 days? Is the stock increasing or decreasing each day?

Step-by-step solution. 1. Daily net change=6−7=-1. 2. Over 5 days, net change=5×-1=-5. 3. End stock=38+(-5)=33.

Answer. 33 units; the stock is decreasing.

Knowledge takeaway. Do not confuse a stored amount with a rate of change; conservation links them.

Formal model. Difference equation S(t+1)=S(t)+inflow−outflow.

Problem 643. Reservoir stock and flow: case 3

Knowledge context. Many geographic and civic quantities are stocks and flows: water in a reservoir, people in a city, or goods in a warehouse.

Given facts. A reservoir starts with 41 water units. Each day 7 units flow in and 6 units flow out. Assume these rates stay constant for 5 days.

Rules. Stock at the end = starting stock + total inflow − total outflow. Flow is an amount per time period; stock is the amount currently stored.

Task. How much water is stored after 5 days? Is the stock increasing or decreasing each day?

Step-by-step solution. 1. Daily net change=7−6=1. 2. Over 5 days, net change=5×1=5. 3. End stock=41+(5)=46.

Answer. 46 units; the stock is increasing.

Knowledge takeaway. Do not confuse a stored amount with a rate of change; conservation links them.

Formal model. Difference equation S(t+1)=S(t)+inflow−outflow.

Problem 644. Reservoir stock and flow: case 4

Knowledge context. Many geographic and civic quantities are stocks and flows: water in a reservoir, people in a city, or goods in a warehouse.

Given facts. A reservoir starts with 44 water units. Each day 8 units flow in and 7 units flow out. Assume these rates stay constant for 5 days.

Rules. Stock at the end = starting stock + total inflow − total outflow. Flow is an amount per time period; stock is the amount currently stored.

Task. How much water is stored after 5 days? Is the stock increasing or decreasing each day?

Step-by-step solution. 1. Daily net change=8−7=1. 2. Over 5 days, net change=5×1=5. 3. End stock=44+(5)=49.

Answer. 49 units; the stock is increasing.

Knowledge takeaway. Do not confuse a stored amount with a rate of change; conservation links them.

Formal model. Difference equation S(t+1)=S(t)+inflow−outflow.

Problem 645. Reservoir stock and flow: case 5

Knowledge context. Many geographic and civic quantities are stocks and flows: water in a reservoir, people in a city, or goods in a warehouse.

Given facts. A reservoir starts with 47 water units. Each day 9 units flow in and 6 units flow out. Assume these rates stay constant for 5 days.

Rules. Stock at the end = starting stock + total inflow − total outflow. Flow is an amount per time period; stock is the amount currently stored.

Task. How much water is stored after 5 days? Is the stock increasing or decreasing each day?

Step-by-step solution. 1. Daily net change=9−6=3. 2. Over 5 days, net change=5×3=15. 3. End stock=47+(15)=62.

Answer. 62 units; the stock is increasing.

Knowledge takeaway. Do not confuse a stored amount with a rate of change; conservation links them.

Formal model. Difference equation S(t+1)=S(t)+inflow−outflow.

N5. Density, capacity, and crowding

Problem 646. Density and capacity in a public space: case 1

Knowledge context. Density compares an amount with the space available. It can reveal crowding even when two places have similar total populations.

Given facts. A park zone covers 4 hectares and currently has 35 visitors. For this exercise, comfortable density is at most 8 visitors per hectare.

Rules. Density = number ÷ area. A zone exceeds comfortable capacity if density is greater than the stated threshold.

Task. What is the visitor density, and is the zone above the stated comfortable capacity?

Step-by-step solution. 1. Density=35÷4=8.75 visitors/hectare. 2. Compare 8.75 with the threshold 8.

Answer. 8.75 visitors/hectare; above capacity.

Knowledge takeaway. Totals and densities answer different questions; always identify the denominator.

Formal model. Ratio density=N/A followed by threshold comparison.

Problem 647. Density and capacity in a public space: case 2

Knowledge context. Density compares an amount with the space available. It can reveal crowding even when two places have similar total populations.

Given facts. A park zone covers 5 hectares and currently has 39 visitors. For this exercise, comfortable density is at most 8 visitors per hectare.

Rules. Density = number ÷ area. A zone exceeds comfortable capacity if density is greater than the stated threshold.

Task. What is the visitor density, and is the zone above the stated comfortable capacity?

Step-by-step solution. 1. Density=39÷5=7.80 visitors/hectare. 2. Compare 7.80 with the threshold 8.

Answer. 7.80 visitors/hectare; within capacity.

Knowledge takeaway. Totals and densities answer different questions; always identify the denominator.

Formal model. Ratio density=N/A followed by threshold comparison.

Problem 648. Density and capacity in a public space: case 3

Knowledge context. Density compares an amount with the space available. It can reveal crowding even when two places have similar total populations.

Given facts. A park zone covers 6 hectares and currently has 43 visitors. For this exercise, comfortable density is at most 8 visitors per hectare.

Rules. Density = number ÷ area. A zone exceeds comfortable capacity if density is greater than the stated threshold.

Task. What is the visitor density, and is the zone above the stated comfortable capacity?

Step-by-step solution. 1. Density=43÷6=7.17 visitors/hectare. 2. Compare 7.17 with the threshold 8.

Answer. 7.17 visitors/hectare; within capacity.

Knowledge takeaway. Totals and densities answer different questions; always identify the denominator.

Formal model. Ratio density=N/A followed by threshold comparison.

Problem 649. Density and capacity in a public space: case 4

Knowledge context. Density compares an amount with the space available. It can reveal crowding even when two places have similar total populations.

Given facts. A park zone covers 7 hectares and currently has 47 visitors. For this exercise, comfortable density is at most 8 visitors per hectare.

Rules. Density = number ÷ area. A zone exceeds comfortable capacity if density is greater than the stated threshold.

Task. What is the visitor density, and is the zone above the stated comfortable capacity?

Step-by-step solution. 1. Density=47÷7=6.71 visitors/hectare. 2. Compare 6.71 with the threshold 8.

Answer. 6.71 visitors/hectare; within capacity.

Knowledge takeaway. Totals and densities answer different questions; always identify the denominator.

Formal model. Ratio density=N/A followed by threshold comparison.

Problem 650. Density and capacity in a public space: case 5

Knowledge context. Density compares an amount with the space available. It can reveal crowding even when two places have similar total populations.

Given facts. A park zone covers 8 hectares and currently has 51 visitors. For this exercise, comfortable density is at most 8 visitors per hectare.

Rules. Density = number ÷ area. A zone exceeds comfortable capacity if density is greater than the stated threshold.

Task. What is the visitor density, and is the zone above the stated comfortable capacity?

Step-by-step solution. 1. Density=51÷8=6.38 visitors/hectare. 2. Compare 6.38 with the threshold 8.

Answer. 6.38 visitors/hectare; within capacity.

Knowledge takeaway. Totals and densities answer different questions; always identify the denominator.

Formal model. Ratio density=N/A followed by threshold comparison.

N6. Land use and spatial compatibility

Problem 651. Land-use compatibility on four plots: case 1

Knowledge context. Land-use planning combines spatial relationships with compatibility rules.

Given facts. Four plots lie in a line A–B–C–D. Assign one different use to each plot: school, factory, park, homes. Constraints: the factory may not be adjacent to the school; the park must be adjacent to the homes. The school must be on an end plot.

Rules. An assignment is valid only if every use is placed exactly once and all adjacency constraints hold.

Task. Give one valid assignment and explain why it satisfies every constraint.

Step-by-step solution. 1. Try the assignment A=school, B=park, C=homes, D=factory. 2. Factory and school are not adjacent. 3. Park and homes are adjacent. The school is on an end plot. 4. Each use appears exactly once.

Answer. A=school, B=park, C=homes, D=factory is a valid assignment.

Knowledge takeaway. A place can be unsuitable not because of its own properties, but because of what would be next to it.

Formal model. Permutation CSP with all-different and adjacency/non-adjacency constraints.

Problem 652. Land-use compatibility on four plots: case 2

Knowledge context. Land-use planning combines spatial relationships with compatibility rules.

Given facts. Four plots lie in a line A–B–C–D. Assign one different use to each plot: school, factory, park, homes. Constraints: the factory may not be adjacent to the school; the park must be adjacent to the homes. The school must be on an end plot.

Rules. An assignment is valid only if every use is placed exactly once and all adjacency constraints hold.

Task. Give one valid assignment and explain why it satisfies every constraint.

Step-by-step solution. 1. Try the assignment A=school, B=homes, C=park, D=factory. 2. Factory and school are not adjacent. 3. Park and homes are adjacent. The school is on an end plot. 4. Each use appears exactly once.

Answer. A=school, B=homes, C=park, D=factory is a valid assignment.

Knowledge takeaway. A place can be unsuitable not because of its own properties, but because of what would be next to it.

Formal model. Permutation CSP with all-different and adjacency/non-adjacency constraints.

Problem 653. Land-use compatibility on four plots: case 3

Knowledge context. Land-use planning combines spatial relationships with compatibility rules.

Given facts. Four plots lie in a line A–B–C–D. Assign one different use to each plot: school, factory, park, homes. Constraints: the factory may not be adjacent to the school; the park must be adjacent to the homes. The school must be on an end plot.

Rules. An assignment is valid only if every use is placed exactly once and all adjacency constraints hold.

Task. Give one valid assignment and explain why it satisfies every constraint.

Step-by-step solution. 1. Try the assignment A=factory, B=park, C=homes, D=school. 2. Factory and school are not adjacent. 3. Park and homes are adjacent. The school is on an end plot. 4. Each use appears exactly once.

Answer. A=factory, B=park, C=homes, D=school is a valid assignment.

Knowledge takeaway. A place can be unsuitable not because of its own properties, but because of what would be next to it.

Formal model. Permutation CSP with all-different and adjacency/non-adjacency constraints.

Problem 654. Land-use compatibility on four plots: case 4

Knowledge context. Land-use planning combines spatial relationships with compatibility rules.

Given facts. Four plots lie in a line A–B–C–D. Assign one different use to each plot: school, factory, park, homes. Constraints: the factory may not be adjacent to the school; the park must be adjacent to the homes. The school must be on an end plot.

Rules. An assignment is valid only if every use is placed exactly once and all adjacency constraints hold.

Task. Give one valid assignment and explain why it satisfies every constraint.

Step-by-step solution. 1. Try the assignment A=factory, B=homes, C=park, D=school. 2. Factory and school are not adjacent. 3. Park and homes are adjacent. The school is on an end plot. 4. Each use appears exactly once.

Answer. A=factory, B=homes, C=park, D=school is a valid assignment.

Knowledge takeaway. A place can be unsuitable not because of its own properties, but because of what would be next to it.

Formal model. Permutation CSP with all-different and adjacency/non-adjacency constraints.

Problem 655. Land-use compatibility on four plots: case 5

Knowledge context. Land-use planning combines spatial relationships with compatibility rules.

Given facts. Four plots lie in a line A–B–C–D. Assign one different use to each plot: school, factory, park, homes. Constraints: the factory may not be adjacent to the school; the park must be adjacent to the homes. The school must be on an end plot. Cross-domain check: researchers collected 11 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. An assignment is valid only if every use is placed exactly once and all adjacency constraints hold. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. Give one valid assignment and explain why it satisfies every constraint. Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. Try the assignment A=school, B=park, C=homes, D=factory. 2. Factory and school are not adjacent. 3. Park and homes are adjacent. The school is on an end plot. 4. Each use appears exactly once. Cross-domain check: 11−4=7 independent reports remain.

Answer. A=school, B=park, C=homes, D=factory is a valid assignment. Cross-domain answer: 7 independent reports.

Knowledge takeaway. A place can be unsuitable not because of its own properties, but because of what would be next to it.

Formal model. Permutation CSP with all-different and adjacency/non-adjacency constraints.

N7. Natural risk: hazard, exposure, protection

Problem 656. Hazard, exposure, and protection: case 1

Knowledge context. Natural risk depends on more than the physical hazard. Exposure and protective measures also affect outcomes.

Given facts. For this classroom model, risk index = hazard × exposure ÷ protection. Data: Site A: hazard=2, exposure=4, protection=1; Site B: hazard=4, exposure=2, protection=2; Site C: hazard=4, exposure=3, protection=2.

Rules. Use the formula exactly as stated. A higher hazard does not necessarily mean higher risk if exposure or protection differs.

Task. Compute the risk index for each site. Which site has the lowest modeled risk?

Step-by-step solution. 1. Compute hazard×exposure÷protection for each site. 2. Results: A=8.00, B=4.00, C=6.00. 3. The lowest value is Site B.

Answer. A=8.00; B=4.00; C=6.00. Lowest: Site B.

Knowledge takeaway. Separating hazard from risk prevents the mistaken idea that the most dangerous process always creates the greatest overall risk.

Formal model. Explicit risk-index function over three variables; compare computed values.

Problem 657. Hazard, exposure, and protection: case 2

Knowledge context. Natural risk depends on more than the physical hazard. Exposure and protective measures also affect outcomes.

Given facts. For this classroom model, risk index = hazard × exposure ÷ protection. Data: Site A: hazard=3, exposure=4, protection=1; Site B: hazard=4, exposure=3, protection=2; Site C: hazard=4, exposure=3, protection=2.

Rules. Use the formula exactly as stated. A higher hazard does not necessarily mean higher risk if exposure or protection differs.

Task. Compute the risk index for each site. Which site has the lowest modeled risk?

Step-by-step solution. 1. Compute hazard×exposure÷protection for each site. 2. Results: A=12.00, B=6.00, C=6.00. 3. The lowest value is Site B.

Answer. A=12.00; B=6.00; C=6.00. Lowest: Site B.

Knowledge takeaway. Separating hazard from risk prevents the mistaken idea that the most dangerous process always creates the greatest overall risk.

Formal model. Explicit risk-index function over three variables; compare computed values.

Problem 658. Hazard, exposure, and protection: case 3

Knowledge context. Natural risk depends on more than the physical hazard. Exposure and protective measures also affect outcomes.

Given facts. For this classroom model, risk index = hazard × exposure ÷ protection. Data: Site A: hazard=4, exposure=4, protection=1; Site B: hazard=4, exposure=4, protection=2; Site C: hazard=4, exposure=3, protection=2.

Rules. Use the formula exactly as stated. A higher hazard does not necessarily mean higher risk if exposure or protection differs.

Task. Compute the risk index for each site. Which site has the lowest modeled risk?

Step-by-step solution. 1. Compute hazard×exposure÷protection for each site. 2. Results: A=16.00, B=8.00, C=6.00. 3. The lowest value is Site C.

Answer. A=16.00; B=8.00; C=6.00. Lowest: Site C.

Knowledge takeaway. Separating hazard from risk prevents the mistaken idea that the most dangerous process always creates the greatest overall risk.

Formal model. Explicit risk-index function over three variables; compare computed values.

Problem 659. Hazard, exposure, and protection: case 4

Knowledge context. Natural risk depends on more than the physical hazard. Exposure and protective measures also affect outcomes.

Given facts. For this classroom model, risk index = hazard × exposure ÷ protection. Data: Site A: hazard=5, exposure=4, protection=1; Site B: hazard=4, exposure=5, protection=2; Site C: hazard=4, exposure=3, protection=2.

Rules. Use the formula exactly as stated. A higher hazard does not necessarily mean higher risk if exposure or protection differs.

Task. Compute the risk index for each site. Which site has the lowest modeled risk?

Step-by-step solution. 1. Compute hazard×exposure÷protection for each site. 2. Results: A=20.00, B=10.00, C=6.00. 3. The lowest value is Site C.

Answer. A=20.00; B=10.00; C=6.00. Lowest: Site C.

Knowledge takeaway. Separating hazard from risk prevents the mistaken idea that the most dangerous process always creates the greatest overall risk.

Formal model. Explicit risk-index function over three variables; compare computed values.

Problem 660. Hazard, exposure, and protection: case 5

Knowledge context. Natural risk depends on more than the physical hazard. Exposure and protective measures also affect outcomes.

Given facts. For this classroom model, risk index = hazard × exposure ÷ protection. Data: Site A: hazard=6, exposure=4, protection=1; Site B: hazard=4, exposure=6, protection=2; Site C: hazard=4, exposure=3, protection=2.

Rules. Use the formula exactly as stated. A higher hazard does not necessarily mean higher risk if exposure or protection differs.

Task. Compute the risk index for each site. Which site has the lowest modeled risk?

Step-by-step solution. 1. Compute hazard×exposure÷protection for each site. 2. Results: A=24.00, B=12.00, C=6.00. 3. The lowest value is Site C.

Answer. A=24.00; B=12.00; C=6.00. Lowest: Site C.

Knowledge takeaway. Separating hazard from risk prevents the mistaken idea that the most dangerous process always creates the greatest overall risk.

Formal model. Explicit risk-index function over three variables; compare computed values.

N8. Ecosystems and indirect effects

Problem 661. Indirect effects in a food chain: case 1

Knowledge context. Ecosystems contain dependency networks. A change at one point can propagate indirectly through several links.

Given facts. Food relation: rabbit eats grass; fox eats rabbit; eagle eats fox. Classroom rule: if a food source sharply decreases and no alternative food is given, the consumer may also decrease. Cross-domain check: a survey team starts work at 12:00 and works for 2 hour(s) without a break. Mixed-domain verification: a regional archive has 68 map sheets and receives 2 additional sheets during the exercise.

Rules. Follow dependency arrows from food to consumer. The rule supports possible directional effects, not exact population numbers. For the cross-domain check, finishing time = starting time + duration. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. If rabbit sharply decreases, which later populations may be indirectly affected according to the rule? Cross-domain check: at what time does the team finish? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. fox depends directly on rabbit, so it may decrease. 2. eagle depends on fox, so it may also be indirectly affected. 3. grass is food for rabbit; the stated rule does not imply it decreases for the same reason. Cross-domain check: 12+2=14, so the team finishes at 14:00. Mixed-domain verification: 68+2=70.

Answer. fox may decrease directly and eagle may be affected indirectly. Cross-domain answer: 14:00. Mixed-domain answer: 70 map sheets.

Knowledge takeaway. Indirect effects require following the direction of dependency and respecting the limits of the stated model.

Formal model. Directed dependency graph; propagate a qualitative decrease along outgoing dependency links.

Problem 662. Indirect effects in a food chain: case 2

Knowledge context. Ecosystems contain dependency networks. A change at one point can propagate indirectly through several links.

Given facts. Food relation: small fish eats algae; large fish eats small fish; heron eats large fish. Classroom rule: if a food source sharply decreases and no alternative food is given, the consumer may also decrease. Cross-domain check: a local committee has 13 members, requires at least 7 present for quorum, and 11 are present.

Rules. Follow dependency arrows from food to consumer. The rule supports possible directional effects, not exact population numbers. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. If small fish sharply decreases, which later populations may be indirectly affected according to the rule? Cross-domain check: is quorum met?

Step-by-step solution. 1. large fish depends directly on small fish, so it may decrease. 2. heron depends on large fish, so it may also be indirectly affected. 3. algae is food for small fish; the stated rule does not imply it decreases for the same reason. Cross-domain check: compare 11 with 7; $11 \geq 7$ is true.

Answer. large fish may decrease directly and heron may be affected indirectly. Cross-domain answer: quorum is met.

Knowledge takeaway. Indirect effects require following the direction of dependency and respecting the limits of the stated model.

Formal model. Directed dependency graph; propagate a qualitative decrease along outgoing dependency links.

Problem 663. Indirect effects in a food chain: case 3

Knowledge context. Ecosystems contain dependency networks. A change at one point can propagate indirectly through several links.

Given facts. Food relation: rabbit eats grass; fox eats rabbit; eagle eats fox. Classroom rule: if a food source sharply decreases and no alternative food is given, the consumer may also decrease. Cross-domain check: researchers collected 9 reports, but 4 are exact duplicate copies that add no new independent information. Mixed-domain verification: a regional archive has 68 map sheets and receives 4 additional sheets during the exercise.

Rules. Follow dependency arrows from food to consumer. The rule supports possible directional effects, not exact population numbers. For the cross-domain check, independent reports = total reports − duplicate copies. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. If rabbit sharply decreases, which later populations may be indirectly affected according to the rule? Cross-domain check: how many non-duplicate reports remain? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. fox depends directly on rabbit, so it may decrease. 2. eagle depends on fox, so it may also be indirectly affected. 3. grass is food for rabbit; the stated rule does not imply it decreases for the same reason. Cross-domain check: 9−4=5 independent reports remain. Mixed-domain verification: 68+4=72.

Answer. fox may decrease directly and eagle may be affected indirectly. Cross-domain answer: 5 independent reports. Mixed-domain answer: 72 map sheets.

Knowledge takeaway. Indirect effects require following the direction of dependency and respecting the limits of the stated model.

Formal model. Directed dependency graph; propagate a qualitative decrease along outgoing dependency links.

Problem 664. Indirect effects in a food chain: case 4

Knowledge context. Ecosystems contain dependency networks. A change at one point can propagate indirectly through several links.

Given facts. Food relation: small fish eats algae; large fish eats small fish; heron eats large fish. Classroom rule: if a food source sharply decreases and no alternative food is given, the consumer may also decrease. Cross-domain check: on a field sketch, 1 cm represents 6 km and two checkpoints are 8 cm apart.

Rules. Follow dependency arrows from food to consumer. The rule supports possible directional effects, not exact population numbers. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. If small fish sharply decreases, which later populations may be indirectly affected according to the rule? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. large fish depends directly on small fish, so it may decrease. 2. heron depends on large fish, so it may also be indirectly affected. 3. algae is food for small fish; the stated rule does not imply it decreases for the same reason. Cross-domain check: 8×6=48 km.

Answer. large fish may decrease directly and heron may be affected indirectly. Cross-domain answer: 48 km.

Knowledge takeaway. Indirect effects require following the direction of dependency and respecting the limits of the stated model.

Formal model. Directed dependency graph; propagate a qualitative decrease along outgoing dependency links.

Problem 665. Indirect effects in a food chain: case 5

Knowledge context. Ecosystems contain dependency networks. A change at one point can propagate indirectly through several links.

Given facts. Food relation: rabbit eats grass; fox eats rabbit; eagle eats fox. Classroom rule: if a food source sharply decreases and no alternative food is given, the consumer may also decrease. Cross-domain check: a survey team starts work at 8:00 and works for 2 hour(s) without a break.

Rules. Follow dependency arrows from food to consumer. The rule supports possible directional effects, not exact population numbers. For the cross-domain check, finishing time = starting time + duration.

Task. If rabbit sharply decreases, which later populations may be indirectly affected according to the rule? Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. fox depends directly on rabbit, so it may decrease. 2. eagle depends on fox, so it may also be indirectly affected. 3. grass is food for rabbit; the stated rule does not imply it decreases for the same reason. Cross-domain check: 8+2=10, so the team finishes at 10:00.

Answer. fox may decrease directly and eagle may be affected indirectly. Cross-domain answer: 10:00.

Knowledge takeaway. Indirect effects require following the direction of dependency and respecting the limits of the stated model.

Formal model. Directed dependency graph; propagate a qualitative decrease along outgoing dependency links.

N9. Renewable resources and sustainable use

Problem 666. Regeneration and harvest: case 1

Knowledge context. Renewable does not mean unlimited. A resource can decline when use exceeds regeneration.

Given facts. A forest resource starts at 130 units. Each year it regenerates 12 units and people harvest 11 units. Assume these amounts stay constant for 3 years and no other changes occur.

Rules. Annual net change = regeneration − harvest. In this simplified model, use is sustainable over the period if stock does not decline.

Task. What is the stock after 3 years, and is the harvest sustainable under this classroom definition?

Step-by-step solution. 1. Annual net change=12−11=1. 2. Three-year change=3. 3. Final stock=130+(3)=133. 4. Compare final stock with starting stock.

Answer. 133 units; sustainable under the stated definition.

Knowledge takeaway. Sustainability is a balance between renewal and extraction over time.

Formal model. Stock-flow model with regeneration and harvest; test non-decreasing stock.

Problem 667. Regeneration and harvest: case 2

Knowledge context. Renewable does not mean unlimited. A resource can decline when use exceeds regeneration.

Given facts. A forest resource starts at 130 units. Each year it regenerates 14 units and people harvest 14 units. Assume these amounts stay constant for 3 years and no other changes occur.

Rules. Annual net change = regeneration − harvest. In this simplified model, use is sustainable over the period if stock does not decline.

Task. What is the stock after 3 years, and is the harvest sustainable under this classroom definition?

Step-by-step solution. 1. Annual net change=14−14=0. 2. Three-year change=0. 3. Final stock=130+(0)=130. 4. Compare final stock with starting stock.

Answer. 130 units; sustainable under the stated definition.

Knowledge takeaway. Sustainability is a balance between renewal and extraction over time.

Formal model. Stock-flow model with regeneration and harvest; test non-decreasing stock.

Problem 668. Regeneration and harvest: case 3

Knowledge context. Renewable does not mean unlimited. A resource can decline when use exceeds regeneration.

Given facts. A forest resource starts at 130 units. Each year it regenerates 16 units and people harvest 17 units. Assume these amounts stay constant for 3 years and no other changes occur.

Rules. Annual net change = regeneration − harvest. In this simplified model, use is sustainable over the period if stock does not decline.

Task. What is the stock after 3 years, and is the harvest sustainable under this classroom definition?

Step-by-step solution. 1. Annual net change=16−17=-1. 2. Three-year change=-3. 3. Final stock=130+(-3)=127. 4. Compare final stock with starting stock.

Answer. 127 units; not sustainable under the stated definition.

Knowledge takeaway. Sustainability is a balance between renewal and extraction over time.

Formal model. Stock-flow model with regeneration and harvest; test non-decreasing stock.

Problem 669. Regeneration and harvest: case 4

Knowledge context. Renewable does not mean unlimited. A resource can decline when use exceeds regeneration.

Given facts. A forest resource starts at 130 units. Each year it regenerates 18 units and people harvest 20 units. Assume these amounts stay constant for 3 years and no other changes occur.

Rules. Annual net change = regeneration − harvest. In this simplified model, use is sustainable over the period if stock does not decline.

Task. What is the stock after 3 years, and is the harvest sustainable under this classroom definition?

Step-by-step solution. 1. Annual net change=18−20=-2. 2. Three-year change=-6. 3. Final stock=130+(-6)=124. 4. Compare final stock with starting stock.

Answer. 124 units; not sustainable under the stated definition.

Knowledge takeaway. Sustainability is a balance between renewal and extraction over time.

Formal model. Stock-flow model with regeneration and harvest; test non-decreasing stock.

Problem 670. Regeneration and harvest: case 5

Knowledge context. Renewable does not mean unlimited. A resource can decline when use exceeds regeneration.

Given facts. A forest resource starts at 130 units. Each year it regenerates 20 units and people harvest 23 units. Assume these amounts stay constant for 3 years and no other changes occur.

Rules. Annual net change = regeneration − harvest. In this simplified model, use is sustainable over the period if stock does not decline.

Task. What is the stock after 3 years, and is the harvest sustainable under this classroom definition?

Step-by-step solution. 1. Annual net change=20−23=-3. 2. Three-year change=-9. 3. Final stock=130+(-9)=121. 4. Compare final stock with starting stock.

Answer. 121 units; not sustainable under the stated definition.

Knowledge takeaway. Sustainability is a balance between renewal and extraction over time.

Formal model. Stock-flow model with regeneration and harvest; test non-decreasing stock.

N10. Migration and population balance

Problem 671. Population balance in a town: case 1

Knowledge context. Population changes through births, deaths, immigration, and emigration.

Given facts. A town begins the year with 1300 residents. During the year: births=20, deaths=13, people moving in=36, people moving out=25.

Rules. End population = start + births − deaths + in-migration − out-migration. Natural change and migration change are separate components.

Task. Compute natural change, migration change, and the year-end population.

Step-by-step solution. 1. Natural change=20−13=7. 2. Migration change=36−25=11. 3. Total change=7+11=18. 4. End population=1300+18=1318.

Answer. Natural change +7; migration change +11; end population 1318.

Knowledge takeaway. Breaking total change into components helps explain why a population grew or shrank.

Formal model. Balance equation P1=P0+births−deaths+in−out.

Problem 672. Population balance in a town: case 2

Knowledge context. Population changes through births, deaths, immigration, and emigration.

Given facts. A town begins the year with 1350 residents. During the year: births=23, deaths=13, people moving in=36, people moving out=29.

Rules. End population = start + births − deaths + in-migration − out-migration. Natural change and migration change are separate components.

Task. Compute natural change, migration change, and the year-end population.

Step-by-step solution. 1. Natural change=23−13=10. 2. Migration change=36−29=7. 3. Total change=10+7=17. 4. End population=1350+17=1367.

Answer. Natural change +10; migration change +7; end population 1367.

Knowledge takeaway. Breaking total change into components helps explain why a population grew or shrank.

Formal model. Balance equation P1=P0+births−deaths+in−out.

Problem 673. Population balance in a town: case 3

Knowledge context. Population changes through births, deaths, immigration, and emigration.

Given facts. A town begins the year with 1400 residents. During the year: births=26, deaths=13, people moving in=36, people moving out=33.

Rules. End population = start + births − deaths + in-migration − out-migration. Natural change and migration change are separate components.

Task. Compute natural change, migration change, and the year-end population.

Step-by-step solution. 1. Natural change=26−13=13. 2. Migration change=36−33=3. 3. Total change=13+3=16. 4. End population=1400+16=1416.

Answer. Natural change +13; migration change +3; end population 1416.

Knowledge takeaway. Breaking total change into components helps explain why a population grew or shrank.

Formal model. Balance equation P1=P0+births−deaths+in−out.

Problem 674. Population balance in a town: case 4

Knowledge context. Population changes through births, deaths, immigration, and emigration.

Given facts. A town begins the year with 1450 residents. During the year: births=29, deaths=13, people moving in=36, people moving out=37.

Rules. End population = start + births − deaths + in-migration − out-migration. Natural change and migration change are separate components.

Task. Compute natural change, migration change, and the year-end population.

Step-by-step solution. 1. Natural change=29−13=16. 2. Migration change=36−37=-1. 3. Total change=16+-1=15. 4. End population=1450+15=1465.

Answer. Natural change +16; migration change -1; end population 1465.

Knowledge takeaway. Breaking total change into components helps explain why a population grew or shrank.

Formal model. Balance equation P1=P0+births−deaths+in−out.

Problem 675. Population balance in a town: case 5

Knowledge context. Population changes through births, deaths, immigration, and emigration.

Given facts. A town begins the year with 1500 residents. During the year: births=32, deaths=13, people moving in=36, people moving out=41.

Rules. End population = start + births − deaths + in-migration − out-migration. Natural change and migration change are separate components.

Task. Compute natural change, migration change, and the year-end population.

Step-by-step solution. 1. Natural change=32−13=19. 2. Migration change=36−41=-5. 3. Total change=19+-5=14. 4. End population=1500+14=1514.

Answer. Natural change +19; migration change -5; end population 1514.

Knowledge takeaway. Breaking total change into components helps explain why a population grew or shrank.

Formal model. Balance equation P1=P0+births−deaths+in−out.

N11. Time intervals and overlapping events

Problem 676. Overlapping historical intervals: case 1

Knowledge context. Historical periods often overlap rather than forming a single neat sequence.

Given facts. Event A lasted from year 100 to 145. Event B lasted from 100 to 134. Treat intervals as continuous for this exercise.

Rules. Two intervals overlap when max(start times) < min(end times). Overlap length = min(end times) − max(start times), if positive.

Task. Did the events overlap? If so, for how many years?

Step-by-step solution. 1. Later start=max(100,100)=100. 2. Earlier end=min(145,134)=134. 3. Difference=134−100=34.

Answer. Yes, overlap 34 years.

Knowledge takeaway. Interval reasoning distinguishes “before/after” from “overlapping in time.”

Formal model. Interval intersection; compute max starts and min ends.

Problem 677. Overlapping historical intervals: case 2

Knowledge context. Historical periods often overlap rather than forming a single neat sequence.

Given facts. Event A lasted from year 120 to 165. Event B lasted from 130 to 164. Treat intervals as continuous for this exercise.

Rules. Two intervals overlap when max(start times) < min(end times). Overlap length = min(end times) − max(start times), if positive.

Task. Did the events overlap? If so, for how many years?

Step-by-step solution. 1. Later start=max(120,130)=130. 2. Earlier end=min(165,164)=164. 3. Difference=164−130=34.

Answer. Yes, overlap 34 years.

Knowledge takeaway. Interval reasoning distinguishes “before/after” from “overlapping in time.”

Formal model. Interval intersection; compute max starts and min ends.

Problem 678. Overlapping historical intervals: case 3

Knowledge context. Historical periods often overlap rather than forming a single neat sequence.

Given facts. Event A lasted from year 140 to 185. Event B lasted from 160 to 194. Treat intervals as continuous for this exercise.

Rules. Two intervals overlap when max(start times) < min(end times). Overlap length = min(end times) − max(start times), if positive.

Task. Did the events overlap? If so, for how many years?

Step-by-step solution. 1. Later start=max(140,160)=160. 2. Earlier end=min(185,194)=185. 3. Difference=185−160=25.

Answer. Yes, overlap 25 years.

Knowledge takeaway. Interval reasoning distinguishes “before/after” from “overlapping in time.”

Formal model. Interval intersection; compute max starts and min ends.

Problem 679. Overlapping historical intervals: case 4

Knowledge context. Historical periods often overlap rather than forming a single neat sequence.

Given facts. Event A lasted from year 160 to 205. Event B lasted from 160 to 194. Treat intervals as continuous for this exercise.

Rules. Two intervals overlap when max(start times) < min(end times). Overlap length = min(end times) − max(start times), if positive.

Task. Did the events overlap? If so, for how many years?

Step-by-step solution. 1. Later start=max(160,160)=160. 2. Earlier end=min(205,194)=194. 3. Difference=194−160=34.

Answer. Yes, overlap 34 years.

Knowledge takeaway. Interval reasoning distinguishes “before/after” from “overlapping in time.”

Formal model. Interval intersection; compute max starts and min ends.

Problem 680. Overlapping historical intervals: case 5

Knowledge context. Historical periods often overlap rather than forming a single neat sequence.

Given facts. Event A lasted from year 180 to 225. Event B lasted from 190 to 224. Treat intervals as continuous for this exercise.

Rules. Two intervals overlap when max(start times) < min(end times). Overlap length = min(end times) − max(start times), if positive.

Task. Did the events overlap? If so, for how many years?

Step-by-step solution. 1. Later start=max(180,190)=190. 2. Earlier end=min(225,224)=224. 3. Difference=224−190=34.

Answer. Yes, overlap 34 years.

Knowledge takeaway. Interval reasoning distinguishes “before/after” from “overlapping in time.”

Formal model. Interval intersection; compute max starts and min ends.

N12. Genealogy and generations

Problem 681. Reasoning across generations: case 1

Knowledge context. Genealogies are relational graphs. Generation distance can be counted through parent links.

Given facts. Maya is a parent of Noah. Noah is a parent of Lina. Maya is also a parent of Owen. Owen is a parent of Sara. Cross-domain check: a survey team starts work at 8:00 and works for 2 hour(s) without a break.

Rules. A parent is one generation above a child. A grandparent is two parent-links above. People who share a parent are siblings under this simplified model. For the cross-domain check, finishing time = starting time + duration.

Task. What family relation between Maya and Lina follows from the stated links? Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. Maya→Noah is one parent link. 2. Noah→Lina is a second parent link. 3. Two links give the grandparent relation. Cross-domain check: 8+2=10, so the team finishes at 10:00.

Answer. Maya is a grandparent of Lina. Cross-domain answer: 10:00.

Knowledge takeaway. Family relations can be derived step by step from a small set of explicit links.

Formal model. Directed parent graph; kinship predicates derived from path length and shared ancestors.

Problem 682. Reasoning across generations: case 2

Knowledge context. Genealogies are relational graphs. Generation distance can be counted through parent links.

Given facts. Maya is a parent of Noah. Noah is a parent of Lina. Maya is also a parent of Owen. Owen is a parent of Sara. Cross-domain check: a local committee has 15 members, requires at least 8 present for quorum, and 14 are present.

Rules. A parent is one generation above a child. A grandparent is two parent-links above. People who share a parent are siblings under this simplified model. For the cross-domain check, quorum exists when present members ≥ the stated threshold.

Task. What family relation between Noah and Lina follows from the stated links? Cross-domain check: is quorum met?

Step-by-step solution. 1. The relation is stated directly. Cross-domain check: compare 14 with 8; 14 ≥ 8 is true.

Answer. Noah is a parent of Lina. Cross-domain answer: quorum is met.

Knowledge takeaway. Family relations can be derived step by step from a small set of explicit links.

Formal model. Directed parent graph; kinship predicates derived from path length and shared ancestors.

Problem 683. Reasoning across generations: case 3

Knowledge context. Genealogies are relational graphs. Generation distance can be counted through parent links.

Given facts. Maya is a parent of Noah. Noah is a parent of Lina. Maya is also a parent of Owen. Owen is a parent of Sara. Cross-domain check: researchers collected 9 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. A parent is one generation above a child. A grandparent is two parent-links above. People who share a parent are siblings under this simplified model. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. What family relation between Noah and Owen follows from the stated links? Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. Maya is parent of both Noah and Owen. 2. Sharing a parent satisfies the simplified sibling rule. Cross-domain check: 9−4=5 independent reports remain.

Answer. Noah and Owen are siblings. Cross-domain answer: 5 independent reports.

Knowledge takeaway. Family relations can be derived step by step from a small set of explicit links.

Formal model. Directed parent graph; kinship predicates derived from path length and shared ancestors.

Problem 684. Reasoning across generations: case 4

Knowledge context. Genealogies are relational graphs. Generation distance can be counted through parent links.

Given facts. Maya is a parent of Noah. Noah is a parent of Lina. Maya is also a parent of Owen. Owen is a parent of Sara. Cross-domain check: on a field sketch, 1 cm represents 6 km and two checkpoints are 7 cm apart.

Rules. A parent is one generation above a child. A grandparent is two parent-links above. People who share a parent are siblings under this simplified model. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. What family relation between Maya and Sara follows from the stated links? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. Maya→Owen and Owen→Sara form two parent links. Cross-domain check: 7×6=42 km.

Answer. Maya is a grandparent of Sara. Cross-domain answer: 42 km.

Knowledge takeaway. Family relations can be derived step by step from a small set of explicit links.

Formal model. Directed parent graph; kinship predicates derived from path length and shared ancestors.

Problem 685. Reasoning across generations: case 5

Knowledge context. Genealogies are relational graphs. Generation distance can be counted through parent links.

Given facts. Maya is a parent of Noah. Noah is a parent of Lina. Maya is also a parent of Owen. Owen is a parent of Sara. Cross-domain check: a survey team starts work at 12:00 and works for 2 hour(s) without a break.

Rules. A parent is one generation above a child. A grandparent is two parent-links above. People who share a parent are siblings under this simplified model. For the cross-domain check, finishing time = starting time + duration.

Task. What family relation between Lina and Sara follows from the stated links? Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. Noah and Owen are siblings because they share parent Maya. 2. Lina is child of Noah; Sara is child of Owen. 3. Children of siblings are cousins in this model. Cross-domain check: 12+2=14, so the team finishes at 14:00.

Answer. Lina and Sara are cousins under the implied family tree. Cross-domain answer: 14:00.

Knowledge takeaway. Family relations can be derived step by step from a small set of explicit links.

Formal model. Directed parent graph; kinship predicates derived from path length and shared ancestors.

N13. Anachronisms and temporal compatibility

Problem 686. Temporal compatibility in a story set in 1400

Knowledge context. Anachronism can be tested as a compatibility problem when the relevant date ranges are supplied.

Given facts. For this problem, availability intervals are: Technology A: 1200–1600; Institution B: 1500–1900; Tool C: 1700–2000; Practice D: 1300–1450. A story is set in 1400 and mentions Technology A and Tool C together. Cross-domain check: a local committee has 10 members, requires at least 6 present for quorum, and 8 are present.

Rules. An item is temporally compatible with a year only if the year lies inside its stated availability interval. An anachronism is an item placed outside its given interval. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. Which mentioned item, if any, is anachronistic in the story? Cross-domain check: is quorum met?

Step-by-step solution. 1. Check 1400 against Technology A interval 1200–1600. 2. Check 1400 against Tool C interval 1700–2000. 3. Items outside their intervals: ['Tool C']. Cross-domain check: compare 8 with 6; $8 \geq 6$ is true.

Answer. Anachronistic: Tool C. Cross-domain answer: quorum is met.

Knowledge takeaway. The reasoning does not require memorizing invention dates if the intervals are part of the problem.

Formal model. Interval membership test for each object-event pair.

Problem 687. Temporal compatibility in a story set in 1550

Knowledge context. Anachronism can be tested as a compatibility problem when the relevant date ranges are supplied.

Given facts. For this problem, availability intervals are: Technology A: 1200–1600; Institution B: 1500–1900; Tool C: 1700–2000; Practice D: 1300–1450. A story is set in 1550 and mentions Technology A and Tool C together. Cross-domain check: researchers collected 13 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. An item is temporally compatible with a year only if the year lies inside its stated availability interval. An anachronism is an item placed outside its given interval. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. Which mentioned item, if any, is anachronistic in the story? Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. Check 1550 against Technology A interval 1200–1600. 2. Check 1550 against Tool C interval 1700–2000. 3. Items outside their intervals: ['Tool C']. Cross-domain check: 13−4=9 independent reports remain.

Answer. Anachronistic: Tool C. Cross-domain answer: 9 independent reports.

Knowledge takeaway. The reasoning does not require memorizing invention dates if the intervals are part of the problem.

Formal model. Interval membership test for each object-event pair.

Problem 688. Temporal compatibility in a story set in 1750

Knowledge context. Anachronism can be tested as a compatibility problem when the relevant date ranges are supplied.

Given facts. For this problem, availability intervals are: Technology A: 1200–1600; Institution B: 1500–1900; Tool C: 1700–2000; Practice D: 1300–1450. A story is set in 1750 and mentions Technology A and Tool C together. Cross-domain check: on a field sketch, 1 cm represents 5 km and two checkpoints are 4 cm apart.

Rules. An item is temporally compatible with a year only if the year lies inside its stated availability interval. An anachronism is an item placed outside its given interval. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Which mentioned item, if any, is anachronistic in the story? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. Check 1750 against Technology A interval 1200–1600. 2. Check 1750 against Tool C interval 1700–2000. 3. Items outside their intervals: ['Technology A']. Cross-domain check: 4×5=20 km.

Answer. Anachronistic: Technology A. Cross-domain answer: 20 km.

Knowledge takeaway. The reasoning does not require memorizing invention dates if the intervals are part of the problem.

Formal model. Interval membership test for each object-event pair.

Problem 689. Temporal compatibility in a story set in 1850

Knowledge context. Anachronism can be tested as a compatibility problem when the relevant date ranges are supplied.

Given facts. For this problem, availability intervals are: Technology A: 1200–1600; Institution B: 1500–1900; Tool C: 1700–2000; Practice D: 1300–1450. A story is set in 1850 and mentions Technology A and Tool C together. Cross-domain check: a survey team starts work at 8:00 and works for 2 hour(s) without a break.

Rules. An item is temporally compatible with a year only if the year lies inside its stated availability interval. An anachronism is an item placed outside its given interval. For the cross-domain check, finishing time = starting time + duration.

Task. Which mentioned item, if any, is anachronistic in the story? Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. Check 1850 against Technology A interval 1200–1600. 2. Check 1850 against Tool C interval 1700–2000. 3. Items outside their intervals: ['Technology A']. Cross-domain check: 8+2=10, so the team finishes at 10:00.

Answer. Anachronistic: Technology A. Cross-domain answer: 10:00.

Knowledge takeaway. The reasoning does not require memorizing invention dates if the intervals are part of the problem.

Formal model. Interval membership test for each object-event pair.

Problem 690. Temporal compatibility in a story set in 1500

Knowledge context. Anachronism can be tested as a compatibility problem when the relevant date ranges are supplied.

Given facts. For this problem, availability intervals are: Technology A: 1200–1600; Institution B: 1500–1900; Tool C: 1700–2000; Practice D: 1300–1450. A story is set in 1500 and mentions Technology A and Tool C together. Cross-domain check: a local committee has 14 members, requires at least 8 present for quorum, and 14 are present.

Rules. An item is temporally compatible with a year only if the year lies inside its stated availability interval. An anachronism is an item placed outside its given interval. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. Which mentioned item, if any, is anachronistic in the story? Cross-domain check: is quorum met?

Step-by-step solution. 1. Check 1500 against Technology A interval 1200–1600. 2. Check 1500 against Tool C interval 1700–2000. 3. Items outside their intervals: ['Tool C']. Cross-domain check: compare 14 with 8; $14 \geq 8$ is true.

Answer. Anachronistic: Tool C. Cross-domain answer: quorum is met.

Knowledge takeaway. The reasoning does not require memorizing invention dates if the intervals are part of the problem.

Formal model. Interval membership test for each object-event pair.

N14. Provenance and chain of custody

Problem 691. Provenance and chain of custody: case 1

Knowledge context. Provenance records where an object came from and how it moved between custodians.

Given facts. An artifact record states: found by Excavator on day 1; transferred to Museum A on day 2; sent from Museum A to Research Lab on day 5; returned by Research Lab to Museum B on day 8. Every transfer must name both sender and receiver. Cross-domain check: researchers collected 7 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. A continuous chain of custody requires each handoff to connect the previous holder to the next. A missing sender or receiver creates a gap that limits certainty about continuous custody. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. Is the stated chain of custody complete under the rule? If not, identify the gap. Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. Read the transfers in time order. 2. Check that every transfer links the previous holder to the next. 3. Every handoff names both sides and connects continuously. Cross-domain check: 7−4=3 independent reports remain.

Answer. The chain is complete under the stated rule. Cross-domain answer: 3 independent reports.

Knowledge takeaway. A strong provenance claim depends on an unbroken sequence of documented handoffs.

Formal model. Time-ordered path through holders; verify continuity of sender/receiver pairs.

Problem 692. Provenance and chain of custody: case 2

Knowledge context. Provenance records where an object came from and how it moved between custodians.

Given facts. An artifact record states: found by Excavator on day 1; transferred to Museum A on day 2; sent from Museum A to Research Lab on day 5; returned by Research Lab to Museum B on day 8. Every transfer must name both sender and receiver. The record for day 5 names Research Lab as receiver but does not name the sender. Cross-domain check: on a field sketch, 1 cm represents 4 km and two checkpoints are 8 cm apart.

Rules. A continuous chain of custody requires each handoff to connect the previous holder to the next. A missing sender or receiver creates a gap that limits certainty about continuous custody. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Is the stated chain of custody complete under the rule? If not, identify the gap. Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. Read the transfers in time order. 2. Check that every transfer links the previous holder to the next. 3. The day-5 record lacks a sender, so the chain has a gap. Cross-domain check: 8×4=32 km.

Answer. The chain is incomplete because the day-5 sender is missing. Cross-domain answer: 32 km.

Knowledge takeaway. A strong provenance claim depends on an unbroken sequence of documented handoffs.

Formal model. Time-ordered path through holders; verify continuity of sender/receiver pairs.

Problem 693. Provenance and chain of custody: case 3

Knowledge context. Provenance records where an object came from and how it moved between custodians.

Given facts. An artifact record states: found by Excavator on day 1; transferred to Museum A on day 2; sent from Museum A to Research Lab on day 5; returned by Research Lab to Museum B on day 8. Every transfer must name both sender and receiver. Cross-domain check: a survey team starts work at 12:00 and works for 2 hour(s) without a break. Mixed-domain verification: a regional archive has 71 map sheets and receives 4 additional sheets during the exercise.

Rules. A continuous chain of custody requires each handoff to connect the previous holder to the next. A missing sender or receiver creates a gap that limits certainty about continuous custody. For the cross-domain check, finishing time = starting time + duration. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. Is the stated chain of custody complete under the rule? If not, identify the gap. Cross-domain check: at what time does the team finish? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. Read the transfers in time order. 2. Check that every transfer links the previous holder to the next. 3. Every handoff names both sides and connects continuously. Cross-domain check: 12+2=14, so the team finishes at 14:00. Mixed-domain verification: 71+4=75.

Answer. The chain is complete under the stated rule. Cross-domain answer: 14:00. Mixed-domain answer: 75 map sheets.

Knowledge takeaway. A strong provenance claim depends on an unbroken sequence of documented handoffs.

Formal model. Time-ordered path through holders; verify continuity of sender/receiver pairs.

Problem 694. Provenance and chain of custody: case 4

Knowledge context. Provenance records where an object came from and how it moved between custodians.

Given facts. An artifact record states: found by Excavator on day 1; transferred to Museum A on day 2; sent from Museum A to Research Lab on day 5; returned by Research Lab to Museum B on day 8. Every transfer must name both sender and receiver. The record for day 5 names Research Lab as receiver but does not name the sender. Cross-domain check: a local committee has 9 members, requires at least 5 present for quorum, and 8 are present. Mixed-domain verification: a regional archive has 71 map sheets and receives 5 additional sheets during the exercise.

Rules. A continuous chain of custody requires each handoff to connect the previous holder to the next. A missing sender or receiver creates a gap that limits certainty about continuous custody. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. Is the stated chain of custody complete under the rule? If not, identify the gap. Cross-domain check: is quorum met? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. Read the transfers in time order. 2. Check that every transfer links the previous holder to the next. 3. The day-5 record lacks a sender, so the chain has a gap. Cross-domain check: compare 8 with 5; $8 \geq 5$ is true. Mixed-domain verification: 71+5=76.

Answer. The chain is incomplete because the day-5 sender is missing. Cross-domain answer: quorum is met. Mixed-domain answer: 76 map sheets.

Knowledge takeaway. A strong provenance claim depends on an unbroken sequence of documented handoffs.

Formal model. Time-ordered path through holders; verify continuity of sender/receiver pairs.

Problem 695. Provenance and chain of custody: case 5

Knowledge context. Provenance records where an object came from and how it moved between custodians.

Given facts. An artifact record states: found by Excavator on day 1; transferred to Museum A on day 2; sent from Museum A to Research Lab on day 5; returned by Research Lab to Museum B on day 8. Every transfer must name both sender and receiver. Cross-domain check: researchers collected 11 reports, but 4 are exact duplicate copies that add no new independent information. Mixed-domain verification: a regional archive has 71 map sheets and receives 6 additional sheets during the exercise.

Rules. A continuous chain of custody requires each handoff to connect the previous holder to the next. A missing sender or receiver creates a gap that limits certainty about continuous custody. For the cross-domain check, independent reports = total reports − duplicate copies. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. Is the stated chain of custody complete under the rule? If not, identify the gap. Cross-domain check: how many non-duplicate reports remain? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. Read the transfers in time order. 2. Check that every transfer links the previous holder to the next. 3. Every handoff names both sides and connects continuously. Cross-domain check: 11−4=7 independent reports remain. Mixed-domain verification: 71+6=77.

Answer. The chain is complete under the stated rule. Cross-domain answer: 7 independent reports. Mixed-domain answer: 77 map sheets.

Knowledge takeaway. A strong provenance claim depends on an unbroken sequence of documented handoffs.

Formal model. Time-ordered path through holders; verify continuity of sender/receiver pairs.

N15. Perspective, interest, and source credibility

Problem 696. Perspective, interest, and source quality: case 1

Knowledge context. Sources have perspectives and interests. Credibility is not a simple label; it depends on the claim being evaluated and how the source could know it.

Given facts. Three sources describe a proposed market tax. Source A is a merchant who would pay the tax and argues it is harmful. Source B is a council report that estimates revenue and lists its calculation. Source C is a traveler repeating a rumor heard in another town. Cross-domain check: on a field sketch, 1 cm represents 3 km and two checkpoints are 5 cm apart.

Rules. Perspective and interest do not automatically make a source false. Evaluate relevance, access to information, explicit evidence, and independence. A source with an interest may still provide useful facts, but its claims should be checked. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Which source is best positioned to estimate the amount of revenue, and which source should be treated most cautiously for that numerical claim? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. Revenue estimation requires access to tax rates and expected transactions. 2. Source B explicitly provides a calculation and is directly about the policy. 3. Source C is a secondhand rumor from elsewhere and has the weakest access to the specific numerical question. 4. Source A has an interest, but interest alone is not enough to discard it. Cross-domain check: 5×3=15 km.

Answer. Source B is best positioned for the revenue estimate; Source C should be treated most cautiously for that numerical claim. Cross-domain answer: 15 km.

Knowledge takeaway. Bias is a reason to inspect evidence, not a magic switch that makes every statement true or false.

Formal model. Source-claim assessment using access, relevance, independence, and evidentiary support.

Problem 697. Perspective, interest, and source quality: case 2

Knowledge context. Sources have perspectives and interests. Credibility is not a simple label; it depends on the claim being evaluated and how the source could know it.

Given facts. Three sources describe a proposed market tax. Source A is a merchant who would pay the tax and argues it is harmful. Source B is a council report that estimates revenue and lists its calculation. Source C is a traveler repeating a rumor heard in another town. Cross-domain check: a survey team starts work at 8:00 and works for 2 hour(s) without a break. Mixed-domain verification: a regional archive has 71 map sheets and receives 8 additional sheets during the exercise.

Rules. Perspective and interest do not automatically make a source false. Evaluate relevance, access to information, explicit evidence, and independence. A source with an interest may still provide useful facts, but its claims should be checked. For the cross-domain check, finishing time = starting time + duration. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. Which source is best positioned to estimate the amount of revenue, and which source should be treated most cautiously for that numerical claim? Cross-domain check: at what time does the team finish? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. Revenue estimation requires access to tax rates and expected transactions. 2. Source B explicitly provides a calculation and is directly about the policy. 3. Source C is a secondhand rumor from elsewhere and has the weakest access to the specific numerical question. 4. Source A has an interest, but interest alone is not enough to discard it. Cross-domain check: 8+2=10, so the team finishes at 10:00. Mixed-domain verification: 71+8=79.

Answer. Source B is best positioned for the revenue estimate; Source C should be treated most cautiously for that numerical claim. Cross-domain answer: 10:00. Mixed-domain answer: 79 map sheets.

Knowledge takeaway. Bias is a reason to inspect evidence, not a magic switch that makes every statement true or false.

Formal model. Source-claim assessment using access, relevance, independence, and evidentiary support.

Problem 698. Perspective, interest, and source quality: case 3

Knowledge context. Sources have perspectives and interests. Credibility is not a simple label; it depends on the claim being evaluated and how the source could know it.

Given facts. Three sources describe a proposed market tax. Source A is a merchant who would pay the tax and argues it is harmful. Source B is a council report that estimates revenue and lists its calculation. Source C is a traveler repeating a rumor heard in another town. Cross-domain check: a local committee has 13 members, requires at least 7 present for quorum, and 11 are present. Mixed-domain verification: a regional archive has 71 map sheets and receives 9 additional sheets during the exercise.

Rules. Perspective and interest do not automatically make a source false. Evaluate relevance, access to information, explicit evidence, and independence. A source with an interest may still provide useful facts, but its claims should be checked. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. Which source is best positioned to estimate the amount of revenue, and which source should be treated most cautiously for that numerical claim? Cross-domain check: is quorum met? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. Revenue estimation requires access to tax rates and expected transactions. 2. Source B explicitly provides a calculation and is directly about the policy. 3. Source C is a secondhand rumor from elsewhere and has the weakest access to the specific numerical question. 4. Source A has an interest, but interest alone is not enough to discard it. Cross-domain check: compare 11 with 7; $11 \geq 7$ is true. Mixed-domain verification: 71+9=80.

Answer. Source B is best positioned for the revenue estimate; Source C should be treated most cautiously for that numerical claim. Cross-domain answer: quorum is met. Mixed-domain answer: 80 map sheets.

Knowledge takeaway. Bias is a reason to inspect evidence, not a magic switch that makes every statement true or false.

Formal model. Source-claim assessment using access, relevance, independence, and evidentiary support.

Problem 699. Perspective, interest, and source quality: case 4

Knowledge context. Sources have perspectives and interests. Credibility is not a simple label; it depends on the claim being evaluated and how the source could know it.

Given facts. Three sources describe a proposed market tax. Source A is a merchant who would pay the tax and argues it is harmful. Source B is a council report that estimates revenue and lists its calculation. Source C is a traveler repeating a rumor heard in another town. Cross-domain check: researchers collected 15 reports, but 4 are exact duplicate copies that add no new independent information. Mixed-domain verification: a regional archive has 71 map sheets and receives 10 additional sheets during the exercise.

Rules. Perspective and interest do not automatically make a source false. Evaluate relevance, access to information, explicit evidence, and independence. A source with an interest may still provide useful facts, but its claims should be checked. For the cross-domain check, independent reports = total reports − duplicate copies. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. Which source is best positioned to estimate the amount of revenue, and which source should be treated most cautiously for that numerical claim? Cross-domain check: how many non-duplicate reports remain? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. Revenue estimation requires access to tax rates and expected transactions. 2. Source B explicitly provides a calculation and is directly about the policy. 3. Source C is a secondhand rumor from elsewhere and has the weakest access to the specific numerical question. 4. Source A has an interest, but interest alone is not enough to discard it. Cross-domain check: 15−4=11 independent reports remain. Mixed-domain verification: 71+10=81.

Answer. Source B is best positioned for the revenue estimate; Source C should be treated most cautiously for that numerical claim. Cross-domain answer: 11 independent reports. Mixed-domain answer: 81 map sheets.

Knowledge takeaway. Bias is a reason to inspect evidence, not a magic switch that makes every statement true or false.

Formal model. Source-claim assessment using access, relevance, independence, and evidentiary support.

Problem 700. Perspective, interest, and source quality: case 5

Knowledge context. Sources have perspectives and interests. Credibility is not a simple label; it depends on the claim being evaluated and how the source could know it.

Given facts. Three sources describe a proposed market tax. Source A is a merchant who would pay the tax and argues it is harmful. Source B is a council report that estimates revenue and lists its calculation. Source C is a traveler repeating a rumor heard in another town. Cross-domain check: on a field sketch, 1 cm represents 2 km and two checkpoints are 2 cm apart.

Rules. Perspective and interest do not automatically make a source false. Evaluate relevance, access to information, explicit evidence, and independence. A source with an interest may still provide useful facts, but its claims should be checked. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Which source is best positioned to estimate the amount of revenue, and which source should be treated most cautiously for that numerical claim? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. Revenue estimation requires access to tax rates and expected transactions. 2. Source B explicitly provides a calculation and is directly about the policy. 3. Source C is a secondhand rumor from elsewhere and has the weakest access to the specific numerical question. 4. Source A has an interest, but interest alone is not enough to discard it. Cross-domain check: 2×2=4 km.

Answer. Source B is best positioned for the revenue estimate; Source C should be treated most cautiously for that numerical claim. Cross-domain answer: 4 km.

Knowledge takeaway. Bias is a reason to inspect evidence, not a magic switch that makes every statement true or false.

Formal model. Source-claim assessment using access, relevance, independence, and evidentiary support.

N16. Uncertain data, intervals, and limits

Problem 701. Reasoning with uncertain intervals: case 1

Knowledge context. Measurements and historical dates can be uncertain. An interval represents a range of possibilities rather than one exact number.

Given facts. Two measurements are uncertain intervals. A is between 8 and 12. B is between 13 and 17. A statement “A is definitely greater than B” is valid only if A’s minimum is greater than B’s maximum.

Rules. For a definite comparison under interval uncertainty, test the worst cases. Overlapping intervals usually prevent a strict guaranteed ordering.

Task. Can we conclude that A is definitely greater than B? What comparison is justified?

Step-by-step solution. 1. A minimum=8, maximum=12; B minimum=13, maximum=17. 2. Test A_min > B_max: 8>17 is False. 3. If false, at least one allowed pair of values defeats the “definitely greater” claim.

Answer. No. The intervals do not justify saying A is definitely greater than B.

Knowledge takeaway. Robust conclusions must remain true across all values allowed by the uncertainty.

Formal model. Interval arithmetic / worst-case comparison.

Problem 702. Reasoning with uncertain intervals: case 2

Knowledge context. Measurements and historical dates can be uncertain. An interval represents a range of possibilities rather than one exact number.

Given facts. Two measurements are uncertain intervals. A is between 9 and 13. B is between 13 and 17. A statement “A is definitely greater than B” is valid only if A’s minimum is greater than B’s maximum.

Rules. For a definite comparison under interval uncertainty, test the worst cases. Overlapping intervals usually prevent a strict guaranteed ordering.

Task. Can we conclude that A is definitely greater than B? What comparison is justified?

Step-by-step solution. 1. A minimum=9, maximum=13; B minimum=13, maximum=17. 2. Test A_min > B_max: 9>17 is False. 3. If false, at least one allowed pair of values defeats the “definitely greater” claim.

Answer. No. The intervals do not justify saying A is definitely greater than B.

Knowledge takeaway. Robust conclusions must remain true across all values allowed by the uncertainty.

Formal model. Interval arithmetic / worst-case comparison.

Problem 703. Reasoning with uncertain intervals: case 3

Knowledge context. Measurements and historical dates can be uncertain. An interval represents a range of possibilities rather than one exact number.

Given facts. Two measurements are uncertain intervals. A is between 10 and 14. B is between 13 and 17. A statement “A is definitely greater than B” is valid only if A’s minimum is greater than B’s maximum.

Rules. For a definite comparison under interval uncertainty, test the worst cases. Overlapping intervals usually prevent a strict guaranteed ordering.

Task. Can we conclude that A is definitely greater than B? What comparison is justified?

Step-by-step solution. 1. A minimum=10, maximum=14; B minimum=13, maximum=17. 2. Test A_min > B_max: 10>17 is False. 3. If false, at least one allowed pair of values defeats the “definitely greater” claim.

Answer. No. The intervals do not justify saying A is definitely greater than B.

Knowledge takeaway. Robust conclusions must remain true across all values allowed by the uncertainty.

Formal model. Interval arithmetic / worst-case comparison.

Problem 704. Reasoning with uncertain intervals: case 4

Knowledge context. Measurements and historical dates can be uncertain. An interval represents a range of possibilities rather than one exact number.

Given facts. Two measurements are uncertain intervals. A is between 11 and 15. B is between 13 and 17. A statement “A is definitely greater than B” is valid only if A’s minimum is greater than B’s maximum.

Rules. For a definite comparison under interval uncertainty, test the worst cases. Overlapping intervals usually prevent a strict guaranteed ordering.

Task. Can we conclude that A is definitely greater than B? What comparison is justified?

Step-by-step solution. 1. A minimum=11, maximum=15; B minimum=13, maximum=17. 2. Test A_min > B_max: 11>17 is False. 3. If false, at least one allowed pair of values defeats the “definitely greater” claim.

Answer. No. The intervals do not justify saying A is definitely greater than B.

Knowledge takeaway. Robust conclusions must remain true across all values allowed by the uncertainty.

Formal model. Interval arithmetic / worst-case comparison.

Problem 705. Reasoning with uncertain intervals: case 5

Knowledge context. Measurements and historical dates can be uncertain. An interval represents a range of possibilities rather than one exact number.

Given facts. Two measurements are uncertain intervals. A is between 12 and 16. B is between 13 and 17. A statement “A is definitely greater than B” is valid only if A’s minimum is greater than B’s maximum.

Rules. For a definite comparison under interval uncertainty, test the worst cases. Overlapping intervals usually prevent a strict guaranteed ordering.

Task. Can we conclude that A is definitely greater than B? What comparison is justified?

Step-by-step solution. 1. A minimum=12, maximum=16; B minimum=13, maximum=17. 2. Test A_min > B_max: 12>17 is False. 3. If false, at least one allowed pair of values defeats the “definitely greater” claim.

Answer. No. The intervals do not justify saying A is definitely greater than B.

Knowledge takeaway. Robust conclusions must remain true across all values allowed by the uncertainty.

Formal model. Interval arithmetic / worst-case comparison.

N17. Counterfactual reasoning

Problem 706. Counterfactual intervention: close the gate

Knowledge context. Counterfactual reasoning asks what the model predicts if one condition were changed while the rest of the causal structure is held fixed.

Given facts. Causal rules for a simplified town model: heavy rain → high river; high river AND unclosed gate → flooded square; flooded square → market closes. Actual facts: heavy rain occurred and the gate was not closed. Cross-domain check: a local committee has 12 members, requires at least 7 present for quorum, and 11 are present.

Rules. A counterfactual intervention changes one specified fact and then recomputes consequences from the rules. Keep all other stated rules the same unless the intervention changes them. For the cross-domain check, quorum exists when present members ≥ the stated threshold.

Task. Under the intervention “close the gate”, what changes in the consequences? Cross-domain check: is quorum met?

Step-by-step solution. 1. Actual chain gives high river, then flood, then market closure. 2. Intervene: set gate=closed. 3. The flood rule requires an unclosed gate, so flooding no longer follows. 4. Without flooding, market closure does not follow. Cross-domain check: compare 11 with 7; 11 ≥ 7 is true.

Answer. With the gate closed, the model no longer implies flooding or market closure. Cross-domain answer: quorum is met.

Knowledge takeaway. A counterfactual is not ordinary storytelling; it is a controlled change followed by recomputation.

Formal model. Structural causal rules with a do-style intervention on one variable.

Problem 707. Counterfactual intervention: remove heavy rain

Knowledge context. Counterfactual reasoning asks what the model predicts if one condition were changed while the rest of the causal structure is held fixed.

Given facts. Causal rules for a simplified town model: heavy rain → high river; high river AND unclosed gate → flooded square; flooded square → market closes. Actual facts: heavy rain occurred and the gate was not closed. Cross-domain check: researchers collected 13 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. A counterfactual intervention changes one specified fact and then recomputes consequences from the rules. Keep all other stated rules the same unless the intervention changes them. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. Under the intervention “remove heavy rain”, what changes in the consequences? Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. Intervene: heavy rain=false. 2. Then high river is not generated by the first rule. 3. Without high river, flooding and market closure are not generated. Cross-domain check: 13−4=9 independent reports remain.

Answer. Without heavy rain, the model generates neither high river, flood, nor market closure. Cross-domain answer: 9 independent reports.

Knowledge takeaway. A counterfactual is not ordinary storytelling; it is a controlled change followed by recomputation.

Formal model. Structural causal rules with a do-style intervention on one variable.

Problem 708. Counterfactual intervention: keep gate open

Knowledge context. Counterfactual reasoning asks what the model predicts if one condition were changed while the rest of the causal structure is held fixed.

Given facts. Causal rules for a simplified town model: heavy rain → high river; high river AND unclosed gate → flooded square; flooded square → market closes. Actual facts: heavy rain occurred and the gate was not closed. Cross-domain check: on a field sketch, 1 cm represents 5 km and two checkpoints are 3 cm apart.

Rules. A counterfactual intervention changes one specified fact and then recomputes consequences from the rules. Keep all other stated rules the same unless the intervention changes them. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Under the intervention “keep gate open”, what changes in the consequences? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. Gate already remains open. 2. Heavy rain gives high river. 3. High river plus open gate gives flood, then market closure. Cross-domain check: 3×5=15 km.

Answer. The original chain remains: flood and market closure occur. Cross-domain answer: 15 km.

Knowledge takeaway. A counterfactual is not ordinary storytelling; it is a controlled change followed by recomputation.

Formal model. Structural causal rules with a do-style intervention on one variable.

Problem 709. Counterfactual intervention: prevent high river

Knowledge context. Counterfactual reasoning asks what the model predicts if one condition were changed while the rest of the causal structure is held fixed.

Given facts. Causal rules for a simplified town model: heavy rain → high river; high river AND unclosed gate → flooded square; flooded square → market closes. Actual facts: heavy rain occurred and the gate was not closed. Cross-domain check: a survey team starts work at 12:00 and works for 2 hour(s) without a break.

Rules. A counterfactual intervention changes one specified fact and then recomputes consequences from the rules. Keep all other stated rules the same unless the intervention changes them. For the cross-domain check, finishing time = starting time + duration.

Task. Under the intervention “prevent high river”, what changes in the consequences? Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. Intervene directly: high river=false. 2. Even with rain, the counterfactual intervention fixes high river as prevented for this question. 3. Flood requires high river, so it does not follow. Cross-domain check: 12+2=14, so the team finishes at 14:00.

Answer. Preventing the high river blocks the flood and market closure in this model. Cross-domain answer: 14:00.

Knowledge takeaway. A counterfactual is not ordinary storytelling; it is a controlled change followed by recomputation.

Formal model. Structural causal rules with a do-style intervention on one variable.

Problem 710. Counterfactual intervention: keep market open

Knowledge context. Counterfactual reasoning asks what the model predicts if one condition were changed while the rest of the causal structure is held fixed.

Given facts. Causal rules for a simplified town model: heavy rain → high river; high river AND unclosed gate → flooded square; flooded square → market closes. Actual facts: heavy rain occurred and the gate was not closed. Cross-domain check: a local committee has 16 members, requires at least 9 present for quorum, and 14 are present.

Rules. A counterfactual intervention changes one specified fact and then recomputes consequences from the rules. Keep all other stated rules the same unless the intervention changes them. For the cross-domain check, quorum exists when present members ≥ the stated threshold.

Task. Under the intervention “keep market open”, what changes in the consequences? Cross-domain check: is quorum met?

Step-by-step solution. 1. The proposed intervention changes the market outcome directly, not the upstream causes. 2. Rain still gives high river and flood. 3. Setting market=open does not erase the flood unless a rule says so. Cross-domain check: compare 14 with 9; 14 ≥ 9 is true.

Answer. The square still floods; the market is held open only by the direct intervention. Cross-domain answer: quorum is met.

Knowledge takeaway. A counterfactual is not ordinary storytelling; it is a controlled change followed by recomputation.

Formal model. Structural causal rules with a do-style intervention on one variable.

N18. Public goods and common contributions

Problem 711. Funding a shared public project: case 1

Knowledge context. Public goods can benefit many people at once, which creates coordination problems about contributions.

Given facts. A community project is built only if total voluntary contributions reach 14 tokens. Contributions from 7 households are [1, 2, 3, 4, 1, 2, 3]. Everyone can use the project if it is built, including households that contribute zero.

Rules. Add contributions and compare with the threshold. A public-good threshold concerns whether the project is produced; it does not by itself decide whether every contribution pattern is fair.

Task. Is the project built? How many additional tokens, if any, are needed?

Step-by-step solution. 1. Total contribution=1+2+3+4+1+2+3=16. 2. Compare 16 with threshold 14. 3. Additional amount needed=max(0,14−16)=0.

Answer. The project is built. Additional tokens needed: 0.

Knowledge takeaway. Threshold provision and fairness are separate questions; first determine whether enough resources exist.

Formal model. Sum contributions; compare with provision threshold; compute shortfall.

Problem 712. Funding a shared public project: case 2

Knowledge context. Public goods can benefit many people at once, which creates coordination problems about contributions.

Given facts. A community project is built only if total voluntary contributions reach 14 tokens. Contributions from 7 households are [2, 3, 4, 1, 2, 3, 4]. Everyone can use the project if it is built, including households that contribute zero.

Rules. Add contributions and compare with the threshold. A public-good threshold concerns whether the project is produced; it does not by itself decide whether every contribution pattern is fair.

Task. Is the project built? How many additional tokens, if any, are needed?

Step-by-step solution. 1. Total contribution=2+3+4+1+2+3+4=19. 2. Compare 19 with threshold 14. 3. Additional amount needed=max(0,14−19)=0.

Answer. The project is built. Additional tokens needed: 0.

Knowledge takeaway. Threshold provision and fairness are separate questions; first determine whether enough resources exist.

Formal model. Sum contributions; compare with provision threshold; compute shortfall.

Problem 713. Funding a shared public project: case 3

Knowledge context. Public goods can benefit many people at once, which creates coordination problems about contributions.

Given facts. A community project is built only if total voluntary contributions reach 14 tokens. Contributions from 7 households are [3, 4, 1, 2, 3, 4, 1]. Everyone can use the project if it is built, including households that contribute zero.

Rules. Add contributions and compare with the threshold. A public-good threshold concerns whether the project is produced; it does not by itself decide whether every contribution pattern is fair.

Task. Is the project built? How many additional tokens, if any, are needed?

Step-by-step solution. 1. Total contribution=3+4+1+2+3+4+1=18. 2. Compare 18 with threshold 14. 3. Additional amount needed=max(0,14−18)=0.

Answer. The project is built. Additional tokens needed: 0.

Knowledge takeaway. Threshold provision and fairness are separate questions; first determine whether enough resources exist.

Formal model. Sum contributions; compare with provision threshold; compute shortfall.

Problem 714. Funding a shared public project: case 4

Knowledge context. Public goods can benefit many people at once, which creates coordination problems about contributions.

Given facts. A community project is built only if total voluntary contributions reach 14 tokens. Contributions from 7 households are [4, 1, 2, 3, 4, 1, 2]. Everyone can use the project if it is built, including households that contribute zero.

Rules. Add contributions and compare with the threshold. A public-good threshold concerns whether the project is produced; it does not by itself decide whether every contribution pattern is fair.

Task. Is the project built? How many additional tokens, if any, are needed?

Step-by-step solution. 1. Total contribution=4+1+2+3+4+1+2=17. 2. Compare 17 with threshold 14. 3. Additional amount needed=max(0,14−17)=0.

Answer. The project is built. Additional tokens needed: 0.

Knowledge takeaway. Threshold provision and fairness are separate questions; first determine whether enough resources exist.

Formal model. Sum contributions; compare with provision threshold; compute shortfall.

Problem 715. Funding a shared public project: case 5

Knowledge context. Public goods can benefit many people at once, which creates coordination problems about contributions.

Given facts. A community project is built only if total voluntary contributions reach 14 tokens. Contributions from 7 households are [1, 2, 3, 4, 1, 2, 3]. Everyone can use the project if it is built, including households that contribute zero. Cross-domain check: researchers collected 11 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. Add contributions and compare with the threshold. A public-good threshold concerns whether the project is produced; it does not by itself decide whether every contribution pattern is fair. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. Is the project built? How many additional tokens, if any, are needed? Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. Total contribution=1+2+3+4+1+2+3=16. 2. Compare 16 with threshold 14. 3. Additional amount needed=max(0,14−16)=0. Cross-domain check: 11−4=7 independent reports remain.

Answer. The project is built. Additional tokens needed: 0. Cross-domain answer: 7 independent reports.

Knowledge takeaway. Threshold provision and fairness are separate questions; first determine whether enough resources exist.

Formal model. Sum contributions; compare with provision threshold; compute shortfall.

N19. Common resources and avoiding depletion

Problem 716. Using a shared renewable resource: case 1

Knowledge context. A common resource can be overused when many individually reasonable withdrawals add up to more than regeneration.

Given facts. A shared fishery has 90 fish at the start of the period and regenerates 10. There are 6 users, each planning to take 3. For this simplified exercise, the stock should not decline below its starting level if use is to be called non-depleting.

Rules. End stock = start + regeneration − total harvest. Total harvest is number of users × harvest per user.

Task. What is the end stock? Is the planned use non-depleting under the stated definition? What equal per-user harvest would exactly match regeneration?

Step-by-step solution. 1. Total harvest=6×3=18. 2. End stock=90+10−18=82. 3. To match regeneration exactly, equal harvest per user would be 10÷6=1.67.

Answer. End stock 82; depleting. Equal regeneration-matching share: 1.67 each.

Knowledge takeaway. The shared-resource problem appears at the total level, so individual plans must be aggregated.

Formal model. Commons stock equation plus equal-share benchmark.

Problem 717. Using a shared renewable resource: case 2

Knowledge context. A common resource can be overused when many individually reasonable withdrawals add up to more than regeneration.

Given facts. A shared fishery has 90 fish at the start of the period and regenerates 11. There are 6 users, each planning to take 4. For this simplified exercise, the stock should not decline below its starting level if use is to be called non-depleting.

Rules. End stock = start + regeneration − total harvest. Total harvest is number of users × harvest per user.

Task. What is the end stock? Is the planned use non-depleting under the stated definition? What equal per-user harvest would exactly match regeneration?

Step-by-step solution. 1. Total harvest=6×4=24. 2. End stock=90+11−24=77. 3. To match regeneration exactly, equal harvest per user would be 11÷6=1.83.

Answer. End stock 77; depleting. Equal regeneration-matching share: 1.83 each.

Knowledge takeaway. The shared-resource problem appears at the total level, so individual plans must be aggregated.

Formal model. Commons stock equation plus equal-share benchmark.

Problem 718. Using a shared renewable resource: case 3

Knowledge context. A common resource can be overused when many individually reasonable withdrawals add up to more than regeneration.

Given facts. A shared fishery has 90 fish at the start of the period and regenerates 12. There are 6 users, each planning to take 5. For this simplified exercise, the stock should not decline below its starting level if use is to be called non-depleting.

Rules. End stock = start + regeneration − total harvest. Total harvest is number of users × harvest per user.

Task. What is the end stock? Is the planned use non-depleting under the stated definition? What equal per-user harvest would exactly match regeneration?

Step-by-step solution. 1. Total harvest=6×5=30. 2. End stock=90+12−30=72. 3. To match regeneration exactly, equal harvest per user would be 12÷6=2.00.

Answer. End stock 72; depleting. Equal regeneration-matching share: 2.00 each.

Knowledge takeaway. The shared-resource problem appears at the total level, so individual plans must be aggregated.

Formal model. Commons stock equation plus equal-share benchmark.

Problem 719. Using a shared renewable resource: case 4

Knowledge context. A common resource can be overused when many individually reasonable withdrawals add up to more than regeneration.

Given facts. A shared fishery has 90 fish at the start of the period and regenerates 13. There are 6 users, each planning to take 6. For this simplified exercise, the stock should not decline below its starting level if use is to be called non-depleting.

Rules. End stock = start + regeneration − total harvest. Total harvest is number of users × harvest per user.

Task. What is the end stock? Is the planned use non-depleting under the stated definition? What equal per-user harvest would exactly match regeneration?

Step-by-step solution. 1. Total harvest=6×6=36. 2. End stock=90+13−36=67. 3. To match regeneration exactly, equal harvest per user would be 13÷6=2.17.

Answer. End stock 67; depleting. Equal regeneration-matching share: 2.17 each.

Knowledge takeaway. The shared-resource problem appears at the total level, so individual plans must be aggregated.

Formal model. Commons stock equation plus equal-share benchmark.

Problem 720. Using a shared renewable resource: case 5

Knowledge context. A common resource can be overused when many individually reasonable withdrawals add up to more than regeneration.

Given facts. A shared fishery has 90 fish at the start of the period and regenerates 14. There are 6 users, each planning to take 7. For this simplified exercise, the stock should not decline below its starting level if use is to be called non-depleting.

Rules. End stock = start + regeneration − total harvest. Total harvest is number of users × harvest per user.

Task. What is the end stock? Is the planned use non-depleting under the stated definition? What equal per-user harvest would exactly match regeneration?

Step-by-step solution. 1. Total harvest=6×7=42. 2. End stock=90+14−42=62. 3. To match regeneration exactly, equal harvest per user would be 14÷6=2.33.

Answer. End stock 62; depleting. Equal regeneration-matching share: 2.33 each.

Knowledge takeaway. The shared-resource problem appears at the total level, so individual plans must be aggregated.

Formal model. Commons stock equation plus equal-share benchmark.

N20. Fairness: equal, proportional, and need-based

Problem 721. Fair allocation by the rule: equal

Knowledge context. Fairness is not one formula. Equal shares, proportional shares, and need-based shares embody different principles.

Given facts. A relief fund has 45 units for three districts A, B, C. Populations are [10, 20, 30]; need scores are [1, 5, 3]. Three fairness principles are defined: equal = same amount each; population-proportional = shares in ratio 10:20:30; need-proportional = shares in ratio 1:5:3.

Rules. Different fairness principles answer different questions. Apply the principle named in the task; do not mix them unless asked to compare.

Task. Allocate the fund equally.

Step-by-step solution. 1. Equal share=45÷3=15.00. 2. Allocation result: A=15.00, B=15.00, C=15.00.

Answer. A=15.00, B=15.00, C=15.00.

Knowledge takeaway. A sound argument states the fairness rule first and then applies it consistently.

Formal model. Allocation function parameterized by a fairness principle and weights.

Problem 722. Fair allocation by the rule: proportional to population

Knowledge context. Fairness is not one formula. Equal shares, proportional shares, and need-based shares embody different principles.

Given facts. A relief fund has 45 units for three districts A, B, C. Populations are [10, 20, 30]; need scores are [2, 5, 4]. Three fairness principles are defined: equal = same amount each; population-proportional = shares in ratio 10:20:30; need-proportional = shares in ratio 2:5:4.

Rules. Different fairness principles answer different questions. Apply the principle named in the task; do not mix them unless asked to compare.

Task. Allocate in proportion to population.

Step-by-step solution. 1. Total population weight=60. 2. Multiply the fund by each population fraction. 3. Allocation result: A=7.50, B=15.00, C=22.50.

Answer. A=7.50, B=15.00, C=22.50.

Knowledge takeaway. A sound argument states the fairness rule first and then applies it consistently.

Formal model. Allocation function parameterized by a fairness principle and weights.

Problem 723. Fair allocation by the rule: proportional to need

Knowledge context. Fairness is not one formula. Equal shares, proportional shares, and need-based shares embody different principles.

Given facts. A relief fund has 45 units for three districts A, B, C. Populations are [10, 20, 30]; need scores are [3, 5, 5]. Three fairness principles are defined: equal = same amount each; population-proportional = shares in ratio 10:20:30; need-proportional = shares in ratio 3:5:5.

Rules. Different fairness principles answer different questions. Apply the principle named in the task; do not mix them unless asked to compare.

Task. Allocate in proportion to need score.

Step-by-step solution. 1. Total need weight=13. 2. Multiply the fund by each need fraction. 3. Allocation result: A=10.38, B=17.31, C=17.31.

Answer. A=10.38, B=17.31, C=17.31.

Knowledge takeaway. A sound argument states the fairness rule first and then applies it consistently.

Formal model. Allocation function parameterized by a fairness principle and weights.

Problem 724. Fair allocation by the rule: minimum guarantee then equal remainder

Knowledge context. Fairness is not one formula. Equal shares, proportional shares, and need-based shares embody different principles.

Given facts. A relief fund has 45 units for three districts A, B, C. Populations are [10, 20, 30]; need scores are [4, 5, 6]. Three fairness principles are defined: equal = same amount each; population-proportional = shares in ratio 10:20:30; need-proportional = shares in ratio 4:5:6.

Rules. Different fairness principles answer different questions. Apply the principle named in the task; do not mix them unless asked to compare.

Task. Give each district a 5-unit minimum guarantee, then divide the remainder equally.

Step-by-step solution. 1. Minimum guarantees use 15 units. 2. Remainder=45-15=30. 3. Each gets an additional 10.00. 4. Allocation result: A=15.00, B=15.00, C=15.00.

Answer. A=15.00, B=15.00, C=15.00.

Knowledge takeaway. A sound argument states the fairness rule first and then applies it consistently.

Formal model. Allocation function parameterized by a fairness principle and weights.

Problem 725. Fair allocation by the rule: compare methods

Knowledge context. Fairness is not one formula. Equal shares, proportional shares, and need-based shares embody different principles.

Given facts. A relief fund has 45 units for three districts A, B, C. Populations are [10, 20, 30]; need scores are [5, 5, 7]. Three fairness principles are defined: equal = same amount each; population-proportional = shares in ratio 10:20:30; need-proportional = shares in ratio 5:5:7.

Rules. Different fairness principles answer different questions. Apply the principle named in the task; do not mix them unless asked to compare.

Task. Compare equal allocation with need-proportional allocation: which gives more to the highest-need district?

Step-by-step solution. 1. Equal allocation gives C 15.00. 2. Need-proportional allocation gives C 18.53. 3. Compare those amounts. 4. Allocation result: A=13.24, B=13.24, C=18.53.

Answer. A=13.24, B=13.24, C=18.53.

Knowledge takeaway. A sound argument states the fairness rule first and then applies it consistently.

Formal model. Allocation function parameterized by a fairness principle and weights.

N21. Negotiation, coalitions, and agreement

Problem 726. Coalition formation: case 1

Knowledge context. In representative bodies, groups may need to cooperate to reach a decision threshold.

Given facts. Three groups have seats A=7, B=3, C=3. A governing coalition needs at least 7 seats. Any two groups may cooperate.

Rules. A coalition is winning if the sum of its members' seats reaches the threshold. Minimal winning means it is winning but removing any member makes it lose.

Task. Which coalitions are minimal winning coalitions?

Step-by-step solution. 1. Majority threshold=7. 2. Add seats for each possible coalition. 3. Minimal winning coalitions: [('A', 7)].

Answer. A with 7 seats.

Knowledge takeaway. A coalition can be winning without being minimal; removing redundant members tests how necessary each partner is.

Formal model. Weighted coalition game; enumerate subsets, test threshold and minimality.

Problem 727. Coalition formation: case 2

Knowledge context. In representative bodies, groups may need to cooperate to reach a decision threshold.

Given facts. Three groups have seats A=7, B=4, C=3. A governing coalition needs at least 8 seats. Any two groups may cooperate.

Rules. A coalition is winning if the sum of its members' seats reaches the threshold. Minimal winning means it is winning but removing any member makes it lose.

Task. Which coalitions are minimal winning coalitions?

Step-by-step solution. 1. Majority threshold=8. 2. Add seats for each possible coalition. 3. Minimal winning coalitions: [('AB', 11), ('AC', 10)].

Answer. AB with 11 seats; AC with 10 seats.

Knowledge takeaway. A coalition can be winning without being minimal; removing redundant members tests how necessary each partner is.

Formal model. Weighted coalition game; enumerate subsets, test threshold and minimality.

Problem 728. Coalition formation: case 3

Knowledge context. In representative bodies, groups may need to cooperate to reach a decision threshold.

Given facts. Three groups have seats A=7, B=5, C=3. A governing coalition needs at least 8 seats. Any two groups may cooperate.

Rules. A coalition is winning if the sum of its members' seats reaches the threshold. Minimal winning means it is winning but removing any member makes it lose.

Task. Which coalitions are minimal winning coalitions?

Step-by-step solution. 1. Majority threshold=8. 2. Add seats for each possible coalition. 3. Minimal winning coalitions: [('AB', 12), ('AC', 10), ('BC', 8)].

Answer. AB with 12 seats; AC with 10 seats; BC with 8 seats.

Knowledge takeaway. A coalition can be winning without being minimal; removing redundant members tests how necessary each partner is.

Formal model. Weighted coalition game; enumerate subsets, test threshold and minimality.

Problem 729. Coalition formation: case 4

Knowledge context. In representative bodies, groups may need to cooperate to reach a decision threshold.

Given facts. Three groups have seats A=7, B=6, C=3. A governing coalition needs at least 9 seats. Any two groups may cooperate.

Rules. A coalition is winning if the sum of its members' seats reaches the threshold. Minimal winning means it is winning but removing any member makes it lose.

Task. Which coalitions are minimal winning coalitions?

Step-by-step solution. 1. Majority threshold=9. 2. Add seats for each possible coalition. 3. Minimal winning coalitions: [('AB', 13), ('AC', 10), ('BC', 9)].

Answer. AB with 13 seats; AC with 10 seats; BC with 9 seats.

Knowledge takeaway. A coalition can be winning without being minimal; removing redundant members tests how necessary each partner is.

Formal model. Weighted coalition game; enumerate subsets, test threshold and minimality.

Problem 730. Coalition formation: case 5

Knowledge context. In representative bodies, groups may need to cooperate to reach a decision threshold.

Given facts. Three groups have seats A=7, B=7, C=3. A governing coalition needs at least 9 seats. Any two groups may cooperate.

Rules. A coalition is winning if the sum of its members' seats reaches the threshold. Minimal winning means it is winning but removing any member makes it lose.

Task. Which coalitions are minimal winning coalitions?

Step-by-step solution. 1. Majority threshold=9. 2. Add seats for each possible coalition. 3. Minimal winning coalitions: [('AB', 14), ('AC', 10), ('BC', 10)].

Answer. AB with 14 seats; AC with 10 seats; BC with 10 seats.

Knowledge takeaway. A coalition can be winning without being minimal; removing redundant members tests how necessary each partner is.

Formal model. Weighted coalition game; enumerate subsets, test threshold and minimality.

N22. Due process, evidence, and right of reply

Problem 731. Procedure, evidence, and a right to respond: case 1

Knowledge context. Fair procedures often combine evidentiary standards with an opportunity for the affected person to answer the case.

Given facts. A school rule says a student may receive a serious penalty only if: (1) at least two independent pieces of evidence support the allegation, and (2) the student is given a chance to respond before the final decision. In a case, one camera record and one independent witness support the allegation, but the student was not allowed to respond. Cross-domain check: researchers collected 7 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. Independence concerns distinct information sources, not merely the number of people repeating one source. Every required procedural condition must be satisfied. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. May the serious penalty be imposed under the stated procedure? Identify the condition that succeeds or fails. Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. Camera and witness are independent, so the evidence condition is met. 2. The student was not given a chance to respond. 3. The procedure requires both conditions, so the penalty is not authorized. Cross-domain check: 7−4=3 independent reports remain.

Answer. No. The right-to-respond condition is not met. Cross-domain answer: 3 independent reports.

Knowledge takeaway. A correct outcome does not excuse skipping a required procedure when the rule explicitly makes procedure part of validity.

Formal model. Conjunctive procedural predicate over evidence independence/count and response opportunity.

Problem 732. Procedure, evidence, and a right to respond: case 2

Knowledge context. Fair procedures often combine evidentiary standards with an opportunity for the affected person to answer the case.

Given facts. A school rule says a serious penalty requires two independent pieces of evidence and a chance to respond. In this case there are two friends repeating the same rumor from one original speaker, plus a chance to respond. Cross-domain check: on a field sketch, 1 cm represents 4 km and two checkpoints are 6 cm apart.

Rules. Independence concerns distinct information sources, not merely the number of people repeating one source. Every required procedural condition must be satisfied. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. May the serious penalty be imposed under the stated procedure? Identify the condition that succeeds or fails. Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. The two friends trace to the same original rumor, so they are not two independent pieces under the rule. 2. The response condition is met. 3. Because the evidence condition fails, the serious penalty is not authorized. Cross-domain check: 6×4=24 km.

Answer. No. The evidence requirement is not met. Cross-domain answer: 24 km.

Knowledge takeaway. A correct outcome does not excuse skipping a required procedure when the rule explicitly makes procedure part of validity.

Formal model. Conjunctive procedural predicate over evidence independence/count and response opportunity.

Problem 733. Procedure, evidence, and a right to respond: case 3

Knowledge context. Fair procedures often combine evidentiary standards with an opportunity for the affected person to answer the case.

Given facts. A school rule says a student may receive a serious penalty only if: (1) at least two independent pieces of evidence support the allegation, and (2) the student is given a chance to respond before the final decision. In a case, one camera record and one independent witness support the allegation, but the student was not allowed to respond. Cross-domain check: a survey team starts work at 12:00 and works for 2 hour(s) without a break. Mixed-domain verification: a regional archive has 75 map sheets and receives 4 additional sheets during the exercise.

Rules. Independence concerns distinct information sources, not merely the number of people repeating one source. Every required procedural condition must be satisfied. For the cross-domain check, finishing time = starting time + duration. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. May the serious penalty be imposed under the stated procedure? Identify the condition that succeeds or fails. Cross-domain check: at what time does the team finish? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. Camera and witness are independent, so the evidence condition is met. 2. The student was not given a chance to respond. 3. The procedure requires both conditions, so the penalty is not authorized. Cross-domain check: 12+2=14, so the team finishes at 14:00. Mixed-domain verification: 75+4=79.

Answer. No. The right-to-respond condition is not met. Cross-domain answer: 14:00. Mixed-domain answer: 79 map sheets.

Knowledge takeaway. A correct outcome does not excuse skipping a required procedure when the rule explicitly makes procedure part of validity.

Formal model. Conjunctive procedural predicate over evidence independence/count and response opportunity.

Problem 734. Procedure, evidence, and a right to respond: case 4

Knowledge context. Fair procedures often combine evidentiary standards with an opportunity for the affected person to answer the case.

Given facts. A school rule says a serious penalty requires two independent pieces of evidence and a chance to respond. In this case there are two friends repeating the same rumor from one original speaker, plus a chance to respond. Cross-domain check: a local committee has 13 members, requires at least 7 present for quorum, and 11 are present. Mixed-domain verification: a regional archive has 75 map sheets and receives 5 additional sheets during the exercise.

Rules. Independence concerns distinct information sources, not merely the number of people repeating one source. Every required procedural condition must be satisfied. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. May the serious penalty be imposed under the stated procedure? Identify the condition that succeeds or fails. Cross-domain check: is quorum met? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. The two friends trace to the same original rumor, so they are not two independent pieces under the rule. 2. The response condition is met. 3. Because the evidence condition fails, the serious penalty is not authorized. Cross-domain check: compare 11 with 7; $11 \geq 7$ is true. Mixed-domain verification: 75+5=80.

Answer. No. The evidence requirement is not met. Cross-domain answer: quorum is met. Mixed-domain answer: 80 map sheets.

Knowledge takeaway. A correct outcome does not excuse skipping a required procedure when the rule explicitly makes procedure part of validity.

Formal model. Conjunctive procedural predicate over evidence independence/count and response opportunity.

Problem 735. Procedure, evidence, and a right to respond: case 5

Knowledge context. Fair procedures often combine evidentiary standards with an opportunity for the affected person to answer the case.

Given facts. A school rule says a student may receive a serious penalty only if: (1) at least two independent pieces of evidence support the allegation, and (2) the student is given a chance to respond before the final decision. In a case, one camera record and one independent witness support the allegation, but the student was not allowed to respond. Cross-domain check: researchers collected 11 reports, but 4 are exact duplicate copies that add no new independent

information. Mixed-domain verification: a regional archive has 75 map sheets and receives 6 additional sheets during the exercise.

Rules. Independence concerns distinct information sources, not merely the number of people repeating one source. Every required procedural condition must be satisfied. For the cross-domain check, independent reports = total reports − duplicate copies. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. May the serious penalty be imposed under the stated procedure? Identify the condition that succeeds or fails. Cross-domain check: how many non-duplicate reports remain? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. Camera and witness are independent, so the evidence condition is met. 2. The student was not given a chance to respond. 3. The procedure requires both conditions, so the penalty is not authorized. Cross-domain check: 11−4=7 independent reports remain. Mixed-domain verification: 75+6=81.

Answer. No. The right-to-respond condition is not met. Cross-domain answer: 7 independent reports. Mixed-domain answer: 81 map sheets.

Knowledge takeaway. A correct outcome does not excuse skipping a required procedure when the rule explicitly makes procedure part of validity.

Formal model. Conjunctive procedural predicate over evidence independence/count and response opportunity.

N23. Samples, surveys, and public opinion

Problem 736. Sampling and opinion: case 1

Knowledge context. Surveys use samples to learn about larger populations. The selection process matters as much as the number of answers.

Given facts. A town wants to estimate how many residents use buses. It surveys only people waiting at the bus station. Cross-domain check: on a field sketch, 1 cm represents 3 km and two checkpoints are 3 cm apart.

Rules. A sample is more representative when selection does not systematically favor people connected to the answer. Random or broad systematic sampling can reduce selection bias; voluntary or topic-specific locations may create bias. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Is the sampling method likely to be biased for the stated question? Explain the direction of the possible bias. Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. Identify the target population. 2. Compare the selection method with that target. 3. People already at the bus station are more likely than average to use buses. 4. Therefore the method is likely biased. Cross-domain check: 3×3=9 km.

Answer. Likely biased. People already at the bus station are more likely than average to use buses. Cross-domain answer: 9 km.

Knowledge takeaway. A large biased sample can still give a misleading estimate; representativeness is a reasoning question about selection.

Formal model. Sampling-frame analysis: compare inclusion mechanism with target population and likely outcome-related selection.

Problem 737. Sampling and opinion: case 2

Knowledge context. Surveys use samples to learn about larger populations. The selection process matters as much as the number of answers.

Given facts. A school wants opinions on lunch. It randomly selects 10 students from each grade. Cross-domain check: a survey team starts work at 8:00 and works for 2 hour(s) without a break.

Rules. A sample is more representative when selection does not systematically favor people connected to the answer. Random or broad systematic sampling can reduce selection bias; voluntary or topic-specific locations may create bias. For the cross-domain check, finishing time = starting time + duration.

Task. Is the sampling method likely to be biased for the stated question? Explain the direction of the possible bias. Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. Identify the target population. 2. Compare the selection method with that target. 3. Random selection within every grade gives broad grade coverage; no obvious topic-specific selection is stated. 4. Therefore the method is not obviously biased for the stated target, within the limits given. Cross-domain check: 8+2=10, so the team finishes at 10:00.

Answer. Not obviously biased under the stated design. Random selection within every grade gives broad grade coverage; no obvious topic-specific selection is stated. Cross-domain answer: 10:00.

Knowledge takeaway. A large biased sample can still give a misleading estimate; representativeness is a reasoning question about selection.

Formal model. Sampling-frame analysis: compare inclusion mechanism with target population and likely outcome-related selection.

Problem 738. Sampling and opinion: case 3

Knowledge context. Surveys use samples to learn about larger populations. The selection process matters as much as the number of answers.

Given facts. A village asks only members of the sports club whether to build a sports field. Cross-domain check: a local committee has 8 members, requires at least 5 present for quorum, and 8 are present.

Rules. A sample is more representative when selection does not systematically favor people connected to the answer. Random or broad systematic sampling can reduce selection bias; voluntary or topic-specific locations may create bias. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. Is the sampling method likely to be biased for the stated question? Explain the direction of the possible bias. Cross-domain check: is quorum met?

Step-by-step solution. 1. Identify the target population. 2. Compare the selection method with that target. 3. Sports-club members are likely to be more favorable to a sports field than the whole village. 4. Therefore the method is likely biased. Cross-domain check: compare 8 with 5; $8 \geq 5$ is true.

Answer. Likely biased. Sports-club members are likely to be more favorable to a sports field than the whole village. Cross-domain answer: quorum is met.

Knowledge takeaway. A large biased sample can still give a misleading estimate; representativeness is a reasoning question about selection.

Formal model. Sampling-frame analysis: compare inclusion mechanism with target population and likely outcome-related selection.

Problem 739. Sampling and opinion: case 4

Knowledge context. Surveys use samples to learn about larger populations. The selection process matters as much as the number of answers.

Given facts. A library surveys every 20th visitor entering over a full week. Cross-domain check: researchers collected 15 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. A sample is more representative when selection does not systematically favor people connected to the answer. Random or broad systematic sampling can reduce selection bias; voluntary or topic-specific locations may create bias. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. Is the sampling method likely to be biased for the stated question? Explain the direction of the possible bias. Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. Identify the target population. 2. Compare the selection method with that target. 3. Sampling every 20th visitor across a full week is systematic within the visitor population; it may represent visitors reasonably, though not non-visitors. 4. Therefore the method is not obviously biased for the stated target, within the limits given. Cross-domain check: 15−4=11 independent reports remain.

Answer. Not obviously biased under the stated design. Sampling every 20th visitor across a full week is systematic within the visitor population; it may represent visitors reasonably, though not non-visitors. Cross-domain answer: 11 independent reports.

Knowledge takeaway. A large biased sample can still give a misleading estimate; representativeness is a reasoning question about selection.

Formal model. Sampling-frame analysis: compare inclusion mechanism with target population and likely outcome-related selection.

Problem 740. Sampling and opinion: case 5

Knowledge context. Surveys use samples to learn about larger populations. The selection process matters as much as the number of answers.

Given facts. An online poll is open to anyone who chooses to click it. Cross-domain check: on a field sketch, 1 cm represents 2 km and two checkpoints are 7 cm apart.

Rules. A sample is more representative when selection does not systematically favor people connected to the answer. Random or broad systematic sampling can reduce selection bias; voluntary or topic-specific locations may create bias. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Is the sampling method likely to be biased for the stated question? Explain the direction of the possible bias. Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. Identify the target population. 2. Compare the selection method with that target. 3. People who choose to click may differ systematically from those who ignore the poll. 4. Therefore the method is likely biased. Cross-domain check: 7×2=14 km.

Answer. Likely biased. People who choose to click may differ systematically from those who ignore the poll. Cross-domain answer: 14 km.

Knowledge takeaway. A large biased sample can still give a misleading estimate; representativeness is a reasoning question about selection.

Formal model. Sampling-frame analysis: compare inclusion mechanism with target population and likely outcome-related selection.

N24. Matching, scheduling, and coverage

Problem 741. Matching people to tasks: case 1

Knowledge context. Scheduling and assignment problems connect people, tasks, times, and constraints.

Given facts. Four students must each receive one different task. Allowed assignments: Ava: ['Map', 'Timeline']; Ben: ['Budget', 'Survey']; Cara: ['Map', 'Survey']; Dion: ['Budget', 'Timeline']. Cross-domain check: a survey team starts work at 12:00 and works for 2 hour(s) without a break. Mixed-domain verification: a regional archive has 76 map sheets and receives 2 additional sheets during the exercise.

Rules. Each student gets exactly one allowed task, and each task is used exactly once. A complete assignment must satisfy both conditions. For the cross-domain check, finishing time = starting time + duration. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. Find one complete valid assignment. If none exists, explain why. Cross-domain check: at what time does the team finish? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. Start with the most restricted student if one exists. 2. Assign an allowed task, then remove that task from the remaining choices. 3. One complete assignment is {'Ava': 'Map', 'Ben': 'Budget', 'Cara': 'Survey', 'Dion': 'Timeline'}. Cross-domain check: 12+2=14, so the team finishes at 14:00. Mixed-domain verification: 76+2=78.

Answer. Valid assignment: Ava→Map, Ben→Budget, Cara→Survey, Dion→Timeline. Cross-domain answer: 14:00. Mixed-domain answer: 78 map sheets.

Knowledge takeaway. A local choice can block a later assignment, so the whole matching must be checked.

Formal model. Bipartite matching / all-different constraint satisfaction.

Problem 742. Matching people to tasks: case 2

Knowledge context. Scheduling and assignment problems connect people, tasks, times, and constraints.

Given facts. Four students must each receive one different task. Allowed assignments: Ava: ['Map', 'Timeline']; Ben: ['Budget', 'Survey']; Cara: ['Survey']; Dion: ['Budget', 'Timeline']. Cross-domain check: a local committee has 12 members, requires at least 7 present for quorum, and 11 are present.

Rules. Each student gets exactly one allowed task, and each task is used exactly once. A complete assignment must satisfy both conditions. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. Find one complete valid assignment. If none exists, explain why. Cross-domain check: is quorum met?

Step-by-step solution. 1. Start with the most restricted student if one exists. 2. Assign an allowed task, then remove that task from the remaining choices. 3. One complete assignment is {'Ava': 'Map', 'Ben': 'Budget', 'Cara': 'Survey', 'Dion': 'Timeline'}. Cross-domain check: compare 11 with 7; 11 ≥ 7 is true.

Answer. Valid assignment: Ava→Map, Ben→Budget, Cara→Survey, Dion→Timeline. Cross-domain answer: quorum is met.

Knowledge takeaway. A local choice can block a later assignment, so the whole matching must be checked.

Formal model. Bipartite matching / all-different constraint satisfaction.

Problem 743. Matching people to tasks: case 3

Knowledge context. Scheduling and assignment problems connect people, tasks, times, and constraints.

Given facts. Four students must each receive one different task. Allowed assignments: Ava: ['Map', 'Timeline']; Ben: ['Budget', 'Survey']; Cara: ['Map', 'Survey']; Dion: ['Budget']. Cross-domain check: researchers collected 9 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. Each student gets exactly one allowed task, and each task is used exactly once. A complete assignment must satisfy both conditions. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. Find one complete valid assignment. If none exists, explain why. Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. Start with the most restricted student if one exists. 2. Assign an allowed task, then remove that task from the remaining choices. 3. One complete assignment is {'Ava': 'Timeline', 'Ben': 'Survey', 'Cara': 'Map', 'Dion': 'Budget'}. Cross-domain check: 9−4=5 independent reports remain.

Answer. Valid assignment: Ava→Timeline, Ben→Survey, Cara→Map, Dion→Budget. Cross-domain answer: 5 independent reports.

Knowledge takeaway. A local choice can block a later assignment, so the whole matching must be checked.

Formal model. Bipartite matching / all-different constraint satisfaction.

Problem 744. Matching people to tasks: case 4

Knowledge context. Scheduling and assignment problems connect people, tasks, times, and constraints.

Given facts. Four students must each receive one different task. Allowed assignments: Ava: ['Map']; Ben: ['Budget', 'Survey']; Cara: ['Map', 'Survey']; Dion: ['Budget', 'Timeline']. Cross-domain check: on a field sketch, 1 cm represents 6 km and two checkpoints are 4 cm apart.

Rules. Each student gets exactly one allowed task, and each task is used exactly once. A complete assignment must satisfy both conditions. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Find one complete valid assignment. If none exists, explain why. Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. Start with the most restricted student if one exists. 2. Assign an allowed task, then remove that task from the remaining choices. 3. One complete assignment is {'Ava': 'Map', 'Ben': 'Budget', 'Cara': 'Survey', 'Dion': 'Timeline'}. Cross-domain check: 4×6=24 km.

Answer. Valid assignment: Ava→Map, Ben→Budget, Cara→Survey, Dion→Timeline. Cross-domain answer: 24 km.

Knowledge takeaway. A local choice can block a later assignment, so the whole matching must be checked.

Formal model. Bipartite matching / all-different constraint satisfaction.

Problem 745. Matching people to tasks: case 5

Knowledge context. Scheduling and assignment problems connect people, tasks, times, and constraints.

Given facts. Four students must each receive one different task. Allowed assignments: Ava: ['Map', 'Timeline']; Ben: ['Budget', 'Survey']; Cara: ['Map', 'Survey']; Dion: ['Budget', 'Timeline']. Cross-domain check: a survey team starts work at 8:00 and works for 2 hour(s) without a break.

Rules. Each student gets exactly one allowed task, and each task is used exactly once. A complete assignment must satisfy both conditions. For the cross-domain check, finishing time = starting time + duration.

Task. Find one complete valid assignment. If none exists, explain why. Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. Start with the most restricted student if one exists. 2. Assign an allowed task, then remove that task from the remaining choices. 3. One complete assignment is {'Ava': 'Map', 'Ben': 'Budget', 'Cara': 'Survey', 'Dion': 'Timeline'}. Cross-domain check: 8+2=10, so the team finishes at 10:00.

Answer. Valid assignment: Ava→Map, Ben→Budget, Cara→Survey, Dion→Timeline. Cross-domain answer: 10:00.

Knowledge takeaway. A local choice can block a later assignment, so the whole matching must be checked.

Formal model. Bipartite matching / all-different constraint satisfaction.

N25. Meta-reasoning: robustness, information, causality

Problem 746. Meta-reasoning: case 1

Knowledge context. Good reasoning includes checking how informative, robust, falsifiable, causal, or dominated a conclusion is.

Given facts. Four equally likely locations A,B,C,D remain possible. Question Q1 asks “Is it A?”; Q2 asks “Is it A or B?” A yes/no question is more balanced when its two answer branches contain more nearly equal numbers of possibilities. Cross-domain check: a local committee has 16 members, requires at least 9 present for quorum, and 14 are present.

Rules. Use branch sizes as the information criterion in this exercise. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. Which question is more balanced and therefore more informative by the stated criterion? Cross-domain check: is quorum met?

Step-by-step solution. 1. Q1 splits the four possibilities 1 versus 3. 2. Q2 splits them 2 versus 2. 3. The 2–2 split is more balanced. Cross-domain check: compare 14 with 9; $14 \geq 9$ is true.

Answer. Q2. Cross-domain answer: quorum is met.

Knowledge takeaway. A result can be numerically correct yet epistemically weak; meta-reasoning evaluates the quality of the inference itself.

Formal model. Meta-level predicate over hypotheses or decisions: information balance, robustness, counterexample, causality, or dominance.

Problem 747. Meta-reasoning: case 2

Knowledge context. Good reasoning includes checking how informative, robust, falsifiable, causal, or dominated a conclusion is.

Given facts. Option A scores between 49 and 53 after measurement error; option B scores between 50 and 52. A ranking is robust only if every allowed value of A remains above every allowed value of B. Cross-domain check: researchers collected 13 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. Test worst cases: robust A>B requires min(A)>max(B). For the cross-domain check, independent reports = total reports − duplicate copies.

Task. Is the ranking A>B robust? Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. min(A)=49; max(B)=52. 2. 49>52 is false. 3. Some allowed values reverse the ranking. Cross-domain check: 13−4=9 independent reports remain.

Answer. No; the ranking is sensitive to the uncertainty. Cross-domain answer: 9 independent reports.

Knowledge takeaway. A result can be numerically correct yet epistemically weak; meta-reasoning evaluates the quality of the inference itself.

Formal model. Meta-level predicate over hypotheses or decisions: information balance, robustness, counterexample, causality, or dominance.

Problem 748. Meta-reasoning: case 3

Knowledge context. Good reasoning includes checking how informative, robust, falsifiable, causal, or dominated a conclusion is.

Given facts. Claim: “Every town with more roads has shorter average travel time.” Town X has 6 roads and average travel time 35 minutes. Town Y has 4 roads and average travel time 20 minutes. Cross-domain check: on a field sketch, 1 cm represents 5 km and two checkpoints are 8 cm apart.

Rules. A universal claim is disproved by one case that satisfies its condition but not its conclusion. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Do X and Y provide a counterexample to the claim? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. X has more roads than Y. 2. Yet X has longer, not shorter, travel time. 3. Therefore the universal claim fails on this pair. Cross-domain check: 8×5=40 km.

Answer. Yes. X and Y form a counterexample. Cross-domain answer: 40 km.

Knowledge takeaway. A result can be numerically correct yet epistemically weak; meta-reasoning evaluates the quality of the inference itself.

Formal model. Meta-level predicate over hypotheses or decisions: information balance, robustness, counterexample, causality, or dominance.

Problem 749. Meta-reasoning: case 4

Knowledge context. Good reasoning includes checking how informative, robust, falsifiable, causal, or dominated a conclusion is.

Given facts. Across several months, ice-cream sales and sunburn cases rise together. Temperature also rises in those months. No intervention is described. Cross-domain check: a survey team starts work at 12:00 and works for 2 hour(s) without a break.

Rules. Correlation between X and Y does not by itself prove X causes Y; a third variable may influence both. For the cross-domain check, finishing time = starting time + duration.

Task. Can we conclude that ice-cream sales cause sunburn? Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. Sales and sunburn are correlated. 2. Temperature is a plausible common factor stated in the data. 3. Without an intervention or stronger causal evidence, direct causation is not established. Cross-domain check: 12+2=14, so the team finishes at 14:00.

Answer. No. The association alone does not establish that ice-cream sales cause sunburn. Cross-domain answer: 14:00.

Knowledge takeaway. A result can be numerically correct yet epistemically weak; meta-reasoning evaluates the quality of the inference itself.

Formal model. Meta-level predicate over hypotheses or decisions: information balance, robustness, counterexample, causality, or dominance.

Problem 750. Meta-reasoning: case 5

Knowledge context. Good reasoning includes checking how informative, robust, falsifiable, causal, or dominated a conclusion is.

Given facts. Two transport plans are compared. A costs 8, takes 20 minutes, and has safety score 9. B costs 10, takes 25 minutes, and has safety score 9. Lower cost/time and higher safety are better. Cross-domain check: a local committee has 11 members, requires at least 6 present for quorum, and 11 are present.

Rules. A Pareto-dominates B if A is at least as good on every stated criterion and strictly better on at least one. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. Does one plan dominate the other? Cross-domain check: is quorum met?

Step-by-step solution. 1. A is cheaper than B. 2. A is faster than B. 3. Safety is equal. 4. A is no worse anywhere and better on two criteria. Cross-domain check: compare 11 with 6; $11 \geq 6$ is true.

Answer. A Pareto-dominates B. Cross-domain answer: quorum is met.

Knowledge takeaway. A result can be numerically correct yet epistemically weak; meta-reasoning evaluates the quality of the inference itself.

Formal model. Meta-level predicate over hypotheses or decisions: information balance, robustness, counterexample, causality, or dominance.

GRADE 4

Difficulty increases through the number of relations and reasoning steps, not through hidden knowledge.

Part A — Foundational reasoning about the world

G1. Positions and orientation

Problem 751. Position chain from Fern to Alder

Knowledge context. A map can be modeled with relative positions. When several moves are given, their effects can be combined rather than memorized as a picture.

Given facts. Meadow is 1 map unit(s) east of Fern; Alder is 1 map unit(s) east of Meadow. A museum in Meadow opens on Saturday.

Rules. Treat east as +x, west as −x, north as +y, and south as −y. Add the moves in order. A fact that does not affect any coordinate is irrelevant.

Task. Where is Alder relative to Fern? What is the straight grid displacement measured as |x|+|y|? Identify any irrelevant fact.

Step-by-step solution. 1. Start Fern at (0,0). 2. Translate each relation into a coordinate move. 3. Adding the moves gives (2,0). 4. That coordinate is east of Fern; the grid displacement is |2|+|0|=2. 5. The museum sentence changes no coordinate, so it is a distractor.

Answer. Alder is east of Fern; grid displacement 2 unit(s). The museum fact is irrelevant.

Knowledge takeaway. Cardinal directions become easier to reason about when they are treated as relations that can be composed.

Formal model. Grid coordinates; each statement adds a vector; sum the vectors and classify the final coordinate.

Problem 752. Position chain from Dover to Linden

Knowledge context. A map can be modeled with relative positions. When several moves are given, their effects can be combined rather than memorized as a picture.

Given facts. Yarrow is 2 map unit(s) north of Dover; Linden is 2 map unit(s) north of Yarrow. A museum in Stone opens on Saturday.

Rules. Treat east as +x, west as −x, north as +y, and south as −y. Add the moves in order. A fact that does not affect any coordinate is irrelevant.

Task. Where is Linden relative to Dover? What is the straight grid displacement measured as |x|+|y|? Identify any irrelevant fact.

Step-by-step solution. 1. Start Dover at (0,0). 2. Translate each relation into a coordinate move. 3. Adding the moves gives (0,4). 4. That coordinate is north of Dover; the grid displacement is |0|+|4|=4. 5. The museum sentence changes no coordinate, so it is a distractor.

Answer. Linden is north of Dover; grid displacement 4 unit(s). The museum fact is irrelevant.

Knowledge takeaway. Cardinal directions become easier to reason about when they are treated as relations that can be composed.

Formal model. Grid coordinates; each statement adds a vector; sum the vectors and classify the final coordinate.

Problem 753. Position chain from Grove to Cedar

Knowledge context. A map can be modeled with relative positions. When several moves are given, their effects can be combined rather than memorized as a picture.

Given facts. Willow is 1 map unit(s) east of Grove; Linden is 1 map unit(s) north of Willow; Cedar is 1 map unit(s) west of Linden. A museum in Elm opens on Saturday.

Rules. Treat east as +x, west as −x, north as +y, and south as −y. Add the moves in order. A fact that does not affect any coordinate is irrelevant.

Task. Where is Cedar relative to Grove? What is the straight grid displacement measured as |x|+|y|? Identify any irrelevant fact.

Step-by-step solution. 1. Start Grove at (0,0). 2. Translate each relation into a coordinate move. 3. Adding the moves gives (0,1). 4. That coordinate is north of Grove; the grid displacement is |0|+|1|=1. 5. The museum sentence changes no coordinate, so it is a distractor.

Answer. Cedar is north of Grove; grid displacement 1 unit(s). The museum fact is irrelevant.

Knowledge takeaway. Cardinal directions become easier to reason about when they are treated as relations that can be composed.

Formal model. Grid coordinates; each statement adds a vector; sum the vectors and classify the final coordinate.

Problem 754. Position chain from Pine to Iris

Knowledge context. A map can be modeled with relative positions. When several moves are given, their effects can be combined rather than memorized as a picture.

Given facts. Alder is 2 map unit(s) west of Pine; Iris is 2 map unit(s) south of Alder. A museum in Cedar opens on Saturday.

Rules. Treat east as +x, west as −x, north as +y, and south as −y. Add the moves in order. A fact that does not affect any coordinate is irrelevant.

Task. Where is Iris relative to Pine? What is the straight grid displacement measured as |x|+|y|? Identify any irrelevant fact.

Step-by-step solution. 1. Start Pine at (0,0). 2. Translate each relation into a coordinate move. 3. Adding the moves gives (-2,-2). 4. That coordinate is south-west of Pine; the grid displacement is |-2|+|-2|=4. 5. The museum sentence changes no coordinate, so it is a distractor.

Answer. Iris is south-west of Pine; grid displacement 4 unit(s). The museum fact is irrelevant.

Knowledge takeaway. Cardinal directions become easier to reason about when they are treated as relations that can be composed.

Formal model. Grid coordinates; each statement adds a vector; sum the vectors and classify the final coordinate.

Problem 755. Position chain from Quartz to Alder

Knowledge context. A map can be modeled with relative positions. When several moves are given, their effects can be combined rather than memorized as a picture.

Given facts. Yarrow is 1 map unit(s) north of Quartz; Kite is 1 map unit(s) east of Yarrow; Grove is 1 map unit(s) south of Kite; Alder is 1 map unit(s) west of Grove. A museum in Juniper opens on Saturday.

Rules. Treat east as +x, west as −x, north as +y, and south as −y. Add the moves in order. A fact that does not affect any coordinate is irrelevant.

Task. Where is Alder relative to Quartz? What is the straight grid displacement measured as |x|+|y|? Identify any irrelevant fact.

Step-by-step solution. 1. Start Quartz at (0,0). 2. Translate each relation into a coordinate move. 3. Adding the moves gives (0,0). 4. That coordinate is the same position of Quartz; the grid displacement is |0|+|0|=0. 5. The museum sentence changes no coordinate, so it is a distractor.

Answer. Alder is the same position of Quartz; grid displacement 0 unit(s). The museum fact is irrelevant.

Knowledge takeaway. Cardinal directions become easier to reason about when they are treated as relations that can be composed.

Formal model. Grid coordinates; each statement adds a vector; sum the vectors and classify the final coordinate.

G2. Roads and routes

Problem 756. The best route from Elm to Willow

Knowledge context. A road map can be modeled as a graph. The best route depends on the stated criterion, not on how short it looks on paper.

Given facts. Elm–Northfield–Willow has segment costs [6, 7] (total 13); Elm–Riverbend–Harbor–Willow has segment costs [5, 2, 6] (total 13); Elm–Birch–Willow has segment costs [9, 4] (total 13).

Rules. A route is feasible only if all of its roads are open. The cost of a route is the sum of its segment costs. Choose the feasible route with the smallest total.

Task. Which route should a traveler use from Elm to Willow, and why?

Step-by-step solution. 1. List the candidate routes and their segment costs. 2. All listed routes are open. 3. Add the segment costs for every feasible route. 4. The smallest feasible total is 13, on Elm–Northfield–Willow.

Answer. Use Elm–Northfield–Willow; total cost 13.

Knowledge takeaway. A shortest path is a reasoning result: first respect constraints, then compare the remaining costs.

Formal model. Weighted graph; delete forbidden edges; minimize the sum of edge weights.

Problem 757. The best route from Oak to Grove

Knowledge context. A road map can be modeled as a graph. The best route depends on the stated criterion, not on how short it looks on paper.

Given facts. Oak–Dover–Grove has segment costs [7, 8] (total 15); Oak–Fern–Juniper–Grove has segment costs [6, 3, 6] (total 15); Oak–Riverbend–Grove has segment costs [10, 5] (total 15). The road Oak–Fern is closed.

Rules. A route is feasible only if all of its roads are open. The cost of a route is the sum of its segment costs. Choose the feasible route with the smallest total.

Task. Which route should a traveler use from Oak to Grove, and why?

Step-by-step solution. 1. List the candidate routes and their segment costs. 2. Remove any route containing a closed road. 3. Add the segment costs for every feasible route. 4. The smallest feasible total is 15, on Oak–Dover–Grove.

Answer. Use Oak–Dover–Grove; total cost 15.

Knowledge takeaway. A shortest path is a reasoning result: first respect constraints, then compare the remaining costs.

Formal model. Weighted graph; delete forbidden edges; minimize the sum of edge weights.

Problem 758. The best route from Linden to Meadow

Knowledge context. A road map can be modeled as a graph. The best route depends on the stated criterion, not on how short it looks on paper.

Given facts. Linden–Dover–Meadow has segment costs [8, 9] (total 17); Linden–Elm–Yarrow–Meadow has segment costs [7, 4, 6] (total 17); Linden–Northfield–Meadow has segment costs [11, 6] (total 17).

Rules. A route is feasible only if all of its roads are open. The cost of a route is the sum of its segment costs. Choose the feasible route with the smallest total.

Task. Which route should a traveler use from Linden to Meadow, and why?

Step-by-step solution. 1. List the candidate routes and their segment costs. 2. All listed routes are open. 3. Add the segment costs for every feasible route. 4. The smallest feasible total is 17, on Linden–Dover–Meadow.

Answer. Use Linden–Dover–Meadow; total cost 17.

Knowledge takeaway. A shortest path is a reasoning result: first respect constraints, then compare the remaining costs.

Formal model. Weighted graph; delete forbidden edges; minimize the sum of edge weights.

Problem 759. The best route from Harbor to Meadow

Knowledge context. A road map can be modeled as a graph. The best route depends on the stated criterion, not on how short it looks on paper.

Given facts. Harbor–Riverbend–Meadow has segment costs [9, 10] (total 19); Harbor–Iris–Yarrow–Meadow has segment costs [8, 5, 6] (total 19); Harbor–Oak–Meadow has segment costs [12, 7] (total 19). The road Harbor–Iris is closed.

Rules. A route is feasible only if all of its roads are open. The cost of a route is the sum of its segment costs. Choose the feasible route with the smallest total.

Task. Which route should a traveler use from Harbor to Meadow, and why?

Step-by-step solution. 1. List the candidate routes and their segment costs. 2. Remove any route containing a closed road. 3. Add the segment costs for every feasible route. 4. The smallest feasible total is 19, on Harbor–Riverbend–Meadow.

Answer. Use Harbor–Riverbend–Meadow; total cost 19.

Knowledge takeaway. A shortest path is a reasoning result: first respect constraints, then compare the remaining costs.

Formal model. Weighted graph; delete forbidden edges; minimize the sum of edge weights.

Problem 760. The best route from Oak to Riverbend

Knowledge context. A road map can be modeled as a graph. The best route depends on the stated criterion, not on how short it looks on paper.

Given facts. Oak–Alder–Riverbend has segment costs [10, 11] (total 21); Oak–Meadow–Dover–Riverbend has segment costs [9, 6, 6] (total 21); Oak–Iris–Riverbend has segment costs [13, 8] (total 21).

Rules. A route is feasible only if all of its roads are open. The cost of a route is the sum of its segment costs. Choose the feasible route with the smallest total.

Task. Which route should a traveler use from Oak to Riverbend, and why?

Step-by-step solution. 1. List the candidate routes and their segment costs. 2. All listed routes are open. 3. Add the segment costs for every feasible route. 4. The smallest feasible total is 21, on Oak–Alder–Riverbend.

Answer. Use Oak–Alder–Riverbend; total cost 21.

Knowledge takeaway. A shortest path is a reasoning result: first respect constraints, then compare the remaining costs.

Formal model. Weighted graph; delete forbidden edges; minimize the sum of edge weights.

G3. Rivers: upstream and downstream

Problem 761. Following water from Silver River

Knowledge context. River systems form directed networks: water has a direction, tributaries join larger streams, and upstream/downstream are relational concepts.

Given facts. Silver River flows into Blue Creek; Clearwater flows into Blue Creek; Blue Creek flows into Meadow River. A canal diverts some water from Clearwater to a reservoir, but the remaining flow still reaches Blue Creek.

Rules. If river X flows into river Y, X is upstream of Y. Upstream relations can be followed through several confluences. Downstream is the reverse direction of flow.

Task. Is Silver River upstream of Meadow River? Can water from Silver River reach Meadow River through the stated network?

Step-by-step solution. 1. Rewrite the facts as arrows: Silver River→Blue Creek, Clearwater→Blue Creek, Blue Creek→Meadow River. 2. Follow arrows starting from Silver River. 3. An arrow path reaches Meadow River, so Silver River is upstream of Meadow River.

Answer. Yes; water from Silver River can reach Meadow River.

Knowledge takeaway. A river network can be reasoned about like any directed network; following the arrows is more reliable than guessing from names.

Formal model. Directed graph; edge X→Y means water can flow from X to Y; query reachability.

Problem 762. Following water from Red Brook

Knowledge context. River systems form directed networks: water has a direction, tributaries join larger streams, and upstream/downstream are relational concepts.

Given facts. Silver River flows into Blue Creek; Red Brook flows into Blue Creek; Blue Creek flows into Pine River. A canal diverts some water from Red Brook to a reservoir, but the remaining flow still reaches Blue Creek.

Rules. If river X flows into river Y, X is upstream of Y. Upstream relations can be followed through several confluences. Downstream is the reverse direction of flow.

Task. Is Red Brook upstream of Pine River? Can water from Red Brook reach Pine River through the stated network?

Step-by-step solution. 1. Rewrite the facts as arrows: Silver River→Blue Creek, Red Brook→Blue Creek, Blue Creek→Pine River. 2. Follow arrows starting from Red Brook. 3. An arrow path reaches Pine River, so Red Brook is upstream of Pine River.

Answer. Yes; water from Red Brook can reach Pine River.

Knowledge takeaway. A river network can be reasoned about like any directed network; following the arrows is more reliable than guessing from names.

Formal model. Directed graph; edge X→Y means water can flow from X to Y; query reachability.

Problem 763. Following water from Pine River

Knowledge context. River systems form directed networks: water has a direction, tributaries join larger streams, and upstream/downstream are relational concepts.

Given facts. Pine River flows into Meadow River; Silver River flows into Meadow River; Meadow River flows into North Fork. A canal diverts some water from Silver River to a reservoir, but the remaining flow still reaches Meadow River.

Rules. If river X flows into river Y, X is upstream of Y. Upstream relations can be followed through several confluences. Downstream is the reverse direction of flow.

Task. Is Pine River upstream of Meadow River? Can water from Pine River reach Meadow River through the stated network?

Step-by-step solution. 1. Rewrite the facts as arrows: Pine River→Meadow River, Silver River→Meadow River, Meadow River→North Fork. 2. Follow arrows starting from Pine River. 3. An arrow path reaches Meadow River, so Pine River is upstream of Meadow River.

Answer. Yes; water from Pine River can reach Meadow River.

Knowledge takeaway. A river network can be reasoned about like any directed network; following the arrows is more reliable than guessing from names.

Formal model. Directed graph; edge X→Y means water can flow from X to Y; query reachability.

Problem 764. Following water from Red Brook

Knowledge context. River systems form directed networks: water has a direction, tributaries join larger streams, and upstream/downstream are relational concepts.

Given facts. Pine River flows into Red Brook; North Fork flows into Red Brook; Red Brook flows into Meadow River. A canal diverts some water from North Fork to a reservoir, but the remaining flow still reaches Red Brook.

Rules. If river X flows into river Y, X is upstream of Y. Upstream relations can be followed through several confluences. Downstream is the reverse direction of flow.

Task. Is Red Brook upstream of Meadow River? Can water from Red Brook reach Meadow River through the stated network?

Step-by-step solution. 1. Rewrite the facts as arrows: Pine River→Red Brook, North Fork→Red Brook, Red Brook→Meadow River. 2. Follow arrows starting from Red Brook. 3. An arrow path reaches Meadow River, so Red Brook is upstream of Meadow River.

Answer. Yes; water from Red Brook can reach Meadow River.

Knowledge takeaway. A river network can be reasoned about like any directed network; following the arrows is more reliable than guessing from names.

Formal model. Directed graph; edge X→Y means water can flow from X to Y; query reachability.

Problem 765. Following water from Pine River

Knowledge context. River systems form directed networks: water has a direction, tributaries join larger streams, and upstream/downstream are relational concepts.

Given facts. Silver River flows into Clearwater; Pine River flows into Clearwater; Clearwater flows into Meadow River. A canal diverts some water from Pine River to a reservoir, but the remaining flow still reaches Clearwater.

Rules. If river X flows into river Y, X is upstream of Y. Upstream relations can be followed through several confluences. Downstream is the reverse direction of flow.

Task. Is Pine River upstream of Clearwater? Can water from Pine River reach Clearwater through the stated network?

Step-by-step solution. 1. Rewrite the facts as arrows: Silver River→Clearwater, Pine River→Clearwater, Clearwater→Meadow River. 2. Follow arrows starting from Pine River. 3. An arrow path reaches Clearwater, so Pine River is upstream of Clearwater.

Answer. Yes; water from Pine River can reach Clearwater.

Knowledge takeaway. A river network can be reasoned about like any directed network; following the arrows is more reliable than guessing from names.

Formal model. Directed graph; edge X→Y means water can flow from X to Y; query reachability.

G4. Neighbours and borders

Problem 766. Crossing the fictional border network of Quartz

Knowledge context. Political maps can be treated as adjacency networks. This lets us reason about border crossings without relying on memorized real-world geography.

Given facts. Countries are represented by the fictional states Quartz, Grove, Harbor, Juniper, Fern, Linden. Shared borders: Quartz–Grove, Grove–Harbor, Harbor–Juniper, Quartz–Fern, Fern–Linden, Linden–Juniper.

Rules. Crossing one border corresponds to traversing one graph edge. A route may use only listed borders. A state is unavoidable only if every feasible route passes through it.

Task. What is the fewest number of border crossings needed to travel from Quartz to Juniper? Is Grove necessarily crossed?

Step-by-step solution. 1. Turn each shared border into an undirected link. 2. A shortest-link search gives distance 3 from Quartz to Juniper. 3. Remove Grove and check whether Juniper is still reachable from Quartz. 4. It is still reachable, so Grove is not unavoidable.

Answer. 3 border crossing(s); Grove is not unavoidable.

Knowledge takeaway. Neighbourhood on a map is a graph relation; shortest routes and unavoidable transit points are different questions.

Formal model. Undirected graph; shortest path length plus reachability after deleting a candidate vertex.

Problem 767. Crossing the fictional border network of Kite

Knowledge context. Political maps can be treated as adjacency networks. This lets us reason about border crossings without relying on memorized real-world geography.

Given facts. Countries are represented by the fictional states Kite, Fern, Northfield, Harbor, Birch, Pine. Shared borders: Kite–Fern, Fern–Northfield, Northfield–Harbor, Kite–Birch, Pine–Harbor.

Rules. Crossing one border corresponds to traversing one graph edge. A route may use only listed borders. A state is unavoidable only if every feasible route passes through it.

Task. What is the fewest number of border crossings needed to travel from Kite to Harbor? Is Fern necessarily crossed?

Step-by-step solution. 1. Turn each shared border into an undirected link. 2. A shortest-link search gives distance 3 from Kite to Harbor. 3. Remove Fern and check whether Harbor is still reachable from Kite. 4. It is not reachable, so Fern is unavoidable.

Answer. 3 border crossing(s); Fern is unavoidable.

Knowledge takeaway. Neighbourhood on a map is a graph relation; shortest routes and unavoidable transit points are different questions.

Formal model. Undirected graph; shortest path length plus reachability after deleting a candidate vertex.

Problem 768. Crossing the fictional border network of Pine

Knowledge context. Political maps can be treated as adjacency networks. This lets us reason about border crossings without relying on memorized real-world geography.

Given facts. Countries are represented by the fictional states Pine, Harbor, Cedar, Alder, Grove, Quartz. Shared borders: Pine–Harbor, Harbor–Cedar, Cedar–Alder, Pine–Grove, Grove–Quartz, Quartz–Alder.

Rules. Crossing one border corresponds to traversing one graph edge. A route may use only listed borders. A state is unavoidable only if every feasible route passes through it.

Task. What is the fewest number of border crossings needed to travel from Pine to Alder? Is Harbor necessarily crossed?

Step-by-step solution. 1. Turn each shared border into an undirected link. 2. A shortest-link search gives distance 3 from Pine to Alder. 3. Remove Harbor and check whether Alder is still reachable from Pine. 4. It is still reachable, so Harbor is not unavoidable.

Answer. 3 border crossing(s); Harbor is not unavoidable.

Knowledge takeaway. Neighbourhood on a map is a graph relation; shortest routes and unavoidable transit points are different questions.

Formal model. Undirected graph; shortest path length plus reachability after deleting a candidate vertex.

Problem 769. Crossing the fictional border network of Willow

Knowledge context. Political maps can be treated as adjacency networks. This lets us reason about border crossings without relying on memorized real-world geography.

Given facts. Countries are represented by the fictional states Willow, Linden, Kite, Pine, Cedar, Juniper. Shared borders: Willow–Linden, Linden–Kite, Kite–Pine, Willow–Cedar, Cedar–Juniper, Juniper–Pine.

Rules. Crossing one border corresponds to traversing one graph edge. A route may use only listed borders. A state is unavoidable only if every feasible route passes through it.

Task. What is the fewest number of border crossings needed to travel from Willow to Pine? Is Linden necessarily crossed?

Step-by-step solution. 1. Turn each shared border into an undirected link. 2. A shortest-link search gives distance 3 from Willow to Pine. 3. Remove Linden and check whether Pine is still reachable from Willow. 4. It is still reachable, so Linden is not unavoidable.

Answer. 3 border crossing(s); Linden is not unavoidable.

Knowledge takeaway. Neighbourhood on a map is a graph relation; shortest routes and unavoidable transit points are different questions.

Formal model. Undirected graph; shortest path length plus reachability after deleting a candidate vertex.

Problem 770. Crossing the fictional border network of Northfield

Knowledge context. Political maps can be treated as adjacency networks. This lets us reason about border crossings without relying on memorized real-world geography.

Given facts. Countries are represented by the fictional states Northfield, Alder, Linden, Riverbend, Iris, Fern. Shared borders: Northfield–Alder, Alder–Linden, Linden–Riverbend, Northfield–Iris, Fern–Riverbend.

Rules. Crossing one border corresponds to traversing one graph edge. A route may use only listed borders. A state is unavoidable only if every feasible route passes through it.

Task. What is the fewest number of border crossings needed to travel from Northfield to Riverbend? Is Alder necessarily crossed?

Step-by-step solution. 1. Turn each shared border into an undirected link. 2. A shortest-link search gives distance 3 from Northfield to Riverbend. 3. Remove Alder and check whether Riverbend is still reachable from Northfield. 4. It is not reachable, so Alder is unavoidable.

Answer. 3 border crossing(s); Alder is unavoidable.

Knowledge takeaway. Neighbourhood on a map is a graph relation; shortest routes and unavoidable transit points are different questions.

Formal model. Undirected graph; shortest path length plus reachability after deleting a candidate vertex.

G5. Maps, legends, and clues

Problem 771. Reading a layered map legend (A1)

Knowledge context. A map legend translates symbols into properties. One location may have several properties at the same time.

Given facts. Legend: tree=forest, wavy line=river, double line=road, arch=bridge, square=village, triangle=hill. Cells: A1 has forest, road; A2 has bridge, river; B1 has forest, river, road; B2 has road, village. Cross-domain check: researchers collected 7 reports, but 4 are exact duplicate copies that add no new independent information. Mixed-domain verification: a regional archive has 79 map sheets and receives 2 additional sheets during the exercise.

Rules. A cell satisfies the task only if it has every required attribute. Extra attributes do not disqualify it unless the task says so. For the cross-domain check, independent reports = total reports − duplicate copies. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. Which cell or cells satisfy all of these conditions: forest, road? Explain by intersecting the clues. Cross-domain check: how many non-duplicate reports remain? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. Convert the requested symbols into the attribute set ['forest', 'road']. 2. Check each cell for every required attribute. 3. The matching set is ['A1', 'B1']. Cross-domain check: 7−4=3 independent reports remain. Mixed-domain verification: 79+2=81.

Answer. A1, B1 Cross-domain answer: 3 independent reports. Mixed-domain answer: 81 map sheets.

Knowledge takeaway. Map reading often means intersecting clues rather than recognizing one symbol in isolation.

Formal model. Set model; each cell has an attribute set; return cells whose set is a superset of the required set.

Problem 772. Reading a layered map legend (A2)

Knowledge context. A map legend translates symbols into properties. One location may have several properties at the same time.

Given facts. Legend: tree=forest, wavy line=river, double line=road, arch=bridge, square=village, triangle=hill. Cells: A1 has forest, road; A2 has bridge, river, village; B1 has forest, river, road; B2 has road, village. Cross-domain check: on a field sketch, 1 cm represents 4 km and two checkpoints are 4 cm apart.

Rules. A cell satisfies the task only if it has every required attribute. Extra attributes do not disqualify it unless the task says so. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Which cell or cells satisfy all of these conditions: bridge, river? Explain by intersecting the clues. Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. Convert the requested symbols into the attribute set ['bridge', 'river']. 2. Check each cell for every required attribute. 3. The matching set is ['A2']. Cross-domain check: 4×4=16 km.

Answer. A2 Cross-domain answer: 16 km.

Knowledge takeaway. Map reading often means intersecting clues rather than recognizing one symbol in isolation.

Formal model. Set model; each cell has an attribute set; return cells whose set is a superset of the required set.

Problem 773. Reading a layered map legend (B1)

Knowledge context. A map legend translates symbols into properties. One location may have several properties at the same time.

Given facts. Legend: tree=forest, wavy line=river, double line=road, arch=bridge, square=village, triangle=hill. Cells: A1 has forest, road; A2 has bridge, river; B1 has forest, river, road; B2 has forest, road, village. Cross-domain check: a survey team starts work at 12:00 and works for 2 hour(s) without a break.

Rules. A cell satisfies the task only if it has every required attribute. Extra attributes do not disqualify it unless the task says so. For the cross-domain check, finishing time = starting time + duration.

Task. Which cell or cells satisfy all of these conditions: forest, river? Explain by intersecting the clues. Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. Convert the requested symbols into the attribute set ['forest', 'river']. 2. Check each cell for every required attribute. 3. The matching set is ['B1']. Cross-domain check: 12+2=14, so the team finishes at 14:00.

Answer. B1 Cross-domain answer: 14:00.

Knowledge takeaway. Map reading often means intersecting clues rather than recognizing one symbol in isolation.

Formal model. Set model; each cell has an attribute set; return cells whose set is a superset of the required set.

Problem 774. Reading a layered map legend (B2)

Knowledge context. A map legend translates symbols into properties. One location may have several properties at the same time.

Given facts. Legend: tree=forest, wavy line=river, double line=road, arch=bridge, square=village, triangle=hill. Cells: A1 has forest, hill, road; A2 has bridge, river; B1 has forest, river, road; B2 has road, village. Cross-domain check: a local committee has 8 members, requires at least 5 present for quorum, and 8 are present.

Rules. A cell satisfies the task only if it has every required attribute. Extra attributes do not disqualify it unless the task says so. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. Which cell or cells satisfy all of these conditions: hill, road? Explain by intersecting the clues. Cross-domain check: is quorum met?

Step-by-step solution. 1. Convert the requested symbols into the attribute set ['hill', 'road']. 2. Check each cell for every required attribute. 3. The matching set is ['A1']. Cross-domain check: compare 8 with 5; $8 \geq 5$ is true.

Answer. A1 Cross-domain answer: quorum is met.

Knowledge takeaway. Map reading often means intersecting clues rather than recognizing one symbol in isolation.

Formal model. Set model; each cell has an attribute set; return cells whose set is a superset of the required set.

Problem 775. Reading a layered map legend (A1)

Knowledge context. A map legend translates symbols into properties. One location may have several properties at the same time.

Given facts. Legend: tree=forest, wavy line=river, double line=road, arch=bridge, square=village, triangle=hill. Cells: A1 has forest, road; A2 has bridge, river; B1 has bridge, forest, river, road; B2 has road, village. Cross-domain check: researchers collected 11 reports, but 4 are exact duplicate copies that add no new independent information. Mixed-domain verification: a regional archive has 79 map sheets and receives 6 additional sheets during the exercise.

Rules. A cell satisfies the task only if it has every required attribute. Extra attributes do not disqualify it unless the task says so. For the cross-domain check, independent reports = total reports − duplicate copies. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. Which cell or cells satisfy all of these conditions: river, road? Explain by intersecting the clues. Cross-domain check: how many non-duplicate reports remain? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. Convert the requested symbols into the attribute set ['river', 'road']. 2. Check each cell for every required attribute. 3. The matching set is ['B1']. Cross-domain check: 11−4=7 independent reports remain. Mixed-domain verification: 79+6=85.

Answer. B1 Cross-domain answer: 7 independent reports. Mixed-domain answer: 85 map sheets.

Knowledge takeaway. Map reading often means intersecting clues rather than recognizing one symbol in isolation.

Formal model. Set model; each cell has an attribute set; return cells whose set is a superset of the required set.

G6. Choosing a settlement site

Problem 776. Choosing a safe settlement site: case 1

Knowledge context. Human settlements depend on several factors at once. Some conditions may be compulsory while others are preferences.

Given facts. Candidate sites: Site A: water=yes, road=yes, flood-zone=no, nearby farmland score=2; Site B: water=yes, road=no, flood-zone=no, nearby farmland score=4; Site C: water=yes, road=yes, flood-zone=yes, nearby farmland score=4; Site D: water=no, road=yes, flood-zone=no, nearby farmland score=6. Cross-domain check: on a field sketch, 1 cm represents 3 km and two checkpoints are 8 cm apart.

Rules. A valid site must have water, road access, and must not be in the flood zone. Among valid sites, choose the one with the highest farmland score. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Which site should be selected under the stated mandatory conditions and preference rule? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. Separate mandatory conditions from the preference. 2. Filtering by water, road, and flood safety leaves ['Site A']. 3. Compare only the farmland scores of feasible sites: Site A=2. 4. Site A has the best score among valid sites. Cross-domain check: 8×3=24 km.

Answer. Site A. Cross-domain answer: 24 km.

Knowledge takeaway. Good decisions first satisfy hard constraints and only then optimize among the remaining options.

Formal model. Constraint-satisfaction problem followed by a score maximization over feasible options.

Problem 777. Choosing a safe settlement site: case 2

Knowledge context. Human settlements depend on several factors at once. Some conditions may be compulsory while others are preferences.

Given facts. Candidate sites: Site A: water=yes, road=yes, flood-zone=no, nearby farmland score=3; Site B: water=yes, road=no, flood-zone=no, nearby farmland score=3; Site C: water=yes, road=yes, flood-zone=yes, nearby farmland score=5; Site D: water=no, road=yes, flood-zone=no, nearby farmland score=5. Cross-domain check: a survey team starts work at 8:00 and works for 2 hour(s) without a break. Mixed-domain verification: a regional archive has 79 map sheets and receives 8 additional sheets during the exercise.

Rules. A valid site must have water, road access, and must not be in the flood zone. Among valid sites, choose the one with the highest farmland score. For the cross-domain check, finishing time = starting time + duration. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. Which site should be selected under the stated mandatory conditions and preference rule? Cross-domain check: at what time does the team finish? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. Separate mandatory conditions from the preference. 2. Filtering by water, road, and flood safety leaves ['Site A']. 3. Compare only the farmland scores of feasible sites: Site A=3. 4. Site A has the best score among valid sites. Cross-domain check: 8+2=10, so the team finishes at 10:00. Mixed-domain verification: 79+8=87.

Answer. Site A. Cross-domain answer: 10:00. Mixed-domain answer: 87 map sheets.

Knowledge takeaway. Good decisions first satisfy hard constraints and only then optimize among the remaining options.

Formal model. Constraint-satisfaction problem followed by a score maximization over feasible options.

Problem 778. Choosing a safe settlement site: case 3

Knowledge context. Human settlements depend on several factors at once. Some conditions may be compulsory while others are preferences.

Given facts. Candidate sites: Site A: water=yes, road=yes, flood-zone=no, nearby farmland score=2; Site B: water=yes, road=no, flood-zone=no, nearby farmland score=4; Site C: water=yes, road=yes, flood-zone=yes, nearby farmland score=4; Site D: water=no, road=yes, flood-zone=no, nearby farmland score=6. Cross-domain check: a local committee has 12 members, requires at least 7 present for quorum, and 11 are present. Mixed-domain verification: a regional archive has 79 map sheets and receives 9 additional sheets during the exercise.

Rules. A valid site must have water, road access, and must not be in the flood zone. Among valid sites, choose the one with the highest farmland score. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. Which site should be selected under the stated mandatory conditions and preference rule? Cross-domain check: is quorum met? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. Separate mandatory conditions from the preference. 2. Filtering by water, road, and flood safety leaves ['Site A']. 3. Compare only the farmland scores of feasible sites: Site A=2. 4. Site A has the best score among valid sites. Cross-domain check: compare 11 with 7; 11 $\geq$ 7 is true. Mixed-domain verification: 79+9=88.

Answer. Site A. Cross-domain answer: quorum is met. Mixed-domain answer: 88 map sheets.

Knowledge takeaway. Good decisions first satisfy hard constraints and only then optimize among the remaining options.

Formal model. Constraint-satisfaction problem followed by a score maximization over feasible options.

Problem 779. Choosing a safe settlement site: case 4

Knowledge context. Human settlements depend on several factors at once. Some conditions may be compulsory while others are preferences.

Given facts. Candidate sites: Site A: water=yes, road=yes, flood-zone=no, nearby farmland score=3; Site B: water=yes, road=no, flood-zone=no, nearby farmland score=3; Site C: water=yes, road=yes, flood-zone=yes, nearby farmland score=5; Site D: water=no, road=yes, flood-zone=no, nearby farmland score=5. Cross-domain check: researchers collected 15 reports, but 4 are exact duplicate copies that add no new independent information. Mixed-domain verification: a regional archive has 79 map sheets and receives 10 additional sheets during the exercise.

Rules. A valid site must have water, road access, and must not be in the flood zone. Among valid sites, choose the one with the highest farmland score. For the cross-domain check, independent reports = total reports − duplicate copies. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. Which site should be selected under the stated mandatory conditions and preference rule? Cross-domain check: how many non-duplicate reports remain? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. Separate mandatory conditions from the preference. 2. Filtering by water, road, and flood safety leaves ['Site A']. 3. Compare only the farmland scores of feasible sites: Site A=3. 4. Site A has the best score among valid sites. Cross-domain check: 15−4=11 independent reports remain. Mixed-domain verification: 79+10=89.

Answer. Site A. Cross-domain answer: 11 independent reports. Mixed-domain answer: 89 map sheets.

Knowledge takeaway. Good decisions first satisfy hard constraints and only then optimize among the remaining options.

Formal model. Constraint-satisfaction problem followed by a score maximization over feasible options.

Problem 780. Choosing a safe settlement site: case 5

Knowledge context. Human settlements depend on several factors at once. Some conditions may be compulsory while others are preferences.

Given facts. Candidate sites: Site A: water=yes, road=yes, flood-zone=no, nearby farmland score=2; Site B: water=yes, road=no, flood-zone=no, nearby farmland score=4; Site C: water=yes, road=yes, flood-zone=yes, nearby farmland score=4; Site D: water=no, road=yes, flood-zone=no, nearby farmland score=6. Cross-domain check: on a field sketch, 1 cm represents 2 km and two checkpoints are 5 cm apart.

Rules. A valid site must have water, road access, and must not be in the flood zone. Among valid sites, choose the one with the highest farmland score. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Which site should be selected under the stated mandatory conditions and preference rule? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. Separate mandatory conditions from the preference. 2. Filtering by water, road, and flood safety leaves ['Site A']. 3. Compare only the farmland scores of feasible sites: Site A=2. 4. Site A has the best score among valid sites. Cross-domain check: 5×2=10 km.

Answer. Site A. Cross-domain answer: 10 km.

Knowledge takeaway. Good decisions first satisfy hard constraints and only then optimize among the remaining options.

Formal model. Constraint-satisfaction problem followed by a score maximization over feasible options.

G7. Relief and movement cost

Problem 781. Terrain and travel cost from Quartz

Knowledge context. Equal-looking distances can require different effort because terrain differs. A simple cost model makes that effect explicit.

Given facts. Terrain energy costs per segment: {'plain': 1, 'forest': 2, 'hill': 3, 'marsh': 4}. Three routes from Quartz to Linden: R1=['plain', 'hill', 'plain']; R2=['forest', 'forest', 'plain']; R3=['plain', 'marsh']. The traveler can spend at most 11 energy units.

Rules. The energy of a route is the sum of its terrain costs. A route above the energy limit is infeasible. Among feasible routes, minimize energy.

Task. Which route is feasible and uses the least energy?

Step-by-step solution. 1. Compute route totals: R1=5, R2=5, R3=5. 2. Discard any total above 11. 3. The smallest remaining total is R1=5.

Answer. Route R1, using 5 energy units.

Knowledge takeaway. Relief changes the cost of movement; “shortest” and “easiest” are not automatically the same.

Formal model. Weighted paths with a feasibility budget; sum costs and choose the minimum feasible path.

Problem 782. Terrain and travel cost from Cedar

Knowledge context. Equal-looking distances can require different effort because terrain differs. A simple cost model makes that effect explicit.

Given facts. Terrain energy costs per segment: {'plain': 1, 'forest': 2, 'hill': 3, 'marsh': 4}. Three routes from Cedar to Dover: R1=['plain', 'hill', 'plain']; R2=['plain', 'forest', 'plain', 'plain']; R3=['plain', 'marsh']. The traveler can spend at most 11 energy units.

Rules. The energy of a route is the sum of its terrain costs. A route above the energy limit is infeasible. Among feasible routes, minimize energy.

Task. Which route is feasible and uses the least energy?

Step-by-step solution. 1. Compute route totals: R1=5, R2=5, R3=5. 2. Discard any total above 11. 3. The smallest remaining total is R1=5.

Answer. Route R1, using 5 energy units.

Knowledge takeaway. Relief changes the cost of movement; “shortest” and “easiest” are not automatically the same.

Formal model. Weighted paths with a feasibility budget; sum costs and choose the minimum feasible path.

Problem 783. Terrain and travel cost from Alder

Knowledge context. Equal-looking distances can require different effort because terrain differs. A simple cost model makes that effect explicit.

Given facts. Terrain energy costs per segment: {'plain': 1, 'forest': 2, 'hill': 3, 'marsh': 4}. Three routes from Alder to Stone: R1=['plain', 'hill', 'plain']; R2=['forest', 'forest', 'plain']; R3=['plain', 'marsh']. The traveler can spend at most 11 energy units.

Rules. The energy of a route is the sum of its terrain costs. A route above the energy limit is infeasible. Among feasible routes, minimize energy.

Task. Which route is feasible and uses the least energy?

Step-by-step solution. 1. Compute route totals: R1=5, R2=5, R3=5. 2. Discard any total above 11. 3. The smallest remaining total is R1=5.

Answer. Route R1, using 5 energy units.

Knowledge takeaway. Relief changes the cost of movement; “shortest” and “easiest” are not automatically the same.

Formal model. Weighted paths with a feasibility budget; sum costs and choose the minimum feasible path.

Problem 784. Terrain and travel cost from Harbor

Knowledge context. Equal-looking distances can require different effort because terrain differs. A simple cost model makes that effect explicit.

Given facts. Terrain energy costs per segment: {'plain': 1, 'forest': 2, 'hill': 3, 'marsh': 4}. Three routes from Harbor to Kite: R1=['plain', 'hill', 'plain']; R2=['plain', 'forest', 'plain', 'plain']; R3=['plain', 'marsh']. The traveler can spend at most 11 energy units.

Rules. The energy of a route is the sum of its terrain costs. A route above the energy limit is infeasible. Among feasible routes, minimize energy.

Task. Which route is feasible and uses the least energy?

Step-by-step solution. 1. Compute route totals: R1=5, R2=5, R3=5. 2. Discard any total above 11. 3. The smallest remaining total is R1=5.

Answer. Route R1, using 5 energy units.

Knowledge takeaway. Relief changes the cost of movement; “shortest” and “easiest” are not automatically the same.

Formal model. Weighted paths with a feasibility budget; sum costs and choose the minimum feasible path.

Problem 785. Terrain and travel cost from Elm

Knowledge context. Equal-looking distances can require different effort because terrain differs. A simple cost model makes that effect explicit.

Given facts. Terrain energy costs per segment: {'plain': 1, 'forest': 2, 'hill': 3, 'marsh': 4}. Three routes from Elm to Zephyr: R1=['plain', 'hill', 'plain']; R2=['forest', 'forest', 'plain']; R3=['plain', 'marsh']. The traveler can spend at most 11 energy units.

Rules. The energy of a route is the sum of its terrain costs. A route above the energy limit is infeasible. Among feasible routes, minimize energy.

Task. Which route is feasible and uses the least energy?

Step-by-step solution. 1. Compute route totals: R1=5, R2=5, R3=5. 2. Discard any total above 11. 3. The smallest remaining total is R1=5.

Answer. Route R1, using 5 energy units.

Knowledge takeaway. Relief changes the cost of movement; “shortest” and “easiest” are not automatically the same.

Formal model. Weighted paths with a feasibility budget; sum costs and choose the minimum feasible path.

G8. Climate from data

Problem 786. Classifying climate data: Station A

Knowledge context. Climate descriptions summarize long-term patterns. In a reasoning problem, explicit thresholds let us classify numerical data without hidden knowledge.

Given facts. Use this classroom classification: cool if mean temperature <12°C; mild if 12–17°C; warm if >17°C. Dry if annual precipitation <500 mm; wet if ≥ 500 mm. Data: Station A: T=8°C, P=520 mm; Station B: T=16°C, P=780 mm; Station C: T=12°C, P=520 mm.

Rules. Apply only the stated thresholds. Temperature and precipitation are separate dimensions, so each station receives two labels.

Task. Classify Station A. Then identify whether any other station has the same two-label class.

Step-by-step solution. 1. Compare T=8 with the temperature thresholds: it is cool. 2. Compare P=520 with 500 mm: it is wet. 3. Compare the combined label with the other stations; matches: none.

Answer. Station A is cool and wet; no other station has the same class.

Knowledge takeaway. Classification is the application of clear predicates to data; the rule must be stated before the label is inferred.

Formal model. Numeric predicates with thresholds; output the conjunction of the temperature and precipitation categories.

Problem 787. Classifying climate data: Station B

Knowledge context. Climate descriptions summarize long-term patterns. In a reasoning problem, explicit thresholds let us classify numerical data without hidden knowledge.

Given facts. Use this classroom classification: cool if mean temperature <12°C; mild if 12–17°C; warm if >17°C. Dry if annual precipitation <500 mm; wet if ≥ 500 mm. Data: Station A: T=9°C, P=520 mm; Station B: T=17°C, P=780 mm; Station C: T=13°C, P=530 mm.

Rules. Apply only the stated thresholds. Temperature and precipitation are separate dimensions, so each station receives two labels.

Task. Classify Station B. Then identify whether any other station has the same two-label class.

Step-by-step solution. 1. Compare T=17 with the temperature thresholds: it is mild. 2. Compare P=780 with 500 mm: it is wet. 3. Compare the combined label with the other stations; matches: ['Station C'].

Answer. Station B is mild and wet; same class: Station C.

Knowledge takeaway. Classification is the application of clear predicates to data; the rule must be stated before the label is inferred.

Formal model. Numeric predicates with thresholds; output the conjunction of the temperature and precipitation categories.

Problem 788. Classifying climate data: Station C

Knowledge context. Climate descriptions summarize long-term patterns. In a reasoning problem, explicit thresholds let us classify numerical data without hidden knowledge.

Given facts. Use this classroom classification: cool if mean temperature <12°C; mild if 12–17°C; warm if >17°C. Dry if annual precipitation <500 mm; wet if ≥ 500 mm. Data: Station A: T=10°C, P=520 mm; Station B: T=18°C, P=780 mm; Station C: T=14°C, P=540 mm.

Rules. Apply only the stated thresholds. Temperature and precipitation are separate dimensions, so each station receives two labels.

Task. Classify Station C. Then identify whether any other station has the same two-label class.

Step-by-step solution. 1. Compare T=14 with the temperature thresholds: it is mild. 2. Compare P=540 with 500 mm: it is wet. 3. Compare the combined label with the other stations; matches: none.

Answer. Station C is mild and wet; no other station has the same class.

Knowledge takeaway. Classification is the application of clear predicates to data; the rule must be stated before the label is inferred.

Formal model. Numeric predicates with thresholds; output the conjunction of the temperature and precipitation categories.

Problem 789. Classifying climate data: Station A

Knowledge context. Climate descriptions summarize long-term patterns. In a reasoning problem, explicit thresholds let us classify numerical data without hidden knowledge.

Given facts. Use this classroom classification: cool if mean temperature <12°C; mild if 12–17°C; warm if >17°C. Dry if annual precipitation <500 mm; wet if ≥ 500 mm. Data: Station A: T=11°C, P=520 mm; Station B: T=19°C, P=780 mm; Station C: T=15°C, P=550 mm.

Rules. Apply only the stated thresholds. Temperature and precipitation are separate dimensions, so each station receives two labels.

Task. Classify Station A. Then identify whether any other station has the same two-label class.

Step-by-step solution. 1. Compare T=11 with the temperature thresholds: it is cool. 2. Compare P=520 with 500 mm: it is wet. 3. Compare the combined label with the other stations; matches: none.

Answer. Station A is cool and wet; no other station has the same class.

Knowledge takeaway. Classification is the application of clear predicates to data; the rule must be stated before the label is inferred.

Formal model. Numeric predicates with thresholds; output the conjunction of the temperature and precipitation categories.

Problem 790. Classifying climate data: Station B

Knowledge context. Climate descriptions summarize long-term patterns. In a reasoning problem, explicit thresholds let us classify numerical data without hidden knowledge.

Given facts. Use this classroom classification: cool if mean temperature <12°C; mild if 12–17°C; warm if >17°C. Dry if annual precipitation <500 mm; wet if ≥ 500 mm. Data: Station A: T=12°C, P=520 mm; Station B: T=20°C, P=780 mm; Station C: T=16°C, P=560 mm.

Rules. Apply only the stated thresholds. Temperature and precipitation are separate dimensions, so each station receives two labels.

Task. Classify Station B. Then identify whether any other station has the same two-label class.

Step-by-step solution. 1. Compare T=20 with the temperature thresholds: it is warm. 2. Compare P=780 with 500 mm: it is wet. 3. Compare the combined label with the other stations; matches: none.

Answer. Station B is warm and wet; no other station has the same class.

Knowledge takeaway. Classification is the application of clear predicates to data; the rule must be stated before the label is inferred.

Formal model. Numeric predicates with thresholds; output the conjunction of the temperature and precipitation categories.

G9. Exchange, routes, and capacities

Problem 791. Bottlenecks in the Willow–Northfield exchange network

Knowledge context. Trade and transport networks have capacities. A single narrow connection can limit a much larger network.

Given facts. Goods move from Willow to Northfield along two branches: Willow→Iris capacity 7, Iris→Northfield capacity 6; Willow→Oak capacity 2, Oak→Northfield capacity 8. The final warehouse at Northfield can accept at most 10 units.

Rules. A branch cannot carry more than its smallest edge capacity. Independent branch capacities can be added, but a shared destination capacity can cap the total.

Task. What is the maximum number of units that can reach Northfield in one period? Identify the bottleneck(s).

Step-by-step solution. 1. Branch 1 is limited by min(7,6)=6. 2. Branch 2 is limited by min(2,8)=2. 3. Together the branches could carry 8 units. 4. The destination accepts at most 10, so the final maximum is min(8,10)=8.

Answer. 8 units.

Knowledge takeaway. A bottleneck is a local limit that constrains the total possible flow.

Formal model. Flow network; branch capacity is a minimum along a path; total is the sum of independent branches subject to shared caps.

Problem 792. Bottlenecks in the Willow–Grove exchange network

Knowledge context. Trade and transport networks have capacities. A single narrow connection can limit a much larger network.

Given facts. Goods move from Willow to Grove along two branches: Willow→Stone capacity 8, Stone→Grove capacity 6; Willow→Riverbend capacity 3, Riverbend→Grove capacity 8. The final warehouse at Grove can accept at most 10 units.

Rules. A branch cannot carry more than its smallest edge capacity. Independent branch capacities can be added, but a shared destination capacity can cap the total.

Task. What is the maximum number of units that can reach Grove in one period? Identify the bottleneck(s).

Step-by-step solution. 1. Branch 1 is limited by min(8,6)=6. 2. Branch 2 is limited by min(3,8)=3. 3. Together the branches could carry 9 units. 4. The destination accepts at most 10, so the final maximum is min(9,10)=9.

Answer. 9 units.

Knowledge takeaway. A bottleneck is a local limit that constrains the total possible flow.

Formal model. Flow network; branch capacity is a minimum along a path; total is the sum of independent branches subject to shared caps.

Problem 793. Bottlenecks in the Cedar–Oak exchange network

Knowledge context. Trade and transport networks have capacities. A single narrow connection can limit a much larger network.

Given facts. Goods move from Cedar to Oak along two branches: Cedar→Stone capacity 9, Stone→Oak capacity 6; Cedar→Meadow capacity 4, Meadow→Oak capacity 8. The final warehouse at Oak can accept at most 10 units.

Rules. A branch cannot carry more than its smallest edge capacity. Independent branch capacities can be added, but a shared destination capacity can cap the total.

Task. What is the maximum number of units that can reach Oak in one period? Identify the bottleneck(s).

Step-by-step solution. 1. Branch 1 is limited by min(9,6)=6. 2. Branch 2 is limited by min(4,8)=4. 3. Together the branches could carry 10 units. 4. The destination accepts at most 10, so the final maximum is min(10,10)=10.

Answer. 10 units.

Knowledge takeaway. A bottleneck is a local limit that constrains the total possible flow.

Formal model. Flow network; branch capacity is a minimum along a path; total is the sum of independent branches subject to shared caps.

Problem 794. Bottlenecks in the Birch–Meadow exchange network

Knowledge context. Trade and transport networks have capacities. A single narrow connection can limit a much larger network.

Given facts. Goods move from Birch to Meadow along two branches: Birch→Northfield capacity 10, Northfield→Meadow capacity 6; Birch→Willow capacity 5, Willow→Meadow capacity 8. The final warehouse at Meadow can accept at most 10 units.

Rules. A branch cannot carry more than its smallest edge capacity. Independent branch capacities can be added, but a shared destination capacity can cap the total.

Task. What is the maximum number of units that can reach Meadow in one period? Identify the bottleneck(s).

Step-by-step solution. 1. Branch 1 is limited by min(10,6)=6. 2. Branch 2 is limited by min(5,8)=5. 3. Together the branches could carry 11 units. 4. The destination accepts at most 10, so the final maximum is min(11,10)=10.

Answer. 10 units.

Knowledge takeaway. A bottleneck is a local limit that constrains the total possible flow.

Formal model. Flow network; branch capacity is a minimum along a path; total is the sum of independent branches subject to shared caps.

Problem 795. Bottlenecks in the Willow–Cedar exchange network

Knowledge context. Trade and transport networks have capacities. A single narrow connection can limit a much larger network.

Given facts. Goods move from Willow to Cedar along two branches: Willow→Iris capacity 11, Iris→Cedar capacity 6; Willow→Meadow capacity 6, Meadow→Cedar capacity 8. The final warehouse at Cedar can accept at most 10 units.

Rules. A branch cannot carry more than its smallest edge capacity. Independent branch capacities can be added, but a shared destination capacity can cap the total.

Task. What is the maximum number of units that can reach Cedar in one period? Identify the bottleneck(s).

Step-by-step solution. 1. Branch 1 is limited by min(11,6)=6. 2. Branch 2 is limited by min(6,8)=6. 3. Together the branches could carry 12 units. 4. The destination accepts at most 10, so the final maximum is min(12,10)=10.

Answer. 10 units.

Knowledge takeaway. A bottleneck is a local limit that constrains the total possible flow.

Formal model. Flow network; branch capacity is a minimum along a path; total is the sum of independent branches subject to shared caps.

G10. Regions, sets, and membership

Problem 796. Nested places: Birch, Eastvale, and Republic B

Knowledge context. Geography uses nested categories: a locality can be inside a region, a region inside a state, and a state inside a continent.

Given facts. Birch is in Eastvale. Eastvale is in Republic B. Republic B is in the continent of Meridian. Another city, Alder, is also in Republic B, but its region is not specified.

Rules. If X is contained in Y and Y is contained in Z, then X is contained in Z. The reverse inference is not valid: knowing that two places are in one country does not prove that they are in the same region.

Task. Is Birch in Republic B?

Step-by-step solution. 1. Follow the chain city→region→country. 2. Containment is transitive, so the city is in the country.

Answer. Yes, Birch is in Republic B.

Knowledge takeaway. Hierarchies support upward transitive inference, but they do not justify arbitrary reverse inferences.

Formal model. Containment relation with transitivity; answer membership queries while rejecting invalid converse inference.

Problem 797. Nested places: Stone, Green Valley, and Republic A

Knowledge context. Geography uses nested categories: a locality can be inside a region, a region inside a state, and a state inside a continent.

Given facts. Stone is in Green Valley. Green Valley is in Republic A. Republic A is in the continent of Meridian. Another city, Dover, is also in Republic A, but its region is not specified.

Rules. If X is contained in Y and Y is contained in Z, then X is contained in Z. The reverse inference is not valid: knowing that two places are in one country does not prove that they are in the same region.

Task. Is Stone in Meridian?

Step-by-step solution. 1. Follow city→region→country→continent. 2. Transitivity carries membership through the chain.

Answer. Yes, Stone is in Meridian.

Knowledge takeaway. Hierarchies support upward transitive inference, but they do not justify arbitrary reverse inferences.

Formal model. Containment relation with transitivity; answer membership queries while rejecting invalid converse inference.

Problem 798. Nested places: Stone, Westmoor, and Republic A

Knowledge context. Geography uses nested categories: a locality can be inside a region, a region inside a state, and a state inside a continent.

Given facts. Stone is in Westmoor. Westmoor is in Republic A. Republic A is in the continent of Aurora. Another city, Iris, is also in Republic A, but its region is not specified.

Rules. If X is contained in Y and Y is contained in Z, then X is contained in Z. The reverse inference is not valid: knowing that two places are in one country does not prove that they are in the same region.

Task. Can we conclude that Republic A is in Stone?

Step-by-step solution. 1. The facts go from smaller units to larger units. 2. No rule allows country→city, so the reverse claim is unsupported.

Answer. No. The containment relation cannot be reversed.

Knowledge takeaway. Hierarchies support upward transitive inference, but they do not justify arbitrary reverse inferences.

Formal model. Containment relation with transitivity; answer membership queries while rejecting invalid converse inference.

Problem 799. Nested places: Zephyr, Highridge, and Republic C

Knowledge context. Geography uses nested categories: a locality can be inside a region, a region inside a state, and a state inside a continent.

Given facts. Zephyr is in Highridge. Highridge is in Republic C. Republic C is in the continent of Meridian. Another city, Yarrow, is also in Republic C, but its region is not specified.

Rules. If X is contained in Y and Y is contained in Z, then X is contained in Z. The reverse inference is not valid: knowing that two places are in one country does not prove that they are in the same region.

Task. Which larger units contain Zephyr?

Step-by-step solution. 1. Read the direct container: Highridge. 2. Propagate upward to Republic C and Meridian.

Answer. Highridge, Republic C, and Meridian.

Knowledge takeaway. Hierarchies support upward transitive inference, but they do not justify arbitrary reverse inferences.

Formal model. Containment relation with transitivity; answer membership queries while rejecting invalid converse inference.

Problem 800. Nested places: Kite, Highridge, and Republic B

Knowledge context. Geography uses nested categories: a locality can be inside a region, a region inside a state, and a state inside a continent.

Given facts. Kite is in Highridge. Highridge is in Republic B. Republic B is in the continent of Meridian. Another city, Cedar, is also in Republic B, but its region is not specified.

Rules. If X is contained in Y and Y is contained in Z, then X is contained in Z. The reverse inference is not valid: knowing that two places are in one country does not prove that they are in the same region.

Task. Does being in Republic B prove two cities share a region?

Step-by-step solution. 1. Two cities can belong to the same country while belonging to different regions. 2. The region relation is therefore not deducible from country membership alone.

Answer. No. Country membership alone is insufficient to identify the region.

Knowledge takeaway. Hierarchies support upward transitive inference, but they do not justify arbitrary reverse inferences.

Formal model. Containment relation with transitivity; answer membership queries while rejecting invalid converse inference.

H1. Ordering events

Problem 801. Ordering the events of Green Valley

Knowledge context. Historical chronology often starts from partial information: some events are ordered relative to others even when exact dates are not known.

Given facts. Founding happened before New school. New school happened before Flood. Bridge opening happened after Founding. Market charter happened after Flood.

Rules. “Before” is transitive: if A is before B and B is before C, then A is before C. If two events are not connected by the facts, their relative order may be undetermined.

Task. What must be true about the order of Founding and Flood? Can the exact order of Bridge opening and New school be determined?

Step-by-step solution. 1. From Founding before New school and New school before Flood, infer Founding before Flood. 2. Bridge opening is only constrained to be after Founding. There is no stated relation between Bridge opening and New school. 3. Therefore the first comparison is forced but the second is underdetermined.

Answer. Founding must be before Flood. The relative order of Bridge opening and New school cannot be determined from the given facts.

Knowledge takeaway. A careful reasoner distinguishes what must be true from what merely could be true.

Formal model. Partial order / directed acyclic graph; use transitive closure and detect incomparable events.

Problem 802. Ordering the events of Eastvale

Knowledge context. Historical chronology often starts from partial information: some events are ordered relative to others even when exact dates are not known.

Given facts. New school happened before Bridge opening. Bridge opening happened before Market charter. Flood happened after New school. Founding happened after Market charter.

Rules. “Before” is transitive: if A is before B and B is before C, then A is before C. If two events are not connected by the facts, their relative order may be undetermined.

Task. What must be true about the order of New school and Market charter? Can the exact order of Flood and Bridge opening be determined?

Step-by-step solution. 1. From New school before Bridge opening and Bridge opening before Market charter, infer New school before Market charter. 2. Flood is only constrained to be after New school. There is no stated relation between Flood and Bridge opening. 3. Therefore the first comparison is forced but the second is underdetermined.

Answer. New school must be before Market charter. The relative order of Flood and Bridge opening cannot be determined from the given facts.

Knowledge takeaway. A careful reasoner distinguishes what must be true from what merely could be true.

Formal model. Partial order / directed acyclic graph; use transitive closure and detect incomparable events.

Problem 803. Ordering the events of Highridge

Knowledge context. Historical chronology often starts from partial information: some events are ordered relative to others even when exact dates are not known.

Given facts. New school happened before Bridge opening. Bridge opening happened before Founding. Flood happened after New school. Market charter happened after Founding.

Rules. “Before” is transitive: if A is before B and B is before C, then A is before C. If two events are not connected by the facts, their relative order may be undetermined.

Task. What must be true about the order of New school and Founding? Can the exact order of Flood and Bridge opening be determined?

Step-by-step solution. 1. From New school before Bridge opening and Bridge opening before Founding, infer New school before Founding. 2. Flood is only constrained to be after New school. There is no stated relation between Flood and Bridge opening. 3. Therefore the first comparison is forced but the second is underdetermined.

Answer. New school must be before Founding. The relative order of Flood and Bridge opening cannot be determined from the given facts.

Knowledge takeaway. A careful reasoner distinguishes what must be true from what merely could be true.

Formal model. Partial order / directed acyclic graph; use transitive closure and detect incomparable events.

Problem 804. Ordering the events of Lake District

Knowledge context. Historical chronology often starts from partial information: some events are ordered relative to others even when exact dates are not known.

Given facts. Flood happened before Founding. Founding happened before Bridge opening. New school happened after Flood. Market charter happened after Bridge opening.

Rules. “Before” is transitive: if A is before B and B is before C, then A is before C. If two events are not connected by the facts, their relative order may be undetermined.

Task. What must be true about the order of Flood and Bridge opening? Can the exact order of New school and Founding be determined?

Step-by-step solution. 1. From Flood before Founding and Founding before Bridge opening, infer Flood before Bridge opening. 2. New school is only constrained to be after Flood. There is no stated relation between New school and Founding. 3. Therefore the first comparison is forced but the second is underdetermined.

Answer. Flood must be before Bridge opening. The relative order of New school and Founding cannot be determined from the given facts.

Knowledge takeaway. A careful reasoner distinguishes what must be true from what merely could be true.

Formal model. Partial order / directed acyclic graph; use transitive closure and detect incomparable events.

Problem 805. Ordering the events of Southplain

Knowledge context. Historical chronology often starts from partial information: some events are ordered relative to others even when exact dates are not known.

Given facts. New school happened before Market charter. Market charter happened before Bridge opening. Flood happened after New school. Founding happened after Bridge opening.

Rules. “Before” is transitive: if A is before B and B is before C, then A is before C. If two events are not connected by the facts, their relative order may be undetermined.

Task. What must be true about the order of New school and Bridge opening? Can the exact order of Flood and Market charter be determined?

Step-by-step solution. 1. From New school before Market charter and Market charter before Bridge opening, infer New school before Bridge opening. 2. Flood is only constrained to be after New school. There is no stated relation between Flood and Market charter. 3. Therefore the first comparison is forced but the second is underdetermined.

Answer. New school must be before Bridge opening. The relative order of Flood and Market charter cannot be determined from the given facts.

Knowledge takeaway. A careful reasoner distinguishes what must be true from what merely could be true.

Formal model. Partial order / directed acyclic graph; use transitive closure and detect incomparable events.

H2. Dates, durations, and centuries

Problem 806. Duration and centuries: 1699–1731

Knowledge context. Dates can be used as numbers, but historical labels such as centuries follow explicit conventions.

Given facts. An institution was founded in 1699 and reorganized in 1731. Use the rule: years 1–100 are the 1st century, 101–200 the 2nd, and so on. Duration is end year minus start year.

Rules. Apply the century rule mechanically; do not infer a century from the first two digits alone. Compute elapsed time by subtraction.

Task. How many years passed between the two events? In which centuries did 1699 and 1731 fall?

Step-by-step solution. 1. Elapsed time: 1731−1699=32 years. 2. Century(1699)=floor((1699−1)/100)+1=17. 3. Century(1731)=18.

Answer. 32 years; 1699 is in the 17th century and 1731 in the 18th century.

Knowledge takeaway. Chronological reasoning combines arithmetic with a naming convention; both should be applied explicitly.

Formal model. Integer arithmetic plus century function c(y)=floor((y−1)/100)+1.

Problem 807. Duration and centuries: 1713–1750

Knowledge context. Dates can be used as numbers, but historical labels such as centuries follow explicit conventions.

Given facts. An institution was founded in 1713 and reorganized in 1750. Use the rule: years 1–100 are the 1st century, 101–200 the 2nd, and so on. Duration is end year minus start year.

Rules. Apply the century rule mechanically; do not infer a century from the first two digits alone. Compute elapsed time by subtraction.

Task. How many years passed between the two events? In which centuries did 1713 and 1750 fall?

Step-by-step solution. 1. Elapsed time: 1750−1713=37 years. 2. Century(1713)=floor((1713−1)/100)+1=18. 3. Century(1750)=18.

Answer. 37 years; 1713 is in the 18th century and 1750 in the 18th century.

Knowledge takeaway. Chronological reasoning combines arithmetic with a naming convention; both should be applied explicitly.

Formal model. Integer arithmetic plus century function c(y)=floor((y−1)/100)+1.

Problem 808. Duration and centuries: 1724–1766

Knowledge context. Dates can be used as numbers, but historical labels such as centuries follow explicit conventions.

Given facts. An institution was founded in 1724 and reorganized in 1766. Use the rule: years 1–100 are the 1st century, 101–200 the 2nd, and so on. Duration is end year minus start year.

Rules. Apply the century rule mechanically; do not infer a century from the first two digits alone. Compute elapsed time by subtraction.

Task. How many years passed between the two events? In which centuries did 1724 and 1766 fall?

Step-by-step solution. 1. Elapsed time: 1766−1724=42 years. 2. Century(1724)=floor((1724−1)/100)+1=18. 3. Century(1766)=18.

Answer. 42 years; 1724 is in the 18th century and 1766 in the 18th century.

Knowledge takeaway. Chronological reasoning combines arithmetic with a naming convention; both should be applied explicitly.

Formal model. Integer arithmetic plus century function c(y)=floor((y−1)/100)+1.

Problem 809. Duration and centuries: 1747–1794

Knowledge context. Dates can be used as numbers, but historical labels such as centuries follow explicit conventions.

Given facts. An institution was founded in 1747 and reorganized in 1794. Use the rule: years 1–100 are the 1st century, 101–200 the 2nd, and so on. Duration is end year minus start year.

Rules. Apply the century rule mechanically; do not infer a century from the first two digits alone. Compute elapsed time by subtraction.

Task. How many years passed between the two events? In which centuries did 1747 and 1794 fall?

Step-by-step solution. 1. Elapsed time: 1794−1747=47 years. 2. Century(1747)=floor((1747−1)/100)+1=18. 3. Century(1794)=18.

Answer. 47 years; 1747 is in the 18th century and 1794 in the 18th century.

Knowledge takeaway. Chronological reasoning combines arithmetic with a naming convention; both should be applied explicitly.

Formal model. Integer arithmetic plus century function c(y)=floor((y−1)/100)+1.

Problem 810. Duration and centuries: 1763–1815

Knowledge context. Dates can be used as numbers, but historical labels such as centuries follow explicit conventions.

Given facts. An institution was founded in 1763 and reorganized in 1815. Use the rule: years 1–100 are the 1st century, 101–200 the 2nd, and so on. Duration is end year minus start year.

Rules. Apply the century rule mechanically; do not infer a century from the first two digits alone. Compute elapsed time by subtraction.

Task. How many years passed between the two events? In which centuries did 1763 and 1815 fall?

Step-by-step solution. 1. Elapsed time: 1815−1763=52 years. 2. Century(1763)=floor((1763−1)/100)+1=18. 3. Century(1815)=19.

Answer. 52 years; 1763 is in the 18th century and 1815 in the 19th century.

Knowledge takeaway. Chronological reasoning combines arithmetic with a naming convention; both should be applied explicitly.

Formal model. Integer arithmetic plus century function c(y)=floor((y−1)/100)+1.

H3. Sources and supported claims

Problem 811. What can one photograph prove?

Knowledge context. Historical sources are evidence, not automatic answers. The strength of a conclusion depends on what the source actually records.

Given facts. A historian has one source: a photograph. The source directly shows that a scene was photographed. A student proposes two claims: C1: a scene was photographed; C2: the whole population agreed with the ruler. Cross-domain check: researchers collected 7 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. A source supports only claims that can reasonably be inferred from what it contains. Absence of evidence for a broad claim is not proof of that claim. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. Which claim is directly supported, and which is not justified by this source alone? Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. The photograph directly contains evidence for C1. 2. C2 is much broader and no stated feature of the source measures the opinion of the whole population. 3. Therefore C1 is supported and C2 is not established. Cross-domain check: 7−4=3 independent reports remain.

Answer. C1 is supported. C2 is not justified by the source alone. Cross-domain answer: 3 independent reports.

Knowledge takeaway. Good historical reasoning matches the scope of a claim to the scope of the evidence.

Formal model. Claim-evidence compatibility relation; accept only claims entailed by the source attributes.

Problem 812. What can one map prove?

Knowledge context. Historical sources are evidence, not automatic answers. The strength of a conclusion depends on what the source actually records.

Given facts. A historian has one source: a map. The source directly shows that a road was mapped. A student proposes two claims: C1: a road was mapped; C2: the whole population agreed with the ruler. Cross-domain check: on a field sketch, 1 cm represents 4 km and two checkpoints are 2 cm apart.

Rules. A source supports only claims that can reasonably be inferred from what it contains. Absence of evidence for a broad claim is not proof of that claim. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Which claim is directly supported, and which is not justified by this source alone? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. The map directly contains evidence for C1. 2. C2 is much broader and no stated feature of the source measures the opinion of the whole population. 3. Therefore C1 is supported and C2 is not established. Cross-domain check: 2×4=8 km.

Answer. C1 is supported. C2 is not justified by the source alone. Cross-domain answer: 8 km.

Knowledge takeaway. Good historical reasoning matches the scope of a claim to the scope of the evidence.

Formal model. Claim-evidence compatibility relation; accept only claims entailed by the source attributes.

Problem 813. What can one tax register prove?

Knowledge context. Historical sources are evidence, not automatic answers. The strength of a conclusion depends on what the source actually records.

Given facts. A historian has one source: a tax register. The source directly shows that taxes were recorded. A student proposes two claims: C1: taxes were recorded; C2: the whole population agreed with the ruler. Cross-domain check: a survey team starts work at 12:00 and works for 2 hour(s) without a break.

Rules. A source supports only claims that can reasonably be inferred from what it contains. Absence of evidence for a broad claim is not proof of that claim. For the cross-domain check, finishing time = starting time + duration.

Task. Which claim is directly supported, and which is not justified by this source alone? Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. The tax register directly contains evidence for C1. 2. C2 is much broader and no stated feature of the source measures the opinion of the whole population. 3. Therefore C1 is supported and C2 is not established. Cross-domain check: 12+2=14, so the team finishes at 14:00.

Answer. C1 is supported. C2 is not justified by the source alone. Cross-domain answer: 14:00.

Knowledge takeaway. Good historical reasoning matches the scope of a claim to the scope of the evidence.

Formal model. Claim-evidence compatibility relation; accept only claims entailed by the source attributes.

Problem 814. What can one photograph prove?

Knowledge context. Historical sources are evidence, not automatic answers. The strength of a conclusion depends on what the source actually records.

Given facts. A historian has one source: a photograph. The source directly shows that a scene was photographed. A student proposes two claims: C1: a scene was photographed; C2: the whole population agreed with the ruler. Cross-domain check: a local committee has 12 members, requires at least 7 present for quorum, and 11 are present.

Rules. A source supports only claims that can reasonably be inferred from what it contains. Absence of evidence for a broad claim is not proof of that claim. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. Which claim is directly supported, and which is not justified by this source alone? Cross-domain check: is quorum met?

Step-by-step solution. 1. The photograph directly contains evidence for C1. 2. C2 is much broader and no stated feature of the source measures the opinion of the whole population. 3. Therefore C1 is supported and C2 is not established. Cross-domain check: compare 11 with 7; $11 \geq 7$ is true.

Answer. C1 is supported. C2 is not justified by the source alone. Cross-domain answer: quorum is met.

Knowledge takeaway. Good historical reasoning matches the scope of a claim to the scope of the evidence.

Formal model. Claim-evidence compatibility relation; accept only claims entailed by the source attributes.

Problem 815. What can one letter prove?

Knowledge context. Historical sources are evidence, not automatic answers. The strength of a conclusion depends on what the source actually records.

Given facts. A historian has one source: a letter. The source directly shows that a named person wrote a message. A student proposes two claims: C1: a named person wrote a message; C2: the whole population agreed with the ruler. Cross-domain check: researchers collected 11 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. A source supports only claims that can reasonably be inferred from what it contains. Absence of evidence for a broad claim is not proof of that claim. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. Which claim is directly supported, and which is not justified by this source alone? Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. The letter directly contains evidence for C1. 2. C2 is much broader and no stated feature of the source measures the opinion of the whole population. 3. Therefore C1 is supported and C2 is not established. Cross-domain check: $11-4=7$ independent reports remain.

Answer. C1 is supported. C2 is not justified by the source alone. Cross-domain answer: 7 independent reports.

Knowledge takeaway. Good historical reasoning matches the scope of a claim to the scope of the evidence.

Formal model. Claim-evidence compatibility relation; accept only claims entailed by the source attributes.

H4. Conflicting sources

Problem 816. Resolving conflicting sources: case 1

Knowledge context. Historical accounts can disagree. Reasoning requires evaluating how a source knows what it claims, not simply counting statements.

Given facts. Source A, written on the day of an event, says 40 carts arrived. Source B, copied 80 years later from an unknown account, says 90 carts arrived. Source C is an independent warehouse record from the same week and lists 42 carts. Cross-domain check: on a field sketch, 1 cm represents 3 km and two checkpoints are 6 cm apart.

Rules. When sources conflict, compare proximity to the event, independence, and whether a source has a reason or mechanism to record the specific fact. Agreement between independent relevant sources increases support but does not create absolute certainty. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Which account is better supported by the evidence given, and why? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. A is contemporary and reports 40. 2. C is an independent operational record and reports 42, close to A. 3. B is much later and has an unknown chain of transmission. 4. The cluster near 40–42 is therefore better supported than 90. Cross-domain check: $6\times3=18$ km.

Answer. An arrival count near 40–42 carts is better supported than 90. Cross-domain answer: 18 km.

Knowledge takeaway. Independent, relevant, well-positioned evidence usually deserves more weight than late repetition with unknown provenance.

Formal model. Evidence graph with source reliability/independence attributes; compare support for competing hypotheses.

Problem 817. Resolving conflicting sources: case 2

Knowledge context. Historical accounts can disagree. Reasoning requires evaluating how a source knows what it claims, not simply counting statements.

Given facts. Source A, an eyewitness diary, says the bridge opened on Monday. Source B, a later textbook, says Tuesday. Source C, an independent dated toll record, begins charging on Monday. Cross-domain check: a survey team starts work at 8:00 and works for 2 hour(s) without a break. Mixed-domain verification: a regional archive has 83 map sheets and receives 8 additional sheets during the exercise.

Rules. When sources conflict, compare proximity to the event, independence, and whether a source has a reason or mechanism to record the specific fact. Agreement between independent relevant sources increases support but does not create absolute certainty. For the cross-domain check, finishing time = starting time + duration. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. Which account is better supported by the evidence given, and why? Cross-domain check: at what time does the team finish? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. A and C are close to the event and independent. 2. Both point to Monday. 3. B is later and conflicts with two stronger, independent pieces of evidence. Cross-domain check: 8+2=10, so the team finishes at 10:00. Mixed-domain verification: 83+8=91.

Answer. Monday is better supported, while remaining a historical conclusion rather than absolute proof. Cross-domain answer: 10:00. Mixed-domain answer: 91 map sheets.

Knowledge takeaway. Independent, relevant, well-positioned evidence usually deserves more weight than late repetition with unknown provenance.

Formal model. Evidence graph with source reliability/independence attributes; compare support for competing hypotheses.

Problem 818. Resolving conflicting sources: case 3

Knowledge context. Historical accounts can disagree. Reasoning requires evaluating how a source knows what it claims, not simply counting statements.

Given facts. Source A, written on the day of an event, says 40 carts arrived. Source B, copied 80 years later from an unknown account, says 90 carts arrived. Source C is an independent warehouse record from the same week and lists 42 carts. Cross-domain check: a local committee has 16 members, requires at least 9 present for quorum, and 14 are present. Mixed-domain verification: a regional archive has 83 map sheets and receives 9 additional sheets during the exercise.

Rules. When sources conflict, compare proximity to the event, independence, and whether a source has a reason or mechanism to record the specific fact. Agreement between independent relevant sources increases support but does not create absolute certainty. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. Which account is better supported by the evidence given, and why? Cross-domain check: is quorum met? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. A is contemporary and reports 40. 2. C is an independent operational record and reports 42, close to A. 3. B is much later and has an unknown chain of transmission. 4. The cluster near 40–42 is therefore better supported than 90. Cross-domain check: compare 14 with 9; $14 \geq 9$ is true. Mixed-domain verification: 83+9=92.

Answer. An arrival count near 40–42 carts is better supported than 90. Cross-domain answer: quorum is met. Mixed-domain answer: 92 map sheets.

Knowledge takeaway. Independent, relevant, well-positioned evidence usually deserves more weight than late repetition with unknown provenance.

Formal model. Evidence graph with source reliability/independence attributes; compare support for competing hypotheses.

Problem 819. Resolving conflicting sources: case 4

Knowledge context. Historical accounts can disagree. Reasoning requires evaluating how a source knows what it claims, not simply counting statements.

Given facts. Source A, an eyewitness diary, says the bridge opened on Monday. Source B, a later textbook, says Tuesday. Source C, an independent dated toll record, begins charging on Monday. Cross-domain check: researchers collected 15 reports, but 4 are exact duplicate copies that add no new independent information. Mixed-domain verification: a regional archive has 83 map sheets and receives 10 additional sheets during the exercise.

Rules. When sources conflict, compare proximity to the event, independence, and whether a source has a reason or mechanism to record the specific fact. Agreement between independent relevant sources increases support but does not

create absolute certainty. For the cross-domain check, independent reports = total reports − duplicate copies. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. Which account is better supported by the evidence given, and why? Cross-domain check: how many non-duplicate reports remain? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. A and C are close to the event and independent. 2. Both point to Monday. 3. B is later and conflicts with two stronger, independent pieces of evidence. Cross-domain check: 15−4=11 independent reports remain. Mixed-domain verification: 83+10=93.

Answer. Monday is better supported, while remaining a historical conclusion rather than absolute proof. Cross-domain answer: 11 independent reports. Mixed-domain answer: 93 map sheets.

Knowledge takeaway. Independent, relevant, well-positioned evidence usually deserves more weight than late repetition with unknown provenance.

Formal model. Evidence graph with source reliability/independence attributes; compare support for competing hypotheses.

Problem 820. Resolving conflicting sources: case 5

Knowledge context. Historical accounts can disagree. Reasoning requires evaluating how a source knows what it claims, not simply counting statements.

Given facts. Source A, written on the day of an event, says 40 carts arrived. Source B, copied 80 years later from an unknown account, says 90 carts arrived. Source C is an independent warehouse record from the same week and lists 42 carts. Cross-domain check: on a field sketch, 1 cm represents 2 km and two checkpoints are 3 cm apart.

Rules. When sources conflict, compare proximity to the event, independence, and whether a source has a reason or mechanism to record the specific fact. Agreement between independent relevant sources increases support but does not create absolute certainty. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Which account is better supported by the evidence given, and why? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. A is contemporary and reports 40. 2. C is an independent operational record and reports 42, close to A. 3. B is much later and has an unknown chain of transmission. 4. The cluster near 40–42 is therefore better supported than 90. Cross-domain check: 3×2=6 km.

Answer. An arrival count near 40–42 carts is better supported than 90. Cross-domain answer: 6 km.

Knowledge takeaway. Independent, relevant, well-positioned evidence usually deserves more weight than late repetition with unknown provenance.

Formal model. Evidence graph with source reliability/independence attributes; compare support for competing hypotheses.

H5. Causes, conditions, and consequences

Problem 821. Cause-and-consequence chain: case 1

Knowledge context. Many explanations can be represented as condition-and-consequence rules. The logical form matters: necessary, sufficient, AND, and OR are not interchangeable.

Given facts. Facts: Rain occurred; The river was already high; The warning siren worked; The gate was closed. Rules: If rain occurs AND the river is already high, flooding occurs. If flooding occurs, the riverside road closes. If the siren works, residents receive a warning. Cross-domain check: a survey team starts work at 12:00 and works for 2 hour(s) without a break.

Rules. Use the rules exactly as written. AND requires all listed conditions; OR requires at least one. A one-way implication cannot automatically be reversed. For the cross-domain check, finishing time = starting time + duration.

Task. Which consequences can be deduced by applying the rules exactly? Do not assume the converse of any rule. Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. Rain and high river satisfy both conditions of the first rule. 2. Infer flooding. 3. Flooding implies road closure. 4. The working siren independently implies a warning. Cross-domain check: 12+2=14, so the team finishes at 14:00.

Answer. Flooding, road closure, and a warning can be deduced. Cross-domain answer: 14:00.

Knowledge takeaway. Causal reasoning becomes clearer when every link is stated and the direction of implication is respected.

Formal model. Forward-chaining rule system over Boolean facts; apply implications until no new facts appear.

Problem 822. Cause-and-consequence chain: case 2

Knowledge context. Many explanations can be represented as condition-and-consequence rules. The logical form matters: necessary, sufficient, AND, and OR are not interchangeable.

Given facts. Facts: Rain occurred; The river was not high; The warning siren worked; The gate was closed. Rules: If rain occurs AND the river is already high, flooding occurs. If flooding occurs, the riverside road closes. If the siren works, residents receive a warning. Cross-domain check: a local committee has 11 members, requires at least 6 present for quorum, and 11 are present.

Rules. Use the rules exactly as written. AND requires all listed conditions; OR requires at least one. A one-way implication cannot automatically be reversed. For the cross-domain check, quorum exists when present members ≥ the stated threshold.

Task. Which consequences can be deduced by applying the rules exactly? Do not assume the converse of any rule. Cross-domain check: is quorum met?

Step-by-step solution. 1. Rain is true but the high-river condition is false. 2. The AND rule for flooding therefore does not fire. 3. Without flooding, road closure is not inferred. 4. The siren rule still gives a warning. Cross-domain check: compare 11 with 6; 11 ≥ 6 is true.

Answer. A warning can be deduced, but flooding and road closure cannot be deduced. Cross-domain answer: quorum is met.

Knowledge takeaway. Causal reasoning becomes clearer when every link is stated and the direction of implication is respected.

Formal model. Forward-chaining rule system over Boolean facts; apply implications until no new facts appear.

Problem 823. Cause-and-consequence chain: case 3

Knowledge context. Many explanations can be represented as condition-and-consequence rules. The logical form matters: necessary, sufficient, AND, and OR are not interchangeable.

Given facts. Facts: Rain occurred; The river was already high; The warning siren worked; The gate was closed; Flooding occurred. Rules: If rain occurs AND the river is already high, flooding occurs. If flooding occurs, the riverside road closes. If the siren works, residents receive a warning. Cross-domain check: researchers collected 9 reports, but 4 are exact duplicate copies that add no new independent information. Mixed-domain verification: a regional archive has 84 map sheets and receives 4 additional sheets during the exercise.

Rules. Use the rules exactly as written. AND requires all listed conditions; OR requires at least one. A one-way implication cannot automatically be reversed. For the cross-domain check, independent reports = total reports − duplicate copies. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. Which consequences can be deduced by applying the rules exactly? Do not assume the converse of any rule. Cross-domain check: how many non-duplicate reports remain? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. Flooding is stated directly, so apply the flooding→road-closure rule. 2. Apply siren→warning. 3. Do not use flooding to infer rain or a high river; that would reverse a rule. Cross-domain check: 9−4=5 independent reports remain. Mixed-domain verification: 84+4=88.

Answer. Road closure and a warning can be deduced; flooding is already given as a fact. Cross-domain answer: 5 independent reports. Mixed-domain answer: 88 map sheets.

Knowledge takeaway. Causal reasoning becomes clearer when every link is stated and the direction of implication is respected.

Formal model. Forward-chaining rule system over Boolean facts; apply implications until no new facts appear.

Problem 824. Cause-and-consequence chain: case 4

Knowledge context. Many explanations can be represented as condition-and-consequence rules. The logical form matters: necessary, sufficient, AND, and OR are not interchangeable.

Given facts. Facts: Rain occurred; The river was already high; The warning siren worked; The gate was closed. Rules: If rain occurs OR the river is already high, the field becomes muddy. If flooding occurs, the riverside road closes. If the siren works, residents receive a warning. Cross-domain check: on a field sketch, 1 cm represents 6 km and two checkpoints are 7 cm apart.

Rules. Use the rules exactly as written. AND requires all listed conditions; OR requires at least one. A one-way implication cannot automatically be reversed. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Which consequences can be deduced by applying the rules exactly? Do not assume the converse of any rule. Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. The mud rule uses OR, so rain alone is enough. 2. Rain is given, therefore mud follows. 3. The siren rule gives a warning. Cross-domain check: 7×6=42 km.

Answer. The field becomes muddy, and the warning follows from the working siren. Cross-domain answer: 42 km.

Knowledge takeaway. Causal reasoning becomes clearer when every link is stated and the direction of implication is respected.

Formal model. Forward-chaining rule system over Boolean facts; apply implications until no new facts appear.

Problem 825. Cause-and-consequence chain: case 5

Knowledge context. Many explanations can be represented as condition-and-consequence rules. The logical form matters: necessary, sufficient, AND, and OR are not interchangeable.

Given facts. Facts: Rain occurred; The river was already high. Rules: If rain AND high river, flooding occurs. If flooding, road closes. If road closes, the bus reroutes. Cross-domain check: a survey team starts work at 8:00 and works for 2 hour(s) without a break.

Rules. Use the rules exactly as written. AND requires all listed conditions; OR requires at least one. A one-way implication cannot automatically be reversed. For the cross-domain check, finishing time = starting time + duration.

Task. Which consequences can be deduced by applying the rules exactly? Do not assume the converse of any rule. Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. Rain and high river jointly imply flooding. 2. Flooding implies road closure. 3. Road closure implies bus rerouting. Cross-domain check: 8+2=10, so the team finishes at 10:00.

Answer. Flooding, road closure, and bus rerouting can all be deduced. Cross-domain answer: 10:00.

Knowledge takeaway. Causal reasoning becomes clearer when every link is stated and the direction of implication is respected.

Formal model. Forward-chaining rule system over Boolean facts; apply implications until no new facts appear.

H6. Travel through time and space

Problem 826. Courier journey from Oak to Zephyr

Knowledge context. Historical travel can be analyzed with ordinary distance, speed, and time reasoning when the needed values are supplied.

Given facts. A courier leaves Oak for Zephyr at 7:00. Distance is 120 km and the stated average speed is 30 km/h. A council meeting in Zephyr begins at 10.0:00. Ignore stops and use travel time = distance ÷ speed.

Rules. Arrival time equals departure time plus travel time. To attend an event from its beginning, arrival must be no later than the event start.

Task. When does the courier arrive, and can the courier attend the meeting from its beginning?

Step-by-step solution. 1. Travel time=120÷30=4.0 hours. 2. Arrival=7.0+4.0=11.0. 3. Compare 11.0 with meeting time 10.0.

Answer. Arrival at about 11.0:00; no, the meeting has already begun.

Knowledge takeaway. Chronology and geography interact: being able to reach a place in time is a constraint, not a fact to memorize.

Formal model. Arithmetic time model; t_arrive=t_depart+d/v; compare with event time.

Problem 827. Courier journey from Juniper to Yarrow

Knowledge context. Historical travel can be analyzed with ordinary distance, speed, and time reasoning when the needed values are supplied.

Given facts. A courier leaves Juniper for Yarrow at 8:00. Distance is 130 km and the stated average speed is 30 km/h. A council meeting in Yarrow begins at 13.3:00. Ignore stops and use travel time = distance ÷ speed.

Rules. Arrival time equals departure time plus travel time. To attend an event from its beginning, arrival must be no later than the event start.

Task. When does the courier arrive, and can the courier attend the meeting from its beginning?

Step-by-step solution. 1. Travel time=130÷30=4.3 hours. 2. Arrival=8.0+4.3=12.3. 3. Compare 12.3 with meeting time 13.3.

Answer. Arrival at about 12.3:00; yes, on time.

Knowledge takeaway. Chronology and geography interact: being able to reach a place in time is a constraint, not a fact to memorize.

Formal model. Arithmetic time model; t_arrive=t_depart+d/v; compare with event time.

Problem 828. Courier journey from Yarrow to Willow

Knowledge context. Historical travel can be analyzed with ordinary distance, speed, and time reasoning when the needed values are supplied.

Given facts. A courier leaves Yarrow for Willow at 9:00. Distance is 140 km and the stated average speed is 30 km/h. A council meeting in Willow begins at 12.7:00. Ignore stops and use travel time = distance ÷ speed.

Rules. Arrival time equals departure time plus travel time. To attend an event from its beginning, arrival must be no later than the event start.

Task. When does the courier arrive, and can the courier attend the meeting from its beginning?

Step-by-step solution. 1. Travel time=140÷30=4.7 hours. 2. Arrival=9.0+4.7=13.7. 3. Compare 13.7 with meeting time 12.7.

Answer. Arrival at about 13.7:00; no, the meeting has already begun.

Knowledge takeaway. Chronology and geography interact: being able to reach a place in time is a constraint, not a fact to memorize.

Formal model. Arithmetic time model; t_arrive=t_depart+d/v; compare with event time.

Problem 829. Courier journey from Dover to Cedar

Knowledge context. Historical travel can be analyzed with ordinary distance, speed, and time reasoning when the needed values are supplied.

Given facts. A courier leaves Dover for Cedar at 10:00. Distance is 150 km and the stated average speed is 30 km/h. A council meeting in Cedar begins at 16.0:00. Ignore stops and use travel time = distance ÷ speed.

Rules. Arrival time equals departure time plus travel time. To attend an event from its beginning, arrival must be no later than the event start.

Task. When does the courier arrive, and can the courier attend the meeting from its beginning?

Step-by-step solution. 1. Travel time=150÷30=5.0 hours. 2. Arrival=10.0+5.0=15.0. 3. Compare 15.0 with meeting time 16.0.

Answer. Arrival at about 15.0:00; yes, on time.

Knowledge takeaway. Chronology and geography interact: being able to reach a place in time is a constraint, not a fact to memorize.

Formal model. Arithmetic time model; t_arrive=t_depart+d/v; compare with event time.

Problem 830. Courier journey from Juniper to Birch

Knowledge context. Historical travel can be analyzed with ordinary distance, speed, and time reasoning when the needed values are supplied.

Given facts. A courier leaves Juniper for Birch at 11:00. Distance is 160 km and the stated average speed is 30 km/h. A council meeting in Birch begins at 15.3:00. Ignore stops and use travel time = distance ÷ speed.

Rules. Arrival time equals departure time plus travel time. To attend an event from its beginning, arrival must be no later than the event start.

Task. When does the courier arrive, and can the courier attend the meeting from its beginning?

Step-by-step solution. 1. Travel time=160÷30=5.3 hours. 2. Arrival=11.0+5.3=16.3. 3. Compare 16.3 with meeting time 15.3.

Answer. Arrival at about 16.3:00; no, the meeting has already begun.

Knowledge takeaway. Chronology and geography interact: being able to reach a place in time is a constraint, not a fact to memorize.

Formal model. Arithmetic time model; t_arrive=t_depart+d/v; compare with event time.

H7. Change and continuity

Problem 831. Change and continuity in a settlement: case 1

Knowledge context. History studies both change and continuity. Comparing two snapshots is a systematic way to separate what persisted from what appeared or disappeared.

Given facts. Period 1 features: footpath, market, well, wooden bridge. Period 2 features: market, road, stone bridge, well. Cross-domain check: researchers collected 7 reports, but 4 are exact duplicate copies that add no new independent information. Mixed-domain verification: a regional archive has 85 map sheets and receives 2 additional sheets during the exercise.

Rules. A continuity is a feature present in both periods. A disappearance is present only in Period 1. An appearance is present only in Period 2. A replacement can be described when two differently named features serve the same stated role. For the cross-domain check, independent reports = total reports − duplicate copies. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. Identify the continuities, disappearances, and appearances. Which conclusions are direct set comparisons rather than interpretations? Cross-domain check: how many non-duplicate reports remain? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. Intersection gives continuities: ['market', 'well']. 2. Period1−Period2 gives disappearances: ['footpath', 'wooden bridge']. 3. Period2−Period1 gives appearances: ['road', 'stone bridge']. 4. Only role-based replacement would require extra information; the set comparisons are direct. Cross-domain check: 7−4=3 independent reports remain. Mixed-domain verification: 85+2=87.

Answer. Continuities: ['market', 'well']; disappearances: ['footpath', 'wooden bridge']; appearances: ['road', 'stone bridge']. Cross-domain answer: 3 independent reports. Mixed-domain answer: 87 map sheets.

Knowledge takeaway. Change is not the opposite of continuity; both can occur in the same community at the same time.

Formal model. Set operations: intersection for continuity, set difference for appearance/disappearance.

Problem 832. Change and continuity in a settlement: case 2

Knowledge context. History studies both change and continuity. Comparing two snapshots is a systematic way to separate what persisted from what appeared or disappeared.

Given facts. Period 1 features: footpath, market, well, wooden bridge. Period 2 features: market, road, school, stone bridge, well. Cross-domain check: on a field sketch, 1 cm represents 4 km and two checkpoints are 8 cm apart.

Rules. A continuity is a feature present in both periods. A disappearance is present only in Period 1. An appearance is present only in Period 2. A replacement can be described when two differently named features serve the same stated role. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Identify the continuities, disappearances, and appearances. Which conclusions are direct set comparisons rather than interpretations? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. Intersection gives continuities: ['market', 'well']. 2. Period1−Period2 gives disappearances: ['footpath', 'wooden bridge']. 3. Period2−Period1 gives appearances: ['road', 'school', 'stone bridge']. 4. Only role-based replacement would require extra information; the set comparisons are direct. Cross-domain check: 8×4=32 km.

Answer. Continuities: ['market', 'well']; disappearances: ['footpath', 'wooden bridge']; appearances: ['road', 'school', 'stone bridge']. Cross-domain answer: 32 km.

Knowledge takeaway. Change is not the opposite of continuity; both can occur in the same community at the same time.

Formal model. Set operations: intersection for continuity, set difference for appearance/disappearance.

Problem 833. Change and continuity in a settlement: case 3

Knowledge context. History studies both change and continuity. Comparing two snapshots is a systematic way to separate what persisted from what appeared or disappeared.

Given facts. Period 1 features: footpath, market, mill, well, wooden bridge. Period 2 features: market, mill, road, stone bridge, well. Cross-domain check: a survey team starts work at 8:00 and works for 2 hour(s) without a break.

Rules. A continuity is a feature present in both periods. A disappearance is present only in Period 1. An appearance is present only in Period 2. A replacement can be described when two differently named features serve the same stated role. For the cross-domain check, finishing time = starting time + duration.

Task. Identify the continuities, disappearances, and appearances. Which conclusions are direct set comparisons rather than interpretations? Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. Intersection gives continuities: ['market', 'mill', 'well']. 2. Period1−Period2 gives disappearances: ['footpath', 'wooden bridge']. 3. Period2−Period1 gives appearances: ['road', 'stone bridge']. 4. Only role-based replacement would require extra information; the set comparisons are direct. Cross-domain check: 8+2=10, so the team finishes at 10:00.

Answer. Continuities: ['market', 'mill', 'well']; disappearances: ['footpath', 'wooden bridge']; appearances: ['road', 'stone bridge']. Cross-domain answer: 10:00.

Knowledge takeaway. Change is not the opposite of continuity; both can occur in the same community at the same time.

Formal model. Set operations: intersection for continuity, set difference for appearance/disappearance.

Problem 834. Change and continuity in a settlement: case 4

Knowledge context. History studies both change and continuity. Comparing two snapshots is a systematic way to separate what persisted from what appeared or disappeared.

Given facts. Period 1 features: footpath, market, well, wooden bridge. Period 2 features: market, road, stone bridge. Cross-domain check: a local committee has 14 members, requires at least 8 present for quorum, and 14 are present.

Rules. A continuity is a feature present in both periods. A disappearance is present only in Period 1. An appearance is present only in Period 2. A replacement can be described when two differently named features serve the same stated role. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. Identify the continuities, disappearances, and appearances. Which conclusions are direct set comparisons rather than interpretations? Cross-domain check: is quorum met?

Step-by-step solution. 1. Intersection gives continuities: ['market']. 2. Period1−Period2 gives disappearances: ['footpath', 'well', 'wooden bridge']. 3. Period2−Period1 gives appearances: ['road', 'stone bridge']. 4. Only role-based replacement would require extra information; the set comparisons are direct. Cross-domain check: compare 14 with 8; 14 ≥ 8 is true.

Answer. Continuities: ['market']; disappearances: ['footpath', 'well', 'wooden bridge']; appearances: ['road', 'stone bridge']. Cross-domain answer: quorum is met.

Knowledge takeaway. Change is not the opposite of continuity; both can occur in the same community at the same time.

Formal model. Set operations: intersection for continuity, set difference for appearance/disappearance.

Problem 835. Change and continuity in a settlement: case 5

Knowledge context. History studies both change and continuity. Comparing two snapshots is a systematic way to separate what persisted from what appeared or disappeared.

Given facts. Period 1 features: footpath, market, wall, well, wooden bridge. Period 2 features: market, park, road, stone bridge, well. Cross-domain check: researchers collected 11 reports, but 4 are exact duplicate copies that add no new independent information. Mixed-domain verification: a regional archive has 85 map sheets and receives 6 additional sheets during the exercise.

Rules. A continuity is a feature present in both periods. A disappearance is present only in Period 1. An appearance is present only in Period 2. A replacement can be described when two differently named features serve the same stated role. For the cross-domain check, independent reports = total reports − duplicate copies. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. Identify the continuities, disappearances, and appearances. Which conclusions are direct set comparisons rather than interpretations? Cross-domain check: how many non-duplicate reports remain? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. Intersection gives continuities: ['market', 'well']. 2. Period1−Period2 gives disappearances: ['footpath', 'wall', 'wooden bridge']. 3. Period2−Period1 gives appearances: ['park', 'road', 'stone bridge']. 4. Only role-based replacement would require extra information; the set comparisons are direct. Cross-domain check: 11−4=7 independent reports remain. Mixed-domain verification: 85+6=91.

Answer. Continuities: ['market', 'well']; disappearances: ['footpath', 'wall', 'wooden bridge']; appearances: ['park', 'road', 'stone bridge']. Cross-domain answer: 7 independent reports. Mixed-domain answer: 91 map sheets.

Knowledge takeaway. Change is not the opposite of continuity; both can occur in the same community at the same time.

Formal model. Set operations: intersection for continuity, set difference for appearance/disappearance.

H8. Layers and dating objects

Problem 836. Reading an archaeological sequence: case 1

Knowledge context. Archaeological layers provide relative chronological information. Dates from objects add constraints but do not automatically make every date exact.

Given facts. An undisturbed archaeological site has four layers from top to bottom: A, B, C, D. In an undisturbed sequence, lower layers were deposited earlier. D contains an object dated 1300, C contains one dated 1450, and B contains one dated 1600. Cross-domain check: on a field sketch, 1 cm represents 3 km and two checkpoints are 5 cm apart.

Rules. In an undisturbed sequence, a lower layer is older than a layer above it. A securely dated object provides a constraint on the layer that contains it, but one object does not date every other object exactly. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Which is older, Layer C or Layer B? Can the exact year of deposition of Layer A be deduced? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. C lies below B, so C is older than B by the stratigraphic rule. 2. No exact dated object or deposition year is given for A. 3. Therefore only relative ordering, not an exact year for A, is justified. Cross-domain check: 5×3=15 km.

Answer. Layer C is older than Layer B. The exact deposition year of Layer A cannot be deduced. Cross-domain answer: 15 km.

Knowledge takeaway. Relative dating and absolute dating answer different questions and should not be confused.

Formal model. Partial order from vertical position plus numeric constraints from dated objects.

Problem 837. Reading an archaeological sequence: case 2

Knowledge context. Archaeological layers provide relative chronological information. Dates from objects add constraints but do not automatically make every date exact.

Given facts. An undisturbed archaeological site has four layers from top to bottom: A, B, C, D. In an undisturbed sequence, lower layers were deposited earlier. D contains an object dated 1305, C contains one dated 1465, and B contains one dated 1620. Cross-domain check: a survey team starts work at 12:00 and works for 2 hour(s) without a break.

Rules. In an undisturbed sequence, a lower layer is older than a layer above it. A securely dated object provides a constraint on the layer that contains it, but one object does not date every other object exactly. For the cross-domain check, finishing time = starting time + duration.

Task. Which is older, Layer C or Layer B? Can the exact year of deposition of Layer A be deduced? Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. C lies below B, so C is older than B by the stratigraphic rule. 2. No exact dated object or deposition year is given for A. 3. Therefore only relative ordering, not an exact year for A, is justified. Cross-domain check: 12+2=14, so the team finishes at 14:00.

Answer. Layer C is older than Layer B. The exact deposition year of Layer A cannot be deduced. Cross-domain answer: 14:00.

Knowledge takeaway. Relative dating and absolute dating answer different questions and should not be confused.

Formal model. Partial order from vertical position plus numeric constraints from dated objects.

Problem 838. Reading an archaeological sequence: case 3

Knowledge context. Archaeological layers provide relative chronological information. Dates from objects add constraints but do not automatically make every date exact.

Given facts. An undisturbed archaeological site has four layers from top to bottom: A, B, C, D. In an undisturbed sequence, lower layers were deposited earlier. D contains an object dated 1310, C contains one dated 1480, and B contains one dated 1640. Cross-domain check: a local committee has 9 members, requires at least 5 present for quorum, and 8 are present.

Rules. In an undisturbed sequence, a lower layer is older than a layer above it. A securely dated object provides a constraint on the layer that contains it, but one object does not date every other object exactly. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. Which is older, Layer C or Layer B? Can the exact year of deposition of Layer A be deduced? Cross-domain check: is quorum met?

Step-by-step solution. 1. C lies below B, so C is older than B by the stratigraphic rule. 2. No exact dated object or deposition year is given for A. 3. Therefore only relative ordering, not an exact year for A, is justified. Cross-domain check: compare 8 with 5; $8 \geq 5$ is true.

Answer. Layer C is older than Layer B. The exact deposition year of Layer A cannot be deduced. Cross-domain answer: quorum is met.

Knowledge takeaway. Relative dating and absolute dating answer different questions and should not be confused.

Formal model. Partial order from vertical position plus numeric constraints from dated objects.

Problem 839. Reading an archaeological sequence: case 4

Knowledge context. Archaeological layers provide relative chronological information. Dates from objects add constraints but do not automatically make every date exact.

Given facts. An undisturbed archaeological site has four layers from top to bottom: A, B, C, D. In an undisturbed sequence, lower layers were deposited earlier. D contains an object dated 1315, C contains one dated 1495, and B contains one dated 1660. Cross-domain check: researchers collected 15 reports, but 4 are exact duplicate copies that add no new independent information. Mixed-domain verification: a regional archive has 85 map sheets and receives 10 additional sheets during the exercise.

Rules. In an undisturbed sequence, a lower layer is older than a layer above it. A securely dated object provides a constraint on the layer that contains it, but one object does not date every other object exactly. For the cross-domain check, independent reports = total reports − duplicate copies. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. Which is older, Layer C or Layer B? Can the exact year of deposition of Layer A be deduced? Cross-domain check: how many non-duplicate reports remain? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. C lies below B, so C is older than B by the stratigraphic rule. 2. No exact dated object or deposition year is given for A. 3. Therefore only relative ordering, not an exact year for A, is justified. Cross-domain check: 15−4=11 independent reports remain. Mixed-domain verification: 85+10=95.

Answer. Layer C is older than Layer B. The exact deposition year of Layer A cannot be deduced. Cross-domain answer: 11 independent reports. Mixed-domain answer: 95 map sheets.

Knowledge takeaway. Relative dating and absolute dating answer different questions and should not be confused.

Formal model. Partial order from vertical position plus numeric constraints from dated objects.

Problem 840. Reading an archaeological sequence: case 5

Knowledge context. Archaeological layers provide relative chronological information. Dates from objects add constraints but do not automatically make every date exact.

Given facts. An undisturbed archaeological site has four layers from top to bottom: A, B, C, D. In an undisturbed sequence, lower layers were deposited earlier. D contains an object dated 1320, C contains one dated 1510, and B contains one dated 1680. Cross-domain check: on a field sketch, 1 cm represents 2 km and two checkpoints are 2 cm apart.

Rules. In an undisturbed sequence, a lower layer is older than a layer above it. A securely dated object provides a constraint on the layer that contains it, but one object does not date every other object exactly. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Which is older, Layer C or Layer B? Can the exact year of deposition of Layer A be deduced? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. C lies below B, so C is older than B by the stratigraphic rule. 2. No exact dated object or deposition year is given for A. 3. Therefore only relative ordering, not an exact year for A, is justified. Cross-domain check: 2×2=4 km.

Answer. Layer C is older than Layer B. The exact deposition year of Layer A cannot be deduced. Cross-domain answer: 4 km.

Knowledge takeaway. Relative dating and absolute dating answer different questions and should not be confused.

Formal model. Partial order from vertical position plus numeric constraints from dated objects.

C1. Rules and exceptions

Problem 841. Applying a rule with an exception: Maya

Knowledge context. Rules often have conditions and exceptions. Logical reading prevents us from applying a general rule when an overriding exception is active.

Given facts. Maya is 13 years old. A library rule says: a child may enter the archive room if age $\geq$ 14 AND an adult permit is present. Exception: during an evacuation drill, nobody may enter. Permit present: yes. Evacuation drill: no.

Rules. Check exceptions first if they override the general rule. Otherwise every condition joined by AND must be true.

Task. May Maya enter the archive room under these facts? Show which conditions succeed or fail.

Step-by-step solution. 1. Age test: 13 ≥ 14 is False. 2. Permit test is True. 3. Because the general rule uses AND, both tests must pass. 4. Therefore entry is not allowed.

Answer. No, Maya may not enter.

Knowledge takeaway. The order of reasoning matters when exceptions override ordinary rules.

Formal model. Boolean rule with priority: exception predicate first, then conjunction of eligibility conditions.

Problem 842. Applying a rule with an exception: Noah

Knowledge context. Rules often have conditions and exceptions. Logical reading prevents us from applying a general rule when an overriding exception is active.

Given facts. Noah is 14 years old. A library rule says: a child may enter the archive room if age $\geq$ 14 AND an adult permit is present. Exception: during an evacuation drill, nobody may enter. Permit present: no. Evacuation drill: no.

Rules. Check exceptions first if they override the general rule. Otherwise every condition joined by AND must be true.

Task. May Noah enter the archive room under these facts? Show which conditions succeed or fail.

Step-by-step solution. 1. Age test: 14 ≥ 14 is True. 2. Permit test is False. 3. Because the general rule uses AND, both tests must pass. 4. Therefore entry is not allowed.

Answer. No, Noah may not enter.

Knowledge takeaway. The order of reasoning matters when exceptions override ordinary rules.

Formal model. Boolean rule with priority: exception predicate first, then conjunction of eligibility conditions.

Problem 843. Applying a rule with an exception: Lina

Knowledge context. Rules often have conditions and exceptions. Logical reading prevents us from applying a general rule when an overriding exception is active.

Given facts. Lina is 15 years old. A library rule says: a child may enter the archive room if age $\geq$ 14 AND an adult permit is present. Exception: during an evacuation drill, nobody may enter. Permit present: yes. Evacuation drill: no.

Rules. Check exceptions first if they override the general rule. Otherwise every condition joined by AND must be true.

Task. May Lina enter the archive room under these facts? Show which conditions succeed or fail.

Step-by-step solution. 1. Age test: 15 ≥ 14 is True. 2. Permit test is True. 3. Because the general rule uses AND, both tests must pass. 4. Therefore entry is allowed.

Answer. Yes, Lina may enter.

Knowledge takeaway. The order of reasoning matters when exceptions override ordinary rules.

Formal model. Boolean rule with priority: exception predicate first, then conjunction of eligibility conditions.

Problem 844. Applying a rule with an exception: Owen

Knowledge context. Rules often have conditions and exceptions. Logical reading prevents us from applying a general rule when an overriding exception is active.

Given facts. Owen is 16 years old. A library rule says: a child may enter the archive room if age ≥ 14 AND an adult permit is present. Exception: during an evacuation drill, nobody may enter. Permit present: no. Evacuation drill: no.

Rules. Check exceptions first if they override the general rule. Otherwise every condition joined by AND must be true.

Task. May Owen enter the archive room under these facts? Show which conditions succeed or fail.

Step-by-step solution. 1. Age test: 16 ≥ 14 is True. 2. Permit test is False. 3. Because the general rule uses AND, both tests must pass. 4. Therefore entry is not allowed.

Answer. No, Owen may not enter.

Knowledge takeaway. The order of reasoning matters when exceptions override ordinary rules.

Formal model. Boolean rule with priority: exception predicate first, then conjunction of eligibility conditions.

Problem 845. Applying a rule with an exception: Sara

Knowledge context. Rules often have conditions and exceptions. Logical reading prevents us from applying a general rule when an overriding exception is active.

Given facts. Sara is 17 years old. A library rule says: a child may enter the archive room if age ≥ 14 AND an adult permit is present. Exception: during an evacuation drill, nobody may enter. Permit present: yes. Evacuation drill: yes.

Rules. Check exceptions first if they override the general rule. Otherwise every condition joined by AND must be true.

Task. May Sara enter the archive room under these facts? Show which conditions succeed or fail.

Step-by-step solution. 1. The evacuation exception applies and overrides ordinary eligibility. 2. Therefore entry is not allowed.

Answer. No, Sara may not enter.

Knowledge takeaway. The order of reasoning matters when exceptions override ordinary rules.

Formal model. Boolean rule with priority: exception predicate first, then conjunction of eligibility conditions.

C2. Rights as limits on decisions

Problem 846. Majority rule and a protected right: case 1

Knowledge context. Democratic procedures use voting, but constitutional systems can place some rights beyond ordinary majority power.

Given facts. A council of 21 members votes 16 to 5 for a proposal that would forbid one named group from the right to speak at a peaceful meeting. The classroom constitution states: (1) ordinary proposals pass by simple majority; (2) an ordinary majority may not remove the listed equal rights from a group.

Rules. A decision must satisfy both the voting procedure and the higher rights constraint. Passing a vote is not sufficient if the content violates a protected right.

Task. Did the proposal receive a majority? May it take effect under the stated constitution?

Step-by-step solution. 1. Vote check: 16>5, so the proposal has a simple majority. 2. Content check: the proposal removes a protected equal right (speak at a peaceful meeting) from a group. 3. The higher rights rule therefore blocks the proposal even though the vote passed.

Answer. It received a majority, but it may not take effect under the stated rights rule.

Knowledge takeaway. A majority answers who won a vote; it does not by itself answer whether every possible decision is lawful or fair under higher rules.

Formal model. Two-stage validity test: procedural predicate AND rights-compliance predicate.

Problem 847. Majority rule and a protected right: case 2

Knowledge context. Democratic procedures use voting, but constitutional systems can place some rights beyond ordinary majority power.

Given facts. A council of 23 members votes 17 to 6 for a proposal that would forbid one named group from the right to use the public library equally. The classroom constitution states: (1) ordinary proposals pass by simple majority; (2) an ordinary majority may not remove the listed equal rights from a group.

Rules. A decision must satisfy both the voting procedure and the higher rights constraint. Passing a vote is not sufficient if the content violates a protected right.

Task. Did the proposal receive a majority? May it take effect under the stated constitution?

Step-by-step solution. 1. Vote check: 17>6, so the proposal has a simple majority. 2. Content check: the proposal removes a protected equal right (use the public library equally) from a group. 3. The higher rights rule therefore blocks the proposal even though the vote passed.

Answer. It received a majority, but it may not take effect under the stated rights rule.

Knowledge takeaway. A majority answers who won a vote; it does not by itself answer whether every possible decision is lawful or fair under higher rules.

Formal model. Two-stage validity test: procedural predicate AND rights-compliance predicate.

Problem 848. Majority rule and a protected right: case 3

Knowledge context. Democratic procedures use voting, but constitutional systems can place some rights beyond ordinary majority power.

Given facts. A council of 25 members votes 18 to 7 for a proposal that would forbid one named group from the right to receive a fair hearing. The classroom constitution states: (1) ordinary proposals pass by simple majority; (2) an ordinary majority may not remove the listed equal rights from a group.

Rules. A decision must satisfy both the voting procedure and the higher rights constraint. Passing a vote is not sufficient if the content violates a protected right.

Task. Did the proposal receive a majority? May it take effect under the stated constitution?

Step-by-step solution. 1. Vote check: 18>7, so the proposal has a simple majority. 2. Content check: the proposal removes a protected equal right (receive a fair hearing) from a group. 3. The higher rights rule therefore blocks the proposal even though the vote passed.

Answer. It received a majority, but it may not take effect under the stated rights rule.

Knowledge takeaway. A majority answers who won a vote; it does not by itself answer whether every possible decision is lawful or fair under higher rules.

Formal model. Two-stage validity test: procedural predicate AND rights-compliance predicate.

Problem 849. Majority rule and a protected right: case 4

Knowledge context. Democratic procedures use voting, but constitutional systems can place some rights beyond ordinary majority power.

Given facts. A council of 27 members votes 19 to 8 for a proposal that would forbid one named group from the right to practice a lawful belief. The classroom constitution states: (1) ordinary proposals pass by simple majority; (2) an ordinary majority may not remove the listed equal rights from a group.

Rules. A decision must satisfy both the voting procedure and the higher rights constraint. Passing a vote is not sufficient if the content violates a protected right.

Task. Did the proposal receive a majority? May it take effect under the stated constitution?

Step-by-step solution. 1. Vote check: 19>8, so the proposal has a simple majority. 2. Content check: the proposal removes a protected equal right (practice a lawful belief) from a group. 3. The higher rights rule therefore blocks the proposal even though the vote passed.

Answer. It received a majority, but it may not take effect under the stated rights rule.

Knowledge takeaway. A majority answers who won a vote; it does not by itself answer whether every possible decision is lawful or fair under higher rules.

Formal model. Two-stage validity test: procedural predicate AND rights-compliance predicate.

Problem 850. Majority rule and a protected right: case 5

Knowledge context. Democratic procedures use voting, but constitutional systems can place some rights beyond ordinary majority power.

Given facts. A council of 29 members votes 20 to 9 for a proposal that would forbid one named group from the right to be protected from arbitrary punishment. The classroom constitution states: (1) ordinary proposals pass by simple majority; (2) an ordinary majority may not remove the listed equal rights from a group.

Rules. A decision must satisfy both the voting procedure and the higher rights constraint. Passing a vote is not sufficient if the content violates a protected right.

Task. Did the proposal receive a majority? May it take effect under the stated constitution?

Step-by-step solution. 1. Vote check: 20>9, so the proposal has a simple majority. 2. Content check: the proposal removes a protected equal right (be protected from arbitrary punishment) from a group. 3. The higher rights rule therefore blocks the proposal even though the vote passed.

Answer. It received a majority, but it may not take effect under the stated rights rule.

Knowledge takeaway. A majority answers who won a vote; it does not by itself answer whether every possible decision is lawful or fair under higher rules.

Formal model. Two-stage validity test: procedural predicate AND rights-compliance predicate.

C3. Voting, majority, and quorum

Problem 851. Quorum and majority in a 23-member committee

Knowledge context. Voting rules often contain more than one threshold. Quorum concerns participation; majority concerns the votes cast according to the stated rule.

Given facts. A committee has 23 members. 23 are present. Votes: yes=11, no=11, abstain=1. Rule: quorum is at least 60% of all members; if quorum exists, a motion passes when yes votes are strictly more than no votes.

Rules. First test quorum using all members as the denominator. Only then compare yes and no; abstentions count for presence but not as no votes under this rule.

Task. Is quorum met, and does the motion pass?

Step-by-step solution. 1. Quorum threshold=ceil(0.60×23)=14. 2. Present=23, so quorum is met. 3. Yes=11, no=11; yes>no is False. 4. Combine the two tests.

Answer. Quorum is met; the motion does not pass.

Knowledge takeaway. Procedural reasoning requires applying tests in the right order and using the correct denominator.

Formal model. Threshold predicates: quorum(present,total) and majority(yes,no); pass = quorum $\wedge$ majority.

Problem 852. Quorum and majority in a 24-member committee

Knowledge context. Voting rules often contain more than one threshold. Quorum concerns participation; majority concerns the votes cast according to the stated rule.

Given facts. A committee has 24 members. 23 are present. Votes: yes=12, no=10, abstain=1. Rule: quorum is at least 60% of all members; if quorum exists, a motion passes when yes votes are strictly more than no votes.

Rules. First test quorum using all members as the denominator. Only then compare yes and no; abstentions count for presence but not as no votes under this rule.

Task. Is quorum met, and does the motion pass?

Step-by-step solution. 1. Quorum threshold=ceil(0.60×24)=15. 2. Present=23, so quorum is met. 3. Yes=12, no=10; yes>no is True. 4. Combine the two tests.

Answer. Quorum is met; the motion passes.

Knowledge takeaway. Procedural reasoning requires applying tests in the right order and using the correct denominator.

Formal model. Threshold predicates: quorum(present,total) and majority(yes,no); pass = quorum $\wedge$ majority.

Problem 853. Quorum and majority in a 25-member committee

Knowledge context. Voting rules often contain more than one threshold. Quorum concerns participation; majority concerns the votes cast according to the stated rule.

Given facts. A committee has 25 members. 23 are present. Votes: yes=13, no=9, abstain=1. Rule: quorum is at least 60% of all members; if quorum exists, a motion passes when yes votes are strictly more than no votes.

Rules. First test quorum using all members as the denominator. Only then compare yes and no; abstentions count for presence but not as no votes under this rule.

Task. Is quorum met, and does the motion pass?

Step-by-step solution. 1. Quorum threshold=ceil(0.60×25)=15. 2. Present=23, so quorum is met. 3. Yes=13, no=9; yes>no is True. 4. Combine the two tests.

Answer. Quorum is met; the motion passes.

Knowledge takeaway. Procedural reasoning requires applying tests in the right order and using the correct denominator.

Formal model. Threshold predicates: quorum(present,total) and majority(yes,no); pass = quorum ∧ majority.

Problem 854. Quorum and majority in a 26-member committee

Knowledge context. Voting rules often contain more than one threshold. Quorum concerns participation; majority concerns the votes cast according to the stated rule.

Given facts. A committee has 26 members. 26 are present. Votes: yes=14, no=11, abstain=1. Rule: quorum is at least 60% of all members; if quorum exists, a motion passes when yes votes are strictly more than no votes.

Rules. First test quorum using all members as the denominator. Only then compare yes and no; abstentions count for presence but not as no votes under this rule.

Task. Is quorum met, and does the motion pass?

Step-by-step solution. 1. Quorum threshold=ceil(0.60×26)=16. 2. Present=26, so quorum is met. 3. Yes=14, no=11; yes>no is True. 4. Combine the two tests.

Answer. Quorum is met; the motion passes.

Knowledge takeaway. Procedural reasoning requires applying tests in the right order and using the correct denominator.

Formal model. Threshold predicates: quorum(present,total) and majority(yes,no); pass = quorum ∧ majority.

Problem 855. Quorum and majority in a 27-member committee

Knowledge context. Voting rules often contain more than one threshold. Quorum concerns participation; majority concerns the votes cast according to the stated rule.

Given facts. A committee has 27 members. 26 are present. Votes: yes=15, no=10, abstain=1. Rule: quorum is at least 60% of all members; if quorum exists, a motion passes when yes votes are strictly more than no votes.

Rules. First test quorum using all members as the denominator. Only then compare yes and no; abstentions count for presence but not as no votes under this rule.

Task. Is quorum met, and does the motion pass?

Step-by-step solution. 1. Quorum threshold=ceil(0.60×27)=17. 2. Present=26, so quorum is met. 3. Yes=15, no=10; yes>no is True. 4. Combine the two tests.

Answer. Quorum is met; the motion passes.

Knowledge takeaway. Procedural reasoning requires applying tests in the right order and using the correct denominator.

Formal model. Threshold predicates: quorum(present,total) and majority(yes,no); pass = quorum ∧ majority.

C4. Representation and seats

Problem 856. Representing three districts: case 1

Knowledge context. Representation can be modeled by rules that convert population into seats. Different allocation rules can produce different results, so the rule must be stated.

Given facts. Three districts have populations A=120, B=120, C=60. They must share 14 seats. Use this stated method: compute each exact quota population/total×seats; give each district the whole-number part; assign remaining seats to the largest fractional remainders.

Rules. Use only the stated largest-remainder method. Seat totals must add exactly to the available number.

Task. How many seats does each district receive?

Step-by-step solution. 1. Total population=300. 2. Exact quotas: A=5.60, B=5.60, C=2.80. 3. Initial whole seats: [5, 5, 2], leaving 2. 4. Give the remaining seat(s) to the largest remainder(s). 5. Final allocation=[5, 6, 3], which sums to 14.

Answer. A=5, B=6, C=3 seats.

Knowledge takeaway. Fair representation is not just arithmetic; it depends on an explicit apportionment procedure.

Formal model. Largest-remainder apportionment: quotas, floors, then rank fractional remainders.

Problem 857. Representing three districts: case 2

Knowledge context. Representation can be modeled by rules that convert population into seats. Different allocation rules can produce different results, so the rule must be stated.

Given facts. Three districts have populations A=140, B=120, C=75. They must share 14 seats. Use this stated method: compute each exact quota population/total×seats; give each district the whole-number part; assign remaining seats to the largest fractional remainders.

Rules. Use only the stated largest-remainder method. Seat totals must add exactly to the available number.

Task. How many seats does each district receive?

Step-by-step solution. 1. Total population=335. 2. Exact quotas: A=5.85, B=5.01, C=3.13. 3. Initial whole seats: [5, 5, 3], leaving 1. 4. Give the remaining seat(s) to the largest remainder(s). 5. Final allocation=[6, 5, 3], which sums to 14.

Answer. A=6, B=5, C=3 seats.

Knowledge takeaway. Fair representation is not just arithmetic; it depends on an explicit apportionment procedure.

Formal model. Largest-remainder apportionment: quotas, floors, then rank fractional remainders.

Problem 858. Representing three districts: case 3

Knowledge context. Representation can be modeled by rules that convert population into seats. Different allocation rules can produce different results, so the rule must be stated.

Given facts. Three districts have populations A=160, B=120, C=90. They must share 14 seats. Use this stated method: compute each exact quota population/total×seats; give each district the whole-number part; assign remaining seats to the largest fractional remainders.

Rules. Use only the stated largest-remainder method. Seat totals must add exactly to the available number.

Task. How many seats does each district receive?

Step-by-step solution. 1. Total population=370. 2. Exact quotas: A=6.05, B=4.54, C=3.41. 3. Initial whole seats: [6, 4, 3], leaving 1. 4. Give the remaining seat(s) to the largest remainder(s). 5. Final allocation=[6, 5, 3], which sums to 14.

Answer. A=6, B=5, C=3 seats.

Knowledge takeaway. Fair representation is not just arithmetic; it depends on an explicit apportionment procedure.

Formal model. Largest-remainder apportionment: quotas, floors, then rank fractional remainders.

Problem 859. Representing three districts: case 4

Knowledge context. Representation can be modeled by rules that convert population into seats. Different allocation rules can produce different results, so the rule must be stated.

Given facts. Three districts have populations A=180, B=120, C=105. They must share 14 seats. Use this stated method: compute each exact quota population/total×seats; give each district the whole-number part; assign remaining seats to the largest fractional remainders.

Rules. Use only the stated largest-remainder method. Seat totals must add exactly to the available number.

Task. How many seats does each district receive?

Step-by-step solution. 1. Total population=405. 2. Exact quotas: A=6.22, B=4.15, C=3.63. 3. Initial whole seats: [6, 4, 3], leaving 1. 4. Give the remaining seat(s) to the largest remainder(s). 5. Final allocation=[6, 4, 4], which sums to 14.

Answer. A=6, B=4, C=4 seats.

Knowledge takeaway. Fair representation is not just arithmetic; it depends on an explicit apportionment procedure.

Formal model. Largest-remainder apportionment: quotas, floors, then rank fractional remainders.

Problem 860. Representing three districts: case 5

Knowledge context. Representation can be modeled by rules that convert population into seats. Different allocation rules can produce different results, so the rule must be stated.

Given facts. Three districts have populations A=200, B=120, C=120. They must share 14 seats. Use this stated method: compute each exact quota population/total×seats; give each district the whole-number part; assign remaining seats to the largest fractional remainders.

Rules. Use only the stated largest-remainder method. Seat totals must add exactly to the available number.

Task. How many seats does each district receive?

Step-by-step solution. 1. Total population=440. 2. Exact quotas: A=6.36, B=3.82, C=3.82. 3. Initial whole seats: [6, 3, 3], leaving 2. 4. Give the remaining seat(s) to the largest remainder(s). 5. Final allocation=[6, 4, 4], which sums to 14.

Answer. A=6, B=4, C=4 seats.

Knowledge takeaway. Fair representation is not just arithmetic; it depends on an explicit apportionment procedure.

Formal model. Largest-remainder apportionment: quotas, floors, then rank fractional remainders.

C5. Institutions and roles

Problem 861. Which institution has the role? Repair a broken public road

Knowledge context. Civic institutions divide responsibilities. A problem can be solved by matching an action to a clearly defined function.

Given facts. Roles are defined as follows: Town Council makes local policy and approves the budget; Public Works Office maintains roads; Local Court decides legal disputes; Election Committee administers elections; School Board sets school policies. A resident asks which institution should repair a broken public road.

Rules. Match the requested action to the stated role. Do not choose an institution merely because it is powerful or familiar.

Task. Should the request go first to the Public Works Office or the Election Committee? Why?

Step-by-step solution. 1. Identify the action: repair a broken public road. 2. Find the role whose definition contains that function: Public Works Office. 3. Election Committee has a different stated role, so it is not the first match.

Answer. The request should go first to the Public Works Office.

Knowledge takeaway. Institutional reasoning is role matching, not guessing which body is “most important.”

Formal model. Bipartite relation between actions and authorized institutional roles.

Problem 862. Which institution has the role? Decide a dispute under law

Knowledge context. Civic institutions divide responsibilities. A problem can be solved by matching an action to a clearly defined function.

Given facts. Roles are defined as follows: Town Council makes local policy and approves the budget; Public Works Office maintains roads; Local Court decides legal disputes; Election Committee administers elections; School Board sets school policies. A resident asks which institution should decide a dispute under law. Cross-domain check: a local committee has 15 members, requires at least 8 present for quorum, and 14 are present.

Rules. Match the requested action to the stated role. Do not choose an institution merely because it is powerful or familiar. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. Should the request go first to the Local Court or the School Board? Why? Cross-domain check: is quorum met?

Step-by-step solution. 1. Identify the action: decide a dispute under law. 2. Find the role whose definition contains that function: Local Court. 3. School Board has a different stated role, so it is not the first match. Cross-domain check: compare 14 with 8; $14 \geq 8$ is true.

Answer. The request should go first to the Local Court. Cross-domain answer: quorum is met.

Knowledge takeaway. Institutional reasoning is role matching, not guessing which body is “most important.”

Formal model. Bipartite relation between actions and authorized institutional roles.

Problem 863. Which institution has the role? Approve the town budget

Knowledge context. Civic institutions divide responsibilities. A problem can be solved by matching an action to a clearly defined function.

Given facts. Roles are defined as follows: Town Council makes local policy and approves the budget; Public Works Office maintains roads; Local Court decides legal disputes; Election Committee administers elections; School Board sets school policies. A resident asks which institution should approve the town budget.

Rules. Match the requested action to the stated role. Do not choose an institution merely because it is powerful or familiar.

Task. Should the request go first to the Town Council or the Election Committee? Why?

Step-by-step solution. 1. Identify the action: approve the town budget. 2. Find the role whose definition contains that function: Town Council. 3. Election Committee has a different stated role, so it is not the first match.

Answer. The request should go first to the Town Council.

Knowledge takeaway. Institutional reasoning is role matching, not guessing which body is “most important.”

Formal model. Bipartite relation between actions and authorized institutional roles.

Problem 864. Which institution has the role? Organize the ballot count

Knowledge context. Civic institutions divide responsibilities. A problem can be solved by matching an action to a clearly defined function.

Given facts. Roles are defined as follows: Town Council makes local policy and approves the budget; Public Works Office maintains roads; Local Court decides legal disputes; Election Committee administers elections; School Board sets school policies. A resident asks which institution should organize the ballot count.

Rules. Match the requested action to the stated role. Do not choose an institution merely because it is powerful or familiar.

Task. Should the request go first to the Election Committee or the Local Court? Why?

Step-by-step solution. 1. Identify the action: organize the ballot count. 2. Find the role whose definition contains that function: Election Committee. 3. Local Court has a different stated role, so it is not the first match.

Answer. The request should go first to the Election Committee.

Knowledge takeaway. Institutional reasoning is role matching, not guessing which body is “most important.”

Formal model. Bipartite relation between actions and authorized institutional roles.

Problem 865. Which institution has the role? Set school calendar rules

Knowledge context. Civic institutions divide responsibilities. A problem can be solved by matching an action to a clearly defined function.

Given facts. Roles are defined as follows: Town Council makes local policy and approves the budget; Public Works Office maintains roads; Local Court decides legal disputes; Election Committee administers elections; School Board sets school policies. A resident asks which institution should set school calendar rules. Cross-domain check: a survey team starts work at 8:00 and works for 2 hour(s) without a break.

Rules. Match the requested action to the stated role. Do not choose an institution merely because it is powerful or familiar. For the cross-domain check, finishing time = starting time + duration.

Task. Should the request go first to the School Board or the Town Council? Why? Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. Identify the action: set school calendar rules. 2. Find the role whose definition contains that function: School Board. 3. Town Council has a different stated role, so it is not the first match. Cross-domain check: 8+2=10, so the team finishes at 10:00.

Answer. The request should go first to the School Board. Cross-domain answer: 10:00.

Knowledge takeaway. Institutional reasoning is role matching, not guessing which body is “most important.”

Formal model. Bipartite relation between actions and authorized institutional roles.

C6. Public budgets and priorities

Problem 866. Choosing a public budget package: case 1

Knowledge context. Public budgets require choices under limited resources. A transparent model separates costs, benefits, and mandatory obligations.

Given facts. A town has a budget of 18 units. Projects (cost, benefit): park=(4,5); bridge repair=(6,9); library books=(3,4); drainage=(5,8). Cross-domain check: a local committee has 10 members, requires at least 6 present for quorum, and 8 are present.

Rules. Choose a subset whose total cost does not exceed the budget. Maximize total benefit. If a mandatory project is stated, every feasible plan must include it. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. Which set of projects gives the greatest total benefit while satisfying every constraint? Cross-domain check: is quorum met?

Step-by-step solution. 1. Budget limit is 18. 2. Evaluate feasible combinations (and keep the mandatory project if any). 3. The best feasible combination has cost 18 and benefit 26: ['park', 'bridge repair', 'library books', 'drainage']. 4. No higher-benefit feasible combination exists under the stated costs. Cross-domain check: compare 8 with 6; $8 \geq 6$ is true.

Answer. Choose park, bridge repair, library books, drainage; cost 18, benefit 26. Cross-domain answer: quorum is met.

Knowledge takeaway. Optimization means comparing feasible alternatives under an explicit objective, not simply choosing the cheapest item.

Formal model. 0/1 knapsack with optional mandatory-item constraint; maximize benefit subject to total cost $\leq$ budget.

Problem 867. Choosing a public budget package: case 2

Knowledge context. Public budgets require choices under limited resources. A transparent model separates costs, benefits, and mandatory obligations.

Given facts. A town has a budget of 19 units. Projects (cost, benefit): tree planting=(5,7); bridge repair=(6,9); library books=(3,4); drainage=(5,8). Cross-domain check: researchers collected 13 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. Choose a subset whose total cost does not exceed the budget. Maximize total benefit. If a mandatory project is stated, every feasible plan must include it. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. Which set of projects gives the greatest total benefit while satisfying every constraint? Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. Budget limit is 19. 2. Evaluate feasible combinations (and keep the mandatory project if any). 3. The best feasible combination has cost 19 and benefit 28: ['tree planting', 'bridge repair', 'library books', 'drainage']. 4. No

higher-benefit feasible combination exists under the stated costs. Cross-domain check: 13−4=9 independent reports remain.

Answer. Choose tree planting, bridge repair, library books, drainage; cost 19, benefit 28. Cross-domain answer: 9 independent reports.

Knowledge takeaway. Optimization means comparing feasible alternatives under an explicit objective, not simply choosing the cheapest item.

Formal model. 0/1 knapsack with optional mandatory-item constraint; maximize benefit subject to total cost ≤ budget.

Problem 868. Choosing a public budget package: case 3

Knowledge context. Public budgets require choices under limited resources. A transparent model separates costs, benefits, and mandatory obligations.

Given facts. A town has a budget of 20 units. Projects (cost, benefit): park=(4,5); bridge repair=(6,9); library books=(3,4); drainage=(5,8).

Rules. Choose a subset whose total cost does not exceed the budget. Maximize total benefit. If a mandatory project is stated, every feasible plan must include it.

Task. Which set of projects gives the greatest total benefit while satisfying every constraint?

Step-by-step solution. 1. Budget limit is 20. 2. Evaluate feasible combinations (and keep the mandatory project if any). 3. The best feasible combination has cost 18 and benefit 26: ['park', 'bridge repair', 'library books', 'drainage']. 4. No higher-benefit feasible combination exists under the stated costs.

Answer. Choose park, bridge repair, library books, drainage; cost 18, benefit 26.

Knowledge takeaway. Optimization means comparing feasible alternatives under an explicit objective, not simply choosing the cheapest item.

Formal model. 0/1 knapsack with optional mandatory-item constraint; maximize benefit subject to total cost ≤ budget.

Problem 869. Choosing a public budget package: case 4

Knowledge context. Public budgets require choices under limited resources. A transparent model separates costs, benefits, and mandatory obligations.

Given facts. A town has a budget of 21 units. Projects (cost, benefit): tree planting=(5,7); bridge repair=(6,9); library books=(3,4); drainage=(5,8).

Rules. Choose a subset whose total cost does not exceed the budget. Maximize total benefit. If a mandatory project is stated, every feasible plan must include it.

Task. Which set of projects gives the greatest total benefit while satisfying every constraint?

Step-by-step solution. 1. Budget limit is 21. 2. Evaluate feasible combinations (and keep the mandatory project if any). 3. The best feasible combination has cost 19 and benefit 28: ['tree planting', 'bridge repair', 'library books', 'drainage']. 4. No higher-benefit feasible combination exists under the stated costs.

Answer. Choose tree planting, bridge repair, library books, drainage; cost 19, benefit 28.

Knowledge takeaway. Optimization means comparing feasible alternatives under an explicit objective, not simply choosing the cheapest item.

Formal model. 0/1 knapsack with optional mandatory-item constraint; maximize benefit subject to total cost ≤ budget.

Problem 870. Choosing a public budget package: case 5

Knowledge context. Public budgets require choices under limited resources. A transparent model separates costs, benefits, and mandatory obligations.

Given facts. A town has a budget of 22 units. Projects (cost, benefit): park=(4,5); bridge repair=(6,9); library books=(3,4); drainage=(5,8). Safety rule: bridge repair must be included.

Rules. Choose a subset whose total cost does not exceed the budget. Maximize total benefit. If a mandatory project is stated, every feasible plan must include it.

Task. Which set of projects gives the greatest total benefit while satisfying every constraint?

Step-by-step solution. 1. Budget limit is 22. 2. Evaluate feasible combinations (and keep the mandatory project if any). 3. The best feasible combination has cost 18 and benefit 26: ['park', 'bridge repair', 'library books', 'drainage']. 4. No higher-benefit feasible combination exists under the stated costs.

Answer. Choose park, bridge repair, library books, drainage; cost 18, benefit 26.

Knowledge takeaway. Optimization means comparing feasible alternatives under an explicit objective, not simply choosing the cheapest item.

Formal model. 0/1 knapsack with optional mandatory-item constraint; maximize benefit subject to total cost ≤ budget.

C7. Collective choice and preferences

Problem 871. Pairwise collective choice: Park versus Library

Knowledge context. Group preferences can be aggregated in different ways. Pairwise majority asks a specific question about two options at a time.

Given facts. Three public projects are considered: Park, Library, Sports Hall. Voter groups and rankings: 4 voters rank Park>Library>Sports Hall; 7 voters rank Library>Sports Hall>Park; 2 voters rank Sports Hall>Park>Library. Use pairwise majority: compare two options at a time; each voter supports the one ranked higher.

Rules. For a pairwise contest, count how many voters prefer each of the two options. Ignore the third option for that comparison.

Task. In a pairwise vote between Park and Library, which option wins?

Step-by-step solution. 1. For each group, check whether Park or Library is ranked higher. 2. Totals: Park=6, Library=7. 3. Therefore the result is Library.

Answer. Library wins 7 to 6.

Knowledge takeaway. A collective decision rule must be explicit; different rules can summarize the same rankings differently.

Formal model. Preference rankings; for pair (a,b), sum weights of voters ranking a>b versus b>a.

Problem 872. Pairwise collective choice: Library versus Sports Hall

Knowledge context. Group preferences can be aggregated in different ways. Pairwise majority asks a specific question about two options at a time.

Given facts. Three public projects are considered: Park, Library, Sports Hall. Voter groups and rankings: 5 voters rank Park>Library>Sports Hall; 7 voters rank Library>Sports Hall>Park; 3 voters rank Sports Hall>Park>Library. Use pairwise majority: compare two options at a time; each voter supports the one ranked higher.

Rules. For a pairwise contest, count how many voters prefer each of the two options. Ignore the third option for that comparison.

Task. In a pairwise vote between Library and Sports Hall, which option wins?

Step-by-step solution. 1. For each group, check whether Library or Sports Hall is ranked higher. 2. Totals: Library=12, Sports Hall=3. 3. Therefore the result is Library.

Answer. Library wins 12 to 3.

Knowledge takeaway. A collective decision rule must be explicit; different rules can summarize the same rankings differently.

Formal model. Preference rankings; for pair (a,b), sum weights of voters ranking a>b versus b>a.

Problem 873. Pairwise collective choice: Park versus Sports Hall

Knowledge context. Group preferences can be aggregated in different ways. Pairwise majority asks a specific question about two options at a time.

Given facts. Three public projects are considered: Park, Library, Sports Hall. Voter groups and rankings: 6 voters rank Park>Library>Sports Hall; 7 voters rank Library>Sports Hall>Park; 4 voters rank Sports Hall>Park>Library. Use pairwise majority: compare two options at a time; each voter supports the one ranked higher.

Rules. For a pairwise contest, count how many voters prefer each of the two options. Ignore the third option for that comparison.

Task. In a pairwise vote between Park and Sports Hall, which option wins?

Step-by-step solution. 1. For each group, check whether Park or Sports Hall is ranked higher. 2. Totals: Park=6, Sports Hall=11. 3. Therefore the result is Sports Hall.

Answer. Sports Hall wins 11 to 6.

Knowledge takeaway. A collective decision rule must be explicit; different rules can summarize the same rankings differently.

Formal model. Preference rankings; for pair (a,b), sum weights of voters ranking a>b versus b>a.

Problem 874. Pairwise collective choice: Library versus Park

Knowledge context. Group preferences can be aggregated in different ways. Pairwise majority asks a specific question about two options at a time.

Given facts. Three public projects are considered: Park, Library, Sports Hall. Voter groups and rankings: 7 voters rank Park>Library>Sports Hall; 7 voters rank Library>Sports Hall>Park; 5 voters rank Sports Hall>Park>Library. Use pairwise majority: compare two options at a time; each voter supports the one ranked higher.

Rules. For a pairwise contest, count how many voters prefer each of the two options. Ignore the third option for that comparison.

Task. In a pairwise vote between Library and Park, which option wins?

Step-by-step solution. 1. For each group, check whether Library or Park is ranked higher. 2. Totals: Library=7, Park=12. 3. Therefore the result is Park.

Answer. Park wins 12 to 7.

Knowledge takeaway. A collective decision rule must be explicit; different rules can summarize the same rankings differently.

Formal model. Preference rankings; for pair (a,b), sum weights of voters ranking a>b versus b>a.

Problem 875. Pairwise collective choice: Sports Hall versus Park

Knowledge context. Group preferences can be aggregated in different ways. Pairwise majority asks a specific question about two options at a time.

Given facts. Three public projects are considered: Park, Library, Sports Hall. Voter groups and rankings: 8 voters rank Park>Library>Sports Hall; 7 voters rank Library>Sports Hall>Park; 6 voters rank Sports Hall>Park>Library. Use pairwise majority: compare two options at a time; each voter supports the one ranked higher.

Rules. For a pairwise contest, count how many voters prefer each of the two options. Ignore the third option for that comparison.

Task. In a pairwise vote between Sports Hall and Park, which option wins?

Step-by-step solution. 1. For each group, check whether Sports Hall or Park is ranked higher. 2. Totals: Sports Hall=13, Park=8. 3. Therefore the result is Sports Hall.

Answer. Sports Hall wins 13 to 8.

Knowledge takeaway. A collective decision rule must be explicit; different rules can summarize the same rankings differently.

Formal model. Preference rankings; for pair (a,b), sum weights of voters ranking a>b versus b>a.

Part B — New and mixed reasoning domains

N1. Scale, proportion, and distance

Problem 876. Map scale 1:10000

Knowledge context. Map scale links a model to the real world through proportional reasoning.

Given facts. A map has scale 1:10000. The measured map distance between two points is 6 cm. Rule: 1 cm on the map represents 10000 cm in reality; 100,000 cm = 1 km.

Rules. Multiply map length by the scale denominator, then convert units. A scale is a ratio, not an added distance.

Task. What is the real straight-line distance in kilometres?

Step-by-step solution. 1. Real centimetres=6×10000=60000. 2. Convert to kilometres: 60000÷100,000=0.6.

Answer. 0.6 km.

Knowledge takeaway. Scale problems are ratio problems: keep units explicit and conversion becomes mechanical.

Formal model. Proportion d_real=d_map×denominator, followed by unit conversion.

Problem 877. Map scale 1:25000

Knowledge context. Map scale links a model to the real world through proportional reasoning.

Given facts. A map has scale 1:25000. The measured map distance between two points is 7 cm. Rule: 1 cm on the map represents 25000 cm in reality; 100,000 cm = 1 km.

Rules. Multiply map length by the scale denominator, then convert units. A scale is a ratio, not an added distance.

Task. What is the real straight-line distance in kilometres?

Step-by-step solution. 1. Real centimetres=7×25000=175000. 2. Convert to kilometres: 175000÷100,000=1.75.

Answer. 1.75 km.

Knowledge takeaway. Scale problems are ratio problems: keep units explicit and conversion becomes mechanical.

Formal model. Proportion d_real=d_map×denominator, followed by unit conversion.

Problem 878. Map scale 1:50000

Knowledge context. Map scale links a model to the real world through proportional reasoning.

Given facts. A map has scale 1:50000. The measured map distance between two points is 8 cm. Rule: 1 cm on the map represents 50000 cm in reality; 100,000 cm = 1 km.

Rules. Multiply map length by the scale denominator, then convert units. A scale is a ratio, not an added distance.

Task. What is the real straight-line distance in kilometres?

Step-by-step solution. 1. Real centimetres=8×50000=400000. 2. Convert to kilometres: 400000÷100,000=4.

Answer. 4 km.

Knowledge takeaway. Scale problems are ratio problems: keep units explicit and conversion becomes mechanical.

Formal model. Proportion d_real=d_map×denominator, followed by unit conversion.

Problem 879. Map scale 1:100000

Knowledge context. Map scale links a model to the real world through proportional reasoning.

Given facts. A map has scale 1:100000. The measured map distance between two points is 9 cm. Rule: 1 cm on the map represents 100000 cm in reality; 100,000 cm = 1 km.

Rules. Multiply map length by the scale denominator, then convert units. A scale is a ratio, not an added distance.

Task. What is the real straight-line distance in kilometres?

Step-by-step solution. 1. Real centimetres=9×100000=900000. 2. Convert to kilometres: 900000÷100,000=9.

Answer. 9 km.

Knowledge takeaway. Scale problems are ratio problems: keep units explicit and conversion becomes mechanical.

Formal model. Proportion d_real=d_map×denominator, followed by unit conversion.

Problem 880. Map scale 1:200000

Knowledge context. Map scale links a model to the real world through proportional reasoning.

Given facts. A map has scale 1:200000. The measured map distance between two points is 10 cm. Rule: 1 cm on the map represents 200000 cm in reality; 100,000 cm = 1 km.

Rules. Multiply map length by the scale denominator, then convert units. A scale is a ratio, not an added distance.

Task. What is the real straight-line distance in kilometres?

Step-by-step solution. 1. Real centimetres=10×200000=2000000. 2. Convert to kilometres: 2000000÷100,000=20.

Answer. 20 km.

Knowledge takeaway. Scale problems are ratio problems: keep units explicit and conversion becomes mechanical.

Formal model. Proportion d_real=d_map×denominator, followed by unit conversion.

N2. Coordinates and constraint-based location

Problem 881. Locating a point by constraints: case 1

Knowledge context. Coordinates turn location into numbers. Multiple spatial clues can be treated as simultaneous constraints.

Given facts. A point P lies on an integer grid with x,y from 0 to 8. It satisfies x+y=8 and x−y=0. It also satisfies x ≥ 3.

Rules. A location is valid only if it satisfies every stated equation and inequality at the same time.

Task. Which coordinate(s) can P occupy? Is the solution unique?

Step-by-step solution. 1. List pairs satisfying x+y=8. 2. Keep only those also satisfying x−y=0. 3. The remaining set is [(4, 4)].

Answer. P=(4, 4); the solution is unique.

Knowledge takeaway. A constraint problem is solved by satisfying all conditions together, not one at a time independently.

Formal model. Constraint satisfaction over integer pairs (x,y).

Problem 882. Locating a point by constraints: case 2

Knowledge context. Coordinates turn location into numbers. Multiple spatial clues can be treated as simultaneous constraints.

Given facts. A point P lies on an integer grid with x,y from 0 to 8. It satisfies x+y=11 and x−y=-1. It also satisfies x ≥ 4.

Rules. A location is valid only if it satisfies every stated equation and inequality at the same time.

Task. Which coordinate(s) can P occupy? Is the solution unique?

Step-by-step solution. 1. List pairs satisfying x+y=11. 2. Keep only those also satisfying x−y=-1. 3. The remaining set is [(5, 6)].

Answer. P=(5, 6); the solution is unique.

Knowledge takeaway. A constraint problem is solved by satisfying all conditions together, not one at a time independently.

Formal model. Constraint satisfaction over integer pairs (x,y).

Problem 883. Locating a point by constraints: case 3

Knowledge context. Coordinates turn location into numbers. Multiple spatial clues can be treated as simultaneous constraints.

Given facts. A point P lies on an integer grid with x,y from 0 to 8. It satisfies x+y=14 and x−y=-2. It also satisfies x ≥ 5.

Rules. A location is valid only if it satisfies every stated equation and inequality at the same time.

Task. Which coordinate(s) can P occupy? Is the solution unique?

Step-by-step solution. 1. List pairs satisfying x+y=14. 2. Keep only those also satisfying x−y=-2. 3. The remaining set is [(6, 8)].

Answer. P=(6, 8); the solution is unique.

Knowledge takeaway. A constraint problem is solved by satisfying all conditions together, not one at a time independently.

Formal model. Constraint satisfaction over integer pairs (x,y).

Problem 884. Locating a point by constraints: case 4

Knowledge context. Coordinates turn location into numbers. Multiple spatial clues can be treated as simultaneous constraints.

Given facts. A point P lies on an integer grid with x,y from 0 to 8. It satisfies x+y=8 and x−y=6. It also satisfies x ≥ 6.

Rules. A location is valid only if it satisfies every stated equation and inequality at the same time.

Task. Which coordinate(s) can P occupy? Is the solution unique?

Step-by-step solution. 1. List pairs satisfying x+y=8. 2. Keep only those also satisfying x−y=6. 3. The remaining set is [(7, 1)].

Answer. P=(7, 1); the solution is unique.

Knowledge takeaway. A constraint problem is solved by satisfying all conditions together, not one at a time independently.

Formal model. Constraint satisfaction over integer pairs (x,y).

Problem 885. Locating a point by constraints: case 5

Knowledge context. Coordinates turn location into numbers. Multiple spatial clues can be treated as simultaneous constraints.

Given facts. A point P lies on an integer grid with x,y from 0 to 8. It satisfies x+y=11 and x−y=5. It also satisfies x ≥ 7.

Rules. A location is valid only if it satisfies every stated equation and inequality at the same time.

Task. Which coordinate(s) can P occupy? Is the solution unique?

Step-by-step solution. 1. List pairs satisfying x+y=11. 2. Keep only those also satisfying x−y=5. 3. The remaining set is [(8, 3)].

Answer. P=(8, 3); the solution is unique.

Knowledge takeaway. A constraint problem is solved by satisfying all conditions together, not one at a time independently.

Formal model. Constraint satisfaction over integer pairs (x,y).

N3. Earth, Sun, seasons, and local time

Problem 886. Earth, sunlight, and local time: case 1

Knowledge context. Earth–Sun relationships explain broad patterns such as day length and local time, but a problem should state the simplified model it expects children to use.

Given facts. For this classroom model, Eastport is 1 time zone(s) east of Westport. Each zone east is one hour later. At Westport it is 12:00.

Rules. Moving east adds one hour per stated time zone; moving west subtracts. Use the given model only.

Task. What local time is it in Eastport?

Step-by-step solution. 1. Time difference is +1 hour(s) because Eastport is east. 2. 12+1=13 (mod 24).

Answer. 13:00.

Knowledge takeaway. A model supports only the conclusions encoded in its rules; seasonal light does not automatically determine daily weather.

Formal model. Rule-based astronomical/time-zone model; apply signed offsets or seasonal relation.

Problem 887. Earth, sunlight, and local time: case 2

Knowledge context. Earth–Sun relationships explain broad patterns such as day length and local time, but a problem should state the simplified model it expects children to use.

Given facts. For this classroom model, Eastport is 2 time zone(s) east of Westport. Each zone east is one hour later. At Westport it is 12:00.

Rules. Moving east adds one hour per stated time zone; moving west subtracts. Use the given model only.

Task. What local time is it in Eastport?

Step-by-step solution. 1. Time difference is +2 hour(s) because Eastport is east. 2. 12+2=14 (mod 24).

Answer. 14:00.

Knowledge takeaway. A model supports only the conclusions encoded in its rules; seasonal light does not automatically determine daily weather.

Formal model. Rule-based astronomical/time-zone model; apply signed offsets or seasonal relation.

Problem 888. Earth, sunlight, and local time: case 3

Knowledge context. Earth–Sun relationships explain broad patterns such as day length and local time, but a problem should state the simplified model it expects children to use.

Given facts. For this classroom model, Eastport is 3 time zone(s) east of Westport. Each zone east is one hour later. At Westport it is 12:00.

Rules. Moving east adds one hour per stated time zone; moving west subtracts. Use the given model only.

Task. What local time is it in Eastport?

Step-by-step solution. 1. Time difference is +3 hour(s) because Eastport is east. 2. 12+3=15 (mod 24).

Answer. 15:00.

Knowledge takeaway. A model supports only the conclusions encoded in its rules; seasonal light does not automatically determine daily weather.

Formal model. Rule-based astronomical/time-zone model; apply signed offsets or seasonal relation.

Problem 889. Earth, sunlight, and local time: case 4

Knowledge context. Earth–Sun relationships explain broad patterns such as day length and local time, but a problem should state the simplified model it expects children to use.

Given facts. Use this simplified model: Earth's axis is tilted. In the Northern Hemisphere, days are generally longer around June than around December. Compare June and December for the same place. Cross-domain check: a survey team starts work at 8:00 and works for 2 hour(s) without a break.

Rules. Use the stated seasonal relation; do not infer daily weather from season. For the cross-domain check, finishing time = starting time + duration.

Task. Which of the two months has the longer daylight period in this model, and can that fact alone tell us whether it will rain? Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. Apply the stated tilt-season relation. 2. June has the longer daylight period in the Northern Hemisphere. 3. The rule contains no information about rainfall, so rainfall cannot be deduced. Cross-domain check: 8+2=10, so the team finishes at 10:00.

Answer. June has longer daylight; rainfall cannot be determined from this fact alone. Cross-domain answer: 10:00.

Knowledge takeaway. A model supports only the conclusions encoded in its rules; seasonal light does not automatically determine daily weather.

Formal model. Rule-based astronomical/time-zone model; apply signed offsets or seasonal relation.

Problem 890. Earth, sunlight, and local time: case 5

Knowledge context. Earth–Sun relationships explain broad patterns such as day length and local time, but a problem should state the simplified model it expects children to use.

Given facts. Use this simplified model: Earth's axis is tilted. In the Northern Hemisphere, days are generally longer around June than around December. Compare June and December for the same place. Cross-domain check: a local committee has 16 members, requires at least 9 present for quorum, and 14 are present.

Rules. Use the stated seasonal relation; do not infer daily weather from season. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. Which of the two months has the longer daylight period in this model, and can that fact alone tell us whether it will rain? Cross-domain check: is quorum met?

Step-by-step solution. 1. Apply the stated tilt-season relation. 2. June has the longer daylight period in the Northern Hemisphere. 3. The rule contains no information about rainfall, so rainfall cannot be deduced. Cross-domain check: compare 14 with 9; 14 ≥ 9 is true.

Answer. June has longer daylight; rainfall cannot be determined from this fact alone. Cross-domain answer: quorum is met.

Knowledge takeaway. A model supports only the conclusions encoded in its rules; seasonal light does not automatically determine daily weather.

Formal model. Rule-based astronomical/time-zone model; apply signed offsets or seasonal relation.

N4. Stocks, flows, and conservation

Problem 891. Reservoir stock and flow: case 1

Knowledge context. Many geographic and civic quantities are stocks and flows: water in a reservoir, people in a city, or goods in a warehouse.

Given facts. A reservoir starts with 40 water units. Each day 5 units flow in and 7 units flow out. Assume these rates stay constant for 6 days.

Rules. Stock at the end = starting stock + total inflow − total outflow. Flow is an amount per time period; stock is the amount currently stored.

Task. How much water is stored after 6 days? Is the stock increasing or decreasing each day?

Step-by-step solution. 1. Daily net change=5−7=-2. 2. Over 6 days, net change=6×-2=-12. 3. End stock=40+(-12)=28.

Answer. 28 units; the stock is decreasing.

Knowledge takeaway. Do not confuse a stored amount with a rate of change; conservation links them.

Formal model. Difference equation S(t+1)=S(t)+inflow−outflow.

Problem 892. Reservoir stock and flow: case 2

Knowledge context. Many geographic and civic quantities are stocks and flows: water in a reservoir, people in a city, or goods in a warehouse.

Given facts. A reservoir starts with 43 water units. Each day 6 units flow in and 8 units flow out. Assume these rates stay constant for 6 days.

Rules. Stock at the end = starting stock + total inflow − total outflow. Flow is an amount per time period; stock is the amount currently stored.

Task. How much water is stored after 6 days? Is the stock increasing or decreasing each day?

Step-by-step solution. 1. Daily net change=6−8=-2. 2. Over 6 days, net change=6×-2=-12. 3. End stock=43+(-12)=31.

Answer. 31 units; the stock is decreasing.

Knowledge takeaway. Do not confuse a stored amount with a rate of change; conservation links them.

Formal model. Difference equation S(t+1)=S(t)+inflow−outflow.

Problem 893. Reservoir stock and flow: case 3

Knowledge context. Many geographic and civic quantities are stocks and flows: water in a reservoir, people in a city, or goods in a warehouse.

Given facts. A reservoir starts with 46 water units. Each day 7 units flow in and 7 units flow out. Assume these rates stay constant for 6 days.

Rules. Stock at the end = starting stock + total inflow − total outflow. Flow is an amount per time period; stock is the amount currently stored.

Task. How much water is stored after 6 days? Is the stock increasing or decreasing each day?

Step-by-step solution. 1. Daily net change=7−7=0. 2. Over 6 days, net change=6×0=0. 3. End stock=46+(0)=46.

Answer. 46 units; the stock is unchanged.

Knowledge takeaway. Do not confuse a stored amount with a rate of change; conservation links them.

Formal model. Difference equation S(t+1)=S(t)+inflow−outflow.

Problem 894. Reservoir stock and flow: case 4

Knowledge context. Many geographic and civic quantities are stocks and flows: water in a reservoir, people in a city, or goods in a warehouse.

Given facts. A reservoir starts with 49 water units. Each day 8 units flow in and 8 units flow out. Assume these rates stay constant for 6 days.

Rules. Stock at the end = starting stock + total inflow − total outflow. Flow is an amount per time period; stock is the amount currently stored.

Task. How much water is stored after 6 days? Is the stock increasing or decreasing each day?

Step-by-step solution. 1. Daily net change=8−8=0. 2. Over 6 days, net change=6×0=0. 3. End stock=49+(0)=49.

Answer. 49 units; the stock is unchanged.

Knowledge takeaway. Do not confuse a stored amount with a rate of change; conservation links them.

Formal model. Difference equation S(t+1)=S(t)+inflow−outflow.

Problem 895. Reservoir stock and flow: case 5

Knowledge context. Many geographic and civic quantities are stocks and flows: water in a reservoir, people in a city, or goods in a warehouse.

Given facts. A reservoir starts with 52 water units. Each day 9 units flow in and 7 units flow out. Assume these rates stay constant for 6 days.

Rules. Stock at the end = starting stock + total inflow − total outflow. Flow is an amount per time period; stock is the amount currently stored.

Task. How much water is stored after 6 days? Is the stock increasing or decreasing each day?

Step-by-step solution. 1. Daily net change=9−7=2. 2. Over 6 days, net change=6×2=12. 3. End stock=52+(12)=64.

Answer. 64 units; the stock is increasing.

Knowledge takeaway. Do not confuse a stored amount with a rate of change; conservation links them.

Formal model. Difference equation S(t+1)=S(t)+inflow−outflow.

N5. Density, capacity, and crowding

Problem 896. Density and capacity in a public space: case 1

Knowledge context. Density compares an amount with the space available. It can reveal crowding even when two places have similar total populations.

Given facts. A park zone covers 4 hectares and currently has 40 visitors. For this exercise, comfortable density is at most 9 visitors per hectare.

Rules. Density = number ÷ area. A zone exceeds comfortable capacity if density is greater than the stated threshold.

Task. What is the visitor density, and is the zone above the stated comfortable capacity?

Step-by-step solution. 1. Density=40÷4=10.00 visitors/hectare. 2. Compare 10.00 with the threshold 9.

Answer. 10.00 visitors/hectare; above capacity.

Knowledge takeaway. Totals and densities answer different questions; always identify the denominator.

Formal model. Ratio density=N/A followed by threshold comparison.

Problem 897. Density and capacity in a public space: case 2

Knowledge context. Density compares an amount with the space available. It can reveal crowding even when two places have similar total populations.

Given facts. A park zone covers 5 hectares and currently has 44 visitors. For this exercise, comfortable density is at most 9 visitors per hectare.

Rules. Density = number ÷ area. A zone exceeds comfortable capacity if density is greater than the stated threshold.

Task. What is the visitor density, and is the zone above the stated comfortable capacity?

Step-by-step solution. 1. Density=44÷5=8.80 visitors/hectare. 2. Compare 8.80 with the threshold 9.

Answer. 8.80 visitors/hectare; within capacity.

Knowledge takeaway. Totals and densities answer different questions; always identify the denominator.

Formal model. Ratio density=N/A followed by threshold comparison.

Problem 898. Density and capacity in a public space: case 3

Knowledge context. Density compares an amount with the space available. It can reveal crowding even when two places have similar total populations.

Given facts. A park zone covers 6 hectares and currently has 48 visitors. For this exercise, comfortable density is at most 9 visitors per hectare.

Rules. Density = number ÷ area. A zone exceeds comfortable capacity if density is greater than the stated threshold.

Task. What is the visitor density, and is the zone above the stated comfortable capacity?

Step-by-step solution. 1. Density=48÷6=8.00 visitors/hectare. 2. Compare 8.00 with the threshold 9.

Answer. 8.00 visitors/hectare; within capacity.

Knowledge takeaway. Totals and densities answer different questions; always identify the denominator.

Formal model. Ratio density=N/A followed by threshold comparison.

Problem 899. Density and capacity in a public space: case 4

Knowledge context. Density compares an amount with the space available. It can reveal crowding even when two places have similar total populations.

Given facts. A park zone covers 7 hectares and currently has 52 visitors. For this exercise, comfortable density is at most 9 visitors per hectare.

Rules. Density = number ÷ area. A zone exceeds comfortable capacity if density is greater than the stated threshold.

Task. What is the visitor density, and is the zone above the stated comfortable capacity?

Step-by-step solution. 1. Density=52÷7=7.43 visitors/hectare. 2. Compare 7.43 with the threshold 9.

Answer. 7.43 visitors/hectare; within capacity.

Knowledge takeaway. Totals and densities answer different questions; always identify the denominator.

Formal model. Ratio density=N/A followed by threshold comparison.

Problem 900. Density and capacity in a public space: case 5

Knowledge context. Density compares an amount with the space available. It can reveal crowding even when two places have similar total populations.

Given facts. A park zone covers 8 hectares and currently has 56 visitors. For this exercise, comfortable density is at most 9 visitors per hectare.

Rules. Density = number ÷ area. A zone exceeds comfortable capacity if density is greater than the stated threshold.

Task. What is the visitor density, and is the zone above the stated comfortable capacity?

Step-by-step solution. 1. Density=56÷8=7.00 visitors/hectare. 2. Compare 7.00 with the threshold 9.

Answer. 7.00 visitors/hectare; within capacity.

Knowledge takeaway. Totals and densities answer different questions; always identify the denominator.

Formal model. Ratio density=N/A followed by threshold comparison.

N6. Land use and spatial compatibility

Problem 901. Land-use compatibility on four plots: case 1

Knowledge context. Land-use planning combines spatial relationships with compatibility rules.

Given facts. Four plots lie in a line A–B–C–D. Assign one different use to each plot: school, factory, park, homes. Constraints: the factory may not be adjacent to the school; the park must be adjacent to the homes. The school must be on an end plot. Cross-domain check: a survey team starts work at 12:00 and works for 2 hour(s) without a break.

Rules. An assignment is valid only if every use is placed exactly once and all adjacency constraints hold. For the cross-domain check, finishing time = starting time + duration.

Task. Give one valid assignment and explain why it satisfies every constraint. Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. Try the assignment A=school, B=park, C=homes, D=factory. 2. Factory and school are not adjacent. 3. Park and homes are adjacent. The school is on an end plot. 4. Each use appears exactly once. Cross-domain check: 12+2=14, so the team finishes at 14:00.

Answer. A=school, B=park, C=homes, D=factory is a valid assignment. Cross-domain answer: 14:00.

Knowledge takeaway. A place can be unsuitable not because of its own properties, but because of what would be next to it.

Formal model. Permutation CSP with all-different and adjacency/non-adjacency constraints.

Problem 902. Land-use compatibility on four plots: case 2

Knowledge context. Land-use planning combines spatial relationships with compatibility rules.

Given facts. Four plots lie in a line A–B–C–D. Assign one different use to each plot: school, factory, park, homes. Constraints: the factory may not be adjacent to the school; the park must be adjacent to the homes. The school must be on an end plot. Cross-domain check: a local committee has 10 members, requires at least 6 present for quorum, and 8 are present.

Rules. An assignment is valid only if every use is placed exactly once and all adjacency constraints hold. For the cross-domain check, quorum exists when present members ≥ the stated threshold.

Task. Give one valid assignment and explain why it satisfies every constraint. Cross-domain check: is quorum met?

Step-by-step solution. 1. Try the assignment A=school, B=homes, C=park, D=factory. 2. Factory and school are not adjacent. 3. Park and homes are adjacent. The school is on an end plot. 4. Each use appears exactly once. Cross-domain check: compare 8 with 6; 8 ≥ 6 is true.

Answer. A=school, B=homes, C=park, D=factory is a valid assignment. Cross-domain answer: quorum is met.

Knowledge takeaway. A place can be unsuitable not because of its own properties, but because of what would be next to it.

Formal model. Permutation CSP with all-different and adjacency/non-adjacency constraints.

Problem 903. Land-use compatibility on four plots: case 3

Knowledge context. Land-use planning combines spatial relationships with compatibility rules.

Given facts. Four plots lie in a line A–B–C–D. Assign one different use to each plot: school, factory, park, homes. Constraints: the factory may not be adjacent to the school; the park must be adjacent to the homes. The school must be on an end plot. Cross-domain check: researchers collected 9 reports, but 4 are exact duplicate copies that add no new independent information.

Rules. An assignment is valid only if every use is placed exactly once and all adjacency constraints hold. For the cross-domain check, independent reports = total reports − duplicate copies.

Task. Give one valid assignment and explain why it satisfies every constraint. Cross-domain check: how many non-duplicate reports remain?

Step-by-step solution. 1. Try the assignment A=factory, B=park, C=homes, D=school. 2. Factory and school are not adjacent. 3. Park and homes are adjacent. The school is on an end plot. 4. Each use appears exactly once. Cross-domain check: 9−4=5 independent reports remain.

Answer. A=factory, B=park, C=homes, D=school is a valid assignment. Cross-domain answer: 5 independent reports.

Knowledge takeaway. A place can be unsuitable not because of its own properties, but because of what would be next to it.

Formal model. Permutation CSP with all-different and adjacency/non-adjacency constraints.

Problem 904. Land-use compatibility on four plots: case 4

Knowledge context. Land-use planning combines spatial relationships with compatibility rules.

Given facts. Four plots lie in a line A–B–C–D. Assign one different use to each plot: school, factory, park, homes. Constraints: the factory may not be adjacent to the school; the park must be adjacent to the homes. The school must be on an end plot. Cross-domain check: on a field sketch, 1 cm represents 6 km and two checkpoints are 3 cm apart.

Rules. An assignment is valid only if every use is placed exactly once and all adjacency constraints hold. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Give one valid assignment and explain why it satisfies every constraint. Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. Try the assignment A=factory, B=homes, C=park, D=school. 2. Factory and school are not adjacent. 3. Park and homes are adjacent. The school is on an end plot. 4. Each use appears exactly once. Cross-domain check: 3×6=18 km.

Answer. A=factory, B=homes, C=park, D=school is a valid assignment. Cross-domain answer: 18 km.

Knowledge takeaway. A place can be unsuitable not because of its own properties, but because of what would be next to it.

Formal model. Permutation CSP with all-different and adjacency/non-adjacency constraints.

Problem 905. Land-use compatibility on four plots: case 5

Knowledge context. Land-use planning combines spatial relationships with compatibility rules.

Given facts. Four plots lie in a line A–B–C–D. Assign one different use to each plot: school, factory, park, homes. Constraints: the factory may not be adjacent to the school; the park must be adjacent to the homes. The school must be on an end plot. Cross-domain check: a survey team starts work at 8:00 and works for 2 hour(s) without a break.

Rules. An assignment is valid only if every use is placed exactly once and all adjacency constraints hold. For the cross-domain check, finishing time = starting time + duration.

Task. Give one valid assignment and explain why it satisfies every constraint. Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. Try the assignment A=school, B=park, C=homes, D=factory. 2. Factory and school are not adjacent. 3. Park and homes are adjacent. The school is on an end plot. 4. Each use appears exactly once. Cross-domain check: 8+2=10, so the team finishes at 10:00.

Answer. A=school, B=park, C=homes, D=factory is a valid assignment. Cross-domain answer: 10:00.

Knowledge takeaway. A place can be unsuitable not because of its own properties, but because of what would be next to it.

Formal model. Permutation CSP with all-different and adjacency/non-adjacency constraints.

N7. Natural risk: hazard, exposure, protection

Problem 906. Hazard, exposure, and protection: case 1

Knowledge context. Natural risk depends on more than the physical hazard. Exposure and protective measures also affect outcomes.

Given facts. For this classroom model, risk index = hazard × exposure ÷ protection. Data: Site A: hazard=2, exposure=4, protection=1; Site B: hazard=4, exposure=2, protection=2; Site C: hazard=3, exposure=3, protection=3.

Rules. Use the formula exactly as stated. A higher hazard does not necessarily mean higher risk if exposure or protection differs.

Task. Compute the risk index for each site. Which site has the lowest modeled risk?

Step-by-step solution. 1. Compute hazard×exposure÷protection for each site. 2. Results: A=8.00, B=4.00, C=3.00. 3. The lowest value is Site C.

Answer. A=8.00; B=4.00; C=3.00. Lowest: Site C.

Knowledge takeaway. Separating hazard from risk prevents the mistaken idea that the most dangerous process always creates the greatest overall risk.

Formal model. Explicit risk-index function over three variables; compare computed values.

Problem 907. Hazard, exposure, and protection: case 2

Knowledge context. Natural risk depends on more than the physical hazard. Exposure and protective measures also affect outcomes.

Given facts. For this classroom model, risk index = hazard × exposure ÷ protection. Data: Site A: hazard=3, exposure=4, protection=1; Site B: hazard=4, exposure=3, protection=2; Site C: hazard=3, exposure=3, protection=3.

Rules. Use the formula exactly as stated. A higher hazard does not necessarily mean higher risk if exposure or protection differs.

Task. Compute the risk index for each site. Which site has the lowest modeled risk?

Step-by-step solution. 1. Compute hazard×exposure÷protection for each site. 2. Results: A=12.00, B=6.00, C=3.00. 3. The lowest value is Site C.

Answer. A=12.00; B=6.00; C=3.00. Lowest: Site C.

Knowledge takeaway. Separating hazard from risk prevents the mistaken idea that the most dangerous process always creates the greatest overall risk.

Formal model. Explicit risk-index function over three variables; compare computed values.

Problem 908. Hazard, exposure, and protection: case 3

Knowledge context. Natural risk depends on more than the physical hazard. Exposure and protective measures also affect outcomes.

Given facts. For this classroom model, risk index = hazard × exposure ÷ protection. Data: Site A: hazard=4, exposure=4, protection=1; Site B: hazard=4, exposure=4, protection=2; Site C: hazard=3, exposure=3, protection=3.

Rules. Use the formula exactly as stated. A higher hazard does not necessarily mean higher risk if exposure or protection differs.

Task. Compute the risk index for each site. Which site has the lowest modeled risk?

Step-by-step solution. 1. Compute hazard×exposure÷protection for each site. 2. Results: A=16.00, B=8.00, C=3.00. 3. The lowest value is Site C.

Answer. A=16.00; B=8.00; C=3.00. Lowest: Site C.

Knowledge takeaway. Separating hazard from risk prevents the mistaken idea that the most dangerous process always creates the greatest overall risk.

Formal model. Explicit risk-index function over three variables; compare computed values.

Problem 909. Hazard, exposure, and protection: case 4

Knowledge context. Natural risk depends on more than the physical hazard. Exposure and protective measures also affect outcomes.

Given facts. For this classroom model, risk index = hazard × exposure ÷ protection. Data: Site A: hazard=5, exposure=4, protection=1; Site B: hazard=4, exposure=5, protection=2; Site C: hazard=3, exposure=3, protection=3.

Rules. Use the formula exactly as stated. A higher hazard does not necessarily mean higher risk if exposure or protection differs.

Task. Compute the risk index for each site. Which site has the lowest modeled risk?

Step-by-step solution. 1. Compute hazard×exposure÷protection for each site. 2. Results: A=20.00, B=10.00, C=3.00. 3. The lowest value is Site C.

Answer. A=20.00; B=10.00; C=3.00. Lowest: Site C.

Knowledge takeaway. Separating hazard from risk prevents the mistaken idea that the most dangerous process always creates the greatest overall risk.

Formal model. Explicit risk-index function over three variables; compare computed values.

Problem 910. Hazard, exposure, and protection: case 5

Knowledge context. Natural risk depends on more than the physical hazard. Exposure and protective measures also affect outcomes.

Given facts. For this classroom model, risk index = hazard × exposure ÷ protection. Data: Site A: hazard=6, exposure=4, protection=1; Site B: hazard=4, exposure=6, protection=2; Site C: hazard=3, exposure=3, protection=3.

Rules. Use the formula exactly as stated. A higher hazard does not necessarily mean higher risk if exposure or protection differs.

Task. Compute the risk index for each site. Which site has the lowest modeled risk?

Step-by-step solution. 1. Compute hazard×exposure÷protection for each site. 2. Results: A=24.00, B=12.00, C=3.00. 3. The lowest value is Site C.

Answer. A=24.00; B=12.00; C=3.00. Lowest: Site C.

Knowledge takeaway. Separating hazard from risk prevents the mistaken idea that the most dangerous process always creates the greatest overall risk.

Formal model. Explicit risk-index function over three variables; compare computed values.

N8. Ecosystems and indirect effects

Problem 911. Indirect effects in a food chain: case 1

Knowledge context. Ecosystems contain dependency networks. A change at one point can propagate indirectly through several links.

Given facts. Food relation: rabbit eats grass; fox eats rabbit; eagle eats fox. Classroom rule: if a food source sharply decreases and no alternative food is given, the consumer may also decrease. Cross-domain check: researchers collected 7 reports, but 4 are exact duplicate copies that add no new independent information. Mixed-domain verification: a regional archive has 93 map sheets and receives 2 additional sheets during the exercise.

Rules. Follow dependency arrows from food to consumer. The rule supports possible directional effects, not exact population numbers. For the cross-domain check, independent reports = total reports − duplicate copies. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. If rabbit sharply decreases, which later populations may be indirectly affected according to the rule? Cross-domain check: how many non-duplicate reports remain? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. fox depends directly on rabbit, so it may decrease. 2. eagle depends on fox, so it may also be indirectly affected. 3. grass is food for rabbit; the stated rule does not imply it decreases for the same reason. Cross-domain check: 7−4=3 independent reports remain. Mixed-domain verification: 93+2=95.

Answer. fox may decrease directly and eagle may be affected indirectly. Cross-domain answer: 3 independent reports. Mixed-domain answer: 95 map sheets.

Knowledge takeaway. Indirect effects require following the direction of dependency and respecting the limits of the stated model.

Formal model. Directed dependency graph; propagate a qualitative decrease along outgoing dependency links.

Problem 912. Indirect effects in a food chain: case 2

Knowledge context. Ecosystems contain dependency networks. A change at one point can propagate indirectly through several links.

Given facts. Food relation: small fish eats algae; large fish eats small fish; heron eats large fish. Classroom rule: if a food source sharply decreases and no alternative food is given, the consumer may also decrease. Cross-domain check: on a field sketch, 1 cm represents 4 km and two checkpoints are 4 cm apart.

Rules. Follow dependency arrows from food to consumer. The rule supports possible directional effects, not exact population numbers. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. If small fish sharply decreases, which later populations may be indirectly affected according to the rule? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. large fish depends directly on small fish, so it may decrease. 2. heron depends on large fish, so it may also be indirectly affected. 3. algae is food for small fish; the stated rule does not imply it decreases for the same reason. Cross-domain check: 4×4=16 km.

Answer. large fish may decrease directly and heron may be affected indirectly. Cross-domain answer: 16 km.

Knowledge takeaway. Indirect effects require following the direction of dependency and respecting the limits of the stated model.

Formal model. Directed dependency graph; propagate a qualitative decrease along outgoing dependency links.

Problem 913. Indirect effects in a food chain: case 3

Knowledge context. Ecosystems contain dependency networks. A change at one point can propagate indirectly through several links.

Given facts. Food relation: rabbit eats grass; fox eats rabbit; eagle eats fox. Classroom rule: if a food source sharply decreases and no alternative food is given, the consumer may also decrease. Cross-domain check: a survey team starts work at 8:00 and works for 2 hour(s) without a break. Mixed-domain verification: a regional archive has 93 map sheets and receives 4 additional sheets during the exercise.

Rules. Follow dependency arrows from food to consumer. The rule supports possible directional effects, not exact population numbers. For the cross-domain check, finishing time = starting time + duration. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. If rabbit sharply decreases, which later populations may be indirectly affected according to the rule? Cross-domain check: at what time does the team finish? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. fox depends directly on rabbit, so it may decrease. 2. eagle depends on fox, so it may also be indirectly affected. 3. grass is food for rabbit; the stated rule does not imply it decreases for the same reason. Cross-domain check: 8+2=10, so the team finishes at 10:00. Mixed-domain verification: 93+4=97.

Answer. fox may decrease directly and eagle may be affected indirectly. Cross-domain answer: 10:00. Mixed-domain answer: 97 map sheets.

Knowledge takeaway. Indirect effects require following the direction of dependency and respecting the limits of the stated model.

Formal model. Directed dependency graph; propagate a qualitative decrease along outgoing dependency links.

Problem 914. Indirect effects in a food chain: case 4

Knowledge context. Ecosystems contain dependency networks. A change at one point can propagate indirectly through several links.

Given facts. Food relation: small fish eats algae; large fish eats small fish; heron eats large fish. Classroom rule: if a food source sharply decreases and no alternative food is given, the consumer may also decrease. Cross-domain check: a local committee has 13 members, requires at least 7 present for quorum, and 11 are present. Mixed-domain verification: a regional archive has 93 map sheets and receives 5 additional sheets during the exercise.

Rules. Follow dependency arrows from food to consumer. The rule supports possible directional effects, not exact population numbers. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. If small fish sharply decreases, which later populations may be indirectly affected according to the rule? Cross-domain check: is quorum met? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. large fish depends directly on small fish, so it may decrease. 2. heron depends on large fish, so it may also be indirectly affected. 3. algae is food for small fish; the stated rule does not imply it decreases for the same reason. Cross-domain check: compare 11 with 7; 11 ≥ 7 is true. Mixed-domain verification: 93+5=98.

Answer. large fish may decrease directly and heron may be affected indirectly. Cross-domain answer: quorum is met. Mixed-domain answer: 98 map sheets.

Knowledge takeaway. Indirect effects require following the direction of dependency and respecting the limits of the stated model.

Formal model. Directed dependency graph; propagate a qualitative decrease along outgoing dependency links.

Problem 915. Indirect effects in a food chain: case 5

Knowledge context. Ecosystems contain dependency networks. A change at one point can propagate indirectly through several links.

Given facts. Food relation: rabbit eats grass; fox eats rabbit; eagle eats fox. Classroom rule: if a food source sharply decreases and no alternative food is given, the consumer may also decrease. Cross-domain check: researchers collected 11 reports, but 4 are exact duplicate copies that add no new independent information. Mixed-domain verification: a regional archive has 93 map sheets and receives 6 additional sheets during the exercise.

Rules. Follow dependency arrows from food to consumer. The rule supports possible directional effects, not exact population numbers. For the cross-domain check, independent reports = total reports − duplicate copies. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. If rabbit sharply decreases, which later populations may be indirectly affected according to the rule? Cross-domain check: how many non-duplicate reports remain? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. fox depends directly on rabbit, so it may decrease. 2. eagle depends on fox, so it may also be indirectly affected. 3. grass is food for rabbit; the stated rule does not imply it decreases for the same reason. Cross-domain check: 11−4=7 independent reports remain. Mixed-domain verification: 93+6=99.

Answer. fox may decrease directly and eagle may be affected indirectly. Cross-domain answer: 7 independent reports. Mixed-domain answer: 99 map sheets.

Knowledge takeaway. Indirect effects require following the direction of dependency and respecting the limits of the stated model.

Formal model. Directed dependency graph; propagate a qualitative decrease along outgoing dependency links.

N9. Renewable resources and sustainable use

Problem 916. Regeneration and harvest: case 1

Knowledge context. Renewable does not mean unlimited. A resource can decline when use exceeds regeneration.

Given facts. A forest resource starts at 140 units. Each year it regenerates 12 units and people harvest 12 units. Assume these amounts stay constant for 3 years and no other changes occur.

Rules. Annual net change = regeneration − harvest. In this simplified model, use is sustainable over the period if stock does not decline.

Task. What is the stock after 3 years, and is the harvest sustainable under this classroom definition?

Step-by-step solution. 1. Annual net change=12−12=0. 2. Three-year change=0. 3. Final stock=140+(0)=140. 4. Compare final stock with starting stock.

Answer. 140 units; sustainable under the stated definition.

Knowledge takeaway. Sustainability is a balance between renewal and extraction over time.

Formal model. Stock-flow model with regeneration and harvest; test non-decreasing stock.

Problem 917. Regeneration and harvest: case 2

Knowledge context. Renewable does not mean unlimited. A resource can decline when use exceeds regeneration.

Given facts. A forest resource starts at 140 units. Each year it regenerates 14 units and people harvest 15 units. Assume these amounts stay constant for 3 years and no other changes occur.

Rules. Annual net change = regeneration − harvest. In this simplified model, use is sustainable over the period if stock does not decline.

Task. What is the stock after 3 years, and is the harvest sustainable under this classroom definition?

Step-by-step solution. 1. Annual net change=14−15=-1. 2. Three-year change=-3. 3. Final stock=140+(-3)=137. 4. Compare final stock with starting stock.

Answer. 137 units; not sustainable under the stated definition.

Knowledge takeaway. Sustainability is a balance between renewal and extraction over time.

Formal model. Stock-flow model with regeneration and harvest; test non-decreasing stock.

Problem 918. Regeneration and harvest: case 3

Knowledge context. Renewable does not mean unlimited. A resource can decline when use exceeds regeneration.

Given facts. A forest resource starts at 140 units. Each year it regenerates 16 units and people harvest 18 units. Assume these amounts stay constant for 3 years and no other changes occur.

Rules. Annual net change = regeneration − harvest. In this simplified model, use is sustainable over the period if stock does not decline.

Task. What is the stock after 3 years, and is the harvest sustainable under this classroom definition?

Step-by-step solution. 1. Annual net change=16−18=-2. 2. Three-year change=-6. 3. Final stock=140+(-6)=134. 4. Compare final stock with starting stock.

Answer. 134 units; not sustainable under the stated definition.

Knowledge takeaway. Sustainability is a balance between renewal and extraction over time.

Formal model. Stock-flow model with regeneration and harvest; test non-decreasing stock.

Problem 919. Regeneration and harvest: case 4

Knowledge context. Renewable does not mean unlimited. A resource can decline when use exceeds regeneration.

Given facts. A forest resource starts at 140 units. Each year it regenerates 18 units and people harvest 21 units. Assume these amounts stay constant for 3 years and no other changes occur.

Rules. Annual net change = regeneration − harvest. In this simplified model, use is sustainable over the period if stock does not decline.

Task. What is the stock after 3 years, and is the harvest sustainable under this classroom definition?

Step-by-step solution. 1. Annual net change=18−21=-3. 2. Three-year change=-9. 3. Final stock=140+(-9)=131. 4. Compare final stock with starting stock.

Answer. 131 units; not sustainable under the stated definition.

Knowledge takeaway. Sustainability is a balance between renewal and extraction over time.

Formal model. Stock-flow model with regeneration and harvest; test non-decreasing stock.

Problem 920. Regeneration and harvest: case 5

Knowledge context. Renewable does not mean unlimited. A resource can decline when use exceeds regeneration.

Given facts. A forest resource starts at 140 units. Each year it regenerates 20 units and people harvest 24 units. Assume these amounts stay constant for 3 years and no other changes occur.

Rules. Annual net change = regeneration − harvest. In this simplified model, use is sustainable over the period if stock does not decline.

Task. What is the stock after 3 years, and is the harvest sustainable under this classroom definition?

Step-by-step solution. 1. Annual net change=20−24=-4. 2. Three-year change=-12. 3. Final stock=140+(-12)=128. 4. Compare final stock with starting stock.

Answer. 128 units; not sustainable under the stated definition.

Knowledge takeaway. Sustainability is a balance between renewal and extraction over time.

Formal model. Stock-flow model with regeneration and harvest; test non-decreasing stock.

N10. Migration and population balance

Problem 921. Population balance in a town: case 1

Knowledge context. Population changes through births, deaths, immigration, and emigration.

Given facts. A town begins the year with 1400 residents. During the year: births=20, deaths=14, people moving in=38, people moving out=25.

Rules. End population = start + births − deaths + in-migration − out-migration. Natural change and migration change are separate components.

Task. Compute natural change, migration change, and the year-end population.

Step-by-step solution. 1. Natural change=20−14=6. 2. Migration change=38−25=13. 3. Total change=6+13=19. 4. End population=1400+19=1419.

Answer. Natural change +6; migration change +13; end population 1419.

Knowledge takeaway. Breaking total change into components helps explain why a population grew or shrank.

Formal model. Balance equation P1=P0+births−deaths+in−out.

Problem 922. Population balance in a town: case 2

Knowledge context. Population changes through births, deaths, immigration, and emigration.

Given facts. A town begins the year with 1450 residents. During the year: births=23, deaths=14, people moving in=38, people moving out=29.

Rules. End population = start + births − deaths + in-migration − out-migration. Natural change and migration change are separate components.

Task. Compute natural change, migration change, and the year-end population.

Step-by-step solution. 1. Natural change=23−14=9. 2. Migration change=38−29=9. 3. Total change=9+9=18. 4. End population=1450+18=1468.

Answer. Natural change +9; migration change +9; end population 1468.

Knowledge takeaway. Breaking total change into components helps explain why a population grew or shrank.

Formal model. Balance equation P1=P0+births−deaths+in−out.

Problem 923. Population balance in a town: case 3

Knowledge context. Population changes through births, deaths, immigration, and emigration.

Given facts. A town begins the year with 1500 residents. During the year: births=26, deaths=14, people moving in=38, people moving out=33.

Rules. End population = start + births − deaths + in-migration − out-migration. Natural change and migration change are separate components.

Task. Compute natural change, migration change, and the year-end population.

Step-by-step solution. 1. Natural change=26−14=12. 2. Migration change=38−33=5. 3. Total change=12+5=17. 4. End population=1500+17=1517.

Answer. Natural change +12; migration change +5; end population 1517.

Knowledge takeaway. Breaking total change into components helps explain why a population grew or shrank.

Formal model. Balance equation P1=P0+births−deaths+in−out.

Problem 924. Population balance in a town: case 4

Knowledge context. Population changes through births, deaths, immigration, and emigration.

Given facts. A town begins the year with 1550 residents. During the year: births=29, deaths=14, people moving in=38, people moving out=37.

Rules. End population = start + births − deaths + in-migration − out-migration. Natural change and migration change are separate components.

Task. Compute natural change, migration change, and the year-end population.

Step-by-step solution. 1. Natural change=29−14=15. 2. Migration change=38−37=1. 3. Total change=15+1=16. 4. End population=1550+16=1566.

Answer. Natural change +15; migration change +1; end population 1566.

Knowledge takeaway. Breaking total change into components helps explain why a population grew or shrank.

Formal model. Balance equation P1=P0+births−deaths+in−out.

Problem 925. Population balance in a town: case 5

Knowledge context. Population changes through births, deaths, immigration, and emigration.

Given facts. A town begins the year with 1600 residents. During the year: births=32, deaths=14, people moving in=38, people moving out=41.

Rules. End population = start + births − deaths + in-migration − out-migration. Natural change and migration change are separate components.

Task. Compute natural change, migration change, and the year-end population.

Step-by-step solution. 1. Natural change=32−14=18. 2. Migration change=38−41=-3. 3. Total change=18+-3=15. 4. End population=1600+15=1615.

Answer. Natural change +18; migration change -3; end population 1615.

Knowledge takeaway. Breaking total change into components helps explain why a population grew or shrank.

Formal model. Balance equation P1=P0+births−deaths+in−out.

N11. Time intervals and overlapping events

Problem 926. Overlapping historical intervals: case 1

Knowledge context. Historical periods often overlap rather than forming a single neat sequence.

Given facts. Event A lasted from year 100 to 150. Event B lasted from 100 to 137. Treat intervals as continuous for this exercise.

Rules. Two intervals overlap when max(start times) < min(end times). Overlap length = min(end times) − max(start times), if positive.

Task. Did the events overlap? If so, for how many years?

Step-by-step solution. 1. Later start=max(100,100)=100. 2. Earlier end=min(150,137)=137. 3. Difference=137−100=37.

Answer. Yes, overlap 37 years.

Knowledge takeaway. Interval reasoning distinguishes “before/after” from “overlapping in time.”

Formal model. Interval intersection; compute max starts and min ends.

Problem 927. Overlapping historical intervals: case 2

Knowledge context. Historical periods often overlap rather than forming a single neat sequence.

Given facts. Event A lasted from year 120 to 170. Event B lasted from 130 to 167. Treat intervals as continuous for this exercise.

Rules. Two intervals overlap when max(start times) < min(end times). Overlap length = min(end times) − max(start times), if positive.

Task. Did the events overlap? If so, for how many years?

Step-by-step solution. 1. Later start=max(120,130)=130. 2. Earlier end=min(170,167)=167. 3. Difference=167−130=37.

Answer. Yes, overlap 37 years.

Knowledge takeaway. Interval reasoning distinguishes “before/after” from “overlapping in time.”

Formal model. Interval intersection; compute max starts and min ends.

Problem 928. Overlapping historical intervals: case 3

Knowledge context. Historical periods often overlap rather than forming a single neat sequence.

Given facts. Event A lasted from year 140 to 190. Event B lasted from 160 to 197. Treat intervals as continuous for this exercise.

Rules. Two intervals overlap when max(start times) < min(end times). Overlap length = min(end times) − max(start times), if positive.

Task. Did the events overlap? If so, for how many years?

Step-by-step solution. 1. Later start=max(140,160)=160. 2. Earlier end=min(190,197)=190. 3. Difference=190−160=30.

Answer. Yes, overlap 30 years.

Knowledge takeaway. Interval reasoning distinguishes “before/after” from “overlapping in time.”

Formal model. Interval intersection; compute max starts and min ends.

Problem 929. Overlapping historical intervals: case 4

Knowledge context. Historical periods often overlap rather than forming a single neat sequence.

Given facts. Event A lasted from year 160 to 210. Event B lasted from 160 to 197. Treat intervals as continuous for this exercise.

Rules. Two intervals overlap when max(start times) < min(end times). Overlap length = min(end times) − max(start times), if positive.

Task. Did the events overlap? If so, for how many years?

Step-by-step solution. 1. Later start=max(160,160)=160. 2. Earlier end=min(210,197)=197. 3. Difference=197−160=37.

Answer. Yes, overlap 37 years.

Knowledge takeaway. Interval reasoning distinguishes “before/after” from “overlapping in time.”

Formal model. Interval intersection; compute max starts and min ends.

Problem 930. Overlapping historical intervals: case 5

Knowledge context. Historical periods often overlap rather than forming a single neat sequence.

Given facts. Event A lasted from year 180 to 230. Event B lasted from 190 to 227. Treat intervals as continuous for this exercise.

Rules. Two intervals overlap when max(start times) < min(end times). Overlap length = min(end times) − max(start times), if positive.

Task. Did the events overlap? If so, for how many years?

Step-by-step solution. 1. Later start=max(180,190)=190. 2. Earlier end=min(230,227)=227. 3. Difference=227−190=37.

Answer. Yes, overlap 37 years.

Knowledge takeaway. Interval reasoning distinguishes “before/after” from “overlapping in time.”

Formal model. Interval intersection; compute max starts and min ends.

N12. Genealogy and generations

Problem 931. Reasoning across generations: case 1

Knowledge context. Genealogies are relational graphs. Generation distance can be counted through parent links.

Given facts. Maya is a parent of Noah. Noah is a parent of Lina. Maya is also a parent of Owen. Owen is a parent of Sara. Cross-domain check: researchers collected 7 reports, but 4 are exact duplicate copies that add no new independent information. Mixed-domain verification: a regional archive has 95 map sheets and receives 2 additional sheets during the exercise.

Rules. A parent is one generation above a child. A grandparent is two parent-links above. People who share a parent are siblings under this simplified model. For the cross-domain check, independent reports = total reports − duplicate copies. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. What family relation between Maya and Lina follows from the stated links? Cross-domain check: how many non-duplicate reports remain? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. Maya→Noah is one parent link. 2. Noah→Lina is a second parent link. 3. Two links give the grandparent relation. Cross-domain check: 7−4=3 independent reports remain. Mixed-domain verification: 95+2=97.

Answer. Maya is a grandparent of Lina. Cross-domain answer: 3 independent reports. Mixed-domain answer: 97 map sheets.

Knowledge takeaway. Family relations can be derived step by step from a small set of explicit links.

Formal model. Directed parent graph; kinship predicates derived from path length and shared ancestors.

Problem 932. Reasoning across generations: case 2

Knowledge context. Genealogies are relational graphs. Generation distance can be counted through parent links.

Given facts. Maya is a parent of Noah. Noah is a parent of Lina. Maya is also a parent of Owen. Owen is a parent of Sara. Cross-domain check: on a field sketch, 1 cm represents 4 km and two checkpoints are 3 cm apart.

Rules. A parent is one generation above a child. A grandparent is two parent-links above. People who share a parent are siblings under this simplified model. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. What family relation between Noah and Lina follows from the stated links? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. The relation is stated directly. Cross-domain check: 3×4=12 km.

Answer. Noah is a parent of Lina. Cross-domain answer: 12 km.

Knowledge takeaway. Family relations can be derived step by step from a small set of explicit links.

Formal model. Directed parent graph; kinship predicates derived from path length and shared ancestors.

Problem 933. Reasoning across generations: case 3

Knowledge context. Genealogies are relational graphs. Generation distance can be counted through parent links.

Given facts. Maya is a parent of Noah. Noah is a parent of Lina. Maya is also a parent of Owen. Owen is a parent of Sara. Cross-domain check: a survey team starts work at 12:00 and works for 2 hour(s) without a break.

Rules. A parent is one generation above a child. A grandparent is two parent-links above. People who share a parent are siblings under this simplified model. For the cross-domain check, finishing time = starting time + duration.

Task. What family relation between Noah and Owen follows from the stated links? Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. Maya is parent of both Noah and Owen. 2. Sharing a parent satisfies the simplified sibling rule. Cross-domain check: 12+2=14, so the team finishes at 14:00.

Answer. Noah and Owen are siblings. Cross-domain answer: 14:00.

Knowledge takeaway. Family relations can be derived step by step from a small set of explicit links.

Formal model. Directed parent graph; kinship predicates derived from path length and shared ancestors.

Problem 934. Reasoning across generations: case 4

Knowledge context. Genealogies are relational graphs. Generation distance can be counted through parent links.

Given facts. Maya is a parent of Noah. Noah is a parent of Lina. Maya is also a parent of Owen. Owen is a parent of Sara. Cross-domain check: a local committee has 15 members, requires at least 8 present for quorum, and 14 are present.

Rules. A parent is one generation above a child. A grandparent is two parent-links above. People who share a parent are siblings under this simplified model. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. What family relation between Maya and Sara follows from the stated links? Cross-domain check: is quorum met?

Step-by-step solution. 1. Maya→Owen and Owen→Sara form two parent links. Cross-domain check: compare 14 with 8; $14 \geq 8$ is true.

Answer. Maya is a grandparent of Sara. Cross-domain answer: quorum is met.

Knowledge takeaway. Family relations can be derived step by step from a small set of explicit links.

Formal model. Directed parent graph; kinship predicates derived from path length and shared ancestors.

Problem 935. Reasoning across generations: case 5

Knowledge context. Genealogies are relational graphs. Generation distance can be counted through parent links.

Given facts. Maya is a parent of Noah. Noah is a parent of Lina. Maya is also a parent of Owen. Owen is a parent of Sara. Cross-domain check: researchers collected 11 reports, but 4 are exact duplicate copies that add no new independent information. Mixed-domain verification: a regional archive has 95 map sheets and receives 6 additional sheets during the exercise.

Rules. A parent is one generation above a child. A grandparent is two parent-links above. People who share a parent are siblings under this simplified model. For the cross-domain check, independent reports = total reports − duplicate copies. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. What family relation between Lina and Sara follows from the stated links? Cross-domain check: how many non-duplicate reports remain? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. Noah and Owen are siblings because they share parent Maya. 2. Lina is child of Noah; Sara is child of Owen. 3. Children of siblings are cousins in this model. Cross-domain check: 11−4=7 independent reports remain. Mixed-domain verification: 95+6=101.

Answer. Lina and Sara are cousins under the implied family tree. Cross-domain answer: 7 independent reports. Mixed-domain answer: 101 map sheets.

Knowledge takeaway. Family relations can be derived step by step from a small set of explicit links.

Formal model. Directed parent graph; kinship predicates derived from path length and shared ancestors.

N13. Anachronisms and temporal compatibility

Problem 936. Temporal compatibility in a story set in 1400

Knowledge context. Anachronism can be tested as a compatibility problem when the relevant date ranges are supplied.

Given facts. For this problem, availability intervals are: Technology A: 1200–1600; Institution B: 1500–1900; Tool C: 1700–2000; Practice D: 1300–1450. A story is set in 1400 and mentions Technology A and Tool C together. Cross-domain check: on a field sketch, 1 cm represents 3 km and two checkpoints are 7 cm apart.

Rules. An item is temporally compatible with a year only if the year lies inside its stated availability interval. An anachronism is an item placed outside its given interval. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Which mentioned item, if any, is anachronistic in the story? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. Check 1400 against Technology A interval 1200–1600. 2. Check 1400 against Tool C interval 1700–2000. 3. Items outside their intervals: ['Tool C']. Cross-domain check: 7×3=21 km.

Answer. Anachronistic: Tool C. Cross-domain answer: 21 km.

Knowledge takeaway. The reasoning does not require memorizing invention dates if the intervals are part of the problem.

Formal model. Interval membership test for each object-event pair.

Problem 937. Temporal compatibility in a story set in 1550

Knowledge context. Anachronism can be tested as a compatibility problem when the relevant date ranges are supplied.

Given facts. For this problem, availability intervals are: Technology A: 1200–1600; Institution B: 1500–1900; Tool C: 1700–2000; Practice D: 1300–1450. A story is set in 1550 and mentions Technology A and Tool C together. Cross-domain check: a survey team starts work at 8:00 and works for 2 hour(s) without a break.

Rules. An item is temporally compatible with a year only if the year lies inside its stated availability interval. An anachronism is an item placed outside its given interval. For the cross-domain check, finishing time = starting time + duration.

Task. Which mentioned item, if any, is anachronistic in the story? Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. Check 1550 against Technology A interval 1200–1600. 2. Check 1550 against Tool C interval 1700–2000. 3. Items outside their intervals: ['Tool C']. Cross-domain check: 8+2=10, so the team finishes at 10:00.

Answer. Anachronistic: Tool C. Cross-domain answer: 10:00.

Knowledge takeaway. The reasoning does not require memorizing invention dates if the intervals are part of the problem.

Formal model. Interval membership test for each object-event pair.

Problem 938. Temporal compatibility in a story set in 1750

Knowledge context. Anachronism can be tested as a compatibility problem when the relevant date ranges are supplied.

Given facts. For this problem, availability intervals are: Technology A: 1200–1600; Institution B: 1500–1900; Tool C: 1700–2000; Practice D: 1300–1450. A story is set in 1750 and mentions Technology A and Tool C together. Cross-domain check: a local committee has 10 members, requires at least 6 present for quorum, and 8 are present.

Rules. An item is temporally compatible with a year only if the year lies inside its stated availability interval. An anachronism is an item placed outside its given interval. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. Which mentioned item, if any, is anachronistic in the story? Cross-domain check: is quorum met?

Step-by-step solution. 1. Check 1750 against Technology A interval 1200–1600. 2. Check 1750 against Tool C interval 1700–2000. 3. Items outside their intervals: ['Technology A']. Cross-domain check: compare 8 with 6; $8 \geq 6$ is true.

Answer. Anachronistic: Technology A. Cross-domain answer: quorum is met.

Knowledge takeaway. The reasoning does not require memorizing invention dates if the intervals are part of the problem.

Formal model. Interval membership test for each object-event pair.

Problem 939. Temporal compatibility in a story set in 1850

Knowledge context. Anachronism can be tested as a compatibility problem when the relevant date ranges are supplied.

Given facts. For this problem, availability intervals are: Technology A: 1200–1600; Institution B: 1500–1900; Tool C: 1700–2000; Practice D: 1300–1450. A story is set in 1850 and mentions Technology A and Tool C together. Cross-domain check: researchers collected 15 reports, but 4 are exact duplicate copies that add no new independent information. Mixed-domain verification: a regional archive has 95 map sheets and receives 10 additional sheets during the exercise.

Rules. An item is temporally compatible with a year only if the year lies inside its stated availability interval. An anachronism is an item placed outside its given interval. For the cross-domain check, independent reports = total reports − duplicate copies. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. Which mentioned item, if any, is anachronistic in the story? Cross-domain check: how many non-duplicate reports remain? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. Check 1850 against Technology A interval 1200–1600. 2. Check 1850 against Tool C interval 1700–2000. 3. Items outside their intervals: ['Technology A']. Cross-domain check: $15-4=11$ independent reports remain. Mixed-domain verification: $95+10=105$.

Answer. Anachronistic: Technology A. Cross-domain answer: 11 independent reports. Mixed-domain answer: 105 map sheets.

Knowledge takeaway. The reasoning does not require memorizing invention dates if the intervals are part of the problem.

Formal model. Interval membership test for each object-event pair.

Problem 940. Temporal compatibility in a story set in 1500

Knowledge context. Anachronism can be tested as a compatibility problem when the relevant date ranges are supplied.

Given facts. For this problem, availability intervals are: Technology A: 1200–1600; Institution B: 1500–1900; Tool C: 1700–2000; Practice D: 1300–1450. A story is set in 1500 and mentions Technology A and Tool C together. Cross-domain check: on a field sketch, 1 cm represents 2 km and two checkpoints are 4 cm apart.

Rules. An item is temporally compatible with a year only if the year lies inside its stated availability interval. An anachronism is an item placed outside its given interval. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Which mentioned item, if any, is anachronistic in the story? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. Check 1500 against Technology A interval 1200–1600. 2. Check 1500 against Tool C interval 1700–2000. 3. Items outside their intervals: ['Tool C']. Cross-domain check: $4\times2=8$ km.

Answer. Anachronistic: Tool C. Cross-domain answer: 8 km.

Knowledge takeaway. The reasoning does not require memorizing invention dates if the intervals are part of the problem.

Formal model. Interval membership test for each object-event pair.

N14. Provenance and chain of custody

Problem 941. Provenance and chain of custody: case 1

Knowledge context. Provenance records where an object came from and how it moved between custodians.

Given facts. An artifact record states: found by Excavator on day 1; transferred to Museum A on day 2; sent from Museum A to Research Lab on day 5; returned by Research Lab to Museum B on day 8. Every transfer must name both sender and receiver. Cross-domain check: a survey team starts work at 12:00 and works for 2 hour(s) without a break. Mixed-domain verification: a regional archive has 96 map sheets and receives 2 additional sheets during the exercise.

Rules. A continuous chain of custody requires each handoff to connect the previous holder to the next. A missing sender or receiver creates a gap that limits certainty about continuous custody. For the cross-domain check, finishing time = starting time + duration. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. Is the stated chain of custody complete under the rule? If not, identify the gap. Cross-domain check: at what time does the team finish? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. Read the transfers in time order. 2. Check that every transfer links the previous holder to the next. 3. Every handoff names both sides and connects continuously. Cross-domain check: 12+2=14, so the team finishes at 14:00. Mixed-domain verification: 96+2=98.

Answer. The chain is complete under the stated rule. Cross-domain answer: 14:00. Mixed-domain answer: 98 map sheets.

Knowledge takeaway. A strong provenance claim depends on an unbroken sequence of documented handoffs.

Formal model. Time-ordered path through holders; verify continuity of sender/receiver pairs.

Problem 942. Provenance and chain of custody: case 2

Knowledge context. Provenance records where an object came from and how it moved between custodians.

Given facts. An artifact record states: found by Excavator on day 1; transferred to Museum A on day 2; sent from Museum A to Research Lab on day 5; returned by Research Lab to Museum B on day 8. Every transfer must name both sender and receiver. The record for day 5 names Research Lab as receiver but does not name the sender. Cross-domain check: a local committee has 14 members, requires at least 8 present for quorum, and 14 are present.

Rules. A continuous chain of custody requires each handoff to connect the previous holder to the next. A missing sender or receiver creates a gap that limits certainty about continuous custody. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. Is the stated chain of custody complete under the rule? If not, identify the gap. Cross-domain check: is quorum met?

Step-by-step solution. 1. Read the transfers in time order. 2. Check that every transfer links the previous holder to the next. 3. The day-5 record lacks a sender, so the chain has a gap. Cross-domain check: compare 14 with 8; $14 \geq 8$ is true.

Answer. The chain is incomplete because the day-5 sender is missing. Cross-domain answer: quorum is met.

Knowledge takeaway. A strong provenance claim depends on an unbroken sequence of documented handoffs.

Formal model. Time-ordered path through holders; verify continuity of sender/receiver pairs.

Problem 943. Provenance and chain of custody: case 3

Knowledge context. Provenance records where an object came from and how it moved between custodians.

Given facts. An artifact record states: found by Excavator on day 1; transferred to Museum A on day 2; sent from Museum A to Research Lab on day 5; returned by Research Lab to Museum B on day 8. Every transfer must name both sender and receiver. Cross-domain check: researchers collected 9 reports, but 4 are exact duplicate copies that add no new independent information. Mixed-domain verification: a regional archive has 96 map sheets and receives 4 additional sheets during the exercise.

Rules. A continuous chain of custody requires each handoff to connect the previous holder to the next. A missing sender or receiver creates a gap that limits certainty about continuous custody. For the cross-domain check, independent reports = total reports − duplicate copies. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. Is the stated chain of custody complete under the rule? If not, identify the gap. Cross-domain check: how many non-duplicate reports remain? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. Read the transfers in time order. 2. Check that every transfer links the previous holder to the next. 3. Every handoff names both sides and connects continuously. Cross-domain check: 9−4=5 independent reports remain. Mixed-domain verification: 96+4=100.

Answer. The chain is complete under the stated rule. Cross-domain answer: 5 independent reports. Mixed-domain answer: 100 map sheets.

Knowledge takeaway. A strong provenance claim depends on an unbroken sequence of documented handoffs.

Formal model. Time-ordered path through holders; verify continuity of sender/receiver pairs.

Problem 944. Provenance and chain of custody: case 4

Knowledge context. Provenance records where an object came from and how it moved between custodians.

Given facts. An artifact record states: found by Excavator on day 1; transferred to Museum A on day 2; sent from Museum A to Research Lab on day 5; returned by Research Lab to Museum B on day 8. Every transfer must name both sender and receiver. The record for day 5 names Research Lab as receiver but does not name the sender. Cross-domain check: on a field sketch, 1 cm represents 6 km and two checkpoints are 8 cm apart.

Rules. A continuous chain of custody requires each handoff to connect the previous holder to the next. A missing sender or receiver creates a gap that limits certainty about continuous custody. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Is the stated chain of custody complete under the rule? If not, identify the gap. Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. Read the transfers in time order. 2. Check that every transfer links the previous holder to the next. 3. The day-5 record lacks a sender, so the chain has a gap. Cross-domain check: 8×6=48 km.

Answer. The chain is incomplete because the day-5 sender is missing. Cross-domain answer: 48 km.

Knowledge takeaway. A strong provenance claim depends on an unbroken sequence of documented handoffs.

Formal model. Time-ordered path through holders; verify continuity of sender/receiver pairs.

Problem 945. Provenance and chain of custody: case 5

Knowledge context. Provenance records where an object came from and how it moved between custodians.

Given facts. An artifact record states: found by Excavator on day 1; transferred to Museum A on day 2; sent from Museum A to Research Lab on day 5; returned by Research Lab to Museum B on day 8. Every transfer must name both sender and receiver. Cross-domain check: a survey team starts work at 8:00 and works for 2 hour(s) without a break. Mixed-domain verification: a regional archive has 96 map sheets and receives 6 additional sheets during the exercise.

Rules. A continuous chain of custody requires each handoff to connect the previous holder to the next. A missing sender or receiver creates a gap that limits certainty about continuous custody. For the cross-domain check, finishing time = starting time + duration. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. Is the stated chain of custody complete under the rule? If not, identify the gap. Cross-domain check: at what time does the team finish? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. Read the transfers in time order. 2. Check that every transfer links the previous holder to the next. 3. Every handoff names both sides and connects continuously. Cross-domain check: 8+2=10, so the team finishes at 10:00. Mixed-domain verification: 96+6=102.

Answer. The chain is complete under the stated rule. Cross-domain answer: 10:00. Mixed-domain answer: 102 map sheets.

Knowledge takeaway. A strong provenance claim depends on an unbroken sequence of documented handoffs.

Formal model. Time-ordered path through holders; verify continuity of sender/receiver pairs.

N15. Perspective, interest, and source credibility

Problem 946. Perspective, interest, and source quality: case 1

Knowledge context. Sources have perspectives and interests. Credibility is not a simple label; it depends on the claim being evaluated and how the source could know it.

Given facts. Three sources describe a proposed market tax. Source A is a merchant who would pay the tax and argues it is harmful. Source B is a council report that estimates revenue and lists its calculation. Source C is a traveler repeating a rumor heard in another town. Cross-domain check: a local committee has 9 members, requires at least 5 present for quorum, and 8 are present.

Rules. Perspective and interest do not automatically make a source false. Evaluate relevance, access to information, explicit evidence, and independence. A source with an interest may still provide useful facts, but its claims should be checked. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. Which source is best positioned to estimate the amount of revenue, and which source should be treated most cautiously for that numerical claim? Cross-domain check: is quorum met?

Step-by-step solution. 1. Revenue estimation requires access to tax rates and expected transactions. 2. Source B explicitly provides a calculation and is directly about the policy. 3. Source C is a secondhand rumor from elsewhere and has the weakest access to the specific numerical question. 4. Source A has an interest, but interest alone is not enough to discard it. Cross-domain check: compare 8 with 5; $8 \geq 5$ is true.

Answer. Source B is best positioned for the revenue estimate; Source C should be treated most cautiously for that numerical claim. Cross-domain answer: quorum is met.

Knowledge takeaway. Bias is a reason to inspect evidence, not a magic switch that makes every statement true or false.

Formal model. Source-claim assessment using access, relevance, independence, and evidentiary support.

Problem 947. Perspective, interest, and source quality: case 2

Knowledge context. Sources have perspectives and interests. Credibility is not a simple label; it depends on the claim being evaluated and how the source could know it.

Given facts. Three sources describe a proposed market tax. Source A is a merchant who would pay the tax and argues it is harmful. Source B is a council report that estimates revenue and lists its calculation. Source C is a traveler repeating a rumor heard in another town. Cross-domain check: researchers collected 13 reports, but 4 are exact duplicate copies that add no new independent information. Mixed-domain verification: a regional archive has 96 map sheets and receives 8 additional sheets during the exercise.

Rules. Perspective and interest do not automatically make a source false. Evaluate relevance, access to information, explicit evidence, and independence. A source with an interest may still provide useful facts, but its claims should be checked. For the cross-domain check, independent reports = total reports − duplicate copies. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. Which source is best positioned to estimate the amount of revenue, and which source should be treated most cautiously for that numerical claim? Cross-domain check: how many non-duplicate reports remain? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. Revenue estimation requires access to tax rates and expected transactions. 2. Source B explicitly provides a calculation and is directly about the policy. 3. Source C is a secondhand rumor from elsewhere and has the weakest access to the specific numerical question. 4. Source A has an interest, but interest alone is not enough to discard it. Cross-domain check: 13−4=9 independent reports remain. Mixed-domain verification: 96+8=104.

Answer. Source B is best positioned for the revenue estimate; Source C should be treated most cautiously for that numerical claim. Cross-domain answer: 9 independent reports. Mixed-domain answer: 104 map sheets.

Knowledge takeaway. Bias is a reason to inspect evidence, not a magic switch that makes every statement true or false.

Formal model. Source-claim assessment using access, relevance, independence, and evidentiary support.

Problem 948. Perspective, interest, and source quality: case 3

Knowledge context. Sources have perspectives and interests. Credibility is not a simple label; it depends on the claim being evaluated and how the source could know it.

Given facts. Three sources describe a proposed market tax. Source A is a merchant who would pay the tax and argues it is harmful. Source B is a council report that estimates revenue and lists its calculation. Source C is a traveler repeating a rumor heard in another town. Cross-domain check: on a field sketch, 1 cm represents 5 km and two checkpoints are 5 cm apart.

Rules. Perspective and interest do not automatically make a source false. Evaluate relevance, access to information, explicit evidence, and independence. A source with an interest may still provide useful facts, but its claims should be checked. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Which source is best positioned to estimate the amount of revenue, and which source should be treated most cautiously for that numerical claim? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. Revenue estimation requires access to tax rates and expected transactions. 2. Source B explicitly provides a calculation and is directly about the policy. 3. Source C is a secondhand rumor from elsewhere and has the weakest access to the specific numerical question. 4. Source A has an interest, but interest alone is not enough to discard it. Cross-domain check: 5×5=25 km.

Answer. Source B is best positioned for the revenue estimate; Source C should be treated most cautiously for that numerical claim. Cross-domain answer: 25 km.

Knowledge takeaway. Bias is a reason to inspect evidence, not a magic switch that makes every statement true or false.

Formal model. Source-claim assessment using access, relevance, independence, and evidentiary support.

Problem 949. Perspective, interest, and source quality: case 4

Knowledge context. Sources have perspectives and interests. Credibility is not a simple label; it depends on the claim being evaluated and how the source could know it.

Given facts. Three sources describe a proposed market tax. Source A is a merchant who would pay the tax and argues it is harmful. Source B is a council report that estimates revenue and lists its calculation. Source C is a traveler repeating a rumor heard in another town. Cross-domain check: a survey team starts work at 12:00 and works for 2 hour(s) without a

break. Mixed-domain verification: a regional archive has 96 map sheets and receives 10 additional sheets during the exercise.

Rules. Perspective and interest do not automatically make a source false. Evaluate relevance, access to information, explicit evidence, and independence. A source with an interest may still provide useful facts, but its claims should be checked. For the cross-domain check, finishing time = starting time + duration. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. Which source is best positioned to estimate the amount of revenue, and which source should be treated most cautiously for that numerical claim? Cross-domain check: at what time does the team finish? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. Revenue estimation requires access to tax rates and expected transactions. 2. Source B explicitly provides a calculation and is directly about the policy. 3. Source C is a secondhand rumor from elsewhere and has the weakest access to the specific numerical question. 4. Source A has an interest, but interest alone is not enough to discard it. Cross-domain check: 12+2=14, so the team finishes at 14:00. Mixed-domain verification: 96+10=106.

Answer. Source B is best positioned for the revenue estimate; Source C should be treated most cautiously for that numerical claim. Cross-domain answer: 14:00. Mixed-domain answer: 106 map sheets.

Knowledge takeaway. Bias is a reason to inspect evidence, not a magic switch that makes every statement true or false.

Formal model. Source-claim assessment using access, relevance, independence, and evidentiary support.

Problem 950. Perspective, interest, and source quality: case 5

Knowledge context. Sources have perspectives and interests. Credibility is not a simple label; it depends on the claim being evaluated and how the source could know it.

Given facts. Three sources describe a proposed market tax. Source A is a merchant who would pay the tax and argues it is harmful. Source B is a council report that estimates revenue and lists its calculation. Source C is a traveler repeating a rumor heard in another town. Cross-domain check: a local committee has 13 members, requires at least 7 present for quorum, and 11 are present. Mixed-domain verification: a regional archive has 97 map sheets and receives 1 additional sheets during the exercise.

Rules. Perspective and interest do not automatically make a source false. Evaluate relevance, access to information, explicit evidence, and independence. A source with an interest may still provide useful facts, but its claims should be checked. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. Which source is best positioned to estimate the amount of revenue, and which source should be treated most cautiously for that numerical claim? Cross-domain check: is quorum met? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. Revenue estimation requires access to tax rates and expected transactions. 2. Source B explicitly provides a calculation and is directly about the policy. 3. Source C is a secondhand rumor from elsewhere and has the weakest access to the specific numerical question. 4. Source A has an interest, but interest alone is not enough to discard it. Cross-domain check: compare 11 with 7; 11 $\geq$ 7 is true. Mixed-domain verification: 97+1=98.

Answer. Source B is best positioned for the revenue estimate; Source C should be treated most cautiously for that numerical claim. Cross-domain answer: quorum is met. Mixed-domain answer: 98 map sheets.

Knowledge takeaway. Bias is a reason to inspect evidence, not a magic switch that makes every statement true or false.

Formal model. Source-claim assessment using access, relevance, independence, and evidentiary support.

N16. Uncertain data, intervals, and limits

Problem 951. Reasoning with uncertain intervals: case 1

Knowledge context. Measurements and historical dates can be uncertain. An interval represents a range of possibilities rather than one exact number.

Given facts. Two measurements are uncertain intervals. A is between 8 and 12. B is between 14 and 18. A statement “A is definitely greater than B” is valid only if A’s minimum is greater than B’s maximum.

Rules. For a definite comparison under interval uncertainty, test the worst cases. Overlapping intervals usually prevent a strict guaranteed ordering.

Task. Can we conclude that A is definitely greater than B? What comparison is justified?

Step-by-step solution. 1. A minimum=8, maximum=12; B minimum=14, maximum=18. 2. Test A_min > B_max: 8>18 is False. 3. If false, at least one allowed pair of values defeats the “definitely greater” claim.

Answer. No. The intervals do not justify saying A is definitely greater than B.

Knowledge takeaway. Robust conclusions must remain true across all values allowed by the uncertainty.

Formal model. Interval arithmetic / worst-case comparison.

Problem 952. Reasoning with uncertain intervals: case 2

Knowledge context. Measurements and historical dates can be uncertain. An interval represents a range of possibilities rather than one exact number.

Given facts. Two measurements are uncertain intervals. A is between 9 and 13. B is between 14 and 18. A statement “A is definitely greater than B” is valid only if A’s minimum is greater than B’s maximum.

Rules. For a definite comparison under interval uncertainty, test the worst cases. Overlapping intervals usually prevent a strict guaranteed ordering.

Task. Can we conclude that A is definitely greater than B? What comparison is justified?

Step-by-step solution. 1. A minimum=9, maximum=13; B minimum=14, maximum=18. 2. Test A_min > B_max: 9>18 is False. 3. If false, at least one allowed pair of values defeats the “definitely greater” claim.

Answer. No. The intervals do not justify saying A is definitely greater than B.

Knowledge takeaway. Robust conclusions must remain true across all values allowed by the uncertainty.

Formal model. Interval arithmetic / worst-case comparison.

Problem 953. Reasoning with uncertain intervals: case 3

Knowledge context. Measurements and historical dates can be uncertain. An interval represents a range of possibilities rather than one exact number.

Given facts. Two measurements are uncertain intervals. A is between 10 and 14. B is between 14 and 18. A statement “A is definitely greater than B” is valid only if A’s minimum is greater than B’s maximum.

Rules. For a definite comparison under interval uncertainty, test the worst cases. Overlapping intervals usually prevent a strict guaranteed ordering.

Task. Can we conclude that A is definitely greater than B? What comparison is justified?

Step-by-step solution. 1. A minimum=10, maximum=14; B minimum=14, maximum=18. 2. Test A_min > B_max: 10>18 is False. 3. If false, at least one allowed pair of values defeats the “definitely greater” claim.

Answer. No. The intervals do not justify saying A is definitely greater than B.

Knowledge takeaway. Robust conclusions must remain true across all values allowed by the uncertainty.

Formal model. Interval arithmetic / worst-case comparison.

Problem 954. Reasoning with uncertain intervals: case 4

Knowledge context. Measurements and historical dates can be uncertain. An interval represents a range of possibilities rather than one exact number.

Given facts. Two measurements are uncertain intervals. A is between 11 and 15. B is between 14 and 18. A statement “A is definitely greater than B” is valid only if A’s minimum is greater than B’s maximum.

Rules. For a definite comparison under interval uncertainty, test the worst cases. Overlapping intervals usually prevent a strict guaranteed ordering.

Task. Can we conclude that A is definitely greater than B? What comparison is justified?

Step-by-step solution. 1. A minimum=11, maximum=15; B minimum=14, maximum=18. 2. Test A_min > B_max: 11>18 is False. 3. If false, at least one allowed pair of values defeats the “definitely greater” claim.

Answer. No. The intervals do not justify saying A is definitely greater than B.

Knowledge takeaway. Robust conclusions must remain true across all values allowed by the uncertainty.

Formal model. Interval arithmetic / worst-case comparison.

Problem 955. Reasoning with uncertain intervals: case 5

Knowledge context. Measurements and historical dates can be uncertain. An interval represents a range of possibilities rather than one exact number.

Given facts. Two measurements are uncertain intervals. A is between 12 and 16. B is between 14 and 18. A statement “A is definitely greater than B” is valid only if A’s minimum is greater than B’s maximum.

Rules. For a definite comparison under interval uncertainty, test the worst cases. Overlapping intervals usually prevent a strict guaranteed ordering.

Task. Can we conclude that A is definitely greater than B? What comparison is justified?

Step-by-step solution. 1. A minimum=12, maximum=16; B minimum=14, maximum=18. 2. Test A_min > B_max: 12>18 is False. 3. If false, at least one allowed pair of values defeats the “definitely greater” claim.

Answer. No. The intervals do not justify saying A is definitely greater than B.

Knowledge takeaway. Robust conclusions must remain true across all values allowed by the uncertainty.

Formal model. Interval arithmetic / worst-case comparison.

N17. Counterfactual reasoning

Problem 956. Counterfactual intervention: close the gate

Knowledge context. Counterfactual reasoning asks what the model predicts if one condition were changed while the rest of the causal structure is held fixed.

Given facts. Causal rules for a simplified town model: heavy rain → high river; high river AND unclosed gate → flooded square; flooded square → market closes. Actual facts: heavy rain occurred and the gate was not closed. Cross-domain check: on a field sketch, 1 cm represents 3 km and two checkpoints are 6 cm apart.

Rules. A counterfactual intervention changes one specified fact and then recomputes consequences from the rules. Keep all other stated rules the same unless the intervention changes them. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Under the intervention “close the gate”, what changes in the consequences? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. Actual chain gives high river, then flood, then market closure. 2. Intervene: set gate=closed. 3. The flood rule requires an unclosed gate, so flooding no longer follows. 4. Without flooding, market closure does not follow. Cross-domain check: 6×3=18 km.

Answer. With the gate closed, the model no longer implies flooding or market closure. Cross-domain answer: 18 km.

Knowledge takeaway. A counterfactual is not ordinary storytelling; it is a controlled change followed by recomputation.

Formal model. Structural causal rules with a do-style intervention on one variable.

Problem 957. Counterfactual intervention: remove heavy rain

Knowledge context. Counterfactual reasoning asks what the model predicts if one condition were changed while the rest of the causal structure is held fixed.

Given facts. Causal rules for a simplified town model: heavy rain → high river; high river AND unclosed gate → flooded square; flooded square → market closes. Actual facts: heavy rain occurred and the gate was not closed. Cross-domain check: a survey team starts work at 12:00 and works for 2 hour(s) without a break.

Rules. A counterfactual intervention changes one specified fact and then recomputes consequences from the rules. Keep all other stated rules the same unless the intervention changes them. For the cross-domain check, finishing time = starting time + duration.

Task. Under the intervention “remove heavy rain”, what changes in the consequences? Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. Intervene: heavy rain=false. 2. Then high river is not generated by the first rule. 3. Without high river, flooding and market closure are not generated. Cross-domain check: 12+2=14, so the team finishes at 14:00.

Answer. Without heavy rain, the model generates neither high river, flood, nor market closure. Cross-domain answer: 14:00.

Knowledge takeaway. A counterfactual is not ordinary storytelling; it is a controlled change followed by recomputation.

Formal model. Structural causal rules with a do-style intervention on one variable.

Problem 958. Counterfactual intervention: keep gate open

Knowledge context. Counterfactual reasoning asks what the model predicts if one condition were changed while the rest of the causal structure is held fixed.

Given facts. Causal rules for a simplified town model: heavy rain → high river; high river AND unclosed gate → flooded square; flooded square → market closes. Actual facts: heavy rain occurred and the gate was not closed. Cross-domain check: a local committee has 12 members, requires at least 7 present for quorum, and 11 are present.

Rules. A counterfactual intervention changes one specified fact and then recomputes consequences from the rules. Keep all other stated rules the same unless the intervention changes them. For the cross-domain check, quorum exists when present members ≥ the stated threshold.

Task. Under the intervention “keep gate open”, what changes in the consequences? Cross-domain check: is quorum met?

Step-by-step solution. 1. Gate already remains open. 2. Heavy rain gives high river. 3. High river plus open gate gives flood, then market closure. Cross-domain check: compare 11 with 7; 11 ≥ 7 is true.

Answer. The original chain remains: flood and market closure occur. Cross-domain answer: quorum is met.

Knowledge takeaway. A counterfactual is not ordinary storytelling; it is a controlled change followed by recomputation.

Formal model. Structural causal rules with a do-style intervention on one variable.

Problem 959. Counterfactual intervention: prevent high river

Knowledge context. Counterfactual reasoning asks what the model predicts if one condition were changed while the rest of the causal structure is held fixed.

Given facts. Causal rules for a simplified town model: heavy rain → high river; high river AND unclosed gate → flooded square; flooded square → market closes. Actual facts: heavy rain occurred and the gate was not closed. Cross-domain check: researchers collected 15 reports, but 4 are exact duplicate copies that add no new independent information. Mixed-domain verification: a regional archive has 97 map sheets and receives 10 additional sheets during the exercise.

Rules. A counterfactual intervention changes one specified fact and then recomputes consequences from the rules. Keep all other stated rules the same unless the intervention changes them. For the cross-domain check, independent reports = total reports − duplicate copies. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. Under the intervention “prevent high river”, what changes in the consequences? Cross-domain check: how many non-duplicate reports remain? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. Intervene directly: high river=false. 2. Even with rain, the counterfactual intervention fixes high river as prevented for this question. 3. Flood requires high river, so it does not follow. Cross-domain check: 15−4=11 independent reports remain. Mixed-domain verification: 97+10=107.

Answer. Preventing the high river blocks the flood and market closure in this model. Cross-domain answer: 11 independent reports. Mixed-domain answer: 107 map sheets.

Knowledge takeaway. A counterfactual is not ordinary storytelling; it is a controlled change followed by recomputation.

Formal model. Structural causal rules with a do-style intervention on one variable.

Problem 960. Counterfactual intervention: keep market open

Knowledge context. Counterfactual reasoning asks what the model predicts if one condition were changed while the rest of the causal structure is held fixed.

Given facts. Causal rules for a simplified town model: heavy rain → high river; high river AND unclosed gate → flooded square; flooded square → market closes. Actual facts: heavy rain occurred and the gate was not closed. Cross-domain check: on a field sketch, 1 cm represents 2 km and two checkpoints are 3 cm apart.

Rules. A counterfactual intervention changes one specified fact and then recomputes consequences from the rules. Keep all other stated rules the same unless the intervention changes them. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Under the intervention “keep market open”, what changes in the consequences? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. The proposed intervention changes the market outcome directly, not the upstream causes. 2. Rain still gives high river and flood. 3. Setting market=open does not erase the flood unless a rule says so. Cross-domain check: 3×2=6 km.

Answer. The square still floods; the market is held open only by the direct intervention. Cross-domain answer: 6 km.

Knowledge takeaway. A counterfactual is not ordinary storytelling; it is a controlled change followed by recomputation.

Formal model. Structural causal rules with a do-style intervention on one variable.

N18. Public goods and common contributions

Problem 961. Funding a shared public project: case 1

Knowledge context. Public goods can benefit many people at once, which creates coordination problems about contributions.

Given facts. A community project is built only if total voluntary contributions reach 16 tokens. Contributions from 8 households are [1, 2, 3, 4, 1, 2, 3, 4]. Everyone can use the project if it is built, including households that contribute zero.

Rules. Add contributions and compare with the threshold. A public-good threshold concerns whether the project is produced; it does not by itself decide whether every contribution pattern is fair.

Task. Is the project built? How many additional tokens, if any, are needed?

Step-by-step solution. 1. Total contribution=1+2+3+4+1+2+3+4=20. 2. Compare 20 with threshold 16. 3. Additional amount needed=max(0,16−20)=0.

Answer. The project is built. Additional tokens needed: 0.

Knowledge takeaway. Threshold provision and fairness are separate questions; first determine whether enough resources exist.

Formal model. Sum contributions; compare with provision threshold; compute shortfall.

Problem 962. Funding a shared public project: case 2

Knowledge context. Public goods can benefit many people at once, which creates coordination problems about contributions.

Given facts. A community project is built only if total voluntary contributions reach 16 tokens. Contributions from 8 households are [2, 3, 4, 1, 2, 3, 4, 1]. Everyone can use the project if it is built, including households that contribute zero.

Rules. Add contributions and compare with the threshold. A public-good threshold concerns whether the project is produced; it does not by itself decide whether every contribution pattern is fair.

Task. Is the project built? How many additional tokens, if any, are needed?

Step-by-step solution. 1. Total contribution=2+3+4+1+2+3+4+1=20. 2. Compare 20 with threshold 16. 3. Additional amount needed=max(0,16−20)=0.

Answer. The project is built. Additional tokens needed: 0.

Knowledge takeaway. Threshold provision and fairness are separate questions; first determine whether enough resources exist.

Formal model. Sum contributions; compare with provision threshold; compute shortfall.

Problem 963. Funding a shared public project: case 3

Knowledge context. Public goods can benefit many people at once, which creates coordination problems about contributions.

Given facts. A community project is built only if total voluntary contributions reach 16 tokens. Contributions from 8 households are [3, 4, 1, 2, 3, 4, 1, 2]. Everyone can use the project if it is built, including households that contribute zero.

Rules. Add contributions and compare with the threshold. A public-good threshold concerns whether the project is produced; it does not by itself decide whether every contribution pattern is fair.

Task. Is the project built? How many additional tokens, if any, are needed?

Step-by-step solution. 1. Total contribution=3+4+1+2+3+4+1+2=20. 2. Compare 20 with threshold 16. 3. Additional amount needed=max(0,16−20)=0.

Answer. The project is built. Additional tokens needed: 0.

Knowledge takeaway. Threshold provision and fairness are separate questions; first determine whether enough resources exist.

Formal model. Sum contributions; compare with provision threshold; compute shortfall.

Problem 964. Funding a shared public project: case 4

Knowledge context. Public goods can benefit many people at once, which creates coordination problems about contributions.

Given facts. A community project is built only if total voluntary contributions reach 16 tokens. Contributions from 8 households are [4, 1, 2, 3, 4, 1, 2, 3]. Everyone can use the project if it is built, including households that contribute zero.

Rules. Add contributions and compare with the threshold. A public-good threshold concerns whether the project is produced; it does not by itself decide whether every contribution pattern is fair.

Task. Is the project built? How many additional tokens, if any, are needed?

Step-by-step solution. 1. Total contribution=4+1+2+3+4+1+2+3=20. 2. Compare 20 with threshold 16. 3. Additional amount needed=max(0,16−20)=0.

Answer. The project is built. Additional tokens needed: 0.

Knowledge takeaway. Threshold provision and fairness are separate questions; first determine whether enough resources exist.

Formal model. Sum contributions; compare with provision threshold; compute shortfall.

Problem 965. Funding a shared public project: case 5

Knowledge context. Public goods can benefit many people at once, which creates coordination problems about contributions.

Given facts. A community project is built only if total voluntary contributions reach 16 tokens. Contributions from 8 households are [1, 2, 3, 4, 1, 2, 3, 4]. Everyone can use the project if it is built, including households that contribute zero. Cross-domain check: a survey team starts work at 12:00 and works for 2 hour(s) without a break.

Rules. Add contributions and compare with the threshold. A public-good threshold concerns whether the project is produced; it does not by itself decide whether every contribution pattern is fair. For the cross-domain check, finishing time = starting time + duration.

Task. Is the project built? How many additional tokens, if any, are needed? Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. Total contribution=1+2+3+4+1+2+3+4=20. 2. Compare 20 with threshold 16. 3. Additional amount needed=max(0,16−20)=0. Cross-domain check: 12+2=14, so the team finishes at 14:00.

Answer. The project is built. Additional tokens needed: 0. Cross-domain answer: 14:00.

Knowledge takeaway. Threshold provision and fairness are separate questions; first determine whether enough resources exist.

Formal model. Sum contributions; compare with provision threshold; compute shortfall.

N19. Common resources and avoiding depletion

Problem 966. Using a shared renewable resource: case 1

Knowledge context. A common resource can be overused when many individually reasonable withdrawals add up to more than regeneration.

Given facts. A shared fishery has 100 fish at the start of the period and regenerates 10. There are 7 users, each planning to take 3. For this simplified exercise, the stock should not decline below its starting level if use is to be called non-depleting.

Rules. End stock = start + regeneration − total harvest. Total harvest is number of users × harvest per user.

Task. What is the end stock? Is the planned use non-depleting under the stated definition? What equal per-user harvest would exactly match regeneration?

Step-by-step solution. 1. Total harvest=7×3=21. 2. End stock=100+10−21=89. 3. To match regeneration exactly, equal harvest per user would be 10÷7=1.43.

Answer. End stock 89; depleting. Equal regeneration-matching share: 1.43 each.

Knowledge takeaway. The shared-resource problem appears at the total level, so individual plans must be aggregated.

Formal model. Commons stock equation plus equal-share benchmark.

Problem 967. Using a shared renewable resource: case 2

Knowledge context. A common resource can be overused when many individually reasonable withdrawals add up to more than regeneration.

Given facts. A shared fishery has 100 fish at the start of the period and regenerates 11. There are 7 users, each planning to take 4. For this simplified exercise, the stock should not decline below its starting level if use is to be called non-depleting.

Rules. End stock = start + regeneration − total harvest. Total harvest is number of users × harvest per user.

Task. What is the end stock? Is the planned use non-depleting under the stated definition? What equal per-user harvest would exactly match regeneration?

Step-by-step solution. 1. Total harvest=7×4=28. 2. End stock=100+11−28=83. 3. To match regeneration exactly, equal harvest per user would be 11÷7=1.57.

Answer. End stock 83; depleting. Equal regeneration-matching share: 1.57 each.

Knowledge takeaway. The shared-resource problem appears at the total level, so individual plans must be aggregated.

Formal model. Commons stock equation plus equal-share benchmark.

Problem 968. Using a shared renewable resource: case 3

Knowledge context. A common resource can be overused when many individually reasonable withdrawals add up to more than regeneration.

Given facts. A shared fishery has 100 fish at the start of the period and regenerates 12. There are 7 users, each planning to take 5. For this simplified exercise, the stock should not decline below its starting level if use is to be called non-depleting.

Rules. End stock = start + regeneration − total harvest. Total harvest is number of users × harvest per user.

Task. What is the end stock? Is the planned use non-depleting under the stated definition? What equal per-user harvest would exactly match regeneration?

Step-by-step solution. 1. Total harvest=7×5=35. 2. End stock=100+12−35=77. 3. To match regeneration exactly, equal harvest per user would be 12÷7=1.71.

Answer. End stock 77; depleting. Equal regeneration-matching share: 1.71 each.

Knowledge takeaway. The shared-resource problem appears at the total level, so individual plans must be aggregated.

Formal model. Commons stock equation plus equal-share benchmark.

Problem 969. Using a shared renewable resource: case 4

Knowledge context. A common resource can be overused when many individually reasonable withdrawals add up to more than regeneration.

Given facts. A shared fishery has 100 fish at the start of the period and regenerates 13. There are 7 users, each planning to take 6. For this simplified exercise, the stock should not decline below its starting level if use is to be called non-depleting.

Rules. End stock = start + regeneration − total harvest. Total harvest is number of users × harvest per user.

Task. What is the end stock? Is the planned use non-depleting under the stated definition? What equal per-user harvest would exactly match regeneration?

Step-by-step solution. 1. Total harvest=7×6=42. 2. End stock=100+13−42=71. 3. To match regeneration exactly, equal harvest per user would be 13÷7=1.86.

Answer. End stock 71; depleting. Equal regeneration-matching share: 1.86 each.

Knowledge takeaway. The shared-resource problem appears at the total level, so individual plans must be aggregated.

Formal model. Commons stock equation plus equal-share benchmark.

Problem 970. Using a shared renewable resource: case 5

Knowledge context. A common resource can be overused when many individually reasonable withdrawals add up to more than regeneration.

Given facts. A shared fishery has 100 fish at the start of the period and regenerates 14. There are 7 users, each planning to take 7. For this simplified exercise, the stock should not decline below its starting level if use is to be called non-depleting.

Rules. End stock = start + regeneration − total harvest. Total harvest is number of users × harvest per user.

Task. What is the end stock? Is the planned use non-depleting under the stated definition? What equal per-user harvest would exactly match regeneration?

Step-by-step solution. 1. Total harvest=7×7=49. 2. End stock=100+14−49=65. 3. To match regeneration exactly, equal harvest per user would be 14÷7=2.00.

Answer. End stock 65; depleting. Equal regeneration-matching share: 2.00 each.

Knowledge takeaway. The shared-resource problem appears at the total level, so individual plans must be aggregated.

Formal model. Commons stock equation plus equal-share benchmark.

N20. Fairness: equal, proportional, and need-based

Problem 971. Fair allocation by the rule: equal

Knowledge context. Fairness is not one formula. Equal shares, proportional shares, and need-based shares embody different principles.

Given facts. A relief fund has 50 units for three districts A, B, C. Populations are [10, 20, 30]; need scores are [1, 6, 3]. Three fairness principles are defined: equal = same amount each; population-proportional = shares in ratio 10:20:30; need-proportional = shares in ratio 1:6:3.

Rules. Different fairness principles answer different questions. Apply the principle named in the task; do not mix them unless asked to compare.

Task. Allocate the fund equally.

Step-by-step solution. 1. Equal share=50÷3=16.67. 2. Allocation result: A=16.67, B=16.67, C=16.67.

Answer. A=16.67, B=16.67, C=16.67.

Knowledge takeaway. A sound argument states the fairness rule first and then applies it consistently.

Formal model. Allocation function parameterized by a fairness principle and weights.

Problem 972. Fair allocation by the rule: proportional to population

Knowledge context. Fairness is not one formula. Equal shares, proportional shares, and need-based shares embody different principles.

Given facts. A relief fund has 50 units for three districts A, B, C. Populations are [10, 20, 30]; need scores are [2, 6, 4]. Three fairness principles are defined: equal = same amount each; population-proportional = shares in ratio 10:20:30; need-proportional = shares in ratio 2:6:4.

Rules. Different fairness principles answer different questions. Apply the principle named in the task; do not mix them unless asked to compare.

Task. Allocate in proportion to population.

Step-by-step solution. 1. Total population weight=60. 2. Multiply the fund by each population fraction. 3. Allocation result: A=8.33, B=16.67, C=25.00.

Answer. A=8.33, B=16.67, C=25.00.

Knowledge takeaway. A sound argument states the fairness rule first and then applies it consistently.

Formal model. Allocation function parameterized by a fairness principle and weights.

Problem 973. Fair allocation by the rule: proportional to need

Knowledge context. Fairness is not one formula. Equal shares, proportional shares, and need-based shares embody different principles.

Given facts. A relief fund has 50 units for three districts A, B, C. Populations are [10, 20, 30]; need scores are [3, 6, 5]. Three fairness principles are defined: equal = same amount each; population-proportional = shares in ratio 10:20:30; need-proportional = shares in ratio 3:6:5.

Rules. Different fairness principles answer different questions. Apply the principle named in the task; do not mix them unless asked to compare.

Task. Allocate in proportion to need score.

Step-by-step solution. 1. Total need weight=14. 2. Multiply the fund by each need fraction. 3. Allocation result: A=10.71, B=21.43, C=17.86.

Answer. A=10.71, B=21.43, C=17.86.

Knowledge takeaway. A sound argument states the fairness rule first and then applies it consistently.

Formal model. Allocation function parameterized by a fairness principle and weights.

Problem 974. Fair allocation by the rule: minimum guarantee then equal remainder

Knowledge context. Fairness is not one formula. Equal shares, proportional shares, and need-based shares embody different principles.

Given facts. A relief fund has 50 units for three districts A, B, C. Populations are [10, 20, 30]; need scores are [4, 6, 6]. Three fairness principles are defined: equal = same amount each; population-proportional = shares in ratio 10:20:30; need-proportional = shares in ratio 4:6:6.

Rules. Different fairness principles answer different questions. Apply the principle named in the task; do not mix them unless asked to compare.

Task. Give each district a 5-unit minimum guarantee, then divide the remainder equally.

Step-by-step solution. 1. Minimum guarantees use 15 units. 2. Remainder=50-15=35. 3. Each gets an additional 11.67. 4. Allocation result: A=16.67, B=16.67, C=16.67.

Answer. A=16.67, B=16.67, C=16.67.

Knowledge takeaway. A sound argument states the fairness rule first and then applies it consistently.

Formal model. Allocation function parameterized by a fairness principle and weights.

Problem 975. Fair allocation by the rule: compare methods

Knowledge context. Fairness is not one formula. Equal shares, proportional shares, and need-based shares embody different principles.

Given facts. A relief fund has 50 units for three districts A, B, C. Populations are [10, 20, 30]; need scores are [5, 6, 7]. Three fairness principles are defined: equal = same amount each; population-proportional = shares in ratio 10:20:30; need-proportional = shares in ratio 5:6:7.

Rules. Different fairness principles answer different questions. Apply the principle named in the task; do not mix them unless asked to compare.

Task. Compare equal allocation with need-proportional allocation: which gives more to the highest-need district?

Step-by-step solution. 1. Equal allocation gives C 16.67. 2. Need-proportional allocation gives C 19.44. 3. Compare those amounts. 4. Allocation result: A=13.89, B=16.67, C=19.44.

Answer. A=13.89, B=16.67, C=19.44.

Knowledge takeaway. A sound argument states the fairness rule first and then applies it consistently.

Formal model. Allocation function parameterized by a fairness principle and weights.

N21. Negotiation, coalitions, and agreement

Problem 976. Coalition formation: case 1

Knowledge context. In representative bodies, groups may need to cooperate to reach a decision threshold.

Given facts. Three groups have seats A=8, B=3, C=2. A governing coalition needs at least 7 seats. Any two groups may cooperate.

Rules. A coalition is winning if the sum of its members' seats reaches the threshold. Minimal winning means it is winning but removing any member makes it lose.

Task. Which coalitions are minimal winning coalitions?

Step-by-step solution. 1. Majority threshold=7. 2. Add seats for each possible coalition. 3. Minimal winning coalitions: [('A', 8)].

Answer. A with 8 seats.

Knowledge takeaway. A coalition can be winning without being minimal; removing redundant members tests how necessary each partner is.

Formal model. Weighted coalition game; enumerate subsets, test threshold and minimality.

Problem 977. Coalition formation: case 2

Knowledge context. In representative bodies, groups may need to cooperate to reach a decision threshold.

Given facts. Three groups have seats A=8, B=4, C=2. A governing coalition needs at least 8 seats. Any two groups may cooperate.

Rules. A coalition is winning if the sum of its members' seats reaches the threshold. Minimal winning means it is winning but removing any member makes it lose.

Task. Which coalitions are minimal winning coalitions?

Step-by-step solution. 1. Majority threshold=8. 2. Add seats for each possible coalition. 3. Minimal winning coalitions: [('A', 8)].

Answer. A with 8 seats.

Knowledge takeaway. A coalition can be winning without being minimal; removing redundant members tests how necessary each partner is.

Formal model. Weighted coalition game; enumerate subsets, test threshold and minimality.

Problem 978. Coalition formation: case 3

Knowledge context. In representative bodies, groups may need to cooperate to reach a decision threshold.

Given facts. Three groups have seats A=8, B=5, C=2. A governing coalition needs at least 8 seats. Any two groups may cooperate.

Rules. A coalition is winning if the sum of its members' seats reaches the threshold. Minimal winning means it is winning but removing any member makes it lose.

Task. Which coalitions are minimal winning coalitions?

Step-by-step solution. 1. Majority threshold=8. 2. Add seats for each possible coalition. 3. Minimal winning coalitions: [('A', 8)].

Answer. A with 8 seats.

Knowledge takeaway. A coalition can be winning without being minimal; removing redundant members tests how necessary each partner is.

Formal model. Weighted coalition game; enumerate subsets, test threshold and minimality.

Problem 979. Coalition formation: case 4

Knowledge context. In representative bodies, groups may need to cooperate to reach a decision threshold.

Given facts. Three groups have seats A=8, B=6, C=2. A governing coalition needs at least 9 seats. Any two groups may cooperate.

Rules. A coalition is winning if the sum of its members' seats reaches the threshold. Minimal winning means it is winning but removing any member makes it lose.

Task. Which coalitions are minimal winning coalitions?

Step-by-step solution. 1. Majority threshold=9. 2. Add seats for each possible coalition. 3. Minimal winning coalitions: [('AB', 14), ('AC', 10)].

Answer. AB with 14 seats; AC with 10 seats.

Knowledge takeaway. A coalition can be winning without being minimal; removing redundant members tests how necessary each partner is.

Formal model. Weighted coalition game; enumerate subsets, test threshold and minimality.

Problem 980. Coalition formation: case 5

Knowledge context. In representative bodies, groups may need to cooperate to reach a decision threshold.

Given facts. Three groups have seats A=8, B=7, C=2. A governing coalition needs at least 9 seats. Any two groups may cooperate.

Rules. A coalition is winning if the sum of its members' seats reaches the threshold. Minimal winning means it is winning but removing any member makes it lose.

Task. Which coalitions are minimal winning coalitions?

Step-by-step solution. 1. Majority threshold=9. 2. Add seats for each possible coalition. 3. Minimal winning coalitions: [('AB', 15), ('AC', 10), ('BC', 9)].

Answer. AB with 15 seats; AC with 10 seats; BC with 9 seats.

Knowledge takeaway. A coalition can be winning without being minimal; removing redundant members tests how necessary each partner is.

Formal model. Weighted coalition game; enumerate subsets, test threshold and minimality.

N22. Due process, evidence, and right of reply

Problem 981. Procedure, evidence, and a right to respond: case 1

Knowledge context. Fair procedures often combine evidentiary standards with an opportunity for the affected person to answer the case.

Given facts. A school rule says a student may receive a serious penalty only if: (1) at least two independent pieces of evidence support the allegation, and (2) the student is given a chance to respond before the final decision. In a case, one camera record and one independent witness support the allegation, but the student was not allowed to respond. Cross-domain check: a survey team starts work at 12:00 and works for 2 hour(s) without a break. Mixed-domain verification: a regional archive has 100 map sheets and receives 2 additional sheets during the exercise.

Rules. Independence concerns distinct information sources, not merely the number of people repeating one source. Every required procedural condition must be satisfied. For the cross-domain check, finishing time = starting time + duration. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. May the serious penalty be imposed under the stated procedure? Identify the condition that succeeds or fails. Cross-domain check: at what time does the team finish? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. Camera and witness are independent, so the evidence condition is met. 2. The student was not given a chance to respond. 3. The procedure requires both conditions, so the penalty is not authorized. Cross-domain check: 12+2=14, so the team finishes at 14:00. Mixed-domain verification: 100+2=102.

Answer. No. The right-to-respond condition is not met. Cross-domain answer: 14:00. Mixed-domain answer: 102 map sheets.

Knowledge takeaway. A correct outcome does not excuse skipping a required procedure when the rule explicitly makes procedure part of validity.

Formal model. Conjunctive procedural predicate over evidence independence/count and response opportunity.

Problem 982. Procedure, evidence, and a right to respond: case 2

Knowledge context. Fair procedures often combine evidentiary standards with an opportunity for the affected person to answer the case.

Given facts. A school rule says a serious penalty requires two independent pieces of evidence and a chance to respond. In this case there are two friends repeating the same rumor from one original speaker, plus a chance to respond. Cross-domain check: a local committee has 9 members, requires at least 5 present for quorum, and 8 are present.

Rules. Independence concerns distinct information sources, not merely the number of people repeating one source. Every required procedural condition must be satisfied. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. May the serious penalty be imposed under the stated procedure? Identify the condition that succeeds or fails. Cross-domain check: is quorum met?

Step-by-step solution. 1. The two friends trace to the same original rumor, so they are not two independent pieces under the rule. 2. The response condition is met. 3. Because the evidence condition fails, the serious penalty is not authorized. Cross-domain check: compare 8 with 5; $8 \geq 5$ is true.

Answer. No. The evidence requirement is not met. Cross-domain answer: quorum is met.

Knowledge takeaway. A correct outcome does not excuse skipping a required procedure when the rule explicitly makes procedure part of validity.

Formal model. Conjunctive procedural predicate over evidence independence/count and response opportunity.

Problem 983. Procedure, evidence, and a right to respond: case 3

Knowledge context. Fair procedures often combine evidentiary standards with an opportunity for the affected person to answer the case.

Given facts. A school rule says a student may receive a serious penalty only if: (1) at least two independent pieces of evidence support the allegation, and (2) the student is given a chance to respond before the final decision. In a case, one camera record and one independent witness support the allegation, but the student was not allowed to respond. Cross-domain check: researchers collected 9 reports, but 4 are exact duplicate copies that add no new independent information. Mixed-domain verification: a regional archive has 100 map sheets and receives 4 additional sheets during the exercise.

Rules. Independence concerns distinct information sources, not merely the number of people repeating one source. Every required procedural condition must be satisfied. For the cross-domain check, independent reports = total reports − duplicate copies. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. May the serious penalty be imposed under the stated procedure? Identify the condition that succeeds or fails. Cross-domain check: how many non-duplicate reports remain? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. Camera and witness are independent, so the evidence condition is met. 2. The student was not given a chance to respond. 3. The procedure requires both conditions, so the penalty is not authorized. Cross-domain check: 9−4=5 independent reports remain. Mixed-domain verification: 100+4=104.

Answer. No. The right-to-respond condition is not met. Cross-domain answer: 5 independent reports. Mixed-domain answer: 104 map sheets.

Knowledge takeaway. A correct outcome does not excuse skipping a required procedure when the rule explicitly makes procedure part of validity.

Formal model. Conjunctive procedural predicate over evidence independence/count and response opportunity.

Problem 984. Procedure, evidence, and a right to respond: case 4

Knowledge context. Fair procedures often combine evidentiary standards with an opportunity for the affected person to answer the case.

Given facts. A school rule says a serious penalty requires two independent pieces of evidence and a chance to respond. In this case there are two friends repeating the same rumor from one original speaker, plus a chance to respond. Cross-domain check: on a field sketch, 1 cm represents 6 km and two checkpoints are 6 cm apart.

Rules. Independence concerns distinct information sources, not merely the number of people repeating one source. Every required procedural condition must be satisfied. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. May the serious penalty be imposed under the stated procedure? Identify the condition that succeeds or fails. Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. The two friends trace to the same original rumor, so they are not two independent pieces under the rule. 2. The response condition is met. 3. Because the evidence condition fails, the serious penalty is not authorized. Cross-domain check: 6×6=36 km.

Answer. No. The evidence requirement is not met. Cross-domain answer: 36 km.

Knowledge takeaway. A correct outcome does not excuse skipping a required procedure when the rule explicitly makes procedure part of validity.

Formal model. Conjunctive procedural predicate over evidence independence/count and response opportunity.

Problem 985. Procedure, evidence, and a right to respond: case 5

Knowledge context. Fair procedures often combine evidentiary standards with an opportunity for the affected person to answer the case.

Given facts. A school rule says a student may receive a serious penalty only if: (1) at least two independent pieces of evidence support the allegation, and (2) the student is given a chance to respond before the final decision. In a case, one camera record and one independent witness support the allegation, but the student was not allowed to respond. Cross-domain check: a survey team starts work at 8:00 and works for 2 hour(s) without a break. Mixed-domain verification: a regional archive has 100 map sheets and receives 6 additional sheets during the exercise.

Rules. Independence concerns distinct information sources, not merely the number of people repeating one source. Every required procedural condition must be satisfied. For the cross-domain check, finishing time = starting time + duration. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. May the serious penalty be imposed under the stated procedure? Identify the condition that succeeds or fails. Cross-domain check: at what time does the team finish? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. Camera and witness are independent, so the evidence condition is met. 2. The student was not given a chance to respond. 3. The procedure requires both conditions, so the penalty is not authorized. Cross-domain check: 8+2=10, so the team finishes at 10:00. Mixed-domain verification: 100+6=106.

Answer. No. The right-to-respond condition is not met. Cross-domain answer: 10:00. Mixed-domain answer: 106 map sheets.

Knowledge takeaway. A correct outcome does not excuse skipping a required procedure when the rule explicitly makes procedure part of validity.

Formal model. Conjunctive procedural predicate over evidence independence/count and response opportunity.

N23. Samples, surveys, and public opinion

Problem 986. Sampling and opinion: case 1

Knowledge context. Surveys use samples to learn about larger populations. The selection process matters as much as the number of answers.

Given facts. A town wants to estimate how many residents use buses. It surveys only people waiting at the bus station. Cross-domain check: a local committee has 13 members, requires at least 7 present for quorum, and 11 are present.

Rules. A sample is more representative when selection does not systematically favor people connected to the answer. Random or broad systematic sampling can reduce selection bias; voluntary or topic-specific locations may create bias. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. Is the sampling method likely to be biased for the stated question? Explain the direction of the possible bias. Cross-domain check: is quorum met?

Step-by-step solution. 1. Identify the target population. 2. Compare the selection method with that target. 3. People already at the bus station are more likely than average to use buses. 4. Therefore the method is likely biased. Cross-domain check: compare 11 with 7; $11 \geq 7$ is true.

Answer. Likely biased. People already at the bus station are more likely than average to use buses. Cross-domain answer: quorum is met.

Knowledge takeaway. A large biased sample can still give a misleading estimate; representativeness is a reasoning question about selection.

Formal model. Sampling-frame analysis: compare inclusion mechanism with target population and likely outcome-related selection.

Problem 987. Sampling and opinion: case 2

Knowledge context. Surveys use samples to learn about larger populations. The selection process matters as much as the number of answers.

Given facts. A school wants opinions on lunch. It randomly selects 10 students from each grade. Cross-domain check: researchers collected 13 reports, but 4 are exact duplicate copies that add no new independent information. Mixed-domain verification: a regional archive has 100 map sheets and receives 8 additional sheets during the exercise.

Rules. A sample is more representative when selection does not systematically favor people connected to the answer. Random or broad systematic sampling can reduce selection bias; voluntary or topic-specific locations may create bias. For the cross-domain check, independent reports = total reports − duplicate copies. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. Is the sampling method likely to be biased for the stated question? Explain the direction of the possible bias. Cross-domain check: how many non-duplicate reports remain? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. Identify the target population. 2. Compare the selection method with that target. 3. Random selection within every grade gives broad grade coverage; no obvious topic-specific selection is stated. 4. Therefore the method is not obviously biased for the stated target, within the limits given. Cross-domain check: 13−4=9 independent reports remain. Mixed-domain verification: 100+8=108.

Answer. Not obviously biased under the stated design. Random selection within every grade gives broad grade coverage; no obvious topic-specific selection is stated. Cross-domain answer: 9 independent reports. Mixed-domain answer: 108 map sheets.

Knowledge takeaway. A large biased sample can still give a misleading estimate; representativeness is a reasoning question about selection.

Formal model. Sampling-frame analysis: compare inclusion mechanism with target population and likely outcome-related selection.

Problem 988. Sampling and opinion: case 3

Knowledge context. Surveys use samples to learn about larger populations. The selection process matters as much as the number of answers.

Given facts. A village asks only members of the sports club whether to build a sports field. Cross-domain check: on a field sketch, 1 cm represents 5 km and two checkpoints are 3 cm apart.

Rules. A sample is more representative when selection does not systematically favor people connected to the answer. Random or broad systematic sampling can reduce selection bias; voluntary or topic-specific locations may create bias. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Is the sampling method likely to be biased for the stated question? Explain the direction of the possible bias. Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. Identify the target population. 2. Compare the selection method with that target. 3. Sports-club members are likely to be more favorable to a sports field than the whole village. 4. Therefore the method is likely biased. Cross-domain check: 3×5=15 km.

Answer. Likely biased. Sports-club members are likely to be more favorable to a sports field than the whole village. Cross-domain answer: 15 km.

Knowledge takeaway. A large biased sample can still give a misleading estimate; representativeness is a reasoning question about selection.

Formal model. Sampling-frame analysis: compare inclusion mechanism with target population and likely outcome-related selection.

Problem 989. Sampling and opinion: case 4

Knowledge context. Surveys use samples to learn about larger populations. The selection process matters as much as the number of answers.

Given facts. A library surveys every 20th visitor entering over a full week. Cross-domain check: a survey team starts work at 12:00 and works for 2 hour(s) without a break.

Rules. A sample is more representative when selection does not systematically favor people connected to the answer. Random or broad systematic sampling can reduce selection bias; voluntary or topic-specific locations may create bias. For the cross-domain check, finishing time = starting time + duration.

Task. Is the sampling method likely to be biased for the stated question? Explain the direction of the possible bias. Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. Identify the target population. 2. Compare the selection method with that target. 3. Sampling every 20th visitor across a full week is systematic within the visitor population; it may represent visitors reasonably, though not non-visitors. 4. Therefore the method is not obviously biased for the stated target, within the limits given. Cross-domain check: 12+2=14, so the team finishes at 14:00.

Answer. Not obviously biased under the stated design. Sampling every 20th visitor across a full week is systematic within the visitor population; it may represent visitors reasonably, though not non-visitors. Cross-domain answer: 14:00.

Knowledge takeaway. A large biased sample can still give a misleading estimate; representativeness is a reasoning question about selection.

Formal model. Sampling-frame analysis: compare inclusion mechanism with target population and likely outcome-related selection.

Problem 990. Sampling and opinion: case 5

Knowledge context. Surveys use samples to learn about larger populations. The selection process matters as much as the number of answers.

Given facts. An online poll is open to anyone who chooses to click it. Cross-domain check: a local committee has 8 members, requires at least 5 present for quorum, and 8 are present.

Rules. A sample is more representative when selection does not systematically favor people connected to the answer. Random or broad systematic sampling can reduce selection bias; voluntary or topic-specific locations may create bias. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. Is the sampling method likely to be biased for the stated question? Explain the direction of the possible bias. Cross-domain check: is quorum met?

Step-by-step solution. 1. Identify the target population. 2. Compare the selection method with that target. 3. People who choose to click may differ systematically from those who ignore the poll. 4. Therefore the method is likely biased. Cross-domain check: compare 8 with 5; $8 \geq 5$ is true.

Answer. Likely biased. People who choose to click may differ systematically from those who ignore the poll. Cross-domain answer: quorum is met.

Knowledge takeaway. A large biased sample can still give a misleading estimate; representativeness is a reasoning question about selection.

Formal model. Sampling-frame analysis: compare inclusion mechanism with target population and likely outcome-related selection.

N24. Matching, scheduling, and coverage

Problem 991. Matching people to tasks: case 1

Knowledge context. Scheduling and assignment problems connect people, tasks, times, and constraints.

Given facts. Four students must each receive one different task. Allowed assignments: Ava: ['Map', 'Timeline']; Ben: ['Budget', 'Survey']; Cara: ['Map', 'Survey']; Dion: ['Budget', 'Timeline']. Cross-domain check: researchers collected 7 reports, but 4 are exact duplicate copies that add no new independent information. Mixed-domain verification: a regional archive has 101 map sheets and receives 2 additional sheets during the exercise.

Rules. Each student gets exactly one allowed task, and each task is used exactly once. A complete assignment must satisfy both conditions. For the cross-domain check, independent reports = total reports − duplicate copies. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. Find one complete valid assignment. If none exists, explain why. Cross-domain check: how many non-duplicate reports remain? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. Start with the most restricted student if one exists. 2. Assign an allowed task, then remove that task from the remaining choices. 3. One complete assignment is {'Ava': 'Map', 'Ben': 'Budget', 'Cara': 'Survey', 'Dion': 'Timeline'}. Cross-domain check: 7−4=3 independent reports remain. Mixed-domain verification: 101+2=103.

Answer. Valid assignment: Ava→Map, Ben→Budget, Cara→Survey, Dion→Timeline. Cross-domain answer: 3 independent reports. Mixed-domain answer: 103 map sheets.

Knowledge takeaway. A local choice can block a later assignment, so the whole matching must be checked.

Formal model. Bipartite matching / all-different constraint satisfaction.

Problem 992. Matching people to tasks: case 2

Knowledge context. Scheduling and assignment problems connect people, tasks, times, and constraints.

Given facts. Four students must each receive one different task. Allowed assignments: Ava: ['Map', 'Timeline']; Ben: ['Budget', 'Survey']; Cara: ['Survey']; Dion: ['Budget', 'Timeline']. Cross-domain check: on a field sketch, 1 cm represents 4 km and two checkpoints are 7 cm apart.

Rules. Each student gets exactly one allowed task, and each task is used exactly once. A complete assignment must satisfy both conditions. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Find one complete valid assignment. If none exists, explain why. Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. Start with the most restricted student if one exists. 2. Assign an allowed task, then remove that task from the remaining choices. 3. One complete assignment is {'Ava': 'Map', 'Ben': 'Budget', 'Cara': 'Survey', 'Dion': 'Timeline'}. Cross-domain check: 7×4=28 km.

Answer. Valid assignment: Ava→Map, Ben→Budget, Cara→Survey, Dion→Timeline. Cross-domain answer: 28 km.

Knowledge takeaway. A local choice can block a later assignment, so the whole matching must be checked.

Formal model. Bipartite matching / all-different constraint satisfaction.

Problem 993. Matching people to tasks: case 3

Knowledge context. Scheduling and assignment problems connect people, tasks, times, and constraints.

Given facts. Four students must each receive one different task. Allowed assignments: Ava: ['Map', 'Timeline']; Ben: ['Budget', 'Survey']; Cara: ['Map', 'Survey']; Dion: ['Budget']. Cross-domain check: a survey team starts work at 8:00 and works for 2 hour(s) without a break.

Rules. Each student gets exactly one allowed task, and each task is used exactly once. A complete assignment must satisfy both conditions. For the cross-domain check, finishing time = starting time + duration.

Task. Find one complete valid assignment. If none exists, explain why. Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. Start with the most restricted student if one exists. 2. Assign an allowed task, then remove that task from the remaining choices. 3. One complete assignment is {'Ava': 'Timeline', 'Ben': 'Survey', 'Cara': 'Map', 'Dion': 'Budget'}. Cross-domain check: 8+2=10, so the team finishes at 10:00.

Answer. Valid assignment: Ava→Timeline, Ben→Survey, Cara→Map, Dion→Budget. Cross-domain answer: 10:00.

Knowledge takeaway. A local choice can block a later assignment, so the whole matching must be checked.

Formal model. Bipartite matching / all-different constraint satisfaction.

Problem 994. Matching people to tasks: case 4

Knowledge context. Scheduling and assignment problems connect people, tasks, times, and constraints.

Given facts. Four students must each receive one different task. Allowed assignments: Ava: ['Map']; Ben: ['Budget', 'Survey']; Cara: ['Map', 'Survey']; Dion: ['Budget', 'Timeline']. Cross-domain check: a local committee has 12 members, requires at least 7 present for quorum, and 11 are present.

Rules. Each student gets exactly one allowed task, and each task is used exactly once. A complete assignment must satisfy both conditions. For the cross-domain check, quorum exists when present members ≥ the stated threshold.

Task. Find one complete valid assignment. If none exists, explain why. Cross-domain check: is quorum met?

Step-by-step solution. 1. Start with the most restricted student if one exists. 2. Assign an allowed task, then remove that task from the remaining choices. 3. One complete assignment is {'Ava': 'Map', 'Ben': 'Budget', 'Cara': 'Survey', 'Dion': 'Timeline'}. Cross-domain check: compare 11 with 7; 11 ≥ 7 is true.

Answer. Valid assignment: Ava→Map, Ben→Budget, Cara→Survey, Dion→Timeline. Cross-domain answer: quorum is met.

Knowledge takeaway. A local choice can block a later assignment, so the whole matching must be checked.

Formal model. Bipartite matching / all-different constraint satisfaction.

Problem 995. Matching people to tasks: case 5

Knowledge context. Scheduling and assignment problems connect people, tasks, times, and constraints.

Given facts. Four students must each receive one different task. Allowed assignments: Ava: ['Map', 'Timeline']; Ben: ['Budget', 'Survey']; Cara: ['Map', 'Survey']; Dion: ['Budget', 'Timeline']. Cross-domain check: researchers collected 11 reports, but 4 are exact duplicate copies that add no new independent information. Mixed-domain verification: a regional archive has 101 map sheets and receives 6 additional sheets during the exercise.

Rules. Each student gets exactly one allowed task, and each task is used exactly once. A complete assignment must satisfy both conditions. For the cross-domain check, independent reports = total reports − duplicate copies. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. Find one complete valid assignment. If none exists, explain why. Cross-domain check: how many non-duplicate reports remain? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. Start with the most restricted student if one exists. 2. Assign an allowed task, then remove that task from the remaining choices. 3. One complete assignment is {'Ava': 'Map', 'Ben': 'Budget', 'Cara': 'Survey', 'Dion': 'Timeline'}. Cross-domain check: 11−4=7 independent reports remain. Mixed-domain verification: 101+6=107.

Answer. Valid assignment: Ava→Map, Ben→Budget, Cara→Survey, Dion→Timeline. Cross-domain answer: 7 independent reports. Mixed-domain answer: 107 map sheets.

Knowledge takeaway. A local choice can block a later assignment, so the whole matching must be checked.

Formal model. Bipartite matching / all-different constraint satisfaction.

N25. Meta-reasoning: robustness, information, causality

Problem 996. Meta-reasoning: case 1

Knowledge context. Good reasoning includes checking how informative, robust, falsifiable, causal, or dominated a conclusion is.

Given facts. Four equally likely locations A,B,C,D remain possible. Question Q1 asks “Is it A?”; Q2 asks “Is it A or B?” A yes/no question is more balanced when its two answer branches contain more nearly equal numbers of possibilities. Cross-domain check: on a field sketch, 1 cm represents 3 km and two checkpoints are 4 cm apart.

Rules. Use branch sizes as the information criterion in this exercise. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Which question is more balanced and therefore more informative by the stated criterion? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. Q1 splits the four possibilities 1 versus 3. 2. Q2 splits them 2 versus 2. 3. The 2–2 split is more balanced. Cross-domain check: 4×3=12 km.

Answer. Q2. Cross-domain answer: 12 km.

Knowledge takeaway. A result can be numerically correct yet epistemically weak; meta-reasoning evaluates the quality of the inference itself.

Formal model. Meta-level predicate over hypotheses or decisions: information balance, robustness, counterexample, causality, or dominance.

Problem 997. Meta-reasoning: case 2

Knowledge context. Good reasoning includes checking how informative, robust, falsifiable, causal, or dominated a conclusion is.

Given facts. Option A scores between 49 and 53 after measurement error; option B scores between 50 and 52. A ranking is robust only if every allowed value of A remains above every allowed value of B. Cross-domain check: a survey team starts work at 12:00 and works for 2 hour(s) without a break.

Rules. Test worst cases: robust A>B requires min(A)>max(B). For the cross-domain check, finishing time = starting time + duration.

Task. Is the ranking A>B robust? Cross-domain check: at what time does the team finish?

Step-by-step solution. 1. min(A)=49; max(B)=52. 2. 49>52 is false. 3. Some allowed values reverse the ranking. Cross-domain check: 12+2=14, so the team finishes at 14:00.

Answer. No; the ranking is sensitive to the uncertainty. Cross-domain answer: 14:00.

Knowledge takeaway. A result can be numerically correct yet epistemically weak; meta-reasoning evaluates the quality of the inference itself.

Formal model. Meta-level predicate over hypotheses or decisions: information balance, robustness, counterexample, causality, or dominance.

Problem 998. Meta-reasoning: case 3

Knowledge context. Good reasoning includes checking how informative, robust, falsifiable, causal, or dominated a conclusion is.

Given facts. Claim: “Every town with more roads has shorter average travel time.” Town X has 6 roads and average travel time 35 minutes. Town Y has 4 roads and average travel time 20 minutes. Cross-domain check: a local committee has 16 members, requires at least 9 present for quorum, and 14 are present.

Rules. A universal claim is disproved by one case that satisfies its condition but not its conclusion. For the cross-domain check, quorum exists when present members $\geq$ the stated threshold.

Task. Do X and Y provide a counterexample to the claim? Cross-domain check: is quorum met?

Step-by-step solution. 1. X has more roads than Y. 2. Yet X has longer, not shorter, travel time. 3. Therefore the universal claim fails on this pair. Cross-domain check: compare 14 with 9; $14 \geq 9$ is true.

Answer. Yes. X and Y form a counterexample. Cross-domain answer: quorum is met.

Knowledge takeaway. A result can be numerically correct yet epistemically weak; meta-reasoning evaluates the quality of the inference itself.

Formal model. Meta-level predicate over hypotheses or decisions: information balance, robustness, counterexample, causality, or dominance.

Problem 999. Meta-reasoning: case 4

Knowledge context. Good reasoning includes checking how informative, robust, falsifiable, causal, or dominated a conclusion is.

Given facts. Across several months, ice-cream sales and sunburn cases rise together. Temperature also rises in those months. No intervention is described. Cross-domain check: researchers collected 15 reports, but 4 are exact duplicate copies that add no new independent information. Mixed-domain verification: a regional archive has 101 map sheets and receives 10 additional sheets during the exercise.

Rules. Correlation between X and Y does not by itself prove X causes Y; a third variable may influence both. For the cross-domain check, independent reports = total reports − duplicate copies. For this verification, the final number of sheets equals starting sheets plus received sheets.

Task. Can we conclude that ice-cream sales cause sunburn? Cross-domain check: how many non-duplicate reports remain? Mixed-domain verification: how many sheets are available after the delivery?

Step-by-step solution. 1. Sales and sunburn are correlated. 2. Temperature is a plausible common factor stated in the data. 3. Without an intervention or stronger causal evidence, direct causation is not established. Cross-domain check: 15−4=11 independent reports remain. Mixed-domain verification: 101+10=111.

Answer. No. The association alone does not establish that ice-cream sales cause sunburn. Cross-domain answer: 11 independent reports. Mixed-domain answer: 111 map sheets.

Knowledge takeaway. A result can be numerically correct yet epistemically weak; meta-reasoning evaluates the quality of the inference itself.

Formal model. Meta-level predicate over hypotheses or decisions: information balance, robustness, counterexample, causality, or dominance.

Problem 1000. Meta-reasoning: case 5

Knowledge context. Good reasoning includes checking how informative, robust, falsifiable, causal, or dominated a conclusion is.

Given facts. Two transport plans are compared. A costs 8, takes 20 minutes, and has safety score 9. B costs 10, takes 25 minutes, and has safety score 9. Lower cost/time and higher safety are better. Cross-domain check: on a field sketch, 1 cm represents 2 km and two checkpoints are 8 cm apart.

Rules. A Pareto-dominates B if A is at least as good on every stated criterion and strictly better on at least one. For the cross-domain check, real distance = sketch distance × kilometres per centimetre.

Task. Does one plan dominate the other? Cross-domain check: what real distance separates the checkpoints?

Step-by-step solution. 1. A is cheaper than B. 2. A is faster than B. 3. Safety is equal. 4. A is no worse anywhere and better on two criteria. Cross-domain check: 8×2=16 km.

Answer. A Pareto-dominates B. Cross-domain answer: 16 km.

Knowledge takeaway. A result can be numerically correct yet epistemically weak; meta-reasoning evaluates the quality of the inference itself.

Formal model. Meta-level predicate over hypotheses or decisions: information balance, robustness, counterexample, causality, or dominance.

Appendix A — Reasoning Skills Trained

• transitivity and relation composition
• classification by necessary properties
• constraint satisfaction
• shortest path and optimization
• flow and bottleneck reasoning
• chronology and partial orders
• evidence evaluation and source comparison
• necessary/sufficient conditions
• threshold and voting logic
• set reasoning
• interval reasoning
• conservation and balance equations
• counterfactual intervention
• uncertainty and robustness
• sampling and bias
• matching and assignment
• fairness principles
• coalition formation
• correlation versus causation
• counterexample and Pareto dominance

Appendix B — Knowledge Fixed Through Use

• cardinal directions; maps; routes; rivers; borders; scale; coordinates; relief; climate; population; settlements; ecosystems; natural risk; renewable resources
• chronology; centuries; historical sources; provenance; archaeological layers; continuity and change; anachronism; causation and counterfactual reasoning
• rules and exceptions; rights; majority and quorum; representation; institutions; budgets; public goods; common resources; fairness; negotiation; due process; surveys and public opinion

Appendix C — How to Turn a Problem into Code

1. Identify entities. List places, people, events, institutions, objects, resources, or options.
2. Encode relations. Use edges, sets, coordinates, Boolean predicates, intervals, costs, capacities, or parent links.
3. Encode rules. Translate “if–then,” thresholds, constraints, and conservation equations into explicit operations.
4. Ask the query. Specify exactly what must be found: reachability, minimum cost, feasibility, ranking, allocation, or justified conclusion.
5. Compute and verify. Run the reasoning procedure and check that every conclusion follows from the stated data only.

Conceptual Index

Geography: orientation, routes, rivers, borders, maps, settlements, relief, climate, trade flows, regions, scale, coordinates, Earth–Sun relations, stocks and flows, density, land use, natural risk, ecosystems, renewable resources, migration. History: chronology, centuries, evidence, conflicting sources, causation, travel constraints, change and continuity, stratigraphy, intervals, genealogy, anachronism, provenance, perspective, uncertainty, counterfactuals. Civics and politics: rules, rights, voting, quorum, representation, institutions, public budgets, collective choice, public goods, commons, fairness, coalitions, due process, sampling and public opinion. Abstract reasoning: graphs, sets, constraints, optimization, thresholds, partial orders, intervals, robustness, counterexamples, causality, dominance.